

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART E PLUMBING SERVICES

(INCLUDING SOLID WASTE MANAGEMENT)

SECTION 1 WATER SUPPLY

BUREAU OF INDIAN STANDARDS

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National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part E/Section 1) covers the requirements of water supply in buildings. The water supply provisions covered in this Section encompass the requirements of water supply, plumbing connected to public water supply, design of water supply systems, principles of conveyance and distribution of water within the premises, hot water supply system, inspection and maintenance of water supply systems. It also covers design of water supply systems in high altitudes and/or sub-zero temperature regions.

In the first version of the Code formulated in 1970, as part of the National Building Code of India, three separate sections of this Part on 'Plumbing Services' were brought out, namely, Section 1 'Water Supply', Section 2 'Drainage and Sanitation', and Section 3 'Gas Supply'. These sections were subsequently revised in 1983.

The major changes incorporated in the first revision in 1983 Section 1 Water Supply, were: Rationalization of definitions and addition of definitions for more terms; addition of universal pipe friction diagram and nomogram of Hazen and Willam's equation for discharge computation, deleting the discharge curves based on Chezy's formula; introduction of a detailed clause giving guidance on the design of water supply system for multi-storeyed buildings; modification to indicate that no separate storage need be provided for flushing and domestic purposes for health reasons and a single storage tank may be provided; modification/amplification of provisions relating to domestic hot water supply installations; introduction of a detailed clause covering recommendations to be considered while planning and designing water supply systems peculiar to high altitude and/or sub-zero temperature regions of the country; and addition of requirements relating to inspection, testing and maintenance applicable to hot water supply system.

As a result of experience gained in implementation of 1983 version of the Code and feedback received as well as revision of some of the standards based on which this Section was formulated, a need to revise this Section was felt then. The second revision of 2005 was therefore formulated to take care of these. In the revision, the erstwhile two sections were merged and a combined and comprehensive section, namely Section 1 Water supply, drainage and sanitation (including solid waste management), was brought out. Gas supply was brought out as Section 2. Following significant changes were incorporated in the second revision of Section 1 on water supply, drainage and sanitation, in respect to water supply: modification of provision of water supply requirement; addition of a new clause on water supply for other than residential purposes; addition of a new clause on quality of water, also including therein a sub-clause on waste water reclamation; modification of the provision regarding storage of water and introduction of guidelines for calculating storage capacity; modification, in the design of distribution system provisions for discharge computation, to include designed consumer pipes based on fixtures unit also taking into account probable simultaneous demand instead of earlier computation based on Reynold's Number; introduction of an alternative option of variable speed drive pumping system to hydro pneumatic system; addition of a new clause on backflow prevention; addition of provision for suitability of galvanized mild steel tanks on the basis of *pH* of the water; detailing of types of hot water heater; and restructuring of the section was done to make it more user friendly.

The second revision in 2005 also incorporated for the first time the provisions on solid waste management.

In the third revision in 2016, to address the various and distinct features related to the plumbing aspects and solid waste management comprehensively, this Part on 'Plumbing Services' was rearranged as follows:

Section 1 Water supply

Section 2 Drainage and sanitation

Section 3 Solid waste management

Section 4 Gas supply

The revision of Section 1 in 2016 incorporated several modifications; provisions on water supply requirements for buildings were related to the estimated occupancy in the chosen type of building; [Table 1](#) on water requirements for buildings other than residences was updated concerning building types and included domestic and flushing requirements separately; water demand for landscaping purposes was included; requirements related

to wastewater treatment and the usage of recycled wastewater were included; materials used for storage water tanks were updated; provisions regarding the quantity of water to be stored were revised; measures for protecting the water supply were included to prevent cross-connection between treated and raw water supplies, ensure backflow prevention, and secure identification of non-potable water supply pipes; the list of materials used for manufacturing pipes was updated; general requirements for the design of distribution systems were introduced; [Table 2](#) on fixture units for different types of fixtures was updated; [Table 3](#) on probable simultaneous demand was revised with demand values covering various intermediate and up to 10 000 fixture units, along with graphs for probable demand; maximum flow rate and flush volumes were specified; provisions concerning inadequate and excessive water pressure were introduced; guidance on water hammer effects was included; distribution system provisions for multi-storeyed buildings were updated with illustrations; provisions related to zoning of distribution systems were included; hot water supply systems were incorporated under the existing clause on hot water supply installations; detailed provisions for swimming pools were included; guidance on allowance for expansion of water pipes was provided; a new clause on colour coding for different types of water pipes was added; provisions were reviewed and updated to ensure accessibility for the elderly and persons with disabilities, duly giving cross-references to **13** of Part 3 'Development Control Rules and General Building Requirements'; certain terminologies were introduced and some were revised; and cross-references to Indian Standards were updated.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

In this revision of the Section 1, the following significant changes have been incorporated:

- a) Population requirements for residential buildings have been updated, in [4.1](#);
- b) Population requirement is introduced for warehouses and industrial occupancies, to arrive at their water demand;
- c) Suggestive breakup of water for general consumption is included, in [4.1.1.1](#);
- d) Table 1 'Water requirements for buildings other than residences' now provides the quantity for various categories of hotels;
- e) Storage requirement for hydro-pneumatic/pressurized water supply system have been included, in [4.4.10](#); and
- f) Terminology has been updated by including drinking water fountain, design flood elevation, demand recirculation water system, vacuum breaker, etc.

This Section is largely based on the following Indian Standards:

IS 1172 : 1993	Code of basic requirements for water supply, drainage and sanitation (<i>fourth revision</i>)
IS 2065 : 1983	Code of practice for water supply in buildings (<i>second revision</i>)
IS 6295 : 1986	Code of practice for water supply and drainage in high altitudes and or sub-zero temperature regions (<i>first revision</i>)
IS 7558 : 1974	Code of practice for domestic hot water installations
IS 12183 (Part 1) : 1987	Code of practice for plumbing in multi-storeyed buildings: Part 1 Water supply

A reference to SP 35 : 1987 'Handbook on water supply and drainage' may be useful, from where also, assistance has been derived.

All standards, whether given herein above or cross-referred to in the main text of this Section, are subject to revision. The parties to agreement based on this Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the formulation of this Section, reference has also been made to the following:

International Plumbing Code 2024, International Code Council, and

Uniform Plumbing Code 2024, International Association of Plumbing and Mechanical Officials

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Section is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Section.

Important Explanatory Note for Users of this NBCS

In any Part/Section of this NBCS, where reference is made to ‘**good practice**’ in relation to **design, constructional procedures or other related information**, and where reference is made to “**accepted standard**” in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Sec shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

a) Good practice [E-1(1)] refers to the Indian Standard(s) given at serial number (1) of the list of standards given at the end of this Part E/Section 1, that is, IS 10446 : 1983 ‘Glossary of terms relating to water supply and sanitation’.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART E PLUMBING SERVICES (INCLUDING SOLID WASTE MANAGEMENT)

SECTION 1 WATER SUPPLY

1 SCOPE

1.1 This Section covers the basic requirements of water supply for residential, business and other types of buildings, including traffic terminal stations. This Section also deals with general requirements of plumbing connected to public water supply and design of water supply systems along with general guidelines about expansion in piping systems, and swimming pools.

1.1.1 This Section does not take into consideration the requirements of water supply for industrial plants and processes, which have to be provided for separately. It also does not provide the requirements of water supply for other purposes, such as firefighting and street cleaning.

2 TERMINOLOGY

For the purpose of this Section, the following definitions shall apply in addition to the definitions given in accepted standard [E-1(1)].

2.1 Definitions Relating to Water Supply

2.1.1 *Access Panel* — A removable panel mounted in a frame, normally secured with screws and mounted in a wall or ceiling, to provide access to concealed valves or items which may require maintenance.

2.1.2 *Air Gap, Water Distribution* — Unobstructed vertical distance through the free atmosphere between the lowest opening from any pipe or faucet conveying water to a tank or plumbing fixture and flood level rim of the receptacle.

2.1.3 *Air Valve* — A valve that releases air from a pipeline automatically without loss of water or introduce air into a line automatically if the internal pressure becomes less than that of the atmosphere.

2.1.4 *Aspirator* — A device supplied with water or other fluid under positive pressure that passes through an integral orifice causing a vacuum. Aspirator is also referred as suction apparatus.

2.1.5 *Authority Having Jurisdiction* — The authority which has been created by a statute and which for the purpose of administering this NBCS/Part may authorize a committee or an official to act on its behalf; hereinafter called the 'Authority'.

2.1.6 *Available Head* — The head of water available at the point of consideration due to mains' pressure or overhead tank or any other source of pressure.

2.1.7 *Anti-siphon* — A device or mechanism to prevent siphonage.

2.1.8 *Backflow*

- a) the flow of water or other liquids, mixtures or substances into the distributing pipes of a system of supply of potable water from any source or sources other than its intended source; and
- b) the flow of a liquid in a direction reverse of that intended.

2.1.9 *Backflow Prevention Device* — Any approved measure or fitting or combination of fittings specifically designed to prevent backflow or back siphonage in a water service.

2.1.10 *Back Pressure Back Flow* — Due to an increased pressure above the supply pressure, which may be due to pumps, other equipment, gravity or other source of pressure.

2.1.11 Back Siphonage — The flowing back of used, contaminated, or polluted water from a plumbing fixture or vessel into a water supply due to a reduced pressure in such pipe (see [2.1.8](#)).

2.1.12 Barrel — This portion of a pipe in which the diameter and wall thickness remain uniform throughout.

2.1.13 Base — The lowest portion or lowest point of a stack of vertical pipe.

2.1.14 Bath Room Group — Group of fixtures consisting of water closet, lavatory, bath tub or shower and other fittings with a floor drain located together.

2.1.15 Bedding — The material on which the pipe is laid and which provides support for the pipe. Bedding can be concrete, granular material or the prepared trench bottom.

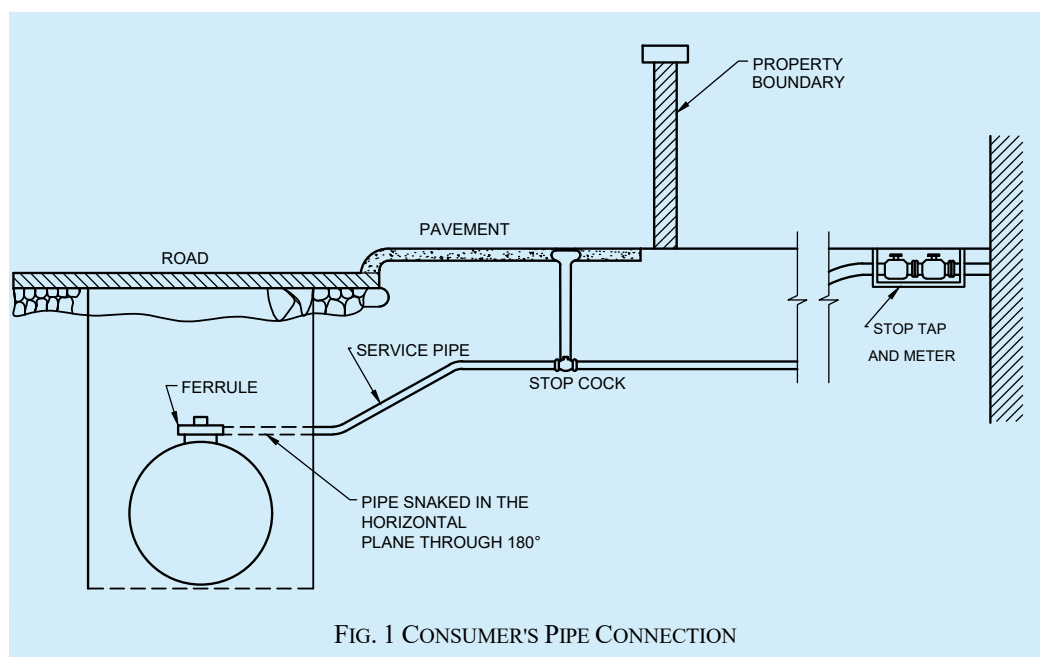
2.1.16 Chair — A bed of concrete or other suitable material on the trench floor to provide a support for the pipes at intervals.

2.1.17 Channel — The open waterway through which sewage, storm water or other liquid wastes flow at the invert of a manhole or an inspection chamber.

2.1.18 Communication Pipe — That part of a service pipe which vests in the water undertakes. It starts at the water main and terminate at a point which differs according to the circumstances of the case.

2.1.19 Consumer — Any person who uses or is supplied water or on whose application such water is supplied by the Authority.

2.1.20 Consumer's Pipe (Service Pipe) — The pipe taking up from the street main originating from the ferrule leading up to the water meter for supply of water to consumer. The portion of pipe used for service connection is not the property of the Authority (see [Fig. 1](#)).



2.1.21 Cover

- a) a removable plate for permitting access to a pipe, fitting, vessel or appliance; and
- b) the vertical distance between the top of the barrel of a buried pipe or other construction and the surface of the ground.

2.1.22 Cross-Connection — Any physical connection or arrangement between the separate piping systems, one of which contains potable water and the other contains water of unknown or questionable safety, whereby water may

flow from one system to the other, the direction of flow depending on the pressure differential between the two systems.

2.1.23 Circulating Hot Water System — A specifically designed hot water distribution system where one or more pumps are operated in the service hot water piping to circulate heated water from the water heating equipment to fixture supply and back to the water heating equipment

2.1.24 Direct Tap — A tap which is connected to a supply pipe and is subject to pressure from the water main.

2.1.25 Demand Recirculation Water System — A water distribution system within the built space where one or more pumps prime the service hot water piping with heated water upon a demand for hot water. Typically, this can be achieved by a timer or sensor.

2.1.26 Design Flood Elevation — The elevation of the ‘design flood’, including wave height, relative to the datum specified on the authorities legally designated flood hazard map. In areas designated as zone, the design flood elevation shall be the elevation of the highest existing grade of the building’s perimeter plus the depth specified in the flood hazard map. In case the depth is not specified in the map the depth shall be taken as 610 mm.

2.1.27 Drinking Water Fountain — A plumbing fixture that is connected to the potable water distribution system and the drainage system. The fixture allows the user to obtain the potable water directly from a stream of flowing water without the use of any accessories.

2.1.28 Down Take Tap — A tap taken off from a storage tank situated at a higher elevation in the building.

2.1.29 Effective Opening — The minimum cross-sectional area at the point of water supply, measured or expressed in terms of,

- a) the diameter of a circle; and
- b) the diameter of a circle of equivalent cross-sectional area, if the opening is not circular.

2.1.30 Feed Cistern — A storage vessel used for supplying cold water to a hot water apparatus, cylinder or tanks.

2.1.31 Fittings — The appurtenances such as coupling, flange, branch, bend, tees, elbows, unions, waste (with or without plug), P or S trap (with or without vent), stop ferrule, bib tap, pillar tap, bath faucet, water meter, garden hydrant, valves and any other article used in connection with water supply, drainage and sanitation.

2.1.32 Fixture Unit — A quantity in terms of which the load producing effects on the plumbing system of different kinds of plumbing fixtures is expressed on some arbitrarily chosen scale.

2.1.33 Float Operated Valve — Ball valves or ball taps and equilibrium valves operated by means of a float.

2.1.34 Flushing Cistern — A cistern provided with a device for rapidly discharging the contained water and used in connection with a sanitary appliance for the purpose of cleaning the appliance and carrying away its contents into a drain.

NOTE — The nominal size of a cistern is the quantity of water discharged per flush.

2.1.35 Flushometer Tank — A tank integrated with an air accumulator vessel that is designed to discharge a predetermined quantity of water to fixtures for flushing purposes.

2.1.36 Flow Pressure — The pressure in the water supply pipe measured near the faucet or water outlet while the faucet or outlet is open or in use.

2.1.37 Formation — The finished level of the excavation at the bottom of a trench or heading prepared to receive the permanent work.

2.1.38 Frost Line — The line joining the points of greatest depths below ground level up to which the moisture in the soil freezes. Reference may be made to good practice [E-1(2)] for more details in this regard.

2.1.39 General Washing Place — A washing place provided with necessary sanitary arrangement and common to more than one tenement.

2.1.40 Geyser — An apparatus for heating water with supply control on the inlet side and delivering it from an outlet.

2.1.41 Gridded Water Distribution System — A water distribution system where every water distribution pipe is interconnected to provide two or more path to each fixture supply line.

2.1.42 Haunching — Outward sloping concrete support to the sides of a pipe or channel above the concrete bedding.

2.1.43 Heel Rest Bend or Duck-Foot Bend — A bend, having a foot formed integrally in its base, used to receive a vertical pipe.

2.1.44 High Altitudes — Elevations higher than 1 500 m above mean sea level (MSL).

2.1.45 Highway Authority — The public body in which is vested, or which is the owner of, a highway repairable by the inhabitants collectively; otherwise the body or persons responsible for the upkeep of the highway.

2.1.46 Horizontal Pipe — Any pipe of fitting which makes an angle of more than 45° with the vertical.

2.1.47 Hot Water Tank — A vessel for storing hot water under pressure greater than atmospheric pressure.

2.1.48 Hydro-pneumatic System — The water supply pumping system which operates automatically in conjunction with pump(s), pressure vessel and pressure switch.

2.1.49 Junction Pipe — A pipe incorporating one or more branches.

2.1.50 Lagging — Thermal insulation on pipes.

2.1.51 Licensed (or Registered) Plumber — A person licensed (or registered) under the provisions of this NBCS.

2.1.52 Offset — A pipe fitting used to connect two pipes whose axes are parallel but not in line.

2.1.53 Period of Supply — The period of the day or night during which water supply is made available to the consumer.

2.1.54 Pipe Work — Any installation of piping with its fittings.

2.1.55 Plumbing

- a) the pipes, fixtures and other apparatus inside a building for bringing in the water supply and removing the liquid and water borne wastes; and
- b) the installation of the foregoing pipes, fixtures and other apparatus.

2.1.56 Plumbing System — The plumbing system shall include the water supply and distribution pipes; plumbing fittings and traps; soil, waste, vent pipes and anti-siphonage pipes; building drains and building sewers including their respective connections, devices and appurtenances within the property lines of the premises; and water-treating or water-using equipment.

2.1.57 Potable Water — Water which is satisfactory for drinking, culinary and domestic purposes and meets the requirements of the Authority.

2.1.58 Premises — Premises shall include passages, buildings and lands of any tenure, whether open or enclosed, whether built on or not, and whether public or private in respect of which a water rate or charge is payable to the Authority or for which an application is made for supply of water.

2.1.59 Pressure Balancing Valve — Mixing valve that senses incoming hot and cold water pressures and compensates for fluctuations.

2.1.60 Residual Head — The head available at any particular point in the distribution system.

2.1.61 Residual Pressure — The pressure available at the fixture after allowance is made for pressure drop due to friction loss and head in the system during maximum demand periods.

2.1.62 Saddle — A purpose made fitting, so shaped as to fit over a hole cut in a sewer or drain used to form connections.

2.1.63 Static Pressure — The pressure exerted by a fluid that is not moving or flowing.

2.1.64 Stop-Cock — A cock fitted in a pipe line for controlling the flow of water.

2.1.65 Stop Tap — Stop tap includes stop-cock, stop valve or any other device for stopping the flow of water in a line or system of pipes at will.

2.1.66 Storage Tank — A container used for storage of water which is connected to the water main or tube-well by means of supply pipe.

2.1.67 Studio Apartment — An apartment unit consisting of a single room and a bathroom, the single room functioning as living room, bedroom and kitchen.

2.1.68 Subsoil Water — Water occurring naturally in the subsoil.

2.1.69 Subsoil Water Drain

- a) a drain intended to collect and carry away subsoil water; and
- b) a drain intended to disperse into the subsoil from a septic tank.

2.1.70 Sub-zero Temperature Regions — Regions where temperatures fall below 0 °C and freezing conditions occur.

2.1.71 Supply Pipe — The water pipe that is owned by the owner of the building and lies within the boundary of the property. It connects via the communication pipe to the water main.

2.1.72 Supports — Hangers and anchors or devices for supporting and securing pipe and fittings to walls, ceilings, floors or structural members.

2.1.73 Surface Water — Natural water from the ground surface, paved areas and roofs.

2.1.74 Surface Water Drain — A drain conveying surface water including storm water.

2.1.75 Thermostatic/Pressure Balancing Valve — Mixing valve that senses outlet temperature and incoming hot and cold water pressure and compensates for fluctuations for stabilization.

2.1.76 Vertical Pipe — Any pipe or fitting which is installed in a vertical position or which makes an angle or not more than 45° with the vertical.

2.1.77 Vacuum Breaker — Type of back flow preventer installed on openings subject to normal atmospheric pressure that prevents back flow by admitting atmospheric pressure through ports to the discharge side of device.

2.1.78 Warning Pipe — An overflow pipe so fixed that its outlet, whether inside or outside a building, is in a conspicuous position where the discharge of any water there from can be readily seen.

2.1.79 Wash-Out Valve — A device located at the bottom of the tank for the purpose of draining a tank for cleaning, maintenance, etc.

2.1.80 Water Hammer Arrestor — A device designed to provide protection against hydraulic shock in the building water supply system.

2.1.81 Water Main (Street Main) — A pipe laid by the water undertakers for the purpose of giving a general supply of water as distinct from a supply to individual consumers and includes any apparatus used in connection with such a pipe.

2.1.82 Water Outlet — A water outlet, as used in connection with the water distributing system, is the discharge opening for the water: (a) to a fitting; (b) to atmospheric pressure (except into an open tank which is part of the water supply system); and (c) to any water-operated device or equipment requiring water to operate.

2.1.83 Water Supply System — Water supply system of a building or premises consists of the water service pipe, the water distribution pipes, and the necessary connecting pipes, fittings, control valves, and all appurtenances in or adjacent to the building or premises.

2.1.84 Waterworks — Waterworks for public water supply include a lake, river, spring, well, pump with or without motor and accessories, reservoir, cistern, tank, duct whether covered or open, sluice, water main, pipe, culvert, engine and any machinery, land, building or a thing used for storage, treatment and supply of water.

2.1.85 Wholesome Water — Water that is safe and suitable for drinking, cooking, and other domestic purposes and free from harmful microorganisms and chemical contaminants, meets regulatory quality standards, and has a pleasant taste.

2.2 Definitions Relating to Swimming Pool

2.2.1 Appurtenance — An accessory facility or feature at a swimming pool, such as a diving board, slide, wading pool, plunge pool, spray pool.

2.2.2 Bather Load — The maximum number of persons that may be allowed in the pool area at one time without creating undue health or safety hazards.

2.2.3 Deep Area — An area of a swimming pool in which the water depth exceeds five feet.

2.2.4 Diving Pool — A pool designed and intended for use exclusively for diving.

2.2.5 Inlet — An opening or fitting through which filtered water enters the pool.

2.2.6 Main Drain — The outlet or outlets for drain provided in the pool.

2.2.7 Make-up Water — The water added to a pool to replace that which is lost.

2.2.8 Pool — A swimming pool or other recreational water basin utilized in conjunction with or as an appurtenance to a swimming pool.

NOTE — The term does not refer to spas and therapy pools not designed or intended for swimming or to basins for individual use that are drained after each use.

2.2.9 Pool Depth — The vertical distance between the pool floor and the water level.

2.2.10 Shallow Area — An area in a swimming pool, in which the water depth does not exceed 1.5 m at any point.

2.2.11 Skimmer — A mechanical device connected to the recirculation piping which is used to skim the pool surface.

2.2.12 Swimming Pool User Load — The number of bathers using the pool in a 24 h period for which a certain per capita capacity is often considered.

NOTE — The capacity of a swimming pool in terms of its user load is dictated by the capacity of water treatment equipment.

2.2.13 Transition Point — A location in a shallow area of a swimming pool where an area, having a floor slope of no more than 300 mm vertical in 3.65 m horizontal, adjoins an area where the floor slope exceeds 1 in 12.

2.2.14 Turnover Period — The time required to recirculate a volume of water equivalent to the water volume of

the pool through the filtration system.

2.2.15 Wading Pool — A pool having a maximum water depth not exceeding 76 cm.

2.2.16 Water Level — The level of the overflow lip of a perimeter overflow system or the mid-level of surge weirs, if present, or the mid-level of the skimmer operating range.

2.2.17 Zero-Depth Edge — That portion of the perimeter of a zero-depth pool where the pool floor intersects the pool water surface.

2.2.18 Zero-Depth Pool — A swimming pool where the pool floor intersects the water surface along a portion of its perimeter.

3 GENERAL

3.1 Basic Principles

The design of water supply takes the following into consideration:

- a) number of occupants;
- b) minimum water requirements for different purposes;
- c) treatment of water based on the quality of water;
- d) quantity of water stored; and
- e) sizing of pipes.

The basic principles of water supply, drainage and sanitation are given below, and the design of water supply should in general be guided by the applicable principles.

3.1.1 Potable Water

All premises intended for human habitation, occupancy, or use shall be provided with supply of potable water. This water supply shall not be connected with unsafe water resources, nor shall it be subject to the hazards of backflow.

3.1.2 Water Provision

Plumbing fixtures, devices and appurtenances shall be provided with water in sufficient volume and at pressures adequate to enable them to function properly and without undue noise under normal conditions of use.

There should be at least a residual head of 0.018 N/mm² (0.18 bar) at the consumer's tap. There may be certain fixtures or appliances in the installation that may require a higher pressure, such as 0.05 N/mm² (0.5 bar) or even higher pressure [such as 0.1 N/mm² (1.0 bar) in case of flush valves], in which case the system shall be designed using pumps, tanks or both to achieve the required minimum pressure.

NOTE — The residual head shall be taken at the highest/farthest outlets in the building.

3.1.3 Water Efficiency

Plumbing system shall be designed, installed and adjusted to use the optimum quantity of water consistent with proper performance and cleaning.

3.1.4 Safety Devices

Plumbing system shall be designed and installed with safety devices to safeguard against dangers from contamination, explosion, overheating, etc.

3.1.5 Minimum Water Amenities

It is recommended that each dwelling unit should have at least one water closet, one lavatory, one kitchen wash place or a sink, and one bathing place or shower to meet the basic requirements of sanitation and personal hygiene.

In case of a group housing, the requirements relating to toilet or sanitary room and kitchen as given in good practice [E-1(3)] shall also be complied with.

3.1.6 Drainage System

The drainage system shall be designed, installed and maintained to guard against fouling, deposit of solids and clogging and with adequate cleanouts so arranged that the pipes may be readily cleaned.

3.1.7 Materials and Workmanship

The plumbing system shall have durable material, free from defective workmanship and so designed and installed as to give satisfactory service for its reasonable expected life. The accessories of the plumbing system should be of such specifications as to meet the functional requirements of the installation, so as to also avoid any inconsistency leading to leakage and resultant seepage.

3.1.8 Fixture Traps and Ventilating Pipes

Each fixture directly connected to the drainage system shall be equipped with a liquid seal trap. Trap seals shall be maintained to prevent sewer gas, other potentially dangerous or noxious fumes, or vermin from entering the building. Further, the drainage system shall be designed to provide an adequate circulation of air in all pipes with no danger of siphonage, aspiration, or forcing of trap seals under conditions of ordinary use by providing ventilating pipes throughout the system.

3.1.9 Foul Air Exhaust

Each vent terminal shall extend to the outer air and be so installed as to minimize the possibilities of clogging and the return of foul air to the building, as it conveys potentially noxious or explosive gases to the outside atmosphere. All ventilating pipes shall be provided with a cowl.

3.1.10 Testing

The plumbing system shall be subjected to required tests to effectively disclose all leaks and defects in the work or the material.

3.1.11 Exclusion from Plumbing System

No substance that will clog or accentuate clogging of pipes, produce explosive mixtures, destroy the pipes or their joints, or interfere unduly with the sewage-disposal process shall be allowed to enter the drainage system.

3.1.12 Light and Ventilation

Wherever water closet or similar fixture is located in a room or compartment, it should be properly lighted and ventilated.

3.1.13 Individual Sewage Disposal Systems

If water closets or other plumbing fixtures are installed in buildings where connection to public sewer is not possible, suitable provision shall be made for acceptable treatment and disposal.

3.1.14 Maintenance

Plumbing systems shall be maintained in a safe and serviceable condition.

3.1.15 Approach for Use and Cleaning

All plumbing fixtures shall be so installed with regard to spacing as to be approachable for their intended use and for cleaning. All doors, windows and any other device needing access within the toilet shall be so located that they have proper approach.

3.1.16 Accessibility for Persons with Disabilities

All doors, windows and fixtures, including WC, urinals, grab bars, washbasin, mirror and all other accessories for use by persons with disabilities shall be so installed/located that they have proper access with appropriate width, height, space, centrelines, and ease of operation {see good practice [E-1(3)]}.

3.1.17 Structural Safety

Plumbing system shall be installed with due regard to preservation of the structural members and prevention of damage to walls and other surfaces.

3.1.18 Protection of Ground and Surface Water

Sewage or other waste shall not be discharged into surface or sub-surface water without acceptable form of treatment.

3.2 Water Supply Connection

3.2.1 Application for Obtaining Supply Connection

Every consumer, requiring a new supply of water or any extension or alteration to the existing supply shall apply in writing in the prescribed form (see [Annex A](#)) to the Authority.

3.2.2 Bulk Supply

In the case of large housing colonies or campuses, or where new services are so situated that it will be necessary for the Authority to lay new mains or extend an existing main, full information about the proposed scheme shall be furnished to the Authority. Information shall also be given regarding their phased requirements of water supply with full justification. Such information shall include site plans, showing the layout of roads, footpaths, building and boundaries and indicating thereon the finished line and level of the roads or footpaths and water supply lines and appurtenances.

3.2.3 Completion Certificate

On completion of the plumbing work for the water supply system, the licensed/registered plumber shall give a completion certificate in the prescribed form (see [Annex B](#)) to the Authority for getting the water connection from the mains.

3.3 Licensing/Registration of Plumbers

3.3.1 Execution of Work

The work which is required to be carried out under the provisions of this Section, shall be executed only by a licensed/registered plumber under the control of the Authority and shall be responsible to carry out all lawful directions given by the Authority. No individual shall engage in the business of plumbing unless so licensed/registered under the provisions of this Section.

3.3.1.1 No individual, firm, partnership or corporation shall engage in the business of installing, repairing or altering plumbing unless the plumbing work performed in the course of such business is under the direct supervision of a licensed/registered plumber.

3.3.2 Examination and Certification

The Authority shall establish standards and procedure for the qualification, examination and licensing/registration

of plumbers and shall issue licences to such persons who meet the qualifications thereof and successfully pass the examination.

3.3.3 For guidelines for registration of plumbers including the minimum standards for qualifications for the grant of licences/registration, reference may be made to good practice [E-1(4)]. The Authority may also utilize the services of the certified plumbers who are certified for the required skill level under the appropriate scheme of the Government.

4 WATER SUPPLY

4.1 Water Supply Requirements for Buildings

The total quantity of water per day is estimated based on the proposed occupancy and activities catered. Designer has to identify all the possible sources for augmenting the shortfall in water supply. The analysis of available water is done to decide the treatment for consumption and treatment process depends on the quality of water and the purpose for which it is used.

Projection of population for each building shall be made on the basis of its usage. Population for each type of building shall be estimated on the basis of information obtained from the users. Alternatively, population may be worked on the following basis, for different type of buildings:

a) Residential buildings

<i>Sl No.</i>	<i>Accommodation</i>	<i>Population Requirements</i>
(1)	(2)	(3)
i)	1 bedroom dwelling unit	3
ii)	2 bedroom dwelling unit	4
iii)	3 bedroom dwelling unit	5
iv)	4 bedroom dwelling unit and above	6

NOTES

1 The above figures consider a domestic household including support personnel, wherever applicable.

2 For plotted development, the population may be arrived at after due consideration of the expected number and type of domestic household units.

3 Dwelling unit under EWS category shall have population requirement of 4 and studio apartment shall have population requirement of 2.

b) Other than residential buildings

<i>Sl No.</i>	<i>Occupancy</i>	<i>Population Requirement</i>
(1)	(2)	(3)
i)	Offices	1 person per 10 m ² of floor area (<i>see</i> Note 1)
ii)	Schools	Strength of school + teaching and non-teaching staff
iii)	Hostels	Number of beds + 4.5 × (warden's residence) + staff
iv)	Hotels	Number of beds + staff + requirement of restaurant seats
v)	Hospitals	Number of beds + staff + patient attendants (generally population density per bed in secondary care hospital is 5, tertiary care is 7 and quaternary care is 9)
vi)	Mercantile	1 person per 3 m ² of street floor and sales basement areas + 1 person per 6 m ² of upper sale floors (Total population may be segregated into 10 percent for fixed and 90 percent for floating/visitors)

<i>Sl No.</i>	<i>Occupancy</i>	<i>Population Requirement</i>
(1)	(2)	(3)
vii)	Traffic terminal stations	Average number of users per day (Total annual passenger traffic/365) + Staff + Vendors
viii)	Ware house	1 person per 20 m ² of floor area (<i>see</i> Note 1)
ix)	Industrial/Hazardous	1 person per 10 m ² of floor area
x)	Storage	1 person per 30 m ² of floor area
xi)	Assembly (<i>maximum should be as per Part F 'Fire and Life Safety' of this NBCS</i>).	
	a) Concentrated use without fixed seating	1 person per 0.65 m ² of floor area
	b) Less concentrated use without fixed seating	1 person per 1.2 m ² of floor area
	c) Fixed seating	1.2 × Number of fixed seats
	d) Dining area and restaurants with seating and tables	1 person per 1.8 m ² of floor area
NOTES		
1 Wherever there are multiple work shifts, the number of users within a 24 h period may be considered as per actuals.		
2 Population of 5 percent to 15 percent, depending on the usage of building, shall be considered for visitors and floating population likely to use the building's facilities.		

4.1.1 Water Supply for Residences

A minimum of 70 litre to 100 litre per head per day may be considered adequate for domestic and flushing needs of urban communities (which varies based on type of building occupancy). As a general rule the following rates per capita per day may be considered for domestic and non-domestic needs:

- For urban communities: 135 LPCD
- For urban communities using water efficient fixtures in buildings: 100 LPCD (*see* Note 1)
- For rural communities, water demand of 70 LPCD may be considered including 15 LPCD as flushing water demand.

NOTES

1 For buildings using water efficient fixtures, reference may be made to accepted standard [E-1(5)].

2 Usage of high flow or high pressure fixtures is not recommended from the point of view of water conservation and sustainable development.

4.1.1.1 Out of the per capita demand as mentioned in [4.1.1 \(a\)](#) and [4.1.1 \(b\)](#), 30 litre per head per day shall be taken for flushing requirements and the remaining quantity for other domestic purposes. The following depicts the general breakdown of the water consumption in residential buildings:

<i>Sl No.</i>	<i>Description</i>	<i>For Urban Communities (Residential)</i>	<i>For Urban Communities Using Water Efficient Fixtures (Residential)</i>
(1)	(2)	(3)	(4)
i)	Drinking	5	5
ii)	Bathing (<i>Suggested cold and hot water is 60:40</i>)	35	25
iii)	Cooking	10	5
iv)	Washing utensils	20	15

<i>Sl No.</i>	<i>Description</i>	<i>For Urban Communities (Residential)</i>	<i>For Urban Communities Using Water Efficient Fixtures (Residential)</i>
(1)	(2)	(3)	(4)
v)	Floor cleaning	10	5
vi)	Washing clothes	25	15
	Total	105	70

4.1.2 Water Supply for Buildings Other than Residences

Minimum requirements for water supply for buildings other than residences shall be in accordance with [Table 1](#).

The water demand for the laboratory facilities will depend on actual requirements based on functional point of view.

Table 1 Water Requirements for Buildings Other than Residences
(Clause [4.1.2](#) and [4.1.3](#))

Sl No.	Type of Building	Domestic per Day	Flushing per Day	Total Consumption per Day
(1)	(2)	litre (3)	litre (4)	litre (5)
i)	Factories including canteen where bathrooms are required to be provided	30 per head	15 per head	45 per head
ii)	Factories including canteen where no bathrooms are required to be provided	20 per head	10 per head	30 per head
iii)	Hospital (excluding laundry and kitchen) (see Note 2):			
	a) Number of beds not exceeding 100	a) 230 per bed	a) 110 per bed	a) 340 per bed
	b) Number of beds exceeding 100	b) 300 per bed	b) 150 per bed	b) 450 per bed
	c) Outpatient department (OPD)	c) 10 per bed	c) 5 per bed	c) 15 per bed
	d) Hospital with secondary care			d) 800 per bed
	e) Super speciality hospital			e) 950 per bed
iv)	Nurses' homes and medical quarters	90 per head	45 per head	135 per head
v)	Hostels	90 per head	45 per head	135 per head
vi)	Hotel (up to 3 star) excluding laundry, kitchen, staff and water bodies without bathtub (see Note 7)	120 per head	60 per head	180 per head
vii)	Hotel (4 Star and above) excluding laundry, kitchen, staff and water bodies and excluding bathtub (see Note 7)	260 per head	60 per head	320 per head
viii)	Hotel with bathtub (see Note 7)	320 per head	60 per head	380 per head
ix)	Banquet area	30 per head	15 per head	45 per head
x)	Health club			
	a) with shower	25 per head	20 per head	45 per head
	b) without shower	15 per head	10 per head	25 per head

Table 1 Water Requirements for Buildings Other than Residences
(Clause 4.1.2 and 4.1.3)

Sl No.	Type of Building	Domestic per Day	Flushing per Day	Total Consumption per Day
(1)	(2)	litre (3)	litre (4)	litre (5)
xi)	Indoor Sports Facility	10 per head	5 per head	15 per head
xii)	Offices (including canteen)	25 per head	20 per head	45 per head
xiii)	Restaurants and food court including water requirement for kitchen:			
	a) Restaurants	55 per seat	15 per seat	70 per seat
	b) Food court	25 per seat	10 per seat	35 per seat
xiv)	Clubhouse	25 per head	20 per head	45 per head
xv)	Cinemas, concert halls and theatres and multiplex	5 per seat	10 per seat	15 per seat
xvi)	Schools/Educational institutions:			
	a) Without boarding facilities	a) 25 per head	a) 20 per head	a) 45 per head
	b) With boarding facilities	b) 90 per head	b) 45 per head	b) 135 per head
xvii)	Shopping and retail (mall)			
	a) Staff	25 per head	20 per head	45 per head
	b) Visitors	5 per head	10 per head	15 per head
xviii)	Traffic terminal stations (see Notes 3 and 4)			
	a) Airports	40 per head	30 per head	70 per head
	b) Railway stations (Junctions) with bathing facility	40 per head	30 per head	70 per head
	c) Railway stations (Junctions) without bathing facility	30 per head	15 per head	45 per head
	d) Railway Stations (Intermediate) with bathing facility	25 per head	20 per head	45 per head
	e) Railway Stations (Intermediate) without bathing facility	15 per head	10 per head	25 per head
	f) Interstate bus terminals	25 per head	20 per head	45 per head
	g) Intrastate Bus Terminals/Metro Stations/ High speed rail stations	10 per head	5 per head	15 per head
NOTES 1 For calculating water demand for visitors, consumption of 15 litre per head per day may be taken. 2 The water demand includes requirements of patients, attendants, visitors and staff. Additional water demand for kitchen, laundry and clinical water shall be computed as per actual requirements. 3 The number of persons shall be determined by the average number of passengers handled by stations, with due consideration given to the staff and vendors who are using these facilities. 4 Consideration should be given to seasonal average peak requirements. 5 The hospitals may be categorized as Category A (25 to 50 beds), Category B (51 to 100 beds), Category C (101 to 300 beds), Category D (301 to 500 beds) and Category E (501 to 750 beds). 6 Total water demand can be reduced by 25 percent if the building is equipped with 100 percent 3-star rated fixtures and fittings as per good practice [E-1(5)]. 7 Bathtub water volume may be considered additionally based on size				

4.1.3 Water Supply Requirements of Traffic Terminal Stations

The water supply requirements of traffic terminal stations (railway stations, bus stations, harbours, airports, etc) include provisions for waiting rooms and waiting halls. They do not, however, include requirements for retiring rooms. Requirements of water supply for traffic terminal stations shall be as per [Table 1](#).

4.1.4 Water Supply for Firefighting Purposes

4.1.4.1 Provisions shall be made to meet the water supply requirements for firefighting in the city/area, depending on the population density and types of occupancy. *See also* Part F 'Fire and Life Safety' of this NBCS.

4.1.4.2 Provisions shall be made for water supply requirements for firefighting purposes within the building, depending upon the height and occupancy of the building. *See also* Part F 'Fire and Life Safety' of this NBCS.

4.1.4.3 Provisions shall be made regarding water supply in storage tanks, capacity of fire pumps, arrangements of wet riser-cum-down comer and wet riser installations for buildings, depending upon the occupancy use and other factors as applicable. *See also* Part F 'Fire and Life Safety' of this NBCS.

4.1.5 Water Supply for Other Purposes

4.1.5.1 Water supply in many buildings is also required for many other applications other than domestic use, which shall be identified in the initial stages of planning so as to provide the requisite water quantity, storage capacity and pressure as required for each application. In such instances information about the water use and the quality required may be obtained from the users. Some typical uses other than domestic use and firefighting purposes are air conditioning, swimming pools and water bodies, and gardening. Treated water from sewage treatment plant, with suitable tertiary treatment, should be used for flushing purpose (with dual piping system), gardening purpose, cooling tower make up, and/or for other non-potable usage.

4.1.5.2 The water demand for landscaping purposes is generally taken as 6 litre/m²/day to 8 litre/m²/day for lawns. For shrubs and trees the above value can be reduced considerably.

4.2 Water Sources and Quality

4.2.1 Sources of Water

The origin of all sources of water is rainfall. Water can be collected as it falls as rain before it reaches the ground; or as surface water when it flows over the ground or is pooled in lakes or ponds; or as ground water when it percolates into the ground and flows or collects as ground water; or from the sea.

Contamination of water supplies can occur in the source water as well as in the distribution system after water treatment has already occurred. There are many sources of water contamination, including naturally occurring chemicals and minerals (for example, arsenic, radon, uranium), local land use practices (fertilizers, pesticides, concentrated animal feeding operations), manufacturing processes, and sewer overflows or wastewater releases. The presence of contaminants in water can lead to adverse health effects, including gastrointestinal illness, reproductive problems, and neurological disorders.

4.2.2 The water supplied shall be free from pathogenic organisms, clear, free from undesirable taste and odour, neither corrosive nor scale forming and free from minerals which could produce undesirable physiological effects. The quality of water to be used for drinking shall be as per accepted standard [E-1(6)].

4.2.3 For purposes other than drinking, water if supplied separately, shall be absolutely safe from bacteriological contamination so as to ensure that there is no danger to the health of the users due to such contaminants.

For purposes other than drinking, where there is an overall risk of legionella growth, it is advisable that for cold water supplies, the temperature does not exceed 20 °C and a minimum temperature of 50 °C for hot water supplies be maintained at all points of network so as to ensure that it is absolutely safe from bacteriological contamination and there is no danger to the health of the users due to such contaminants.

4.2.4 Waste Water Reclamation

Treated sewage or other waste water of the community may be utilized for non-domestic purposes such as water for flushing, landscape irrigation, cooling towers of HVAC system, in fountains and recreational lakes where swimming is not allowed, and for certain industrial purposes after its necessary treatment to suit the nature of the use. This supply system shall be allowed in residences only if proper provision is made to avoid any cross-connection of this treated waste water with domestic water supply system for which it is recommended to use dual piping system.

4.2.4.1 Treatment of waste water and usage of recycled water

Waste water is generated by residential and other establishments like institutional, business, mercantile and industrial. It includes household waste liquid from toilets, baths, showers, kitchens and sinks that is disposed of *via* sewers. Waste water treatment is the process of removing contaminants from wastewater, including household sewage and runoff (effluents). It includes physical, chemical, and biological processes to remove contaminants. Treatment of waste water and usage of recycled waste water may be done to make it usable for appropriate applications. The objective is to produce an environmentally safe fluid waste stream (or treated effluent) and a solid waste (or treated sludge) suitable for disposal or reuse.

Separation of household waste into grey water and black water (and draining of black water into sewerage system) is becoming more common with grey water being permitted to be used for watering plants or recycled for flushing toilets after proper treatment. Waste water collection and treatment is typically subject to statutory regulations. Treatment depends on the characteristics of influent and the treatment requirements that are needed for treating the same. Waste water treatment generally involves the following three stages:

- a) *Primary treatment* — It consists of temporarily holding the wastewater for settlement of heavy solids at the bottom while oil, grease and lighter solids float to the surface;
- b) *Secondary treatment* — It removes dissolved and suspended biological matter; and
- c) *Tertiary treatment* — It is more intensive treatment done in order to allow rejection into a highly sensitive or fragile ecosystem. The tertiary treatment is generally followed by disinfection.

4.2.5 Water Conservation, Water Balance and Use of Recycled Water

Water conservation encompasses the policies, strategies and activities to manage fresh water as a sustainable resource, to protect the water environment and to meet current and future demand. Population, household size, and growth and affluence all affect the quantity of water used. Water balance studies should be carried out to study the availability of water from different sources and its usage for different purposes.

4.2.6 Whenever a building is used after long intervals, the water quality of the stored water shall be checked so as to ensure that the water is safe for use as per water quality requirements specified in this NBCS.

4.3 Estimate of Demand Load

4.3.1 Estimates of total water supply requirements for buildings shall be based on the occupant load consistent with the provisions of [4.1](#).

4.3.2 In making assessment of water supply requirements of large complexes, the future occupant load shall be kept in view. The following methods may be used for estimating future requirements:

- a) demographic method of population projection;
- b) arithmetic progression method;
- c) geometrical progression method;
- d) method of varying increment or incremental increase;
- e) logistic method;
- f) graphical projection method; or
- g) graphical comparison method.

4.4 Storage of Water

4.4.1 In a building, provision is required to be made for storage of water for the following reasons:

- a) to provide against interruptions of the supply caused by repairs to mains, etc;
- b) to reduce the maximum rate of demand on the mains;
- c) to tide over periods of intermittent supply; and
- d) to maintain storage for the firefighting requirement of the building (taking into consideration the water for sprinklers and hydrants separately, apart from domestic water requirements).

4.4.2 The water may be stored in overhead tanks (OHT) and/or underground tanks (UGT).

4.4.3 Materials Used

Reservoirs and tanks for the storage of water shall be constructed of reinforced concrete, brick masonry, ferrocement, mild steel, stainless steel, or plastic or glass reinforced panels.

4.4.3.1 Tanks made of steel may be of welded, riveted or panel/pressed construction. The metal shall be galvanized or coated externally with a good quality anti-corrosive weather-resisting paint. Lead-based paint shall not be used in the tank. Lead-lined tanks shall not be used. Rectangular pressed steel tanks shall conform to good practice [E-1(7)].

4.4.4 Each tank shall be provided with the following:

- a) *Manholes* — Adequate number of manholes for access and repair. The manholes shall be made of corrosion resistant material (for example, cast iron, reinforced cement concrete, steel fibre reinforced concrete, galvanized steel, high density polyethylene, fibre glass reinforced plastic or such other materials) acceptable to the Authority. Manholes shall be provided with locking arrangement to avoid misuse and tampering;
- b) *Ladders* — Tanks higher than 900 mm deep shall be provided with corrosion resistant ladders according to the depth to enable a person to reach the bottom of the tank;
- c) *Overflow pipe* — Each tank shall be provided with an overflow pipe terminating above the ground/terrace level to act as a 'Warning Pipe' to indicate overflow conditions. The size of the overflow pipe shall be adequate to accept the flow. Normally the overflow pipe size shall be one size higher than the inlet pipe. When the inlet pipe diameter is large, two or more overflow pipes of equivalent cross-section may be provided;
- d) *Vent pipes* — Tanks larger than 5 000 litre capacity shall be provided with vent pipes to prevent development pressure in the tank which might result in 'NO FLOW' condition or inward collapse of the tank;
- e) *Scour pipe* — Each tank shall be provided with a scour pipe with an accessible valve for emptying the tank
- f) *Connection of overflow and scour pipe* — Under no circumstances tank overflow and scour pipe shall be connected to any drain, gully trap or manhole to prevent back flow and contamination of the water. All such connections shall be discharged over a grating with an air gap of 50 mm. All overflow and vent pipes shall be provided with a mosquito proof brass grating to prevent ingress of mosquito, vermin and other insects;
- g) The top slab of the tank shall be suitably sloped away from its centre for proper drainage of the rainwater; and
- h) Tanks on terraces and above ground shall be supported by appropriate structural members so as to transfer the load of the tank and the water directly on the structural members of the building.

4.4.5 Every storage tank shall be easily accessible and placed in such a position as to enable thorough inspection and cleaning to be carried out. If the storage capacity required is more than 5 000 litre, it is advantageous to arrange it in a series of tanks so interconnected that each tank can be isolated for cleaning and inspection without

interfering with the supply of water. In large storage tanks, the outlet shall be at the end opposite the inlet to avoid stagnation of the water.

4.4.6 The outlet pipe shall be fixed 50 mm to 75 mm above the bottom of the tank and fitted with a strainer, preferably of brass.

4.4.7 In the case of underground storage tanks, the design of the tank shall be such as to provide for the draining of the tank when necessary and water shall not be allowed to collect around the tank. The tank shall be perfectly water-proof and shall be provided with a cement concrete cover, having a manhole opening, with a properly fitting hinged cast iron cover on a leak-proof cast iron frame.

The underground tanks should not be located in low lying areas or near any public or private sewer, septic tank, leaching pool or soakage pit to prevent any contamination. The overflow of the tank should be well above (preferably 600 mm) the external surface level and terminate as a warning pipe with a mosquito proof grating. Care shall be taken to prevent backflow of local surface water into the tank in case of local flooding. Otherwise the overflow shall be terminated in a safer manner as per the site conditions. For tanks with at least one side exposed to a basement, it is safer to discharge the overflow into the basement level.

The tank top slab shall also be designed to carry the load due to fire tender movement where anticipated as in the case of an extended basement.

There should be no common wall between the tanks storing safe water and tanks storing water from unsafe sources.

4.4.8 In case of overhead tanks, bottom of the tanks shall be placed clear off the terrace slab such that the elevation difference between the outlet pipe of the tank and the highest fixture at the top floor of the building is minimum 2 m, which shall also prevent leakage into the structural slab. In tall buildings, the top of the tank shall be provided with the safe ladder or staircase. The top slab shall be provided with railing or a parapet wall.

4.4.9 For jointing steel pipe to a storage tank, the end of the pipe shall be screwed, passed through a hole in the tank and secured by backnuts, both inside and outside. The pipe end shall be flush with the face of the inside backnut. For jointing copper pipe to steel or copper tank, a connector of non-ferrous material shall be used. The connector shall have a shoulder to bear on the outside of the tank and shall be secured by a backnut inside.

4.4.10 The quantity of water to be stored shall be calculated taking into account the following factors:

- a) hours of supply at sufficiently high pressure to fill up the overhead storage tanks;
- b) frequency of replenishment of overhead tanks, during the 24 h;
- c) rate and regularity of supply; and
- d) consequences of exhausting storage particularly in case of public buildings like hospitals.

In case of intermittent water supply, the following guidelines should be adopted for calculation of capacity:

- 1) when the supply is from main OHT, such as in case of plotted development, the capacity of individual OHT may be taken as half a day demand;
- 2) in case of all the other buildings (apartments, hotels, hospitals, and commercial), the capacity of UGT and OHT may be taken as one and a half days and half a day demand. Wherever raw water and treated water are stored in separate UGTs, the combined storage capacity shall be of one and a half day's demand; and
- 3) in case of sewage treatment plant, for treated water storage in respect of flushing, one-day flushing demand shall be stored in UGT or treated effluent storage tank, which is part of the plant, and half a day demand shall be stored in OHT.

NOTES

Minimum requirements for calculation of capacity of these storage tanks are as follows:

- 1 In case only OHT is provided, it may be taken as 33.33 to 50 percent of one day's requirement.

- 2 In case of combined storage, that is UGT and OHT with gravity water supply system is provided, it may be taken as 150 percent of one day's requirement for UGT and 50 percent of one day's requirement for OHT.
- 3 In the case of hydro-pneumatic/pressurized water supply system from UGT, the storage of water in UGT to be considered as 200 percent of one day's requirement.

For additional requirement of water storage for firefighting purposes, reference may be made to Part F 'Fire and Life Safety' of this NBCS.

4.4.11 Where the water supply distribution system is catering for separate potable water and flushing water supply, and where reclaimed or recycled treated water is being used within the building, it shall be considered as dual water supply system. In such cases, storages for flushing/reclaimed/recycled water shall be separated proportionately. Adequate measures and precautions shall be taken for physical segregation for sanitary purposes and to avoid cross-contamination in the distribution system and to prevent back contamination of water supply sources (see [4.5](#)).

4.4.12 When only one communication pipe is provided for water supply to a building, it is not necessary to have separate storage for flushing and sanitary purposes for health reasons. In such cases when only one storage tank has been provided, tapping of water may be done at two different levels (the lower tapping for flushing) so that a part of the water will be exclusively available for flushing purposes.

4.5 Protection of Water Supply

4.5.1 General

The water supply system shall be designed, installed and maintained in such a manner so as to prevent contamination from non-potable liquids, solids or gases being introduced into the potable water supply system through cross-connections or any other connection to the system.

4.5.2 Plumbing Fixtures

The water supply lines and fittings for plumbing fixtures shall be installed so as to prevent back flow and shall provide required back flow protections in accordance with [4.11](#).

The devices, appurtenance intended for special function such as sterilization, processing, distillation, etc, shall be provided with back flow protection devices.

The water supply for hospital fixtures shall be protected against backflow with a reduced pressure principle back flow assembly, an atmospheric or spill resistant vacuum breaker assembly, or an air gap. Vacuum breakers for bed pan washer hoses shall not be located less than 1 525 mm above floor. Vacuum breakers for hose connections in health care or laboratory areas shall not be less than 1 800 mm above floor.

4.5.3 Cross-Connection Control

Cross-connections shall be prohibited, except where approved back flow prevention assemblies/devices are installed to protect the potable water supply (see [4.11](#)).

Potable water outlets and combination stop and waste valves shall not be installed underground or below grade. Freeze proof yard hydrants that drain the riser into the ground are considered to be stop and waste valves.

Back flow prevention can be achieved by means of providing proper air gap, reduced pressure principle back flow prevention assemblies, back flow preventer with intermediate atmospheric vent, barometric loop, pressure vacuum breaker assemblies, atmospheric type vacuum breakers, double check back flow prevention assemblies, spill resistant pressure vacuum breaker, dual check back flow preventer, etc.

4.5.4 Identification of Non-potable Water Systems

Where non-potable water systems are installed, the piping conveying the non-potable water shall be identified either by colour marking, metal tags or tapes in accordance with the relevant standards and good engineering practices.

4.6 Materials, Fittings and Appliances

4.6.1 Standards for Materials, Fittings and Appliances

All materials, water fittings and appliances shall conform to Part B 'Building Materials' of this NBCS.

4.6.2 Materials for Pipes

Pipes may be of any of the following materials:

- a) Cast iron, vertically cast or centrifugally (spun) cast;
- b) Steel (internally lined or coated with bitumen or a bituminous composition, and out-coated with cement concrete or mortar, where necessary);
- c) Ductile iron, internally lined;
- d) Reinforced concrete;
- e) Prestressed concrete;
- f) Galvanized mild steel tubes;
- g) Copper;
- h) Brass;
- j) Wrought iron;
- k) Stainless steel;
- m) Polyethylene;
- n) High density polyethylene (HDPE);
- p) Unplasticized PVC;
- q) Chlorinated PVC;
- r) Polypropylene-random copolymer (PPR);
- s) Composite pipes (PE-AL-PE) or any other combination;
- t) Cross-linked polyethylene (PEX); or
- u) Polybutylene pipe.

4.6.2.1 The material chosen shall be resistant to corrosion, both inside and outside or shall be suitably protected against corrosion.

4.6.2.2 Polyethylene and unplasticized PVC pipes shall not be installed near hot water pipes or near any other heat sources. For temperature limitations in the use of polyethylene and unplasticized PVC pipes to convey water, reference may be made to accepted standards [E-1(8)].

4.7 Design of Distribution Systems

4.7.1 General

For designing the distribution system, the following guidelines, in addition to those given in [4.7.2](#) to [4.7.6](#) shall be followed:

- a) All plumbing systems in buildings shall conform to the general requirements given in [3.1](#);
- b) Peak factor for calculation in case of intermittent flows may generally be adopted in design as 2 to 3; and
- c) The residual head at consumer's tap shall be as per [3.1.2](#).

4.7.2 Rate of Flow

One of the important items that needs to be determined before the sizes of pipes and fittings for any part of the

water piping system may be decided upon, is the rate of flow in the service pipe which in turn depends upon the number of hours for which the supply is available at sufficiently high pressure. If the number of hours for which the supply is available is less, there will be large number of fittings in use simultaneously and the rate of flow will be correspondingly large.

The data required for determining the size of the communication and service pipes are:

- the maximum rate of discharge required;
- the length of the pipe; and
- the head loss by friction in pipes, fittings and meters.

4.7.3 Discharge Computation

4.7.3.1 Design of consumer's pipes based on fixture units

The design of the consumers' pipes or the supply pipe to the fixtures is based on:

- the number and kind of fixtures installed;
- the fixture unit flow rate; and
- the probable simultaneous use of these fixtures.

The rates at which water is desirably drawn into different types of fixtures are known. These rates become whole numbers of small size when they are expressed in fixture unit.

The water supply fixture units (WSFU) for different sanitary appliances or groups of appliances are given in [Table 2](#).

4.7.3.2 Probable simultaneous demand

The possibility that all water supply taps in any system in domestic and commercial use will draw water at the same time is extremely remote. Designing the water mains for the gross flow will result in bigger and uneconomical pipe mains and may not be necessary. A probability study made by Hunter suggests the relationship shown in [Fig. 2](#) and [Table 3](#). In the absence of similar studies in India, the curves based on Hunter's study may be followed. In making use of these curves, special allowances are made as follows:

- demands for service sinks are ignored in calculating the total fixture demand;
- demands of supply outlets such as hose connections and air conditioners through which water flows more or less continuously over a considerable length of time shall be added to the probable flow rather than the fixture demand; and
- fixtures supplied with both hot and cold water exert reduced demands upon main hot water and cold water branches (not fixture branches).

**Table 2 Water Supply Fixture Units (WSFU) for Different Fixtures
with Minimum Pipe Sizes**

(Clause [4.7.3.1](#))

Sl No.	Type of Fixture	Application		Minimum Pipe Size mm
		Private	Public	
(1)	(2)	(3)	(4)	(6)
i)	Bath tub	4	—	15
ii)	Ablution faucet/Bidet	1	1	15
iii)	Clothes washer	4	4 (see Note 7)	15

**Table 2 Water Supply Fixture Units (WSFU) for Different Fixtures
with Minimum Pipe Sizes**

(Clause 4.7.3.1)

SI No.	Type of Fixture	Application		Minimum Pipe Size mm
		Private	Public	
(1)	(2)	(3)	(4)	(6)
iv)	Dishwasher	1.5	1.5	15
v)	Drinking fountain	—	0.5 (0.75)	15
vi)	Hose bib	2.5	2.5	15
vii)	Wash basin (with metered faucet)	1	1	15
viii)	Wash basin (with standard faucet)	1.5	1.5 (2)	15
ix)	Service sink	—	3	15
x)	Kitchen sink	2	4	15
xi)	Surgical sink	—	2	15
xii)	Scrub station in hospital (per outlet)	—	3	15
xiii)	Shower	2	3	15
xiv)	Bathroom group (flush tank)	5	6	20
xv)	Bathroom group (flush valve)	8	10	25/32
xvi)	Urinal (flush valve)	3	5 (6)	20
xvii)	Urinal (flush tank)	2	2 (3)	15
xviii)	Urinal (sensor operated)	2	2 (3)	15
xix)	Water closet (flush valve)	5	5(8)	25/32
xx)	Water closet (flush tank)	2	3 (5)	15
xxi)	Combination fixture (faucet)	3	—	15
xxii)	Laundry trays (faucet)	3	—	15

NOTES

1 The above table is based on Hunter's method.

2 Hunter's method of estimating load in plumbing systems is based on assigning a fixture unit (FU) weight to the plumbing fixtures and then converting these to equivalent litres per minute, based on the theory of probability of usage and based on the observation that all fixtures are not used simultaneously.

3 The fixture unit concept is a method of calculating maximum probable water demand within large buildings based on theory of probability. The method is based on assigning a fixture unit (FU) value to each type of fixture based on its rate of water consumption, on the length of time it is normally in use and on the average period between successive uses.

4 The values of probable demand will not change in respect of systems with flush valves and flush tanks for fixture units more than 1 000.

5 The fixtures or appliances which are not included in the above table may be sized referring to fixtures having similar flow rate and frequency of usage.

6 The minimum supply branch pipe sizes for individual fixtures are nominal sizes.

7 The clothes washer for public does not include large washer extractors, and in such cases the pipe sizing shall be determined as per manufacturer's recommendations.

8 For more information on bathroom groups, reference may be made to specialist literature.

9 The fixture units listed in the above table represent the load for cold water service. The separate cold and hot water fixture unit value for fixtures having both hot and cold water connections may each be taken as three quarter of the listed total value of fixture.

10 A shower head over a bath tub does not increase the fixture unit value.

11 The values given in parantheses pertain to such public use buildings where an enhanced requirement is expected to be encountered as compared to the normal maximum use in public use buildings.

Table 3 Probable Simultaneous Demand

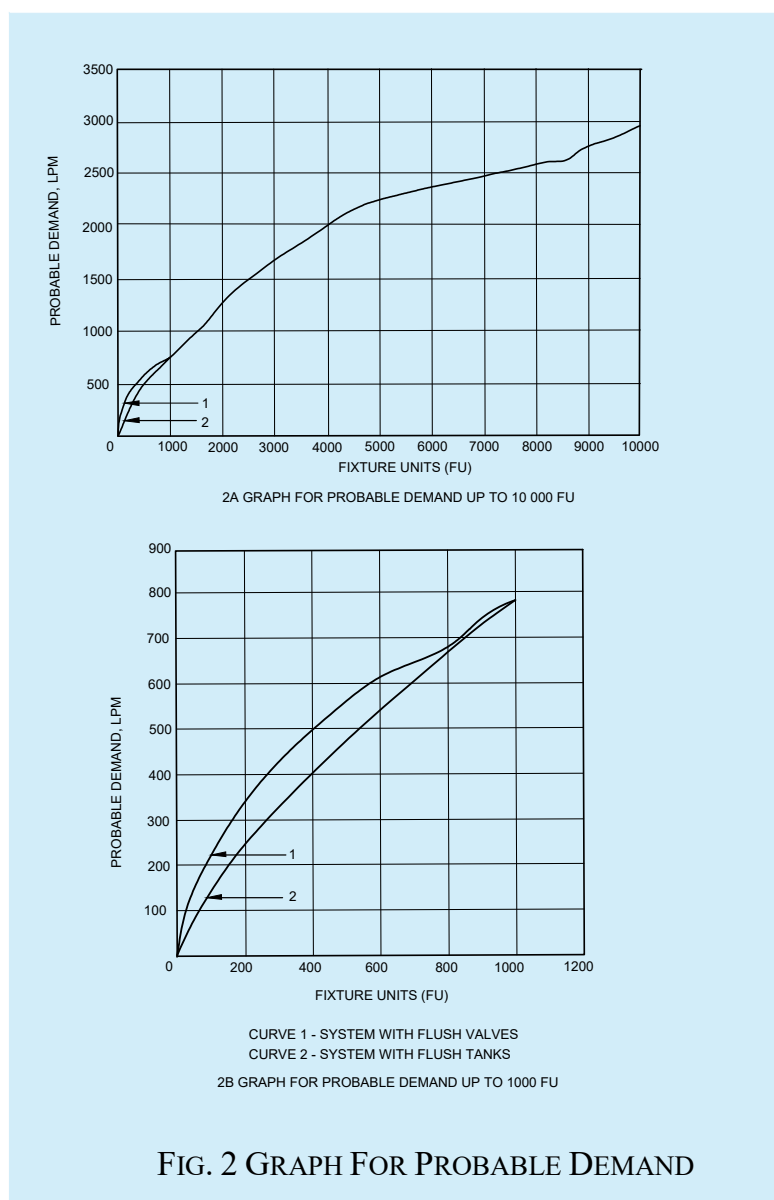
(Clause [4.7.3.2](#))

SI No.	Demand in Fixture Units	Demand with Flush Tanks litre/min	Demand with Flush Valves litre/min
(1)	(2)	(3)	(4)
i)	1	0	—
ii)	2	3.8	—
iii)	3	11.4	—
iv)	4	15.1	—
v)	5	22.7	—
vi)	6	25.5	—
vii)	8	28.1	—
viii)	10	30.3	102.20
ix)	20	53.0	132.48
x)	30	75.7	155.19
xi)	40	94.6	177.90
xii)	50	109.8	196.82
xiii)	60	121.1	208.18
xiv)	70	132.5	223.32
xv)	80	143.8	234.67
xvi)	90	155.2	246.03
xvii)	100	166.5	257.38
xviii)	140	200.6	295.23
xix)	180	230.9	329.30
xx)	200	246.0	348.22
xxi)	250	283.9	382.29
xxii)	300	321.7	416.35
xxiii)	400	397.4	476.91
xxiv)	500	473.1	537.47
xxv)	750	643.5	673.73
xxvi)	1 000	787.3	787.28
xxvii)	1 250	908.4	908.40
xxviii)	1 500	1 010.6	1 010.60
xxix)	1 750	1 112.8	1 112.79
xxx)	2 000	1 215.0	1 214.99
xxxi)	2 500	1 419.4	1 419.38
xxxii)	3 000	1 635.1	1 635.12
xxxiii)	3 500	1 811.1	1 811.12
xxxiv)	4 000	1 987.1	1 987.13
xxxv)	4 500	2 115.8	2 115.82
xxxvi)	5 000	2 244.5	2 244.51
xxxvii)	5 500	2 312.6	2 312.64
xxxviii)	6 000	2 380.8	2 380.77
xxxix)	6 500	2 411.0	2 411.05
xl)	7 000	2 479.2	2 479.18
xli)	7 500	2 547.3	2 547.31
xlii)	8 000	2 615.4	2 615.44

Table 3 (Concluded)			
Sl No.	Demand in Fixture Units	Demand with Flush Tanks litre/min	Demand with Flush Valves litre/min
(1)	(2)	(3)	(4)
xlili)	8 500	2 683.6	2 683.57
xliv)	9 000	2 751.7	2 751.70
xliv)	9 500	2 831.2	2 831.18
xlvi)	1 000 0	2 910.7	2 910.67
NOTE — For any water line other than flushing line, refer flush tank column to arrive at flow.			

4.7.3.3 The maximum flow rate and flush volumes shall be as below:

<i>Sl No.</i>	<i>Plumbing Fixtures/Fittings</i>	<i>Baseline (Max) Flow Rate</i>	<i>Max. Flow Rate for 1 or 2- or 3- Star Water Efficient Fixture/Fitting [accepted standards [E-1(5)]]</i>
(1)	(2)	(3)	(4)
i)	Water closets	6 litre/flush	6.0/4.8/4.0 litre/full flush and 3.0/2.8/2.0 litre/reduced flush
ii)	Urinals	3.8 litre/flush	3.0/2.0/1.0 litre/flush
iii)	Lavatory, metered faucet (Public)	1 litre/use	1.0/0.8/0.6 litre/use
iv)	Lavatory, faucet (Private)	8 litre/min	8.0/6.0/3.0 litre/min
v)	Sink, faucet	8 litre/min	8.0/6.0/4.5 litre/min
vi)	Bidet, hand-held spray	8 litre/min	6.0/5.0/4.0 litre/min
vii)	Shower head	10 litre/min	10.0/8.0/6.8 litre/min
NOTE — The maximum flow rates of plumbing fixtures and fittings provided are at the pressure of 0.42 N/mm ² . Water closets with dual flush cisterns and urinals with reduced flush volumes are recommended. Users/designers are encouraged to use low-flow fixtures.			



4.7.4 Pipe Size Computation

Commercially available standard sizes of pipes are only to be used against the sizes arrived at by actual design. Therefore, several empirical formulae are used, even though they give less accurate results. The Hazen and William's formula and the charts based on the same may be used without any risk of inaccuracy in view of the fact that the pipes normally to be used for water supply are of smaller sizes. Nomogram of Hazen and William's equation has been provided in [Annex C](#).

4.7.5 Adequate and Excessive Water Pressure

The requirements shall be as follows:

- Adequate water pressure* — A minimum water pressure in accordance with [3.1.2](#) shall be ensured in the distribution system. The maximum pressure shall be maintained at 0.42 N/mm^2 . Also, while designing, the maximum/minimum pressure required for operation of special fixtures and fittings is to be maintained as per the technical data sheet; and
- Excessive water pressure* — Whenever pressure exceeds 0.42 N/mm^2 , and the distribution system may not be able to withstand the same, pressure reducing valves shall be installed for reduction of pressure.

4.7.6 Water Hammer Effects

Water hammer is a pressure wave phenomenon created by a relatively sudden changes in the water velocity or its direction. It is also observed when valves are shut off. Water hammer arrestors may be used to absorb high pressures resulting from sudden closure of valves/taps.

4.8 Distribution Systems in Multi-Storeyed Buildings

4.8.1 There are following four basic methods of distribution of water to a multi-storeyed building:

- a) direct supply system from mains – public or private;
- b) gravity distribution system;
- c) pressurized distribution system (Hydro-pneumatic pumping system); and
- d) combined distribution system.

4.8.2 Direct Supply System from Mains — Public or Private

This system is adopted when adequate pressure is available in the mains to supply water at adequate pressure at the topmost floor. With limited pressure available in most city mains, water from direct supply is normally not available above two or three floors. However, in gated communities or large campuses, this system can be adopted for taller buildings by incorporating design parameters such as elevated centralized water tank(s) or central hydro-pneumatic pumping system(s). For details of this system, reference may be made to good practice [E-1(9)] may be referred.

4.8.3 Gravity Distribution System

This is the most common water distribution system. The system comprises pumping water to one or more overhead water tanks. Water transferred to overhead tank(s) is distributed by gravity to various parts of the building by the system of piping network.

4.8.4 Pressurized Distribution System (Hydro-pneumatic Pumping System)

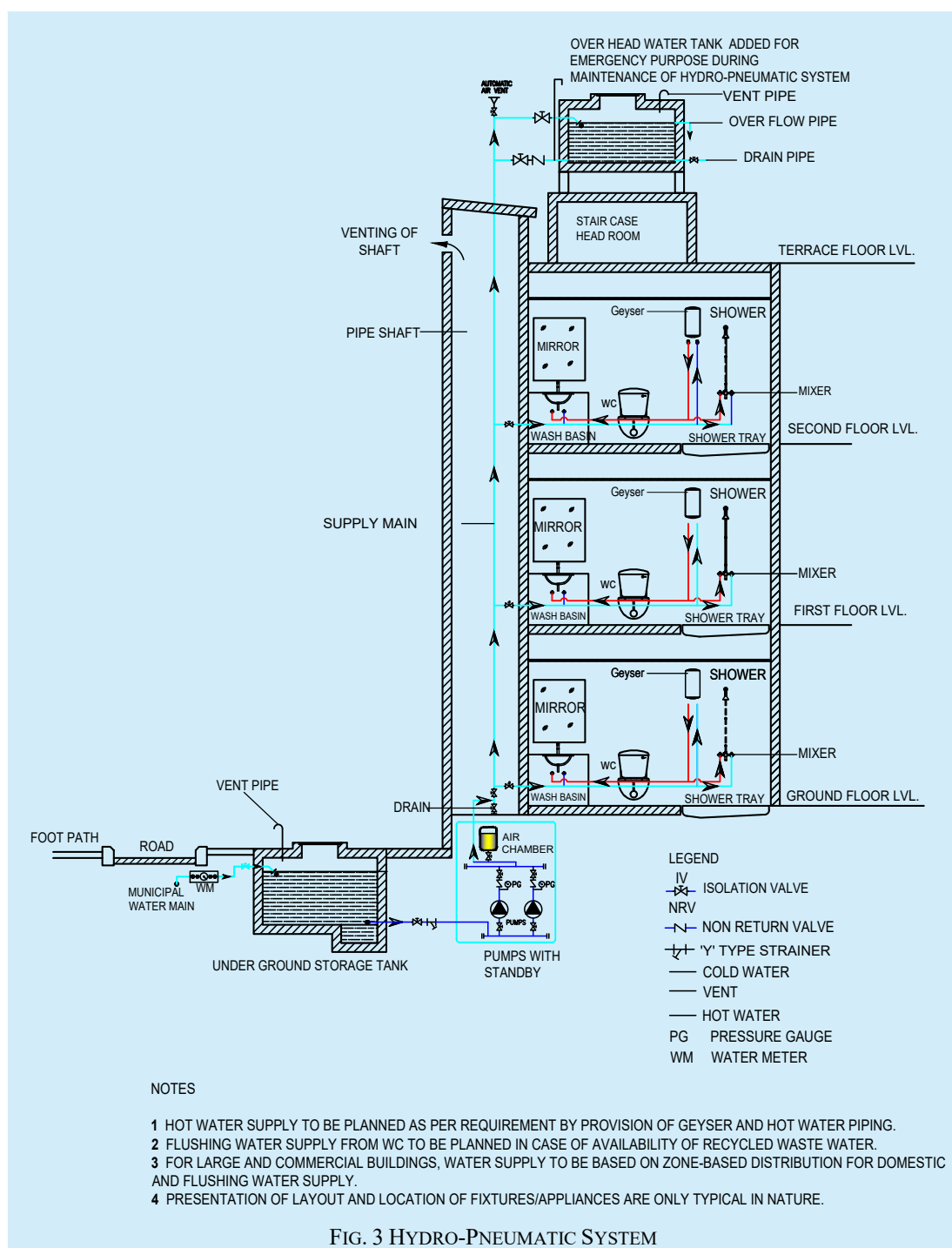
4.8.4.1 Pressurized distribution system is a direct pumping system incorporating a recharge diaphragm vessel.

4.8.4.2 The system may incorporate multiple pumps with suction and discharge manifolds and a control panel to facilitate automatic operation. Total discharge capacity required may be shared by a number of pumps, where the pumps operate in duty, assist and standby configuration.

4.8.4.3 The system shall also incorporate automatic sequencing of pumps to ensure even wear and tear also a low level cut-off, to prevent dry run of the pumps. The system shall be provided with continuous power supply with provision of emergency power backup.

4.8.4.4 Modern hydro-pneumatic systems are available with variable frequency drive, where the pump is efficiently used to deliver water at rates of flow as required by the system, by varying its speed with the assistance of an electronic device, thereby meeting the demand flow through variation in speed of the motor. With this arrangement, the same pump can deliver water at required pressure and flow as required at different times of the day. The system consumes energy in proportion to the work done and helps in controlling the water surge in the distribution line.

4.8.4.5 Hydro-pneumatic system generally eliminates the need of an overhead tank. As a good engineering practice and to take care of emergencies, an overhead of smaller capacity should be provided which feeds by gravity to the system (see [Fig. 3](#)).



4.8.5 Combined Distribution System

4.8.5.1 In this system, a combination of gravity and pressurized distribution is adopted. A few upper floors are provided with a pressure booster pumping system to achieve the desired residual pressure, while the lower floors are fed by gravity supply.

4.8.5.2 Water collected in the overhead tank is distributed to the various parts of the building. To achieve required residual pressure for top 2 to 4 floors for proper functioning of the fixtures, a pressure booster pumping system is installed on the dedicated outlet from overhead tank with its own distribution piping serving the top 2 floors to 4 floors. For lower floors, water is distributed by gravity system.

4.8.5.3 Water distribution is accomplished by providing down take pipes in the shaft from the terrace ring mains (see Fig. 4).

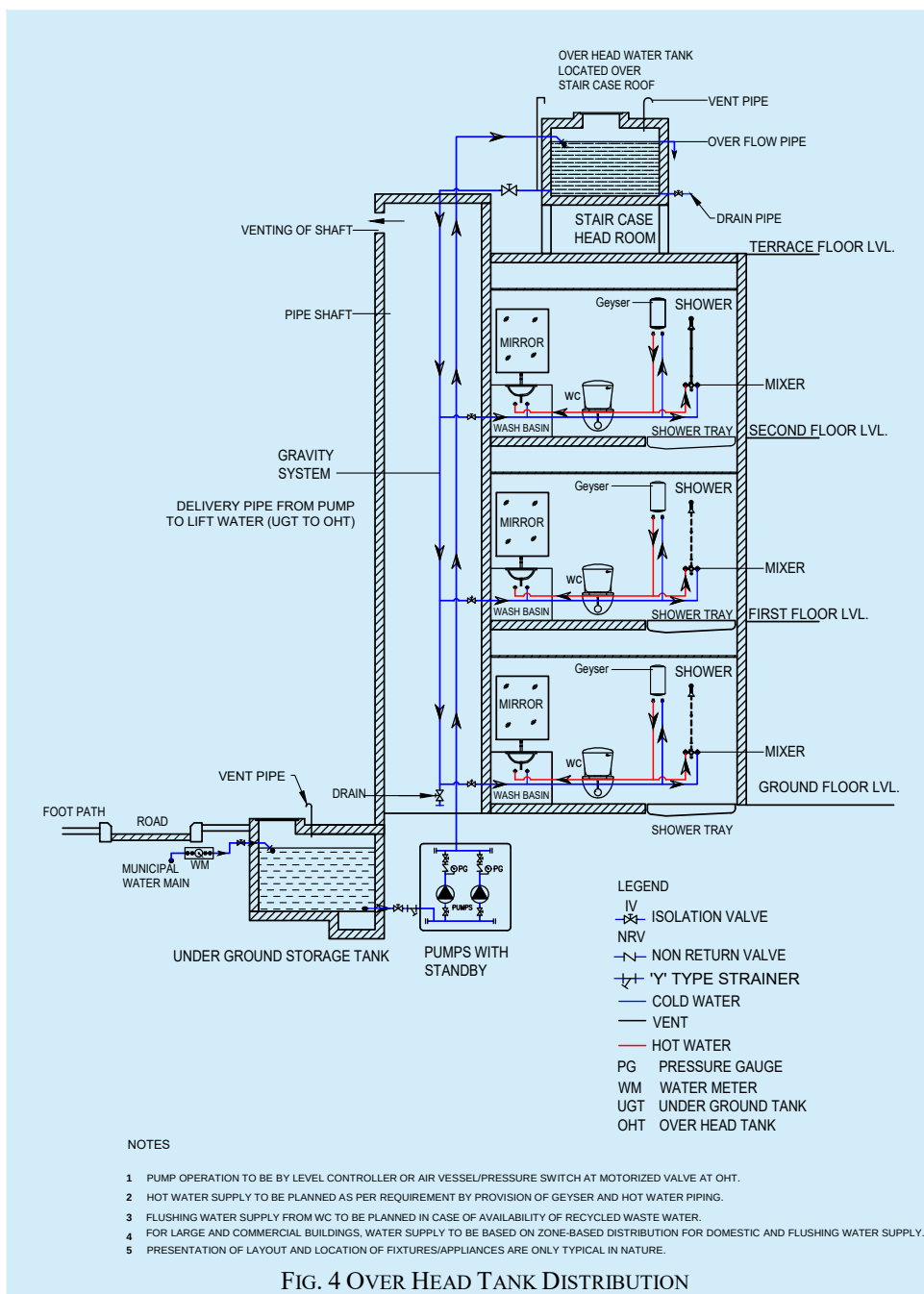
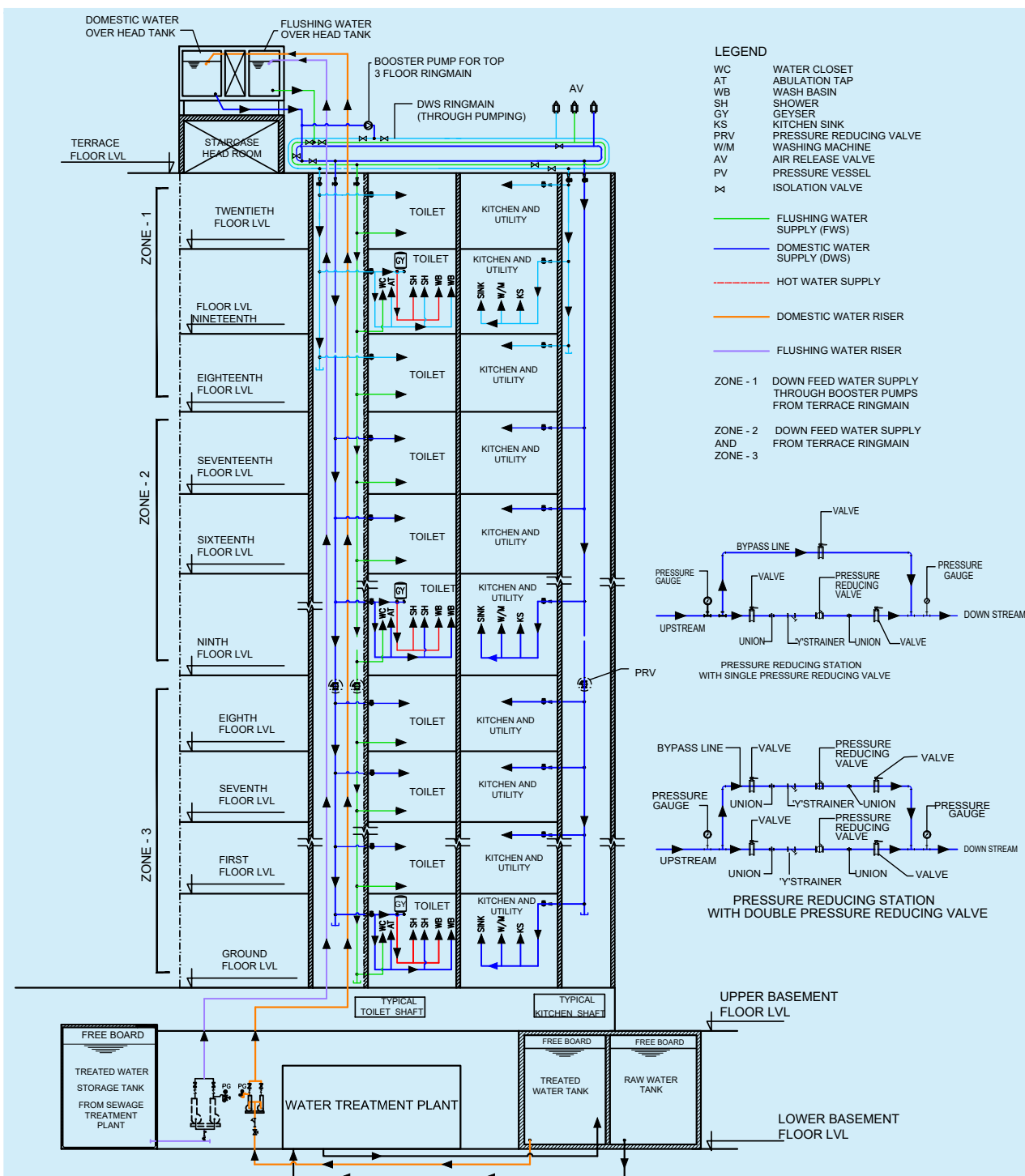


FIG. 4 OVER HEAD TANK DISTRIBUTION

4.8.6 Zoning of Distribution Systems

The zoning of water distribution network may be adopted for 7 floors to 9 floors, while conforming to the adequate pressure requirements and excessive pressure limitations in the hydraulic design (see 4.7.5). See Fig. 5 and Fig. 6 for zoning of gravity distribution system and pressurized distribution system (hydro-pneumatic pumping system), respectively.

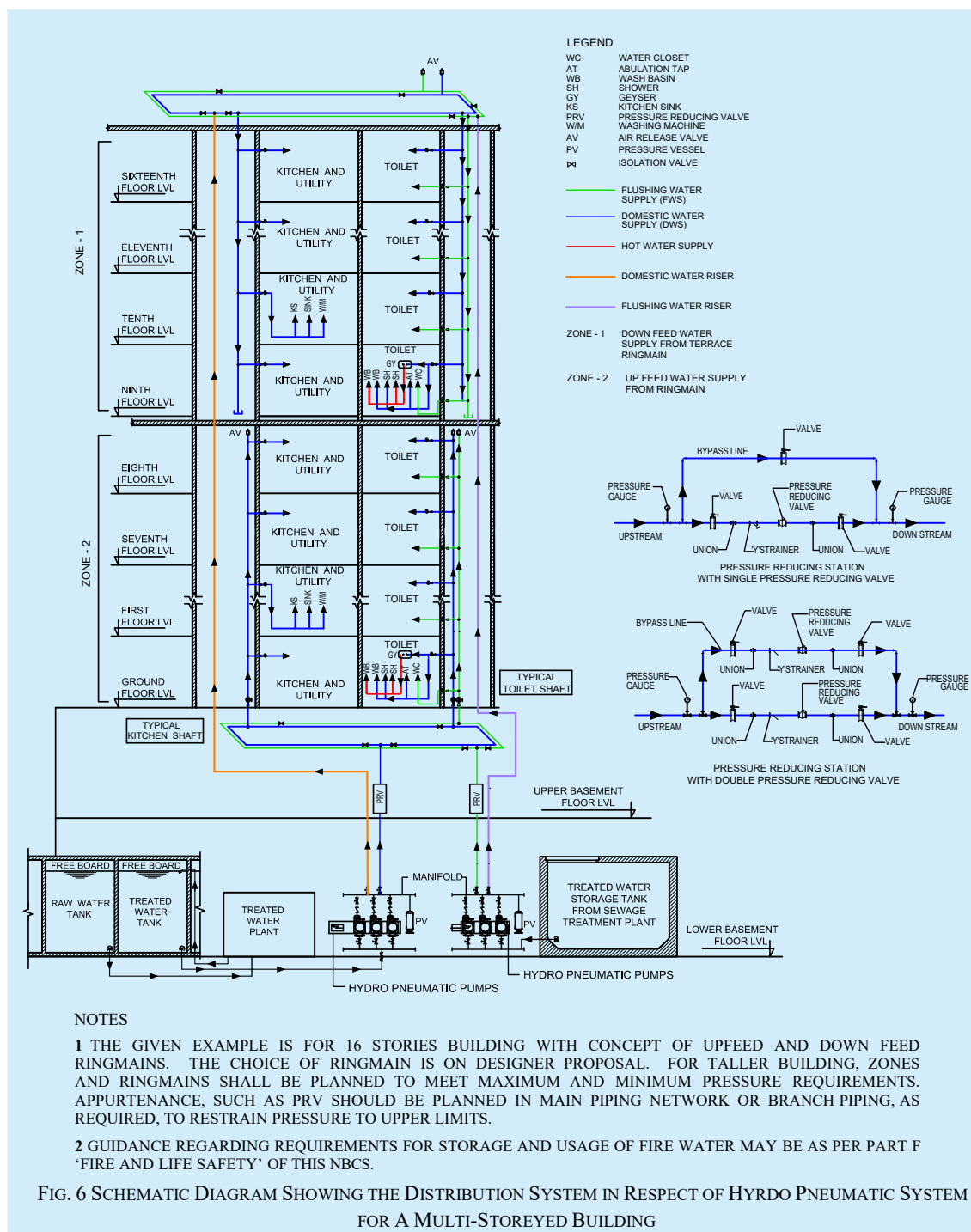
The recommended maximum permissible velocity is 2.4 m/s for water distribution. In case of hot water distribution through copper pipes, the velocity is restricted to 1.5 m/s due to concern of erosion of the piping material.



NOTES

- 1 APPURTENANCE, SUCH AS PRV SHOULD BE PLANNED IN MAIN PIPING NETWORK OR BRANCH PIPING, AS REQUIRED, TO RESTRAIN PRESSURE TO UPPER LIMITS.
- 2 REQUIREMENTS FOR STORAGE AND USAGE OF FIRE WATER SHOULD BE AS PER PART F 'FIRE AND LIFE SAFETY' OF THIS NBCS.

FIG. 5 SCHEMATIC DIAGRAM SHOWING THE DISTRIBUTION SYSTEM IN RESPECT OF GRAVITY SYSTEM FOR A MULTI-STOREYED BUILDING



4.9 General Requirements for Pipe Work

4.9.1 Mains

The following principles shall apply for the mains:

- a) Service mains shall be of adequate size to give the required rate of flow.
- b) Mains shall be divided into sections by the provisions of sluice valves and other valves so that water may be shut off for repairs.
- c) To avoid dead ends, the mains shall be arranged in a grid formation or in a network.
- d) Where dead ends are unavoidable, a hydrant shall be provided to act as a wash-out.
- e) Wash-out valve shall not discharge directly into a drain or sewer, or into a manhole or chamber directly connected to it; an effectively trapped chamber shall be interposed, into which the wash-out shall discharge.
- f) Air valves shall be provided at all summits, and wash-out at low points between summits.
- g) Mains need not be laid at unvarying gradients, but may follow the general contour of the ground. They shall, however, fall continuously towards the washout and rise towards the air valves. The gradient shall be such that there shall always be a positive pressure at every point under working conditions.
- h) Cover for the mains shall be at least 900 mm under roadways and 750 mm in the case of footpaths. This cover shall be measured from the top of the pipe to the surface of the ground.
- j) Mains shall be located sufficiently away from other service lines like electric and telegraph cables to ensure safety and where the mains cannot be located away from such lines, suitable protective measures shall be accorded to the mains. As far as possible, water supply and sewerage mains should be isolated from each other with certain minimum horizontal distance as per good engineering practice.

4.9.2 Communication Pipes

- a) Every premises that is supplied with water by the Authority shall have its own separate communication pipe. In the case of a group or block of premises belonging to the same owner the same communication pipe may supply water to more than one premises with the prior permission of the Authority.
- b) The communication pipe between the water main and the stop-cock at the boundary of the premises shall be laid by the Authority.
- c) Connections up to 50 mm diameter may be made on the water main by means of screwed ferrules, provided the size of the connections does not exceed one third the size of the water main. In all other cases, the connection shall be made by a T-branch off the water main.
- d) As far as practicable, the communication pipe and the underground service pipe shall be laid at right angles to the main and in approximately straight lines to facilitate location for repairs. It is also recommended that the communication pipe be laid in a pipe in pipe sleeve of larger diameter made of non-corrosive material to protect the communication pipe.
- e) Every communication pipe shall have a stop-cock and meter inserted in it. The waterway of each such fitting shall not be less than the internal sectional area of the communication pipe and the fittings shall be located within the premises at a conspicuous place accessible to the Authority which shall have exclusive control over it.

4.9.3 Consumer Pipes

- a) No consumer pipe shall be laid in the premises to connect the communication pipe without the approval of the Authority.
- b) The consumer pipe within the premises shall be laid underground with a suitable cover to safeguard against damage from traffic and extremes of weather.
- c) To control the branch pipe to each separately occupied part of a building supplied by a common service pipe, a stop valve shall be fixed to minimize the interruption of the supply during repairs. All such stop valves shall be fixed in accessible positions and properly protected. To supply water for drinking or for

culinary purposes, direct taps shall be provided on the branch pipes connected directly to the consumer pipe. In the case of multi-storeyed buildings, down take taps shall be supplied from overhead tanks.

- d) Pumps shall not be allowed on the service pipe, as they cause a drop in pressure on the suction side, thereby affecting the supply to the adjoining properties. In cases where pumping is required, a properly protected storage tank of adequate capacity shall be provided to feed the pump.
- e) No direct boosting (by booster pumps) shall be allowed from the service pipes (communication and consumer pipes).
- f) Consumer pipes shall be so designed and constructed as to avoid air-locks. Draining taps shall be provided at the lowest points from which the piping shall rise continuously to draw-off taps.
- g) Consumer pipes shall be so designed as to reduce the production and transmission of noise as much as possible.
- h) Consumer pipes in roof spaces and unventilated air spaces under floors or in basements shall be protected against corrosion.
- j) Consumer pipes shall be so located that they are not unduly exposed to accidental damage and shall be fixed in such positions as to facilitate cleaning and avoid accumulations of dirt.
- k) All consumer pipes shall be so laid as to permit expansion and contraction or other movements.

4.9.4 Prohibited Connections

- a) A service pipe shall not be connected into any distribution pipe; such connection may permit the backflow of water from a cistern into the service pipe, in certain circumstances, with consequent danger of contamination and depletion of storage capacity. It might also result in pipes and fittings being subjected to a pressure higher than that for which they are designed, and in flooding from overflowing cisterns.
- b) No pipe for conveyance or in connection with water supplied by the Authority shall communicate with any other receptacle used or capable of being used for conveyance other than water supplied by the Authority.
- c) Where storage tanks are provided, no person shall connect or be permitted to connect any service pipe with any distributing pipe.
- d) No service or supply pipe shall be connected directly to any water-closet or a urinal. All such supplies shall be from flushing cisterns/flush valves which shall be supplied from storage tank.
- e) No service or supply pipe shall be connected directly to any hot water system or to any other apparatus used for heating other than through a feed cistern thereof.

4.10 Jointing of Pipes

4.10.1 Cast Iron Pipes

Jointing may be done by any of the following methods:

- a) spigot and socket joints, or
- b) flanged joints.

in accordance with good practice [E-1(10)]. Alternative jointing materials which are found to be equally effective, may be used in place of lead joints, with the approval of the Authority.

4.10.2 Steel Pipes

Plain-ended steel pipes may be joined by welding. Electrically welded steel pipes shall be jointed in accordance with good practice [E-1(11)].

4.10.3 Wrought Iron and Steel Screwed Pipes

Screwed wrought iron or steel piping may be jointed with screwed and socketed joints. Care shall be taken to

remove any burr from the end of the pipes after screwing. A jointing compound approved by the Authority and containing no red lead composition shall be used. Screwed wrought iron or steel piping may also be jointed with screwed flanges.

4.10.4 Asbestos Cement Pipes

Asbestos cement pipes may be jointed in accordance with good practice [E-1(12)].

4.10.5 Copper Pipes

Copper pipes shall be jointed by internal solder ring joint, end-brazing joint or by use of compression fitting. The flux used shall be non-toxic and the solder used shall be lead free. The use of dezincification fittings shall be made in case of jointing of copper pipe and steel pipe. The jointing technology shall be used as per good engineering practice and as per manufacturers recommendations.

4.10.6 Concrete Pipes

Concrete pipes shall be jointed in accordance with good practice [E-1(13)].

4.10.7 Polyethylene and Unplasticized PVC Pipes

Polyethylene and unplasticized PVC pipes shall be jointed in accordance with good practice [E-1(14)].

4.11 Backflow Prevention

4.11.1 The installation shall be such that water delivered is not liable to become contaminated or that contamination of the public water supply does not occur.

4.11.2 The various types of piping and mechanical devices acceptable for backflow protection are:

- a) Barometric loop,
- b) Air gap,
- c) Atmosphere vacuum breaker,
- d) Pressure vacuum breaker,
- e) Double check valve,
- f) Reduced pressure backflow device.

4.11.3 The installation shall not adversely affect drinking water:

- a) by materials in contact with the water being unsuitable for the purpose;
- b) as a result of backflow of water from water fittings, or water using appliances into pipework connected to mains or to other fittings and appliances;
- c) by cross-connection between pipes conveying water supplied by the water undertaker with pipes conveying water from some other source; and
- d) by stagnation, particularly at high temperatures.

4.11.4 No pump or similar apparatus, the purpose of which is to increase the pressure in or rate of flow from a supply pipe or any fitting or appliance connected to a supply pipe, shall be connected unless the prior written permission of the water supplier has been obtained in each instance.

The use of such a pump or similar apparatus is likely to lead to pressure reduction in the upstream pipe work which, if significant, increase the risk of backflow from other fittings.

4.11.5 The water shall not come in contact with unsuitable materials of construction.

4.11.6 No pipe or fitting shall be laid in, on or through land fill, refuse, an ash pit, sewer, drain, cesspool or refuse chute or any manhole connected with them.

4.11.7 No pipe susceptible to deterioration by contact with any substance shall be laid or installed in a place where such deterioration is likely to occur. No pipe that is permeable to any contaminant shall be laid or installed in any position where permeation is likely to occur.

4.11.8 If a liquid (other than water) is used in any type of heating primary circuit, which transfers heat to water for domestic use, the liquid shall be non-toxic and non-corrosive.

4.11.9 A backflow prevention device shall be arranged or connected at or as near as practicable to each point of delivery and use of water. Appliances with built-in backflow prevention shall be capable of passing the test. All backflow prevention devices shall be installed so that they are accessible for examination, repair or replacement. Such devices shall be capable of being tested periodically by the Authority to ensure that the device is functioning efficiently and no backflow is occurring at any time.

4.12 Conveyance and Distribution of Water within the Premises

4.12.1 Basic Principles

Wholesome water supply provided for drinking and culinary purposes shall not be liable to contamination from any less satisfactory water. There shall, therefore, be no cross-connection whatsoever between the distribution system for wholesome water and any pipe or fitting containing unwholesome water, or water liable to contamination, or of uncertain quality, or water which has been used for any other purpose. The provision of reflux or non-return valves or closed and sealed stop valves shall not be construed as a permissible substitute for complete absence of cross-connection.

4.12.2 The design of the pipe work shall be such that there is no possibility of backflow towards the source of supply from any cistern or appliance, whether by siphonage or otherwise. Reflux non-return valves shall not be relied upon to prevent such backflow.

4.12.3 Where a supply of less satisfactory water than wholesome water becomes inevitable as an alternative or is required to be mixed with the latter, it shall be delivered only into a cistern and by a pipe or fitting discharging into the air gap at a height above the top edge of the cistern equal to twice its nominal bore and in no case less than 150 mm. It is necessary to maintain a definite air gap in all appliances or taps used in water-closets.

4.12.4 All pipe work shall be so designed, laid or fixed and maintained as to remain completely water-tight, thereby avoiding wastage, damage to property and the risk of contamination.

4.12.5 No water supply line shall be laid or fixed so as to pass into or through any sewer, scour outlet or drain or any manhole connected therewith, nor through any ash pit or manure pit or any material of such nature that is likely to cause undue deterioration of the pipe.

4.12.5.1 Where the laying of any pipe through corrosive soil or pervious material is unavoidable, the piping shall be properly protected from contact with such soil or material by being carried through an exterior pipe sleeves as approved by the Authority. Any existing piping or fitting laid or fixed, which does not comply with the above requirements, shall be removed immediately by the consumer and re-laid by him in conformity with the above requirements and to the satisfaction of the Authority.

4.12.5.2 Where lines have to be laid in close proximity to electric cables or in corrosive soils, adequate precautions/protection should be taken to avoid corrosion.

4.12.6 Underground piping shall be laid at such a depth that it is unlikely to be damaged by frost or traffic loads and vibrations. It shall not be laid in ground liable to subsidence, but where such ground cannot be avoided, special precautions shall be taken to avoid damage to the piping. Where piping has to be laid across recently disturbed ground, the ground shall be thoroughly consolidated so as to provide a continuous and even support. In the case of frost conditions, the pipes shall be laid below the frost line with suitable insulation to the pipes.

4.12.7 In designing and planning the layout of the pipe work, due attention shall be given to the maximum rate of discharge required, economy in labour and materials, protection against damage and corrosion, water hammer, protection from frost, if required, and avoidance of airlocks, noise transmission and unsightly arrangement.

4.12.8 To reduce frictional losses, piping shall be as smooth as possible inside. Methods of jointing shall be such as to avoid internal roughness and projection at the joints, whether of the jointing materials or otherwise.

4.12.9 Change in diameter and in direction shall preferably be gradual rather than abrupt to avoid undue loss of head. No bend or curve in piping shall be made which is likely to materially diminish or alter the cross-section.

4.12.10 No boiler for generating steam or closed boilers of any description or any machinery shall be supplied direct from a service or supply pipe. Every such boiler or machinery shall be supplied from a feed cistern.

4.13 Laying of Mains and Pipes on Site

4.13.1 The mains and pipes on site shall be laid in accordance with good practices [E-1(15)]. The pipes laid in buildings situated in seismic zone and across large building expansion joints should be installed with suitable expansion bellows and expansion loops to mitigate expansion of piping system for its intended function without failures.

4.13.2 *Excavation and Refilling*

The bottoms of the trench excavations shall be so prepared that the barrels of the pipes, when laid, are well bedded for their whole length on a firm surface and are true to line and gradient. In the refilling of trenches, the pipes shall be surrounded with fine selected material, well rammed so as to resist subsequent movement of the pipes. No stones shall be in contact with the pipes; when resting on rock, the pipes shall be bedded on fine-selected material or (especially where there is a steep gradient) on a layer of concrete.

4.13.2.1 The pipes shall be carefully cleared of all foreign matter before being laid.

4.13.3 *Laying Underground Mains*

Where there is a gradient, pipe laying shall proceed in 'uphill' direction to facilitate joint making.

4.13.3.1 Anchor blocks shall be provided to withstand the hydraulic thrust.

4.13.4 Iron surface boxes shall be provided to give access to valves and hydrants and shall be supported on concrete or brickwork which shall not be allowed to rest on pipes.

4.13.5 *Laying Service Pipes*

4.13.5.1 Service pipes shall be connected to the mains by means of right-hand screw down ferrule or T-branches. The ferrules shall conform to accepted standard [E-1(16)].

4.13.5.2 Precaution against contamination of the mains shall be taken when making a connection and, where risk exists, the main shall be subsequently disinfected. The underground water service pipe and the building sewer or drain shall be kept at a sufficient distance apart so as to prevent contamination of water. Water service pipes or any underground water pipes shall not be run or laid in the same trench as the drainage pipe. Where this is unavoidable, the following conditions shall be fulfilled:

- a) the bottom of the water service pipe, at all points, shall be at least 300 mm above the top of the sewer line at its highest point;
- b) the water service pipe shall be placed on a solid shelf excavated on one side of the common trench;
- c) the number of joints in the service pipe shall be kept to a minimum; and
- d) the materials and joints of sewer and water service pipe shall be installed in such a manner and shall possess such necessary strength and durability as to prevent the escape of solids, liquids and gases there from under all known adverse conditions, such as corrosion strains due to temperature changes, settlement, vibrations and superimposed loads.

4.13.5.3 The service pipe shall pass into or beneath the buildings at a depth of not less than 750 mm below the outside ground level and, at its point of entry through the structure, it shall be accommodated in a sleeve which shall have previously been solidly built into the wall of the structure. The space between the pipe and the sleeve shall be filled with bituminous or other suitable material for a minimum length of 150 mm at both ends.

4.13.6 *Pipes Laid through Ducts, Chases, Notches or Holes*

Ducts or chases in walls for piping shall be provided during the building of the walls. If they are cut into existing walls, they shall be finished sufficiently smooth and large enough for fixing the piping.

4.13.6.1 Piping laid in notches or holes shall not be subjected to external pressure.

4.13.7 *Lagging of Pipes*

Where lagged piping outside buildings is attached to walls, it shall be entirely covered all round with water-proof and fire insulating material and shall not be in direct contact with the wall. Where it passes through a wall, the lagging shall be continued throughout the thickness of the wall.

4.14 Hot Water Supply Installations

4.14.1 *Design Consideration*

4.14.1.1 *General*

In electric water heating practice for domestic purposes, the accepted method is to use storage heaters in which water is steadily heated up to a predetermined temperature and stored until required for use. The heating by electricity of a large quantity of water, such as water required for a hot bath, within the time normally taken to run the water into the bath, requires a heater of too high a rating to be practicable in normal domestic premises. It should be ensured that such vents and pipes are above reach/inaccessible during daily use to avoid accidents such as accidental contact and burning.

4.14.1.2 In modern hotels and apartment blocks and service apartments, centralized storage and distribution systems are adopted, where other energy sources such as oil, gas, solar panels, etc, may be used for the generation of hot water as these options prove more economical and convenient in heating large volumes of water for storage.

4.14.1.3 The implementation of a hybrid heating system, integrating solar-based heating with electrical geysers, has demonstrated effectiveness in extreme cold conditions where water freezing occurs, and prolonged use of electric geysers presents challenges. When selecting such hybrid systems, the design shall ensure operational safety during both the heating process and the extraction of hot water.

4.14.1.4 When water supplied to the buildings contain dissolved salts resulting in hardness of water, measures such as installation of water softening plants, etc, shall be taken to avoid formation of scales in the hot water installations.

4.14.2 *Storage Temperature*

4.14.2.1 The design of hot water supply system and its appliances shall be based on the temperatures at which water is normally required for the various uses, namely:

Scalding	:	65 °C
Sink	:	60 °C
Hot bath	:	43 °C as run for use at 41 °C
Warm bath	:	37 °C
Tepid bath	:	29.5 °C

4.14.2.2 In order to minimize the danger of scalding, precipitation of scale from hard water, standing heat losses, risk of steam formation and the possibility of damage to porcelain or other fittings and to surface finishes, a storage

temperature of 60 °C is recommended. If storage capacity is limited, a higher temperature up to 65 °C may be adopted when soft water is used.

4.14.3 Storage Capacity

The size of the storage vessel is governed by the maximum short time demand of the domestic premises. Depending on local conditions this shall be 50 litre to 75 litre at 60 °C in a dwelling with a bath tub and 25 litre at 60 °C for a shower or a tap (for bucket supply). The capacity of the storage vessel shall not be less than 20 percent in excess of the required maximum short time demand. In larger houses where a single hot water heater is intended to supply hot water to more than one bathroom or kitchen or both, the maximum short time demand shall be estimated and the capacity decided accordingly. Small electric or gas storage heaters of 15 litre to 25 litre capacity may be used to supply one or two points of draw off depending on the use of hot water.

4.14.4 Rate of Flow

With storage type installation, the recommended maximum rates of flow for different types of fixtures are given in [Table 4](#).

Table 4 Rate of Hot Water Flow (Clause 4.14.4)		
SI No. (1)	Fixtures (2)	Rate of Flow litre/min (3)
i)	Kitchen sink	5
ii)	Wash basin	5
iii)	Shower (spray type)	6.5

4.14.4.1 Hot water supply systems

Identification of water supply system is critical for safe functioning and protection of occupants. The first step is correct labelling of various water systems in the building. The requirements of identification/labelling shall be adhered on every installation where different water systems are provided.

While designing centralized hot water systems in major projects, the following points should be considered:

- Fuel to be used,
- Location and dimension of boiler house and calorifier chambers,
- Location and dimension of fuel storage tank,
- Means of ash disposal (in case of solid fuel),
- Location and dimension of chimney,
- Location of cold-water points,
- Drainage facilities during emptying,
- Quality of water supply,
- Provision for air combustion and ventilation.

In laying hot water piping systems, the pressures of hot and cold water should be made equal at each fixture, especially where mixing faucets/thermostatic mixer are to be used. Otherwise, there would be imbalance of pressure where one pressure would be more than the other. These conditions should be avoided by proper design of the system.

Return circulation systems are recommended for energy conservation and user comfort.

In the inverted systems, hot water heaters and tanks are located at the highest point of water supply system and supply and return risers below the level of hot water source.

Environmental and consumer requirements should be considered when planning the installation of a solar hot water system. Factors affecting the performance of a system and decisions about how the system should be installed, include:

- a) the climate zone of the site and possibility of:
 - 1) shading;
 - 2) frost and freezing;
 - 3) wind;
 - 4) dust;
 - 5) hail; and
 - 6) corrosion and scaling.
- b) the ambient air temperature;
- c) the cold-water temperature;
- d) the availability of space and pitch of a suitable south-facing roof;
- e) the presence and location of an existing hot water service;
- f) the available energy sources (for example, gas or electricity);
- g) the householder's hot water usage; and
- h) the householder's budget.

4.14.5 Design of Storage Vessel

Storage tanks shall be oblong or cylindrical in shape and shall be installed, preferably with the long side vertical in order to assist the effective stratification or 'layering' of hot or cold water. The ratio of height to width or diameter shall not be less than 2 : 1. An inlet baffle should preferably be fitted near the cold inflow pipe in order to spread the incoming cold water.

4.14.6 Materials for Storage Vessel and Pipes

4.14.6.1 Under no circumstances shall ungalvanized (black) mild steel pipes and fittings, such as sockets, bushes, etc, be used in any part of domestic hot water installation, including the cold feed pipe and the vent pipe. Materials resistant to the chemical action of water supplied shall be used in construction of vessels and pipes. Each installation shall be restricted to one type of metal only, such as all copper or all galvanized mild steel. When water supplied is known to have appreciable salt content, galvanized iron vessels and pipes shall not be used. However, it is advisable to avoid use of lead pipes in making connection to wash basins. Where required it is also advisable to use vessels lined internally with glass, stainless steel, etc.

4.14.6.2 In general, tinned copper and other metals such as monel metal, etc, are suitable for most types of water. The suitability of galvanized mild steel for storage tanks depends upon the pH value of the water and the extent of its temporary hardness. For values of pH 7.2 or less, galvanized mild steel should not be used. For values of pH 7.3 and above, galvanized mild steel may be used provided the corresponding temporary hardness is not lower than those given below:

<i>Sl No.</i>	<i>pH Value</i>	<i>Minimum Temporary Hardness Required mg/l</i>
(1)	(2)	(3)
i)	7.3	210
ii)	7.4	150
iii)	7.5	140
iv)	7.6	110
v)	7.7	90
vi)	7.8	80
vii)	7.9 to 8.5	70

4.14.7 Location of Storage Vessel

The loss of heat increases in proportion to the length of pipe between the storage vessel and the hot water outlet since each time the water is drawn, the pipe fills with hot water which then cools. The storage vessel shall therefore be so placed that the pipe runs to the most frequently used outlets are as short as possible.

4.14.8 Immersion Heater Installation

4.14.8.1 If a domestic storage vessel is to be adopted to electric heating by the provision of an immersion heater and thermostat, the following precautions shall be observed:

- a) *Location of immersion heaters* — The immersion heater shall be mounted with its axis horizontal, except in the case of the circulation type which is normally mounted with its axis approximately vertical.
- b) In a tank with a flat bottom, a space of not less than 75 mm below the immersion heater and 50 mm below the cold feed connection shall be provided to allow for accumulation of sludge and scale, where it will not affect the working of the immersion heater.
- c) In a cylindrical storage vessel with inwardly dished bottom, the inlet pipe shall be so arranged that the incoming cold water is not deflected directly into the hot water zone. The lowest point of the immersion heater shall be 25 mm above the centre line of the cold feed inlet, which, in turn, is usually 100 mm above the cylinder rim.
- d) *Location of thermostat* — Where the thermostat does not form an integral part of the immersion heater, it shall be mounted with its axis horizontal, at least 50 mm away from and not lower than the immersion heater.
- e) *Dual heater installations* — If desired, the principle of the dual heater may be adopted. In this case, one heater and its thermostat shall be installed at a low level as indicated in (b) and (c). The second heater and its thermostat shall be similarly disposed in the upper half of the cylinder at a level depending on the reserve of hot water desired for ordinary domestic use. The bottom heater shall be under separate switch control.
- f) *Clearance around storage vessel* — Adequate clearance shall be provided between the tank and the cupboard, door or walls to allow convenient insertion and adjustment of the immersion heater and thermostat and to give space for thermal insulation.

4.14.8.2 Rating of immersion heaters

The rating of an immersion heater shall be determined according to the following factors:

- a) proposed hot water storage capacity (the maximum with cold water as indicated in [4.14.3](#) shall be taken into account);
- b) rate of utilization (draw off frequency);
- c) permissible recovery period; and
- d) inlet water temperature.

For details regarding rated input of water, refer to good practice [E-1(17)].

4.14.9 Thermal Insulation

The hot water storage vessel and pipes shall be adequately insulated, wherever necessary to minimize heat loss.

The whole external surface of the storage vessel including the cover to the man hole shall also be duly insulated.

Insulation may be provided by wrapping storage vessel and pipes with fibre glass, mineral wool or closed cell flexible elastomeric foam based materials of desired thickness, covered with cladding for protection from damage or from weather in case of external use.

4.14.10 Cold Water Supply to Heaters

4.14.10.1 A storage water heater (pressure type) shall be fed from a cold water storage tank and under no circumstances connected directly to the water main, except the type which incorporates a feed tank with ball valves and overflow pipe arrangement (cistern type heaters) or non-pressure type heaters.

4.14.10.2 Storage cisterns

4.14.10.2.1 The storage capacity of a cold water tank shall be at least twice the capacity of the hot water heater. The capacity of the storage tank may, however, be 1.5 times when the number of heaters connected to one common tank exceeds 10.

4.14.10.2.2 The storage tank for supply of cold water to hot water heaters shall be separate, if practicable. In the case of a common tank which also supplies cold water to the fixtures, this cold water supply connection shall be so arranged that 50 percent of the net capacity, worked out as in [4.14.10.2.1](#), shall be available for supply to the hot water heaters.

4.14.10.2.3 In the case of multi-storeyed buildings where a common overhead tank over the stair/lift well is generally installed, it is advisable to have one or more local tanks for supply to the hot water heaters or a compartment in the tank with required storage always available for hot water heater.

4.14.10.2.4 In tall multi-storeyed buildings where the static pressure increases with the height, the total static pressure on the hot water heaters on the lowest floor shall not exceed the rated working pressure of the hot water heater installed. Should the height of the building so require, additional tanks shall be provided on the intermediate floors to restrict the static head to permissible limits, or the pressure shall be reduced to working pressure by providing pressure reducing valves.

4.14.11 Cold Water Feed

4.14.11.1 The feed pipe connecting cold water tank with the hot water heater shall not be of less than 20 mm bore and it shall leave the cold water tank at a point not less than 50 mm above the bottom of the tank and shall connect into the hot water heater near its bottom. The feed pipe shall not deliver cold water to any other connection, but into the hot water cylinders only.

4.14.11.2 In the case of multi-storeyed buildings, a common cold water feed pipe may be installed, but each hot water heater shall be provided with a check valve (horizontal type check valve shall be preferred to vertical type for easy maintenance).

4.14.11.3 Care shall be taken in installing the piping to prevent air locks in the piping and negative pressure in the hot water heater. Cold water feed pipe shall not be cross-connected with any other source of supply under pressure.

4.14.12 Hot Water Piping

4.14.12.1 Expansion pipe or vent pipe

4.14.12.1.1 Each non-pressure type hot water heater or cylinder shall be provided with a vent pipe of not less than 20 mm bore. The vent pipe shall rise above the water line of the cold water tank by at least 150 mm plus 10 mm for every 300 mm height of the water line above the bottom of the heater. The vent shall discharge at a level higher than the cold water tank and preferably in the cold water tank supplying the hot water heaters. Care shall be taken to ensure that any accidental discharge from the vent does not hurt or scald any passer-by or persons in the vicinity.

It shall be ensured that such vents and pipes are so located so as not to encroach into the minimum maneuvering space and should be suitably insulated to avoid accidental contact and burns.

NOTE — Pressure type water heaters are very commonly used and non-pressure type is slowly phased out.

4.14.12.1.2 The vent pipe shall be connected to the highest point of the heater vessel and it shall not project downwards inside it, as otherwise air may be trapped inside, resulting in surging and consequent noises.

4.14.12.1.3 At no point, after leaving the vessel, shall the vent pipe dip below the level of its connection with the vessel.

4.14.12.1.4 A vent pipe may, however, be used for supply of hot water to any point between the cold water tank and the hot water heaters.

4.4.12.1.5 The vent pipe shall not be provided with any valve or check valves.

4.4.12.2 Hot water heaters

4.4.12.2.1 The common hot water delivery pipe shall leave the hot water heater near its top and shall be of not less than 20 mm bore generally, not less than 25 mm bore if hot water taps are installed on the same floor as that on which the hot water heater is situated.

4.14.12.2.2 Hot water taps shall be of such design as would cause the minimum friction. Alternatively, oversized tap may be provided, such as a 20 mm tap on a 15 mm pipe.

4.14.12.2.3 The hot water distributing system shall be so designed as to ensure that the time lag between opening of the draw-off taps and discharge of hot water is reduced to the minimum to avoid wastage of an undue amount of water which may have cooled while standing in the pipes when the taps are closed. With this end in view, a secondary circulation system with flow and return pipes from the hot water tank shall be used where justified. Whether such a system is used or not, the length of pipe to a hot water draw-off tap, measured along the pipe from the tap to the hot water tank or the secondary circulation pipe, shall not exceed the lengths given in [Table 5](#).

Table 5 Maximum Permissible Lengths of Hot Water Draw-Off Pipes <i>(Clause 4.14.12.2.3)</i>		
SI No.	Largest Internal Diameter of Pipe	Length m
(1)	(2)	(3)
i)	Not exceeding 20 mm	12
ii)	Exceeding 20 mm but not exceeding 25 mm	7.5
iii)	Exceeding 25 mm	3.0
NOTE — In the case of a composite pipe of different diameters, the largest diameter is to be taken into consideration for the purpose of this table.		

4.14.12.2.4 Wherever mixing of hot and cold water is done by a mixing fitting, that is, hot and cold stop-cocks deliver to a common outlet of mixed water (that is, showers, basin or bath supply fittings), the pressure in the cold and hot water systems shall be equal. This can be achieved by connecting the cold water supply from an overhead tank at the same static height as the overhead tank supplying cold water to the hot water heaters. In case this is not possible, hot and cold water should be supplied to the fixtures by separate supply taps.

The hot and cold water thermostatic mixers/diverters/faucets for public use should have 'hot' and 'cold' clearly marked in the form of colour coding and tactile information. The functioning of hot and cold water from the thermostatic mixer/diverter/faucet should be consistent, for example, a clockwise turn should result in discharge of hot water and counter clockwise for cold. It is recommended that a thermostat be installed to limit the temperature of the hot water to a maximum of 40 °C in order to prevent scalding. All faucets/mixers/diverts in public use shall be automatic (sensor operated) or lever type.

4.14.13 Types of Hot Water Heaters

The various types of water heaters used for preparation of hot water are as follows:

- a) *Electric storage heaters:*
 - 1) Non-pressure or open outlet type,

- 2) Pressure type,
- 3) Cistern type,
- 4) Dual heater type.
- b) *Gas water heaters:*
 - 1) Instantaneous type {see accepted standard [E-1(18)]},
 - 2) Storage type.
- c) *Solar heating systems:*
 - 1) Independent roof mounted heating units,
 - 2) Centrally banked heated system.
- d) *Central hot water system:*
 - 1) Oil fired,
 - 2) Gas fired,
 - 3) Electrical coil type,
 - 4) Heat Pump.

4.14.13.1 The quality and construction of the different types of hot water heaters shall be in accordance with good practice [E-1(19)].

4.14.13.2 For solar water heating system and their installation reference shall be made to good practice [E-1(20)].

4.14.13.3 Requirements in regard to inspection and maintenance of hot water supply installations shall be in accordance with [4.15.1](#) to [4.15.4](#).

4.15 Inspection and Testing

4.15.1 Testing of Mains before Commencing Work

All pipes, fittings and appliances shall be inspected, before delivery at the site to see whether they conform to accepted standards. All pipes and fittings shall be inspected and tested by the manufacturers at their factory and shall comply with the requirements of this Section. They shall be tested hydraulically under a pressure equal to twice this maximum permissible working pressure or under such greater pressure as may be specified. The pipes and fittings shall be inspected on site before laying and shall be sounded to disclose cracks. Any defective items shall be clearly marked as rejected and forthwith removed from the site.

4.15.2 Testing of Mains after Laying

After laying and jointing, the main shall be slowly and carefully charged with water by providing a 25 mm inlet with a stop-cock, so that all air is expelled from the main. The main is then allowed to stand full of water for a few days if time permits, and then tested under pressure. The test pressure shall be 0.5 N/mm² or double the maximum working pressure, whichever is greater. The pressure shall be applied by means of a manually operated test pump, or, in the case of long mains or mains of a large diameter, by a power-driven test pump, provided the pump is not left unattended. In either case, due precaution shall be taken to ensure that the required test pressure is not exceeded. Pressure gauges shall be accurate and shall preferably have been recalibrated before the test. The pump having been stopped, the test pressure shall maintain itself without measurable loss for at least 5 min. The mains shall be tested in sections as the work of laying proceeds; it is an advantage to have the joints exposed for inspection during the testing. The open end of the main may be temporarily closed for testing under moderate pressure by fitting a water-tight expanding plug of which several types are available. The end of the main and the plug shall be secured by struts or otherwise, to resist the end thrust of the water pressure in the mains.

4.15.2.1 If the section of the main tested terminates into a sluice valve, the wedge of the valve shall not be used to retain the water; instead the valve shall be temporarily fitted with a blank flange, or, in the case of a socketed valve, with a plug, and the wedge placed in the open position while testing. End support shall be given as in [4.15.2](#).

4.15.3 Testing of Service Pipes and Fittings

When the service pipe is complete, it shall be slowly and carefully charged with water, allowing all air to escape, care being taken to avoid all shock or water hammer. The service pipe shall then be inspected under working conditions of pressure and flow. When all draw-offs taps are closed, the service pipe shall be absolutely water-tight. All piping, fittings and appliances shall be checked for satisfactory support, and protection from damage, corrosion and frost. Because of the possibility of damage in transit, cisterns shall be re-tested for water-tightness on arrival at the site, before fixing.

4.15.4 In addition to the provisions given in [4.15.1](#), provisions given in [4.15.4.1](#) to [4.15.4.3](#) shall also apply to hot water supply installations in regard to inspection and testing.

4.15.4.1 Testing of the system after installation

After the hot water system, including the hot water heaters, has been installed, it shall be carefully charged with water, so that all air is expelled from the system. The entire system shall then be hydraulically tested to a pressure of 0.5 N/mm^2 or twice the working pressure, whichever is greater, for a period of at least 30 min after a steady state is reached. The entire installation shall then be inspected visually for leakages, and sweating. All defects found shall be rectified by removing and remaking the particular section. Caulking of threads, hammering and welding of leaking joints shall not be allowed.

4.15.4.2 Hot water testing

After the system has been proved water-tight, the hot water heaters shall be commissioned by connecting the same to the electrical supply. The system shall then be observed for leakage in pipes due to expansion or overheating. The temperature of water at outlets shall be recorded. The thermostats of the appliances shall be checked and adjusted to temperatures specified in [4.14.2.1](#).

4.15.4.3 Electrical connection

For relevant provisions regarding general and safety requirements for household and similar electrical appliances, reference may be made to good practice [E-1(17)]. The metal work of the water heating appliances and installation other than current carrying parts shall be bonded and earthed in conformity with the good practice [E-1(17)]. It should be noted that screwing of an immersion heater into a tank or cylinder cannot be relied upon to effect a low resistance earth connection, a satisfactory separate earthing of heater should be effected.

4.16 Cleaning and Disinfection of the Supply System

4.16.1 All water mains communications pipes, service pipes and pipes used for distribution of water for domestic purposes shall be thoroughly and efficiently disinfected before being taken into use and also after every major repair. The method of disinfection shall be subject to the approval of the Authority. The pipes shall also be periodically cleaned at intervals, depending upon the quality of water, communication pipes and the storage cisterns shall be thoroughly cleaned at least once every year in order to remove any suspended impurities that may have settled in the pipes or the tanks.

4.16.2 Disinfection of Storage Tanks and Down Take Distribution Pipes

The storage tanks and pipes shall first be filled with water and thoroughly flushed out. The storage tank shall then be filled with water again and a disinfecting chemical containing chlorine added gradually while the tanks are being filled, to ensure thorough mixing. Sufficient quantities of chemicals shall be used to give the water a dose of 50 parts of chlorine to one million parts of water. If ordinary bleaching powder is used, the proportions will be 150 g of powder to 1 000 litre of water. The powder shall be mixed with water to a creamy consistency before being added to the water in the storage tank. When the storage tank is full, the supply shall be stopped and all the taps on the distributing pipes opened successively working progressively away from the storage tank. Each tap shall be closed when the water discharged begins to smell of chlorine. The storage tank shall then be topped up with water from the supply pipe and with more disinfecting chemical in the recommended proportions. The storage tank and pipes shall then remain charged for at least 3 h. Finally, the tank and pipes shall be thoroughly flushed out before any water is used for domestic purposes.

4.17 Water Supply Systems in High Altitudes and/or Sub-zero Temperature Regions

4.17.1 Selection and Source

In general, the site selected for a water source shall be such as to minimize the length of transmission line so as to reduce the inspection and upkeep. Attempt shall be made, where feasible, to locate the source near the discharge of waste heat, such as of power plants provided it does not affect the potability of water.

4.17.2 Pumping Installation

Pump and pumping machinery shall be housed inside well-insulated chambers. Where necessary, arrangements shall be made for heating the inside of pump houses. Pump houses, as far as possible, should be built directly above the water intake structures.

4.17.3 Protection of Storage Water and Treatment

Where ambient temperatures are so low as to cause danger of freezing, proper housing, insulation and protection shall be provided for all processes and equipment. If necessary, means shall be provided for proper heating of the enclosure.

4.17.4 Transmission and Distribution

Freezing of the buried pipe may be avoided primarily by laying the pipe below the level of the frost line; well consolidated bedding of clean earth or sand, under, around or over the pipe should be provided. For the efficient operation and design of transmission and distribution work, the available heat in the water shall be economically utilized and controlled. If the heat which is naturally present in water is made equate to satisfy heat losses from the system, the water shall be warmed. Where economically feasible, certain faucets on the distribution system may be kept in a slightly dripping condition so as to keep the fluid in motion and thus prevent is freezing. If found unsuitable for drinking purposes, such water may be used for heating purposes. Heat losses shall be reduced by insulation, if necessary. Any material that will catch, absorb or hold moisture shall not be used for insulation purposes. Adequate number of break pressure water tanks and air release valves shall be provided in the distribution system.

NOTE — The level of frost line is generally found to be between 0.9 m and 1.2 m below ground level in the northern regions of India, wherever freezing occurs.

4.17.4.1 Materials for pipes

Distribution pipes shall be made of any of the following materials conforming to Part B 'Building Materials' of this NBCS:

- a) High density polyethylene pipes;
- b) Asbestos cement pipes;
- c) Galvanized iron pipes;
- d) Cast iron pipes;
- e) Copper pipes;
- f) Chlorinated PVC pipes;
- g) Unplasticized PVC pipes (where it is laid before frost line); and
- h) Stainless steel pipes.

4.17.4.2 Materials for insulation of pipes

Insulation of pipes may be provided by wrapping the pipe with fibre glass, mineral wool or closed cell flexible elastomeric foam based insulation materials of desired thickness covered with cladding for protection from

damage or from weather in case of external use/pipes laid in exposed conditions; other materials, like 85 percent magnesia, preformed pipe sections, etc, may also be used.

4.17.4.3 Distribution methods

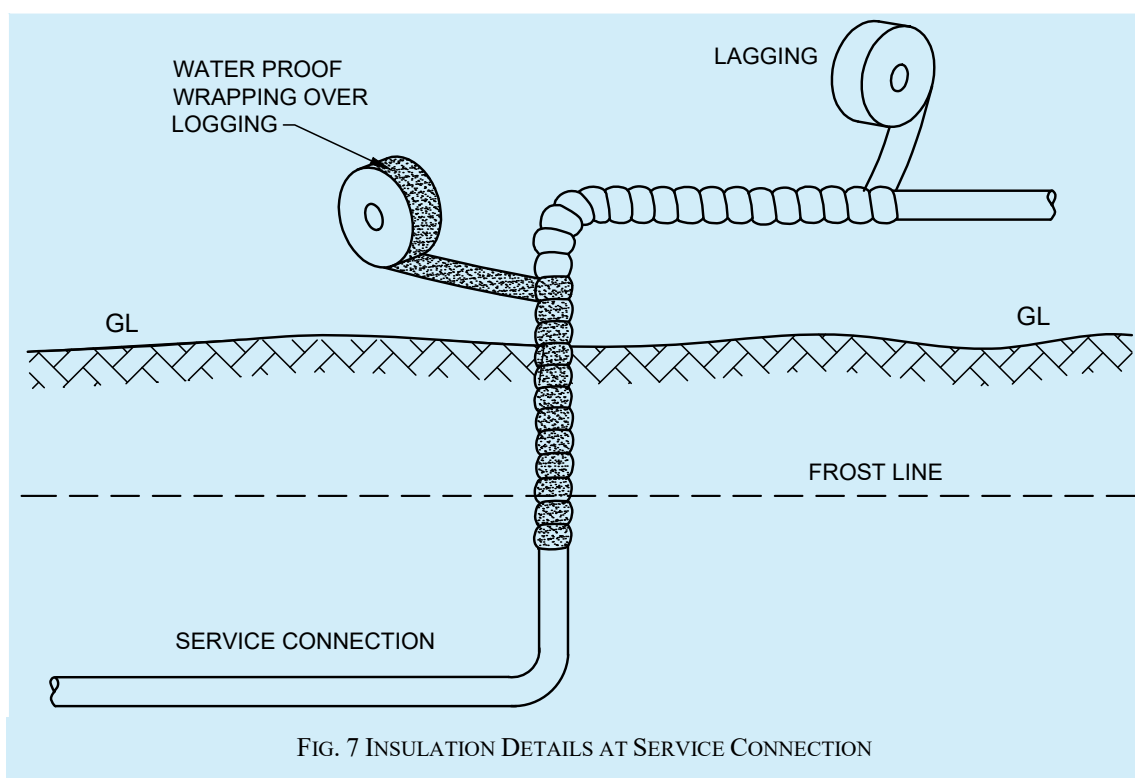
Distribution by barrels or tank trucks shall be employed, where the water requirements are temporary and small. Utmost care shall be exercised for preventing the water from being contaminated by maintaining a residual of disinfecting agent at all times. Hoses, pails and the tank shall be kept free from dust and filth during all period of operation. Where winter temperatures are low, making frost penetration depths greater during the winter and where adequate facilities for heating the water in the distribution system do not exist, the use of tank trucks or barrels for delivery of water shall be considered only for cold weather; during the warm weather, piping system for seasonal use may be supplemented.

4.17.4.4 In the conventional distribution system involving the use of a network of pipelines requiring no auxiliary heat, it is essential that the pipelines are buried well below the frost line. Adequate facilities for draining the pipelines shall be provided where there is a danger of frost.

4.17.4.5 House service connections

House service connections shall be kept operative by the use of adequate insulation at exposed places extending below the frost line. [Figure 7](#) shows a typical arrangement for providing insulation for house service connections.

4.17.5 For detailed information on planning and designing water supply system peculiar to high altitudes and/or sub-zero temperature regions of the country, reference may be made to good practice [E-1(2)].



4.18 Guidelines to Maintenance

4.18.1 Storage tanks shall be regularly inspected and shall be cleaned out periodically, if necessary. Tanks showing signs of corrosion shall be emptied, thoroughly wire brushed to remove loose material (but not scraped), cleaned and coated with suitable bituminous compositions or other suitable anti-corrosive material not liable to impart taste or odour or otherwise contaminate the water. Before cleaning the cistern, the outlets shall be plugged to prevent debris from entering the pipes. Tanks shall be examined for metal wastage and water tightness after cleaning.

4.18.2 Record drawings showing pipe layout and valve positions shall be kept up to date and inspection undertaken to ensure that any maintenance work has not introduced cross-connections or any other undesirable feature. Any addition or alterations to the systems shall be duly recorded from time to time.

4.18.3 Any temporary attachment fixed to a tap or outlet shall never be left in such a position that back-siphonage of polluted water may occur into the supply system.

4.18.4 All valves shall periodically be operated to maintain free movement of the working parts.

4.18.5 All taps and ball valves shall be watertight, glands shall be made good, washers shall be replaced and the mechanism of spring operated taps and ball valves shall be repaired where required.

4.18.6 All overflow pipes shall be examined and kept free from obstructions.

4.18.7 The electrical installation shall be checked for earth continuity and any defects or deficiencies corrected in the case of hot water supply installations.

4.19 Swimming Pools

4.19.1 General

A swimming pool is a container that is filled with water to enable swimming or such other leisure activities. Pools can be sunk into the ground or built above ground (as a freestanding construction or as part of a building or other larger structure), and are also a standard feature aboard ocean liners and cruise ships. In-ground pools are most commonly constructed from materials such as concrete, natural stone, metal, plastic or fiberglass, and can be of a custom size and shape or built to a standardized size.

All pools open to public shall take care of the barrier-free design for accessibility and use by persons with disabilities. Reference shall also be made to good practice [E-1(3)] for applicable requirements.

4.19.1.1 Swimming pools shall be of three types, namely:

<i>Type</i>	<i>Characteristic</i>
Fill and draw	Clear water of potable quality is retained till it becomes turbid or unfit for use. Thereafter, the pool is drained, cleaned and refilled with clear water. This type is not recommended considering water conservation
Flow through	Clear water of hygienic quality flows continuously. This type requires more water for replenishment and so cautious decision of usage of such pools should be made; it is not recommended considering water conservation
Recirculating	Recirculation system shall be provided to minimize water wastage and disinfection shall be done to ensure hygiene. The recirculation system shall be based on the nature of usage such as private, public, wading and competition pools. The entire pool water should be filtered based on the turnover period for various types of pools.

4.19.1.2 Turnover rate

It is the amount of time it would take for the volume of the pool to pass through the filtration system one time. Turnover rate is measured in hours and is expressed as:

$$\text{Turnover rate, in hours} = \frac{\text{Pool volume (in litres)}}{\text{Flow (in litre/h)}}$$

NOTE — Pool volume comprises volume of the pool water, balancing tank, overflow gutter and circulation pipe.

Typical turnover rates for the following pools are:

<i>Sl No.</i>	<i>Type of Pool</i>	<i>Turnover rate</i> <i>h</i>
(1)	(2)	(3)
i)	Swimming pool:	
	a) Private (outdoor);	6
	b) Private (indoor);	10
	c) Public; and	6
	d) Competition.	6
ii)	Wading Pool	1
iii)	Spa (public)	½
	Spa (private)	1

4.19.1.3 Swimming pool user load

The maximum load for the pool shall be:

- a) For pool depth up to 1m : 2.2 m² surface area per user
- b) For pool depth up to 1.5m : 2.7 m² surface area per user
- c) For pool depth more than 1.5m : 4.0 m² surface area per user

4.19.1.4 Recommended dimensions of the swimming pools shall be:

- a) Half sized olympic pools : 25 m (length) × 12 to 13 m (width) × 1.8 m (depth)
- b) Full size olympic pools : 50 m (length) × 25 m (wide) × 2 m (depth)
- c) Diving depth : 3.5 m for 1 m spring board; and
5 m for a 10 m platform
- d) Maximum depth : See [4.19.2.7](#)
- e) Ladders : 2 treads for 1.2 m depth;
3 treads for 1.2 m to 1.5 m depth; and
4 treads for 1.5 m depth
- f) Displacement : 60 litre per user for deeper pools; and
45 litre for shallow pools

4.19.2 Design

4.19.2.1 The pool should be so designed to withstand all anticipated hydraulic structural loadings for both full and empty conditions. All appurtenances to the pool, such as diving boards and slides, shall be designed to carry the anticipated load. Any obstruction creating a safety hazard shall not extend into or above the pool, or shall not protrude from the floor of the pool. Designers may refer good practice [E-1(21)] for providing the guidance on the procedures and the precautions to be taken during construction of swimming pool to ensure the water tightness of the structure.

4.19.2.2 Material

Any suitable material that is non-toxic and provides a rigid watertight shell with a smooth, impervious, light colour finish should be used to construct the pool. The floor of shallow areas shall have a slip-resistant finish. Sand or earth shall not be permitted to use as an interior finish in a swimming pool.

4.19.2.3 Dimensions

The shape and size of a pool largely depend on the usage. Shape should be considered from the standpoint of safety and circulation of the pool water.

NOTES

- 1 A long and rectangular pool may be ideal for sports and exercise as it gives length and breadth, but the shape of a pool for recreation largely depend on the choice of the owners, available space for making the pool and the design of the house.
- 2 In sports, normally the swimming pools are 50 m long, minimum 21 m wide and 1.8 m overall depth.

4.19.2.4 Floor slopes

Slope of the floor of the pool should be made downward toward the main drain. All slopes should be uniform. The slope in shallow areas should not exceed 300 mm vertical in 3.6 m horizontal except for a slope directed downward from a transition point, which shall not exceed 300 mm vertical in 1m horizontal. In portions of the pool with a depth greater than 1.5 m, the front slope of the deep area shall not be steeper than 300 mm in 1 m.

4.19.2.5 Transition point

Transition points should be marked with a stripe on the pool floor having a width of at least 100 mm and a colour that contrasts with that of the floor, and with a buoyed safety rope with colour buoys, installed at least 300 mm on the shallow side of the transition point. In other pools having adjoining shallow and deep areas, a safety rope with colour buoys shall be installed where the water depth reaches 1.5 m.

4.19.2.6 Pool walls

Where the pool depth is 1 m or less, pool walls shall be vertical to the floor and the junction of the wall with the floor shall consist of a cove with a radius not exceeding 150 mm. Where the pool depth exceeds 1 m, pool walls shall meet one of the following criteria:

- a) The wall shall be vertical for a depth of at least 1.5 m below the water level, below which the wall may angle to the floor; or
- b) The wall shall be vertical for a depth of at least 1 m below the water level, below which the wall shall form a curve to the floor. The curve shall be tangent to the pool wall.

4.19.2.7 Water depth

The depth of a swimming pool depends on the purpose of the pool, and whether it is open to the public or strictly for private use. If it is a private casual, relaxing pool, it may go from 1.0 m to 1.5 m deep. If it is a public pool designed for diving, it may slope from 3.5 m to 5.0 m in the deep end. A children's play pool may be from 0.3 m to 0.6 m deep. Public pools may have differing depths to accommodate different swimmer requirements. Water depths may be clearly marked on the pool walls.

The width between handrails of the pool stairs should be between 500 mm and 600 mm.

4.19.2.8 Walkways and deck areas

Pools shall be completely surrounded by a deck that is at least 1.2 m in width and extends completely around and adjacent to the pool. There shall be no obstructions or interruptions of the pool deck within the 1.2 m adjacent to the pool other than necessary structural supports, or appurtenances such as diving boards, slides, perimeter overflow systems, or handrails. A clear, unobstructed walkway at least 1.2 m in width shall be maintained at such obstructions or interruptions. A wheelchair turning space of minimum 1.5m × 1.5 m shall be provided at key places in the walkway.

Structural supports located within the minimum required deck width or within 1.2 m of the swimming pool shall be no closer than 3 m apart measured parallel to the adjacent perimeter of the pool, with the dimension of any single support in a plane parallel to the adjacent pool perimeter not greater than 1 m and the sum of all such support dimensions not greater than 10 percent of the pool perimeter.

The deck between two adjacent swimming pools shall be at least 2.5 m wide. All decks and walkways shall have an unobstructed overhead clearance of at least 2 m.

Synthetic material which meets the following criteria may be installed for deck coverings:

- a) Non-fibrous and allows drainage such that it will not remain wet or retain moisture;
- b) Inert and will not support bacterial or fungal growth;
- c) Durable;
- d) Cleanable; and
- e) Provides a slip-resistant finish.

The decks and walkways shall have a paved surface. The surface of the pool deck and other surfaces used for foot contact, such as gratings of perimeter overflow systems, shall be slip-resistant.

The outer perimeter of the deck for outdoor pools shall be at least 10 cm higher than the surrounding ground surface except where access is provided to adjacent turf areas.

The pool shall have an accessible entry and exit that is accessible to persons with disabilities. Accessible entry/exit may require provision of a transfer wall or/and a sloped entry with an aquatic chair. Installing a pool lifts may be another option for accessible entry/exit.

The transfer wall should have a minimum clear deck space of 1.5 m × 1.5m, with a slope not steeper than 1 : 48 at the base of the transfer wall. The height of the transfer wall should be 400 mm to 480 mm from the deck floor. The transfer wall should be 300 mm to 400 mm wide. The length should be minimum 1.5 m centered on the clear deck space. Surfaces of transfer walls shall not be sharp and shall have rounded edges.

A grab bar should be provided on the transfer wall. Grab bars shall be perpendicular to the pool wall and shall extend the full depth of the transfer wall. The top of the gripping surface shall be 100 mm to 150 mm above transfer walls. Clearance of 610 mm should be available on both sides of the grab bar.

4.19.2.9 Starting platforms

Starting platforms may be from 0.50 m to 0.75 m above the surface of water. The maximum height of the platform above the water shall be 0.75 m where the water depth is 1.2 m or greater, and 0.50 m when the water depth is less than 1.2 m. The surface area of each platform shall be 0.5 m × 0.5 m with a maximum slope of not more than 10°. Surface of each block shall be covered with non-slip material and with back stroke hand grip facility.

4.19.3 Electrical Installation

4.19.3.1 Lighting

Artificial lighting shall be provided at all indoor pools and at all outdoor pools that are open for use after sunset in accordance with one of the following:

- a) Underwater lighting of at least 8.35 lumens or 5.5 W per square metre of pool water surface area, located to provide illumination of the entire pool floor; plus area lighting of at least 10 lumens or 6.6 W per square metre of deck area; and
- b) If underwater lights are not provided, at least 33.5 lumens or 2.2 W per square metre of pool water surface area and deck area.

Where portable electric vacuum cleaning equipment is used, electrical receptacles with ground-fault circuit interrupter protection shall be provided. Separation between receptacles shall be a maximum of 30 m. All receptacles installed in the swimming pool area shall have waterproof covers and ground-fault circuit interrupter protection. Lighting controls should not be accessible to the public.

4.19.3.2 Ventilation

Adequate ventilation shall be provided in facilities to prevent objectionable odor.

4.19.3.3 Shower and bathroom

Separate shower, dressing booth and sanitary facilities shall be provided for each gender. This may not be applicable for schools and other institutional use where a pool may be open to one gender at a time. The rooms should be well lit, drained, ventilated, and of good construction, using impervious materials. They should be developed and planned to ensure maintenance of good sanitation throughout the building at all times. Floors should have a slip-resistant surface and sufficiently smooth to ensure ease in cleaning.

Each shower and toilet block provided per pool, open for use by public, shall have a unisex accessible shower, dressing-cum-toilet facility for persons with disabilities. Design specifications of this accessible facility shall be as per good practice [E-1(3)].

4.19.4 Water Treatment System

4.19.4.1 General

A water treatment system shall be provided to filter, chemically balance and disinfect the swimming pool water.

4.19.4.2 Hair and lint strainer

A hair and lint strainer shall be installed on the suction side of the pump except on vacuum filter systems. The strainer basket shall be easily removable. Valves shall be installed to allow the flow to be shut off during cleaning, switching baskets or inspection.

4.19.4.3 Inlets

Inlets for filtered water shall be located and directed suitably to produce uniform circulation of water to facilitate the maintenance of a uniform disinfectant residual throughout the entire pool without the existence of dead spots, and to produce surface flow patterns that effectively assist skimming.

4.19.4.4 Outlets

Pools shall be provided with a minimum of two drains (outlets) at the deepest point. Centre-to-centre distance between drains shall not exceed 2.0 m. Drains shall not be more than 3.0 m away from the pool walls. The main drain may be connected to the recirculation system. Openings shall be covered by grating which cannot be removed without the use of tools. Openings of the grating shall be at least four times the area of the main drain pipe or have an open area. The maximum width of grate openings shall not exceed 8 mm. Main drains and all other suction outlets installed in the pool shall be designed to prevent bather entrapment and shall be of anti-vortex type. The velocity at outlet pipe shall not be more than 0.3 m/s.

4.19.4.5 Velocities

Maximum permissible velocities for various components of the pools are as follows:

- a) Suction pipe: less than 1.5 m/s;
- b) Return pipe: 1.5 to 2 m/s; and
- c) Return/inlet fittings: 2.4 m/s to 2.75 m/s in private pools and 1.5 m/s to 2 m/s in public pools.

4.19.4.6 Balancing (surge) tank

Overflow system shall be designed for effective surge capacity. Balancing tank should be provided of capacity to accommodate surge storage and storage for makeup water. The volume of the balancing tank shall be the total of the above with equal proportions.

4.19.4.7 Make-up water

Make-up water shall be added through a fixed air gap of at least 150 mm to the pool, surge tank, vacuum filter tank, or other receptacle. When make-up water is added directly to the pool, the fill-spout should be located under a low diving board or immediately adjacent to a ladder rail, grab rail, or fixed lifeguard chair.

4.19.4.8 Filtration

The design filtration rate in the particular application in which the filter is utilized shall not exceed the maximum design filtration rate for which the filter was installed. Wash or backwash water from diatomaceous earth filters shall be passed through a separation tank designed for removal of suspended diatomaceous earth and solids, prior to disposal.

NOTE — Filtration rates for various type of filters:

- a) Low rate filter — $10 \text{ m}^3/\text{m}^2/\text{h}$.
- b) Medium rate filter — $11 \text{ to } 30 \text{ m}^3/\text{m}^2/\text{h}$.
- c) High rate filter — $31 \text{ to } 50 \text{ m}^3/\text{m}^2/\text{h}$.
- d) Minimum flow rate for filter backwash — $30 \text{ m}^3/\text{m}^2/\text{h}$.

4.19.4.9 Disinfection

The pool water shall be continuously disinfected by suitable disinfecting agent that imparts easily measured residual. Gaseous chlorine, chlorine compounds, bromine compounds or other bactericidal agents should be used to maintain the quality parameters of water.

4.19.5 Water Quality

4.19.5.1 Disinfectant residual

Where chlorine is used as a disinfectant, the chlorine residual shall be maintained between 1.0 ppm and 4.0 ppm as free chlorine residual. A free chlorine residual of at least 2.0 ppm shall be maintained when the pool water temperature exceeds 30°C .

Where bromine is used as a disinfectant, a bromine residual shall be maintained between 2.0 ppm and 8.0 ppm as total bromine. A bromine residual of at least 4.0 ppm shall be maintained when the pool water temperature exceeds 30°C .

Where chlorinated cyanurates are used, the cyanuric acid concentration shall not exceed 100 ppm.

Where silver/copper or copper ion generators are used, the concentration of copper shall not exceed 1.3 ppm and the concentration of silver shall not exceed 0.05 ppm.

Where ozone is used, the ambient air ozone concentration shall be less than 0.1 ppm at all times either in the vicinity of the ozonator or at the pool water surface.

For all other physical, chemical and bacteriological parameters, the quality of water used in swimming pools in continuous circulation type shall conform to good practice [E-1(22)]. For safety aspects related to public swimming pools, the good practice [E-1(23)] shall be referred.

4.20 Allowance for Expansion

4.20.1 The allowances for expansion of the water pipes are recommended as given below:

- a) All pipes should be installed at ambient temperature;
- b) Pipes carrying hot fluids such as water or steam operate at higher temperatures. It follows that they expand, especially in length, with an increase from ambient to working temperatures. This will create stress upon certain areas within the distribution system, such as pipe joints, which, in the extreme, could cause fracture;
- c) The pipe work system shall be sufficiently flexible to accommodate the movements of the components as they expand. In many cases, the flexibility of the pipe work system, due to the length of the pipe and number of bends and supports, means that no undue stresses are imposed. In other installations, however, it will be necessary to incorporate some means of achieving this required flexibility;
- d) The expansion fitting is one method of accommodating expansion. These fittings are placed within a line and are designed to accommodate the expansion without the total length of the line changing. They are commonly called expansion bellows, due to the bellows construction of the expansion sleeve; and

- e) Other expansion fittings can be made from the pipe work itself. This can be a cheaper way to solve the problem, but more space is needed to accommodate the pipe.

4.20.2 Full Loop

- a) This is simply one complete turn of the pipe and, on steam pipe work, should preferably be fitted in a horizontal rather than a vertical position to prevent condensate accumulating on the upstream side;
- b) The downstream side passes below the upstream side and great care shall be taken that it is not fitted wrong way round, as condensate can accumulate in the bottom. When full loops are to be fitted in a confined space, care shall be taken to specify that wrong-handed loops are not supplied; and
- c) The full loop does not produce a force in opposition to the expanding pipe work as in some other types, but with steam pressure inside the loop, there is a slight tendency to unwind, which puts an additional stress on the flanges.

4.21 Colour Codes for Different Types of Water Pipes

The following colour codes are recommended for pipes {see also good practice [E-1(24)]}:

<i>Sl No.</i>	<i>Contents</i>	<i>Ground Colour</i>	<i>First Colour Band</i>	<i>Second Colour Band</i>
(1)	(2)	(3)	(4)	(5)
i)	Cooling water	Sea green	French blue	—
ii)	Boiler feed water	Sea green	Gulf red	—
iii)	Condensate	Sea green	Light brown	—
iv)	Drinking water	Sea green	French blue	Signal red
v)	Treated water	Sea green	Light orange	—
vi)	Fire water	Fire red	Crimson red	—
vii)	Central heating below 60 °C	Sea green	Canary yellow	—
viii)	Central heating 60 °C to 100 °C	Sea green	Dark violet	—
ix)	Central heating above 100 °C	Sea green	Dark violet	Signal red
x)	Cold water down service from storage tanks	Sea green	French blue	Canary yellow
xi)	Domestic, hot	Sea green	Light grey	—
xii)	Hydraulic power	Sea green	Black	—
xiii)	Sea, river, untreated	Sea green	White	—
xiv)	Filtered water	Sea green	Light brown	—
xv)	Soft water	Sea green	Light brown	Signal red
xvi)	Warm water	Sea green	Light grey	Canary yellow
xvii)	Chilled water	Sea green	Black	Canary yellow
xviii)	Sprinkle and hydrant water	Sea green	White	Signal red
xix)	Waste water	Sea green	Canary yellow	Signal red
xx)	Demineralized water	Sea green	Gulf red	—
xxi)	Process water	Sea green	Oxide red	—

<i>Sl No.</i>	<i>Contents</i>	<i>Ground Colour</i>	<i>First Colour Band</i>	<i>Second Colour Band</i>
(1)	(2)	(3)	(4)	(5)
xxii)	Wash water	Sea green	Canary yellow	—
xxiii)	Quench water	Sea green	Dark grey	—

ANNEX A

(Clause [3.2.1](#))

APPLICATION FORM FOR TEMPORARY/PERMANENT SUPPLY OF WATER/FOR ADDITIONS AND/OR ALTERATIONS FOR SUPPLY OF WATER

I/Wehereby make application to the* for the temporary/permanent supply of water for the following additions and/or alterations to the water supply requirements and water fittings at the premises..... Ward No.Street No.Road/Street known asfor the purpose described below and agreed to pay such charges as the Authority may from time to time be entitled to make and to conform to all their byelaws and regulations.....licensed/registered plumber, has been instructed by me/us to carry out the plumbing work.

Description of the premises:.....
Address :.....
Purpose for which water is required.....
.....

The connection/connections taken by me/us for temporary use, shall not be used by me/us for permanent supply unless such a permission is granted to me/us in writing by the Authority.

I/We hereby undertake to give the*..... due notice of any additions or alterations to the above mentioned supply which I/we may desire to make.

My/Our requirements of water supply are as under:

- a) I/We request that one connection be granted for the whole of the premises.
- b) I/We request that separate connections may be granted for each floor and I/we undertake to pay the cost of the separate connections.
- c) My/Our probable requirements for trade purpose arelitre per day and for domestic purposes are.....litre per day.
- d) Our existing supply islitre per day. Our additional requirement of supply islitre per day.
- e) The details as regards proposed additions and alterations in fittings are as follows:
.....
.....

Signature of the licensed/registered plumber.....
Name and address of the licensed/registered
Plumber.....

Signature of the applicant.....
Name and address of the
Applicant

Date

Date

NOTES

1 Please strike out whatever is not applicable.

2 The application should be signed by the owner of the premises or his constituted attorney and shall be countersigned by the licensed/registered plumber.

* Insert here the name of the Authority.

ANNEX B

(Clause 3.2.3)

FORM FOR LICENSED/REGISTERED PLUMBER'S COMPLETION CERTIFICATE

Certified that I/we have completed the plumbing work of water connection No..... for the premises as detailed below. This may be inspected and connection given.

Ward No..... Road/Street.....

Locality.....

Block No..... House No.....

Existing water connection No. (if any).....

Owned by.....

Owner's address.....

Applicant's name..... son of.....

Address.....

Situation.....

Size of main.....on..... street

Where main is situated.....

Size of service pipe.....

Size of ferrule.....

No. of taps..... No. of closets.....

No. of other fittings and appliances.....

Road cutting and repairing fee

Paid Rs.....(Receipt No.....dated.....) (receipt enclosed)

Dated

Signature of licensed/registered plumber.....
Name and address of the licensed/registered plumber
.....

The Authority's Report

Certified that the communication and distribution pipes and all water fittings have been laid, applied and executed in accordance with the provisions of bye-laws, and satisfactory arrangements have been made for draining off waste water.

Connection will be made on.....

Date.....

The Authority.....

ANNEX C

(Clause [4.7.4](#))

NOMOGRAM OF HAZEN AND WILLIAM'S EQUATION

C-1 Examples of the use of nomogram are given below:

Example 1

Find the total friction loss in 25 mm diameter G.I. Pipe discharging 0.25 l/s in a total length of 300 m.

Procedure

Discharge, $Q = 0.25$ l/s

Pipe diameter = 25 mm

Frictional loss (from nomogram) = 30 m per 1 000 m

Total friction loss in 300 m length

$$= \frac{30}{1\,000} \times 300 = 9\,m$$

Example 2

Find suitable diameter pipe to carry 15 l/s from service line to overhead tank.

Total length of service main = 200 m

Residual pressure available at the take off point on supply line is 15 m.

Procedure

Available head = 15 m

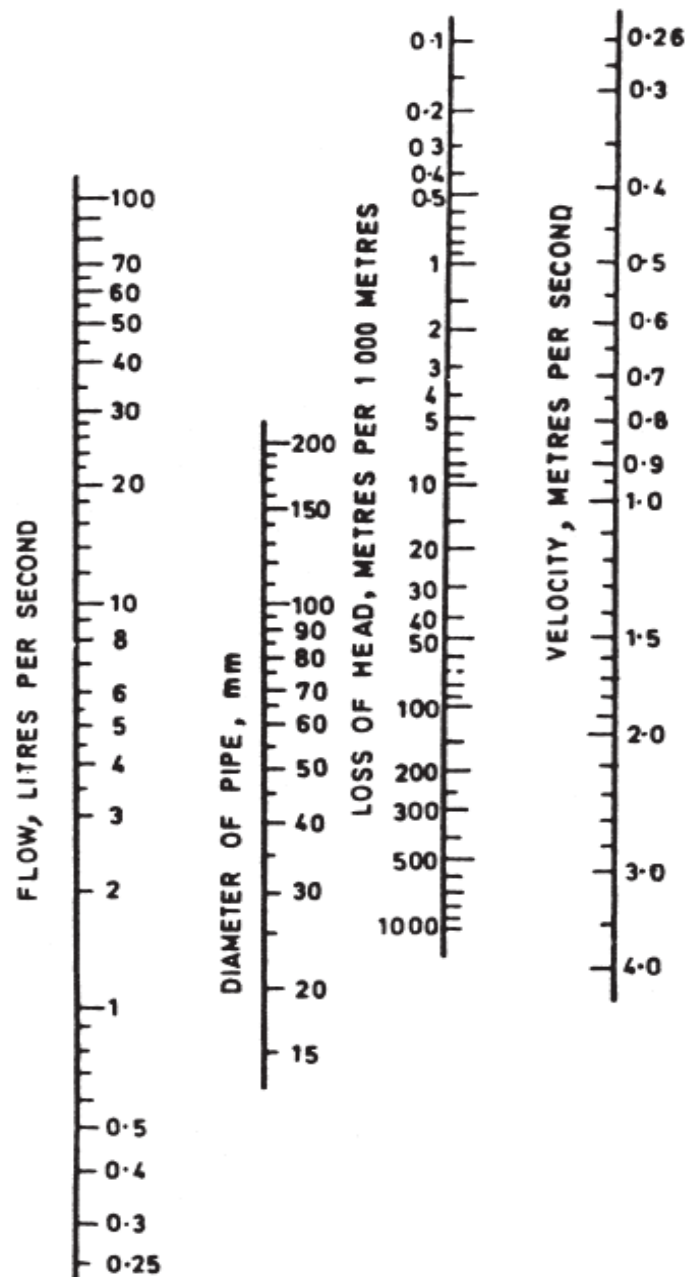
Deduct residual head = 2 m

Deduct 10 percent for losses in bends and specials = 1.3 m

Friction head available for loss in pipe of 1 000 m

$$= \frac{11.7}{200} \times 1\,000 = 58.5\,m\,per\,1\,000\,m$$

From the following nomogram, for a discharge of 15 l/s and friction loss of 58.5 m per 1 000 m, nearest commercial size (diameter) of pipe is 100 mm.



*Nomogram of Hazen & Williams
Equation ($c = 100$)*

LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfilment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used as a guide in conformance with the requirements of the referred clauses in this NBCS.

	<i>IS No.</i>	<i>Title</i>
(1)	IS 10446 : 1983	Glossary of terms relating to water supply and sanitation
(2)	IS 6295 : 1986	Code of practice for water supply and drainage in high altitudes and/or sub-zero temperature regions (<i>first revision</i>)
(3)	IS 4963 : 2025	Accessibility in built environment for older adults and persons with disabilities — Requirements (<i>second revision</i>)
(4)	IS 11208 :1985	Guidelines for registration of plumbers
(5)	IS 17650	Water efficient plumbing products — Requirements
	(Part 1) : 2021	Part 1 Sanitaryware
	(Part 2) : 2021	Part 2 Sanitary fittings
(6)	IS 10500 : 2012	Specification for drinking water (<i>second revision</i>)
(7)	IS 2041 : 2024	Steel plates and strips for pressure vessels used at moderate and low temperature — Specification (<i>fourth revision</i>)
	IS 804 : 1967	Specification for rectangular pressed steel tanks (<i>first revision</i>)
(8)	IS 4984 : 2016	Polyethylene pipes for water supply — Specification (<i>fifth revision</i>)
	IS 4985 : 2021	Unplasticized PVC pipes for water supplies — Specification (<i>fourth revision</i>)
	IS 15778 : 2007	Chlorinated polyvinyl chloride (CPVC) pipes for portable hot and cold water distribution supplies — Specification
(9)	IS 2065 : 1983	Code of practice for water supply in buildings (<i>second revision</i>)
(10)	IS 3114 : 1994	Code of practice for laying of cast iron pipes (<i>second revision</i>)
(11)	IS 5822 : 1994	Code of practice for laying of electrically welded steel pipes for water supply (<i>second revision</i>)
(12)	IS 6530 : 1972	Code of practice for laying of asbestos cement pressure pipes
(13)	IS 783 : 1985	Code of practice for laying of concrete pipes (<i>first revision</i>)
(14)	IS 7634	Code of practice for plastics pipe work for potable waters supplies
	(Part 1) : 1975	Choice of materials and general recommendations
	(Part 2) : 2012	Laying and jointing of polyethylene (PE) pipes (<i>first revision</i>)
	(Part 3) : 2003	Laying and jointing of UPVC pipes (<i>first revision</i>)
(15)	IS 783 : 1985	Code of practice for laying of concrete pipes (<i>first revision</i>)
	IS 3114 : 1994	Code of practice for laying of cast iron pipes (<i>second revision</i>)
	IS 5822 : 1994	Code of practice for laying of welded steel pipes for water supply (<i>second revision</i>)
	IS 6530 : 1972	Code of practice for laying of asbestos cement pressure pipes
	IS 7634	Code of practice for plastics pipe work for potable water supplies
	(Part 1) : 1975	Choice of materials and general recommendations
	(Part 2) : 2012	Laying and jointing of polyethylene (PE) pipes (<i>first revision</i>)
	(Part 3) : 2003	Laying and jointing of UPVC pipes (<i>first revision</i>)

	<i>IS No.</i>	<i>Title</i>
(16)	IS 2692 : 1989	Ferrules for water services — Specification (<i>second revision</i>)
(17)	IS 302 (Part 1) : 2024 /IEC 60335-1 : 2020	Household and similar electrical appliances — Safety: Part 1 General requirements (<i>seventh revision</i>)
	IS 2082 : 2018	Stationary storage type electric water heaters — Specification (<i>fifth revision</i>)
(18)	IS 17150 : 2025	Mini domestic water heater for use with piped natural gas (PNG) — Specification (<i>first revision</i>)
(19)	IS 7558 : 1974	Code of practice for domestic hot water installations
	IS 16093 : 2013	Installation of gas based instantaneous water heater — Code of practice
(20)	IS 12976 : 2023	Solar water heating systems — Code of practice (<i>first revision</i>)
(21)	IS 6494 : 1988	Code of practice for water-proofing of underground water reservoirs and swimming pools (<i>first revision</i>)
(22)	IS 3328 : 2026	Quality requirements of water for swimming pools (<i>second revision</i>)
(23)	IS 16508 : 2017	Swimming pool for public — Code of safety
(24)	IS 2379 : 2024	Colour code for pipeline identification — Code of practice (<i>second revision</i>)

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART E PLUMBING SERVICES (INCLUDING SOLID WASTE MANAGEMENT)

SECTION 2 DRAINAGE AND SANITATION

BUREAU OF INDIAN STANDARDS

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National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part E/Section 2) covers the drainage and sanitation requirements of buildings, design, layout, construction and maintenance of drains inside buildings and from the buildings up to the connection to public sewer, private sewer, individual sewage disposal system, cesspool, or to other approved point of disposal/treatment work. It also covers drainage systems peculiar to high altitudes and/or sub-zero temperature regions of the country.

In the first version of the Code formulated in 1970, as part of the National Building Code of India, three separate sections of this Part on 'Plumbing Services', were brought out, namely, Section 1 'Water Supply', Section 2 'Drainage and Sanitation', and Section 3 'Gas Supply'. These sections were subsequently revised in 1983.

The major changes incorporated in the first revision in Section 2 'Drainage and Sanitation' were rationalization of definitions; modification of the requirements for fitments for drainage and sanitation in the case of buildings other than residences; addition of table for sanitation facilities in fruit and vegetable markets; introduction of table giving detailed guidance regarding the selection of plumbing system, depending on the nature of drainage load in buildings and height of buildings; amplification of provision relating to safeguards to be adopted in single stack system; modification of the values of gradients, pipe sizes and the corresponding discharges; rationalization of sizes of manholes/inspection chambers; modification of the sizing of rainwater pipe for roof drainage, to take into account rainfall intensities and recommend sizes on a more rational basis; addition of provisions for drainage and sanitation system peculiar to high altitudes and/or sub-zero temperature regions of the country; and inclusion of requirements of the refuse chute system.

As a result of experience gained in implementation of 1983 version of the Code and feedback received as well as revision of some of the standards based on which this Section was prepared, a need to revise this Section was felt. The second revision of 2005 was therefore prepared to take care of these. In the revision, the erstwhile two Sections were merged and a combined and comprehensive section, namely Section 1 Water Supply, Drainage and Sanitation (including solid waste management), was brought out. Gas supply was brought out as Section 2. Following significant changes were incorporated in the second revision of Section 1 on Water Supply, Drainage and Sanitation, in respect to drainage and sanitation: rationalization and addition of new definitions under terminology; enunciation of certain basic principles for water supply and drainage; addition of a new clause on sanitary appliances; updation of Table 1 to Table 14 of the existing version, regarding drainage and sanitation requirement; addition of additional requirements under layout clause of design considerations; modification and rationalization of provisions regarding choice of plumbing systems; addition of new clause on drain appurtenances having details on trap, floor drain and cleanout; incorporation of provisions on indirect wastes, special wastes (covering laboratory wastes, infected wastes, research laboratory wastes, etc), grease traps, oil interceptors, radio-active wastes, etc; revision of manhole details on size and enhancement of construction clause; inclusion of provisions on rainwater harvesting; revision of the minimum rainfall intensity which is drain design basis for discharge of storm water drain into a public storm water drain, to 50 mm/h; modification of the table for sizing of rainwater pipes for roof drainage, with inclusion of rainfall data which were not available in the earlier version; inclusion of figure on detail of subsoil drainage; and addition of details on support/protection of pipes. This revision also incorporated for the first time the provisions on solid waste management.

In the third revision in 2016, to comprehensively address the various and distinct features related to the plumbing aspects, this Part on 'Plumbing Services' was rearranged as follows:

- Section 1 Water supply
- Section 2 Drainage and sanitation
- Section 3 Solid waste management
- Section 4 Gas supply

In the 2016 version of this Section 2 on Drainage and Sanitation, several significant updates were incorporated to address contemporary needs and enhance the comprehensiveness of the provisions. Various tables on the requirements for fitments for drainage and sanitation across different occupancies were updated, including a new

table specifically for shopping malls and retail buildings. Enabling provisions were introduced for the use of corrugated pipes, low-noise pipes and under-slung pipes, alongside information on bio-toilets. Typical mounting arrangements for plumbing fixtures and drainage systems were illustrated and a new table for single-stack sizing was included. Key updates also focused on drainage and sanitation system layouts, venting systems for high-rise buildings with detailed tables, and the gradients of drainage pipes. Fixture unit tables and maximum connections to branches, stacks, and sewers were revised, and new provisions for high-rise building drainage systems were added. Provisions for manhole covers, stormwater runoff estimation, rainwater pipe sizing, siphonic drainage, rainwater harvesting and artificial groundwater recharge were included, along with recommendations for septic tanks.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

In this revision, following significant changes have been incorporated:

- a) Key terms along with their definition, such as emergency floor drain, flood level rim and yoke vent, have been included to enhance clarity and technical precisions in plumbing systems;
- b) Tables on requirements for fitments in drainage and sanitation in various building occupancies, have been updated;
- c) Provisions have been added for managing the horizontal and vertical distances of sanitary fixtures and connections to drainage stacks to ensure efficiency and prevent blockages;
- d) Guidelines for packaged sewage treatment plant (PSTP) have been included to facilitate on-site treatment of grey and black water, particularly in areas without centralized sewer systems; and
- e) Biodigesters have been included as a sustainable, non-sewer on-site waste treatment solution.

This Section is largely based on the following Indian Standards:

IS 1742 : 1983	Code of practice for building drainage (<i>second revision</i>)
IS 4111 (Part 1) : 1986	Code of practice for ancillary structures in sewage system: Part 1 Manholes (<i>first revision</i>)
IS 5329 : 1983	Code of practice for sanitary pipe work above ground for buildings (<i>first revision</i>)
IS 6295 : 1986	Code of practice for water supply and drainage in high altitudes and/or sub-zero temperature regions (<i>first revision</i>)

A reference to SP 35 : 1987 'Handbook on Water Supply and Drainage' may be useful, from where also, assistance has been derived.

All standards, whether given herein above or cross-referred to in the main text of this Section, are subject to revision. The parties to agreement based on this section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the formulation of this Section, reference has also been made to the following:

International Plumbing Code 2024, International Code Council, and

Uniform Illustrated Plumbing Code – India (Volume I and Volume II) 2022, IAPMO India Private Limited

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Section is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Section.

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Important Explanatory Note for Users of this NBCS

In any Part/Section of this NBCS, where reference is made to ‘**good practice**’ in relation to **design, constructional procedures or other related information**, and where reference is made to “**accepted standard**” in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

- a) Good practice [E-2(1)] refers to the Indian Standard(s) given at serial number (1) of the list of standards given at the end of this Part E/Section 2, that is, IS 10446 : 1983 ‘Glossary of terms relating to water supply and sanitation’.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART E PLUMBING SERVICES (INCLUDING SOLID WASTE MANAGEMENT)

SECTION 2 DRAINAGE AND SANITATION

1 SCOPE

1.1 This NBCS (Part E/Section 2) covers the design, layout, construction and maintenance of drains for foul water, surface water, subsoil water and sewage together with all ancillary works, such as connections, manholes and inspection chambers used within the building and from building to the connection to a public sewer, private sewer, individual sewage-disposal system, cess-pool, soakaway or to other approved point of disposal/treatment work.

NOTE — A sanitary drainage system consists of a building sewer, a building drain, a soil and/or waste stack, horizontal branches or fixture drain and vents. The sanitary drainage of a large building may have a number of primary and secondary branches and several soil and/or waste stacks, each of them in turn may have a number of horizontal branches.

2 TERMINOLOGY

For the purpose of this Section, the following definitions shall apply in addition to the definitions given in accepted standards [E-2(1)].

2.1 Air Admittance Valve — One way valves designed to allow air to enter drainage system when negative pressure develops in the system. The purpose is to provide a method of allowing air to enter the drainage system for prevention of siphonage of traps.

2.2 Air Break — Physical separation which may be a low inlet into the indirect waste receptor from the fixture or device indirectly connected.

2.3 Air Gap, Drainage — Unobstructed vertical distance through the free atmosphere between the lowest opening from any pipe, plumbing fixture or appurtenance conveying waste to flood level of the receptor.

2.4 Back Up — A condition where the waste water may flow back into another fixture or compartment but not back into the potable water system.

2.5 Back Water Valve — Device installed in a drainage system to prevent reverse flow.

2.6 Bath Room Group — Group of fixtures consisting of water closet, lavatories bath tub or shower and other fittings with a floor drain located together.

2.7 Battery of Fixtures — Any group of two or more similar adjacent fixtures which discharge into a common horizontal waste or soil pipe.

2.8 Bed Pan Washer — A washer which is to be used for disposing human waste and sterilizing the bed pan for reuse. Bed pan washer requires water connection and electrical heating arrangement. This is to be considered as soil appliance.

2.9 Benching — Sloping surfaces constructed on either side of channels at the base of a manhole or inspection chamber for the purpose of confining the flow of sewage, avoiding the accumulation of deposits and providing a safe working platform.

2.10 Black Water — Waste water from water closet, bidet, urinals, kitchen sink and bed pan sink or similar contaminated sources.

2.11 Branch

- a) Special form of sewer pipe used for making connections to a sewer or water main. The various types are called 'T', 'Y', 'T-Y', double Y and V branches, according to their respective shapes; and
- b) Any part of a piping system other than a main or stack.

2.12 Branch Soil Pipe (BSP) — A pipe connecting one or more soil appliances to the main soil pipe.

2.13 Branch Soil Waste Pipe (BSWP) — A pipe connecting one or more soil and/or waste appliances to the main soil waste pipe (one pipe system).

2.14 Branch Vent Pipe (BVP) — A pipe, one end of which is connected to the system adjacent to the trap of an appliance and the other to a main vent pipe or a drain-vent pipe. It is fitted to prevent loss of water seal from a trap owing to partial vacuum, back-pressure, or surging caused by air movement within the pipe system. It also provides ventilation for the branch waste pipe.

2.15 Branch Waste Pipe (BWP) — A pipe connecting one or more waste appliances to the main waste pipe.

2.16 Building Drain-Combined — A building drain which conveys both sewage and storm water or other drainage.

2.17 Building Drain-Sanitary — A building drain which conveys sewage and sullage only.

2.18 Building Drain-Storm — A building drain which conveys storm water or other drainage but no sewage or sullage.

2.19 Building Sewer — That part of the horizontal piping of a drainage system which extends from the end of the building drain and which receives the discharge of the building drain and conveys it to a public sewer, private sewer, individual sewage-disposal system or approved point of disposal.

2.20 Building Sub-Drain — That portion of a drainage system which cannot drain by gravity in the building sewer.

2.21 Building Trap — A device, fitting or assembly of fittings installed in the building drain to prevent circulation of air between the drainage of the building and the building sewer. It is usually installed as running trap.

2.22 Cesspool

- a) An underground chamber for the reception and storage of foul water, the contents of which are periodically removed for disposal; and
- b) A box-shaped receiver constructed in a roof or gutter for collecting rainwater which then passes into a rainwater pipe connected thereto.

2.23 Cleaning Eye — An access opening in a pipe or pipe fitting arranged to facilitate the cleaning of obstructions and fitted with removable cover.

2.24 Clear Waste Water — Cooling water and condensate drainage from refrigeration and air conditioning equipment, cooled condensate from steam heating systems, cooled boiler blow-down water, waste water drainage from equipment rooms and other areas where water is used without an appreciable addition of oil, gasoline, solvent, acid, etc, and treated effluent in which impurities have been reduced below a minimum concentration considered harmful.

2.25 Collection Chamber — A compartment situated at the lower end of the chute for collecting and housing the refuse during the period between two successive cleanings.

2.26 Connection — The junction of a foul water drain, surface water drain or sewer from building or building with public sewer treatment works, public sewer, private sewer, individual sewage-disposal system, cess-pool, soakaway or to other approved point of disposal/treatment work.

- 2.27 Consumer** — Any person who discharges waste or sewage to the public sewer installed by the Authority.
- 2.28 Crown of Trap** — The topmost point of the inside of a trap outlet.
- 2.29 Deep Manhole** — A manhole of such depth that an access shaft is required in addition to the working chamber.
- 2.30 Depth of Manhole** — The vertical distance from the top of the manhole cover to the outgoing invert of the main drain channel.
- 2.31 Developed Length** — The length measured along the centre line of a pipe and fittings.
- 2.32 Diameter** — The nominal internal diameter of pipes and fittings.
- 2.33 Drain** — A conduit, channel or pipe for the carriage of storm water, sewage, waste water or other water-borne wastes in a building drainage system.
- 2.34 Drain Vent Pipe (DVP)** — A pipe installed to provide flow of air to or from a drain to prevent undue concentration of foul air in the drain. The main soil pipe or main waste pipe may serve as drain vent pipe wherever their upper portions, which do not receive discharges, are extended to the roof level and let open to air.
- 2.35 Drainage** — The removal of any liquid by a system constructed for the purpose.
- 2.36 Drainage Fixture Unit (DFU)** — A measure of probable discharge into the drainage system by various types of plumbing fixtures. The drainage fixture unit value for a particular fixture depends on its volume rate of drainage discharge, on the time duration of a single drainage operation and on the average time between successive operations.
- 2.37 Drainage Work** — The design and construction of a system of drainage.
- 2.38 Drop Connection** — A length of conduit installed vertically immediately before its connection to a sewer or to another drain.
- 2.39 Drop Manhole** — A manhole installed in a sewer where the elevation of the incoming sewer considerably exceeds that of the outgoing sewer; a vertical waterway outside the manhole is provided to divert the waste from the upper to the lower level so that it does not fall freely into the manhole except at peak rate of flow.
- 2.40 Eco-Toilet (or Bio-toilet)** — The dry toilet that uses a predominantly aerobic processing system to treat human excreta, by composting or managed-aerobic decomposition. It generally uses little to no water and may be used as an alternative to flush toilets.
- 2.41 Emergency Floor Drain** — A floor drain that does not receive the discharge of any drain or indirect waste pipe, and that protects against damage from accidental spills, fixture overflows and leakage.
- 2.42 Fittings** — The appurtenances such as coupling, flange, branch, bend, tees, elbows, unions, waste (with or without plug), P or S trap (with or without vent), stop ferrule, bib tap, pillar tap, bath faucet, water meter, garden hydrant, valves and any other article used in connection with water supply, drainage and sanitation.
- 2.43 Fixture Unit** — A quantity in terms of which the load producing effects on the plumbing system of different kinds of plumbing fixtures is expressed on some arbitrarily chosen scale.
- 2.44 Formation** — The finished level of the excavation at the bottom of a trench or heading prepared to receive the permanent work.
- 2.45 French Drain or Rubble Drain** — A shallow trench filled with coarse rubble, clinker, or similar material with or without field drain pipes.
- 2.46 Frost Line** — The line joining the points of greatest depths below ground level up to which the moisture in the soil freezes.

2.47 Flood Level Rim — The edge of the receptacle from which water overflows.

2.48 Grease Interceptor (or Grease Trap) — A chamber, on the line of a drain or waste pipe, for preventing grease from passing into the drainage system. It reduces non-petroleum fats, oils and grease in effluent by separation and volume reduction to improve sewage treatment efficiency.

As it is used to intercept fats, oils and grease (FOG) from a waste water discharge, it is also known as FOG interceptor.

2.49 Grey Water — Waste water that has not encountered water closet, kitchen sink waste, or similar contaminated sources. Grey water includes waste water from bathtubs, showers, lavatories, clothes washers, dishwashers and domestic reverse osmosis (RO) reject water.

2.50 Gully Chamber — The chamber built of masonry round a gully trap for housing the same.

2.51 Gully Trap — A trap provided in a drainage system with a water seal fixed in a suitable position to collect wastewater from the scullery, kitchen sink, wash basins, baths and rainwater pipes.

2.52 High Altitudes — Elevations higher than 1 500 m above mean sea level (MSL).

2.53 Highway Authority — The public body in which is vested, or which is the owner of, a highway repairable by the inhabitants collectively; otherwise the body or persons responsible for the upkeep of the highway.

2.54 Horizontal Pipe — Any pipe of fitting which makes an angle of more than 45° with the vertical.

2.55 Indirect Waste Pipe — The pipe that does not connect directly with the drainage system but conveys liquid wastes by discharging into a plumbing fixture/interceptor that is directly connected to the drainage system.

2.56 Inlet Fittings — An arrangement of connecting the internal waste branch pipe from wash basin, sinks and shower drains to the main deep seal trap with the help of hopper extension.

2.57 Inlet Hopper — A receptacle fitting for receiving refuse from each floor and dropping it into the chute.

2.58 Insanitary — Condition that is contrary to sanitary principles or is injurious to health.

2.59 Inspection Chamber — A water-tight chamber constructed in any house-drainage system which takes wastes from gully traps and disposes to manhole with access for inspection and maintenance.

2.60 Interceptor — A device designed and installed so as to separate and retain deleterious, hazardous or undesirable matter from normal wastes and permit normal sewage or liquid wastes to discharge into the disposal terminal by gravity.

2.61 Interceptor Manhole or Interceptor Chamber — A manhole incorporating an intercepting trap and providing means of access thereto.

2.62 Invert — The lowest point of the internal surface of a pipe or channel at any cross section.

2.63 Junction Pipe — A pipe incorporating one or more branches.

2.64 Lagging — Thermal insulation or pipes.

2.65 Licensed (or Registered) Plumber — A person licensed (or registered) under the provisions of this NBCS.

2.66 Main Soil Pipe (MSP) — A pipe connecting one or more branch soil pipes to the drain.

2.67 Main Soil and Waste Pipe (MSWP) — A pipe connecting one or more branch soil and waste pipes to the drain.

2.68 Main Vent Pipe (MVP) — A pipe which receives a number of branch vent pipes.

2.69 Main Waste Pipe (MWP) — A pipe connecting one or more branch waste pipes to the drain.

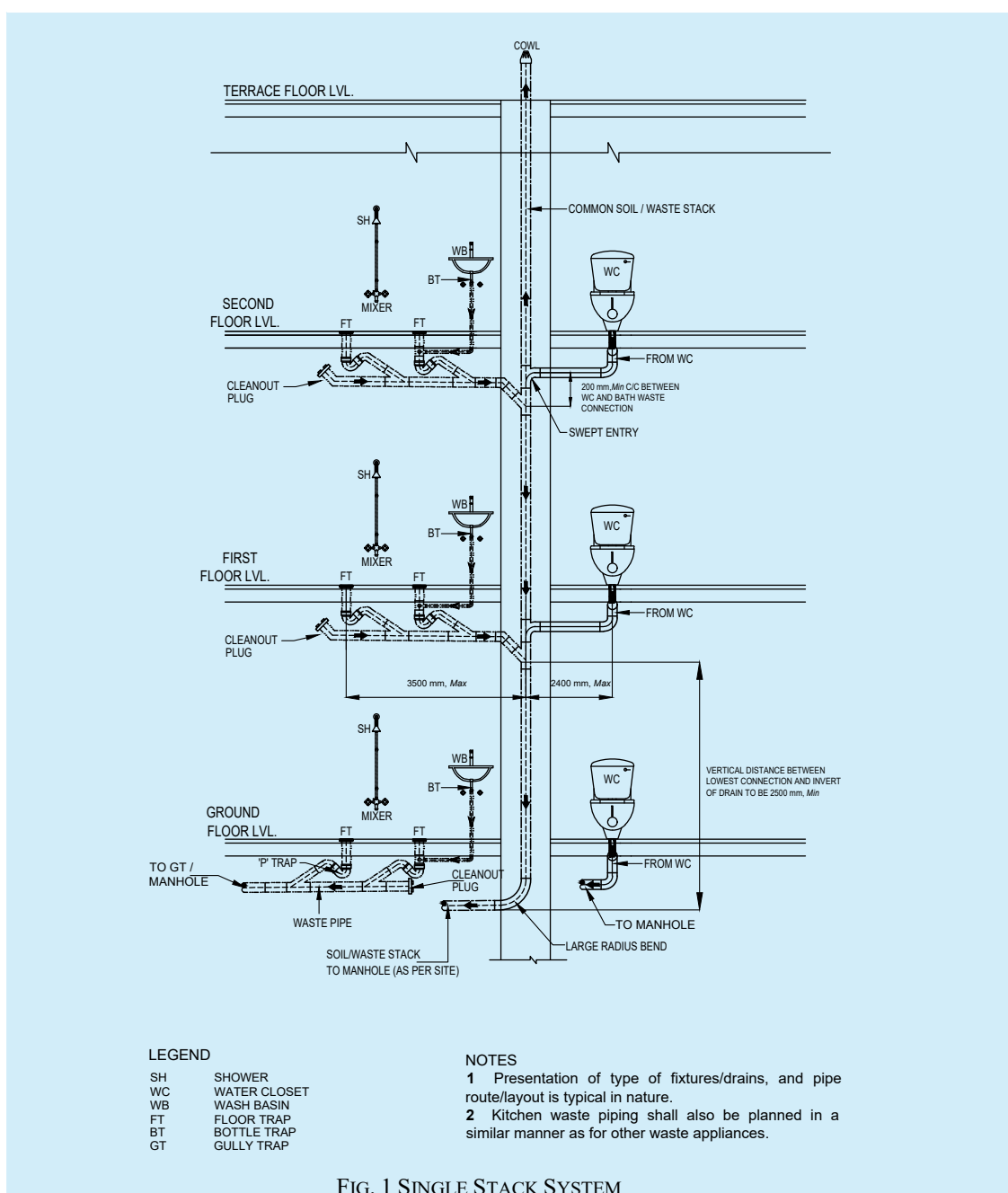
2.70 Manhole — An opening by which a man may enter or leave a drain, a sewer or other closed structure for inspection, cleaning and other maintenance operations, fitted with suitable cover.

2.71 Manhole Chamber — A Chamber constructed on a drain or sewer so as to provide access thereto for inspection, testing or clearance of obstruction.

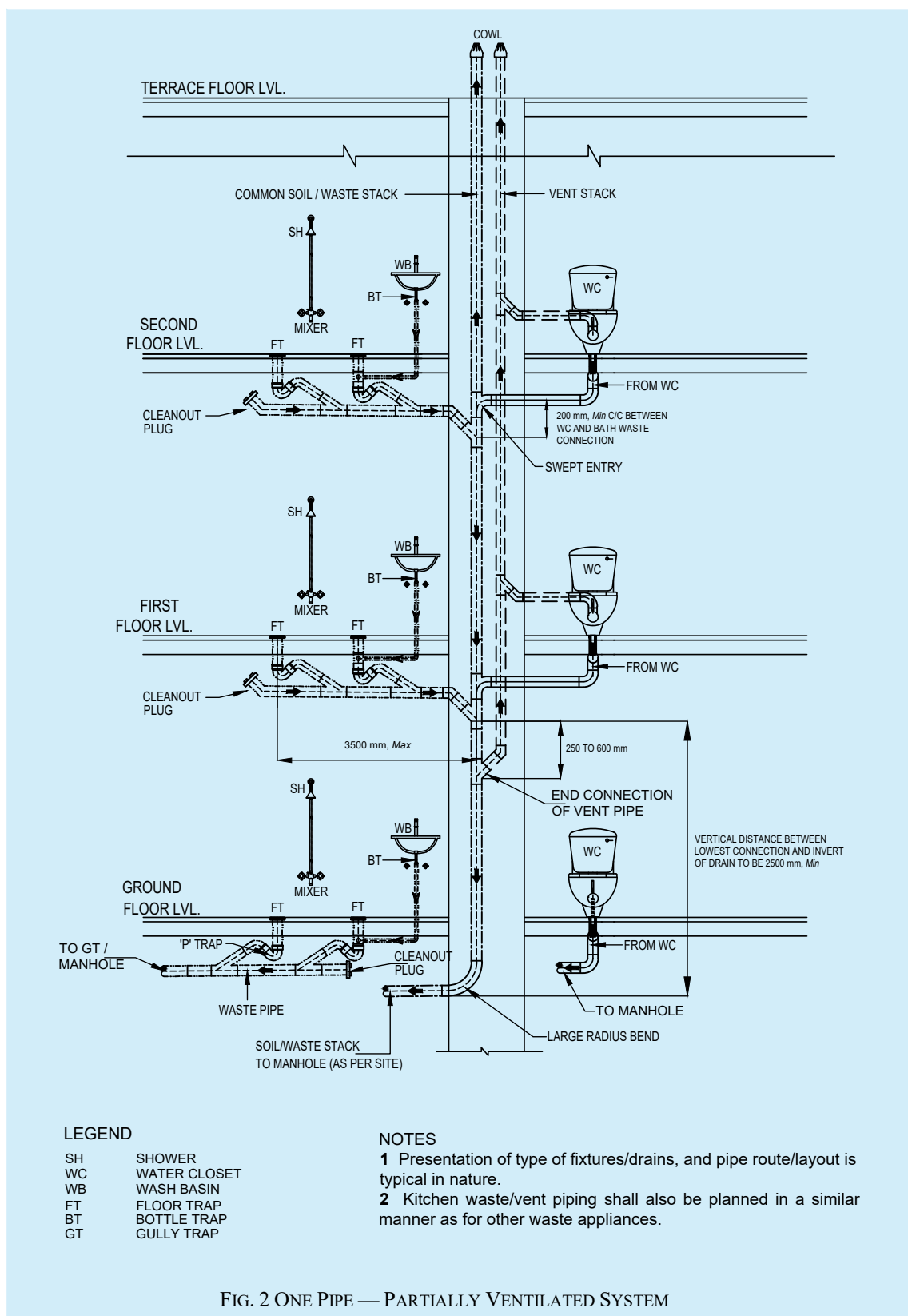
2.72 Offset — A pipe fitting used to connect two pipes whose axes are parallel but not in line.

2.73 Pipe System — The system to be adopted will depend on the type and planning of the building in which it is to be installed and will be one of the following:

- a) *Single stack system* (see [Fig. 1](#)) — The piping system in which there is no trap ventilation and the stack itself acts as vent through roof. In this system care shall be taken for proper sizing of the pipes and the trap arm distance, specially the horizontal distance of sanitary fixtures from the drainage stack and vertical distance between connection of branches from fixtures to drainage stack;



- b) *One pipe — Partially ventilated system* (see Fig. 2) — The piping system in which soil and waste pipes are connected to a single vertical stack [as explained in 2.73(a)] with additional vent pipe for ventilation of traps of water closets;



- c) *One pipe — Fully ventilated system (see Fig. 3)* — The piping system in which the waste pipes from the sinks, baths and wash basins, and the soil pipe from water closet are connected to a single vertical stack. The traps of the water closets, waste appliances, etc, are individually ventilated and connected with a separate stack to preserve the water seal;

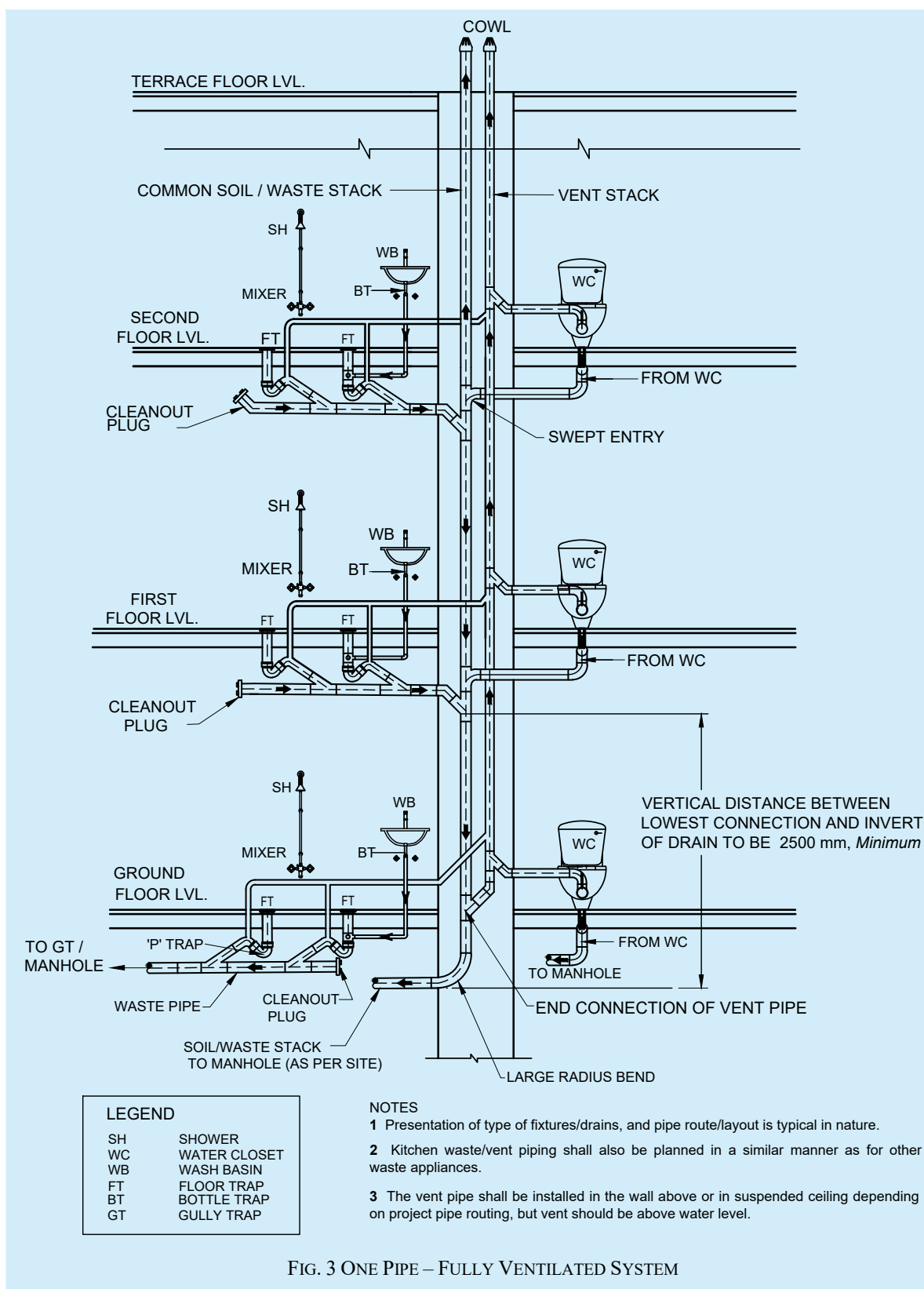


FIG. 3 ONE PIPE – FULLY VENTILATED SYSTEM

d) *Two pipe system:*

- 1) *Two pipe system with common vent pipe* (see Fig. 4A) — The piping system in which all soil appliances such as water closets, urinals, bidet and bedpan sinks are connected to soil stack and all other plumbing fixtures such as baths, sinks, and basins are connected to waste stack through deep seal trap. In addition, a common vent stack is provided for ventilation of soil and waste stacks; and
- 2) *Two pipe system with independent vent pipes* (see Fig. 4B) — The piping system in which all soil appliances such as water closets, urinals, bidet and bedpan sinks are connected to soil stack, and all other plumbing fixtures such as baths, sinks, and basins are connected to waste stack through deep seal trap. In addition, there are independent a vent stacks provided for respective ventilation of soil and waste stacks.

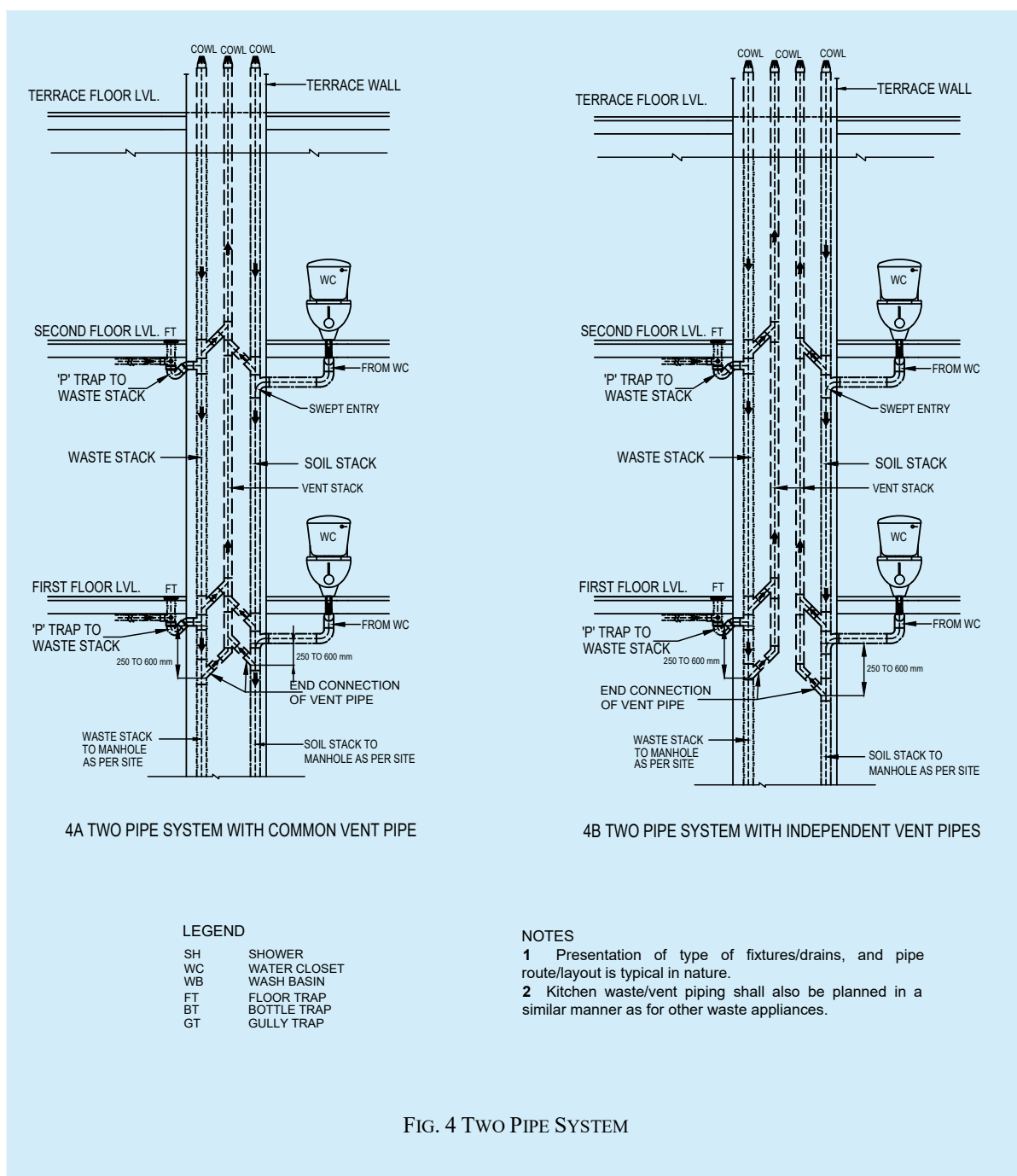


FIG. 4 TWO PIPE SYSTEM

2.74 Plumbing

- a) The pipes, fixtures and other apparatus inside a building for bringing in the water supply and removing the liquid and water borne wastes; and
- b) The installation of the foregoing pipes, fixtures and other apparatus.

2.75 Plumbing System ð The plumbing system shall include the water supply and distribution pipes; plumbing fittings and traps; soil, waste, ventilating pipes and anti-siphonage pipes; building drains and building sewers including their respective connections, devices and appurtenances within the property lines of the premises; and water-treating or water-using equipment.

2.76 Premises ð Premises shall include passages, buildings and lands of any tenure, whether open or enclosed, whether built on or not, and whether public or private in respect of which a water rate or charge is payable to the authority or for which an application is made for supply of water and disposal of waste to the public sewer.

2.77 Puff Ventilation ð The ventilation provided for waste traps in two-pipe system, in order to preserve the water seal.

2.78 Relief Vent ð An additional ventilating pipe connected to a discharge pipe at any point where excessive pressure fluctuation is likely to occur.

2.79 Saddle ð A purpose made fitting, so shaped as to fit over a hole cut in a sewer or drain used to form connections.

2.80 Sanitary Appliances ð The appliances for the collection and discharge of soil or waste matter.

2.81 Sewer ð A pipe or conduit, generally closed, but normally not flowing full for carrying sewage and/or other waste liquids.

2.82 Slop Hopper (Slop Sink) ð A hopper shaped sink, with a flushing run and outlet similar to those of a WC pan, for the reception and discharge of human excreta.

2.83 Soakaway ð A pit, dug into permeable ground lined to form a covered perforated chamber or filled with hard-core, to which liquid is led, and from which it may soak away into the ground.

2.84 Soffit (Crown) ð The highest point of the internal surface of a sewer or culvert at any cross section.

2.85 Soil Appliances ð A sanitary appliance for the collection and discharge of excretory matter.

2.86 Soil Pipe ð A pipe that conveys the discharge of water closets or fixtures having similar functions, with or without the discharges from other fixtures.

2.87 Soil Waste ð The discharge from water closets, urinals, slop hopper, stable yard or cowshed gullies and similar appliances.

2.88 Stack Vent ð The extension of a soil or waste stack above the highest horizontal drain connected to the stack.

2.89 Sub Soil Water ð Water occurring naturally in the subsoil.

2.90 Sub Soil Water Drain

- a) A drain intended to collect and carry away subsoil water; and
- b) A drain intended to disperse into the subsoil from a septic tank.

2.91 Sullage ð See [2.104](#).

2.92 Supply Pipe ð So much of any service pipe as is not a communication pipe.

2.93 Supports — Hangers and anchors or devices for supporting and securing pipe and fittings to walls, ceilings, floors or structural members.

2.94 Surface Water — Natural water from the ground surface, paved areas and roofs.

2.95 Surface Water Drain — A drain conveying surface water including storm water.

2.96 Systems of Drainage

- a) *Combined system* — A system in which foul water (sewage) (black and grey water) and surface water are conveyed by the same sewers and drains;
- b) *Separate system* — A system in which foul water (sewage) (black and grey water) and surface water are conveyed by the separate sewers and drains; and
- c) *Partially separate system* — A modification of the separate system in which part of the surface water is conveyed by the foul (sanitary) sewers and drains.

2.97 Trade Effluent — Any liquid either with or without particles of matter in suspension which is wholly or in part produced in the course of any trade or industry, at trade premise. It includes farm wastes but does not include domestic sewage.

2.98 Trap — A fittings or device so designed and constructed as to provide, when properly vented, a liquid seal which will prevent the back passage of air without materially affecting the flow of sewage or waste water through it.

2.99 Vertical Pipe — Any pipe or fitting which is installed in a vertical position or which makes an angle or not more than 45° with the vertical.

2.100 Vent Stack/Vent Pipe — A vertical vent-pipe installed primarily for the purpose of proving circulation of air to and from any part of the drainage system. It also protects trap seals from excessive pressure fluctuation.

2.101 Vent System — A pipe or pipes installed to provide a flow of air to or from a drainage system or to provide a circulation of air with in such system to protect traps seals from siphonage and back-pressure.

2.102 Waste Appliance — A sanitary appliance for the collection and discharge of water after use for ablutionary, culinary and other domestic purpose.

2.103 Waste Pipe — In plumbing, any pipe that receives the discharge of any fixtures, except water closets or similar fixtures and conveys the same to the house drain or soil or waste stack. When such pipe does not connect directly with a house drain or soil stack, it is called an indirect waste pipe.

2.104 Waste Water (Sullage) — The discharge from wash basins, sinks and similar appliances, which does not contain human or animal excreta.

2.105 Water Closet — A water flushed plumbing fixture designed to receive human excrement directly from the user of the fixture. The term is used sometimes to designate the room or compartment in which the fixture is placed.

2.106 Water Seal — The water in a trap, which acts as a barrier to the passage of air through the trap.

2.107 Yoke Vent (Cross-Vent) — A short relief vent between the main soil or waste pipe and a main ventilating pipe.

3 GENERAL

3.1 Basic Principles

The basic principles of water supply, drainage and sanitation are given below and the design of drainage and sanitation should in general be guided by the applicable principles.

3.1.1 Potable Water

All premises intended for human habitation, occupancy, or use shall be provided with supply of potable water. This water supply shall not be connected with unsafe water resources, nor shall it be subject to the hazards of backflow.

3.1.2 Water Provision

Plumbing fixtures, devices and appurtenances shall be provided with water in sufficient volume and at pressures adequate to enable them to function properly and without undue noise under normal conditions of use.

There should be at least a residual head of 0.018 N/mm^2 at the consumer's tap. There may be certain fixtures or appliances in the installation that may require a higher pressure, such as 0.05 N/mm^2 or even higher, in which case the system shall be designed using pumps, tanks or both to achieve the required minimum pressure.

NOTE — The residual head shall be taken at the highest/farthest outlets in the building.

3.1.3 Water Efficiency

Plumbing system shall be designed, installed and adjusted to use the optimum quantity of water consistent with proper performance and cleaning.

3.1.4 Safety Devices

Plumbing system shall be designed and installed with safety devices to safeguard against dangers from contamination, explosion, overheating, etc.

3.1.5 Minimum Amenities

Each dwelling unit on premises (abutting on a sewer or with a private sewage disposal system) shall have at least one water closet, one kitchen wash place or a sink and one bathing place or shower to meet the basic requirements of sanitation and personal hygiene.

In case of a group housing, the requirements relating to toilet or sanitary room and kitchen as given in good practice [E-2(2)] shall also be complied with.

3.1.6 Drainage System

The drainage system shall be designed, installed and maintained to guard against fouling, deposit of solids and clogging and with adequate cleanouts so arranged that the pipes may be readily cleaned.

3.1.7 Materials and Workmanship

The plumbing system shall have durable material, free from defective workmanship and so designed and installed as to give satisfactory service for its reasonable expected life. The accessories of the plumbing system should be of such specifications as to meet the functional requirements of the installation, so as to also avoid any inconsistency leading to leakage and resultant seepage.

3.1.8 Fixture Traps and Vent Pipes

Each fixture directly connected to the drainage system shall be equipped with a liquid seal trap. Trap seals shall be maintained to prevent sewer gas, other potentially dangerous or noxious fumes, or vermin from entering the building. Further, the drainage system shall be designed to provide an adequate circulation of air in all pipes with no danger of siphonage, aspiration, or forcing of trap seals under conditions of ordinary use by providing vent pipes throughout the system.

3.1.9 Foul Air Exhaust

Each vent terminal shall extend to the outer air and be so installed as to minimize the possibilities of clogging and the return of foul air to the building, as it conveys potentially noxious or explosive gases to the outside atmosphere. All vent pipes shall be provided with a cowl.

3.1.10 Testing

The plumbing system shall be subjected to required tests to effectively disclose all leaks and defects in the work or the material.

3.1.11 Exclusion from Plumbing System

No substance that will clog or accentuate clogging of pipes, produce explosive mixtures, destroy the pipes or their joints, or interfere unduly with the sewage-disposal process shall be allowed to enter the drainage system.

3.1.12 Light and Ventilation

Wherever water closet or similar fixture is located in a room or compartment, it should be properly lighted and ventilated.

3.1.13 Individual Sewage Disposal Systems

If water closets or other plumbing fixtures are installed in buildings where connection to public sewer is not possible, suitable provision shall be made for acceptable treatment and disposal.

3.1.14 Maintenance

Plumbing systems shall be maintained in a safe and serviceable condition.

3.1.15 Approach for Use and Cleaning

All plumbing fixtures shall be so installed with regard to spacing as to be approachable for their intended use and for cleaning. All doors, windows and any other device needing access within the toilet shall be so located that they have proper approach.

3.1.16 Accessibility for Persons with Disabilities

All doors, windows and fixtures, including WC, urinals, grab bars, washbasin, mirror and all other accessories for use by persons with disabilities shall be so installed/located that they have proper access with appropriate width, height, space, centerlines and ease of operation {see good practice [E-2(2)]}.

3.1.17 Structural Safety

Plumbing system shall be installed with due regard to preservation of the structural members and prevention of damage to walls and other surfaces.

3.1.18 Protection of Ground and Surface Water

Sewage or other waste shall not be discharged into surface or sub-surface water without acceptable form of treatment.

3.2 Drainage and Sanitation

3.2.1 Preparation and Submission of Plan

No person shall install or carry out any water-borne sanitary installation or drainage installation or any works in connection with anything existing or new buildings or any other premises without obtaining the prior sanction of the Authority.

The owner shall make an application in the prescribed form (see [Annex A](#)) to the Authority to carry out such a work.

3.2.2 Site Plan

A site plan of the premises on which the building is to be situated or any such work is to be carried out shall be prepared drawn to a scale not smaller than 1 in 500. The site plan of the building premises shall show:

- a) The adjoining plots and streets with their names;
- b) The position of the municipal sewer and the direction of flow in it;
- c) The invert level of the municipal sewer, the road level, and the connection level of the proposed drain connecting the building in relation to the sewer,
- d) The angle at which the drain from the building joints the sewer; and
- e) The alignment, sizes and gradients of all drains and also of surface drains, if any.

A separate site plan is not necessary if the necessary particulars to be shown in such a site plan are already shown in the drainage plan.

3.2.3 Drainage Plan

The application (see [3.3.1](#)) shall be accompanied by a drainage plan drawn to a scale of not smaller than 1 in 100 and furnished along with the building plan. The plans shall show the following:

- a) Every floor of the building in which the pipes or drains are to be used;
- b) The position, forms, level and arrangement of the various parts of such building, including the roof thereof;
- c) All new drains as proposed with their sizes and gradients;
- d) Invert levels of the proposed drains with corresponding ground levels;
- e) The position of every manhole, gully, soil and waste pipe, ventilating pipe, rainwater pipe, water closet, urinal, latrine, bath, lavatory, sink, trap or other appliances in the premises proposed to be connected to any drain and the following colours are recommended for indicating sewers, waste water pipes, rainwater pipes an existing work:

<i>Sl No.</i>	<i>Description of Work</i>	<i>Colour</i>
(1)	(2)	(3)
i)	Sewers	Red
ii)	Waste water pipes and rainwater pipes	Blue
iii)	Existing work	Black

- f) The position of refuse chute, inlet hopper and collection chamber.

3.2.3.1 In the case of an alteration or addition to an existing building, this clause shall be deemed to be satisfied if the plans as furnished convey sufficient information for the proposals to be readily identified with previous sanctioned plans and provided the locations of tanks and other fittings are consistent with the structural safety of the building.

3.2.3.2 The plans for the building drainage shall in every case be accompanied by specifications for the various items of work involved. This information shall be supplied in the prescribed form given in [Annex B](#).

3.2.4 In respect of open drains, cross-sectional details shall be prepared to a scale not smaller than 1 in 50 showing the ground and invert levels and any arrangement already existing or proposed for the inclusion of any or exclusion of all storm water from the sewers.

3.2.5 Completion Certificate

At the completion of the plumbing installation work, the licensed/registered plumber shall give a completion certificate in the prescribed form, which is given in [Annex C](#).

3.3 Licensing/Registration of Plumbers

3.3.1 Execution of Work

The work which is required to be carried out under the provisions of this Section, shall be executed only by a licensed/registered plumber under the control of the Authority and shall be responsible to carry out all lawful directions given by the Authority. No individual shall engage in the business of plumbing unless so licensed/registered under the provisions of this Section.

3.3.1.1 No individual, firm, partnership or corporation shall engage in the business of installing, repairing or altering plumbing unless the plumbing work performed in the course of such business is under the direct supervision of a licensed/registered plumber from approved Authority.

3.3.2 Examination and Certification

The Authority shall establish standards and procedure for the qualification, examination and licensing/registration of plumbers and shall issue licences to such persons who meet the qualifications thereof and successfully pass the examination.

3.3.3 For guidelines for registration of plumbers including the minimum standards for qualifications for the grant of licences, reference may be made to good practice [E-2(3)]. The Authority may also utilize the services of the certified plumbers who are certified for the required skill level under the appropriate scheme of the Government.

4 DRAINAGE AND SANITATION

4.1 Types of Sanitary Appliances

4.1.1 Soil Appliances

4.1.1.1 Water closet

It shall essentially consist of a closet consisting of a bowl to receive excretory matter, trap and a flushing apparatus. It is recommended to provide ablution tap adjacent to the water closet, preferably on right hand side wall. The various types/style of water closets may be:

- a) Squatting Indian type water closet;
- b) Washdown type water closet;
- c) Siphonic washdown type water closet; and
- d) Universal or Anglo-Indian water closet.

4.1.1.2 Bidet

Bidet is an ablution appliance for cleansing excretory organs with water for personnel hygiene. It may be pedestal type or wall hung with a shallow bowl. Hot and cold water supplied through a mixer is discharged from a spray installed over the rim of the appliance. The bidet outlet should essentially connect to soil pipe in a system.

4.1.1.3 Urinal

It is a soil appliance for urination and is connected to soil pipe after a suitable trap. Urinal should have adequate provision of flushing apparatus. The various types/style of urinal may be:

- a) bowl type urinal: flat back or angle back;
- b) slab (single) type urinal;
- c) stall (single) type urinal;
- d) squatting plate type urinal;
- e) syphon jet urinal with integral trap; and
- f) water less (non-water) urinal

4.1.1.4 Slop sink

Slop sink is a large sink, generally of square shape. The appliance is used in hospitals and is installed in the dirty utility room, sluice room and similar locations for disposal of excreta and other foul waste and for washing bed pans and urine bottles/pans. It is provided with a flushing mechanism. It is also provided in Janitor's closet where it is used for cleaning the housekeeping mops, etc.

4.1.1.5 Bed pan sink

Bed pan sink, also known as bedpan washer and disinfecter, is a part of medical equipment inventory which is used to clean manually or automatically, the hospital bedpans, urine bottles and other containers used for collecting body fluids. It is provided in dirty utility room, sluice room, similar locations for disposal of excreta and other foul waste and for washing bed pans, urine bottles/pans. It is a soil appliance and is connected to soil pipe after a suitable trap. In manual models, it has a flushing arrangement.

Provision for installing a bedpan washer and disinfecter should be made in all intensive care units in hospitals.

4.1.2 Waste Appliances

4.1.2.1 Wash basin

It is of one-piece construction having a combined overflow and preferably should have soap holding recess or recesses that should properly drain into the bowl. Each basin shall have circular waste hole through which the liquid content of the basin shall drain.

4.1.2.2 Wash-trough

It is a linear trough for simultaneous use by number of persons.

4.1.2.3 Sink

It is used in kitchen and laboratory for the purpose of cleaning utensils/ apparatus and also serve the purpose of providing water for general usage. The sink may be made with or without overflow arrangement. The sink shall be of one-piece construction including combined over flow, where provided. The sink shall have a circular waste hole into which the interiors of the sink shall drain.

4.1.2.4 Bath tub

Bath tub may be of enameled steel, cast iron, gel-coated, glass fibre reinforced plastic or may be cast in-situ. It shall be stable, comfortable, easy to get in and out, water tight, with anti-skid base, and easy to install and maintain. The bath tub shall be fitted with overflow and waste pipe of nominal diameter of not less than 32 mm and 40 mm, respectively.

4.1.2.5 Drinking fountain

It is a bowl fitted with a push button tap and a water bubbler or a tap with a swan neck outlet fitting. It has a waste fitting, a trap and is connected to the waste pipe.

4.1.3 The requirements of various soil appliances and waste appliances shall be in accordance with accepted standards [E-2(4)].

4.2 Drainage and Sanitation Requirements

4.2.1 General

There should be at least one water tap and arrangement for drainage in the vicinity of each water closet or group of water closet in all the buildings.

4.2.2 Each dwelling unit on premises (abutting on a sewer or with a private sewage disposal system) shall have at least one water closet, one kitchen wash place or a sink, and one bathing place or shower to meet the basic requirements of sanitation and personal hygiene.

In case of a group housing, the requirements relating to toilet or sanitary room and kitchen as given in good practice [E-2(2)] shall also be complied with.

4.2.3 All other structures for human occupancy or use on premises (abutting on a sewer or with a private sewage disposal system) shall have adequate sanitary facilities, but in no case less than one water closet and one other fixture for cleaning purposes.

4.2.4 For Residences

4.2.4.1 Dwelling with individual convenience shall have at least the following fitments:

- a) One bathroom provided with a tap and a floor trap;
- b) One water closet with flushing apparatus with an ablution tap; and
- c) One tap with a floor trap or a sink in kitchen or wash place.

4.2.4.1.1 Where only one water closet is provided in a dwelling, it is desirable to have the bath and water closet separately accommodated.

4.2.4.2 Dwellings without individual conveniences shall have the following fitments:

- a) One water tap with floor trap in each tenement;
- b) One water closet with flushing apparatus and one ablution tap bath for every two tenements; and
- c) One bath with water tap and floor trap for every two tenements

4.2.5 For Buildings other than Residences

4.2.5.1 The requirements for fitments for drainage and sanitation in the case of buildings other than residences shall be in accordance with [Table 1](#) to [Table 17](#).

The accessibility requirements for provision of these facilities for persons with disabilities shall be in accordance with good practice [E-2(2)].

The following shall be, in addition, taken into consideration:

- a) The figures shown are based upon one (1) fixture being the minimum required for the number of persons indicated or part thereof;
- b) Building categories not included in the tables shall be considered separately by the Authority;
- c) Drinking fountains shall not be installed in the toilets;
- d) Where there is the danger of exposure to skin contamination with poisonous, infectious or irritating material, washbasin with eye wash jet and an emergency shower (safety shower with eye wash unit) located in an area accessible at all times with the passage/right of way suitable for access to a wheel chair, shall be provided;
- e) When applying the provision of these tables for providing the number of fixtures, consideration shall be given to the accessibility of the fixtures. Using purely numerical basis may not result in an installation suited to the need of a specific building. For example, schools should be provided with toilet facilities on each floor. Similarly, toilet facilities shall be provided for temporary workmen employed in any establishment according to the needs; and in any case one WC and one washbasin shall be provided;
- f) All buildings used for human habitation for dwelling, work, occupation, medical care or any purpose detailed in the various tables, abutting a public sewer or a private sewage disposal system, shall be provided with minimum sanitary facilities as per the schedule in the tables. In case the disposal facilities are not available, they shall be provided as a part of the building design for ensuring high standards of sanitary conditions in accordance with this Section;

- g) Workplaces where crèches are provided, they shall be provided with one WC for 10 persons or part thereof, one wash basin for 15 persons or part thereof, one kitchen sink with floor trap for preparing food/milk preparations. The sink provided shall with a drinking water tap;
- h) In all types of buildings, individual toilets and pantry should be provided for executives, and for meeting/seminar/conference rooms, etc as per the user requirement;
- j) Where food is consumed indoors, water stations and handwashing points may be provided in place of drinking water fountains; and
- k) All public areas where floating population is expected need to provide low-mounted urinals and washbasins for children/dwarfs, at least 1 percent or 10 percent of the total.

Table 1 Office Buildings

(Clause [4.2.5.1](#))

Sl No.	Fixtures	Public Toilets		Staff Toilets	
		Males	Females	Males	Females
(1)	(2)	(3)	(4)	(5)	(6)
Executive rooms and conference halls in office buildings					
i)	Toilet suite comprising one WC, one washbasin (with optional shower stall if building is used round the clock at user's option) Pantry optional as per user requirement	Unit could be common for male/female or separate depending on the number of user of each facility		For individual officer rooms	
Main office toilets for staff and visitors					
ii)	Water closets	See Note		1 per 25	1 per 15
iii)	Ablution tap with each water closet	← 1 in each water closet →			
iv)	Urinals	See Note	–	Nil up to 6 1 for 7 to 20 2 for 21 to 45 3 for 46 to 70 4 for 71 to 100 From 101 to 200, add @ 3 percent For over 200, add @ 2.5 percent	–
v)	Wash basins	See Note		1 per 25	1 per 25
vi)	Drinking water fountain	See Note		1 per 100	1 per 100
vii)	Cleaner's sink	← 1 per floor →			
NOTES					
1 Staff and public toilet utilities are generally common in office buildings. Where public toilets are to be provided independently, similar requirements as that of staff toilet may be provided.					
2 Space for one janitor per toilet core shall to be planned.					
3 The male population may be assumed as half and the female population as half.					

Table 2 Factories

(Clause 4.2.5.1)

Sl No.	Fixtures	Offices/Visitors		Workers	
		Males	Females	Males	Females
(1)	(2)	(3)	(4)	(5)	(6)
i)	Water closets (workers and staff)	1 for up to 25	1 for up to 15	1 for up to 15	1 for up to 12
		2 for 26 to 35	2 for 16 to 25	2 for 16 to 35	2 for 13 to 25
		3 for 36 to 65	3 for 26 to 40	3 for 36 to 65	3 for 26 to 40
		4 for 66 to 100	4 for 41 to 57	4 for 66 to 100	4 for 41 to 57
			5 for 58 to 77		5 for 58 to 77
			6 for 78 to 100		6 for 78 to 100
	For persons 101 to 200 add	3 percent	5 percent	3 percent	5 percent
	For persons over 200 add	2.5 percent	4 percent	2.50 percent	4 percent
ii)	Ablution tap	1 in each water closet	1 in each water closet	1 in each water closet	1 in each water closet
iii)	Urinals	Nil up to 6	—	Nil up to 6	—
		1 for 7 to 20		1 for 7 to 20	
		2 for 21 to 45		2 for 21 to 45	
		3 for 46 to 70		3 for 46 to 70	
		4 for 71 to 100		4 for 71 to 100	
		From 101 to 200, add @ 3 percent		From 101 to 200, add @ 3 percent	
	For over 200, add @ 2.5percent		For over 200, add @ 2.5 percent		
iv)	Wash basins	1 per 25 or part thereof	1 per 25 or part thereof	1 per 25 or part thereof	1 per 25 or part thereof
	Wash basins in rows or troughs and taps spaced 750 mm c/c				
v)	Drinking water fountain	1 per every 100 or part thereof with minimum one on each floor		1 per every 100 or part thereof with minimum one on each floor	
vi)	Cleaner’s sink	1 on each floor	1 on each floor	1 on each floor	1 on each floor
vii)	Showers/bathing rooms	← As per trade requirements →			
viii)	Emergency shower and eye wash fountain	—	—	1 per every shop floor per 500 persons	
NOTES					
1 For factories requiring workers to be engaged in dirty and dangerous operations or requiring them to being extremely clean and sanitized conditions additional and separate (if required so) toilet facilities and if required by applicable Industrial and Safety Laws and the <i>Factories Act</i> shall be provided in consultation with the user.					
2 Depending on the type of disability of a person and the hazard posed by the type of activities in the factory for a person with disabilities, if a person with disabilities is decided to be engaged for a particular activity, the requirements of accessibility shall be guided by the provisions given in good practice [E-2(2)].					

Table 3 Warehouse and Storage Area

(Clause [4.2.5.1](#))

Sl No.	Fixtures	Offices/Visitors		Workers	
		Males	Females	Males	Females
(1)	(2)	(3)	(4)	(5)	(6)
i)	Water closets (workers and staff)	1 for up to 25	1 for up to 15	1 for up to 15	1 for up to 12
	For persons 101 to 200 add	3 percent	5 percent	4 percent	5 percent
	For persons over 200 add	2.5 percent	4 percent	2.50 percent	4 percent
ii)	Ablution tap	1 in each water closet	1 in each water closet	1 in each water closet	1 in each water closet
iii)	Urinals	Nil up to 6	—	Nil up to 6	—
		1 for 7 to 20	—	1 for 7 to 20	—
		2 for 21 to 45	—	2 for 21 to 45	—
		3 for 46 to 70	—	3 for 46 to 70	—
		4 for 71 to 100	—	4 for 71 to 100	—
		From 101 to 200, add @ 3 percent	—	From 101 to 200, add @ 3 percent	—
		For over 200, add @ 2.5 percent	—	For over 200, add @ 2.5 percent	—
iv)	Wash basins	1 per 25 or part thereof	1 per 15 or part thereof	1 per 25 or part thereof	1 per 25 or part thereof
	Wash basins in rows or troughs and taps spaced 750 mm c/c				
v)	Drinking water fountain	1 per every 100 or part thereof with minimum one on each floor		1 per every 100 or part thereof with minimum one on each floor	
vi)	Cleaner’s sink	1 on each floor	1 on each floor	1 on each floor	1 on each floor
vii)	Showers/bathing rooms	← As per trade requirements →			
viii)	Emergency shower and eye wash fountain	—	—	1 per every shop floor per 500 persons	
NOTE — Emergency shower and eye wash as per trade requirement to be provided.					

Table 4 Cinema, Multiplex Cinema, Concerts and Convention Halls, Theatres and Stadia

(Clause 4.2.5.1)

Sl No.	Fixtures	Public		Staff	
		Males	Females	Males	Females
(1)	(2)	(3)	(4)	(5)	(6)
i)	Water closets	1 per 100 up to 400	4 per 100 up to 200	1 for up to 15 2 for 16 to 35	1 for up to 12 2 for 13 to 25
		Over 400 add at 1 per 250 or part thereof	Over 200 add at 2 per 100 or part thereof		
ii)	Ablution tap	1 in each water closet	1 in each water closet	1 in each water closet	1 in each water closet
		1 water tap with draining arrangements shall be provided for every 50 persons or part thereof in the vicinity of water closets and urinals			
iii)	Urinals	1 per 25 or part thereof	—	Nil up to 6 1 for 7 to 20 2 for 21 to 45	—
iv)	Wash basins	1 per 200 or part thereof		1 for up to 15 2 for 16 to 35	1 for up to 12 2 for 13 to 25
v)	Drinking water fountain	← 1 per 100 persons or part thereof →			
vi)	Cleaner's sink	← 1 per floor →			
vii)	Showers/bathing rooms	← As per trade requirements →			
NOTES					
1 Some WCs may be Indian style if desired.					
2 Male population may be assumed as half and female population as half.					

Table 5 Art Galleries, Libraries, Indoor Activity/Sports Rooms and Museums

(Clause 4.2.5.1)

Sl No.	Fixtures	Public		Staff	
		Males	Females	Males	Females
(1)	(2)	(3)	(4)	(5)	(6)
i)	Water closets	1 per 200 up to 400	1 per 100 up to 200	1 for up to 15	1 for up to 12
		Over 400 add at 1 per 250 or part thereof	Over 200 add at 1 per 150 or part thereof	2 for 16 to 35	2 for 13 to 25
ii)	Ablution tap	One in each water closet	One in each water closet	One in each water closet	One in each water closet
		1 water tap with draining arrangements shall be provided for every 50 persons or part thereof in the vicinity of water closets and urinals			
iii)	Urinals	1 per 50	—	Nil up to 6	—
				1 per 7 to 20	
				2 per 21 to 45	
iv)	Wash basins	1 for every 200 or part thereof. For over 400, add at 1 per 250 persons or part thereof	1 for every 200 or part thereof. For over 200, add at 1 per 150 persons or part thereof	1 for up to 15	1 for up to 12
				2 for 16 to 35	2 for 13 to 25
v)	Drinking water fountain	← 1 per 100 persons or part thereof →			
vi)	Cleaner's sink	← 1 per floor (Minimum) →			
vii)	Showers/bathing rooms	← As per requirements →			

NOTES

1 Some WCs may be Indian style if desired.

2 Male population may be assumed as half and female population as half.

3 Board game rooms, indoor sports and club house for community in residential premises can be considered as indoor activity areas.

Table 6 Hospitals with Indoor Patient Wards

(Clause 4.2.5.1)

Sl No.	Fixtures	Patient Toilets		Staff Toilets	
		Males	Females	Males	Females
(1)	(2)	(3)	(4)	(5)	(6)
i)	Toilet suite comprising one WC and one washbasin and shower stall	Private room with up to 4 patients		For individual doctor's/officer's rooms	
For General Wards, Hospital Staff and Visitors					
ii)	Water closets	1 per 5 beds or part thereof	1 per 5 beds or part thereof	1 for up to 15	1 for up to 12
				2 for 16 to 35	2 for 13 to 25
iii)	Ablution tap	One in each water closet	One in each water closet	One in each water closet	One in each water closet
1 water tap with draining arrangements shall be provided for every 50 persons or part thereof in the vicinity of water closets and urinals					
iv)	Urinals	1 per 15 beds	—	Nil up to 6	—
				1 for 7 to 20	
				2 for 21 to 45	
v)	Wash basins	2 for every 30 beds or part thereof. Add 1 per additional 30 beds or part thereof		1 for up to 15	1 for up to 12
				2 for 16 to 35	2 for 13 to 25
vi)	Drinking water fountain	1 per ward		1 per 100 persons or part thereof	
vii)	Cleaner's sink	1 per ward		—	
viii)	Bed pan sink	1 per ward		—	
ix)	Kitchen sink	1 per ward		—	
NOTES					
1 Some WCs may be of Indian style if desired.					
2 Male population may be assumed as half and female population as half.					
3 Provision for additional and special hospital fittings where required shall be made.					
4 Drinking water fountains are not recommended for hospitals for reasons of infection control. This to be decided by the health authority recommendation.					

Table 7 Hospitals with Outdoor Patient Department

(Clause 4.2.5.1)

Sl No.	Fixtures	Patient Toilets		Staff Toilets	
		Males	Females	Males	Females
(1)	(2)	(3)	(4)	(5)	(6)
i)	Toilet suite comprising one WC and one washbasin (with optional shower stall if building used for 24 h)	For up to 4 patients		For individual doctor's/officer's rooms	
ii)	Water closets	1 per 100 persons or part thereof	2 per 100 persons or part thereof	1 for up to 15 2 for 16 to 35	1 for up to 12 2 for 13 to 25
iii)	Ablution tap	One in each water closet	One in each water closet	One in each water closet	One in each water closet
		1 water tap with draining arrangements shall be provided for every 50 persons or part thereof in the vicinity of water closets and urinals			
iv)	Urinals	1 per 50 persons or part thereof	—	Nil up to 6 1 for 7 to 20 2 for 21 to 45	—
v)	Wash basins	2 per 100 persons of part thereof	2 per 100 persons or part thereof	1 for up to 15 2 for 16 to 35	1 for up to 12 2 for 13 to 25
vi)	Drinking water fountain	See Note 2		1 per 100 persons or part thereof	
NOTES					
1 Some WCs may be Indian style if desired.					
2 Drinking water fountains are not recommended for hospitals for reasons of infection control. This to be decided by the health authority recommendation.					
3 The WCs shall be provided keeping in view the location of main OPD waiting hall and sub-waiting halls, floor wise, so as to serve the people effectively. The number of patients shall be calculated floor wise. The OPD population shall include patient attendants @ at least 1 per patient.					
4 Male population may be assumed as half and female population as half.					
5 Provision for additional and special hospital fittings where required shall be made.					

Table 8 Hospitals, Administrative Buildings

(Clause [4.2.5.1](#))

Sl No.	Fixtures	Staff Toilets	
		Males	Females
(1)	(2)	(3)	(4)
i)	Toilet suite comprising one WC, one urinal and one washbasin (with optional shower stall if building used for 24 h)	For individual doctor's/officer's rooms	
ii)	Water closets	1 per 25 persons or part thereof	1 per 15 persons or part thereof
iii)	Ablution tap	One in each water closet	One in each water closet
		1 water tap with draining arrangements shall be provided for every 50 persons or part thereof in the vicinity of water closets and urinals	
iv)	Urinals	1 for 6 to 15	—
		2 for 16 to 50	—
v)	Wash basins	1 per 25 persons or part thereof	1 per 25 persons or part thereof
vi)	Drinking water fountain	1 per 100 persons or part thereof (See Note 2)	
vii)	Cleaner's sink	1 per floor, <i>Min</i>	
viii)	Kitchen sink	1 per floor, <i>Min</i>	
NOTES			
1 Some WCs may be Indian style if desired.			
2 Drinking water fountains to be provided only when it is a separate block and patients will not use it.			

Table 9 Hospitals Staff Quarters and Nurses Homes

(Clause [4.2.5.1](#))

Sl No.	Fixtures	Staff Quarters		Nurses Homes	
		Males	Females	Males	Females
(1)	(2)	(3)	(4)	(5)	(6)
i)	Water closets	1 per 4 persons or part thereof	1 per 4 persons or part thereof	1 per 4 persons or part thereof	1 per 4 persons or part thereof
				2 for 5 to 35	2 for 5 to 25
ii)	Ablution tap	One in each water closet	One in each water closet	One in each water closet	One in each water closet
		1 water tap with draining arrangements shall be provided for every 50 persons or part thereof in the vicinity of water closets and urinals			
iii)	Wash basins	1 per 8 persons or part thereof		1 per 8 persons or part thereof	
iv)	Bath (showers)	1 per 4 persons or part thereof		1 per 4 to 6 persons or part thereof	
v)	Drinking water fountain	1 per 100 persons or part thereof, <i>Min</i> 1 per floor		1 per 100 persons or part thereof, <i>Min</i> 1 per floor	
vi)	Cleaner’s sink	1 per Floor		1 per Floor	
NOTES					
1 Some WCs may be Indian style if desired.					
2 For independent housing units, fixtures shall be provided as for residences.					

Table 10 Hotels

(Clause [4.2.5.1](#))

Sl No.	Fixtures	Public Rooms		Non-Residential Staff	
		Males	Females	Males	Females
(1)	(2)	(3)	(4)	(5)	(6)
i)	Toilet suite comprising one WC, wash basin with Shower or a bathtub	Individual guest rooms with attached toilets		—	
Guest rooms with common facilities, and public areas like reception, health club, waiting lounge, business centre, recreation area					
ii)	Water closets	1 per 100 persons up to 300	2 per 100 persons up to 200	1 for up to 15	1 for up to 12
		Over 300 add at 1 per 200 or part thereof	Over 200 add at 1 per 100 or part thereof	2 for 16 to 35 3 for 36 to 65 4 for 66 to 100	2 for 13 to 25 3 for 26 to 40 4 for 41 to 57 5 for 58 to 77 6 for 78 to 100
iii)	Ablution tap	One in each water closet	One in each water closet	One in each water closet	One in each water closet
		1 water tap with draining arrangements shall be provided for every 50 persons or part thereof in the vicinity of water closets and urinals			
iv)	Urinals	1 per 50 persons or part thereof	—	Nil up to 6 1 for 7 to 20 2 for 21 to 45 3 for 46 to 70 4 for 71 to 100	—
v)	Wash basins	1 per WC/urinal	1 per WC	1 for up to 15 2 for 16 to 35 3 for 36 to 65 4 for 66 to 100	1 for up to 12 2 for 13 to 25 3 for 26 to 40 4 for 41 to 57
vi)	Bath (showers)	1 per 10 persons or part thereof		—	—
vii)	Cleaner’s sink	1 per 30 rooms, <i>Min</i> 1 per floor			
viii)	Kitchen sink	1 per kitchen			
NOTES					
1 Some WCs may be Indian style if desired.					
2 Male population may be assumed as half and female population as half.					
3 Provision for additional and special fittings where required shall be made based on end user or operator guidelines.					

Table 11 Restaurants

(Clause [4.2.5.1](#))

Sl No.	Fixtures	Public Rooms		Non Residential Staff	
		Males	Females	Males	Females
(1)	(2)	(3)	(4)	(5)	(6)
i)	Water closets	1 per 50 seats up to 200	2 per 50 seats up to 200	1 for up to 15	1 for up to 12
		Over 200 add at 1 per 100 or part thereof	Over 200 add at 1 per 100 or part thereof	2 for 16 to 35 3 for 36 to 65 4 for 66 to 100	2 for 13 to 25 3 for 26 to 40 4 for 41 to 57 5 for 58 to 77 6 for 78 to 100
ii)	Ablution tap	One in each water closet	One in each water closet	One in each water closet	One in each water closet
		1 water tap with draining arrangements shall be provided for every 50 persons or part thereof in the vicinity of water closets and urinals			
iii)	Urinals	1 per 50 persons or part thereof	—	Nil up to 6 1 for 7 to 20 2 for 21 to 45 3 for 46 to 70 4 for 71 to 100	—
iv)	Wash basins	1 per WC	1 per WC	1 per WC	1 per WC
v)	Cleaner's sink		1 per each restaurant		
vi)	Kitchen sink/dish washer		1 per kitchen		
NOTES					
1 Some WCs may be Indian style if desired.					
2 Male population may be assumed as half and female population as half.					
3 Provision for additional and special fittings where required shall be made.					

Table 12 Microbrewery, Special Restaurants and Food Court Area

(Clause [4.2.5.1](#))

Sl No.	Fixtures	Public Rooms		Staff	
		Males	Females	Males	Females
(1)	(2)	(3)	(4)	(5)	(6)
i)	Water closets	1 per 50 seats up to 300 Over 300 add at 1 per 100 or part thereof	2 per 50 seats up to 200 Over 200 add at 1 per 100 or part thereof	3 percent for every 100	4 percent for every 100
ii)	Ablution tap	One in each water closet	One in each water closet	One in each water closet	One in each water closet
		1 water tap with draining arrangements shall be provided for every 50 persons or part thereof in the vicinity of water closets and urinals			
iii)	Urinals	3 percent for every 100	—	2.5 percent for every 100	—
iv)	Wash basins	1 per WC	1 per WC	1 per WC	1 per WC
v)	Cleaner's sink		1 per each floor		
NOTES					
1 Some WCs may be Indian style if desired.					
2 Male population may be assumed as half and female population as half.					
3 Provision for additional and special fittings where required shall be made.					
4 Kitchen sinks and washers shall be provided as per the kitchen design layout.					

Table 13 Schools, Colleges and Educational Institutions

(Clause [4.2.5.1](#))

Sl No.	Fixtures	Nursery School	Non-Residential		Residential	
			Boys	Girls	Boys	Girls
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Water closets	1 per 15 pupils or part thereof	1 per 40 pupils or part thereof	1 per 25 pupils or part thereof	1 per 8 pupils or part thereof	1 per 6 pupils or part thereof
ii)	Ablution tap	One in each water closet	One in each water closet	One in each water closet	One in each water closet	One in each water closet
		1 water tap with draining arrangements shall be provided for every 50 persons or part thereof in the vicinity of water closets and urinals				
iii)	Urinals	—	1 per 20 pupils or part thereof	—	1 per 25 pupils or part thereof	—
iv)	Wash basins	1 per 15 pupils or part thereof	1 per 60 pupils or part thereof	1 per 40 pupils or part thereof	1 per 8 pupils or part thereof	1 per 6 pupils or part thereof
v)	Bath/showers	1 per 40 pupils or part thereof	—	—	1 per 8 pupils or part thereof	1 per 6 pupils or part thereof
vi)	Drinking water fountain or taps	1 per 50 pupils or part thereof	1 per 50 pupils or part thereof	1 per 50 pupils or part thereof	1 per 50 pupils or part thereof	1 per 50 pupils or part thereof
vii)	Cleaner's sink			1 per each floor		
NOTES						
1 Some WCs may be Indian style if desired.						
2 For teaching staff, the schedule of fixtures to be provided shall be the same as in case of office building.						
3 Male population may be assumed as half and female population as half.						

Table 14 Hostels

(Clause [4.2.5.1](#))

Sl No.	Fixtures	Resident		Non-Resident		Visitor/Common Rooms	
		Males	Females	Males	Females	Males	Females
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Water closets	1 per 8 or part thereof	1 per 6 or part thereof	1 for up to 15 2 for 16 to 35 3 for 36 to 65 4 for 66 to 100	1 for up to 12 2 for 13 to 25 3 for 26 to 40 4 for 41 to 57 5 for 58 to 77 6 for 78 to 100	1 per 100 up to 400 Over 400 add at 1 per 250	2 per 100 up to 200 Over 200 add at 1 per 100
ii)	Ablution tap	One in each water closet	One in each water closet	One in each water closet	One in each water closet	One in each water closet	One in each water closet
1 water tap with draining arrangements shall be provided for every 50 persons or part thereof in the vicinity of water closets and urinals							
iii)	Urinals	1 per 25 or part thereof	—	Nil up to 6 1 for 7 to 20 2 for 21 to 45 3 for 46 to 70 4 for 71 to 100	—	1 per 50 or part thereof	—
iv)	Wash basins	1 per 8 persons or part thereof	1 per 6 persons or part thereof	1 for up to 15 2 for 16 to 35 3 for 36 to 65 4 for 66 to 100	1 for up to 12 2 for 13 to 25 3 for 26 to 40 4 for 41 to 57 5 for 58 to 77 6 for 78 to 100	1 per WC/ Urinal	1 per WC
v)	Bath/showers	1 per 8 persons or part thereof	1 per 6 persons or part thereof	—	—	—	—
vi)	Cleaner’s sink	1 per each floor					
NOTE — Some WCs may be Indian style if desired.							

Table 15 Fruit and Vegetable Markets

(Clause [4.2.5.1](#))

Sl No.	Fixtures	Shop Owners		Common Toilets in Market Building		Public Toilet for Floating Population	
		Males	Females	Males	Females	Males	Females
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Water closets	1 per 8 or part thereof		1 for up to 15 2 for 16 to 35 3 for 36 to 65 4 for 66 to 100	1 for up to 12 2 for 13 to 25 3 for 26 to 40 4 for 41 to 57 5 for 58 to 77 6 for 78 to 100	1 per 50 (Min 2)	1 per 50 (Min 2)
ii)	Ablution tap	One in each water closet	One in each water closet	One in each water closet	One in each water closet	One in each water closet	One in each water closet
		1 water tap with draining arrangements shall be provided in receiving/ sale area of each shop and for every 50 persons or part thereof in the vicinity of water closets and urinals					
iii)	Urinals	–	–	Nil up to 6 1 for 7 to 20 2 for 21 to 45 3 for 46 to 70 4 for 71 to 100	–	1 per 50	–
iv)	Wash basins	1 per 8 or part thereof		1 for up to 15 2 for 16 to 35 3 for 36 to 65 4 for 66 to 100	1 for up to 12 2 for 13 to 25 3 for 26 to 40 4 for 41 to 57	–	–
v)	Bath/showers	1 per 8 persons or part thereof	1 per 6 persons or part thereof	–	–	1 per 50 persons	1 per 50 persons
NOTES							
1 Toilet facilities for individual buildings in a market should be taken same as that for office buildings.							
2 Common toilets in the market buildings provide facilities for persons working in shops and their regular visitors.							
3 Special toilet facilities for a large floating population of out of town buyers/sellers, labour, drivers of vehicles for whom special toilet (public toilets).							

Table 16 Bus Stations, Airports and Railway Stations

(Clause [4.2.5.1](#))

Sl No.	Fixtures	Junction Stations, Intermediate Stations and Bus Stations		Terminal Railway and Bus Stations		Domestic and International Airports	
		Males	Females	Males	Females	Males	Females
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Water closets	3 for up to 1 000 Add 1 per additional 1 000 or part thereof	4 for up to 1 000 Add 1 per additional 1 000 or part thereof	4 for up to 1 000 Add 1 per additional 1 000 or part thereof	5 for up to 1 000 Add 1 per additional 1 000 or part thereof	Min 2 For 200 : 5 For 400 : 9 For 600 : 12 For 800 : 16 For 1 000 : 18	Min 2 For 200 : 8 For 400 : 15 For 600 : 20 For 800 : 26 For 1 000 : 29
ii)	Ablution tap	One in each water closet	One in each water closet	One in each water closet	One in each water closet	One in each water closet	One in each water closet
		1 water tap with draining arrangements shall be provided for every 50 persons or part thereof in the vicinity of water closets and urinals					
iii)	Urinals	4 for up to 1 000 Add 1 per additional 1 000	—	6 for up to 1 000 Add 1 per additional 1 000	—	1 per 40 or part thereof	—
iv)	Wash basins	1 per WC/Urinal	1 per WC	1 per WC/Urinal	1 per WC	1 per WC/Urinal	1 per WC
v)	Bath/showers	2 per 1 000		3 per 1 000		4 per 1 000	
vi)	Drinking water fountain or taps (in common lobby for male/female)	2 per 1 000 or part thereof		3 per 1 000 or part thereof		4 per 1 000 or part thereof	
vii)	Cleaner's sink	1 per toilet compartment with 3 WC's	1 per toilet compartment with 3 WC's	1 per toilet compartment with 3 WC's	1 per toilet compartment with 3 WC's	1 per toilet compartment with 3 WC's	1 per toilet compartment with 3 WC's

NOTES

- 1 Some WCs may be Indian style if desired.
- 2 Male population may be assumed as three-fifth and female population as two-fifth.
- 3 Separate provision shall be made for staff and workers.

Table 17 Shopping Malls and Retail Buildings

(Clause 4.2.5.1)

Sl No.	Fixtures	Staff Toilets in Shopping Building		Public Toilet for Floating Population	
		Males	Females	Males	Females
(1)	(2)	(5)	(6)	(7)	(8)
i)	Water closets	1 for up to 15	1 for up to 12	1 per 50	1 per 50
		2 for 16 to 35	2 for 13 to 25	(2, <i>Min</i>)	(2, <i>Min</i>)
		3 for 36 to 65	3 for 26 to 40		
		4 for 66 to 100	4 for 41 to 57		
			5 for 58 to 77		
			6 for 78 to 100		
ii)	Ablution tap	One in each water closet	One in each water closet	One in each water closet	One in each water closet
iii)	Urinals	Nil up to 6	–	1 per 50	–
		1 for 7 to 20			
		2 for 21 to 45			
		3 for 46 to 70			
		4 for 71 to 100			
iv)	Wash basins	1 for up to 15	1 for up to 12	1 per 50	1 per 50
		2 for 16 to 35	2 for 13 to 25	(2, <i>Min</i>)	(2, <i>Min</i>)
		3 for 36 to 65	3 for 26 to 40		
		4 for 66 to 100	4 for 41 to 57		
v)	Bath/showers	1 per 100 persons	1 per 100 persons	–	–
NOTES					
1 Toilet facilities for individual facilities in a shopping building should be taken same as that for office buildings.					
2 Staff toilets in the shopping buildings provide facilities for persons working in shops and building, as well as for maintenance staff.					
3 The number of persons against the fixture bath/showers represent the maintenance staff requiring such facility.					
4 Public toilet facilities are provided for a large floating population for buyers and visitors.					
5 Drinking water facility should be planned at two locations per floor, with water quality fit for drinking as per the applicable codes/standards.					

4.3 Materials, Fittings and Appliances

4.3.1 Standards for Materials, Fittings and Sanitary Appliances

All materials, fittings and sanitary appliances shall conform to Part B ‘Building Materials’ of this NBCS.

4.3.2 Choice of Material for Pipes

4.3.2.1 Salt glazed stoneware pipe

For all sewers and drains in all soils, except where supports are required as in made-up ground, glazed stoneware pipe shall be used as far as possible in preference to other types of pipes. These pipes are particularly suitable

where acid effluents or acid subsoil conditions are likely to be encountered. Salt glazed stoneware pipes shall conform to accepted standards [E-2(5)].

4.3.2.2 Cement concrete pipes

When properly ventilated, cement concrete pipes with spigot and socket or collar joints present an alternative to glazed stoneware sewers of over 150 mm diameter. These shall not be used to carry acid effluents or sewage under conditions favourable for the production of hydrogen sulphide and shall not be laid in those subsoils that are likely to affect adversely the quality or strength of concrete. However, where these pipes are employed for conveying liquid containing sulphates, the pipes used shall be manufactured using sulphate resisting Portland cement. Owing to the longer lengths of pipes available, the joints would be lesser in the case of cements concrete pipes. These pipes may be used for surface water drains in all diameters. Cement concrete pipes shall conform to accepted standards [E-2(6)].

4.3.2.3 Cast iron pipes

4.3.2.3.1 These pipes shall be used in the following situation:

- a) In bed or unstable ground where soil movement is expected;
- b) In made-up or tipped ground;
- c) To provide for increased strength where a sewer is laid at insufficient depth, where it is exposed or where it has to be carried on piers or above ground;
- d) Under buildings and where pipes are suspended in basements and like situations;
- e) In reaches where the velocity is more than 2.4 m/s; and
- f) For crossings of watercourses.

NOTE — In difficult foundation condition such as in the case of black cotton soil, the cast iron pipes shall be used only when suitable supporting arrangements are made.

4.3.2.3.2 It shall be noted that cast iron pipes even when given a protective paint are liable to severe external corrosion in certain soils; among such soils are:

- a) Soils permeated by peaty waters; and
- b) Soils in which the subsoil contains appreciable concentrations of sulphates. Local experiences shall be ascertained before cast iron pipes are used where corrosive soil conditions are suspected. Where so used, suitable measures for the protection of the pipes may be resorted to as an adequate safeguard.

4.3.2.3.3 Cast iron pipes shall conform to accepted standards [E-2(7)].

4.3.2.4 Asbestos cement pipes

Asbestos cement pipes are commonly used for house drainage systems and they shall conform to accepted standards [E-2(8)]. They are not recommended for underground situations. However, asbestos cement pressure pipes conforming to accepted standards [E-2(8)] may be used in underground situations also, provided they are not subject to heavy superimposed loads. These shall not be used to carry acid effluents or sewage under conditions favourable for the production of hydrogen sulphide and shall not be laid in those subsoils which are likely to affect adversely the quality or strength of asbestos cement pipes. Where so desired, the life of asbestos cement pipes may be increased by lining inside of the pipe with suitable coatings like epoxy/polyester resins, etc.

4.3.2.5 PVC pipes

Unplasticized PVC pipes may be used for drainage purposes; however, where hot water discharge is anticipated, the wall thickness shall be minimum 3 mm irrespective of the size and flow load.

PVC and HDPE pipes shall conform to accepted standards [E-2(9)].

NOTE — Where possible, high density polyethylene pipes (HDPE) and PVC pipes may be used for drainage and sanitation purposes, depending upon the suitability.

4.3.2.6 Corrugated pipes

These pipes (externally corrugated) are used for sewerage and drainage applications. They are light weight and have long life. The leakage and infiltration at joints are less. The operational cost is low and can easily withstand natural settlements without suffering cracks or leakages. They consume fewer raw material and have less carbon dioxide emission.

4.3.2.7 Low noise pipes

Wastewater systems encompass the system of drainage and pressure relief pipes within a building and terminate 0.5 m outside the external wall. Waste water systems are based on the primary pressure-relief system in which water and air-flow occurs in the same pipe. The wastewater system shall be separated from the roof drainage system.

Noise is a variance in air pressure that spreads like a wave. If quick changes in pressure occur between 20 and 20 000 times a second (frequency 20 Hz and 20 kHz), they are audible to humans. The loudness of noise is determined by the amplitude of the wave, which is measured in decibels (dB).

The main cause of noise in indoor drainage systems (primarily focused on the downpipe) are the choice of the pipe system, the bracket type and the design of drainage system. Optimizing these factors will therefore have the best influence on noise reduction.

4.3.2.8 Under slung pipes

In under slung plumbing, the toilet slab is built at the same level as the slabs outside the toilet. Holes/core cuts are punctured through the slab wherever pipes have to pass through, and the plumbing is clamped to the bottom of the slab. It is then concealed above a false ceiling, which is accessible above the false ceiling for routine maintenance. In this type of plumbing, any leaks will drip onto the false ceiling which is easily detectable thus allow quick maintenance without much damage to the structure or occupied premises.

4.4 Preliminary Data for Design

4.4.1 General

Before the drainage system for a building or group of buildings is designed and constructed, accurate information regarding the site conditions is essential. This information may vary with the individual scheme but shall, in general, be covered by the following:

- a) Site plan (see [3.2.2](#));
- b) Drainage plan (see [3.2.3](#));
- c) *Use* — A description of the use for which the building is intended and periods of occupation in order that peak discharges may be estimated;
- d) *Nature of waste* — While dealing with sewage from domestic premises, special problems under this head may not arise; however, note shall be taken of any possibility of trade effluents being discharged into the pipes at a future date;
- e) *Outlet connection* — The availability of sewers or other outlets;
- f) *Cover* — The depth (below ground) of the proposed sewers and drains and the nature and weight of the traffic on the ground above them;
- g) *Subsoil condition*
 - 1) The approximate level of the subsoil water, and any available records of flood levels shall be ascertained, as also the depth of the water table relative to all sewer connections, unless it is known to be considerably below the level of the latter;
 - 2) In the case of deep manholes, this information will influence largely the type of construction to be adopted. The probable safe bearing capacity of the subsoil at invert level may be ascertained in the case of a deep manhole;

- 3) Where work of any magnitude is to be undertaken, trial pits or boreholes shall be put at intervals along the line of the proposed sewer or drain and the data therefrom tabulated, together with any information available from previous works carried out in the vicinity. In general, the information derived from trial pits is more reliable than that derived from boreholes. For a long length of sewer or drain, information derived from a few trial pits at carefully chosen points may be supplemented by that obtained from number of intermediate boreholes;

Much useful information is often obtained economically and quickly by the use of a soil auger;
- 4) The positions of trial pits or boreholes shall be shown on the plans, together with sections showing the strata found and the dates on which water levels are recorded;
- h) *Location of other services* — The position, depth and size of all other pipes, mains, cables, or other services, in the vicinity of the proposed work, may be ascertained from the Authority, if necessary;
- j) *Reinstatement of surfaces* — Information about the requirements of the highway authority is necessary where any part of the sewer or drain is to be taken under a highway. Those responsible for the sewer or drain shall be also responsible for the maintenance of the surface until permanently reinstated. The written consent of the highway authority to break up the surface and arrangement as to the charges thereof and the method and type of surface reinstatement shall always be obtained before any work is commenced;
- k) *Diversion and control of traffic*
 - 1) In cases where sewers cross roads or foot-paths, cooperation shall be maintained with the police and Authorities regarding the control and diversion of vehicular and/or pedestrian traffic as may be necessary. Access to properties along the road shall always be maintained and adequate notice shall be given to the occupiers of any shops or business premises, particularly if obstruction is likely; and
 - 2) During the period of diversion, necessary danger lights, red flags, diversion boards, caution boards, watchmen, etc, shall be provided as required by the Authority.
- m) *Way-leaves (easements)* — The individual or authority carrying out the work is responsible for negotiating way-leaves where the sewer crosses land in other ownership. The full extent and conditions of such way-leaves shall be made known to the contractor and his employees, and prior notice of commencement of excavation shall always be given to the owners concerned, and cooperation with them shall be maintained at all stages, where sewers run across fields or open ground, the exact location of manholes shall be shown on way-leaves or easement plans. The right of access to manhole covers and the right to maintain the sewer shall be specifically included in any way-leave or easement arrangements which may be made with the owner of the land; and
- n) *Damage to buildings and structures* — When sewer trenches have to be excavated near buildings or walls a joint inspection with the owners of the property shall be made to establish whether any damage or cracks exist before starting the work, and a properly authenticated survey and record of the condition of buildings likely to be affected shall be made. Tell tales may be placed across outside cracks and dated, and kept under observation. Un-retouched photographs taken by an independent photographer may provide useful evidence.

4.4.2 Drainage into a Public Sewer

Where public sewerage is available, the following information is particularly necessary and may be obtained from the Authority:

- a) The position of the public sewer or sewers in relation to the proposed buildings;
- b) The invert level of the public sewer;
- c) The system on which the public sewers are designed (combined, separate or partially separate), the lowest level at which connection may be made to it, and the Authority in which it is vested;
- d) The material of construction and condition of the sewer if connection is not to be made by the Authority;
- e) The extent to which surcharge in the sewer may influence the drainage scheme;
- f) Whether the connection to the public sewer is made, or any part of the drain laid, by the Authority, or whether the owner is responsible for this work; if the latter, whether the Authority imposes any special conditions;

- g) Whether an intercepting trap is required by the Authority on the drain near the boundary of the curtilage; and
- h) Where manholes are constructed under roads, the approval of the Highway Authority for the type of cover to be fitted shall be obtained.

4.4.3 Other Methods of Disposal of Sewage

4.4.3.1 Where discharge into a public sewer is not possible, the drainage of the building/building campus shall be on a separate system. Foul water shall be disposed of by adequate treatment approved by the Authority on the site. The effluent from the plant shall be discharged after meeting the norms specified by the statutory authority into a natural watercourse or on the surface of the ground or disposed of subsoil dispersion preferably draining to a suitable outlet channel.

4.4.3.2 In the case of dilution into a natural stream course, the quality of the effluent shall conform and the requirements of the Authority controlling the prevention of pollution of streams.

4.4.3.3 In the case of subsoil dispersion, the requirements of the Authority for water supply shall be observed to avoid any possible pollution of local water supplies or wells.

4.4.3.4 The general subsoil water level and the subsoil conditions shall be ascertained, including the absorptive capacity of the soil.

4.4.3.5 A subsoil dispersion is not desirable near a building or in such positions that the ground below the foundations is likely to be affected.

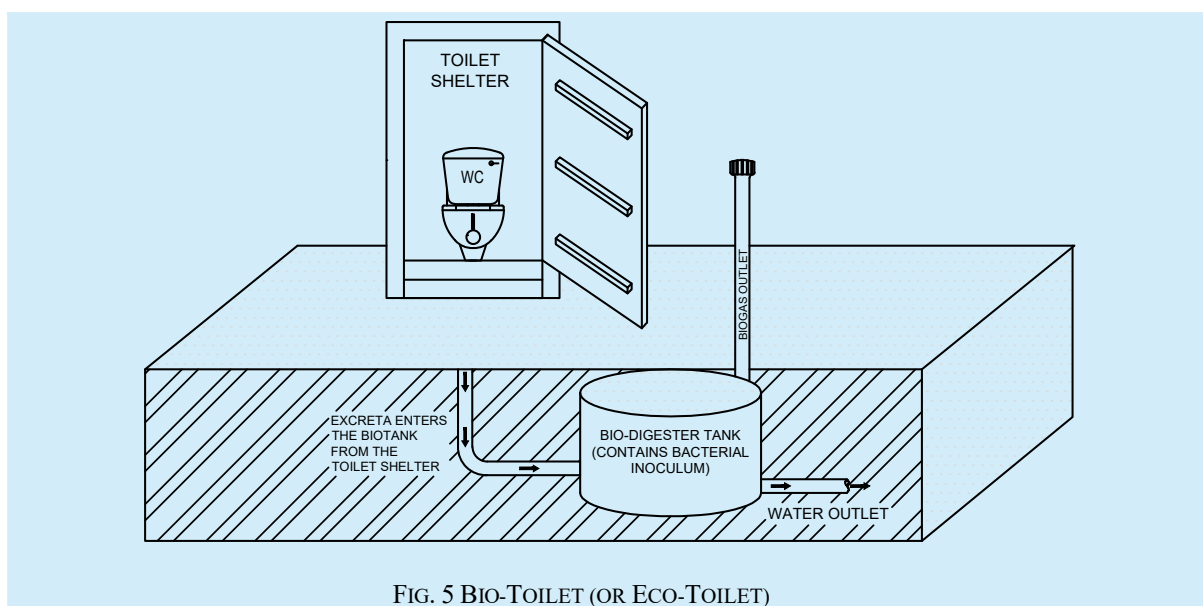
4.4.3.6 Where no other method of disposal is possible, foul water may be diverted to cesspools and arrangements made with the Authority for satisfactory periodical removal and conveyance to a disposal works.

4.4.3.7 Under the separate system, drainage of the building shall be done through septic tanks of different sizes or by stabilization ponds or by any other treatment methods such as extended aeration activated process, sequential batch process, fluidized bio-reactors, membrane bio-reactor, submerged aerobic fixed film, rotating biological contactor, electrolyte process, etc, as approved by the Authority.

For detailed information on the design and construction of septic tanks and waste stabilization ponds, sewage treatment plants reference may be made to good practice [E-2(10)].

4.4.3.8 Biodigester or bio-toilet (or eco-toilet)

Refer [4.13.3\(k\)](#), (see [Fig. 5](#)).



4.4.3.9 Packaged sewage treatment plant

Refer [4.13.3\(i\)](#).

4.4.4 Disposal of Surface and Subsoil Waters

All information which may influence the choice of methods of disposal of surface and/or subsoil waters shall be obtained. In the absence of surface water drainage system and if practicable and permissible, disposal into a natural water-course or soakaway may be adopted. The location and flood levels of the water course as also the requirements of the Authority controlling the river or the waterway shall be ascertained.

4.5 Planning and Design Considerations

4.5.1 Aim

The efficient disposal of foul and surface water from a building is of great importance to public health and is an essential part of the construction of the building. In designing a drainage system for an individual building or a housing colony, the aim shall be to provide a system of self-cleaning conduits for the conveyance of foul, waste, surface or subsurface waters and for the removal of such wastes speedily and efficiently to a sewer or other outlet without risk of nuisance and hazard to health. Also, intent should be to take into consideration that, waste water, storm water and solid waste are potential resources.

4.5.1.1 To achieve this aim, a drainage system shall satisfy the following requirements:

- a) Rapid and efficient removal of liquid wastes without leakage;
- b) Prevention of access of foul gases to the building and provision for their escape from the system.
- c) Adequate and easy access for clearing obstructions;
- d) Prevention of undue external or internal corrosion, or erosion of joints and protection of materials of construction; and
- e) Avoidance of air locks, siphonage, proneness to obstruction, deposit and damage.

4.5.1.2 The realization of an economical drainage system is added by compact grouping of fitments in both horizontal and vertical directions. This implies that if care is taken and ingenuity brought into play when designing the original building or buildings to be drained, it is possible to group the sanitary fittings and other equipment requiring drainage; both in vertical and horizontal planes, as to simplify the drainage system and make it most economical.

4.5.1.3 Efficient and an economical plumbing system can be achieved by planning the toilets in compact grouping with the layout of the bathrooms and observing the following guidelines:

- a) Placing of plumbing fixtures around an easily accessible pipe shaft; in high rise buildings the pipe shafts may have to be within the building envelope and easy provision for access panels and doors should be planned in advance, in such cases so as not to cause inconvenience during maintenance;
- b) Adopting repetitive layout of toilets in the horizontal and vertical directions;
- c) In planning for malls, the required public and toilet facilities shall be located not more than one storey above or below the space to be served with such facilities and the path of travel to such facilities shall not exceed 91 m;

In other occupancies, the required public and toilet facilities shall be located not more than one storey above or below the space to be served with such facilities and the path of travel to such facilities shall not exceed 152 m.

However, in the above buildings, toilet rooms shall not open directly into a room used for preparation of food for service to public.

- d) Avoiding any conflict with the reinforced cement concrete structure by avoiding embedding pipes in it, avoiding pipe crossings in beams, columns and major structural elements;
- e) Identifying open terraces and areas subject to ingress of rainwater directly or indirectly and providing for location of inlets at each level for down takes for disposal at ground levels;

- f) Avoiding crossing of services of individual property through property of other owners;
- g) Planning to avoid accumulation of rainwater or any backflow from sewers particularly in planned low elevation areas in a building;
- h) *Setting of sanitary fixtures* — A water closet, urinal, lavatory or bidet shall not be set closer than 380 mm from the centre to any side wall partition, vanity or other obstruction, or closer than 760 mm centre-to-centre between adjacent fixtures. There shall not be less than 530 mm clearance in front of the water closet, urinal, lavatory or bidet to any wall, fixture or door. Water closet compartment shall not be less than 760 mm in width and 1 520 mm in depth for floor mounted closets, and not less than 760 mm in width and 1 420 mm in depth for wall hung water closets;

The urinal partitions shall begin at a height not greater than 305 mm from and extend not less than 1 520 mm above the finished floor surface. The walls or partitions shall extend from the wall surface at each side of the urinal not less than 460 mm or to a point not less than 150 mm beyond the outermost front lip of the urinal measured from the finished back wall surface, whichever is greater.
- j) Usually, the vertical distance between two horizontal connections to a vertical drain should be more than 200 mm in order to avoid back flow.

4.5.1.4 Typical mounting arrangements for various plumbing fixtures, including drainage systems and ventilation are illustrated in [Annex D](#).

4.5.2 Layout

4.5.2.1 General

Rainwater should preferably be dealt separately from sewage and sullage. Sewage and sullage shall be connected to sewers. However, storm water from the courtyard may be connected to the sewer where it is not possible to drain otherwise; after obtaining permission of the authority.

4.5.2.2 Additional requirements

The following requirements are suggested to be considered in the design of drainage system:

- a) The layout shall be as simple and direct as practicable;
- b) The pipes should be laid in straight lines, as far as possible, in both vertical and horizontal planes;
- c) Anything that is likely to cause irregularity of flow, such as abrupt changes of direction, shall be avoided;
- d) The pipes should be non-absorbent, durable, smooth in bore and of adequate strength;
- e) The pipes should be adequately supported without restricting movement;
- f) Drains should be well ventilated, to prevent the accumulation of foul gases and fluctuation of air pressure within the pipe, which could lead to unsealing (siphoning) of gully or water closet traps;
- g) All the parts of the drainage system should be accessible for feasibility of inspection and practical maintenance;
- h) No bends and junctions whatsoever shall be permitted in sewers except at manholes and inspection chambers;
- j) Sewer drain shall be laid for self-cleaning velocity of 0.75 m/s and generally should not flow more than half-full;
- k) Pipes crossing in walls and floors shall be through mild steel sleeves of diameter leaving an annular space of 5 mm around the outer diameter of the pipe crossing the wall;
- m) Pipes should not be laid close to building foundation;
- n) Pipes should not pass near large trees because of possibility of damage by the roots;
- p) Branch connections should be swept in the direction of flow;
- q) Sewer pipes should be at least 900 mm below road and at least 600 mm below fields and gardens;

- r) Pipes should not pass under a building unless absolutely necessary. Where it is necessary to lay pipes under a building, the following conditions shall be observed:
- 1) Pipes shall be centrifugally cast (spun) iron pressure pipe as per good practice [E-2(11)];
 - 2) The pipe shall be laid in straight line and at uniform gradient;
 - 3) Means of access in form of manholes/inspection chamber shall be provided at each end, immediately outside the building;
 - 4) In case the pipe or any part of it is laid above the natural surface of the ground, it shall be laid on concrete supports, the bottom of which goes at least 150 mm below the ground surface.
- NOTE — It is desirable that pipe/drains should not be taken through a living room or kitchen and shall preferably be taken under a staircase room or passage.
- s) Consideration shall be given to alternative layouts so as to ensure that the most economical and practical solution is adopted. The possibility of alterations shall be avoided by exercising due care and forethought.

4.5.2.3 Protection against vermin and dirt

The installation of sanitary fittings shall not introduce crevices which are not possible to inspect and clean readily.

Pipes, if not embedded, shall be run well clear of the wall. Holes through walls to lay pipes shall be made good on both sides to prevent entry of insects. Materials used for embedding pipes shall be rodent-proof. Passage of rodents from room-to-room or from floor-to-floor shall be prevented by suitable sealing. The intermediate lengths of ducts and chases shall be capable of easy inspection. Any unused drains, sewers, etc, shall be demolished or filled in to keep them free from rodents.

All pipe shafts shall be plastered before any pipes are installed in the shaft. It is advisable to lay pipes on the steel supports with adequate gap between plastered wall and support structure. This will provide a smooth surface and prevent location for survival of insects and vermins.

4.5.2.4 Choice of plumbing system

4.5.2.4.1 In selecting one or more of the type of piping systems, the building and the layout of toilets/fixtures, relationship with other services, acceptability to the Authority, and any special requirements of users, shall be studied.

- a) *Single stack system* [see [2.73\(a\)](#)]
- 1) The single stack system is ideal when the toilet/fixture layouts are repetitive and there is less space for pipes on the wall;
 - 2) In any system so selected there should be not more than two toilet connections per floor;
 - 3) The system requires minimum 100 mm diameter stack for a maximum of 5 floors in a building;
 - 4) In this system, care shall be taken of the horizontal distance of sanitary fixtures from the drainage stack and vertical distance between connections of branches from fixtures to drainage stack; and
 - 5) All the safeguards for the use of this system given in [4.5.2.4.2](#) shall be complied with.
- b) *One pipe – partially ventilated system* [see [2.73\(b\)](#)]

The system and the applicable safeguards under this system are the same as for single stack system. The prime modification is to vent the soil appliance(s).

- c) *One pipe – fully ventilated system* [see [2.73\(c\)](#)]
- 1) This system is suitable for buildings where the toilet/fixture layouts and the shafts are repetitive. It requires less shaft space and is economical;
 - 2) Continuous flow of water in the pipe from waste appliances makes it less prone to blockage and makes the system more efficient;

- 3) The system eliminates the need for a gully trap which requires constant cleaning;
 - 4) This system requires individual vent pipes installed either in suspended ceiling or in the wall above the floor level in the toilet, provided construction details of the building allow provision of vent pipes for individual fixtures and routing of vent pipes to remote shafts where main vent stacks may have been located;
 - 5) The system requires minimum 100 mm diameter stack;
 - 6) The system is ideal when the main pipes run at the ceiling of the lowest floor or in a service floor.
- d) *Two pipe system* [see [2.73\(d\)](#)]
- 1) This system provides safety and flexibility in layouts for larger toilets and for buildings having higher requirement of performance due to its height and type of occupancy;
 - 2) In large buildings and houses with open ground and gardens, the sullage water from the waste system can be suitably treated and usefully utilized for gardening and agriculture; and
 - 3) In larger and multi-storeyed buildings, the sullage is treated within the building for reuse as makeup water for cooling towers for air conditioning system and is also used for flushing water closets provided it has absolutely no connection with any water supply line, tank or system used for domestic and drinking supply.
- e) *Vacuum sewage system* — Vacuum based sewage disposal system is efficient and environmentally friendly alternative. This system is easier to construct compared to conventional systems. The system operates with a closed, leak-proof vacuum network and requires shallower excavation compared to conventional systems. It is suitable for flat terrain, high water table areas, seaside locations, and environmentally sensitive zones.

The system comprises the following:

- 1) House gravity connection;
- 2) Collection chamber with interface unit;
- 3) Vacuum connection and piping;
- 4) Vacuum station; and
- 5) Odour fitter and ring main.

One of the characteristic features of this system is the presence of the collection chamber. The collection chamber serves or reserve storage for a day, during any power failure.

Sewage flows by gravity into a collection chamber and remains for a short time with reserve storage availability. When the level rises, a starter valve activates the vacuum piston valve to open and to allow the sewage water with air to be derived in the vacuum network. Multiple connections can be linked to a single collection chamber.

The system can be used in the following applications:

- i) Septic tank replacements;
- ii) Ribbon development and new residential layouts;
- iii) Villages;
- iv) Industrial areas;
- v) Islands;
- vi) Rocky terrain; and
- vii) Environmentally sensitive areas.

For detailed information regarding design and installation of soil, waste and vent pipes, reference may be made to good practice [E-2(12)].

4.5.2.4.2 Safeguards for single stack system

- a) As far as practicable, the fixtures on a floor shall be connected to stack in order of increasing discharge rate in the downward direction;
- b) The size of the vent pipe (terminating to the outdoor) shall be same as the size of drainage stack in the case of single stack system;
- c) Water closets shall be connected to the stack through a sanitary tee; the maximum horizontal distance from the stack being 2 400 mm;
- d) For fixtures other than water closets, the maximum horizontal distance from the stack shall be 3 500 mm.
- e) The vertical distance between the waste branch (from floor trap or from the individual appliance) and the soil branch connection, when soil pipe is connected to stack above the waste pipe, shall be not less than 200 mm;
- f) Depth of water seal traps from different fixtures shall be as follows:

Water closets : 50 mm

Floor traps : 50 mm

Other fixtures directly connected to the stack:

- 1) Where attached to branch : 40 mm waste pipes of 75 mm dia or more
- 2) Where attached to branch : 75 mm waste pipes of less than 75 mm dia

NOTE — When connection is made through floor trap, no separate seals are required for individual fixtures.

- g) Branches and stacks which receive discharges from WC pans should not be less than 100 mm, except where the outlet from the siphonic water closet is 80 mm, in which case a branch pipe of 80 mm may be used. For outlet of floor traps 75 mm dia pipes may be used;
- h) The horizontal branch distance for fixtures from stack, bend(s) at the foot of stack to avoid back pressure as well as vertical distance between the lowest connection and the invert of drain shall be as good practice [E-2(12)]; and
- j) For tall buildings, ground floor appliances are recommended to be connected directly to manhole/ inspection chamber.

4.5.2.4.2.1 Drainage stack shall be sized in accordance with [Table 18](#). The drainage stack and branch piping shall be the vents for the drainage system. Stack shall be uniformly sized based on the total connected drainage fixture unit load.

Table 18 Single Stack Sizing

(Clause [4.5.2.4.2.1](#))

Sl No.	Stack Size mm	Maximum Connected Drainage Fixture Units (DFU) ²⁾		
		Stacks Less than 23 m in Height	Stacks 23 m to Less than 49 m in Height	Stacks 49 m and Greater in Height ¹⁾
(1)	(2)	(3)	(4)	(5)
i)	75	24	—	—
ii)	100	225	24	—
iii)	125	480	225	24
iv)	150	1 015	480	225
v)	200	2 320	1 015	480
vi)	250	4 500	2 320	1 015
vii)	300	8 100	4 500	2 320
viii)	375	13 600	8 100	4 500

¹⁾ To be considered for one pipe – partially ventilated system only.

²⁾ 1 DFU may be taken as equivalent to 0.47 L/s.

4.5.2.4.2.2 An alternative design approach for designing the single stack system is also used in some of the countries, which may be followed if approved by the Authority. Such an approach may utilize the following, and may also require reference to available specialist literature:

- a) The size of the vent pipe (terminating to the outdoor) shall be same as the size of drainage stack in the case of single stack system;
- b) Using branch discharge pipes, self and external siphonage of the water seal of traps can be avoided;
- c) In respect of branch discharge pipes, the entire developed length including fittings up to farthest connection is 4.0 m;
- d) A 45° bend shall be used for the horizontal branch pipe with a drop connection after the connection from the appliance;
- e) To match the crowns of small and larger pipes, an eccentric reducer is used with straight portion on top;
- f) A non-ventilated discharge pipe shall be connected to discharge stack with a sweep junction having an angle of 87° to 88.5°;
- g) In respect of vertical discharge stacks, concentric or eccentric reducers may be permitted;
- h) In respect of stacks not exceeding 10 m in height, from the upper most connection to change of direction, no branch discharge pipe shall be connected to the stack or horizontal side for a distance of 1.0 m. In respect of second change in direction, no branch discharge pipe shall be connected within a distance of 0.5 m after the change of direction. The pipe stack shall not be connected to sanitary appliances from the lowest floor;
- j) In respect of stacks exceeding 10 m in height, from the upper most connection to change of direction, no branch discharge pipe shall be connected to the stack or horizontal side for a distance of 2.0 m. The connection of the ventilated branch to the main stack shall not be within 2.0 m from the change of direction. In respect of second change in direction, no branch discharge pipe shall be connected within a distance of 0.5 m after the change of direction. The pipe stack shall not be connected to sanitary appliances from the lowest floor;
- k) The connection of branch discharge pipes to the vertical stack shall be such that back flow will not cross flow into fixtures on the opposite ends; and
- m) The single stack system shall be designed and installed based on the principles of sanitary appliance usage, proper dimensioning, and installation guidelines.

4.5.3 Drainage (Soil, Waste and Vent) Pipes

4.5.3.1 General considerations

4.5.3.1.1 Drainage pipes shall be kept clear of all other services. Provisions shall be made during the construction of the building for the entry of the drainage pipes. In most cases this may be done conveniently by installing sleeves or conduit pipes into or under the structure in appropriate positions. This will facilitate the installation and maintenance of the services.

4.5.3.1.2 Horizontal drainage piping should be so routed as not to pass over any equipment or fixture where leakage from the line could possibly cause damage or contamination. Drainage piping shall never pass over switch-gear or other electrical equipment. If it is impossible to avoid these areas and piping shall run in these locations, then a pan or drain tray should be installed below the pipe to collect any leakage or condensation. A drain line should run from this pan to a convenient floor drain or service sink.

4.5.3.1.3 All vertical soil, waste, ventilating and anti-siphonage pipes shall be covered on top with a copper or heavily galvanized iron wire dome or cast iron terminal guards. All cast iron pipes, which are to be painted periodically, shall be fixed to give a minimum clearance of 50 mm clear from the finished surface of the wall by means of a suitable clamp.

NOTE — Asbestos cement cowls may be used in case asbestos cement pipes are used as soil pipes.

4.5.3.1.4 Drainage pipes shall be carried to a height above the buildings as specified for ventilating pipe (see [4.5.3.4](#)).

4.5.3.2 Soil pipes

A soil pipe, conveying to a drain, any solid or liquid filth, shall be circular and shall have a minimum diameter of 100 mm.

4.5.3.2.1 Except where it is impracticable, the soil pipe shall be situated outside the building or in suitably designed pipe shafts and shall be continued upwards without diminution of its diameter, and (except where it is unavoidable) without any bend or angle, to such a height and position as to afford by means of its open end a safe outlet for foul air. The position of the open end with its covering shall be such as to comply with the conditions set out in [4.5.3.4](#) relating to ventilating pipe. Even if the pipes are laid externally, the soil pipes shall not be permitted on a wall abutting a street unless the Authority is satisfied that it is unavoidable. Where shafts for pipes are provided, the cross section area of the shaft shall be suitable to allow free and unhampered access to the pipes and fittings proposed to be installed in the shaft. However, in no case cross section area of the shaft shall be less than a square of one-meter side. All pipe shafts shall be provided with an access door at ground level and facilities for shaft ventilation.

4.5.3.2.2 Soil pipes, whether inside or outside the building, shall not be connected with any rainwater pipe and there shall not be any trap in such soil pipe or between it and any drain with which it is connected.

4.5.3.2.3 The soil pipe shall be provided with heel rest bend which shall rest on sound footing, if terminating at firm ground level. When the stack is terminating at the ceiling of a floor, the bend shall be provided with sufficient structural support to cater for the stack dead weight and the thrust developed from the falling soil/waste. Vertical stack shall be fixed at least 50 mm clear of the finished surface of the wall by means of a suitable clamp of approved type.

4.5.3.3 Waste pipes

Every pipe in a building for carrying off the waste or overflow water from every bath, wash basin or sink to a drain shall be of 32 mm to 50 mm diameter, and shall be trapped immediately beneath such wash basins or sink by an efficient siphon trap – P trap with adequate means for inspection and cleaning. Such P traps shall be ventilated into the external air whenever such ventilation is necessary to preserve the seal of the trap. Waste pipes, P traps, etc, shall be constructed of iron, lead, brass, PVC, engineering plastics, stoneware, asbestos cement or other approved material. The overflow pipe from wash basin, sinks, etc, shall be connected with the waste pipe immediately above the trap. Vertical pipes carrying off waste water shall have a minimum diameter of 75 mm.

NOTE — Whenever wash basins and sinks have in-built overflow arrangements, there is no need to provide overflow pipes in such cases.

4.5.3.3.1 Every pipe in a building for carrying off waste water to a drain shall be taken through an external wall of the building by the shortest practicable line, and shall discharge below the grating or surface box of the chamber but above the inlet of a properly trapped gully. The waste pipe shall be continued upwards without any diminution in its diameter and (except when unavoidable) without any bend or angle to such a height and position as to afford by means of the open end of the waste pipe, a safe outlet for foul air, the position of the open end and its covering being such as to comply with the conditions.

4.5.3.3.2 Except where it is impracticable, the common waste pipe shall be situated outside the building and shall be continued upwards without diminution of its diameter (except where it is unavoidable) without any bend or angle being formed to such a height and position as to avoid by means of the open end a safe outlet for foul air, the position of the open end and the covering threat being such as to comply with the conditions set out in [4.5.3.4](#) relating to ventilating pipe.

4.5.3.3.3 If the waste pipe is of cast iron, it shall be firmly attached 50 mm clear of the finished surface of the wall by means of a suitable clamps or with properly fixed holder bats or equally suitable and efficient means.

4.5.3.4 Vent pipes

Ventilating pipes should be so installed that water cannot be retained in them. They should be fixed vertically. Whenever possible, horizontal runs should be avoided. Ventilating pipe shall be carried to such a height and in such a position as to afford by means of the open end of such pipe or vent shaft, a safe outlet for foul air with the least possible nuisance.

4.5.3.4.1 The upper end of the main ventilating pipe may be continued to the open air above roof level as a separate pipe, or it may join the MSP and/or MWP above the floor level of the highest appliance. Its lower end may be carried down to join the drain, at a point where air relief may always be maintained.

4.5.3.4.2 Branch ventilating pipes should be connected to the top of the BSP and BWP between 75 mm and 450 mm from the crown of the trap.

4.5.3.4.3 The ventilating pipe shall always be taken to a point 1 500 mm above the level of the eaves or flat roof or terrace parapet whichever is higher or the top of any window within a horizontal distance of 3 m. The least dimension shall be taken as a minimum and local conditions shall be taken into account. The upper end of every ventilating pipe shall be protected by means of a cowl.

4.5.3.4.4 In case the adjoining building is taller, the ventilating pipe shall be carried higher than the roof of the adjacent building, wherever it is possible.

4.5.3.4.5 The building drain intended for carrying waste water and sewage from a building shall be provided with at least one ventilating pipe situated as near as practicable to the building from an inspection chamber and as far away as possible from the point at which the drain empties into the sewer or other carrier.

4.5.3.4.6 *Size of ventilating pipe*

- a) The building drain ventilating pipe shall be of not less than 75 mm diameter. When, however, it is used as MSP or MWP, the upper portion, which does not carry discharges, shall not be of lesser diameter than the remaining portion;
- b) The diameter of the main ventilating pipe in any case should not be less than 50 mm;
- c) A branch ventilating pipe on a waste pipe in both one and two-pipe systems shall be of not less than two-thirds the diameter of the branch waste pipe, subject to a minimum of 25 mm;
- d) A branch ventilating pipe on a soil pipe in both one and two-pipe systems shall be not less than 32 mm in diameter; and
- e) Ventilating pipes to be sized/designed considering the drainage fixture units of individual fixtures/appliances as per good engineering practices (see [4.5.3.4.7](#)).

4.5.3.4.7 *Venting system for high rise buildings*

In the case of fully ventilated system, suitable sized pipe stacks carry soil and waste drainage, wherein each sanitary fixture is individually vented. The practical way of implementation of this system is by venting each water closet and floor drain. A fully ventilated one pipe drainage system is most popular in advanced plumbing installations.

Properly sized ventilating pipe would ensure maintenance of atmospheric pressure within gravity drainage pipes.

Sizing of ventilating pipe is based on cumulative drainage fixture units of all fixtures served by the pipe. Sizing also depends on maximum permissible lengths. Maximum lengths are also subject to the limitation that one-third of length shall be horizontal. Horizontal pipes always rise towards termination point, avoiding vertical loops.

The minimum required diameter of stack vents and vent stacks shall be determined from the developed length and the total drainage fixture units (DFU) connected thereto, in accordance with [Table 19](#). However, in no case, the diameter shall be less than half the diameter of the drain served or less than 32 mm.

NOTE — In case, if fully ventilated one-pipe system is not implemented then it shall indicate the case of two-pipe system along with common ventilating pipe for which the guideline for sizing shall be provided. Also, it shall provide a sketch for two-pipe system along with connection of branch pipes from wash basin to the deep seal trap through inlet fitting as per prevailing conditions as approved by Authority.

Table 19 Size and Developed Length of Stack Vents and Vent Stacks

(Clause [4.5.3.4.7](#))

Sl No.	Diameter of Soil and/or Waste Stack mm	Total Fixture Units Being Vented (DFU)	Maximum Developed Length of Vent for the Diameter of Vent m										
			32 mm	40 mm	50 mm	65 mm	75 mm	100 mm	125 mm	150 mm	200 mm	250 mm	300 mm
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
i)	32	2	9	–	–	–	–	–	–	–	–	–	–
ii)	40	8	15	45	–	–	–	–	–	–	–	–	–
iii)	40	10	9	30	–	–	–	–	–	–	–	–	–
iv)	50	12	–	23	61	–	–	–	–	–	–	–	–
v)	50	20	9	15	45	–	–	–	–	–	–	–	–
vi)	65	42	8	9	30	91	–	–	–	–	–	–	–
vii)	75	10	–	13	45	110	317	–	–	–	–	–	–
viii)	75	21	–	10	34	82	247	–	–	–	–	–	–
ix)	75	53	–	8	29	70	207	–	–	–	–	–	–
x)	75	102	–	–	26	64	189	–	–	–	–	–	–
xi)	100	43	–	8	11	26	76	299	–	–	–	–	–
xii)	100	140	–	–	8	20	61	229	–	–	–	–	–
xiii)	100	320	–	–	7	17	52	195	–	–	–	–	–
xiv)	100	540	–	–	6	15	46	177	–	–	–	–	–
xv)	125	190	–	–	–	8.5	25	98	302	–	–	–	–
xvi)	125	490	–	–	–	6.4	19	76	232	–	–	–	–
xvii)	125	940	–	–	–	5.5	16	64	204	–	–	–	–
xviii)	125	1 400	–	–	–	4.9	15	58	180	–	–	–	–
xix)	150	500	–	–	–	–	10	40	122	305	–	–	–
xx)	150	1 100	–	–	–	–	8	30	94	238	–	–	–
xxi)	150	2 000	–	–	–	–	7	26	79	201	–	–	–

Table 19 (Concluded)

Sl No.	Diameter of Soil and/or Waste Stack mm	Total Fixture Units Being Vented (DFU)	Maximum Developed Length of Vent for the Diameter of Vent m										
			32 mm	40 mm	50 mm	65 mm	75 mm	100 mm	125 mm	150 mm	200 mm	250 mm	300 mm
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
xxii)	150	2 900	—	—	—	—	6	23	73	183	—	—	—
xxiii)	200	1 800	—	—	—	—	—	9	29	73	287	—	—
xxiv)	200	3 400	—	—	—	—	—	7	22	58	222	—	—
xxv)	200	5 600	—	—	—	—	—	6	19	49	186	—	—
xxvi)	200	7 600	—	—	—	—	—	5	17	43	171	—	—
xxvii)	250	4 000	—	—	—	—	—	—	9	24	94	293	—
xxviii)	250	7 200	—	—	—	—	—	—	7	18	73	226	—
xxix)	250	11 000	—	—	—	—	—	—	6	16	61	192	—
xxx)	250	15 000	—	—	—	—	—	—	5	14	55	174	—
xxxi)	300	7 300	—	—	—	—	—	—	—	9	37	116	287
xxxii)	300	13 000	—	—	—	—	—	—	—	7	29	91	219
xxxiii)	300	20 000	—	—	—	—	—	—	—	6	24	76	186
xxxiv)	300	26 000	—	—	—	—	—	—	—	5	22	70	152
xxxv)	375	15 000	—	—	—	—	—	—	—	—	12	40	94
xxxvi)	375	25 000	—	—	—	—	—	—	—	—	9	29	73
xxxvii)	375	38 000	—	—	—	—	—	—	—	—	8	25	61
xxxviii)	375	50 000	—	—	—	—	—	—	—	—	7	23	55

NOTE — The developed length shall be measured from the vent connection to the open air.

4.5.3.5 Design of drainage pipes

A stack is the main vertical pipe that carries away discharge from water closets and urinals (soil stack) or other clear wastewater from equipment (waste stack) with adequate suitable fittings, which may be a long-turn, tee-wye or short-turn or sanitary tee. Special fittings may also be used, provided that their drainage capacity has been certified in accordance with the relevant Indian Standards or by approved laboratories. Depending on the rate of flow into the drain stack, the diameter of the stack, the type of stack fittings and the flow down the stack from higher levels (if any), the discharge from the fixture drain may or may not fill the cross section of the stack at the level of entry. In any event, as soon as the water enters the stack, the force of gravity rapidly accelerates it downward and before it travels very far, it assumes the form of a sheet around the wall of the stack, leaving the centre of the pipe open for the flow of air.

This sheet of water continues to accelerate until the frictional force exerted by the wall of the falling sheet of water equals the gravitational force. If the distance the water travel is sufficient enough and provided that no flow enters the stack at lower levels to interfere the sheet, the sheet remains unchanged in thickness and velocity until it reaches the bottom of the stack. The ultimate vertical velocity the sheet attains is called the 'terminal velocity'. The distance the sheet should fall to attain this terminal velocity is called the 'terminal length'.

Following formulae may be used for calculating the terminal velocity and terminal length:

$$V_t = 3.0 (Q/d)^{2/5}$$

$$L_t = 0.052 V_t^2$$

where

V_t = terminal velocity in the stack, m/s;

Q = quantity rate of flow, l/s;

d = diameter of stack, mm; and

L_t = terminal length below the point of flow entry, m;

At the centre of the stack is a core of air that is dragged along with the water by friction. A supply source of air shall be provided to avoid excessive pressures in the stack. The usual means of supplying this air are through the stack vent or vent stack. The entrained air in the stack causes a pressure reducing inside the stack, which is caused by the frictional effect of the falling sheet of water dragging the core of air with it.

4.5.3.5.1 Estimation of maximum flow of sewer

a) Simultaneous discharge flow

- 1) The maximum flow in a building drain or a stack depends on the probable maximum number of simultaneous discharging appliances. For the calculation of this peak flow certain loading factors have been assigned to appliances in terms of fixture units, considering their probability and frequency of use. These fixture unit values are given in [Table 20](#);
- 2) For any fixtures not covered under [Table 20](#), [Table 21](#) may be referred to for deciding their fixture unit rating depending on their drain or trap size;
- 3) From [Table 20](#) and [Table 21](#), the total load on any pipe in terms of fixtures units may be calculated knowing the number and type of appliances connected to this pipe;
- 4) For converting the total load in fixture units to the peak flow in litre per minute, [Fig. 6](#) is to be used.
- 5) The maximum number of fixture units that are permissible for various recommended pipe size in the drainage system are given in [Table 22](#) and [Table 23](#); and
- 6) Results should be checked to see that the soil, waste and building sewer pipes are not reduced in diameter in the direction of flow. Where appliances are to be added in fixture, these should be taken into account in assessing the pipe sizes by using the fixture units given in [Table 20](#) and [Table 21](#).

- b) *Maximum discharge flow* — The maximum rate of *discharge* flow shall be taken as thrice the average rate; allowance being made in addition for any exceptional peak discharges. A good average rule is to allow for a flow of liquid wastes from buildings at the rate of 3 litre per minute per 10 persons.

4.5.3.5.2 Gradients

4.5.3.5.2.1 The discharge of water through a domestic drain is intermittent and limited in quantity and, therefore, small accumulations of solid matter are liable to form in the drains between the building and the public sewer. There is usually a gradual shifting of these deposits as discharges take place. Gradients should be sufficient to prevent these temporary accumulations building up and blocking the drains:

- a) *Drainage loads* — Single family dwellings contain plumbing fixtures, such as one or more bathroom groups, each consisting of a toilet, wash basin and bathtub or shower unit, a kitchen sink, dishwasher and washing machine. Large buildings also have other fixtures, slop sinks and drinking water coolers. The important characteristic of these fixtures is that they are not used continuously. Rather, they are used with irregular frequencies that vary greatly during the day. In addition, the various fixtures have quite different discharge characteristics regarding both the average flow rate per use and the duration of a single discharge. Consequently, the probability of all the fixtures in the building operating simultaneously is small; and
- b) *Stack capacities* — The criterion of flow capacities in drainage stacks is based on the limitation of the water occupied cross section to a specified fraction of the cross section of the stack where terminal velocity exists, as suggested by earlier investigations.

Flow capacity can be expressed in terms of the stack diameter and the water cross section, as follows:

$$Q = 27.8 \times r_s^{5/3} \times D^{8/3}$$

where

Q = capacity, l/s;

r_s = ratio of the cross-sectional area of the sheet of water to the cross-sectional area of the stack; and

D = diameter of the stack, mm.

Table 20 Drainage Fixture Units (DFU) for Different Fixtures with Minimum Pipe Sizes

(Clause 4.5.3.5.1)

Sl No.	Type of Fixture	Application		Minimum Trap Size mm
		Private	Public	
(1)	(2)	(3)	(4)	(6)
i)	Bathroom group (water closet, wash basin, bidet and tub or shower):			
	a) Water closet (flush valve)	8	—	—
	b) Water closet (flush tank)	6	—	—
ii)	Bathtub	3	—	40
iii)	Bar sink	1	2	40
iv)	Ablution faucet/Bidet	1	2	40
v)	Clothes washer	3	3	50
vi)	Dishwasher	2	2	40
vii)	Drinking fountain	—	0.5 (0.75)	32
viii)	Floor drain	1	2	50

Table 20 (Concluded)

Sl No.	Type of Fixture	Application		Minimum Trap Size mm
		Private	Public	
(1)	(2)	(3)	(4)	(6)
ix)	Wash basin	1	1	32
x)	Service or mop basin/sink	—	3	50
xi)	Kitchen sink	2	2	40
xii)	Shower	2	2	50
xiii)	Laundry sink	2	2	40
xiv)	Clinical or surgeon's scrub sink	—	6	80
xv)	Urinal (with flush valve)	2	2	50
xvi)	Urinal (with flush tank)	2	2	50
xvii)	Urinal with sensor operated	2	2	50
xviii)	Water closet (flush valve)	4	6 (8)	80
xix)	Water closet (flush tank)	3	4 (6)	80
xx)	Combination fixture (faucet)	1	2	40

NOTES

- 1 A shower head over a bath tub does not increase the fixture unit value.
- 2 Size of floor trap shall be determined by the area of surface water to be drained.
- 3 Wash basins with 32 mm and 40 mm trap have the same load value.
- 4 Trap size shall be consistent with the fixture outlet size.
- 5 For fixtures added to bathroom group, the DFU value is added to those additional fixtures to the bathroom group fixture count.
- 6 No floor traps to be provided inside operating rooms, procedure rooms, AIIR isolation rooms and PE isolation rooms.
- 7 The values given in parentheses pertain to such public use buildings where an enhanced requirement is expected to be encountered as compared to the normal maximum use in public use buildings.

**Table 21 Fixture Unit Values for Fixtures Based on Fixture
Drain on Trap Size
(Clause [4.5.3.5.1](#))**

Sl No.	Fixture Drain on Trap Size	Fixture Unit Value
(1)	(2)	(3)
i)	32 mm and smaller	1
ii)	40 mm	2
iii)	50 mm	3
iv)	65 mm	4
v)	80 mm	5
vi)	100 mm	6

Table 22 Horizontal Fixture Branches and Stacks¹⁾

(Clause [4.5.3.5.1](#))

SI No.	Diameter of Pipe mm	Maximum Number of Drainage Fixture Units ²⁾ (DFU) that can be Connected to			
		Branch Total for Horizontal Branch	Stacks ³⁾		
			Total Discharge into One Branch Interval	Total for Stack of Three Branch Intervals or Less	Total for Stack Greater than Three Branch Intervals
(1)	(2)	(3)	(4)	(5)	(6)
i)	32	1	2	2	2
ii)	40	3	2	4	8
iii)	50	6	6	10	24
iv)	65	12	9	20	42
v)	75	20	20	48	72
vi)	100	160	90	240	500
vii)	125	360	200	540	1 100
viii)	150	620	350	960	1 900
ix)	200	1 400	600	2 200	3 600
x)	250	2 500	1 000	3 800	5 600
xi)	300	3 900	1 500	6 000	8 400
xii)	375	7 000	See Note	See Note	See Note

NOTE — Sizing load based on design criteria.

¹⁾ Does not include branches of the building sewer.

²⁾ Depending upon the probability of simultaneous use of appliances considering the frequency of use and peak discharge rate.

³⁾ Stacks shall be sized based on the total accumulated connected load at each story or branch interval. As the total accumulated connected load decreases, stacks are permitted to be reduced in size. Stack diameters shall not be reduced to less than one-half of the diameter of the larger stack size required.

Table 23 Building Drains and Sewers¹⁾

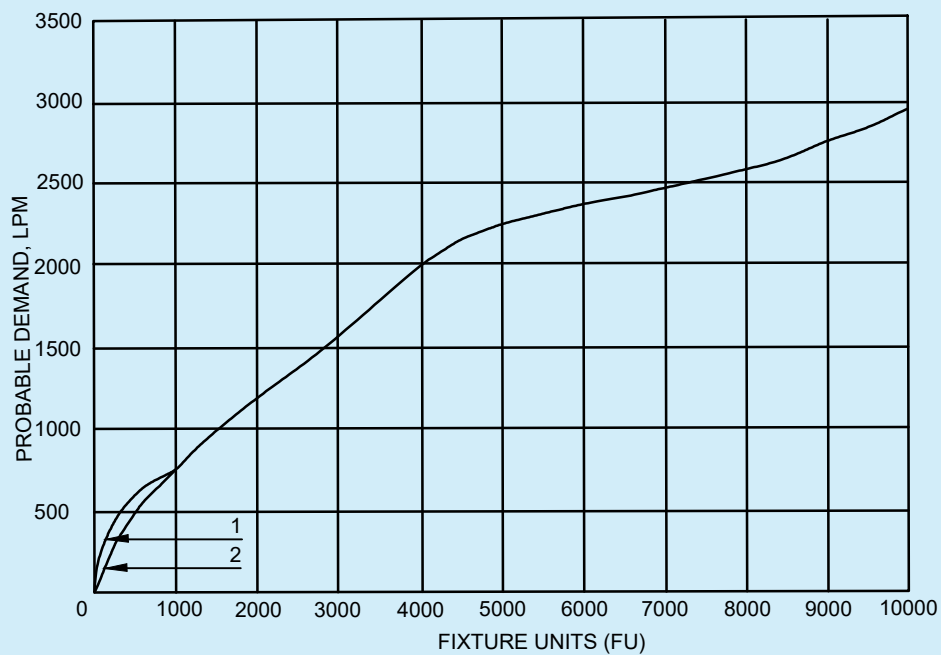
(Clause [4.5.3.5.1](#))

Sl No.	Diameter of Pipe mm	Maximum Number of Drainage Fixture Units (DFU) that can be Connected to any Portion of the Building Drain or the Building Sewer Including Branches of the Building Drain ²⁾ , for Slope			
		1/200	1/100	1/50	1/25
(1)	(2)	(3)	(4)	(5)	(6)
i)	32	—	—	1	1
ii)	40	—	—	3	3
iii)	50	—	—	21	26
iv)	65	—	—	24	31
v)	75	—	36	42	50
vi)	100	—	180	216	250
vii)	125	—	390	480	575
viii)	150	—	700	840	1 000
ix)	200	1 400	1 600	1 920	2 300
x)	250	2 500	2 900	3 300	4 200
xi)	300	3 900	4 600	5 600	6 700
xii)	375	7 000	8 300	10 000	12 000

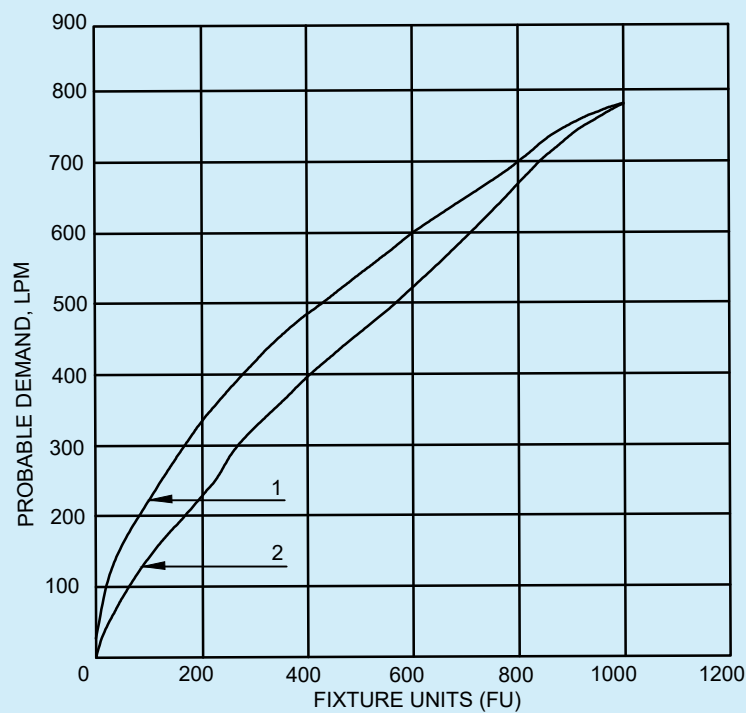
¹⁾ Includes branches of the building sewer.

²⁾ The minimum size of any building drain serving a water closet shall be 75 mm.

- c) *Design requirement for high rise buildings drainage system* — Drainage pipe stacks are sized for one third of their carrying capacity. Plumbing codes provides values of drainage fixture units for each fixture. Different values for fixture units are based on the nature of occupancy and the place of installation. Sizing of drainage pipes is based on the cumulative values of drainage fixture units connected to the pipe. A vertical pipe shall always have larger carrying capacity when compared to horizontal pipe of same size. Carrying capacity of horizontal pipe is dependent on gradient provided. Carrying capacities of vertical pipes are restricted by their maximum permissible lengths. This restriction does not apply to horizontal pipes. Irrespective of drainage fixture units, the minimum size for the fixture shall be adopted. The minimum size of horizontal pipe is critical to ensure self cleansing velocities in sewers.



6A GRAPH FOR PROBABLE DEMAND UP TO 10 000 FU



6B GRAPH FOR PROBABLE DEMAND UP TO 1000 FU

CURVE 1 - SYSTEM WITH FLUSH VALVES
CURVE 2 - SYSTEM WITH FLUSH TANKS

FIG. 6 GRAPH FOR PROBABLE DEMAND

4.5.3.5.2.2 When flow occurs in drain piping, it should not entirely fill the cross section of the pipe under flow condition. If the pipe were to flow full, pressure fluctuations would occur which could possibly destroy the seal of the traps within the building. Normally, the sewer shall be designed for discharging the peak flow as given in [4.5.3.5.1](#), flowing half-full with a minimum self-cleansing velocity of 0.75 m/s. The approximate gradients which give this velocity for the sizes of pipes likely to be used in building drainage and the corresponding discharges when flowing half-full are given in [Table 24](#).

4.5.3.5.2.3 In cases where it is practically not possible to conform to the ruling gradients, a flatter gradient may be used, but the minimum velocity in such cases shall on no account be less than 0.61 m/s and adequate flushing should be done.

NOTE — Where gradients are restricted, the practice of using a pipe of larger diameter than that required by the normal flow, in order to justify laying at a flatter gradient does not result in increasing the velocity of flow, further this reduces the depth of flow and thus for this reasons the above mentioned practice should be discouraged.

4.5.3.5.2.4 On the other hand, it is undesirable to employ gradients giving a velocity of flow greater than 2.4 m/s. Where it is unavoidable, cast iron pipes shall be used. The approximate gradients, which give a velocity of 2.4 m/s for pipes of various sizes and the corresponding discharge when flowing half-full are given in [Table 24](#).

4.5.3.5.2.5 The discharge values corresponding to nominal diameter and gradient given in [Table 24](#) are based on Manning's formula ($n = 0.015$).

NOTE — Subject to the minimum size of 100 mm, the sizes of pipes shall be decided in relation to the estimated quantity of flow and the available gradient.

Table 24 Different Diameter Pipes Giving Velocity and Corresponding Discharge at Minimum and Maximum Gradient <i>(Clauses 4.5.3.5.2.2, 4.5.3.5.2.4 and 4.5.3.5.2.5)</i>					
SI No.	Diameter (mm)	Minimum Gradient (Velocity : 0.75 m/s)	Discharge at the Minimum Gradient m ³ /min	Maximum Gradient (Velocity : 2.4 m/s)	Discharge at the Maximum Gradient m ³ /min
(1)	(2)	(3)	(4)	(5)	(6)
i)	100	1 in 57	0.18	1 in 5.6	0.59
ii)	150	1 in 100	0.42	1 in 9.7	1.32
iii)	200	1 in 145	0.73	1 in 14	2.40
iv)	230	1 in 175	0.93	1 in 17	2.98
v)	250	1 in 195	1.10	1 in 19	3.60
vi)	300	1 in 250	1.70	1 in 24.5	5.30

4.5.3.6 Drain appurtenances

4.5.3.6.1 Trap

All traps shall be protected against siphonage and back pressure ensuring access to atmospheric air for air circulation and preserving the trap seal in all conditions.

4.5.3.6.1.1 A trap may be formed as an integral trap with the appliance during manufacture or may be a separate fitting called an attached trap which may be connected to the waste outlet of the appliance.

4.5.3.6.1.2 Traps should always be of a self-cleansing pattern. A trap, which is not an integral part of an appliance, should be directly attached to its outlet and the pipe should be uniform throughout and have a smooth surface.

4.5.3.6.1.3 The trap should have minimum size of outlet/exit, same as that of largest waste inlet pipe.

4.5.3.6.1.4 Traps for use in domestic waste installations and all other traps should be conveniently accessible and provided with cleansing eyes or other means of cleaning.

4.5.3.6.1.5 The minimum internal diameter for sanitary appliances shall be as follows:

<i>Sl No.</i>	<i>Sanitary Appliance</i>	<i>Minimum Internal Diameter of Waste Outlet mm</i>
(1)	(2)	(3)
i)	<i>Soil appliances:</i>	
	a) Indian and European type water closets	100
	b) Bed pan washers and slop sinks	100
	c) Urinal with integral traps	75
	d) Stall urinals (with not more than 120 mm of channel drainage)	50
	e) Lipped urinal small/large	40
ii)	<i>Waste appliances:</i>	
	a) Drinking fountain	25
	b) Wash basin	32
	c) Bidets	32
	d) Domestic sinks and baths	40
	e) Shower bath trays	40
	f) Domestic bath tubs	50
	g) Hotel and canteen sinks	50
	h) Floor traps (outlet diameter)	75

4.5.3.6.2 *Floor drains*

All toilets/bathrooms in a building desirably should be provided with floor drains to facilitate cleaning.

4.5.3.6.2.1 Floor drains shall connect into a trap so constructed that it can be readily cleaned and of a size to serve efficiently the purpose for which it is intended. The trap shall be either accessible from the floor drain or by a separate cleanout within the drain.

4.5.3.6.2.2 Floor drain also receives, waste piping which does not connect to the sanitary system, known as indirect waste. This discharge from an indirect waste should be conveyed into a water supplied, trapped and vented floor drain.

4.5.3.6.2.3 Floor drain should be provided in mechanical equipment rooms, where pumps, boilers, water chillers, heat exchangers and other air conditioning equipment are periodically drained for maintenance and repair. Boiler requires drain at safety relief valve discharge.

4.5.3.6.2.4 Strategically floor drains are required to be located in buildings with wet fire protection sprinkler systems to drain water in case of activation of sprinkler heads.

4.5.3.6.2.5 The minimum diameter for floor drains outlets before connecting to floor trap is 75 mm.

4.5.3.6.3 *Cleanouts*

The cleanout provides access to horizontal and vertical lines and stacks to facilitate inspection and means to remove obstructions common to all piping systems, such as solid objects, greasy wastes, hair and the like.

4.5.3.6.3.1 Cleanouts in general should be gas and water tight, provide quick and easy plug removal, allow ample space for rodding tools, have means of adjustments to finished floor level, be attractive and be designed to support whatever load is directed over them.

4.5.3.6.3.2 Waste lines are normally laid beneath the floor slab at a sufficient distance to provide adequate back-fill over the joints. Cleanouts are then brought up to floor level grade by pipe extension pieces.

4.5.3.6.3.3 The size of the cleanout within a building should be the same size as the piping up to 100 mm. For larger size piping 100 mm cleanouts are adequate for their intended purpose.

4.5.3.6.3.4 Cleanouts are suggested to be provided at the following locations:

- a) Inside the building at a point of exit. Use a wye branch or a trap;
- b) At every change of direction greater than 45°;
- c) At the base of all stacks; and
- d) At the horizontal header, receiving vertical stacks and serving the purpose of offset header.

4.5.3.6.3.5 *Supports for drainage and sewerage pipes*

The supports for the above pipes and fittings shall be in accordance with manufacturer's recommendations and shall comply with the applicable standards.

4.5.4 *Indirect Wastes*

4.5.4.1 *General*

Waste, overflow and drain pipes from the following types of equipment shall not be connected into any drainage system directly to prevent backflow from the drainage system into the equipment/installation:

- a) *Plumbing and kitchen appliances:*
 - 1) Underground or overhead water tanks;
 - 2) Drinking water fountains;
 - 3) Dishwashing sinks and culinary sinks used for soaking and preparation of food;
 - 4) Cooling counters for food and beverages;
 - 5) Kitchen equipment for keeping food warm; and
 - 6) Pressure drainage connections from equipment.
- b) *Air conditioning, heating and other mechanical equipment:*
 - 1) Air handling equipment;
 - 2) Cooling tower and other equipment;
 - 3) Condensate lines from equipment;
 - 4) Storage tanks;
 - 5) Condensate lines;
 - 6) Boiler blow down lines; and
 - 7) Steam trap drain lines.
- c) *Laboratories and other areas:*
 - 1) Water stills;
 - 2) Waste from laboratory in specified sinks;
 - 3) Sterilizers and similar equipment; and
 - 4) Water purification equipment.

4.5.4.2 Indirect waste receptors

All plumbing fixtures or other receptors receiving the discharge of indirect waste pipes shall be of such shape and capacity as to prevent splashing or flooding and shall be located where they are readily accessible for inspection and cleaning.

4.5.4.3 Pressure drainage connections

Indirect waste connections shall be provided for drains, overflows or relief vents from the water supply system, and no piping or equipment carrying wastes or producing wastes or other discharges under pressure shall be directly connected to any part of the drainage system.

The above shall not apply to any approved sump pump or to any approved plumbing fixture discharging pressurized waste or device when the Authority has been satisfied that the drainage system has the capacity to carry the waste from the pressurized discharge.

An indirect waste is required for any type of fixture or equipment that may come in contact with the food. The purpose is to isolate the fixture or equipment from drainage system waste.

Indirect waste piping shall be a minimum of 25 mm in size, but not smaller than drain of the equipment or fixture. There is no limitation on the length of indirect waste piping.

4.5.5 Special Wastes

4.5.5.1 General

Wastes having characteristics which may be detrimental to the pipes in which it is disposed as well as to the persons handling it. Such wastes used in a building need to be specially identified and a suitable and safe method of its disposal installed to ensure that the piping system is not corroded nor the health and safety of the occupants is affected in any way.

Whenever the occupant or the user of any wastes is unaware of the dangers of the consequences of disposing the waste, he shall be made aware of the dangers of his action along with providing suitable warning and instruction for correct disposal be provided to him.

Piping system for all special wastes should be separate and independent for each type of waste and should not be connected to the building drainage system. Other applicable provisions for installation of soil and waste pipe system shall be however be followed.

4.5.5.2 Laboratory wastes

A study of the possible chemical and corrosive and toxic properties of wastes handled and disposed of in a laboratory need to be ascertained in advance. The relevant statutory rules and regulation regarding the method of disposal of strong and objectionable wastes shall be followed.

All sinks, receptacles, traps, pipes, fittings and joints shall be of materials resistant to the liquids disposed of in the system.

In laboratories for educational, research and medical institutions, handling mildly corrosive and toxic wastes, they may be neutralized in chambers using appropriate neutralizing agents. The chamber shall be provided with chambers at inlet and outlet for collecting samples of the incoming and outgoing waste for monitoring its characteristics.

4.5.5.3 Infected wastes

Infected liquid wastes are generated in hospitals from patient excreta, operation theatres, laboratories testing samples of stools, urine, blood, flesh, etc, which shall not be disposed of into the drainage system. Such waste shall be collected separately and pre-treated, and sterilized, if required, before disposal into the building drainage system.

Soiled linen from infectious patients needs to be collected from the respective areas of the hospital in separate linen bins and pre-washed in dirty utility room in a sluice tub and sterilized in the laundry (sluice machine) before its regular washing in the hospital laundry. Liquid wastes from the washing operations shall be neutralized to prevent any cross contamination before discharge in the building's drainage system.

4.5.5.4 Research laboratory wastes

Research laboratories conducting research in all areas of science and technology, for example, chemical industry, pharmacy, metallurgy, bio-sciences, agriculture, atomic energy, medicine, etc, shall follow the established procedures laid down by statutory bodies to handle, treat and dispose wastes which are highly toxic, corrosive, infectious, inflammable, explosive and having bacterial cultures, complex organic and inorganic chemicals. Such wastes shall not be disposed of in a building drainage system or the city sewerage system unless they are pre-treated and meet the disposal criteria in accordance with the relevant rules/regulations.

It requires a liquid waste decontamination system which consists of at least one reactor or sterilizer plus a holding tank. An ideal system would consist of two or more sterilizers to provide redundancy with two or more tanks. There should be a sampling tank between this system and building drainage. The final treated waste water should flow to ETP and not STP.

4.5.6 Grease Traps

Oil and grease are found in wastes generated from kitchens in hotels, industrial canteens, restaurant, food and beverage areas, butcheries, some laboratories and manufacturing units having a high content of oil and greases in their final waste.

Waste exceeding temperature of 60 °C should not be allowed in the grease trap. When so encountered, it may be allowed to cool in a holding chamber before entering the grease trap.

Oil and greases tend to solidify as they cool within the drainage system. The solidified matter clogs the drains and the other matter in the waste stick to it due to the adhesion properties of the grease. Oil and greases are lighter than water and tend to float on the top of the wastewater.

Grease traps shall be installed in building having the above types of wastes. In principle the grease laden water is allowed to retain in a grease trap which enables any solids to be settled or separated for manual disposal. The retention time allows the incoming waste to cool and allow the grease to solidify. The clear waste is then allowed to discharge into the building's drainage system.

4.5.7 Oil Interceptors

Oils and lubricants are found in wastes from vehicle service stations, workshops manufacturing units whose waste may contain high content of oils. Oils for example, petroleum, kerosene and diesel used as fuel, cooking, lubricant oils and similar liquids are lighter than water and thus float on water in a pipe line or in a chamber when stored. Such oils have a low ignition point and are prone to catch fire if exposed to any flame or a spark and may cause explosion inside or outside the drainage system. The flames from such a fire spread rapidly if not confined or prevented at the possible source. Lighter oils and lubricants are removed from the system by passing them through an oil interceptor/petrol gully. They are chambers in various compartments which allow the solids to settle and allow the oils to float to the top. The oil is then decanted in separate containers for disposal in an approved manner. The oil free waste collected from the bottom of the chamber is disposed in the building drainage system.

4.5.8 Radioactive Waste

Scientific research institutions, hospital and many types of manufacturing processes use radioactive material in the form of radio-isotopes and other radioactive sources for their activities. Manufacture, sale, use and disposal of radioactive material is regulated by the statutory rules and regulation. Proposal for usage and disposal of radioactive materials shall be done in consultation with and prior permission of the Authority by the users of the materials. No radioactive material shall be disposed off in any building drainage system without the authorization of the Authority.

NOTE — All procedures followed and precautions taken shall be in accordance with AERB regulations, including regarding employing delay and decay tanks. Where applicable, the toilets may be designated as 'hot toilets' and should be clearly indicated. All drainage plumbing needs to be isolated and connected to delay and decay tanks (usually required for PET CT scan room, SPECT CT scan room and iodine therapy ward, and may be also required for cyclotron liquid waste).

4.5.9 Special Situations of Waste Water Disposal

Buildings may generate uncontaminated waste water from various sources continuously, intermittently or in large volumes for a short time for example, emptying any water tanks or pools, testing fire and water lines for flow conditions, etc. Connections from all such sources shall be made to the building drainage system indirectly through a trap. It should be ensured in advance that the building drain or a sump with a pump has the capacity to receive to rate of flow. In case the capacity is less the rate of discharge from the appliances should be regulated to meet the capacity of the disposal. Under no circumstances shall any waste water described above shall be disposed of in any storm water drains.

4.5.10 Manholes

4.5.10.1 General

A manhole or inspection chamber shall be capable of sustaining the loads which may be imposed on it, exclude subsoil water and be water tight. The size of the chamber should be sufficient to permit ready access to the drain or sewer for inspection, cleaning and rodding and should have a removable cover of adequate strength, constructed of suitable and durable material. Where the depth of the chamber so requires, access rungs, step irons, ladders or other means should be provided to ensure safe access to the level of the drain or sewer. If the chamber contains an open channel, benching should be provided having a smooth finish and formed so as to allow the foul matter to flow towards the pipe and also ensure a safe foothold.

No manhole or inspection chamber shall be permitted inside a building or in any passage therein. The minimum depth of the manhole shall not be less than 800 mm to facilitate gully trap connection. Further, ventilating covers shall not be used for domestic drains. At every change of alignment, gradient or diameter of a drain, there shall be a manhole or inspection chamber. Bends and junctions in the drains shall be grouped together in manholes as far as possible.

Alternative materials of manholes using RCC rings and PVC in sewer lines are being used in some sites. These may be considered subject to ensuring their proper design, keeping in view the lateral and vertical loads at the place of installation.

The holes over the surfaces of covers/gratings of various manholes, gullies and other chambers/drains shall comply with the requirements given in good practice [E-2(2)].

4.5.10.2 Spacing of manholes

The spacing of manholes for a given pipe size should be as follows:

Sl No.	Pipe Diameter mm	Spacing of Manhole m
(1)	(2)	(3)
i)	Up to 300	45
ii)	301 to 500	75
iii)	501 to 900	90
iv)	Beyond 900	Spacing shall depend upon local condition and shall be gotten approved by the Authority

Where the diameter of a drain is increased, the crown of the pipes shall be fixed at the same level and the necessary slope given in the invert of the manhole chamber. In exceptional cases and where unavoidable, the crown of the branch sewer may be fixed at a lower level, but in such cases the peak flow level of the two sewers shall be kept the same.

4.5.10.3 Size of manhole

The manhole or chamber shall be of such size as will allow necessary examination or clearance of drains. The size of shall be adjusted to take into account any increase in the number of entries into the chamber.

4.5.10.3.1 Manholes may be rectangular, arch or circular type. The minimum internal size of manholes, chambers (between faces of masonry) shall be as follows:

a) Rectangular manholes:

- 1) For depths less than 0.90 m : 900 mm × 800 mm
- 2) For depths from 0.90 m and up to 2.5 m : 1 200 mm × 900 mm

NOTE — For depths up to 0.60 m, 600 mm × 600 mm manhole may be used.

b) Arch type manholes:

For depths of 2.5 m and above : 1 400 mm × 900 mm

NOTE — The width of manhole chamber shall be suitably increased more than 900 mm on bends, junctions or pipes with diameter greater than 450 mm so that benching width in either side of channel is minimum 200 mm.

c) Circular manholes:

- 1) For depths above 0.90 m and up to 1.65 m : 900 mm diameter
- 2) For depths above 1.65 m and up to 2.30 m : 1 200 mm diameter
- 3) For depths above 2.30 m and up to 9.00 m : 1 500 mm diameter
- 4) For depths above 9.00 m and up to 14.00 m : 1 800 mm diameter

NOTES

1 In adopting the above sizes of chambers, it should be ensured that these sizes accord with full or half bricks with standard thickness of mortar joints so as to avoid wasteful cutting of bricks.

2 The sizes of the chambers/manhole may be adjusted to suit the availability of local building materials, economics of construction and to meet local authority approval.

3 The access shaft shall be corbelled inwards on three sides at the top to reduce its size to that of the cover frame to be fitted or alternatively the access shaft shall be covered over by a reinforced concrete slab of suitable dimensions with an opening for manhole cover and frame.

4 The minimum sewer pipe diameter is 200 mm based on good practice [E-2(13)].

4.5.10.4 Construction

4.5.10.4.1 Excavation

The manhole shall be excavated true to dimensions and levels as shown on the plan. The excavation of deep manholes shall be accompanied with safety measures like timbering, staging, etc. In areas where necessary, appropriate measures for dewatering should be made.

4.5.10.4.2 Bed concrete

The manhole shall be built on a bed of concrete 1 : 4 : 8 (1 cement : 4 coarse sand : 8 graded stone aggregate 40 mm nominal size). The thickness of bed concrete shall be at least 150 mm for manholes up to 0.9 m in depth, at least 200 mm for manholes from 0.90 m up to 2.5 m depth and at least 300 mm for manholes of greater depth, unless the structural design demands higher thickness.

This thickness may be verified considering the weight of wall, cover, the wheel loads, impact of traffic which are transmitted through cover and the shaft walls and for water pressure, if any. In case of weak soil, special foundation as suitable shall be provided

4.5.10.4.3 Brickwork

The thickness of walls shall be designed depending upon its shape and taking into account all loads coming over it, including earth pressure and water pressure.

Generally, the brickwork shall be with first class bricks in cement mortar 1 : 5 (1 cement : 5 coarse sand). All brickwork in manhole chambers and shafts shall be carefully built in English Bond, the jointing faces of each

brick being well 'buttered' with cement mortar before laying, so as to ensure a full joint. The construction of walls in brickwork shall be done in accordance with good practice [E-2(14)].

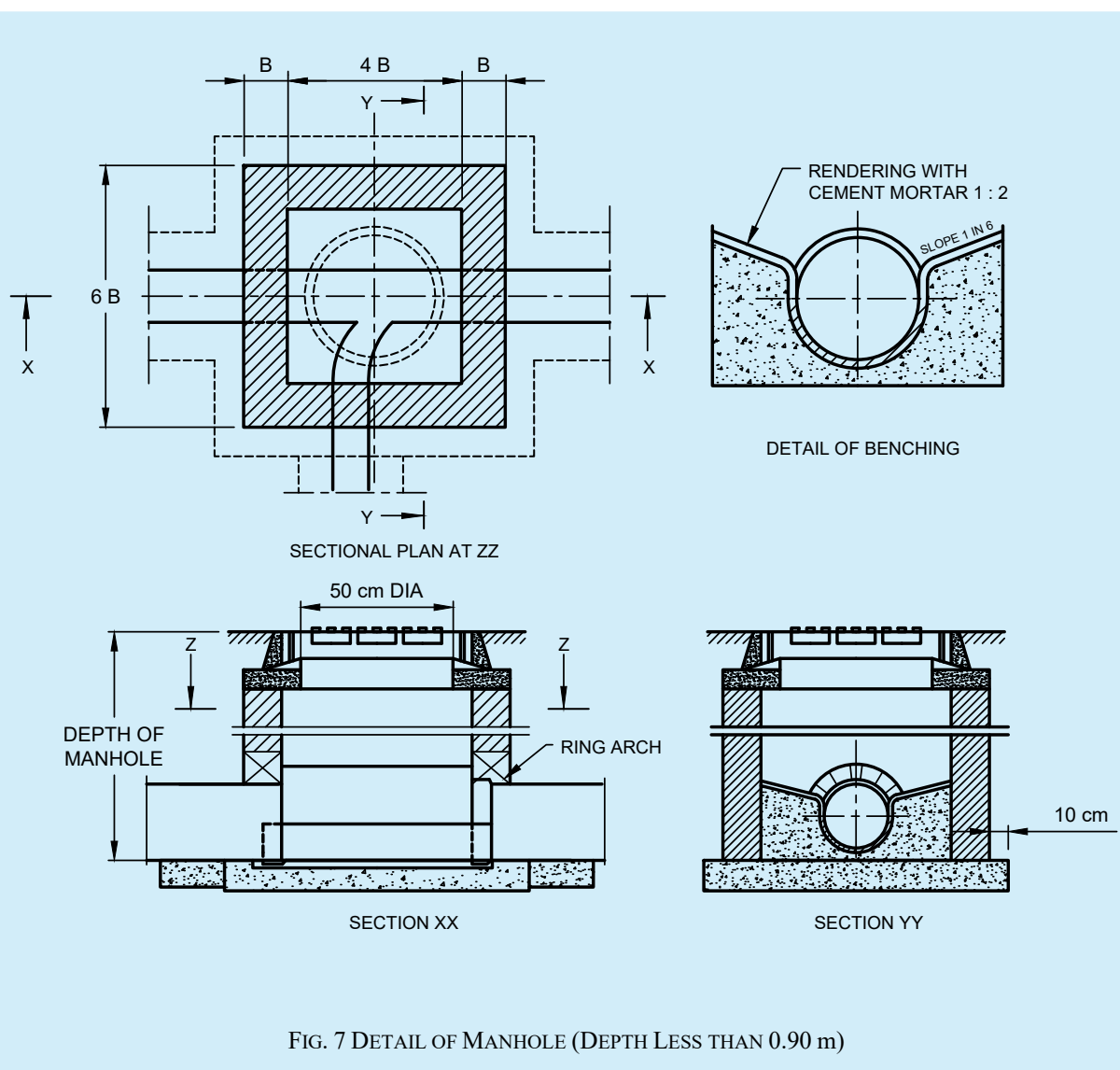
For various depths the recommended thickness of wall may be as follows:

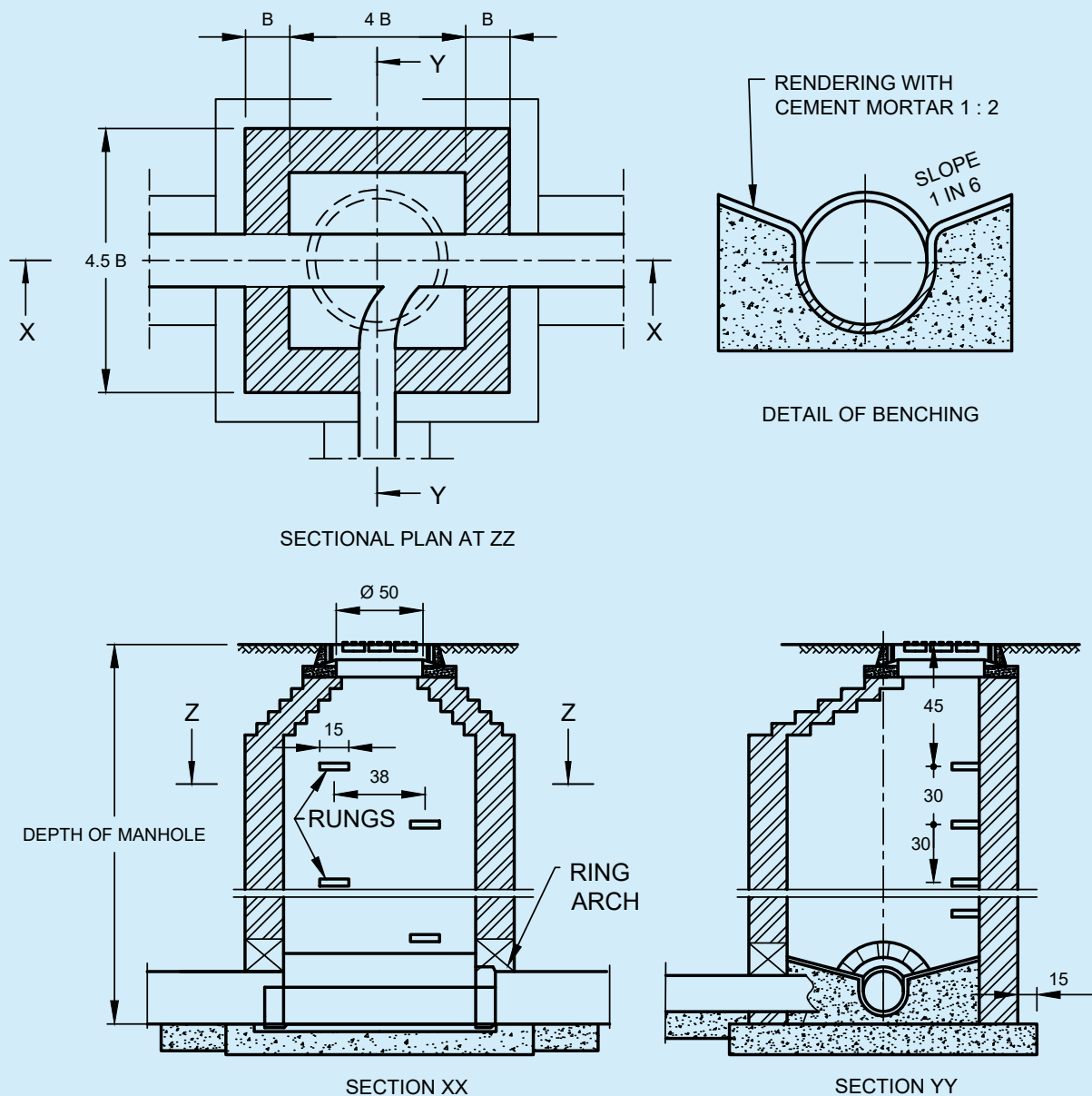
<i>Depth of the Chamber</i>	<i>Thickness of Wall</i>
a) Up to 2.25 m	: 200 mm (one brick length)
b) From 2.25 m up to 3.0 m	: 300 mm (one and half brick length)
c) From 3.00 m up to 5.0 m	: 400 mm (two brick length)
d) From 5.00 m up to 9.0 m	: 500 mm (two and half brick length)
e) Above 9.00 m	: 600 mm (three brick length)

The actual thickness in any case shall be calculated on the basis of engineering design. Typical sections of the manholes are illustrated in [Fig. 7](#), [Fig. 8](#) and [Fig. 9](#).

NOTES

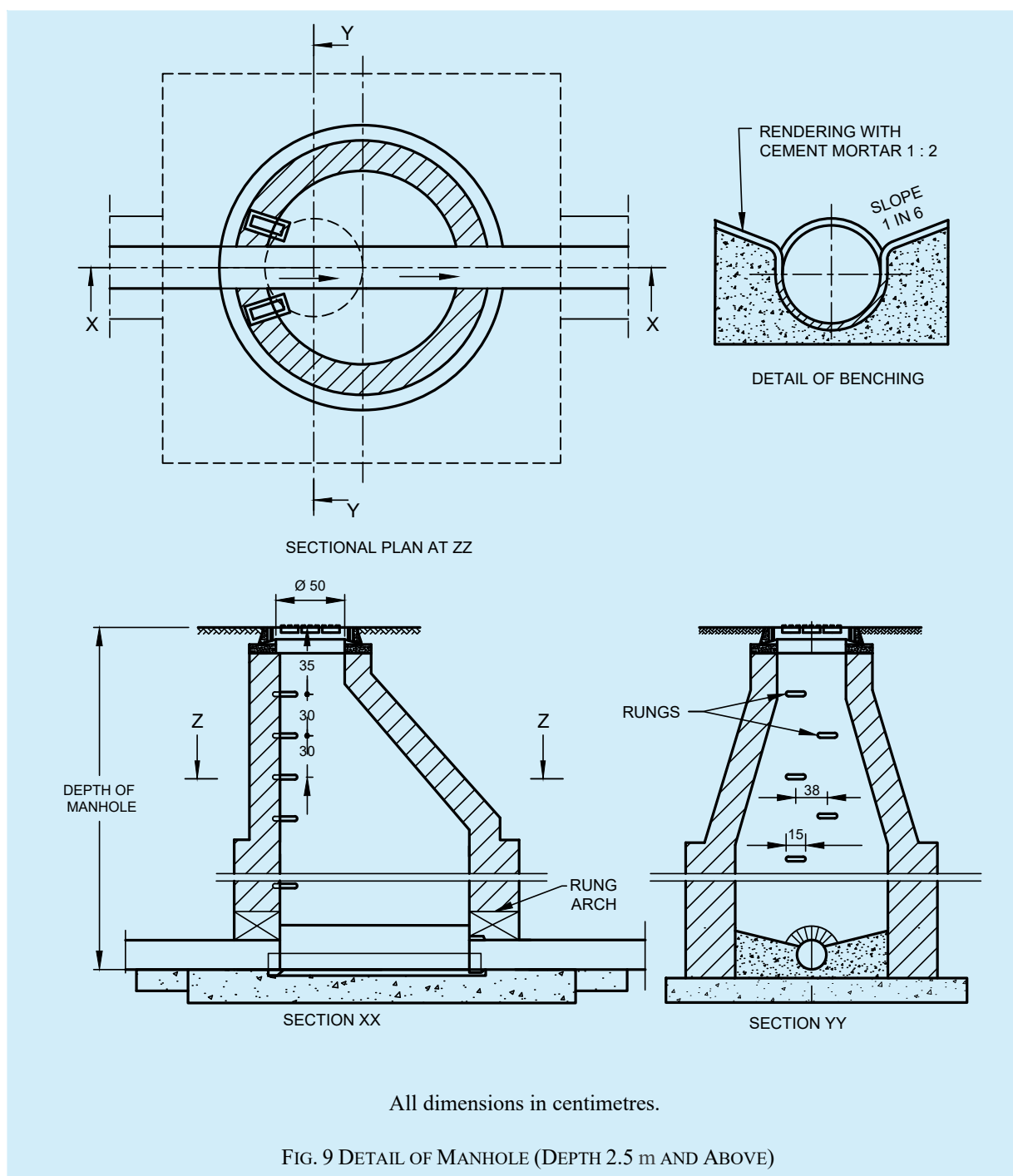
- 1 Rich mix of cement mortar, not weaker than 1 : 3, should be used in brick masonry, where subsoil water conditions are encountered.
- 2 For arched type of manholes, the brick masonry in arches and arching over pipes shall be in cement mortar 1 : 3.





All dimensions in centimetres.

FIG. 8 DETAIL OF MANHOLE (DEPTH FROM 0.9 m AND UP TO 2.5 m)



4.5.10.4.4 Plastering

The wall shall be plastered (15 mm, min) both inside and outside with cement mortar 1 : 3 and finished smooth with a coat of neat cement. Where subsoil water conditions exist, a richer mix may be used and it shall further be waterproofed with addition of approved waterproofing compound in a quantity as per manufacturer specifications.

All manholes shall be so constructed as to be water-tight under test.

All angles shall be rounded to 75 mm radius and all rendered internal surface shall have hard impervious finish obtained using a steel trowel.

4.5.10.4.5 Channels and benching

These shall be semi-circular in the bottom half and of diameter equal to that of the sewer. Above the horizontal diameter, the sides shall be extended vertically 50 mm above the crown of sewer pipe and the top edge shall be suitably rounded off. The branch channels shall also be similarly constructed with respect to the benching, but at their junction with the main channel an appropriate fall, if required suitably rounded off in the direction of flow in the main channel shall be given.

The channel/drain and benching at the bottom of the chamber shall be done in cement concrete 1 : 2 : 4 and subsequently plastered with cement mortar of 1 : 2 proportion or weaker cement mortar with a suitable waterproofing compound and finished smooth, to the grade (where required). The benching at the sides shall be carried up in such a manner as to provide no lodgment for any splashing in case of accidental flooding of the chamber.

Channels shall be rendered smooth and benchings shall have slopes towards the channel.

4.5.10.4.6 Rungs

Rungs shall be provided in all manholes over 0.8 m in depth and shall be of cast iron, or PVC encapsulated or composite non-corrosive materials, and of suitable dimensions, conforming to accepted standards [E-2(15)]. These rungs may be set staggered in two vertical rungs which may be 300 mm apart horizontally as well as vertically and shall project a minimum of 100 mm beyond the finished surface of the manhole wall. The top rung shall be 450 mm below the manhole cover and the lowest not more than 300 mm above the benching.

4.5.10.4.7 Manhole covers and frames

The size of manhole covers shall be such that there shall be a clear opening of at least 500 mm in diameter for manholes exceeding 0.90 m in depth. The manhole covers and frames used shall conform to accepted standards [E-2(16)].

The frame of manhole shall be firmly embedded to concrete alignment and level in plain concrete on the top of masonry.

4.5.10.5 Drop manhole

Where it is uneconomic or impracticable to arrange the connection within 600 mm height above the invert of the manholes, the connection shall be made by constructing a vertical shaft outside the manhole chamber, as shown in [Fig. 10](#). If the difference in level between the incoming drain and the sewer does not exceed 600 mm and there is sufficient room in the manhole, the connecting pipe may be directly brought through the manhole wall and the fall accommodated by constructing a ramp in the benching of the manhole.

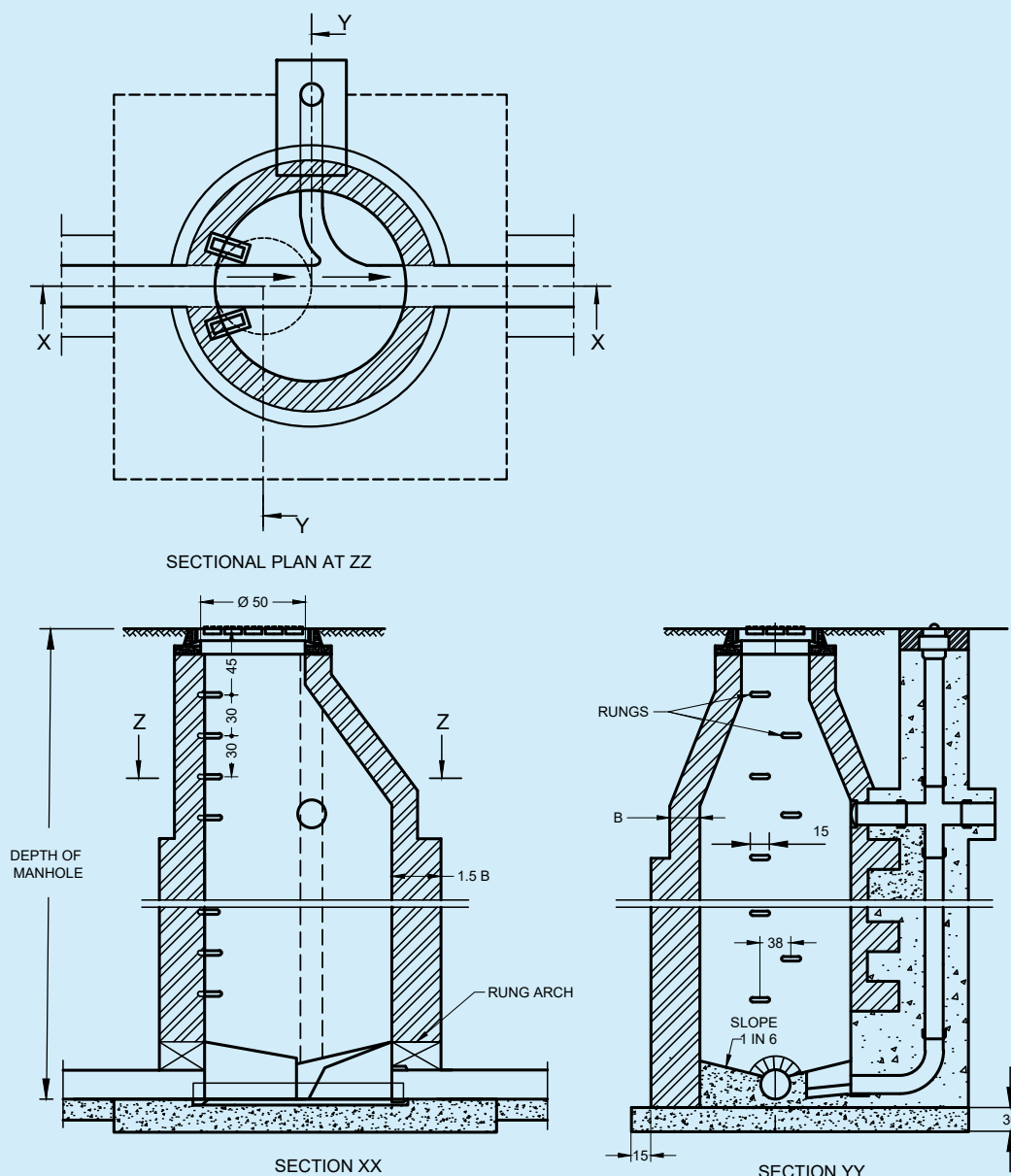
For detailed information regarding manholes in sewerage system, reference may be made to good practice [E-2(17)].

4.5.10.6 Manhole covers and recommended locations

Manhole covers were traditionally and presently manufactured from concrete, steel fibre reinforced concrete, cast iron and ductile iron or PVC materials and these materials are used based on the load carrying capacity and for the following type of applications:

- a) Inspection chambers for sewerage;
- b) Underground electrical cabling;
- c) Telecom cabling;
- d) Water, gas and petroleum installations; and
- e) Beautification of gardens and landscapes.

Recommended locations conforming to load capacity shall be as per accepted standard [E-2(16)].



NOTE – Wall thickness have been indicated in brick length to provide for use of modular bricks or traditional bricks.
In the figure, B = one brick length, 1.5 B = one and a half brick length, etc.

All dimensions in centimetres.

FIG. 10 DROP MANHOLE

4.5.11 Storm Water Drainage

4.5.11.1 General

The object of storm water drainage is to collect and carry, the rainwater collected within the premises of the building, for suitable disposal.

4.5.11.2 Design factors

Estimate of the quantity that reaches the storm water drain (runoff) depends on the following factors:

- Type of soil and its absorption capacity determined by its soil group;

- b) Ground slope and the time in which the area is drained;
- c) Intensity of the rainfall for a design period; and
- d) Duration of the rain/storm.

The runoff reaching the sewer may be given by the expression:

$$Q = 10 C \times i \times A$$

where

Q = runoff, m³/h;

C = coefficient of runoff;

i = intensity of rainfall, mm/h; and

A = area of the drainage district, hectares.

Coefficient of runoff for various surfaces may be taken as:

<i>Sl No.</i>	<i>Type of Surface</i>	<i>Coefficient of Runoff</i>
(1)	(2)	(3)
i)	Concrete roof area	0.9
ii)	Paved podium areas and asphalted roads	0.8
iii)	Unpaved ground	0.3
iv)	Lawns and parks	0.15

4.5.11.2.1 Imperviousness

The soil conditions and the ground slope determine the impermeability factor. Impermeability factor is the proportion of the total rainfall received on the surface which will be discharging into a storm water drain after allowing for initial abstraction (in local pond and lakes), ground absorption by evaporation, vegetation and other losses. The net flow reaching the storm water drain is called runoff.

The percentage of imperviousness of the drainage area may be obtained from available data for a particular area. In the absence of such data, the following figures may serve as a guide:

<i>Sl No.</i>	<i>Type of Area</i>	<i>Imperviousness Factor Percent</i>
(1)	(2)	(3)
i)	Commercial and industrial areas	70 to 90
ii)	Residential areas (high density)	60 to 75
iii)	Residential areas (low density)	35 to 60
iv)	Parks and underdeveloped areas	10 to 20

4.5.11.2.2 Terrain modelling

Areas planned for urbanization from agricultural land, forest or low grade land for example, low lying areas prone to flooding, marshy or abandoned quarries, etc, need detailed and careful consideration with respect to its drainage. A detailed contour survey shall be carried out not only with respect to the site but also the surrounding areas to verify the quantity/area contributing runoff, presence of any low lying and natural water body acting as holding pond or any natural drain passing through the area and beyond whose filling up or diversion may cause water logging problem on the site or to the surrounding areas.

The planning of the area should ensure that:

- a) All areas become self draining by gravity with respect to the high flood level of the area or the drainage channels passing, whichever is higher;
- b) As far as possible, natural drainage pattern with respect to the whole area be maintained except when low lying areas need to be filled up for grading purposes;
- c) The drainage in the area shall be planned in accordance with the natural slopes; and
- d) Levels of the main highway or road connecting to the property shall be determined to ensure proper drainage and protection of the site.

The formation levels of the entire area shall be prepared to determine proposed formation levels by preparing a terrain model which will show the proposed site contours, ground and road levels and connections to all services including storm water disposal system.

4.5.11.2.3 Rainfall intensity

The intensity of rainfall is the amount of rain falling in unit time and is expressed in centimetre or millimetre per hour. For design purposes, the mean rainfall intensity is collected for 50 years.

NOTE — For storm water drainage design, the hourly maximum rainfall may be referred from the relevant Indian Roads Congress guidelines.

4.5.11.2.4 Design frequency

Storm water drainage system for an urbanized area is planned on the basis of the design frequency of the storm which shall be determined by the designer. Frequency is the period in which the selected design intensity recurs in a given period of time in years.

NOTE — The rainfall intensity varies with the period, which includes good, bad and average years of rainfall. Meteorological department has long term data regarding the rainfall and have graphs for different periods, and the designers are advised to consult the department before designing any scheme.

4.5.11.2.5 Time of concentration

Time of concentration is the time required for the rainwater to flow to reach the farthest point of the drainage system or the outfall under consideration. Time of concentration is equal to the inlet time plus the time required for the flow to reach the main or branch drain. The inlet time is the time dependent on the distance of the farthest point in the drainage area to the inlet of the manhole and the surface slopes, etc, and will vary between 5 min and 30 min.

In highly developed sections for example with impervious surfaces it may be as low as 3 min or lower (with good slopes) as in building terraces and paved areas. Correspondingly the design intensity for the drainage for such areas will be much higher. Rainwater pipes have to be designed for an intensity for a very low time of concentration.

4.5.11.2.6 Natural infiltration

In planning any area with buildings, layout with paved and non-permeable surfaces, care should be taken to allow maximum discharge of the rainwater to flow directly or indirectly to permeate into the ground for enabling the ground water to be recharged. Some of the techniques which allow infiltration that may be considered are:

- a) Use of brick paved open jointed storm water drains;
- b) Providing bore holes in the storm water drains; and
- c) Using paving tiles with open joints which enable water to percolates as it flows on it.

4.5.11.3 Combined system

A combined system of drainage is one which carries the sewerage as well as the runoff from the storm water drainage. Relevant applicable statutory rules/regulations may not allow such system in new areas and the sewerage and the storm water drainage have to be separate and independent of each other. Such systems are

however existing in many old cities and the storm water may have to be discharged into the combined drainage system.

Where levels do not permit for connection to a public storm water drain, storm water from courtyards of buildings may be connected to the public sewer, provided it is designed to or has the capacity to convey combined discharge. In such cases, the surface water shall be admitted to the soil sewer through trapped gullies in order to prevent the escape of foul air.

4.5.11.4 Discharging into a watercourse

It may often be convenient to discharge surface water to a nearby stream or a watercourse. The invert level of the outfall shall be about the same as the normal water level in the watercourse or ideally should be above the highest flood level of the watercourse. The out-fall shall be protected against floating debris by a screen.

4.5.11.5 Discharge to a public storm water drain

Where it is necessary to connect the discharge rainwater into a public storm water drain, such drains shall be designed for the intensity of rain based on local conditions, but in no case shall they be designed for intensity of rainfall of less than 50 mm/h. Rainwater from each building plot shall be connected to the storm water drainage through a separate pipe or an open public drain directly. No trap shall be installed before the connection.

4.5.11.6 Rainwater pipes for roof drainage

4.5.11.6.1 The roofs of a building shall be so constructed or framed as to permit effectual drainage of the rainwater therefrom by means of a sufficient number of rainwater pipes of adequate size so arranged, jointed and fixed as to ensure that the rainwater is carried away from the building without causing dampness in any part of the walls or foundations of the building or those of an adjacent building.

4.5.11.6.2 The rainwater pipes shall be fixed to the outside of the external walls of the building or in recesses or chases cut or formed in such external wall or in such other manner as may be approved by the Authority.

4.5.11.6.3 Rainwater pipes conveying rainwater shall discharge directly or by means of a channel into or over an inlet to a surface drain or shall discharge freely in a compound, drained to surface drain but in no case shall it discharge directly into any closed drain.

4.5.11.6.4 Whenever it is not possible to discharge a rainwater pipe into or over an inlet to a surface drain or in a compound or in a street drain within 30 m from the boundary of the premises, such rainwater pipe shall discharge into a gully trap which shall be connected with the street drain for storm water and such a gully-trap shall have a screen and a silt catcher incorporated in its design.

4.5.11.6.5 If such streets drain is not available within 30 m of the boundary of the premises, a rainwater pipe may discharge directly into the kerb drain and shall be taken through a pipe outlet across the foot path, if any, without obstructing the path.

4.5.11.6.6 A rainwater pipe shall not discharge into or connect with any soil pipe or its vent pipe or any waste pipe or its vent pipe nor shall it discharge into a sewer unless specifically permitted to do so by the Authority, in which case such discharge into a sewer shall be intercepted by means of a gully trap.

4.5.11.6.7 Rainwater pipes shall be constructed of cast iron, PVC, asbestos cement, galvanized sheet or other equally suitable material and shall be securely fixed.

4.5.11.6.8 The factors that decide the quantity of rainwater entering are:

- a) Intensity of rainfall; and
- b) Time of concentration selected for rainwater pipe.

A bell mouth inlet at the roof surface is found to give better drainage effect, provided proper slopes are given to the roof surface. The spacing of rainwater pipes depends on the locations available for the down takes and the area which each pipe serves. The spacing will also be determined by the amount of slopes that can be given to the roof. The recommended slopes for the flat roofs with smooth finish would be 1 : 150 to 1 : 133, with rough

stone/tiles 1 : 100 and for gravel set in cement or loosely packed concrete finish 1 : 75 to 1 : 66. The effective strainer area should preferably be 1.5 times to 2 times the area of pipe to which it connects to considerably enhance the capacity of rainwater pipes.

The rainwater pipes of cast iron (coefficient of roughness 0.013) shall normally be sized on the basis of roof areas according to [Table 25](#). The vertical down take rainwater pipes, having a bell mouth inlet on the roof surface with effective cross sectional area of grating 1.5 to 2 times the rainwater pipe area, may be designed by considering the outlet pipe as weir.

For full circumference of pipe acting as weir, the roof area (RA) for drainage may be worked out by using:

$$RA = 0.084 \times d^{5/2}/I$$

where

RA = roof area, m² (see Note);

d = pipe diameter, mm; and

I = intensity of rainfall, mm/h.

NOTES

1 For one adjacent wall, add 50 percent of the wall area to the roof.

2 For two adjacent walls, add 35 percent of the wall area to the roof.

Table 25 Sizing of Rainwater Pipes for Roof Drainage

(Clause [4.5.11.6.8](#))

Sl No.	Diameter of Pipe mm	Roof Area, in m ² for Average Rate of Rainfall in mm/h					
		50	75	100	125	150	200
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	50	29.70	19.80	14.85	11.88	9.90	7.42
ii)	65	57.23	38.15	28.61	22.89	19.08	14.31
iii)	75	81.84	54.56	40.92	32.74	27.28	20.46
iv)	100	168.00	112.00	84.00	67.20	56.00	42.00
v)	125	293.48	195.66	146.74	117.39	97.83	73.37
vi)	150	462.95	308.64	231.48	185.18	154.32	115.74

NOTE — For rainwater pipes of other materials, the roof areas shall be multiplied by (0.013/coefficient of roughness of surface of that material). For example, for rainwater pipes of PVC (coefficient of roughness = 0.009), the above values of roof area shall be multiplied by 0.013/0.009 = 1.44.

4.5.11.6.9 The storm water may be led off in a suitable open drain to a watercourse. The open drain, if not a Pucca masonry throughout, shall be so at least where there is either a change in direction or gradient.

4.5.11.6.10 Flat roof should have an emergency drain as secondary drain, in the form of down take or spout. Horizontal outlets should be considered for 100 percent flow and side gratings to be considered for 50 percent flow (see [Fig. 11](#)).

NOTE — Care should be taken to design the roof for the additional load due to ponded water.

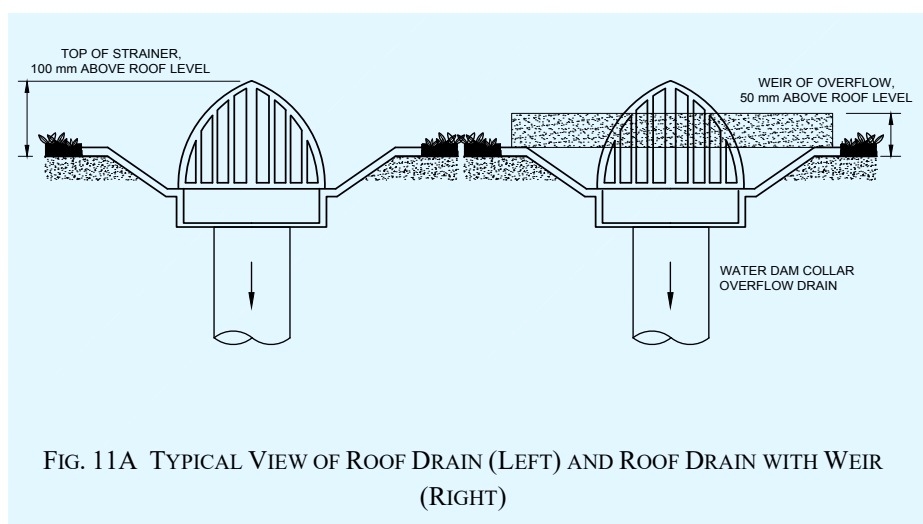


FIG. 11 ROOF DRAIN IN FLAT ROOFS

4.5.12 Rainwater Harvesting

4.5.12.1 General

To supplement the ever growing shortage of protected, pure and safe water supply for human consumption, rainwater is an ideal source which can be conserved and used in a useful manner by the people. The amount of rainfall available varies from region to region. Each area has to develop its own method and system to conserve, store and use it to suit its requirement and local conditions. There are several methods by which rainwater can be stored, used and conserved. Each system depends on the amount of precipitation, the period in which the rainfall occurs in a year and the physical infrastructure, for example space available to store the water, etc.

There are several techniques available for catching and storing the rainwater. Most of the techniques are applicable for large open areas, farms, sloping grounds, etc, with a low population base. Two major systems that are ideal for urban and semi-urban developed areas are:

- a) Artificial ground water recharge; and
- b) Roof top rainwater harvesting.

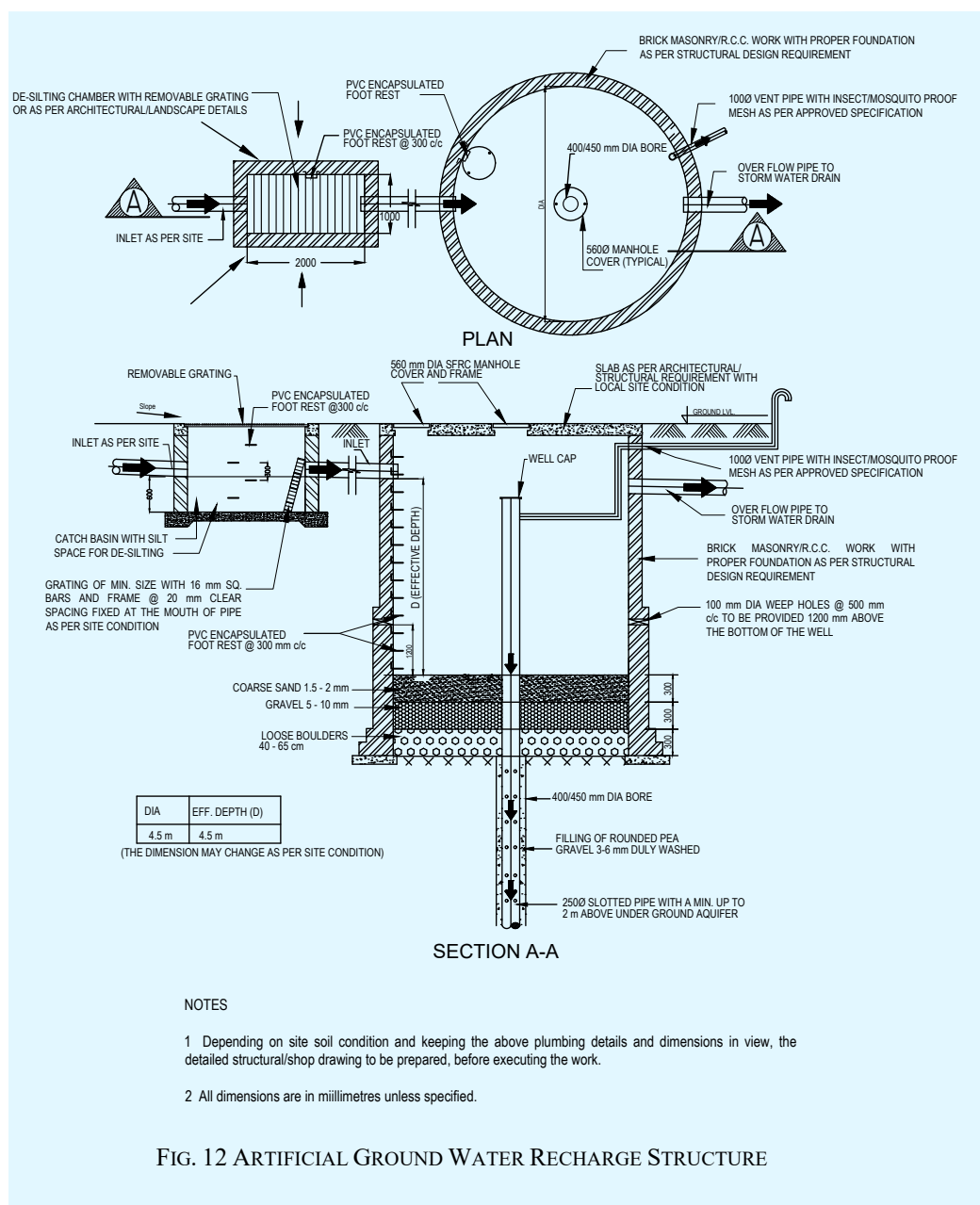
NOTE — Type and number of recharge pits and roof top rainwater harvesting capacity shall be provided as per the local/state bye-laws.

4.5.12.2 Artificial ground water recharge

With increase in the impermeable surfaces in modern built up areas, there is gross reduction in the original catchment area. This prevents the natural percolation that would have recharged the area in the normal course if the ground was in its natural condition, for example a farm, open ground, forest, etc. This results in a large quantity of water, normally percolating into the ground, running off to the natural drains and into the rivers, causing increased runoff and flooding of downstream areas.

It is therefore essential to catch the runoff and use it for augmentation of ground water reservoir by modifying the natural movement of surface water by recharging it by artificial means, for example construction of recharge structures (see Fig. 12). The main objectives achieved may be:

- Enhancement of sustainable yield in areas where over development and depletion of the aquifers;
- Conservation and storage of excess surface water in the aquifers;
- Improve the quality of the existing ground water through dilution;
- Remove bacteriological and suspended impurities during the surface water transition within the subsoil; and
- Maintain the natural balance of the ground water and its usage as the rainwater is a renewable supply source. A well managed and controlled tapping of the aquifers will provide constant, dependable and safe water supply.



In planning and designing the ground water recharge structures, following should be taken into consideration:

- a) Annual rainfall (for estimating approximately rainwater recharge per year);
- b) Peak intensity and duration of each storm;
- c) Type of soil and subsoil conditions and their permeability factor;
- d) Ground slopes and run off which cannot be caught;
- e) Location of recharge structures and its overflow outfall; and
- f) Rainwater measuring devices for finding the flow of water in the system.

For artificial recharge to ground water, good practice [E-2(18)] may be referred.

NOTE — Rainwater harvesting with underground recharging will depend from region to region as per the climatic, topography, soil conditions and data about intensity of rainfall and retention time for calculating the capacity of recharge wells to be obtained from reference documents from the Authority.

4.5.12.3 Roof top rainwater harvesting

4.5.12.3.1 Harvesting in regular rainfall areas

In areas having rainfall over a large period in a year, for example in hilly areas, coastal regions, etc, constant and regular rainfall can be usefully harvested and stored in suitable water tanks. Water shall be collected through roof gutters and down take pipes. Provision should be made to divert the 1st rainfall after a dry spell so that ant dust, soot and leaves, etc are drained away before the water is collected into the water tank. The capacity of the water tank should be enough for storing water required for consumption between two dry spells. The water tank shall be located in a well protected area and shall not be exposed to any hazards of water contamination from any other sources. The water shall be chlorinated using chlorine tablets or solution to maintain a residual chlorine of approximately 1 mg/l. The tank shall have an overflow leading to natural water courses or to any additional tanks (see [Table 26](#)).

4.5.12.3.2 Harvesting in limited rainfall areas

In areas with the rainfall limited during the monsoon period (usually from 15 days to 90 days), roof top rainwater can be stored and used for non-potable purposes, after proper treatment, in the premises itself as mentioned above; excess water, if any, is best used for recharging the ground water. For individual properties and plots the roof top rainwater can be diverted to underground or above ground water storage tank(s), pretreated, stored and used for non-potable purposes. Excess water, if any, may be discharged to existing open or abandoned tube wells. In a well-planned building complex, a system should be laid out so that the runoff is collected in underground or above ground water storage tank(s), pretreated, stored, and used for non-potable purposes. Excess water, if any, shall be discharged in bore-wells as per designs specified by the Central Ground Water Board of the Government of India.

For roof top rainwater harvesting, reference may be made to good practice [E-2(19)].

4.5.12.4 Care to be taken in rainwater harvesting

Water conservation technique discussed above shall be constructed with due care taking following precautions:

- a) No sewage or waste water should be admitted into the system;
- b) No waste water from areas likely to have oil, grease or other pollutants should be connected to the system;
- c) Each structure/well shall have an inlet chamber with a silt trap to prevent any silt from finding its way into the subsoil water;
- d) The wells should be terminated at least 5 m above the natural static subsoil water at its highest level so that the incoming flow passes through the natural ground condition and prevent contamination hazards; and
- e) No recharge structure or a well shall be used for drawing water for any purpose.

Table 26 Rainwater Available from Roof Top Harvesting

(Clause [4.5.12.3.1](#))

Rain fall, mm	100	200	300	400	500	600	700	800	900	1 000	1 100	1 200	1 300	1 400	1 500	1 600	1 700	1 800	1 900	2 000
Roof Top Area m ²	Harvested Water from Roof Tops, m ³ (80 percent of gross precipitation)																			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)
20	2	3	5	6	8	10	11	13	14	16	18	19	21	22	24	26	27	29	30	32
30	2	5	7	10	12	14	17	19	22	24	26	29	31	34	36	38	41	43	46	48
40	3	6	10	13	16	19	22	26	29	32	35	38	42	45	48	51	54	58	61	64
50	4	8	12	16	20	24	28	32	36	40	44	48	52	56	60	64	68	72	76	80
60	5	10	14	19	24	29	34	38	43	48	53	58	62	67	72	77	82	86	91	96
70	6	11	17	22	28	34	39	45	50	56	62	67	73	78	84	90	95	101	106	112
80	6	13	19	26	32	38	45	51	58	64	70	77	83	90	96	102	109	115	122	128
90	7	14	22	29	36	43	50	58	65	72	79	86	94	101	108	115	122	130	137	144
100	8	16	24	32	40	48	56	64	72	80	88	96	104	112	120	128	136	144	152	160
110	9	18	26	35	44	53	62	70	79	88	97	106	114	123	132	141	150	158	167	176
120	10	19	29	38	48	58	67	77	86	96	106	115	125	134	144	154	163	173	182	192
130	10	21	31	42	52	62	73	83	94	104	114	125	135	146	156	166	177	187	198	208
140	11	22	34	45	56	67	78	90	101	112	123	134	146	157	168	179	190	202	213	224
150	12	24	36	48	60	72	84	96	108	120	132	144	156	168	180	192	204	216	228	240
200	16	32	48	64	80	96	112	128	144	160	176	192	208	224	240	256	272	288	304	320
250	20	40	60	80	100	120	140	160	180	200	220	240	260	280	300	320	340	360	380	400
300	24	48	72	96	120	144	168	192	216	240	264	288	312	336	360	384	408	432	456	480
400	32	64	96	128	160	192	224	256	288	320	352	384	416	448	480	512	544	576	608	640
500	40	80	120	160	200	240	280	320	360	400	440	480	520	560	600	640	680	720	760	800
1 000	80	160	240	320	400	480	560	640	720	800	880	960	1 040	1 120	1 200	1 280	1 360	1 440	1 520	1 600
2 000	160	320	480	640	800	960	1 120	1 280	1 440	1 600	1 760	1 920	2 080	2 240	2 400	2 560	2 720	2 880	3 040	3 200
3 000	240	480	720	960	1 200	1 440	1 680	1 920	2 160	2 400	2 640	2 880	3 120	3 360	3 600	3 840	4 080	4 320	4 560	4 800

4.5.12.5 Siphonic drainage system

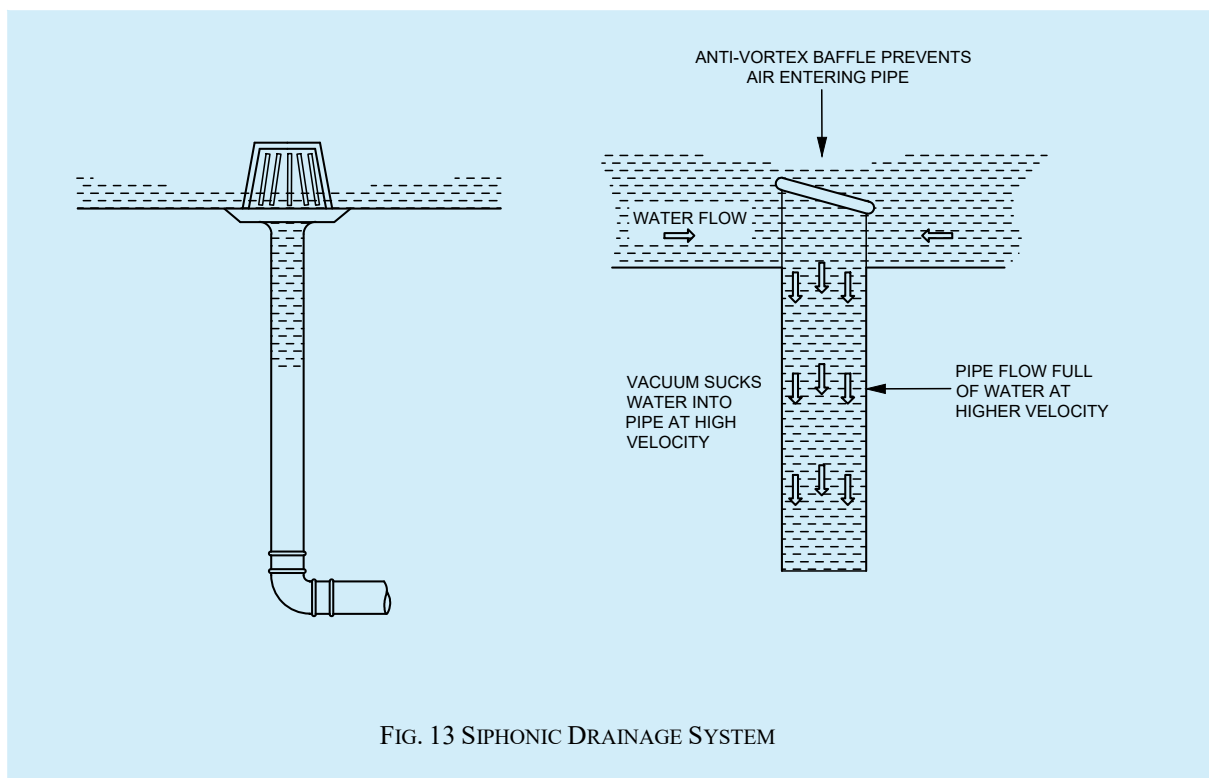
It is an installation generally used for special situations such as roofs with large spans for structures like hangars, airport terminals, stadia and industrial sheds, where the number of down pipes has to be limited. These are designed for full flow of pipes and the roof outlets are different from conventional ones.

Siphonic roof drainage systems are engineered on the concept of a full bore (a fill rate of 100 percent). This implies that rainwater flows at high speed through small diameter pipe work, at normally zero gradient.

This siphonic effect is created by the (kinetic) energy derived from the hydraulic head, caused by the difference in height between the roof outlet and the discharge point in a building. Specialist roof outlets prevent air being sucked into the system (see [Fig. 13](#)).

The engineering principle of siphonic roof drainage design is based on the Bernoulli energy equation for a steady flow of an incompressible fluid with constant density. In order to balance the equation, and to guarantee the required siphonic effect according to the rainfall intensity, the ideal pipe dimensions per flow path need to be determined.

In case of syphon system, the outlets of drainage system prevent vortex formation, thereby preventing air from entering the pipeline ensuring the entire system is running at full flow. It uses design that complies with Bernoulli's principle where equal energy is created under positive and negative pressure.



4.5.12.6 Rainwater harvesting for plotted/group housing developments

The rainwater harvesting methods adopted for plotted and group housing are through collection of rooftop rainwater and surface runoff harvesting.

A network of storm water drains in the entire residential area is used for harvesting rooftop rainwater and surface runoff. Appropriate number of recharge wells measuring $1\text{ m} \times 1\text{ m} \times 2\text{ m}$ may be constructed in the storm water drain for facilitating groundwater recharge. The quality of runoff, which passes through the borewell installed inside the recharge well, is ensured through a filter bed of pebbles (see [Fig. 12](#)).

4.5.12.7 Other methods of rainwater harvesting

- a) *Creation of artificial reservoirs/lakes for utilization of available storm water* — This shall be based on estimation of amount of runoff volume based on rainfall data (for a period of 10 years) considering the percolation and evaporation losses and efforts to be made to collect all available runoff for proper utilization; and
- b) *Water balancing methods* — This refers to optimum utilization of available water from different sources, namely ground water, recycled waste water, storm water and municipal supply and its adoption would ensure that there is no wastage of water.

4.5.13 Subsoil Water Drainage

4.5.13.1 General

Subsoil water is that portion of the rainfall which is absorbed into the ground.

The drainage of subsoil water may be necessary for the following reasons:

- a) To increase the stability of the surface;
- b) To avoid surface flooding;
- c) To alleviate or to avoid causing dampness in the building, especially in the cellars;
- d) To reduce the humidity in the immediate vicinity of the building; and
- e) To increase the workability of the soil.

4.5.13.2 Depth of water table

The standing level of the subsoil water will vary with the season, the amount of rainfall and the proximity and level of drainage channels. Information regarding this level may be obtained by means of boreholes or trial pits, preferably the latter. It is desirable though not always practicable to ascertain the level of the standing water over a considerable period so as to enable the seasonal variations to be recorded and in particular the high water level. The direction of flow of the subsoil water may usually be judged by the general inclination of the land surface and the main lines of the subsoil drains shall follow the natural falls, wherever possible.

4.5.13.3 Precautions

Subsoil drains shall be so sited as not to endanger the stability of the buildings or earthwork. In some portions of the drain, it may be necessary to use non-porous jointed pipes.

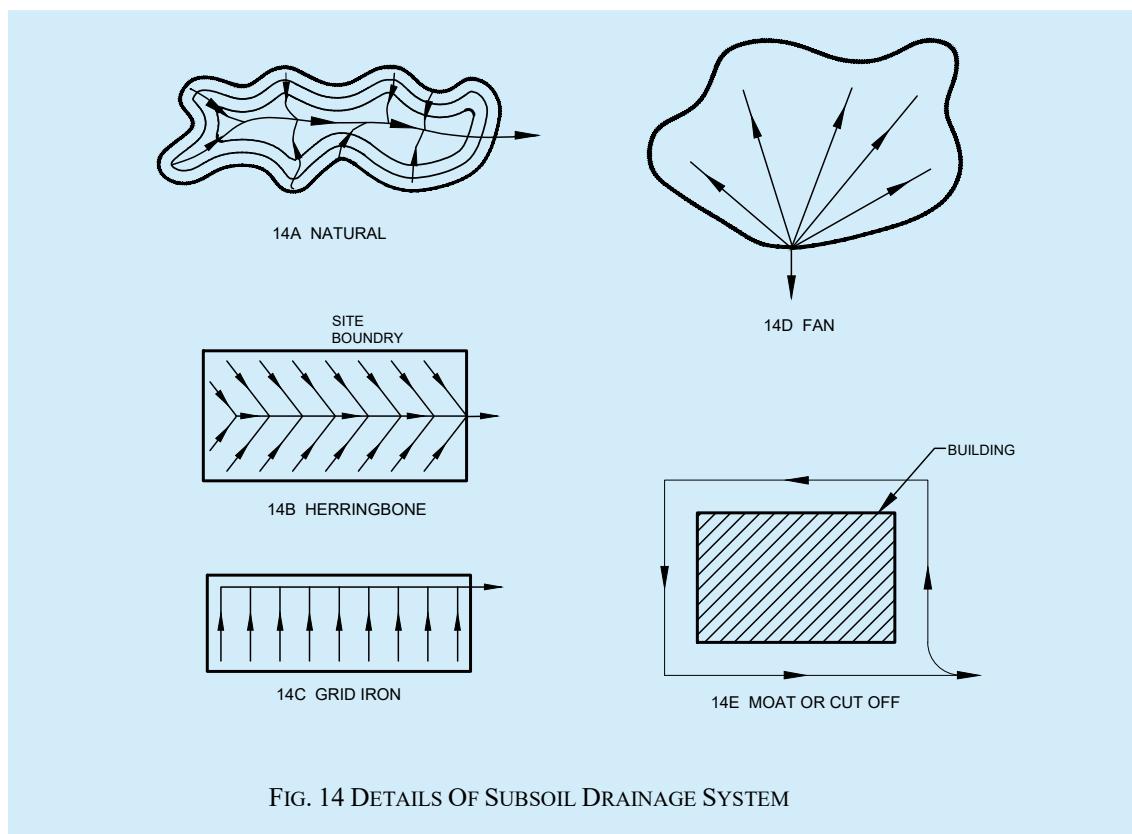
4.5.13.3.1 No field pipe shall be laid in such a manner or in such a position as to communicate directly with any drain constructed or adopted to be used for conveying sewage, except where absolutely unavoidable and in such case a suitable efficient trap shall be provided between subsoil drain and such sewer.

4.5.13.4 Systems of subsoil drainage

Clay or concrete porous field drain pipes may be used and shall be laid in one of the following ways (*see also* [Fig. 14](#)):

- a) *Natural* — The pipes are laid to follow the natural depressions or valleys of the site; branches discharge into the main as tributaries do into a river;
- b) *Herringbone* — The system consists of a number of drains into which discharges from both sides smaller subsidiary branch drains parallel to each other, but at an angle to the mains forming a series of herringbone pattern. Normally these branch drains should not exceed 30 m in length;
- c) *Grid* — A main or mains drain is laid to the boundaries of the site into which subsidiary branches discharge from one side only;
- d) *Fan-shaper* — The drains are laid converging to a single outlet at one point on the boundary of a site, without the use of main or collecting drains; and

- e) *Moat or cut-off system* — This system consists of drains laid on one or more sides of a building to intercept the flow of subsoil water and carry it away, thereby protecting the foundations of a building.



The choice of one or more of these systems will naturally depend on the local conditions of the site. For building sites, the mains shall be not less than 75 mm in diameter and the branches not less than 65 mm in diameter but normal practice tends towards the use of 100 mm and 75 mm, respectively. The pipes shall generally be laid at 60 cm to 90 cm depth, or to such a depth to which it is desirable to lower the water-table and the gradients are determined rather by the fall of the land than by considerations of self-cleansing velocity. The connection of the subsidiary drain to the main drain is best made by means of a clayware or concrete junction pipe. The outlet of a subsoil system may discharge into a soakaway or through a catch pit into the nearest ditch or watercourse. Where these are not available, the subsoil drains may be connected, with the approval of the Authority, through an intercepting trap to the surface water drainage system.

NOTE — Care shall be taken that there is no backflow from sub-surface drains during heavy rains.

4.5.13.5 Deep well/borewell recharging

Recharge well is one which pushes back surface water into ground water system. The recharge well may be of 1 m in diameter and 6 m deep lined with concrete rings having perforations.

Direct recharge is recommended when the well has gone dry and is yielding negligible amount of water. Indirect recharge is adopted for functioning borewells.

4.5.14 Waste Disposal Systems in High Altitudes and/or Sub-zero Temperature Regions

4.5.14.1 In general, all the care to be exercised regarding water supply systems shall also be applicable in the case of waste disposal systems. The biological and chemical reduction of organic material proceeds slowly under low temperature conditions, consequently affecting the waste disposal systems. The waste disposal methods given in [4.5.14.2](#), [4.5.14.3](#) and [4.5.14.4](#) shall be used only where it is not practical to install water carriage system.

4.5.14.2 Box and can system

Where box and can systems are employed, adequate arrangements shall be made for the cleaning and disinfection of the can after it is emptied of its contents. The excrement from the can shall be disposed of by burial in isolated spots far from habitation or by incineration, where feasible. The can shall be fitted with a tight fitting lid for use when it is carried for emptying.

4.5.14.3 Trench or pit latrines

Trench or pit latrines shall be used only where soil and subsoil conditions favour their use. Whenever they are used, they shall not be closer than 18 m from any source of drinking water, such as well, to eliminate the possibility of bacterial pollution of water.

4.5.14.4 Chemical toilets

For the successful functioning of chemical toilets, they shall preferably be installed in heated rooms or enclosures.

NOTE — Chemical toilet essentially consists of small cylindrical tanks with a water-closer seat for the use of 8 persons to 10 persons. A ventilation pipe is fitted to the seat. A strong solution of caustic soda is used as a disinfectant. It kills bacteria, liquefies the solids and thus checks the decomposition of organic matter. The tank is provided with a drain plug for which liquid runs to a soak pit at the time of disposal.

4.5.14.5 Water-borne sanitation systems

Water-borne sanitation systems shall be used, where practicable. Sanitation systems for the collection of sewage should be constructed in such a manner that maximum heat is retained by insulation, if necessary.

4.5.14.5.1 Sewerage laying

Under normal circumstances, sewers shall be laid below the frost line. Manholes shall be made of air-tight construction so as to prevent the cold air from gaining access inside and freezing the contents. The trenches for sewers shall be loosely filled with earth after laying sewers, since loose soil is a better insulator than compacted soil. Consequently, sewers laid under traffic ways and other places where soil compaction may be expected are required to be given adequate insulation. Where feasible, sewers shall be so located that the trench line is not in shadow, when the sun is shining. Concrete, cast iron and stoneware pipes conduct heat relatively rapidly and as such should be adequately insulated.

4.5.14.5.2 Septic tanks

Septic tanks can function only when it can be ensured that the contents inside these do not freeze at low temperature. For this purpose, the septic tanks shall be located well below the frost line. The location of manhole openings shall be marked by staves. Fencing around the septic tanks shall be provided for discouraging traffic over them. As the rate of biological activity is reduced by 50 percent for every 10 °C fall in temperature, the capacity of septic tanks shall be increased by 100 percent for operation at 10 °C over that for operation at 20 °C.

The construction of septic tanks is preferred in rural and fringe areas of suburban and isolated buildings where underground system may neither be feasible nor economical. Septic tanks are only recommended for small communities and institutions whose contributory population does not exceed 300.

For other details, refer good practice [E-2(10)].

4.5.14.5.3 Seepage pits

Seepage pits can function only when the soil and subsoil conditions are favourable. Frozen soil extending to a great depth would preclude the use of such disposal devices in view of the lower water absorption capacity. The discharge of effluent should be made below the frost line.

4.5.14.5.4 Sewage treatment plants

Suitable design modifications for sedimentation, chemical and biological processes shall be applied to sewage treatment plants for satisfactory functioning (see also [4.13](#)).

NOTE — Lavatories and bathrooms shall be kept heated to avoid freezing of water inside traps and flushing cisterns.

4.6 Construction Relating to Conveyance of Sanitary Wastes

4.6.1 Excavation

4.6.1.1 General

The safety provisions as given in relevant guidelines/best practices (Handbook on ‘Construction Management, Practices and Safety in Buildings’) should be referred for guidance purpose.

4.6.1.2 Turf, topsoil or other surface material shall be set aside, turf being carefully rolled and stacked for use in reinstatement. All suitable broken surface material and hard-core shall be set on one side for use in subsequent reinstatement.

4.6.1.3 Excavated material shall be stacked sufficiently away from the edge of the trench and the size of the spoil bank shall not be allowed to become such as to endanger the stability of the excavation. Spoil may be carried away and used for filling the trench behind the work.

4.6.1.4 Excavation shall proceed to within about 75 mm of the finished formation level. This final 75 mm is to be trimmed and removed as a separate operation immediately prior to the laying of the pipes or their foundations.

4.6.1.5 Unless specified otherwise by the Authority, the width at bottom of trenches for pipes of different diameters laid at different depths shall be as given below:

- a) For all diameters, up to an average depth of 1 200 mm,
Width of trench, in mm = Diameter of pipe + 300 mm;
- b) For all diameters for depths above 1 200 mm,
Width of trench, in mm = Diameter of pipe + 400 mm; and
- c) Notwithstanding (a) and (b), the total width of trench at the top should not be less than 750 mm for depths exceeding 900 mm.

4.6.1.6 Excavation in roads shall be so arranged, in agreement with the proper authority, as to cause the minimum obstruction to traffic. The methods to be adopted shall depend on local circumstances.

4.6.1.7 All pipes, ducts, cables, mains or other services exposed in the trench shall be effectively supported by timber and/or chain or rope-slings.

4.6.1.8 All drainage sumps shall be sunk clear of the work outside the trench or at the sides of manholes. After the completion of the work, any pipes or drains leading to such sumps or temporary subsoil drains under permanent work shall be filled in properly with sand and consolidated.

4.6.2 Laying of Pipes

Laying of pipes shall be done in accordance with good practice [E-2(20)].

4.6.3 Jointing

All soil pipes, waste pipes, vent pipes and other such pipes above ground shall be gas-tight. All sewers and drains laid below the ground shall be water-tight. Jointing shall be done in accordance with good practice [E-2(20)].

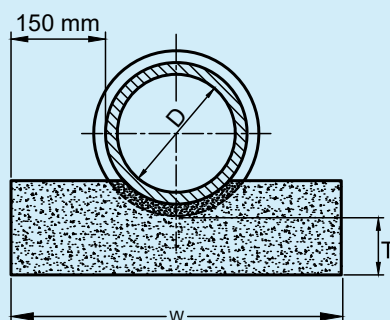
4.6.4 Support or Protection for Pipes

4.6.4.1 General

It may be necessary to support or surround pipe sewers or drains by means of concrete in certain circumstances. Some of the suggested methods are given in [4.6.4.2](#) to [4.6.4.4](#).

4.6.4.2 Bedding

Bedding (see Fig. 15) shall be rectangular in section and shall extend laterally at least 150 mm beyond and on both sides of the projection of the barrel of the pipe. The thickness of the concrete below the barrel of the pipe shall be not less than 100 mm for pipes under 150 mm diameter and 150 mm for pipes 150 mm and over in diameter. Where bedding is used alone, the concrete shall be brought up at least to the invert level of the pipe to form a cradle and to avoid line contact between the pipe and the bed.



$$W = D + 300 \text{ mm}$$

where D is external diameter of the pipe

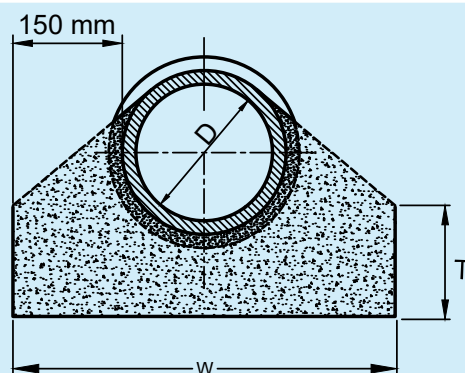
$$T = \begin{cases} 100 \text{ mm for pipes under 150 mm} \\ \text{nominal dia} \\ 150 \text{ mm for pipes of 150 mm} \\ \text{nominal dia and over} \end{cases}$$

FIG. 15 BEDDING

4.6.4.3 Haunching

Concrete haunching (see Fig. 16) shall consist of:

- a concrete bed as described for bedding (see 4.6.4.2);
- the full width of the bed carried up to the level of the horizontal diameter of the pipe; and
- splays from this level carried up on both sides of the pipe, from the full width of the bed to meet the pipe barrel tangentially.



$$W = D + 300 \text{ mm}$$

where D is external diameter of the pipe

$$T = \begin{cases} 100 \text{ mm for pipes under 150 mm} \\ \text{nominal dia} \\ 150 \text{ mm for pipes of 150 mm} \\ \text{nominal dia and over} \end{cases}$$

FIG. 16 HAUNCHING

4.6.4.4 Surround or encasing

The surround or encasing (see Fig. 17) shall be similar to haunching up to the horizontal diameter of the pipe and the top portion over this shall be finished in a semicircular form to give a uniform encasing for the top half of the pipe.

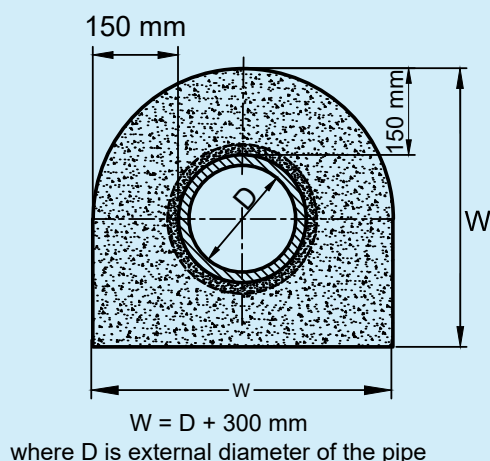


FIG. 17 SURROUND OR ENCASING

4.6.5 Connection to Existing Sewers

The connection to an existing sewer shall, as far as possible, be done at the manholes. Where it is unavoidable to make connection in between two manholes, the work of breaking into the existing sewer and forming the connection shall be carried out by the Authority or under its supervision.

4.6.5.1 Breaking into the sewer shall be effected by the cautious enlargement of a small hole and every precaution shall be taken to prevent any material from entering the sewer. No connection shall be formed in such a way as to constitute a projection into the sewer or to cause any diminution in its effective size.

4.6.6 Back-Filling

4.6.6.1 Filling of the trench shall not be commenced until the length of pipes therein has been tested and passed (see 4.10.2).

4.6.6.2 All timber which may be withdrawn with safety shall be removed as filling proceeds.

4.6.6.3 Where the pipes are unprotected by concrete haunching, the first operation in filling shall be carefully done to hand-pack and tamp selected fine material around the lower half of the pipes so as to buttress them to the sides of the trench.

4.6.6.4 The filling shall then be continued to 150 mm over the top of the pipe using selected fine hand-packed material, watered and rammed on both sides of the pipe with a wooden rammer. On no account shall material be tipped into the trench until the first 150 mm of filling has been completed. The process of filling and tamping shall proceed evenly so as to maintain an equal pressure on both sides of the pipeline.

4.6.6.5 Filling shall be continued in layers not exceeding 150 mm in thickness, each layer being watered and well rammed.

4.6.6.6 In roads, surface materials previously excavated shall be replaced as the top layer of the filling, consolidated and maintained satisfactorily till the permanent reinstatement of the surface is made by the Authority.

4.6.6.7 In gardens, the top soil and turf, if any, shall be carefully replaced.

4.7 Construction Relating to Conveyance of Rain or Storm Water

4.7.1 Roof Gutters

Roof gutters shall be of any material of suitable thickness. All junctions and joints shall be water-tight.

4.7.2 Rainwater Pipes

Rainwater pipes shall conform to the accepted standards [E-2(21)].

4.7.3 Subsoil Drain Pipes

4.7.3.1 Field drain pipes

Suitable pipes for this purpose are plain cylindrical glazed water pipes, or concrete porous pipes, though the latter may prove unsuitable where subsoil water carries sulphates or is acidic owing to the presence of peat. Trenches for these pipes need be just wide enough at the bottom to permit laying the pipes, which shall be laid with open joints to proper lines and gradients.

It is advisable to cover the pipes with clinker free from fine ash, brick ballast or other suitable rubble, or a layer of inverted turf, brush-wood or straw before refilling the trench, in order to prevent the infiltration of silt through the open joints. Where the subsoil drain is also to serve the purpose of collecting surface water, the rubble shall be carried up to a suitable level and when required for a lawn or playing field, the remainder of the trench shall be filled with pervious top soil. When refilling the trenches, care shall be taken to prevent displacement of pipes in line of levels. When they pass near trees or through hedges, socket pipes with cement or bitumen joints shall be used to prevent penetration by roots.

4.7.3.2 French drain

A shallow trench is excavated, the bottom neatly trimmed to the gradient and the trench filled with broken stone, gravel or clinker, coarse at the bottom and finer towards the top.

4.8 Selection and Installation of Sanitary Appliances

Selection, installation and maintenance of sanitary appliances shall be done in accordance with good practice [E-2(22)].

4.9 Refuse Chute System

The refuse chute system provided shall be in accordance with Part E 'Plumbing Services, Section 3 Solid Waste Management' of this NBCS.

4.10 Inspection and Testing

4.10.1 Inspection

4.10.1.1 All sanitary appliances and fitments shall be carefully examined for defects before they are installed and also on the completion of the work.

4.10.1.2 Pipes are liable to get damaged in transit and, notwithstanding tests that may have been made before despatch, each pipe shall be carefully examined on arrival on the site. Preferably, each pipe shall be rung with a hammer or mallet and those that do not ring true and clear shall be rejected. Sound pipes shall be carefully stored to prevent damage. Any defective pipes shall be segregated, marked in a conspicuous manner and their use in the works prevented.

4.10.1.3 Cast iron pipes shall be carefully examined for damage to the protective coating. Minor damage shall be made good by painting over with hot tar or preferably bitumen. But if major defects in coating exist, the pipes shall not be used unless recoated. Each pipe shall be carefully re-examined for soundness before laying.

4.10.1.4 Close inspection shall be maintained at every stage in the work, particularly as to the adequacy of timber supports used in excavation and the care and thoroughness exercised in filling.

4.10.1.4.1 Careful note shall be kept of the condition of any sewer, manhole or other existing work which may be uncovered and any defects evident shall be pointed out immediately to the Authority.

4.10.1.4.2 No work shall be covered over or surrounded with concrete until it has been inspected and approved by the Authority.

4.10.2 *Testing*

4.10.2.1 Comprehensive tests of all appliances shall be made by simulating conditions of use. Overflow shall be examined for obstructions.

4.10.2.2 *Smoke test*

All soil pipes, waste pipes, and vent pipes and all other pipes when above ground shall be approved gas-tight by a smoke test conducted under a pressure of 25 mm of water and maintained for 15 min after all trap seals have been filled with water. The smoke is produced by burning only waste or tar paper or similar material in the combustion chamber of a smoke machine. Chemical smokes are not satisfactory.

4.10.2.3 *Water test*

4.10.2.3.1 *For pipes other than cast iron*

Glazed and concrete pipes shall be subjected to a test pressure of at least 1.5 m head of water at the highest point of the section under test. The tolerance figure of 2 litre/cm of diameter/km may be allowed during a period of 10 min. The test shall be carried out by suitably plugging the low end of the drain and the ends of connections, if any, and filling the system with water. A knuckle bend shall be temporarily jointed in at the top end and a sufficient length of the vertical pipe jointed to it so as to provide the required test head, or the top end may be plugged with a connection to a hose ending in a funnel which could be raised or lowered till the required head is obtained and fixed suitably for observation.

Subsidence of the test water may be due to one or more of the following causes:

- a) Absorption by pipes and joints;
- b) Sweating of pipes or joints;
- c) Leakage at joints or from defective pipes; and
- d) Trapped air.

Allowance shall be made for (a) by adding water until absorption has ceased after which the test proper should commence. Any leakage will be visible and the defective part of the work should be cut out and made good. A slight amount of sweating which is uniform may be overlooked, but excessive sweating from a particular pipe or joint shall be watched for and taken as indicating a defect to be made good. A slight amount of sweating which is uniform may be overlooked, but excessive sweating from a particular pipe or joint shall be watched for and taken as indicating a defect to be made good.

NOTE — This test will not be applicable to sanitary pipe work above ground level.

4.10.2.3.2 *For cast iron pipes*

Cast iron sewers and drains shall be tested as for glazed and concrete pipes. The drain plug shall be suitably strutted to prevent their being forced out of the pipe during the test.

4.10.2.4 Tests for straightness and obstruction

The following tests shall be carried out:

- a) By inserting at the high end of the sewer or drain a smooth ball of a diameter 13 mm less than the pipe bore. In the absence of obstruction, such as yarn or mortar projecting through the joints, the ball should roll down the invert of the pipe and emerge at the lower end; and
- b) By means of a mirror at one end of the line and lamp at the other. If the pipeline is straight, the full circle of light may be observed. If the pipe line is not straight, this will be apparent. The mirror will also indicate obstruction in the barrel.

4.10.2.5 Test records

Complete records shall be kept of all tests carried out on sewers and drains both during construction and after being put into service.

4.11 Maintenance

4.11.1 General

Domestic drainage system shall be inspected at regular intervals. The system shall be thoroughly cleaned out at the same time and any defects discovered shall be made good.

4.11.2 Cleaning of Drainage System

4.11.2.1 Sewer maintenance crews, when entering a deep manhole or sewer where dangerous gas or oxygen deficiencies may be present, shall follow the following procedures:

- a) Allow no smoking or open flames and guard against sparks;
- b) Erect warning signs;
- c) Use only safety gas-proof, electric lighting equipment;
- d) Test the atmosphere for noxious gases and oxygen deficiencies (presence of hydrogen sulphide is detected using lead acetate paper and that of oxygen by safety lamps);
- e) If the atmosphere is normal, workmen may enter with a safety belt attached and with two men available at the top. For extended jobs, the gas tests shall be repeated at frequent intervals, depending on circumstances;
- f) If oxygen deficiency or noxious gas is found, the structure shall be ventilated with pure air by keeping open at least one manhole cover each on upstream and downstream side for quick exit of toxic gases or by artificial means. The gas tests shall be repeated and the atmosphere cleared before entering. Adequate ventilation shall be maintained during this work and the tests repeated frequently;
- g) If the gas or oxygen deficiency is present and it is not practicable to ventilate adequately before workers enter, a hose mask shall be worn and extreme care taken to avoid all sources of ignition. Workers shall be taught how to use the hose equipment. In these cases, they shall always use permissible safety lights (not ordinary flash lights), rubber boots or non-sparking shoes and non-sparking tools;
- h) Workmen descending a manhole shaft to inspect or clean sewers shall try each ladder step or rung carefully before putting the full weight on it to guard against insecure fastening due to corrosion of the rung at the manhole wall. When work is going on in deep sewers, at least two men shall be available for lifting workers from the manhole in the event of serious injury; and
- j) Portable air blowers, for ventilating manhole, are recommended for all tank, pit or manhole work where there is a question as to the presence of noxious gas, vapours or oxygen deficiency. The motors for these shall be of weather proof and flame-proof types; compression ignition diesel type (without sparking plug) may be used. When used, these shall be placed not less than 2 m away from the opening and on the leeward side protected from wind, so that they will not serve as a source of ignition for any inflammable gas which might be present. Provision should be made for ventilation and it should be of the forced type which can be provided by a blower located at ground level with suitable flexible ducting to displace out air from the manhole.

4.11.2.2 The following operations shall be carried out during periodical cleaning of a drainage system:

- a) The covers of inspection chambers and manholes shall be removed and the side benching and channels scrubbed;
- b) The interceptive trap, if fitted, shall be adequately cleaned and flushed with clean water. Are shall be taken to see that the stopper in the rodding arm is securely replaced;
- c) All lengths of main and branch drains shall be rodded by means of drain rods and a suitable rubber or leather plunger. After rodding, the drains shall be thoroughly flushed with clean water. Any obstruction found shall be removed with suitable drain cleaning tools and the system thereafter shall be flushed with clean water;
- d) The covers of access plates to all gullies shall be removed and the traps plunged and flushed out thoroughly with clean water. Care shall be taken not to flush the gully deposit into the system;
- e) Any defects revealed as a result of inspection or test shall be made good;
- f) The covers or inspection chambers and gullies shall be replaced, bedding them in suitable grease or other materials; and
- g) Painting of ladders/rings in deep manholes and external painting of manhole covers shall be done with approved paints.

4.11.3 All surface water drains shall be periodically rodded by means of drain rods and a suitable rubber or leather plunger. After rodding, they shall be thoroughly flushed with clean water. Any obstruction found shall be removed with suitable drain cleaning tools.

4.11.4 All subsoil drains shall be periodically examined for obstruction at the open joints due to the roots of plants or other growths.

4.12 Pumping of Sewage

4.12.1 In the design of sewerage system, it is necessary to collect the sewage of a low lying area at some convenient point from which it shall be lifted by pumps. At the treatment plant also, lifting of sewage may be necessary to provide head for the flow by gravity of sewage.

4.12.2 Sewage Pump Stations, Sizing of Sumps and Pumps

They are required as onsite pump stations to cater to sewage from toilets and kitchen. The stations would be located in basement floors and leading the sewage to onsite treatment plants. Submersible centrifugal pumps are used for pumping the sewage. The sump capacity depends on effective holding capacity considering the flow from drainage fixtures. Usually, retention period of 15 min is adopted. The size of sump depends on the availability of space. The size of the discharge pipe shall be one nominal size higher than the design size or the pump outlet, with a velocity of 1 m/s. Pumps are sized for 100 percent design flow.

NOTE — For other details reference to good practice [E-2(13)] shall be made.

4.13 Sewage Treatment Systems

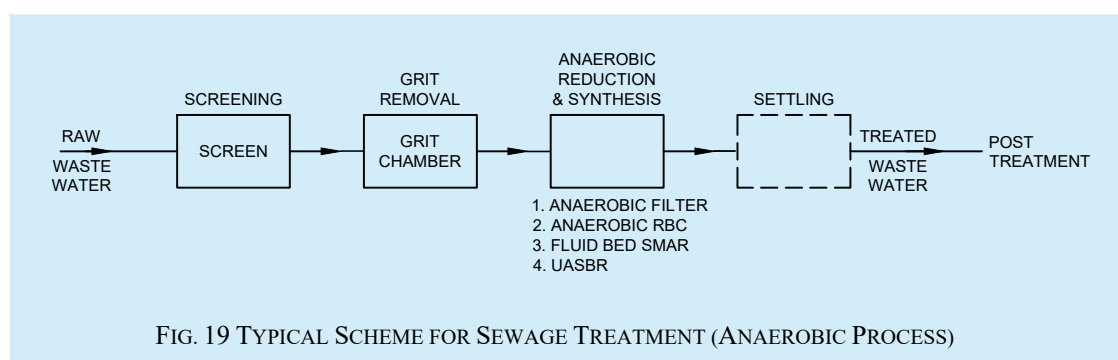
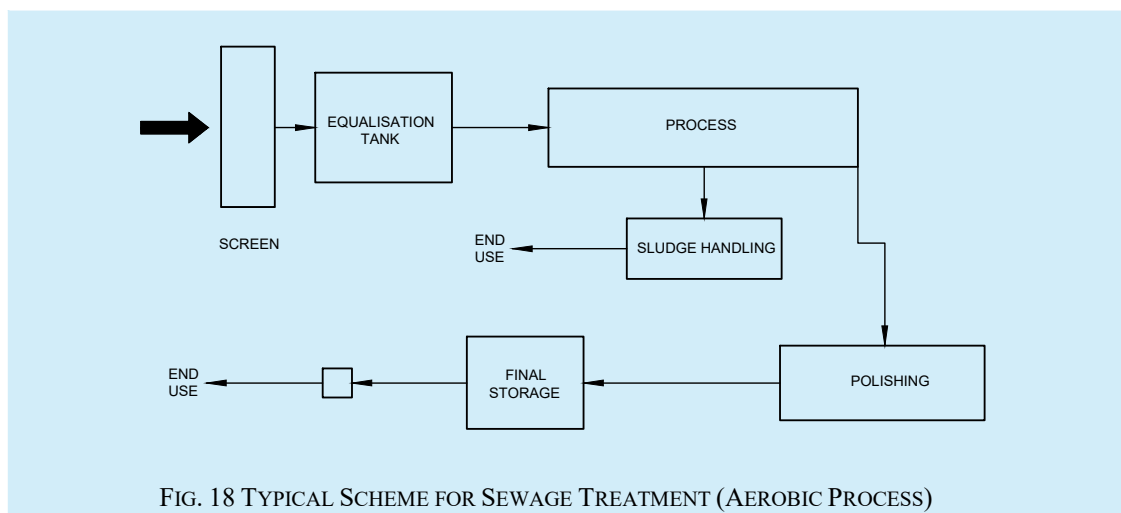
4.13.1 Sewage treatment is the process of removing contaminants from waste water, primarily from domestic sewage. It includes physical, chemical and biological processes to remove these contaminants and produce environmentally safe treated water (or treated effluent). Sewage treatment is normally carried out by aerobic bacteria (aerobic process; see [Fig. 18](#)) or anaerobic bacteria (anaerobic process; see [Fig. 19](#)), which metabolize the organic matter, multiply, and are subsequently settled out and disposed of as sludge.

The by-product of sewage treatment is usually a semisolid waste or slurry, called sewage sludge, that has to undergo further treatment before being suitable for disposal or further application.

4.13.2 The main objective of treating waste water is to stabilize decomposable organic matter present in the sewage to produce treated effluent and sludge, which can be disposed of in the environment without causing health hazards or nuisance.

The microbial population in treatment process can be cultured using the following systems:

- Suspended growth system* — In this system, microorganisms remain in suspension, and their concentration is related to mixed liquor suspended solids (MLSS);
- Attached growth system* — In this system, microorganisms are developed over mobile or immobile solid covered in biomass (slime);
- Combined growth system* — In this system, both systems are used to attain the shock loads.



4.13.3 Technologies

Following technologies for sewage treatment shall be considered based on the requirement:

- Extended aeration with activated sludge process (EAASP)* — Activated sludge process is the aerobic biological process by which non-settleable substances occurring in dissolved and colloidal forms are converted into settleable sludge which is removed from the liquid carrier (water);

This process is well suitable for completely populated community and does not require skilled labour for handling the plant. It is time tested and requires more number of units to achieve biological oxygen demand (BOD) less than 10 ppm, like coagulants required for better quality;

- Sequential batch reactor (SBR)* — It is an activated sludge process designed to operate under non-steady state conditions. An SBR operates in a true batch mode with aeration and sludge settlement both occurring in the same tank. There is a degree of flexibility associated with working in a time rather than in a space sequence;

Plant for this process requires less space due to batch process, well suitable for modular growth. It can handle smallest flows and the desired quality can be achieved without coagulants. It requires good quality of decanters and skilled labour as the process is timer-based automation;

- c) *Membrane bio-reactor (MBR)* — It is a continuous process of activated sludge treatment, but solids separation is done through pressurised system or through media. As there is physical barrier for solid separation, quality of treatment is good;

It is suitable for any type of growth pattern and can handle smaller flows. It requires less space, and skilled labours to handle.
- d) *Rotating bio bed reactor (RBBR)* — It is combined growth process (aerobic and anaerobic). Bacteria are allowed to grow on media, which is exposed to atmospheric air. It is also known as rotating biological contactor (RBC);

It is suitable for smaller capacities and does not require skilled labour to handle. The quality of treatment is good.
- e) *Fluidized bio bed reactor (FBBR)* — In this process, cells are immobilized in small particles which move with the fluid. The small particles create a large surface area for cells to stick and enable a high rate of transfer of oxygen and nutrients to the cells. It is also known as moving bed bio reactor (MBBR). It is suitable for all type of communities, does not require skilled labour and requires less space. It can be installed for all capacities, and the quality of treatment is good;
- f) *Submerged aerobic fixed film reactor (SAFF)* — It is a combined growth process in which bio-media is fixed and bacteria are allowed to grow on media, which has air supply from below;

It does not require skilled labour. Supporting structure for media should be anti-corrosive to enable media fixing.
- g) *Trickling filters* — It is an attached growth process where media is used to grow bacteria by trickling sewage over the media. It is an old concept and is not now widely used;
- h) *Oxidation ponds/lagoons* — Oxidation ponds, also called lagoons or stabilization ponds are large shallow ponds designed to treat waste water through the interaction of sunlight, bacteria and algae;
- j) *Packaged sewage treatment plants (PSTPs)* — Packaged sewage treatment plants enable on-site treatment of domestic waste water, including grey and black water. These systems are designed as turnkey solutions, catering to the advanced treatment needs of individual communities or decentralized community setups. PSTPs are particularly effective in areas lacking centralized sewerage systems or where such systems are impractical. The design, structural integrity, installation, operation, and maintenance of PSTPs shall comply with the provisions of accepted standard [E-2(23)];
- k) *Biodigester* — Biodigester is a non-sewer on-site human waste treatment system that operates under anaerobic conditions to breakdown organic waste into simpler molecules, water, and biogas. It primarily consists of a specially designed biodigester tank and a bacterial consortium, known as anaerobic microbial inoculum (AMI). The anaerobic digestion process significantly reduces organic matter, minimizes sludge generation, and lowers pathogen transmission. Effluent from biodigesters can also be introduced into centralized sewer systems, reducing the organic load on sewage treatment plants (STPs) and improving their efficiency. The design, construction, installation and effluent standards of biodigester systems shall adhere to good practices specified in [E-2(24)]; and
- m) Another option for on-site sanitation (OSS) systems in both urban and rural areas is the use of septic tanks. Septic tanks may be either prefabricated or constructed on site using suitable materials of construction. Precast concrete septic tanks have been in use in the country. Precast concrete septic tank have been in use in the country another type of septic tank is the rotationally moulded polyethylene septic tank (RMPST). RMPST could be single/multiple piece factory made unit, including inlet and outlet level openings which leaves the factory completed, controlled and ready for installation. These tanks are manufactured using rotational moulding technique which is a method for producing hollow plastic objects by placing finely divided plastic powder particles in a hollow mould that is rotated about two axes or given rock and roll type of motion, exposing it to heat and then to cold. RMPST is thus suitable for installation at standard temperature and pressure conditions. These septic tanks shall meet the accepted standards [E-2(25)].

4.14 Treatment of Wastewater and Usage of Recycled Water

See [4.2.4](#) of Part E 'Plumbing Services, Section 1 Water Supply' of this NBCS.

ANNEX A

(Clause [3.2.1](#))

APPLICATION FOR DRAINAGE OF PREMISES

I/We hereby make application to the *.....
..... for permission to drain the premises.....Ward
No..... Street No..... Road/Street known as.....

The sanitary arrangement and drains of the said premises are shown in the accompanying plans and a description of the specification of the work/material used is also appended ([Annex B](#)).

I/We undertake to carry out the work in accordance with Part E 'Plumbing Services, Section 2 Drainage and Sanitation' of the National Building Construction Standards (NBCS).

.....	
Signature of the licensed/registered plumber	Signature of the owner
Name and address of the	Name and address.....
.....	
.....	

Date.....

Date.....

NOTE — The application should be signed by the owner of the premises and shall be countersigned by the licensed/registered plumber.

* Insert the name of the Authority.

ANNEX B

(Clause [3.2.3.2](#) and [Annex A](#))

FORM FOR DETAILED DESCRIPTION OF WORK AND SPECIFICATION OF MATERIALS

- a) Separation of rainwater and foul water.....
- b) Rainwater drains, curbs and points of discharge.....
- c) Rainwater gutters, pipes or spouts where discharging.....
- d) Open-full-water drains, materials, sizes, curbs and other means places, verandahs, latrines
- e) Silt-catcher and grating, size and position.....
- f) Drains.....
 - 1) Main sewage drains: Fall
Size.....
 - 2) Branch drains : Fall
Size.....
 - 3) Materials.....
 - 4) Method of jointing.....
- g) Bedding of pipes:
 - 1) Method of bedding.....
 - 2) Thickness and width of beds of concrete.....
 - 3) Thickness of concrete round pipes.....
- h) Protection of drain laid under wall.....
- j) Traps, description and interceptor:
 - 1) Lavatory waste pipes.....
 - 2) Bath waste pipes.....
 - 3) Sink.....
 - 4) Gully-traps.....
 - 5) Water closet traps.....
 - 6) Grease traps.....
 - 7) Slop sink.....
 - 8) Urinal.....
 - 9) Others.....
- k) Manholes and inspection chambers:
 - 1) Thickness of walls.....
 - 2) Description of bricks.....
 - 3) Description of rendering.....
 - 4) Description of invert channels.....

- 5) Depth of chambers.....
- 6) Size and description of cover and manner of fixing.....
- m) Ventilation of drain:
 - 1) Position – Height above nearest ground level.....
 - 2) Outlet shaft position of terminal at top.....
- n) Soil pipe, waste pipe and ventilating pipe connections:
 - 1) Lead and iron pipes.....
 - 2) Lead pipe of trap with cast iron pipe.....
 - 3) Stoneware pipe or trap with lead pipe.....
 - 4) Lead soil pipe or trap with stoneware pipe or trap.....
 - 5) Cast iron pipe with stoneware drain.....
 - 6) Stoneware trap with cast iron soil pipe.....
- p) Ventilation of water closet trap sink, lavatory and other traps material and supports.
- q) Water closets (apartments):
 - 1)
 - i) At or above ground level.....
 - ii) Approached from.....
 - iii) Floor material.....
 - iv) Floor fall towards door.....
 - v) Size of window opening in wall made to open.....
 - vi) Position of same.....
 - vii) Means of constant ventilation.....
 - viii) Position of same.....
 - 2) Water closet apparatus:
 - i) Description of pan, basin, etc.....
 - ii) Kind.....
 - iii) Flushing cistern.....
 - iv) Material of flushing pipe.....
 - v) Internal diameter.....
 - vi) Union with basin.....
- r) Sanitary fittings, water storage tank, etc:
 - 1) Number and description of sanitary fittings in room and rooms in which they are to be installed.....
 - 2) Capacity and position of water storage tanks.....
 - 3) Size and number of draw off taps and whether taken off storage tanks or direct from main supply.....
 - 4) Details of draw off taps, that is, whether they are of plain screw down pattern or ‘waste not’ and description of any other sanitary work to be carried out not included under above headings.....

s) Depth of sewer below surface of street.....

t) Level of invert of house drain at point of junction:

1) With sewer.....

2) Level of invert of sewer at point of junction with house drain.....

3) Distance of nearest manhole on sewer from the point at which the drain leaves the premises.....

u) Schedule of pipes:

<i>Sl No.</i>	<i>Description of Pipe/Drain</i>	<i>Materials</i>	<i>Diameter</i>	<i>Weight</i>	<i>Method of Jointing</i>
(1)	(2)	(3)	(4)	(5)	(6)
i)	Subsoil drains				
ii)	Main sewage drains				
iii)	Branch sewage drains				
iv)	Soil pipes				
v)	Ventilating pipes other than soil pipes				
vi)	Waste pipes				
vii)	Rainwater pipes				
viii)	Anti-syphon pipes				

Signature of the licensed/registered plumber.....

Name and address of the licensed/registered plumber.....

.....

.....

.....

Date.....

ANNEX C

(Clause 3.2.5)

FORM FOR LICENSED/REGISTERED PLUMBER'S COMPLETION CERTIFICATE

Certified that I/we have completed the plumbing work of drainage and sanitation system for the premises as detailed below. This may be inspected, approved and connection given.

Ward No.....

Street.....

Locality.....

Block No.....

House No.....

Details of work.....
.....
.....

The work was sanctioned by the Authority*
vide

.....

Signature of the owner

Signature of the licensed/registered plumber

Name and address.....

Name and address

.....

Date.....

The Authority's Report

Certified that the plumbing work of drainage and sanitation system for the premises, have been laid, applied, executed in accordance with Part E 'Plumbing Services, Section 2 Drainage and Sanitation' of the National Building Construction Standards (NBCS).

Drainage Connection to the main sewer will be made on.....

Date.....

The Authority.....

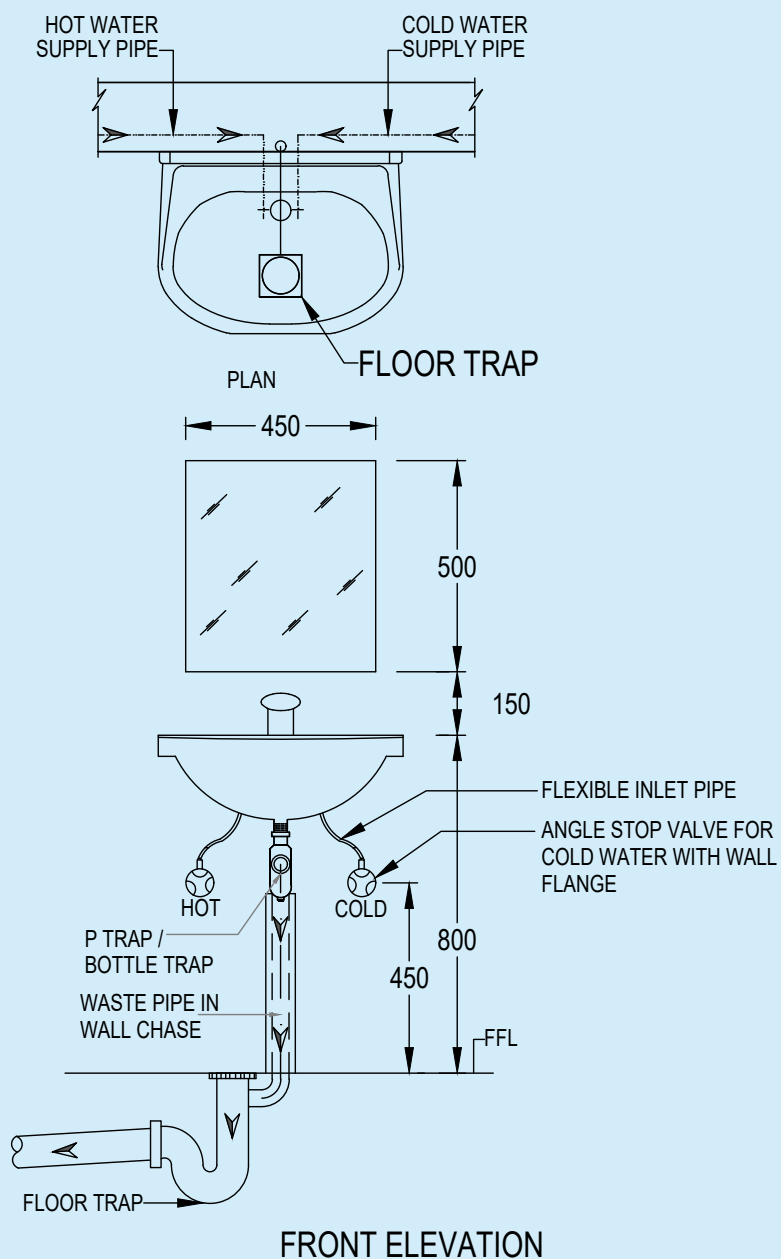
* Insert the name of the Authority

ANNEX D

(Clause 4.5.1.4)

TYPICAL MOUNTING ARRANGEMENTS FOR VARIOUS PLUMBING FIXTURES, INCLUDING DRAINAGE SYSTEMS AND VENTILATION

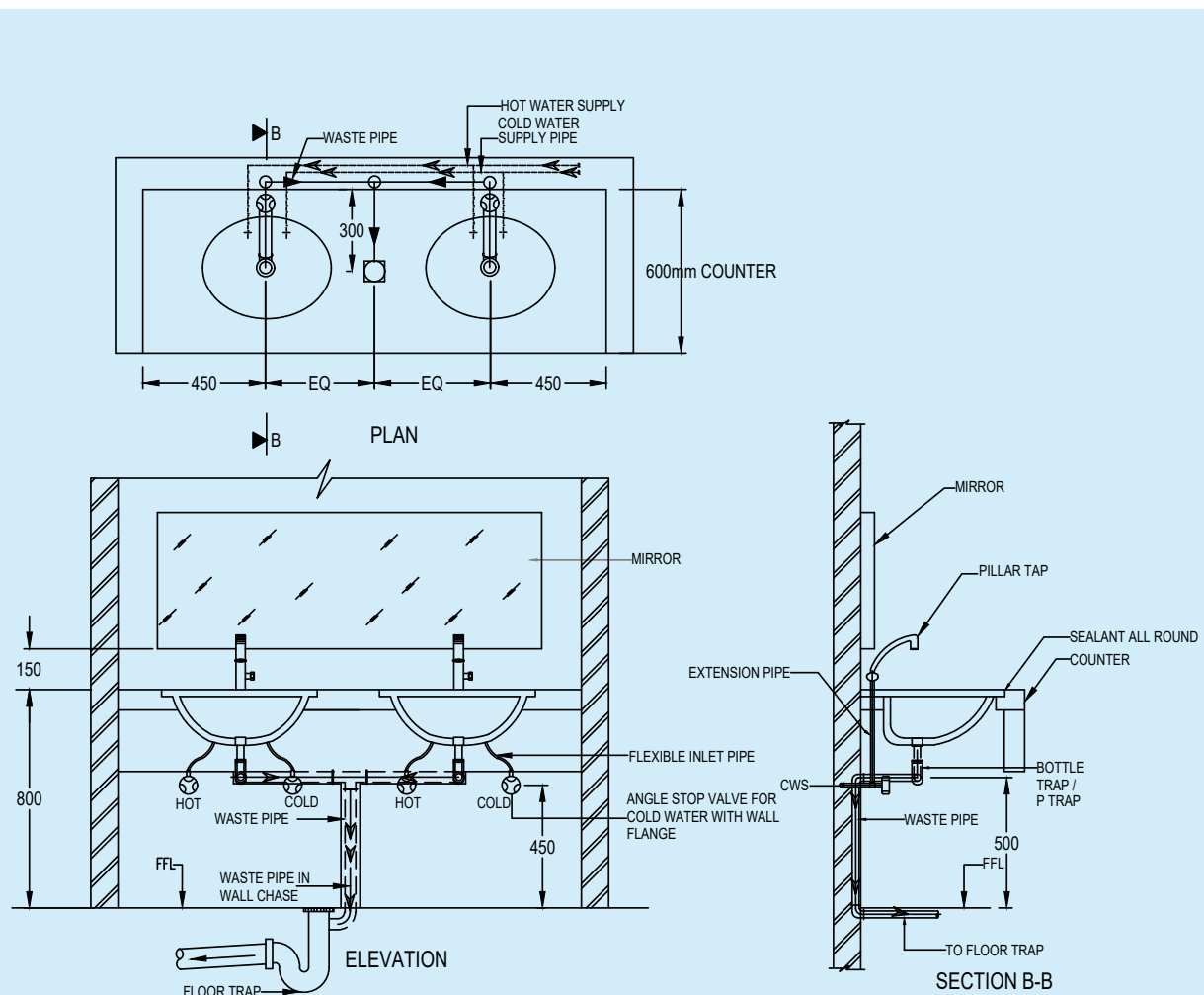
D-1 ARRANGEMENT FOR RECTANGULAR WASH BASIN — See Fig. 20



All dimensions in millimetres.

FIG. 20 RECTANGULAR WASH BASIN

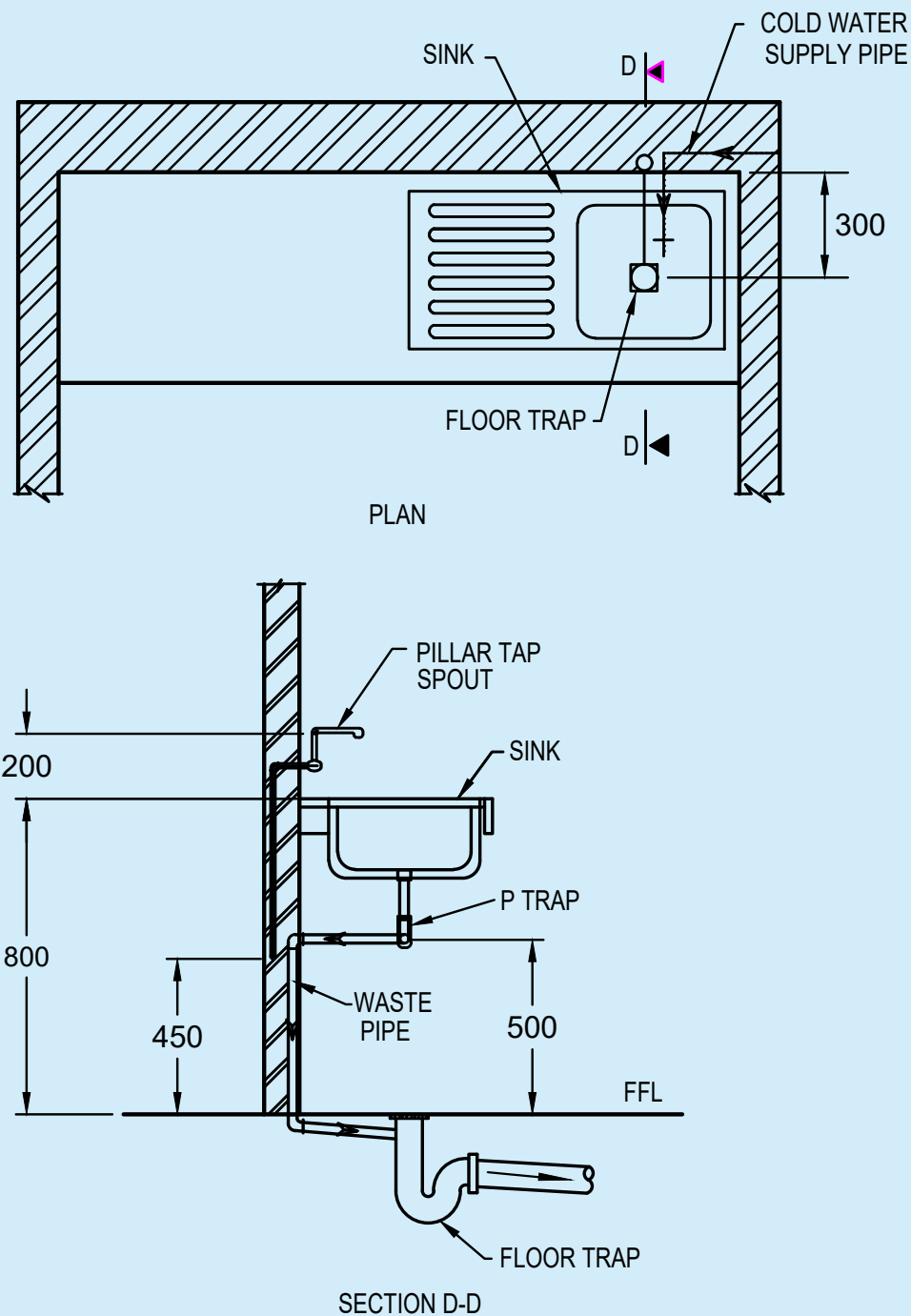
D-3 ARRANGEMENT FOR OVAL WASH BASIN (BELOW COUNTER) — See [Fig. 22](#)



All dimensions in millimetres.

FIG. 22 OVAL WASH BASIN (BELOW COUNTER)

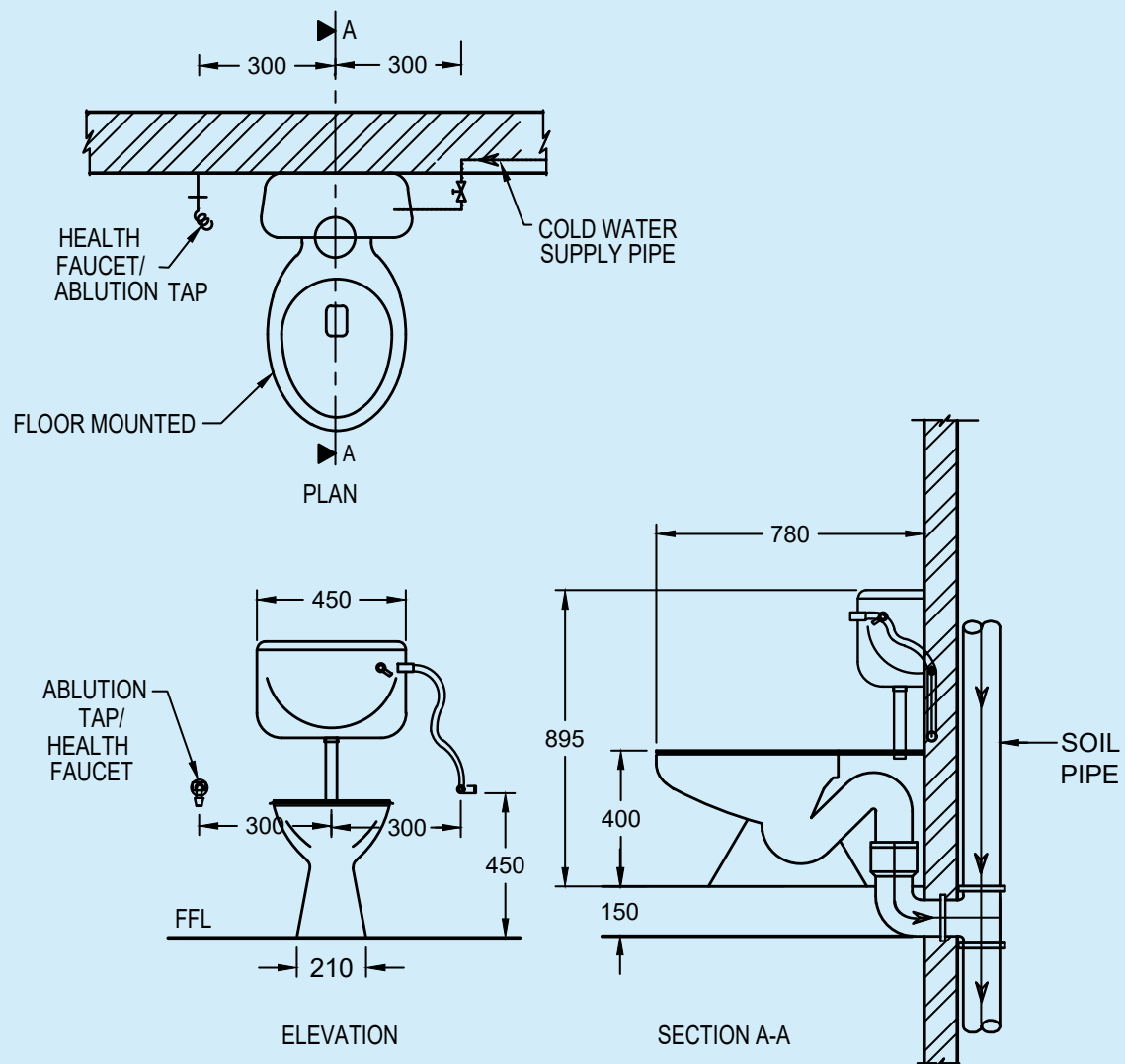
D-4 ARRANGEMENT FOR SINK — See [Fig. 23](#)



All dimensions in millimetres.

FIG. 23 SINK

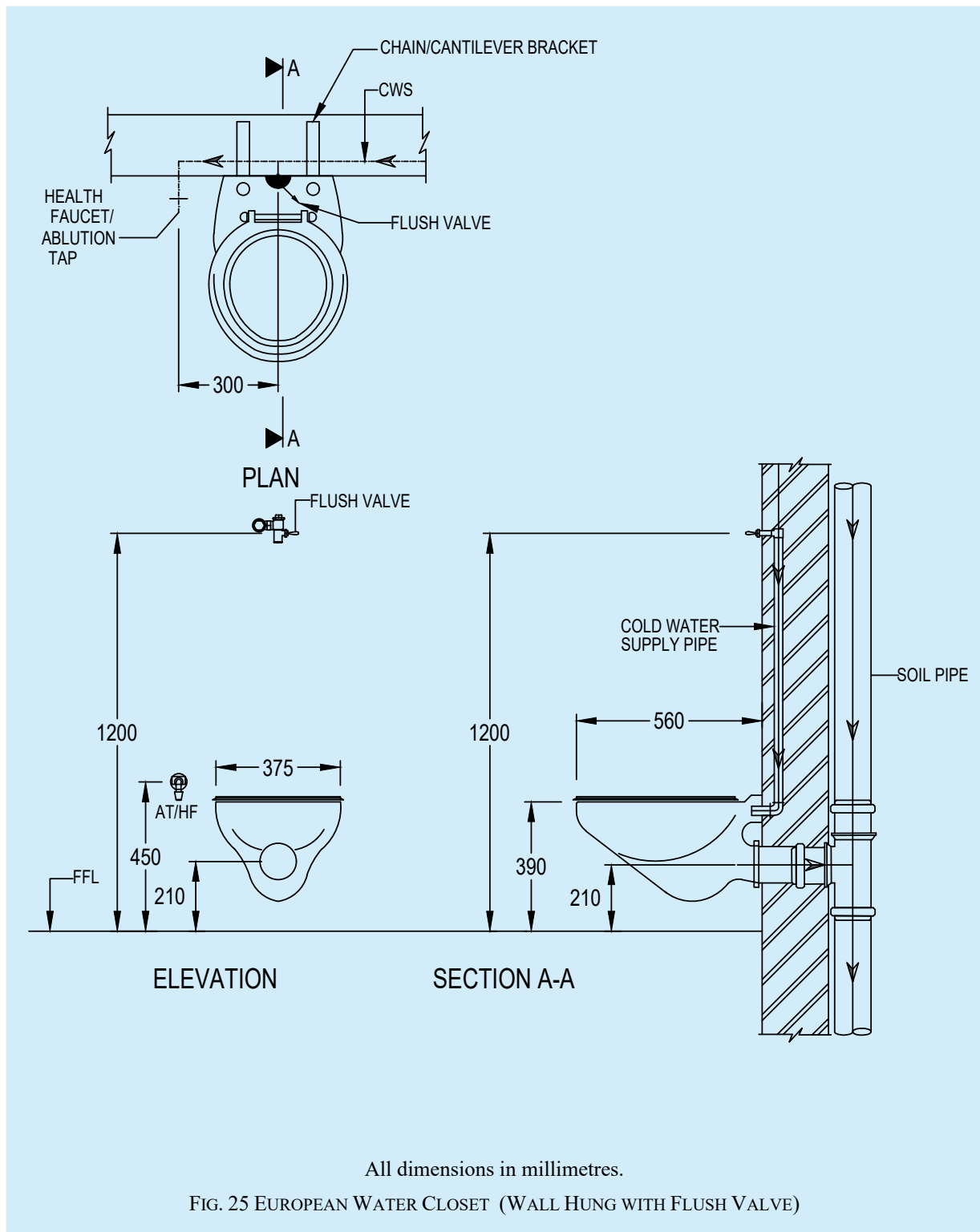
D-5 ARRANGEMENT FOR EUROPEAN WATER CLOSET (FLOOR MOUNTED WITH FLUSH TANK AND S-TRAP) — See [Fig. 24](#)



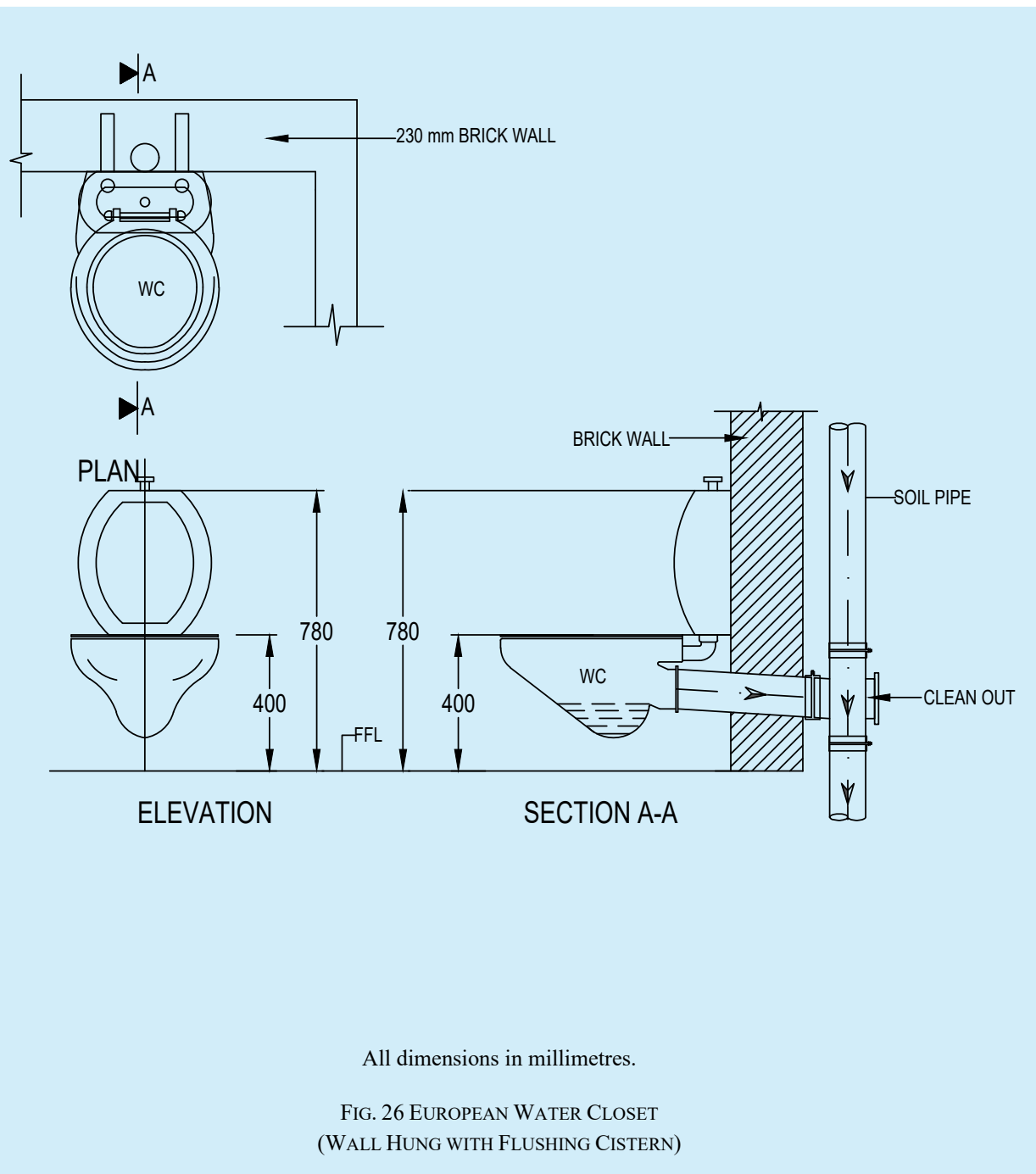
All dimensions in millimetres.

FIG. 24 EUROPEAN WATER CLOSET
(FLOOR MOUNTED WITH FLUSH TANK AND S-TRAP)

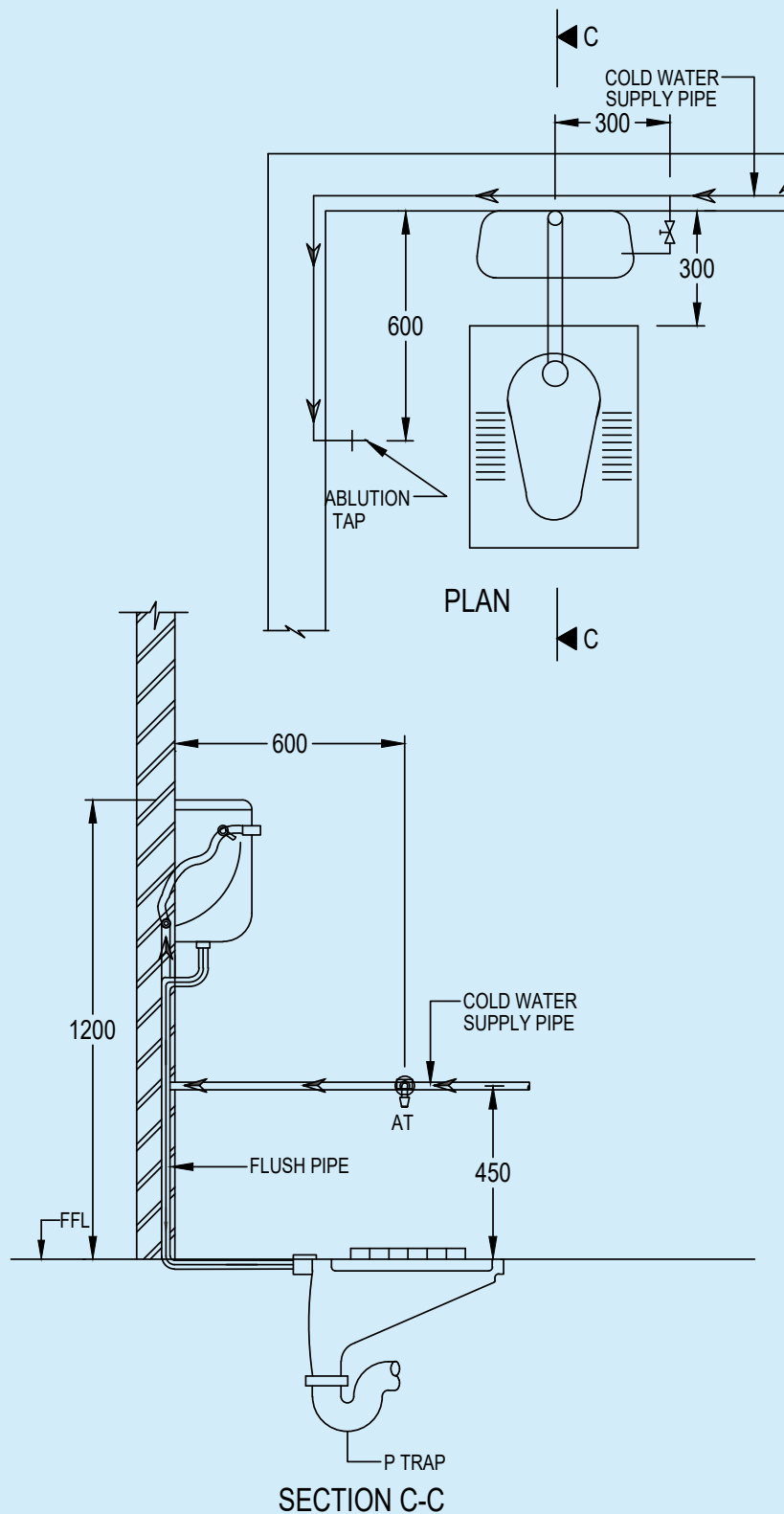
D-6 ARRANGEMENT FOR EUROPEAN WATER CLOSET (WALL HUNG WITH FLUSH VALVE) —
See [Fig. 25](#)



D-7 ARRANGEMENT FOR EUROPEAN WATER CLOSET (WALL HUNG WITH FLUSHING CISTERN) — See [Fig. 26](#)



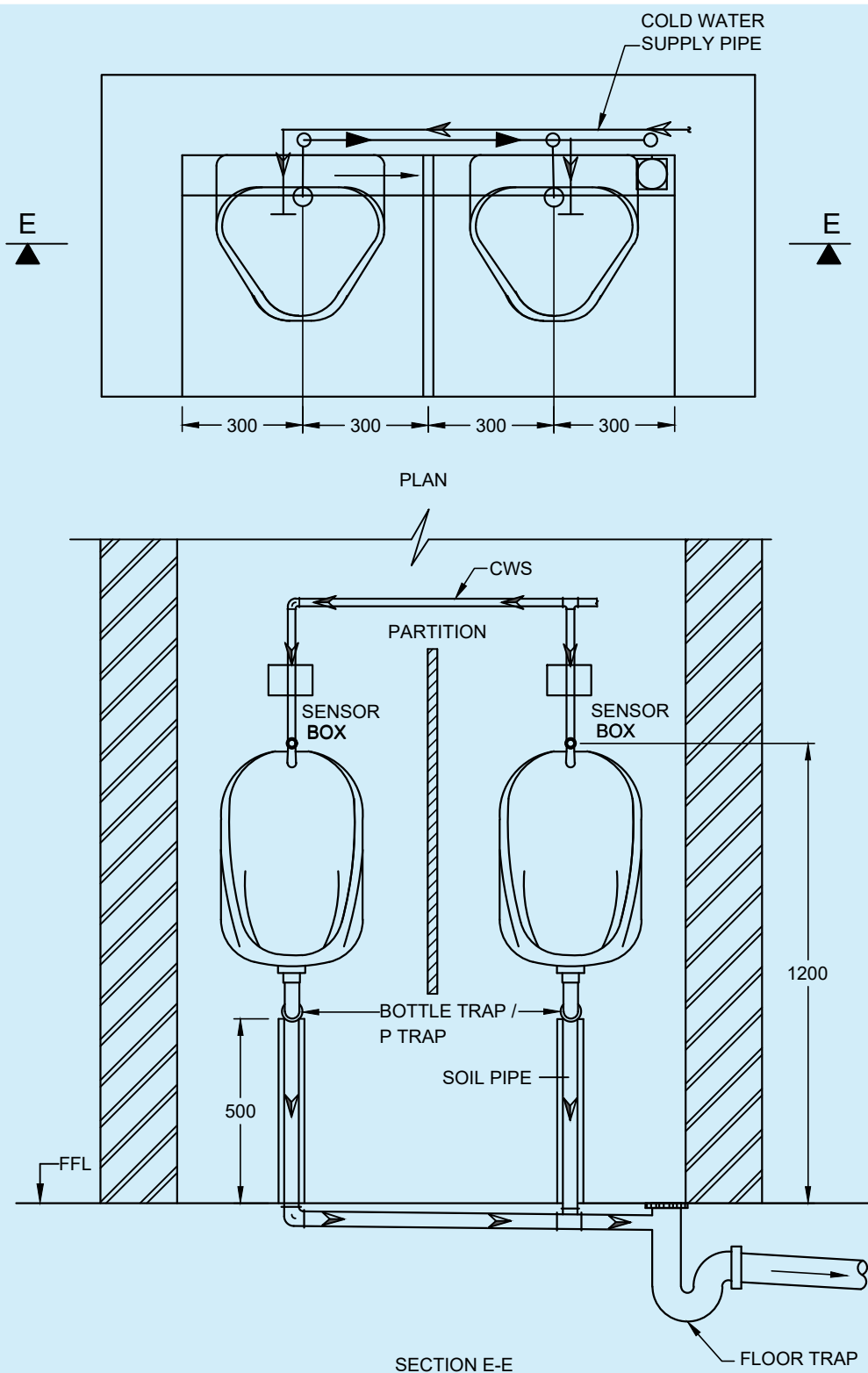
D-8 ARRANGEMENT FOR ORISSA PAN INDIAN WATER CLOSET — See [Fig. 27](#)



All dimensions in millimetres.

FIG. 27 ORISSA PAN INDIAN WATER CLOSET

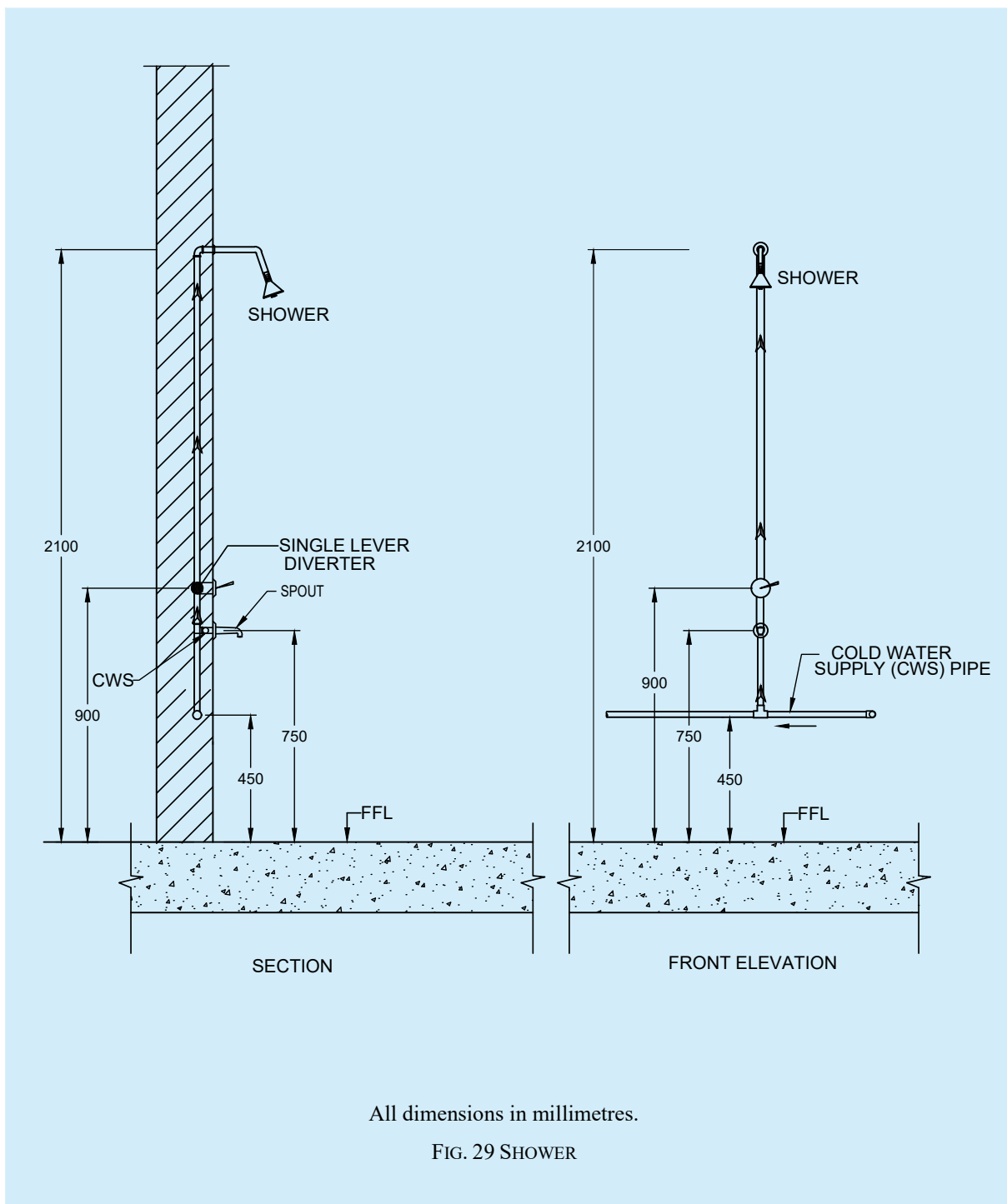
D-9 ARRANGEMENT FOR URINAL — See [Fig. 28](#)



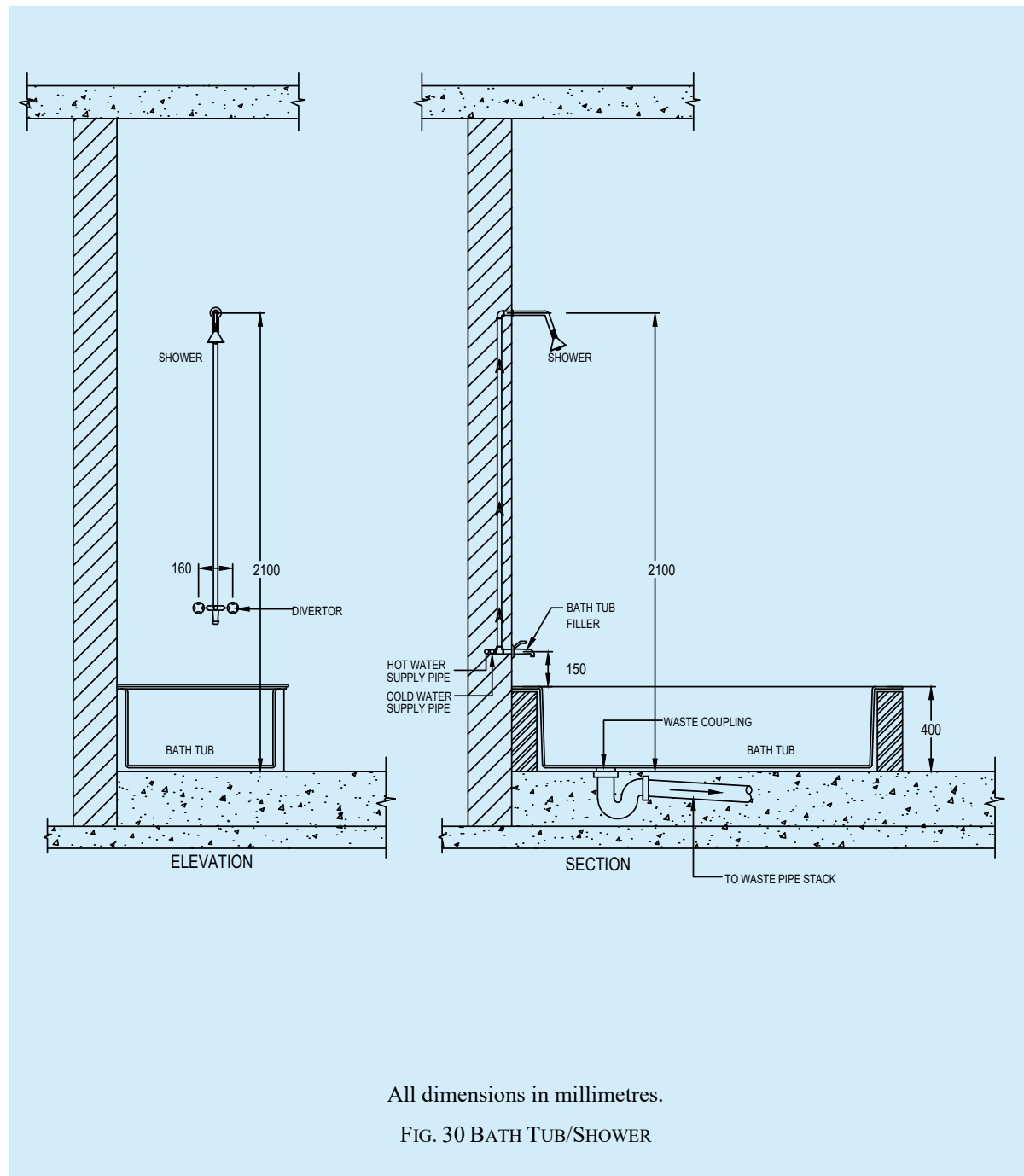
All dimensions in millimetres.

FIG. 28 URINAL

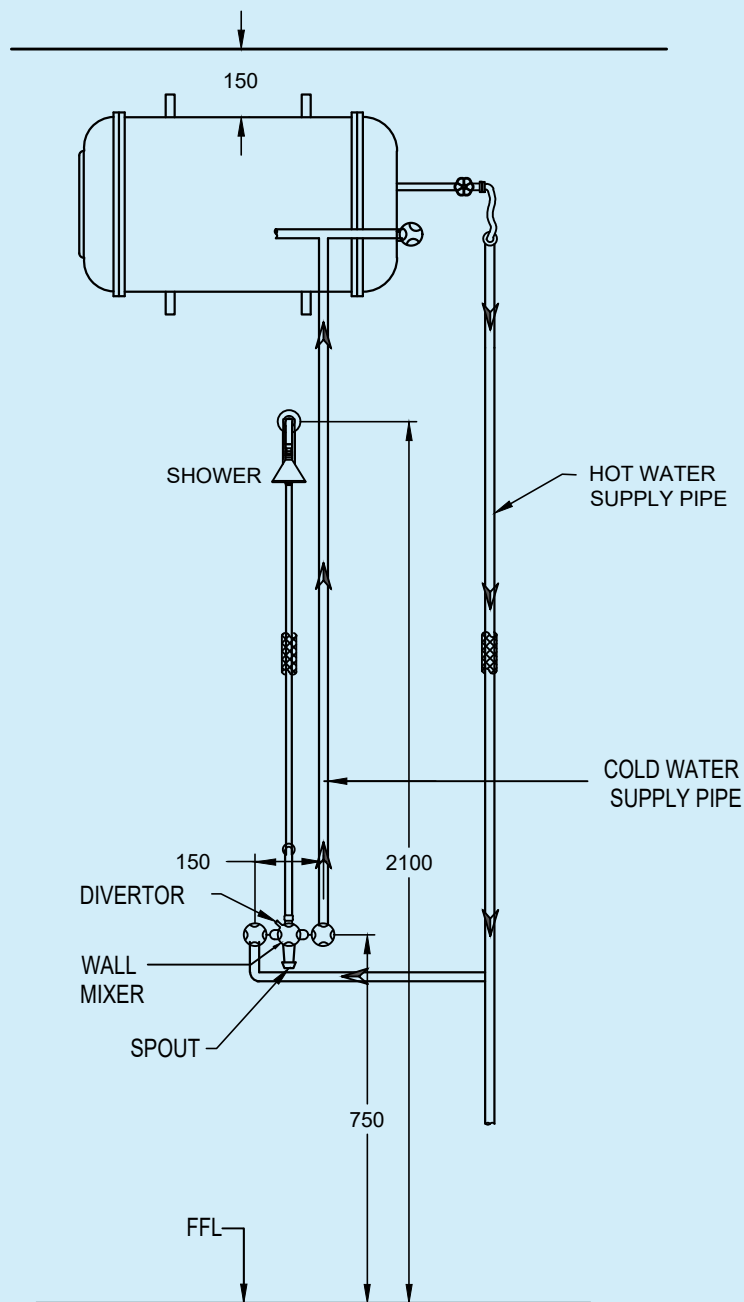
D-10 ARRANGEMENT FOR SHOWER — See [Fig. 29](#)



D-11 ARRANGEMENT FOR BATH TUB/SHOWER — See [Fig. 30](#)



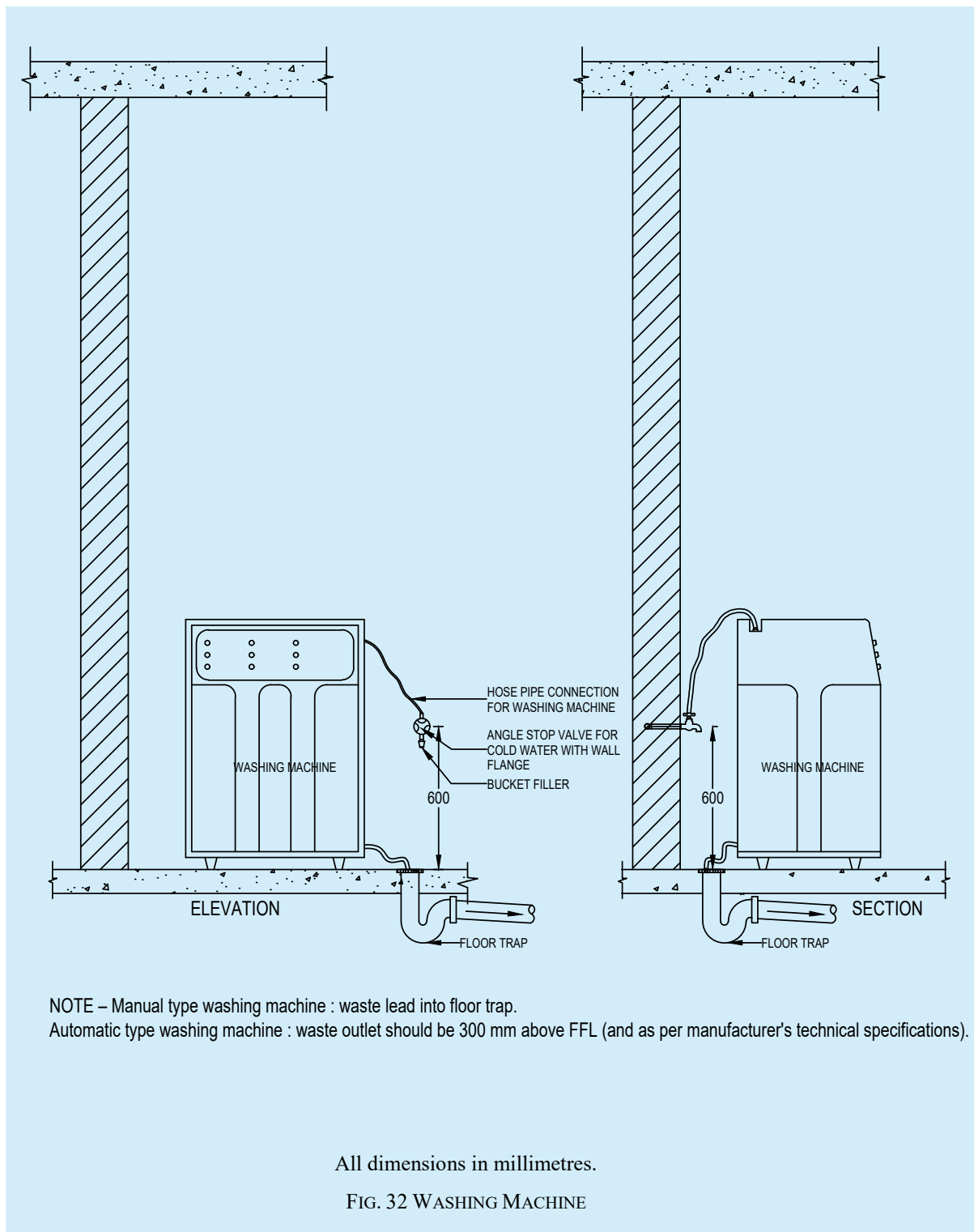
D-12 ARRANGEMENT FOR SHOWER AND ELECTRIC WATER HEATER (HORIZONTAL MOUNTED TYPE) — See [Fig. 31](#)



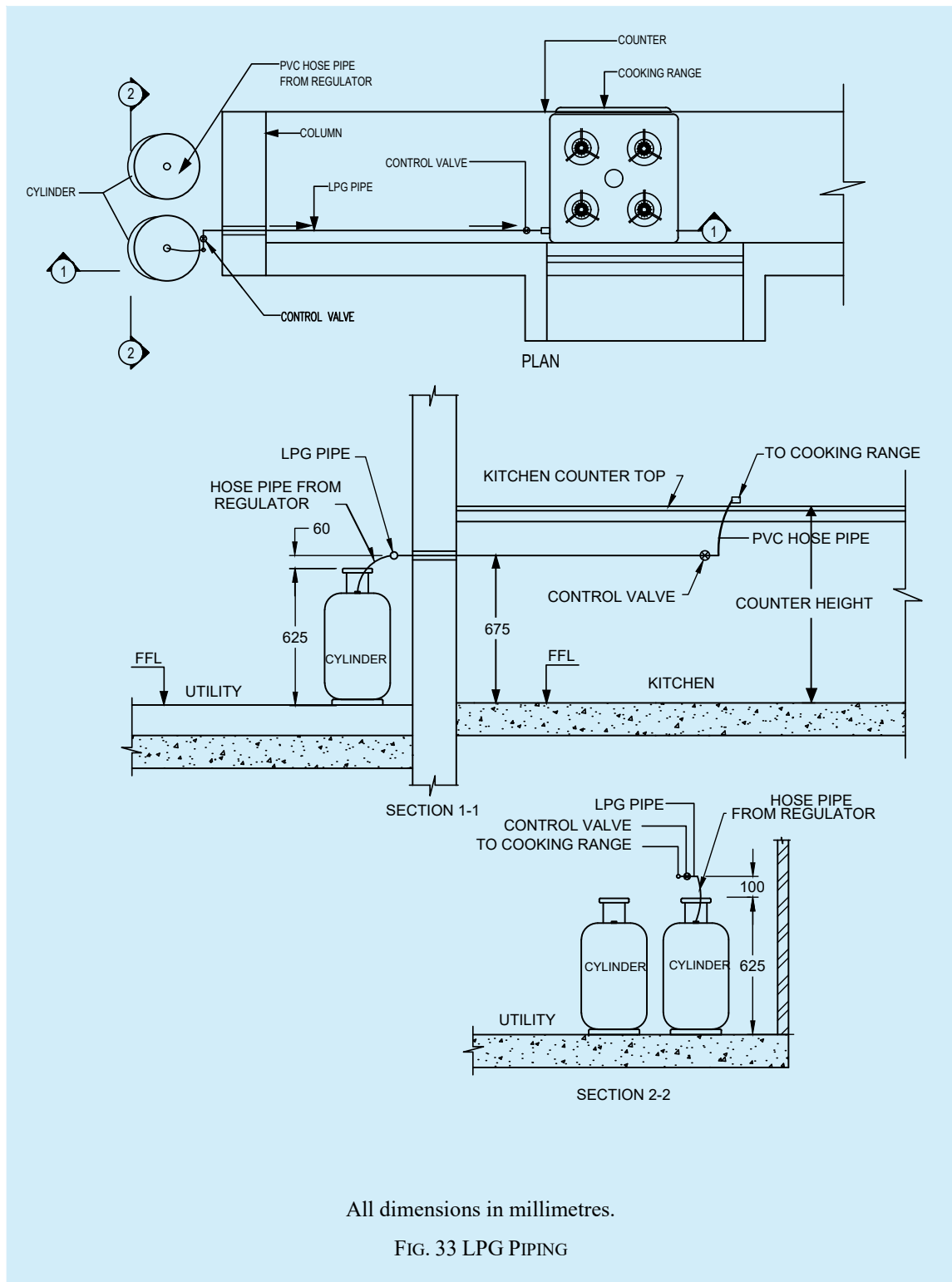
All dimensions in millimetres.

FIG. 31 SHOWER AND ELECTRIC WATER HEATER (HORIZONTAL MOUNTED TYPE)

D-13 ARRANGEMENT FOR WASHING MACHINE — See [Fig. 32](#)



D-14 ARRANGEMENT FOR LPG PIPING — See [Fig. 33](#)



LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfillment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used as a guide in conformance with the requirements of the referred clauses in this NBCS.

<i>IS No.</i>	<i>Title</i>
(1) IS 10446 : 1983	Glossary of terms relating to water supply and sanitation
(2) IS 4963 : 2025	Accessibility in built environment for older adults and persons with disabilities — Requirements (<i>second revision</i>)
(3) IS 11208 : 1985	Guidelines for registration of plumbers
(4) IS 774 : 2021	Ceramic (vitreous china) flushing cisterns for water closets and urinals — Specification (<i>sixth revision</i>)
IS 2548	Plastic seats and covers for water-closets — Specification:
(Part 1) : 1996	Thermoset seats and covers (<i>fifth revision</i>)
(Part 2) : 1996	Thermo plastic seats and covers (<i>fifth revision</i>)
IS 2556	Vitreous china sanitary appliances — Specification:
(Part 1) : 2021	General requirements (<i>fourth revision</i>)
(Part 2) : 2024	Specific requirements of washdown water closets (<i>sixth revision</i>)
(Part 3) : 2024	Specific requirements of squatting pans (<i>sixth revision</i>)
(Part 4) : 2004	Specific requirements of wash basins (<i>fourth revision</i>)
(Part 5) : 1994	Specific requirements of laboratory sinks (<i>third revision</i>)
(Part 6) : 2021	Specific requirements of urinals and partition plates (<i>fifth revision</i>)
(Part 7) : 1995	Specific requirements of accessories for sanitary appliances (<i>third revision</i>)
(Part 8) : 2021	Specific requirements of close-coupled and one-piece pedestal washdown and syphonic water closets (<i>sixth revision</i>)
(Part 9) : 2004	Specific requirements of pedestal type bidets (<i>fifth revision</i>)
(Part 14) : 2024	Specific requirements of integrated squatting pans (<i>second revision</i>)
(Part 15) : 2024	Specific requirements of universal water closets (<i>third revision</i>)
(Part 16) : 2024	Specific requirements of washdown wall mounted water closets (<i>first revision</i>)
(Part 17) : 2001	Specific requirements for wall mounted bidets
IS 6411 : 1985	Specification for gel-coated glass fibre reinforced polyester resin bath tubs (<i>first revision</i>)
IS 7231 : 2021	Plastic flushing cisterns for water closets and urinals — Specification (<i>third revision</i>)
IS 11246 : 1992	Glass fibre reinforced polyester resins (GRP) squatting pans — Specification (<i>first revision</i>)
IS 13983 : 1994	Stainless steel sinks for domestic purposes — Specification
(5) IS 651 : 2007	Glazed stoneware pipes and fittings — Specification (<i>sixth revision</i>)
IS 3006 : 1979	Specification for chemically resistant glazed stoneware pipes and fittings (<i>first revision</i>)

	<i>IS No.</i>	<i>Title</i>
(6)	IS 458 : 2021	Precast concrete pipes (with and without reinforcement) — Specification (<i>fifth revision</i>)
	IS 784 : 2019	Prestressed concrete pipes (including specials) — Specification (<i>third revision</i>)
	IS 1916 : 2018	Steel cylinder pipes with concrete lining and coating (<i>second revision</i>) — Specification
	IS 4350 : 1967	Specification for concrete porous pipes for under drainage
	IS 7319 : 1974	Specification for perforated concrete pipes
(7)	IS 1536 : 2023	Centrifugally cast (spun) iron pressure pipes for water, gas and sewage — Specification (<i>fifth revision</i>)
	IS 1537 : 1976	Specification for vertically cast iron pressure pipes for water, gas and sewage (<i>first revision</i>)
	IS 1538 : 1993	Cast iron fittings for pressure pipes for water, gas and sewage — Specification (<i>third revision</i>)
	IS 3989 : 2024	Centrifugally cast (spun) spigot and socket pipes, fittings and accessories — Specification (<i>fourth revision</i>)
	IS 7181 : 1986	Specification for horizontally cast iron double flanged pipes for water, gas and sewage (<i>first revision</i>)
(8)	IS 1592 : 2003	Asbestos cement pressure pipes and joints — Specification (<i>fourth revision</i>)
	IS 1626	Asbestos cement building pipes and pipe fittings, gutters and gutter fittings, and roofing fittings — Specification:
	(Part 1) : 1994	Pipes and pipe fittings (<i>second revision</i>)
	(Part 2) : 1994	Gutters and gutter fittings (<i>second revision</i>)
	(Part 3) : 1994	Roofing fittings (<i>second revision</i>)
	IS 6908 : 1991	Asbestos cement pipes and fittings for sewerage and drainage — Specification (<i>first revision</i>)
(9)	IS 13592 : 2013	Unplasticized polyvinyl chloride (PVC-U) pipes for soil and waste discharge system for inside and outside buildings including ventilation and rain water System — Specification (<i>first revision</i>)
	IS 14333 : 2022	Polyethylene pipes for sewerage and industrial chemicals and effluent — Specification (<i>first revision</i>)
	IS 14735 : 1999	Unplasticized polyvinyl chloride (UPVC) injection moulded fittings for soil and waste discharge system for inside and outside buildings including ventilation and rain water system — Specification
	IS 15328 : 2003	Unplasticized non-pressure polyvinyl chloride (PVC-U) pipes for use in underground drainage and sewerage systems — Specification
(10)	IS 2470	Code of practice for installation of septic tanks:
	(Part 1) : 1985	Design criteria and construction (<i>second revision</i>)
	(Part 2) : 1985	Secondary treatment and disposal of septic tank effluent (<i>second revision</i>)
(11)	IS 1536 : 2023	Centrifugally cast (spun) iron pressure pipes for water, gas and sewage — (<i>fifth revision</i>)
(12)	IS 5329 : 1983	Code of practice for sanitary pipe work above ground for buildings (<i>first revision</i>)
(13)	SP 35 : 1987	Handbook on water supply and drainage with special emphasis on plumbing

	<i>IS No.</i>	<i>Title</i>
(14)	IS 2212 : 1991	Brick works — Code of practice (<i>first revision</i>)
(15)	IS 5455 : 1969	Specification for cast iron steps for manholes
(16)	IS 1726 : 1991	Cast iron manhole covers and frames — Specification (<i>third revision</i>)
	IS 12592 : 2002	Precast concrete manhole covers and frames — Specification (<i>first revision</i>)
(17)	IS 4111 (Part 1) : 1986	Code of practice for ancillary structures in sewerage system : Part 1 Manholes (<i>first revision</i>)
(18)	IS 15792 : 2008	Artificial recharge to ground water — Guidelines
(19)	IS 14961 : 2024	Surface water management in hilly areas (including rainwater harvesting) — Guidelines (<i>first revision</i>)
	IS 15797 : 2008	Roof top rainwater harvesting — Guidelines
(20)	IS 783 : 1985	Code of practice for laying of concrete pipes (<i>first revision</i>)
	IS 1742 : 1983	Code of practice for building drainage (<i>second revision</i>)
	IS 3114 : 1994	Code of practice for laying of cast iron pipes (<i>second revision</i>)
	IS 4127 : 2023	Laying of glazed stoneware pipes — Code of practice (<i>second revision</i>)
	IS 6530 : 1972	Code of practice for laying of asbestos cement pressure pipes
(21)	IS 2527 : 1984	Code of practice for fixing rainwater gutters and downpipes for roof drainage (<i>first revision</i>)
	IS 1729 : 2023	Sand cast iron spigot and socket pipes, fittings and accessories — Specification (<i>third revision</i>)
	IS 1626 (Part 1) : 1994	Asbestos cement building pipes and pipe fittings, gutters and gutter fittings and roofing fittings: Part 1 Pipe and pipe fittings (<i>second revision</i>)
	IS 1239 (Part 1) : 2004	Steel tubes, tubulars and other wrought steel fittings: Part 1 Steel tubes — Specification (<i>sixth revision</i>)
	IS 13592 : 2013	Unplasticized polyvinyl chloride (PVC-U) pipes for soil and waste discharge systems inside buildings including ventilation and rainwater system — Specification (<i>first revision</i>)
(22)	IS 2064 : 1993	Selection, installation and maintenance of sanitary appliances — Code of practice (<i>second revision</i>)
(23)	IS 18797 : 2024	Packaged sewage treatment plant — Specification
(24)	IS 18150 : 2023	Non-sewered on-site sanitation system — Design, fabrication and installation of biodigesters for human waste decomposition — Code of practice
(25)	IS 9872 : 1981	Specification for precast concrete septic tanks
	IS 18666 : 2024	Rotationally moulded polyethylene septic tanks — Specification

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART E PLUMBING SERVICES (INCLUDING SOLID WASTE MANAGEMENT)

SECTION 3 SOLID WASTE MANAGEMENT

BUREAU OF INDIAN STANDARDS

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5 ASSESSMENT OF PER CAPITA WASTE QUANTITY	... 13
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National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part E/Section 3) covers the solid waste management systems, assessment of per solid waste quantities and treatment of solid waste within the building, building complexes and their built environments.

In the first version of the Code formulated in 1970, as part of the National Building Code of India, three separate sections of this Part on 'Plumbing Services', were brought out, namely, Section 1 Water Supply, Section 2 Drainage and Sanitation, and Section 3 Gas Supply. These sections were subsequently revised in 1983 and in 2005.

In the second revision in 2005, the Part 9 was renamed as 'Plumbing Services (Including solid waste management)' and provisions on solid waste management were included for the first time under Section 1 which contained in it, Water supply, drainage and sanitation including solid waste management. Gas Supply was covered in Section 2 of Part 9 of the Code.

In the third revision in 2016, to comprehensively address the various and distinct features related to the plumbing aspects, this Part 9 was rearranged as follows:

Section 1 Water supply

Section 2 Drainage and sanitation

Section 3 Solid waste management

Section 4 Gas supply

The significant changes included in the 2016 version of NBC included: Elaborative provisions on solid waste management systems; inclusions of the provisions for persons with disabilities towards operation of refuse chutes; recommendation of electrically operated dumb waiters for 5 storied (and above); inclusion of assessment of per capita waste quantity; referencing the latest statutory rules and regulations; inclusion of detailed provisions on treatment methods; inclusion of annexes namely [Annex A](#) on municipal solid waste generation, [Annex B](#) on treatment of food wastes; and review of terminology.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

The significant changes incorporated in this revision include:

- a) References to the latest rules and regulations have been made;
- b) Terminology clause has been reviewed and updated;
- c) Circularity aspects of solid waste management has been included;
- d) Domestic waste now covers all the remaining types of wastes generated;
- e) Caution with respect to plastic waste by avoiding their down cycling is included;
- f) Primary segregation and secondary segregation have been detailed in sorting at source;
- g) Additional options towards efficient use of refuse chutes are included;
- h) Treatment options namely aerobic processes, vermi-composting have been updated taking care of the best practices in these activities;
- j) Provisions for the colour coding of waste bins have been incorporated to ensure proper segregation of solid waste;
- k) Provisions have been incorporated for the treatment of biodegradable waste by household and commercial facilities; and
- m) References to all the concerned Indian Standards have been updated.

Reference may also be made to the Municipal Solid Waste Management Manual — Part I — An Overview and Part II — The Manual, 2016, CPHEEO, Ministry of Urban Development, Government of India.

All Indian Standards, cross-referred to in the main text of this Section, are subject to revision. The parties to agreement based on this Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Section is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Section.

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Important Explanatory Note for Users of this NBCS

In any Part/Section of this NBCS, where reference is made to **‘good practice’** in relation to **design, constructional procedures or other related information**, and where reference is made to **‘accepted standard’** in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

a) Good practice [E-3(1)] refers to the Indian Standard(s) given at serial number (1) of the list of standards given at the end of this Part E/Section 3, that is, IS 9569 : 2022 ‘Solid waste management — Glossary (*first revision*)’.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART E PLUMBING SERVICES (INCLUDING SOLID WASTE MANAGEMENT)

SECTION 3 SOLID WASTE MANAGEMENT

1 SCOPE

This NBCS (Part E/Section 3) covers the solid waste management systems, assessment of per capita solid waste quantities and treatment of solid waste within the building, building complexes and their built environments.

2 TERMINOLOGY

For the purpose of this Section, the following definitions shall apply, and for other terms those given in the accepted standard [E-3(1)] shall apply.

2.1 Anaerobic Digestion — A controlled process involving microbial decomposition of organic matter in the absence of oxygen.

2.2 Authorization — The consent given by the State Pollution Control Board or Pollution Control Committee, as the case may be, to the operator of a facility.

2.3 Biodegradable Substance — A substance that can be degraded by microorganisms into simpler stable compounds.

2.4 Biomethanation — A process which entails enzymatic decomposition of the organic matter by microbial action to produce methane rich biogas.

2.5 Collection — Lifting and removal of solid wastes from door step, collection points or any other designated position in segregated manner as per *Solid Waste Management Rules*/local bylaws.

2.6 Composting — A controlled process involving microbial decomposition of organic matter.

2.7 Construction and Demolition Waste — Wastes from building materials debris and rubble resulting from construction, re-modelling, repair and demolition operation.

2.8 Disposal — Final disposal of non-biodegradable and non-recyclable part of municipal solid wastes in accordance with the specified measures to prevent contamination of groundwater, surface water and ambient air quality.

2.9 Generator of Wastes — Persons or establishments generating municipal solid wastes.

2.10 Landfilling — Disposal of residual solid wastes on land in a facility designated and with protective measures against pollution of groundwater, surface water and air fugitive dust, wind-blown litter, bad odour, fire hazard, bird menace, pests or rodents, greenhouse gas emissions, slope instability and erosion.

2.11 Leachate — Liquid that seeps through solid wastes or other medium and has extracts of dissolved or suspended material from it.

2.12 Municipal Authority — Municipal Corporation, Municipality, Nagar Palika, Nagar Nigam, Nagar Panchayat, Municipal Council including notified area committee (NAC) or any other local body constituted under the relevant statutes and, where the management and handling of municipal solid waste is entrusted to such agency.

2.13 Municipal Solid Waste — Commercial and residential wastes generated in municipal or notified areas in either solid or semi-solid form, excluding industrial hazardous wastes and construction and demolition waste but including treated bio- medical wastes.

2.14 Operator of a Facility — A person who owns or operates a facility for collection, segregation, storage, transportation, processing and disposal of municipal solid wastes and also includes any other agency appointed as such by the municipal authority for the management and handling of municipal solid wastes in the respective areas.

2.15 Pelletization — A process whereby pellet are prepared which are small cubes or cylindrical pieces made out of solid wastes and includes fuel pellets which are also referred as refuse derived fuel.

2.16 Processing — The process by which solid wastes are transformed into new or recycled products.

2.17 Recycling — The process of transforming segregated solid wastes into raw materials for producing new products, which may or may not be similar to the original products.

2.18 Segregation — To separate the municipal solid wastes into the groups of bio-degradable, recyclables, hazardous and inert wastes.

2.19 Storage — The temporary containment of municipal solid wastes at a designated place in a manner so as to prevent littering, attraction to vectors, stray animals and excessive foul odour.

2.20 Transportation — Conveyance of municipal solid wastes from place to place hygienically and covered through specially designed transport system so as to prevent foul odour, littering, unsightly conditions and accessibility to vectors.

2.21 Vermi-composting — A process of using earthworms for conversion of biodegradable wastes into compost.

3 GENERAL

3.1 Solid waste is generated from various human activities and is normally disposed as useless and unwanted by the primary user because of end-of-life condition.

Broadly, depending upon the type of building and built environment, it includes solid or semi-solid domestic waste, such as sanitary waste, commercial waste, office waste, catering and market waste and other non-residential wastes, street sweepings, silt removed or collected from the surface drains, horticulture waste and construction and demolition waste. Further, there may be industrial hazardous waste, bio-medical waste and e-waste.

The primary objective of solid waste management is to collect, treat and dispose of solid wastes in an environmentally and socially satisfactory manner using the most economical means available. Ideally, solid waste should not contain faecal matter or urine but despite all precautions, some pathogens and chemical residues inevitably may be present in the waste. Uncontrolled or poorly managed intermediate decomposition of organic components of solid waste can contaminate air, water and soil resources. Implementation of integrated solid waste management practices benefits substantially, both the public health and the quality of the environment. It is also an integral part of an efficient solid waste management (SWM) system to create circularity by ensuring the maximum recycling and recovery of solid waste generated and avoiding/reducing of landfillable waste and dumping.

3.2 Based on source of generation, solid waste can be broadly categorized as follows:

- a) *Domestic waste* — It is generated in residential units, commercial and educational entities; and consists of food wastes, paper, cardboard, plastics, textiles, leather, yard wastes, wood, glass, metals, ashes, special wastes (for example, bulky items, consumer electronics, white goods, batteries, oil and tyres), and household hazardous wastes including domestic sanitary, expired/discarded household usage chemicals, paints, automobiles batteries, household cleaners, swimming pool chemicals, and domestic e-waste;
- b) *Industrial waste* — It is generated in manufacturing units, fabrication plants, construction sites, power and chemical plants; and consists of packaging, hazardous wastes, ashes, special wastes, wood, steel, concrete, dirt, etc;
- c) *Institutional and commercial waste* — It is generated in schools, hospitals, prisons, government centers, stores, hotels, restaurants, markets, office buildings, etc; and consists of domestic wastes, paper, cardboard, plastics, wood, glass, metals, special wastes and hazardous wastes; and

- d) *Horticulture waste* — It is generated in landscaping, parks, orchards, dairies, and consists of street sweepings; landscape and tree trimmings; etc.

3.3 Based on suitability for handling and disposal, solid waste can also be categorized as follows:

- a) *Garbage* — Garbage is the term applied to animal and vegetable waste resulting from the handling, storage, sale, cooking and serving food. Such wastes contain putrescible organic matter, which produces strong odours and therefore attracts rats, flies and other vermin. It requires immediate attention in its storage, handling and disposal;
- b) *Municipal waste* — Municipal waste includes waste resulting from municipal activities and services such as street wastes, dead animals, market wastes and abandoned vehicles. However, the term is commonly applied in a wider sense to incorporate domestic wastes and commercial wastes;
- c) *Hazardous wastes* — Hazardous wastes may be defined as wastes of industrial, institutional or consumer origin which because of their physical, chemical or biological characteristics are potentially dangerous to human and the environment. It includes any waste that exhibits one of the hazardous characteristics, such as ignitability, corrosivity, reactivity, or toxicity, whether alone or when in contact with other wastes or substances. In some cases although the active agents may be liquid or gaseous, they are classified as solid waste because they are confined in solid containers. Typical examples are solvents, paints and pesticides whose spent containers are frequently mixed with municipal wastes. Good waste management should ensure that hazardous wastes are stored, collected, transported and disposed of separately, preferably after suitable treatment to render them innocuous;

A sub-category of household hazardous waste, is post-consumer utilization waste which qualifies as hazardous waste when discarded. It includes household chemicals and other consumer products used in home care, personal care, automotive care, pest control, etc. These products exhibit characteristics of reactivity, ignitability, corrosivity, toxicity, or persistence;

- d) *Bio-medical waste* — Bio-medical waste is any waste, which is generated during the diagnosis, treatment or immunization of human beings or animals or in research activities pertaining thereto or in the production or testing of biologicals. This waste is highly infectious and can be a serious threat to human health if not managed in a scientific and discriminate manner. It may include wastes like sharps, soiled waste, disposables, anatomical waste, cultures, discarded medicines, chemical wastes, etc. It has been estimated that hospitals generate total waste at a rate of 1.5 kg per bed per day, out of which approximately 25 percent is bio-medical waste. Generally, these wastes are handed over to common bio-medical treatment and disposal facilities for suitable treatment and final disposal.

Biological means any preparation made from organisms or microorganisms or product of metabolism and biochemical reactions intended for use in the diagnosis, immunization or the treatment of human beings or animals or in research activities pertaining thereto.

Bio-medical waste treatment facility means any facility wherein treatment, disposal of bio-medical waste or processes incidental to such treatment and disposal is carried out, and includes common treatment facilities. The common bio-medical treatment facilities are set up based on the need for ensuring environmentally sound management of bio-medical waste keeping in view the techno-economic feasibility and viable operation of the facility with minimal impacts on human health and environment.

All healthcare establishments including hospitals, research facilities and laboratories generate solid waste in course of performing their intended functions. 75 percent to 90 percent of such solid waste is non-risk healthcare waste, comparable to domestic waste. It requires minimal treatment and is safely handled by housekeeping staff. The remaining 10 percent to 25 percent of healthcare waste is designated as bio-hazardous waste and is designated bio-medical waste. It carries a variety of health risks in its handling and disposal. It is handled by trained staff and many safety devices and apparels are deployed. In addition, bio-medical waste includes the waste originating from minor or scattered sources, such as that produced in the course of healthcare undertaken in the home (dialysis, insulin injections, etc);

- e) *Plastics* — Plastics, due to their non-biodegradable nature, versatility in use and impact on environment, can be grouped under a different category of solid waste. Significant quantity of the plastic waste

generated in our country is recycled and used in the manufacture of various plastic products. Careless disposal of plastic in various forms chokes drains, blocks the porosity of the soil, causes problems for groundwater recharge, disturbs the soil microbe activity, contaminates foodstuff and when ingested by animals, it can kill them; apart from impacting the marine life when they end up in river/water bodies. Down cycling of plastics degrades their quality in each cycle and hence the same should be least preferred; and

- f) *E-waste* — It refers to discarded electrical and electronic equipment and their components that are no longer in use or functional. It encompasses a wide range of household appliances, office equipment, and related devices. E-waste contains hazardous substances such as lead, mercury, cadmium and beryllium, which can contaminate soil, water and air, thereby adversely affecting ecosystems and human health. Proper e-waste management through formal recycling processes enables recovery of valuable materials, reduces environmental pollution, and minimizes health hazards.

3.4 Efficient collection and disposal of domestic garbage from a building or activity area is of significant importance to public health and environmental sanitation and, therefore, an essential part of the built environment. Notwithstanding the provisions given herein, the solid waste management shall have to comply with relevant statutory Rules and Regulations in force from time-to-time. The provisions of the following shall, as amended from time to time, govern the procedures for handling, treatment, etc, of solid wastes as applicable to the concerned building occupancy:

- a) *The Environment (Protection) Act*, 1986;
- b) *The Solid Waste Management Rules*, 2016;
- c) *The Bio-Medical Waste Management Rules*, 2016;
- d) *The E-Waste (Management) Rules*, 2022;
- e) *The Battery Waste Management Rules*, 2022;
- f) *The Hazardous and Other Wastes (Management and Transboundary Movement) Rules*, 2016;
- g) *Plastic Waste Management (Amendment) Rules*, 2024; and
- h) *Environment (Construction and Demolition) Waste Management Rules*, 2025.

NOTE — Radioactive wastes are covered under the provisions of *The Atomic Energy Act*, 1962 and rules made thereunder and the enforcement agency for these is Atomic Energy Regulatory Commission (AERC).

3.5 The provisions relating to solid waste management given in [4](#) are applicable to municipal solid waste and specifically exclude the hazardous chemical wastes, bio- medical waste and radioactive waste.

4 SOLID WASTE MANAGEMENT SYSTEMS

4.1 In designing a system dealing with collection of domestic garbage for a built premises/community/environment, the aim shall be to provide speedy and efficient conveyance as an essential objective for design of the system. The various available systems may be employed in accordance with [4.2](#) to [4.4](#), which may be adopted individually or in combination as appropriate in specific situations.

4.1.1 The basic stages of the municipal solid waste management system include the following:

- a) Waste sorting at source, collection and transportation;
- b) Resource recovery through centralized sorting and recycling;
- c) Resource recovery through waste processing, that is, recovery of energy; and
- d) Waste transformation without recovery.

4.1.2 The functional elements of municipal solid waste management system include the following:

- a) *Waste generation* — Activity in which the waste gets generated as material no longer having any value to the primary consumer;
- b) *Waste handling, sorting, storage and processing at site* — Waste handling and sorting involves the activities associated with management of waste until they are placed in storage container for collection;

- c) *Collection* — Gathering of solid wastes in segregated form including transportation; and
- d) *Transfer and transport* — Transfer of waste from smaller collection vehicle to larger transport equipment and subsequent transport of wastes over long distances to a designated processing or disposal site.

4.2 Sorting at Source

Sorting shall be planned based on the existing markets for recyclable materials. It is preferable to have coloured-bin system based on the usage of the facility and the types of the waste likely to be generated in the facility. Colour code for bins/bags shall be as follows:

- a) Green — for bio-degradable waste;
- b) White or Blue — for dry waste (excluding biomedical waste/hazardous waste); and
- c) Red — for bio-medical waste and rejects.

All bins shall be covered with lid. Reference shall be made to good practice [E-3(2)] for detailed provisions on the colour coding of bins.

Common sorting methods involve sorting into;

- a) dry recyclable materials, for example, glass, paper, plastics, metals, etc;
- b) bio-waste and garden waste;
- c) bulky waste;
- d) hazardous material in household waste;
- e) construction and demolition waste; and
- f) mingled MSW (mixed waste).

In case of sorting not done at source, it is required to plan centralized sorting. The types of sorting that may be adopted, are:

- a) Hand sorting from a raised picking belt; and
- b) Mechanized sorting facilities using magnetic and electric field separation, density separation, etc.

Sorting at source (also known as primary segregation) normally recovers most of the recyclable materials for reuse. However, a small fraction of such materials may escape the sorting process. Sorting may therefore be also undertaken just prior to waste processing (secondary segregation), waste transformation or landfilling to recover such recyclable materials.

Wherever manual sorting is adopted, care shall be taken to ensure that sorters are protected from all disease pathways and work in hygienic and safe conditions using necessary safety gear including aprons, protective mask, gloves and boots/shoes. The movement of waste shall be planned by suitable mechanism and building shall be planned for easy movement of waste by means of garbage/material lift, ramps, etc, meeting the good engineering practices and minimum handling of waste as well as no contact with ground during the whole process or littering.

4.3 Refuse Chute System

4.3.1 General

Refuse chute system is a convenient and safe mode of transportation and collection of domestic solid wastes from buildings exceeding 5 storeys from floors at different heights. The refuse is received from the successive floor through the inlets located on the vertical system of pipes that convey refuse through it and discharge it into the collecting chamber, located at ground level, from where the refuse is cleared at suitable intervals. This system has got four functionally important components, namely, the chutes, the inlet hopper, the collection chamber and the cleaning and disinfecting arrangements.

The chute may be carried through service shafts meant for carrying drainage pipes. However, the location shall be mostly determined by the position of the inlet hopper and the collecting chamber that is most convenient for the user. It should also be considered to locate the chute away from living rooms in order to avoid noise and smell

nuisance. The internal diameter of the chute shall be at least 300 mm.

In individual chute system, the inlet hopper shall be located in the passage near the kitchen and in the common chute system towards the end of the common passage. Natural ventilation should be adequate to prevent any possible odour nuisance. There should be adequate lighting at this location. For ground floor (floor 1), the inlet hoppers may be placed at a higher level, but should be easily accessible. The access to the refuse chute shall be provided from well ventilated and well illuminated common corridor or lobby and preferably it should not be located opposite or adjacent to entry of individual flats or lift.

4.3.2 Opening for Feeding of Refuse Chute

Opening, with top or bottom hinged shutters with appropriate lockable latch, shall be provided for convenient accessing of the refuse chute by users. It facilitates total building garbage collection at one single point and provides separate collection for dry and wet garbage. The opening of the refuse chute may have separate receptacles for both wet and dry waste. Mixed waste disposal may not be allowed. For wet waste, there should be a provision for collection of leachate. Chute system shall have the following features:

- a) Sanitation system for internal cleaning of chutes, with sprinklers at every floor intake level, with disinfectant cleaning, which may preferably be automatic in operation;
- b) Exhaust system with 25 to 35 air changes per hour;
- c) Discharge end fire resistant door;
- d) Chute may be of masonry, R.C.C pipe, asbestos cement pipe or suitable non- corrosive material, preferably stainless steel. It should be rigid with smooth internal finish, high ductility and alkali/acid resistant properties {see also good practice [E-3(3)]};
- e) For fire rating, reference should be made to relevant clauses of Part F 'Fire and Life Safety' of this NBCS;
- f) Fire sprinklers shall be provided inside chute;
- g) Vent shall be installed at roof level to remove foul smell and gases generated;
- h) Chute should discharge to trolley directly;
- j) Necessary automation of the system may be planned to avoid malfunction by manual operation; and
- k) All care shall be taken for ensuring accessibility to the chute for persons with disabilities. The opening of the chute shall not be higher than 700 mm and there shall be a clear space of minimum 1 500 mm × 1 500 mm in front of the chute opening to gain easy access even for persons using wheelchairs. There may be high colour contrast provided around the chute opening to make it easy to identify it. The flooring in front of the chute opening may be texturally highly different from the rest of the floor.

4.3.3 Refuse Collection Chamber

The collection chamber may be located in ground floor or basement level, provided appropriate arrangement is made for drainage of the collection pit by gravity flow to ensure its dryness, an appropriate ramp access is provided for convenient removal of garbage from the collection pit, and satisfactory ventilation is provided for escape of gas and odour.

The floor of the chamber shall be provided with drainage through a 100 mm diameter trap and screen to prevent any solid matters flowing into the drain and the drain shall be connected to the sewer line. The floor shall be finished with smooth hard surface for convenient cleaning.

The height of the collection chamber and vertical clearance under the bottom level of garbage chute shall be such that the garbage trolley can be conveniently placed.

The collection chamber shall be provided with appropriate shutter to prevent access of scavenging animals like cattle, dogs, cats and rats.

Building owners and associations should be encouraged about proper usage of this system which otherwise can be clogged and malfunction. They may also be encouraged to have a single agent/person for door to door collection having effective segregation at source and ensuring hygienic conditions and authorizing only such

agent/person operate the chute. This system is quite effective in industrial and commercial buildings which are professionally controlled all the time and also typically having one type of waste or largely dry waste only.

4.3.4 Refuse Collection Room

The refuse collection room should be planned in ground or basement level with appropriate ventilation and proper drainage. The room should have access for vehicle or trolley transfer of garbage.

4.3.5 See also accepted standards [E-3(4)].

4.4 Dumb-Waiter or Service Lift

In high rise buildings with more than 5 storeys, electrically operated dumb-waiters may be used for carrying domestic garbage in packets or closed containers. For handling of garbage by dumb-waiters in a building, a garbage chamber shall have to be provided either at ground floor or basement level and the provisions of garbage collection chamber for chute as given in [4.3](#) shall apply.

5 ASSESSMENT OF PER CAPITA WASTE QUANTITY

5.1 For purposes of this Section, the following municipal refuse generation rates are recommended:

- | | | |
|-------------------------|---|---|
| a) Residential refuse | : | 0.3 kg/capita/day to 0.6 kg/capita/day |
| b) Commercial refuse | : | 0.2 kg/capita/day to 0.6 kg/capita/day |
| c) Street sweepings | : | 0.34 kg/capita/day to 0.7 kg/capita/day |
| d) Institutional refuse | : | 0.25 kg/capita/day to 0.6 kg/capita/day |

Out of the total solid waste generated, 40 percent may be considered inorganic waste, which includes a proportion of inert waste, and 60 percent as organic waste. The knowledge of chemical characteristics of waste is important for selecting and designing the waste processing and disposal facilities.

5.2 These generation rates are subject to considerable site-specific factors, such as degree of urbanization, economic level, geographical terrain and other site-specific factors and are required to be supported by field data studies. The waste contains a high percentage of ash and fine earth. The calorific value of Indian solid waste generally varies between 800 kcal/kg and 1 000 kcal/kg and the density varies between 300 kg/m³ and 500 kg/m³.

5.3 Other than municipal solid waste, the following types of waste may also be generated in urban centres:

- Industrial waste* — Hazardous and non-hazardous waste from industrial areas within municipal limits;
- Bio-medical waste* — Waste from hospitals, slaughter houses, etc;
- Thermal power plant waste* — Fly ash from coal-based electricity generating plant within municipal limits;
- Effluent treatment plant waste* — Sludge from sewage treatment plants and industrial effluent treatment plants;
- Construction and demolition waste* — Generated from demolition, remodelling of existing buildings and new construction; and
- Other wastes* — Special wastes from non-conforming areas or special units.

All waste streams shall be managed by their own waste management system.

6 TREATMENT

6.1 Garbage collected at one location may be treated for organic portion, and inorganic portion shall be handed over to vendors carrying out recycling. Organic waste generated in building/premises/community/environment may be treated and reused as manure and as per *The Solid Waste Management Rules, 2016*, preferably should be recycled at source site itself for bulk waste generators. The norms prescribed by the pollution board authorities shall be followed, keeping hygiene and safe operation for the working/operating people.

Biological or thermal treatment of waste can result in recovery of useful products such as compost or energy.

6.2 Biological Processes

Biological treatment involves using microorganisms to decompose the biodegradable components of waste. The following two types of processes may be typically used, the former being used more commonly:

- a) *Aerobic processes* — It may comprise composting, aerated static pile composting and in-vessel composting; vermi-culture, etc. In the aerobic process the utilizable product is compost and high microbial liquid drained out which can be recollected and used as liquid manure after dilution or as for sprinkling over new manure to quicken the process as bioenzyme;
- b) *Anaerobic processes* — It may comprise low-solids anaerobic digestion (wet process), high solids anaerobic digestion (dry process) and combined processes. In the anaerobic process the utilizable product is methane gas (for energy recovery).

6.2.1 Composting

Decomposition and stabilization of organic waste matter is a natural phenomenon. Composting is an organized method of producing compost manure by adopting this natural phenomenon.

- a) *Manual composting* — It is an anaerobic method conventionally carried out in pits. Initially the waste is anaerobically stabilized in pits where alternate layers of solid waste and night soil are laid. The pit is completely filled and a final soil layer is laid to prevent fly breeding, entry of rain water into the pit and for conservation of the released energy. The material is allowed to decompose for 4 to 6 months after which the stabilized material is taken out and used as compost;
- b) *Mechanical composting* — Mechanical processes are preferred where higher labour costs and limitations of space exist. It is the process using a combination of aerobic and anaerobic decomposition in enclosed containers. Organic wet waste composter is used where raw refuse is composted in large vessel, which are turned at intervals by mobile pedals or plates. Several other processes have now been developed, using different methods of processing of solid waste, using different designs of digester.

Latest technology uses electrically operated rapid processing organic waste convertor with different capacities as mechanical composters. The organic wet and dry waste is separated and shredder is used to homogenize the materials and blenders are used under controlled temperature and bio-enzyme addition in closed containers. These output manure may be used for horticultural/agriculture works; and

- c) *Vermi-composting* — Vermicompost treatment shall be provided to the organic wastes in composting pits located in shade. The pits shall be used to receive the garbage in a predetermined (periodic) cyclic order (for example, 5 pits to receive garbage in 5 days and these 5 pits together accepting daily load of garbage). The gross area of the composting pits may be about 0.1 m² per person.

The site for vermi-composting shall be enclosed from all sides with appropriate fencing (for keeping scavenging animals away) and provided with a small door for accessing the enclosed premises.

Composting pits shall be constructed either under the shade of trees (except Neem tree) or created under sheeting or shade-net, so as to keep the pits under shade. The pits shall be easily accessible for convenience of receiving of garbage through trolleys.

The composting pits shall be made in a manner that the pits do not have the risk of inundation by water. This may be achieved by appropriately raising the base level of the pit and providing weep holes from sides. Height of side walls of compost pits shall be 0.6 m to 0.75 m high. However, the same can be lined to avoid any contamination of ground waste or ground by providing proper slope and drainage arrangement; optionally, empty green coconuts (after the removal of coconut water) is used as bottom layer considering their long life and ability to decompose at a very slow pace.

Initiation of composting pits shall be done by providing a 75 mm thick layer of cow dung (fresh or partially decomposed), spreading 1 kg of vermi-compost and covering it with 75 mm to 100 mm thick layer of dry leaves/grass, etc, and sprinkling of water and allowing to decompose naturally for about 10 to 15 days.

Sorted garbage free from inert and toxic matters shall be applied in the composting pit in layers of 75 mm and spread, and covered with a layer of 75 mm thick dry leaves, followed by sprinkling of water or, by sprinkling cow urine or cow dung plus water slurring or bioenzymes prepared by mixing cow waste and jaggery and kept covered for a few days for fermentation. Apart from these, microbial culture application may also be considered for odour control and rapid biodegradation.

The compost may be removed from the bottom of the compost pit after intervals of 3 months to 6 months. The compost so made needs to be taken out and aired for 1 to 2 days post that it stabilizes fully and is ready to be used after sieving it. May be used in appropriate horticultural and related applications.

See also accepted standard [E-3(5)] for manure grade municipal solid waste compost.

6.3 Thermal Processes

Thermal treatment involves conversion of waste into gaseous, liquid and solid conversion products with concurrent or subsequent release of heat energy.

Three types of systems can be adopted, namely:

- a) *Combustion Systems (Incinerators)* — Thermal processing with excess amounts of air;
- b) *Pyrolysis Systems* — Thermal processing in complete absence of oxygen (low temperature); and
- c) *Gasification Systems* — Thermal processing with less amount of air (high temperature).

6.4 Land Filling

6.4.1 Disposal of waste on lands identified for the purpose is typical of this method of waste disposal. The depth of filling depends upon the site topography and location and varies from deep to shallow filling. A municipal solid waste landfill plant is an engineered facility used for disposal of solid waste. Landfills in which municipal waste is placed are designated as 'MSW landfills'.

The main considerations in respect of landfill sites are:

- a) Specifications for landfill sites;
- b) Site selection;
- c) Facilities at site; and
- d) Specification of land filling.

6.4.2 Improper landfill generally impacts the environment in the following ways:

- a) Ground water contamination by the leachate generated by the waste dump;
- b) Surface water contamination by the run-off from the waste dump;
- c) Bad odour, pests, rodents and wind-blown litter in and around the waste dump;
- d) Generation of inflammable gas (for example, methane) within the waste dump;
- e) Bird menace above the waste dump which affects flight of aircraft;
- f) Fires within the waste dump;
- g) Erosion and stability problems relating to slopes of the waste dump;
- h) Epidemics through stray animals;
- j) Acidity to surrounding soil; and
- k) Release of greenhouse gas.

6.5 Some recycling/treatment methods for food waste are given in [Annex B](#) for guidance.

6.6 See also accepted standards [E-3(6)] for solid waste management, and for analysis and testing of solid wastes.

6.7 Households and communities shall be highly encouraged to practice various methods such as home composting, bio-methanation, etc, listed in Annex C of good practice [E-3(7)] for treatment of bio-degradable waste. In case of treatment of bio-degradable waste at community level, conventional facilities/ modern technologies such as composting, biogas plants, biomethanation facilities, etc, as provided by Urban Local Bodies/Regulatory Authorities shall be used.

6.8 Commercial facilities like markets, malls, where group of wet waste generators are clustered, shall be highly encouraged to practice various methods such as community composting/bio-methanation etc, listed in Annex B of good practice [E-3(8)] for treatment of bio-degradable waste. In case of treatment of bio-degradable waste at local level, conventional facilities/modern technologies such as composting, biogas plants, bio-methanation facilities or any other suitable technology as provided by urban local bodies/regulatory authorities shall be used.

6.9 Other/special wastes shall be dealt with in accordance with the statutory provisions in respect of the same (see [3.4](#)). See various other good practice [E-3(9)] for segregation, collection and treatment of different types of wastes.

ANNEX A

(Foreword)

MUNICIPAL SOLID WASTE GENERATION

A-1 WASTE GENERATION

A-1.1 The typical type of wastes generated may be as follows:

<i>Sl No.</i>	<i>Types of Solid Waste</i>	<i>Description</i>	<i>Sources</i>
(1)	(2)	(3)	(4)
i)	Food waste (garbage)	Wastes from the preparation, cooking, and serving of food. Market refuse, waste from the handling, storage, and sale of produce and meats and vegetable	Households, institutions and commercial such as hotels, stores, restaurants, markets, etc
ii)	Rubbish	a) <i>Combustible (primary organic)</i> — Paper, cardboard, cartons wood, boxes, plastics, rags, cloth, bedding, leather, rubber, grass, leaves, yard trimmings b) <i>Non-combustible (primary inorganic)</i> — Metals, tin cans, metal foils dirt, stones, bricks, ceramics, crockery, glass bottles, other mineral refuse	-do-
iii)	Ashes and residues	Residue from fires used for cooking and for heating buildings, cinders, clinkers, thermal power plants	-do-
iv)	Bulky waste	Large auto parts, tyres, stoves refrigerators, others large, appliances, furniture, large crates, trees, branches, palm fronts, stumps	-do-
v)	Street waste	Street sweepings, dirt, leaves, catch basin dirt, animal droppings, contents of litter receptacles, dead animals	Streets, sidewalks, alleys, vacant lots, etc
vi)	Dead animals	<i>Small animals</i> — Cats, dogs, poultry, etc <i>Large animals</i> — Horses, cows, etc	-do-
vii)	Construction and demolition waste	Lumber, roofing and sheathing scraps, crop residues, rubble, broken concrete, plaster, conduit pipe, wire, insulation, etc	Construction and demolition sites, remodelling, repairing sites
viii)	Industrial waste and sludge	Solid wastes resulting from industry processes and manufacturing operations, such as food processing wastes and boiler House cinders, wood, plastic and metal scraps and shaving, etc. Effluent treatment plant sludge of industries and sewage treatment plant sledges, coarse screening, grit and septic tank	Factories, power plants, treatment plants, etc
ix)	Hazardous wastes	Pathological waste, explosives, radioactive material, toxic waste, etc	Households, hospitals, institution, stores, industry, etc
x)	Horticulture wastes	Tree-trimmings, leaves, waste from parks and gardens, etc	Parks, gardens, roadside trees, etc

A-1.2 The typical quantity of waste generated measured per capita in urban areas may be as follows:

<i>Sl No.</i>	<i>Population Range</i> <i>Million</i>	<i>Average Per Capita Value</i> <i>kg/capita/day</i>
(1)	(2)	(3)
i)	0.1 to 0.5	0.21
ii)	0.5 to 1.0	0.25
iii)	1.0 to 2.0	0.27
iv)	2.0 to 5.0	0.35
v)	More than 5.0	0.50

A-2 INDUSTRIAL WASTE

The major generators are thermal power plants, integrated iron and steel mills, non-ferrous industries, pulp and paper industries and fertilizer and allied industries. Following are the source and quantum of generation of some major industrial wastes.

<i>Sl No.</i>	<i>Name of Industry</i>	<i>Quantity</i> <i>Million tonne per annum</i>
(1)	(2)	(3)
i)	Steel and blast	35.0
ii)	Brine mud	0.02
iii)	Copper slag	0.016 4
iv)	Fly ash	30.0
v)	Kiln dust	1.6
vi)	Lime sludge	3.0
vii)	Mica scraper waste	0.005
viii)	Phospho-gypsum	4.5
ix)	Red mud/bauxite	3.0

A-3 SLAUGHTER HOUSE WASTE

Slaughtering of animals generates waste consisting of non-edible offal (lungs, large intestines, etc), stomach/intestinal contents, dung, sludge from waste water and bones, etc. These have to be disposed of by methods like rendering/controlled incineration/burial/composting/anaerobic digestion, etc. The estimated waste generation may be as per the following:

<i>Sl No.</i>	<i>Type of Slaughter House</i>	<i>Annual Capacity</i>	<i>Dry Waste Generated</i> <i>(tonne per day)</i>
(1)	(2)	(3)	(4)
i)	Large	Large animals more than 40 000 Small animals more than 600 000	6 to 7
ii)	Medium	Large animals — 10 001 to 40 000 Small animals — 100 001 to 600 000	2 to 6
iii)	Small	Large animals up to 10 000 Small animals up to 100 000	0.5 to 1.0

NOTES

1 Animals are commonly classified as 'large' or 'small' according to species, which affects handling, processing, and facility requirements.

2 Large animals typically include bovines such as cattle and buffalo, as well as horses and camels.

3 Small animals generally refer to species like sheep, goats, and calves. Processing these animals is less equipment-intensive than processing large animals.

A-4 BIO-MEDICAL WASTE

The waste generated from medical activities can be hazardous, toxic and even lethal because of their high potential for disease transmission.

The components of bio-medical waste include:

- a) Human anatomical waste (tissues, organs, body parts, etc);
- b) Animal waste (as above, generated during research/experimentation, from veterinary hospitals, etc);
- c) Microbiology and biotechnology waste, such as, laboratory cultures, microorganisms, human and animal cell cultures, toxins, etc;
- d) Waste sharps, such as hypodermic needles, syringes, scalpels, broken glass, etc;
- e) Discarded medicines and cyto-toxic drugs;
- f) Soiled waste, such as dressing, bandages, plastercasts, material contaminated with blood, etc;
- g) Solid waste (disposable items like tubes, catheters, etc, excluding sharps);
- h) Liquid waste generated from any of the infected areas;
- j) Incineration ash; and
- k) Chemical waste.

The estimated generation of total waste is 1.5 kg/bed and the estimated generation of bio-medical waste at 25 percent of total waste generation.

ANNEX B

(*Foreword and Clause 6.5*)

TREATMENT OF FOOD WASTE

B-1 FOOD WASTE

Food waste represents a large percentage of the solid waste. For establishments that serve food, the waste generated per meal varies significantly; however, on average, 60 g of food waste is generated per meal served when both pre-service and post-service waste is considered. The organic nature of food waste makes it unsuitable for disposal by landfills, though it is bio-degradable.

In order to use the nutrient and calorific value of the food waste, it is collected, stored and then recycled. When collecting, it should be ensured that food waste does not enter the drainage system. When providing temporary storage, the risk of physical contamination of kitchen working areas should be avoided. It should be a separate ventilated room with 6 total air changes per hour. In hot climates, it shall be cooled to a temperature of 24 °C to minimize odour generation. The storage area should be cleaned at least once every day.

A brief overview of some of the ways of food waste recycling/treatment is given in [B-2](#) to [B-4](#). Before recycling/treatment, it requires non-organic materials to be removed from food waste.

B-2 FEEDING ANIMALS

Food discards can be made available for feeding animals. This may involve provisions for farmers, zoos, piggeries (very common) and many other applications. Converting food discards to animal feed and pet food is common and a suitable option for recycling food scraps. Vegetable market waste to cow shelters, slaughter house waste to dog shelters, food waste from events, weddings, etc to piggeries, fish market waste to ducks, etc are suggested aiding to circularity.

B-3 INDUSTRIAL USES

Food waste can be used in the rendering industry for converting these materials to soaps, cosmetics and biodiesel fuel.

B-4 ANAEROBIC DIGESTION

This process involves breaking down organic matter in an oxygen-free environment in order to generate biogas, which is a combination of methane and carbon dioxide. The methane is burned for energy. The material that remains after digestion (digestate) should then be composted aerobically to complete the process and produce a valuable soil amendment.

Food waste drying system can provide a fast, simple and user friendly onsite process for decomposing and dehydrating food scraps into rich soil sediments. These systems do not require microorganisms, enzymes, fresh water or other additives.

Food cycler sterilizes, deodorizes and turns potentially harmful food scraps into safe, sterilized compost that can be reused as a soil amendment. It is a multi-phase, onsite organic food waste recycling machine that dehydrates and converts daily food waste into a sterile biomass. The sterilization process takes place at 82 °C and reduces the waste volume by 85 percent to 93 percent. After sterilization, the food waste is converted into compostable humus-rich, high-carbon organic particles. The by-products of the process can be reused as soil amendment, compost accelerant, biofuels, animal feed, pet food, etc. Further benefits can be obtained by mixing the water produced from the food cycler units with potable or non-potable supplies to reduce the amount of water used from mains for use in gardens and landscaping.

LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfillment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used for conformance with the requirements of the referred clauses in this NBCS.

<i>IS No.</i>	<i>Title</i>
(1) IS 9569 : 2022	Solid waste management — Glossary (<i>first revision</i>)
(2) IS 16557 : 2016	Solid waste management — Segregation, collection and utilization at household/community level — Guidelines
IS18520 : 2024	Solid waste management — Segregation, collection and utilization at commercial facilities (shops, markets, malls etc) — Guidelines
(3) IS 6924 : 1973	Code of practice for the construction of refuse chutes in multi-storeyed buildings
(4) IS 12402	Specification for mobile containers for solid waste:
(Part 1) : 1988	General characteristics
(Part 2) : 1988	Methods of test
IS 12647 : 2021	Solid waste management system collection equipment – Requirements and Guidelines (<i>first revision</i>)
IS 12662	Guidelines for use of vehicles for collection and transportation of solid wastes:
(Part 1) : 2021	Selection of vehicles (<i>first revision</i>)
(Part 2) : 2025	Guidelines for maintenance (<i>first revision</i>)
(5) IS 16556 : 2016	Specification for municipal solid waste compost, manure grade
(6) IS 9234	Methods of sampling and test for solid waste:
(Part 1) : 2021	Preparation of samples for analysis (<i>first revision</i>)
(Part 2) : 2021	Determination of moisture content (excluding industrial solid wastes) (<i>first revision</i>)
(Part 3) : 2023	Chemical analysis (excluding industrial solid wastes) (<i>first revision</i>)
IS 16557 : 2016	Solid waste management — Segregation, collection and utilization at household/community level — Guidelines
IS 18520 : 2024	Solid waste management — Segregation, collection and utilization at commercial facilities (shops, markets, malls etc) — Guidelines
(7) IS 16557 : 2016	Solid waste management — Segregation, collection and utilization at household/community level — Guidelines
(8) IS 18520 : 2024	Solid waste management — Segregation, collection and utilization at commercial facilities (shops, markets, malls etc) — Guidelines
(9) IS 17862 : 2022	Storage, collection, dismantling and recycling of E-waste — Guidelines
IS 14534 : 2023	Plastics — Recovery and recycling of plastics waste — Guidelines (<i>second revision</i>)

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NATIONAL BUILDING CONSTRUCTION STANDARDS

PART E PLUMBING SERVICES (INCLUDING SOLID WASTE MANAGEMENT)

SECTION 4 GAS SUPPLY

BUREAU OF INDIAN STANDARDS

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National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part E/Section 4) covers the safe use of gases for purposes like fuel, lighting and medical in buildings.

The use of gases for fuel and lighting purposes in buildings is becoming more and more common in different parts of the country, and with the advent of new petroleum complexes and piped gas system, community gas supply is bound to become one of the important services like electricity and water supply in buildings.

The use of liquefied petroleum gas supplied in containers and cylinders is already quite popular. On release of pressure, by opening the valve, they readily convert into the gaseous phase. In this state they present a hazard comparable to any inflammable natural or manufactured gas, except that being heavier than air, low level ventilation is necessary to avoid inflammable concentration of gas. Similarly, a series of precautions and regulations are necessary for laying of natural gas pipelines from city gas distribution entity to building premises and finally in the use location. Of late, the use of medical gases in hospitals through pipelines has become quite prevalent due to associated advantages; however, this needs to be supplemented with required care and precautions.

A minimum set of safety provisions are, therefore, laid down to safeguard the gas piping installation and the mode of operation in the interest of public safety.

In the first version formulated in 1970, as part of the National Building Code of India, three separate sections of this Part on 'Plumbing Services' were brought out, namely, Section 1 Water Supply, Section 2 Drainage and Sanitation, and Section 3 Gas Supply. These sections were subsequently revised in 1983.

In the first revision of the Code, in 1983, the safe distance between gas piping and electrical wiring system was modified as well as between gas piping and steam piping was incorporated. Additional information regarding the handling, use, storage and transportation of LPG in cylinders exceeding 500 ml water capacity were included. Provisions relating to LPG cylinders, installations regarding some aspects, such as jointing compound used at joints, painting of gas piping, details of fire extinguishers, total quantity of LPG at stationary and portable installations in proportion to the floor area were added. Also, some provisions of LPG bulk storage installations were introduced.

In the second revision of the Code in 2005, the Part 9 was renamed as 'Plumbing Services (Including Solid Waste Management)' and provisions on solid waste management were included for the first time. In the second revision, provisions with regard to pressure regulations were modified; in the provision of service shut-off valves, number of additional shut-off valves were specified; in the provision of installation of gas pipe, new materials for pipes were mentioned; the minimum diameter for gas pipe was reduced to 8 mm; the colour for pipe line for supplying natural gas was specified; the provisions regarding protection against the corrosion were modified; the process of installation of meters were clarified; and additional method for detection of leakage of gas was recommended.

In the third revision of the Code in 2016, to comprehensively address the various and distinct features related to the plumbing aspects, this Part 9 was rearranged as follows:

Section 1 Water supply

Section 2 Drainage and sanitation

Section 3 Solid waste management

Section 4 Gas supply

In the revision in 2016, in Section 4, several key updates were made to enhance the safety and clarity of the guidelines. Certain terminologies were included and updated to reflect current practices, while a provision for the use of fire stops/sleeves at openings was introduced to ensure better fire safety. The clause on the use of liquefied petroleum gas was completely revised for improved accuracy, and a new detailed clause on the use of piped natural gas was added to address emerging needs. Additionally, a new clause on the medical gas pipeline system was

included to ensure proper installation and safety measures in healthcare facilities. Further, cross-referenced standards were updated to align with the latest industry practices and regulations.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

Based on the experience gained in the use of this Section, the provisions have been modified in this revision. The significant changes incorporated in this revision include the following:

- a) New provisions have been introduced for the installation of sloped piping and drips to prevent gas system blockages, ensure proper condensate drainage, and facilitate cleaning, especially in systems handling wet gas.
- b) Comprehensive guidelines for the installation of shutoff valves have been added to ensure proper control, easy accessibility, and maintenance, with specific provisions for individual shutoff valves for each meter and tenant in multi-tenant buildings.
- c) The inclusion of over-pressure protection devices for gas systems has been mandated to ensure safe operation and prevent appliance damage due to excessive pressure.
- d) Detailed provisions for the installation and safe operation of equipment powered by internal combustion engines and gas turbines have been added, with guidelines for flexible gas supply connections to prevent system stress from equipment movement.
- e) References to all the concerned Indian Standards have been updated.

It may also be noted that the following Indian Standards have also been formulated on design and installation of natural gas pipelines:

IS 15663	Code of practice for design and installation of natural gas pipelines:
(Part 1) : 2006	Laying of pipelines
(Part 2) : 2006	Laying of pipelines in crossings
(Part 3) : 2006	Pre-commissioning and commissioning of pipelines

While implementing this NBCS, compliance with statutory regulations shall be ensured.

The information regarding the use of liquefied petroleum gas has been largely based on the following Indian Standards:

IS 6044	Liquefied petroleum gas storage installations — Code of practice:
(Part 1) : 2018	Residential, commercial and industrial cylinder installations (<i>third revision</i>)
(Part 2) : 2021	Commercial, industrial and domestic bulk storage installations (<i>second revision</i>)

All standards, whether given herein above or cross-referred to in the main text of this Section, are subject to revision. The parties to agreement based on this Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Section is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Section.

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Important Explanatory Note for Users of this NBCS

In any Part/Section of this NBCS, where reference is made to ‘**good practice**’ in relation to **design, constructional procedures or other related information**, and where reference is made to “**accepted standard**” in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

Accepted standards [E-4(1)] refers to the Indian Standard(s) given at serial number (1) of the list of standards given at the end of this Part E/Section 4, that is, IS 14885 : 2022 ‘Specification for polyethylene pipe for the supply of gaseous fuel’ (*first revision*) and IS 17544 : 2022 Multilayer (PE-AL-PE) plastic piping system for indoor gas installations with a maximum operating pressure up to and including 500 kPA (5 bar) — Specification.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART E PLUMBING SERVICES (INCLUDING SOLID WASTE MANAGEMENT)

SECTION 4 GAS SUPPLY

1 SCOPE

This NBCS (Part E/Section 4) covers the requirements regarding safety of persons and property for all piping uses and for all types of gases for purposes like fuel, lighting and medical in buildings.

1.2 This Section does not cover safety rules for gas burning appliances.

2 TERMINOLOGY

For the purpose of this Section, the following definitions shall apply.

2.1 Appliance Valve — A device that will shut-off the gas supply to the burner(s).

2.2 Approved Agency — Person or agency or a corporate body approved by competent authority or distribution company to execute the job of designing, erection and maintenance of multi cylinder installation.

2.3 Authority Having Jurisdiction — The authority which has been created by a statute and which, for the purpose of administering this NBCS /Part, may authorize a committee or an official to act on its behalf; hereinafter called the 'Authority'.

2.4 Competent Authority — The Authority designated or otherwise recognized under *Gas Cylinder Rules*, 2004 for approving LPG cylinders, valves and regulators.

2.5 Customer's/Consumer's Connection — Piping tapped on riser to supply each individual customer/consumer.

2.6 Distribution/Distributing Company — The company which is in the field of marketing LPG and PNG, and is the owner of LPG cylinders.

2.7 Gas Fitter — A qualified personnel of the authorized gas supplying organization.

2.8 Installation — A designated premise in an establishment where the complete multi-cylinder-system comprising cylinder, piping manifold, vaporizers, etc, is installed.

2.9 Manifold — A pipe header provided with several opening to which the cylinders are connected by using suitable pipe fittings.

2.10 Pilot — A small flame which is utilized to ignite the gas at the main burner(s).

2.11 Pressure Regulator — A device designed to lower the pressure of gas coming from the distribution main and to maintain it practically constants downstream. This normal operation pressure shall be practically in all cases that of the gas appliances used.

2.12 Purge — To free a gas conduit of air or gas or a mixture of gas and air.

2.13 Qualified Installing Agency — An individual, firm or agency which either in person or through a representative is engaged in and is responsible for the installation or replacement of gas piping on the outlet side of the gas meter, or the connection, installation or repair of gas supply piping and appliances within a building, and who is experienced in such work, familiar with all precautions required, and who has complied with all the requirements as to qualification, registration, licensing, etc, of the Authority.

2.14 Riser — Piping usually vertical on most of its length that supplies gas from the service to the various storeys

of the building.

2.15 Service Pipe — Pipe that runs between the distribution main in the street and the riser in the case of multistoried building or the meter in the case of an individual house.

2.16 Service Shut-Off Valve (Isolation Valve) — A device installed outside the premises to cut-off the main supply of gas from pipeline by the supplier.

2.17 Vent Pipe — A safety device to which certain regulators are connected to evacuate outside gas that may escape from the normal circuit when some part of system gets damaged or malfunctions or a safety valve is open.

3 PRESSURE REGULATIONS

3.1 Pressure regulation is required to economize the sizing of piping system. Where the pressure of gas supplied to domestic system or other low pressure gas piping system in buildings is in excess of the pressure to be used in the appliance, a gas pressure regulator of suitable specification shall be installed in service pipe of each system to prevent excess pressure reaching the appliance. The pressure regulators to be used can be from 400 kN/m² upstream pressure to 2.1 kN/m² for domestic consumers and 10 kN/m², 30 kN/m², 200 kN/m² for commercial consumers, as the case may be.

3.1.1 In some place the reduction of pressure from main distribution source of 400 kN/m² to intermediate pressure (say 7 kN/m²) and then to operating pressure of 2.1 kN/m² is achieved.

3.1.2 Whereas in most of the other places the reduction of pressure from main distribution source of 400 kN/m² to directly operating pressure (say 2.1 kN/m², 10 kN/m², 30 kN/m², 200 kN/m²) is achieved in single stage pressure reduction.

3.2 If located inside a building, the required regulator shall comply with the following:

- a) If any of the diaphragms of the regulator ruptures, the gas shall be sent to an outlet vent pipe made of brass or plastic in order to ventilate or drain the gas out of the building. The vent pipe will, however, lead to outer air about 1 m above the topmost storey of the building. Means shall be employed to prevent water from entering this pipe and also to prevent stoppage of it by insects or other foreign bodies;
- b) If the gas pressure at the outlet of the regulator falls below 50 percent of the operating gas pressure or rises above twice the operating gas pressure, the gas input to the pressure regulator shall be cut off; and
- c) In the event of malfunctioning of this safety device, a supplementary device shall connect the low pressure circuit to the outlet circuit (vent pipe) as soon as the exit pressure reaches 7 kN/m².

3.3 It shall also be ensured by the supply authority that the calorific value and supply pressure of gas shall not exceed the values for the type of gas used.

4 SERVICE SHUT-OFF VALVES

4.1 Service shut-off valves shall be installed on all new services including replacements in a readily accessible location.

4.2 Service shut-off valves shall be located upstream of the meter if there is no regulator or upstream of the regulator, if there is one.

4.2.1 Service shut-off valves shall be located in the upstream of the meter, if a single regulator is supplying more than one consumer and each such stream shall have one additional shut-off valve upstream of regulator.

4.3 All gas services operating at pressure greater than 7 kN/m² shall be equipped with an approved service shut-off valve located on the service pipe outside the building.

4.4 Underground shut-off valves shall be located in a covered durable curb box, manhole, vault or stand pipe which is designed to permit ready operation of the valve and the covers of which shall be clearly marked 'Gas'.

5 EXISTING WORK

Nothing herein shall prohibit the continued use of existing system of the gas piping without further inspection or test, unless the Authority has reason to believe that defects which make the system dangerous to life or property exist.

6 RULES FOR TURNING GAS ON

6.1 No person, unless is the employ of the gas company or having permission from the gas company, shall turn on the gas at a service shut-off valve or at any valve that controls the supply of gas to more than one consumer.

6.2 Gas shall not be turned on at any meter valve without specific permission from the gas company or other authority if any of the following conditions exists:

- a) If the gas piping appliances or meter supply through the meter valve are known to leak or otherwise to be defective (*see 10*);
- b) If required inspection of the piping or appliance has not been made;
- c) If the gas company or other authority has requested that the gas be left turned off; and
- d) If the meter valve is found shut-off for some reason not known to the gas fitter.

The gas shall not be turned on in the event of fire.

6.3 Gas shall not be turned on at any branch line valve if any of the conditions specified in **6.2** prevails. Where a branch line valve is found closed, a gas fitter shall again turn the gas on at such valve only if proper precautions to prevent leakage are taken and no other unsafe conditions are created thereby.

6.4 Gas shall not be turned on at either the meter valve or service line unless all gas keys or valves provided on all outlets in the piping system are closed or all outlets in the piping system are capped or plugged.

7 RULES FOR SHUTTING OFF THE GAS

7.1 The gas fitter shall put the gas off to any appliance, pipe or piping system and shall leave the gas turned off, until the causes for interrupting the supply has been removed in any one of the following cases:

- a) If ordered to do so by the Authority;
- b) If leakage of gas is noted, which appears to be sufficient to cause fire, explosion or asphyxiation;
- c) If an installation of some gas appliance is found to be such as to cause a serious hazard to persons or property; and
- d) If any condition exists which threatens interruption of gas supply which may cause burner outage or otherwise prove dangerous.

7.2 It shall be the duty of the installing agency when the gas supply is to be turned off to notify all affected consumers.

7.3 Before turning off the gas at the meter, for the purpose of installation, repair, replacement or maintenance of piping or appliance, all burner and pilot valves on the premises supplied with gas through the meter shall be turned off and the meter test hand observed for a sufficient length of time to ascertain that there is no gas passing through the meter. Where there is more than one meter on the premises, precaution shall be exercised to ensure that the concerned meter is turned off.

8 INSTALLATION OF GAS PIPES

8.1 Installation, repair and replacement of gas piping or appliances shall be performed only by a qualified installing agency.

8.2 Piping

8.2.1 Piping shall be of wrought iron, steel, copper, stainless steel or cast iron when the gas pressure is less than 7 kN/m²; with higher gas pressure use of cast iron shall be prohibited.

8.2.1.1 SS 316/304/321 and flexible PE coated flexible pipe in rolls shall be permitted in low pressure system provided the pipe meets the required standard, to avoid the bends, fittings and leakages from the joint which are potential leakage points. Also, reference may be made to accepted standards [E-4(1)]. Heavy rubber flexible tube shall be permitted only as direct connection to burner from appliance valve.

8.2.1.2 For details regarding medical gas pipeline systems, a reference may be made to good practice [E-4(2)].

8.2.2 Size of Gas Piping

Gas piping shall be of such size and so installed as to provide supply of gas sufficient to meet the maximum demand without undue loss of pressure between the meter or service regulator when a meter is not provided, and the appliance(s).

8.2.2.1 The size of gas piping depends upon the following factors:

- a) Allowable loss in pressure from meter or service regulator, when a meter is not provided, to appliance;
- b) Maximum consumption to be provided;
- c) Length of piping and number of fittings; and
- d) Specific gravity of gas.

8.2.2.2 No gas pipe smaller than 8 mm shall be used.

8.2.3 As far as possible, straight lengths of piping should be used. Where there are bends in the pipe line, these should have a radius of at least five times the diameter of the pipe.

8.2.4 For any thread joint proper sealant shall be used on male threads only.

8.3 The gas piping shall be of the colour stipulated by explosive authority to distinguish it from other piping and the piping shall be painted silver grey with red band of 150 mm width. The gas pipeline shall be painted canary yellow in case of natural gas.

8.4 Piping Underground

8.4.1 Protection of Piping

Piping shall be buried to a minimum depth of 1 m or covered in a manner so as to protect the piping from physical damage.

8.4.2 Protection Against Corrosion

Generally, all the piping within the premises where it has to run on the wall shall be exposed and should not be in contact with wall to ensure that no corrosion takes place. Epoxy sealant or polyethylene conduit shall be used to ensure no contact of pipe with the wall in the situation of pipe crossing the wall. Underground or concealed gas pipeline in contact with earth or other materials which may corrode the piping shall be protected against corrosion by application of adequate corrosion resistant coating backed up by cathodic protection system.

8.5 The building shall not be weakened by the installation of any gas piping.

8.6 Gas piping in building shall be supported with pipe hooks, metal pipe straps, bonds or hangers suitable for the size of piping and of adequate strength and quality and located at proper intervals so that the piping may not be moved accidentally from the installed position.

8.7 Pipe Entrance to Buildings

Where gas pipe enters a building through a wall or floor of masonry or concrete, any gas piping or other piping entering the walls or floors shall be suitably sealed against the entrance of water/moisture or gas.

The openings or imperfection of fit or design are also source of possible fire and smoke passing through them. Fire stops shall be provided to fill the openings around penetrating items such as gas pipes, etc, through the wall

or floor openings. Non-combustible sleeving may also be used as an alternative to proprietary seals for penetration of pipes of lead, aluminium, aluminium alloy, fibre cement or uPVC up to a specified nominal internal diameter. Proprietary fire stopping and sealing systems which have been shown by test to maintain the fire resistance of the wall or other elements, are available and may be used. Other fire stopping materials include cement mortar; gypsum-based plaster; cement or gypsum vermiculite/perlite mixes; glass fibre, crushed rock, blast furnace slag, or ceramic based products (with or without resin binders), and intumescent mastics. These may be used in situations appropriate to the particular material. Not all of them will be suitable in every situation. When sleeving is used, its length should be greater than the thickness of wall or floor. The sleeve shall be properly grouted to maintain fire/smoke separation.

Regarding protection of openings in walls or floors, from fire, reference may be made to Part F 'Fire and Life Safety' of this NBCS.

8.7.1 Piping in Floors

Piping in solid floors, such as concrete, shall be laid in channels in the floor suitably covered to permit access to the piping with a minimum damage to the building.

8.7.2 Single pipe without joint shall be used for wall crossing in any building.

8.8 Gas pipe shall not be bent. Fittings shall be used when making turns in gas pipe.

8.9 Generally concealed piping shall not be allowed. However, if it is necessary then it shall be under the **8.4** of underground piping and all protection such as coating, cathodic protection shall be done.

8.10 A drip shall be provided in the gas distribution system, if the moisture contents in the gas is likely to reach saturation point at any stretch of pipe line in the system; a drip shall, however, be provided at any suitable point in the line of the pipe where condensate may collect and from where it can be easily removed. This drip should be so installed as to constitute a trap where in an accumulation of condensate will shut-off the flow of gas before it will run back into the meter.

8.10.1 Drip has to be provided in the case of gas consisting moisture content.

8.11 Prohibited Devices

No device shall be placed inside the gas piping or fittings that will reduce the cross-sectional area or otherwise obstruct the free flow of gas.

8.12 Piping shall be electrically continuous throughout its length and properly earthed except in stretches where cathodic protection system is used for protection against corrosion. It shall not, however, be used to earth any electrical equipment.

8.12.1 The distance between gas piping and electrical wiring system shall be at least 60 mm and, where necessary, they shall be securely fixed to prevent contact due to movement. The gas piping should run above the electrical wiring. In this type of installation in the event of any leakage of natural gas, the gas would move up (natural gas being lighter than air) and would not come directly in contact with the electrical wiring. If the gas to be supplied is heavier than the air then the gas piping should run below the electrical wiring.

8.13 The distance between the gas piping and steam piping, if running parallel, shall be at least 150 mm. The gas piping should preferably run below the steam piping.

8.14 Piping installation shall be thoroughly gastight.

8.15 Smoking shall not be permitted when working on piping which contains or has contaminated gas.

8.16 Meters shall be installed in such a way that there shall be no load transfer from the pipeline to the inlet/outlet of the meter and shall be easily accessible.

8.17 Sloped Piping and Drips

Piping systems designed for gas flow, particularly those handling wet gas, should include provisions to ensure the smooth and safe operation of the system by adhering to the guidelines given in [8.17.1](#) to [8.17.3](#).

8.17.1 Slopes

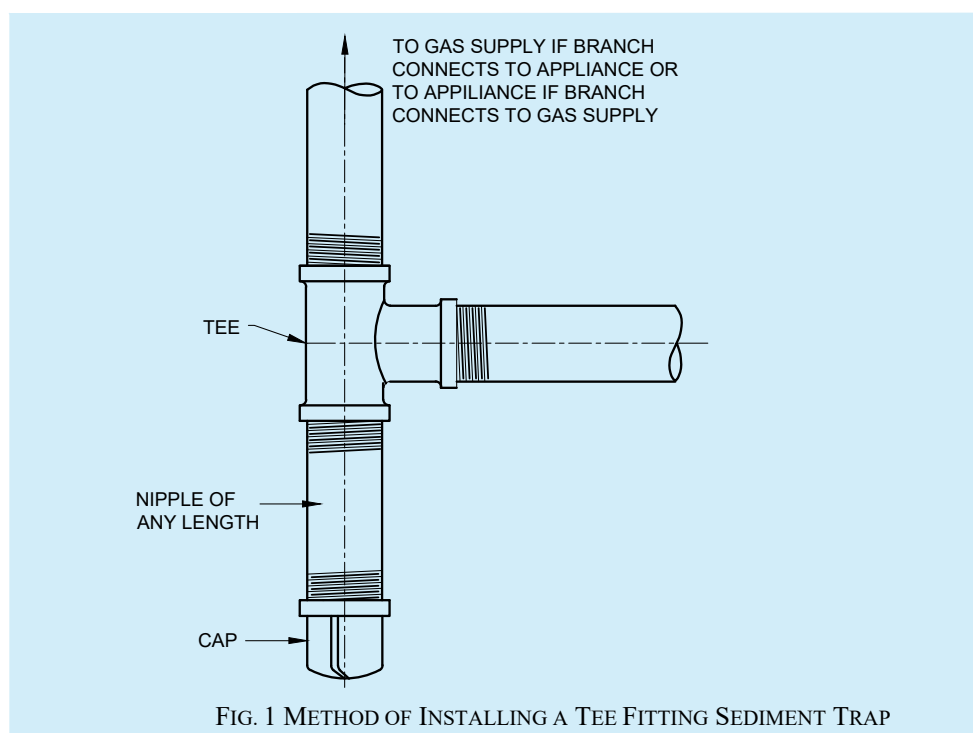
Pipes shall be installed with a slope of not less than 1/6.3 mm per 4 572 mm to prevent the formation of traps.

8.17.2 Drips

Where conditions involve wet gas, drips shall be installed at points where condensate may accumulate, such as the outlet of meters, to act as traps. These traps should allow for the collection of condensate and prevent its backflow into the meter or other critical components. Drips shall be located in accessible areas to facilitate cleaning or removal of condensate, and they shall be positioned to avoid freezing conditions that may compromise functionality.

8.17.3 Sediment Trap

Where a sediment trap is not incorporated as part of the appliance, a sediment trap shall be installed down-stream of the appliance shutoff valve as close to the inlet of the appliance as practical. The sediment trap shall be either a tee fitting having a capped nipple of any length installed vertically in the bottommost opening of the tee as illustrated in [Fig. 1](#) or other device approved as an effective sediment trap. Illuminating appliances, ranges, clothes dryers, decorative vented appliances for installation in vented fireplaces, gas fireplaces and outdoor grills need not be so equipped.



8.18 Shutoff Valves

Piping systems shall be equipped with shutoff valves to ensure proper control and isolation of the system as required. The following guidelines shall be followed for the installation and use of shutoff valves in piping systems:

- Shutoff valves shall be of a suitable type and constructed from materials that are compatible with the piping system. They shall meet the necessary standards for the pressure and specific application of the system;
- Shutoff valves shall not be installed in concealed spaces, furnace plenums, or locations where access may

be hindered;

- c) Shutoff valves shall be installed in locations that allow easy access for operation. They should also be protected from potential damage;
- d) Each meter installation shall include a shutoff valve on the supply side of the meter to allow isolation of the gas supply;
- e) For systems supplying gas to multiple buildings or tenants from a single meter, each building or tenant shall have an individual shutoff valve;
- f) In buildings with multiple tenants, a shutoff valve shall be provided for each tenant, ensuring they have access to the valve controlling their space;
- g) Each shutoff valve serving a house line shall be clearly marked with an identification tag to facilitate easy identification of the systems it controls;
- h) Each appliance shall be equipped with a shutoff valve installed in close proximity, allowing easy isolation of the appliance;
- j) Laboratories with multiple fuel gas outlets shall be provided with a dedicated shutoff valve. This valve shall be readily accessible, located within the laboratory, and clearly identified with signage; and
- k) Valves installed in tubing systems shall be securely supported independently of the tubing to maintain system integrity.

8.19 Over-pressure Protection Devices

Over-pressure protection devices are essential for ensuring the safe operation of gas systems by preventing excessive pressure that could damage appliances or pose safety risks. The guidelines cover when and where these devices are required, their pressure limitation capabilities, and the materials and construction standards to ensure reliable performance. Proper installation and maintenance of over-pressure protection devices are critical to ensure compliance with safety standards and protect both equipment and personnel.

The following steps and guidelines shall be followed for the installation and use of over-pressure protection devices in gas piping systems:

- a) Over-pressure protection devices shall be installed in piping systems serving appliances designed to operate at a gas pressure of 3.5 kPa or less, when the serving gas supplier delivers gas at a pressure greater than 13.8 kPa. Piping systems serving equipment with higher inlet pressures shall also be equipped with over-pressure protection devices as specified by the appliance manufacturer.
- b) The pressure limitation for over-pressure protection devices shall comply with the relevant guidelines for systems designed to operate at specific gas supply pressures, as follows:
 - 1) *Pressure under 3.5 kPa* — Over-pressure protection devices shall limit the gas pressure to 13.8 kPa or less when the line pressure regulator fails;
 - 2) *Pressure over 3.5 kPa* — Over-pressure protection devices shall limit the gas pressure to the requirements specified by the appliance manufacturer's installation instructions when the system operates with pressures above 3.5 kPa;
 - 3) *Device capability* — Over-pressure protection devices shall be capable of independently limiting the gas pressure to the appliances as required, without relying on other pressure control equipment in the system; and
 - 4) *Failure detection* — Systems requiring over-pressure protection shall be designed so that any failure of the primary pressure control device is easily detectable. If a relief valve is used for over-pressure protection, it shall maintain pressure within safe limits under all failure conditions, including the failure of the line pressure regulator.
- c) *Types of Over-Pressure Protection Devices* — The over-pressure protection device shall be one of the following:
 - 1) Pressure relief valve;
 - 2) Monitoring regulator;
 - 3) Series regulator; and

4) Automatic shutoff device.

These devices shall either be integrated with the service or line pressure regulator or installed separately, in compliance with the relevant guidelines.

- i) *Construction and installation* — Over-pressure protection devices shall be constructed from materials that prevent impairment due to corrosion and shall be designed for easy operation, testing, and examination for leakage when closed;
- ii) *External control piping* — Control piping shall be installed to avoid damage that could render both the regulator and the over-pressure protection device inoperative;
- iii) *Setting* — The over-pressure protection devices shall be set to ensure the gas pressure supplied to connected appliances does not exceed the limits established by the relevant pressure guidelines;
- iv) *Unauthorized operation* — Measures shall be taken to prevent unauthorized operation of shutoff valves that could disable the over-pressure protection device. This may include locking the valve in the open position or installing duplicate relief valves;
- v) *Vents* — Discharge stacks, vents, and outlets of over-pressure protection devices shall be directed outdoors, designed to prevent blockages, and sized to match the outlet of the relieving device; and
- vi) *Size of fittings, pipe, and openings* — All fittings, pipes, and openings between the protected system and the over-pressure device shall be properly sized to prevent hammering and ensure adequate relief capacity.

8.20 Engine and Gas Turbine Powered Equipment

The requirements for the installation and operation of equipment powered by internal combustion engines and gas turbines shall be followed towards reducing fire hazards both during installation and operation. Proper installation is essential to ensure safe operation, minimize risks, and enhance the equipment's performance and longevity. The following guidelines cover the installation practices and considerations necessary for integrating these systems safely into the overall setup.

- a) *Powered Equipment* — Permanently installed equipment powered by internal combustion engines and turbines shall be installed according to the manufacturer's instructions. This ensures that the equipment operates as intended, with all safety and operational specifications followed.
- b) *Gas Supply Connection* — Equipment powered by internal combustion engines and turbines shall not be rigidly connected to the gas supply piping. Flexible connections shall be used to prevent stress or strain on the piping system caused by movement or vibration of the equipment, ensuring safe and reliable operation.

9 INSPECTION OF SERVICES

9.1 No person shall use or permit the use of a new system or an extension of an old system of gas piping in a building or structure before the same has been inspected and tested to ensure the tightness of the system, and a certificate has been issued by the Authority.

9.1.1 Test of Piping for Tightness

Before any system of gas piping is finally put in service, it shall be carefully tested to ensure that it is gastight. Where any part of the system is to be enclosed or concealed, this test should precede the work of closing in. To test for tightness, the piping may be filled with city gas, air or inert gas but not with any other gas or liquid. In no case shall oxygen be used. The piping shall stand a pressure of at least 20 kN/m² measured with a manometer or slope gauge, for a period of not less than 10 min without showing any drop in pressure.

9.1.2 When the gas pressure exceeds 7 kN/m², the piping shall withstand a pressure of 0.6 MN/m² for 4 h (this test is for piping designed for working pressure less than 0.4 MN/m²).

9.2 The Authority shall, within a reasonable time after being requested to do so, inspect and test a system of gas piping that is ready for such inspection and test, and if the work is found satisfactory and test requirements are complied with, it shall issue the certificate.

10 LEAKAGE CHECK

10.1 Before turning gas under pressure into any piping, all openings from which gas may escape shall be closed.

10.2 Checking for Gas Leakage

No matches, flame or other sources of ignition shall be employed to check for gas leakage from meters, piping or appliances. Checking for gas leakage with soap and water solution is recommended.

10.3 Use of Lights

Artificial illumination used in connection with a search of gas leakage shall be restricted to electric hand flash lights (preferably of the safety type) or approved safety lamps. In searching for leaks, electric switches should not be operated. If electric lights are already turned on, they should not be turned off.

10.4 Checking for Leakage with Meter

Immediately after turning gas into the piping, the system shall be checked to ascertain that no gas is escaping. This may be done by carefully watching the test dial of the meter to determine whether gas is passing through the meter. In no case should a leakage test be made using a gas meter unless immediately prior to the test it has been determined that the meter is in operating condition.

10.5 Checking of Leakage Without Using a Meter

This may be done by attaching to an appliance, orifice or a manometer or equivalent device and momentarily turning on the gas supply and deservng the gauging device for pressure drop with the gas supply shut-off. No discernible drop in pressure shall occur during a period of 3 min.

10.6 After piping has been checked, all gas piping shall be fully purged. Piping shall not be purged into the combustion chamber of an appliance. A suggested method for purging the gas piping to an appliance is to disconnect the pilot piping at the outlet of the pilot valve.

10.7 After the gas piping has been effectively purged, all appliances shall be purged and the pilots lighted.

10.8 In addition to the checking of gas leakage with soap and water solution, a suitable gas detector is also recommended for use.

11 USE OF LIQUEFIED PETROLEUM GAS (LPG)

11.1 The cylinders used for the storage and transportation of liquefied petroleum gas (LPG) shall conform to accepted standards [E-4(3)] approved by the statutory authority.

11.2 The handing, use, storage and transportation of liquefied petroleum gas in cylinders exceeding 500 ml water capacity shall be done in accordance with good practice [E-4(4)].

11.3 LPG Cylinder Installation

The following recommendations apply to installation in commercial, industrial, educational and institutional premises.

11.3.1 General Recommendations

11.3.1.1 Those responsible for the installation of cylinders, equipment and piping should understand the characteristics of LPG and be trained in good practice of handling, installing and maintaining installations.

11.3.1.2 The jointing compound used at different joints in the system shall be decided by the Qualified Installing Agency. Hemp and similar materials shall not be used at the joint. In any joint in which the thread provides a gastight seal, jointing compound shall be used only on the male thread.

11.3.1.3 Fire extinguishers of dry powder type or carbon dioxide type conforming to accepted standard [E-4(5)]

shall be provided in places where LPG cylinder installations are situated and shall be located near such installations. Two buckets filled with sand and two with water shall also be installed nearby. The number, type and size of the fire extinguishers shall be as follows:

	<i>Number</i>	<i>Type</i>	<i>Capacity</i>
a) For installations with LPG 40 kg to 200 kg	2	Dry powder	9 kg
b) For installations with LPG more than 200 kg and up to 320 kg	3	Dry powder	9 kg
c) For installations with LPG more than 320 kg and up to 1 000 kg	4	Dry powder	9 kg

NOTE — For electrical installations, one number CO₂ fire extinguisher (4.5 kg capacity) shall be provided.

11.3.1.4 Liquefied petroleum gas shall not be transferred from the cylinders in which it is received to any other container.

11.3.2 *Cylinder Location*

11.3.2.1 *Stationary installations*

- a) Stationary installation not exceeding 50 kg of LPG may be installed indoors on any floor. It is recommended to have a minimum floor area of 5 m² for such an installation.
- b) Stationary installations each not exceeding 50 kg of LPG may be installed indoors on any floor and within the same workspace provided the minimum distance between two such installations is 3 m, the proportion of such installations to floor area is one installation per 5 m² and the aggregate quantity of gas of all such installations does not exceed 200 kg.
- c) Stationary installation not exceeding 100 kg of LPG may be installed indoors on any floor provided the floor area for such an installation is not less than 12 m².
- d) Stationary installations each not exceeding 100 kg of LPG may be installed indoors on any floor and within the same workspace provided the minimum distance between two such installations is 3 m, the proportion of such installations to floor area is one installation per 12 m² and the aggregate quantity of gas of all such installations does not exceed 200 kg.
- e) Stationary installation not exceeding 400 kg of LPG may be installed indoors in an enclosed section of a building or a room reserved exclusively for this purpose and ventilated at low level directly to the outside air.
- f) Stationary installations above 400 kg [200 kg in case provision as in (e) is not possible] but not exceeding 1 000 kg shall be installed outdoors on ground floor level only. A minimum distance of 3 m shall be maintained between an installation and any building, public place, roadways, and other surroundings. The installation shall be protected from excessive weathering by sun, rain, etc, and from tampering by unauthorized persons. A lean-to roof with expanded metal on angle-iron framework on the sides is considered suitable for this purpose. In any case, adequate ventilation at ground level to the outside air shall be provided.
- g) If the storage per installation is more than 1 000 kg, installations in multiples of 1 000 kg with manifold, safety devices, change over mechanism may be provided with minimum inter distance of 3 m from all sides. Total number of such installations shall not exceed 4 including stand by installation. If more than two installations are used, then number of cylinders per installations should not exceed 30.
- h) For storage installations of 1 000 kg or installations in multiple of 1 000 kg the number of cylinders to be connected per manifold shall not exceed 15 and for each 1 000 kg installation two such manifolds shall be provided.
- j) Cylinders shall be installed upright with the valves uppermost.
- k) Cylinder containing more than 20 kg of gas shall not be located on floors above ground level.
- m) Cylinders shall be located on a concrete or brick floor, preferably raised in case of outdoor installations.

- n) In order to prevent the hazardous collection of gas, cylinders shall be placed at least 1 m away from culverts, depressions, or openings leading to below ground level compartment and drains.
- p) Cylinders which have safety relief valves or similar devices incorporated in them shall be so positioned so that if the relief device operates, escaping gas is not hazardous;.
- q) Cylinder installation for commercial/industrial/domestic/residential complexes, etc, or at any public place shall meet the following requirements:
 - 1) For commercial and industrial cylinder installation for any public places like mall, buildings, industries, hotels, etc, the sum total of all stationery installations inside the building shall not exceed 400 kg of LPG in any case. In case sum total of net weight of all the cylinders installed indoors exceed 400 kg provision for outdoor installations shall be made. For installations where the sum total is less than 400 kg of LPG provisions of (b) to (e) shall apply;
 - 2) For commercial and industrial cylinder installation for any public place, industry, etc, if the net weight of all the cylinders installed is more than 400 kg but less than 1 000 kg, than the installation shall be provided outdoors in line with (f). The cylinder installation shall be provided in a covered industrial shed but open from all sides for proper ventilation. In case the installed capacity increases more than 1 000 kg, than the installation shall be provided in line with (g). Two separate installations shall be provided and distance between each of the installations shall be minimum 3 m (this distance shall be between the outer edge of the two sheds);
 - 3) For multistoried buildings, flats housing society the reticulated installation of any capacity shall be provided at a safe place especially earmarked for this purpose. The installation shall not be provided by the side of road where there is continuous movement of vehicles or is approachable to residents especially children. The installation should be secured and should be provided in a covered shed open from all sides for ventilation. The area shall be demarcated by 1.5 m high chain link fencing having one gate. There should be minimum 3 m clear space all around the fencing and the space between the fencing and the edge of the shed should be minimum 2 m all around;
 - 4) If the requirement is more than 1 000 kg, installations in multiples of 1 000 kg with manifold, safety devices, change over mechanism, not exceeding 4 such installations (including stand by) may be provided with minimum distance of 3 m from all sides. The number of cylinders per installations should not exceed 30; and
 - 5) Location of cylinder bank:
 - i) The site for LPG cylinder manifold shall be located away from the kitchen/LPG utility area/residential building. Installation shall be slightly raised minimum 100 mm from surrounding ground level;
 - ii) Cylinders installation should not be located in inaccessible location, under a stairway, basements, cellars, where air movement across cylinders is very low/not present, to prevent the cylinders from getting inadequate latent heat and unapproachable during emergency;
 - iii) Cylinders installation should not be close to steam pipes, boilers, transformers, DG sets, etc, to prevent cylinders from getting affected due to radiant heat;
 - iv) Cylinders shall not be installed at a place where they are likely to cause obstruction, suffer damage or be exposed to conditions likely to affect safety;
 - v) Cylinders installation should not be located along the drive way. Under unavoidable conditions, minimum distance of 3 m from drive way shall be maintained. Brick masonry (minimum 230 mm thick)/concrete (minimum 100 mm thick) walls or metallic shields/barriers (minimum 1.63 mm sheet) and of minimum 2 m height should be erected between drive way and cylinder installation for the safety and security of the installation;
 - vi) Cylinders shall be located on a concrete or brick floor that is firm, at level, smooth, drained in case of outdoor installation; and
 - vii) The cylinder shall be installed in upright position with the valve pointing upwards and minimum 1 m away from any combustible materials.
- r) Cylinder storage room:
 - 1) The cylinder storage room shall be made out of non-flammable material that is concrete (minimum 100 mm thick) or brick masonry walls (minimum 230 mm thick) or steel structures made out of minimum 5 mm thick MS angle iron structure and 1.63 mm MS wire mesh of size minimum 11;

- 2) Since LPG is heavier than air, ventilation shall be provided at floor level, that is 100 mm above inside cylinder bank room, open to atmosphere. The ventilators shall be provided with 2 layers' copper or non-corroding metal wire mesh not less than 11 to the linear centimetre. The size of the ventilators should be minimum 500 mm long and 300 mm height;
- 3) The storage room shall be well ventilated, that is openings/ventilators shall be provided in the walls of cylinder storage covering minimum 25 percent of wall area. It should also have adequate lighting. Both exteriors and interiors should be painted with weather proof paints. FLP fittings, approved by statutory authorities shall be provided for electrical use;
- 4) The doors of the room where cylinders are installed shall open outwards and shall have louvers/wire mesh to ensure visibility and ventilation; and
- 5) Flammable materials like wood and plastic shall not be used. The cylinder bank storage room shall not use for storing any other materials.

11.3.2.2 Portable installations

When portability of cylinders is necessary, the following requirements shall be fulfilled:

- a) The sum total of capacity of the cylinders connected to each manifold shall not exceed 100 kg of LPG. The total quantity of gas thus installed in a workspace shall not exceed 200 kg.
- b) If cylinders are mounted on a trolley, the trolley shall be stable. Where necessary, the cylinders shall be secured to prevent them from falling.
- c) The regulator shall be connected directly to the cylinder valve or to a manifold which shall be connected to the cylinder valves by means of rigid connections to give adequate support to the regulator. The only exception to this requirement is where cylinders are mounted on a trolley and the manifold is rigidly supported on the trolley. In such a case flexible or semi-flexible connections may be used between the cylinder valves and the manifold but not between the manifold and the regulator.
- d) Anytime the total quantity of gas at portable installations shall be in proportion to the floor area as specified in [11.3.2.1 \(a\) to \(f\)](#).
- e) At any time, the provisions at [11.3.2.1 \(a\) to \(r\)](#) shall be ensured for all installations.

11.3.3 Cylinder Manifolds

11.3.3.1 All materials, fittings, etc, used in cylinder manifold systems shall comply with the statutory provisions or relevant Indian Standards. In absence of any such provisions or Indian Standards, equivalent international norms shall be followed.

11.3.3.2 The individual component parts of manifolds, that is piping, fittings, pigtails, etc, which are subject to cylinder pressure shall be capable of withstanding a test pressure without bursting of 2.5 N/mm² or one and a half times the developed pressure at 65 °C, whichever is more.

11.3.3.3 Where cylinder installations are made up with service and reserve batteries of cylinders, suitable change-over devices or valves shall be incorporated in the manifold header to prevent undue escape of the gas when cylinders are changed.

11.3.3.4 In case pressure regulators, manifold headers and automatic change-over devices are connected to cylinder by semi-flexible connectors, these shall be rigidly supported. Copper tube pigtails are considered to be semi-flexible for this purpose.

11.3.3.5 It is recommended that joints in manifold headers which do not have to be broken in normal use should be welded or brazed using a material which shall have a melting point of at least 540 °C.

11.3.3.6 All joints between manifold headers and cylinder connectors shall be readily accessible.

11.3.3.7 All joints in the manifold should be welded (except for valve fixation) and be easily accessible for inspection/repairs, etc.

11.3.3.8 Each manifold arm shall be fabricated in such a way that minimum joints are provided. Only seamless pipes are recommended for use in manifolds. There should be a minimum gap of 300 mm between the manifold

and valve protection ring of the LPG cylinders.

11.3.3.9 The cylinders connected to a manifold shall be safely spaced for easy and safe replacement of cylinder when empty. A minimum distance of 400 mm shall be maintained between two nipples provided on the manifold for connecting the cylinders to the manifold. In case the diameter of the cylinder is higher than 400 mm then the distance between the two nipples shall be diameter of the cylinder plus 100 mm.

11.3.3.10 A pressure gauge of 100 mm dial shall be provided in the manifold to indicate the gas pressure in the manifold. The pressure gauge shall be suitable for a pressure range 0 to 1.0 N/mm².

11.3.4 Pressure Regulators

11.3.4.1 Pressure regulators and other devices used to control the gas shall comply with the distributing company's stipulations and accepted standards [E-4(6)].

11.3.4.2 Pressure regulator fitted with a safety valve shall be either,

- a) Installed in the open air; or
- b) Vented to the open by means of a metal vent pipe connected to the safety valve outlet.

11.3.4.3 Care shall be taken that safety valve outlets do not become choked with dust or other foreign matter.

11.3.4.4 If the regulator is fitted with a relief valve, care should be taken in positioning the regulator to avoid unnecessary hazards if the relief valve functions.

11.3.4.5 Pressure regulators and other control devices shall be adequately supported.

11.3.5 Instructions to Consumers

Consumers shall be instructed by the distribution company on the following, through appropriate written instructions:

- a) Operation of the whole system;
- b) How to recognize gas leaks;
- c) Action to be taken in case of leakage;
- d) Action to be taken in case of fire; and
- e) Action to be taken in case of damage to, or failure of, any part of the installation.

11.3.6 For detailed information regarding installation of LPG cylinders in commercial, industrial, educational and institutional premises, reference may be made to good practice [E-4(7)].

11.4 LPG Bulk Storage Installations

The following recommendations apply to LPG bulk storage installations where storage tanks over 450 ℓ water capacity are used at industrial, commercial and domestic consumers' premises.

The maximum capacity of an individual tank and group of tanks at industrial, commercial and domestic premises shall be as follows:

<i>Sl No.</i>	<i>Premises</i>	<i>Maximum Water Capacity of an Individual Tank</i>	<i>Maximum Water Capacity of Group of Tanks</i>
(1)	(2)	ℓ (3)	ℓ (4)
i)	Industrial	130 000	260 000
ii)	Commercial	40 000	80 000
iii)	Domestic	20 000	80 000

11.4.1 Location and Spacing of Storage Tanks

11.4.1.1 Storage tanks shall be located outside the buildings and shall not be installed one above the other.

11.4.1.2 Each individual tank shall be located with respect to the nearest important building or group of buildings or line of adjoining property which may be built in accordance with [Table 1](#). The distances given refer to the horizontal distance in plan between the nearest point of the storage tank and building/property line.

11.4.1.3 In heavily populated or congested areas the authority may determine the need for other reasonable protective methods to be taken, such as provision of fire walls, etc. If fire walls are to be provided, the authority may determine the extent to which the safety distances for above ground tanks may be reduced.

11.4.1.4 No LPG tank(s) shall be located within the bonded enclosures of any petroleum installation. The minimum distance of separation between LPG storage tanks and any petroleum installation shall be as prescribed under the *Petroleum Rules*, 1976 or as specified in [Table 1](#), whichever is more.

11.4.1.5 The number of storage tanks in one storage installation shall not exceed six. In case there are more than one storage installations, the safety distance between two installations shall be the same as the distance between the tanks and the property line in accordance with [Table 1](#).

Table 1 Minimum Safety Distances

(Clauses [11.4.1.2](#), [11.4.1.4](#) and [11.4.1.5](#))

Sl No.	LPG Storage Water Capacity of Individual Tank	Distance from Building/ Property Line		Distance between Tanks	
		Above Ground	Under Ground	Above Ground	Under Ground
	ℓ	m	m	m	m
(1)	(2)	(3)	(4)	(5)	(6)
i)	Up to 2 000	5	5	1	1.5
ii)	Above 2 000 and up to 10 000	10	7.5	1	1.5
iii)	Above 10 000 and up to 20 000	15	10	1.5	1.5
iv)	Above 20 000 and up to 40 000 adjacent	20	15	2	0.25 × diameter of vessel or 1.5 m (<i>Min</i>)
v)	Above 40 000 and above adjacent	30	15	2	0.25 × diameter of vessel or 1.5 m (<i>Min</i>)

NOTE — If the aggregate water capacity of a multi-tank installation is 40 000 ℓ or greater, the above minimum safety distances shall apply to the aggregate storage capacity rather than the capacity per individual storage tank.

11.4.2 Bunding

Since LPG is heavier than air, storage tank shall not be enclosed within bund walls. The accumulation of flammable liquid under LPG tanks shall be prevented by suitably slopping the ground.

11.4.3 Protection

11.4.3.1 To prevent trespassing or tampering, the area which includes tanks, direct fired vaporizers, pumping equipment and loading and unloading facilities shall be closed by an industrial type fence at least 2 m high along the perimeter of the safety zone. Any fence shall have at least two means of exit. Gates shall open outwards and shall not be self-locking.

11.4.3.2 When damage to LPG systems from the LPG tank lorry is a possibility, precautions against such damage shall be taken.

11.4.3.3 Underground tanks shall be protected from above ground loading by providing a suitable kerb to prevent a possible accidental damage to the tank and its fittings by LPG tank lorry.

11.4.4 Grass and Weed Removal

Road ignitable material, such as weeds, long grass or any combustible material shall be removed from an area within 3 m from the shell of any LPG tank of up to 2 000 litre water capacity, and within 6 m from the shell of larger tanks. If weed killers are used, chemicals which are a potential source of fire hazard shall not be selected for this purpose.

11.4.5 Warning Signs

No smoking or naked flames shall be permitted within the safety zone of the installation. Prominent notices to this effect shall be posted at access point. From accessibility point of view, the signs shall be in accordance with good practice [E-4(8)].

11.4.6 Fire Protection

The possibility of a major fire outbreak, leading to direct flame impingement of the storage tank, shall be minimized by sound engineering in plant design and layout, good operating practice, and proper education and training of personnel on both routine operations and on action to be taken in an emergency.

11.4.6.1 Water supply

Provision shall be made for an adequate supply of water and fire protection in the storage area according to the local hoses and mobile equipment, fixed monitors or by fixed spray systems which may be automatic. Control of water flow should be possible from outside any danger area.

11.4.6.2 Fire extinguishers

At least two dry chemical powder type fire extinguishers of 9 kg capacity each, conforming to the quality requirements in accordance with the accepted standard [E-4(5)], shall be installed at points of access to the storage installations. The fire extinguishers should be mounted keeping in view the ease of accessibility in accordance with good practice [E-4(9)] and Part F 'Fire and Life Safety' of this NBCS.

11.4.7 For detailed information regarding LPG bulk storage installations, reference may be made to good practice [E-4(10)].

12 USE OF PIPED NATURAL GAS (PNG)

12.1 The PNG infrastructure shall be in complete compliance with all the requirements of *Petroleum and Natural Gas Regulatory Board (Technical Standards and Specifications including Safety Standards for City or Local Natural Gas Distribution Network) Regulations, 2008*.

12.1.1 The user-group/housing society concerned (referred hereinafter in these clauses as 'society') should contact the local city gas distribution (CGD) entity at the stage of the design of the building. This ensures that all the adequate provisions are incorporated in the design itself, as also the compliance with all the safety standards stipulated/required by the regulations/gas entities, as applicable from time-to-time. This ensures avoidance/mitigation of any rework at a later stage, or even situations where incorporation of any changes may become impractical or not possible, thereby rendering the building devoid of PNG infrastructure.

12.1.2 The entire gas pipeline infrastructure proposed to be laid in the premises of the society shall be easily accessible in future by the representatives of the CGD entity. Provision should be made by the builder for the same. Builder/society should ensure communication of the same to the future customer and should incorporate it in their agreements with the customer.

12.1.3 No permanent/temporary structure shall be erected on the ground along the paths where gas pipeline exists underground.

12.2 The PNG infrastructure inside the premises of societies primarily comprises the following:

- a) Underground medium pressure (MP) pipeline networks, comprising mainly polyethylene pipes and fittings;
- b) Underground low pressure (LP) pipeline networks, comprising mainly polyethylene (PE) pipes and fittings;
- c) Service regulator modules;
- d) Above ground riser and lateral system, comprising mainly electric resistance welded GI pipes and fittings; and
- e) Pipeline inside premise and/or kitchen of individual customer, comprising various components like meter regulator, diaphragm gas meter, copper pipe, brass fittings, brass valves, etc.

The guidelines mentioned under **12.3** to **12.6** give the minimum requirements to be complied with.

12.3 Guidelines for MP and LP Pipeline Networks

12.3.1 Gas pipelines are required to be laid at a minimum top cover of 1 m from the finished surface level. Underground PE pipeline shall not be laid through the basement.

12.3.2 A minimum clearance of at least 300 mm is required to be maintained between the gas pipelines and any other obstacles/services.

12.3.3 Gas pipelines should not be laid closer to properties than as stipulated below:

<i>Sl No.</i>	<i>Pressure</i>	<i>Minimum Proximity</i>
(1)	(2)	(3)
i)	Up to 0.01 N/mm ²	0.25 m
ii)	Up to 0.4 N/mm ²	3.00 m

12.3.4 Laying of underground natural gas pipelines within the premises of society shall be in accordance with *Petroleum and Natural Gas Regulatory Board (Technical Standards and Specifications including Safety Standards for City or Local Natural Gas Distribution Network) Regulations, 2008*.

12.3.5 A dedicated utility corridor may be provided in the society for natural gas pipelines. However, if provision of dedicated utility corridor for natural gas pipeline is not feasible inside the society premises due to space constraints, then, provisions pertaining to safety inter-distances of natural gas pipelines with other utilities as mentioned in *Petroleum and Natural Gas Regulatory Board (Technical Standards and Specifications including Safety Standards for City or Local Natural Gas Distribution Network) Regulations, 2008* shall be followed.

12.3.6 In case of contradiction between the provisions of this NBCS and *Petroleum and Natural Gas Regulatory Board (Technical Standards and Specifications including Safety Standards for City or Local Natural Gas Distribution Network) Regulations, 2008*, then, provision of *Petroleum and Natural Gas Regulatory Board (Technical Standards and Specifications including Safety Standards for City or Local Natural Gas Distribution Network) Regulations, 2008* shall prevail.

12.4 Guidelines for Service Regulator Modules

The user-group/society should provide adequate open spaces in the premises for installation of gas equipment like regulating stations, which will be required to facilitate supply of PNG to the residents of the building/premise.

The identified location shall be such that it does not obstruct any vehicular movement and is at a safe location in the premise.

12.5 Guidelines for Above Ground Riser and Lateral System

12.5.1 The riser and lateral system supplying gas to multistoried residential buildings shall be in compliance with

the requirements of the PNGRB regulations.

12.5.2 Multi-occupancy buildings having eight floors or more should have purpose- built utility shafts/ducts for accommodating riser and lateral system, which should be preferred route for constructing/locating the riser.

12.5.3 Provision shall be made for safe access to the riser and lateral system by the representatives of the CGD entity for future maintenance and repair. In the case of a continuous shaft, a concrete slab or similar flooring shall be provided for maintenance work at each floor.

12.5.4 Ventilation

12.5.4.1 Adequate provision for natural ventilation shall be provided in the building where PNG infrastructure is laid, so as to prevent any potential gas leaks from accumulating in the atmosphere to an unsafe level. Mechanical ventilation shall not be used to achieve the required ventilation levels. The minimum levels of ventilation in riser's ducts/shafts shall be as given below:

<i>Sl No.</i>	<i>Cross Sectional Area (CSA) of Riser Duct</i>	<i>Minimum Free Area of Each High and Low Level Ventilation (Open Ventilator)</i>	<i>Typical Air Duct Size</i>	<i>Typical Ventilator Size</i>
	m ²	m ²	mm	mm
(1)	(2)	(3)	(4)	(5)
i)	Less than 0.01	0.002	Ø50	Ø70
ii)	0.01 to 0.375	0.010	Ø100	100 × 100
iii)	0.375 to 0.5	0.02	Ø150	100 × 200
iv)	0.5 to 7.5	0.05	Ø250	200 × 250
v)	More than 7.5	1 percent CSA of the duct	—	—

12.5.4.2 Where the duct has a single ventilator that is part of the architectural design of the building, the free area of the ventilator to outside area shall be a minimum of 0.04 m² or 4 percent of the CSA of the duct, whichever is greatest.

12.5.4.3 Pipe work shall be installed in an unventilated void.

12.5.4.4 The pipe may enter the building to gain access to the riser shaft through a basement or car parking area, provided the area is well ventilated.

12.5.4.5 Any duct containing a riser or lateral shall be ventilated either directly to outside air, or indirectly to outside air via an area that is normally occupied and is itself ventilated to outside.

12.5.4.6 The provision of ventilation should be through permanent ventilators within the area/duct containing pipe-work.

12.5.4.7 All flammable release sources shall be at a lower elevation than the bottom of the highest ventilation opening.

12.5.4.8 Pipes may run in the same duct (subject to availability of adequate clearance from other utilities and ventilation) as most other services, including hot and cold water services, heating pipes, electrical conduits and cables and pipes containing other fuels. However, the following services shall not be installed in the same duct as gas pipes:

- Ventilation ducts and vacuum pipes that operate at sub-atmospheric pressure and that are not of all-welded or all-brazed construction; and
- Services containing oxidizing or corrosive fluids.

12.5.5 There shall be provision for installing risers and laterals such that a minimum distance of 300 mm is ensured

from any electrical equipment or installations and associated excess current control boxes of fuse boxes, and 25 mm is ensured from the electricity supply and distribution cables and other metal piped services.

12.5.6 In case where the gas pipeline is required to cross through a wall, provisions (like core cutting) or holes of adequate size drilled through the wall/building structure, to accommodate an easy passage of the gas pipelines through them, shall be made. The elevation of these provisions/holes shall be near the roof and not near the floor of the room.

12.6 Guidelines for Pipeline Inside Premise and/or Kitchen of Individual Customer

12.6.1 Meters will normally be located inside the property on an outside wall with a minimum length of inlet pipe-work inside the property. Kitchen is the preferred room to locate the meter, thereby minimizing the length of the outlet pipe work.

12.6.2 The gas meter shall be installed in an area with adequate ventilation and at a proper location to facilitate safe and convenient meter reading and maintenance activities in future. A minimum ventilation area of 2 percent of the internal floor area of the room shall be provided in the room where the gas meter is located. The gas pipeline inside the kitchen shall be laid along the walls of the kitchen and in a ventilated area.

12.6.3 In case pipe work is required to be laid in solid floors inside the kitchen, adequate provisions (in the shortest possible route and a minimum of 75 mm screed to provide an approximate cover of at least 50 mm over the gas pipe) shall be provided to protect the pipeline against corrosion and failure caused by movement. This method of construction consists of a base concrete raft foundation onto which a cement screed is subsequently laid in order to obtain a satisfactory surface to accept the floor finish. Provision of preformed ducts with protective covers should be made to facilitate laying of gas pipeline. It is advisable to contact the CGD entity at the designing stage so that adequate provisions are incorporated in the design of the kitchen/floor plan to facilitate easy and safe installation of PNG system.

12.6.4 Installation pipe work shall not be placed within the cavity of a cavity wall. Every pipe passing through a cavity wall shall take the shortest possible practicable route and shall be sleeved.

12.6.5 In case of provision of sleeve, the same should be of a diameter that provides a sliding fit to the pipe allowing normal pipe expansion/contraction.

13 USE OF MEDICAL GAS PIPELINE SYSTEM (MGPS)

13.1 General

A medical gas pipeline system (MGPS) is installed to provide a safe, convenient and cost effective system for the provision of medical gases to the clinical and nursing staff at the point of use. It reduces the problems associated with the use of gas cylinders such as safety, portage, storage and noise.

MGPS generally handles distribution of oxygen, nitrous oxide, medical air, surgical air, medical vacuum only; and sometimes provides an aesthetic gas scavenging systems. All other gases used in hospitals have low volume use and are provided through cylinders at the point of use.

It is essential to ensure that there is no possibility of a cross-connection between any system and that all parts of each system to which connections can be made by users are gas specific.

Separate installations are provided for pathology, general laboratories and special laboratories. These are designated Laboratory Gases (Pipeline) Supply System (LGSS). LGSS is provided with valve controlled outlets and shall never be provided with medical gas terminal units.

Use and storage of medical gases is governed by *Indian Drugs and Cosmetics Act, 1940*.

13.2 Design and Installation

13.2.1 Pipelines Systems for Compressed Medical Gases and Vacuum

The design, installation, performance, documentation, testing and commissioning of pipeline systems for

compressed medical gases, gases for driving surgical tools and vacuum in healthcare facilities to ensure continuous delivery of the correct gas and the provision of vacuum from the pipeline system, shall be in accordance with the good practice [E-4(11)]. This is inclusive of the requirements for supply systems, pipeline distribution systems, control systems, monitoring and alarm systems and non-interchangeability between components of different gas systems.

13.2.2 Anaesthetic Gas Scavenging Disposal Systems

The design, installation, performance, documentation, testing and commissioning of an aesthetic gas scavenging disposal systems to ensure patient safety and to minimize exposure of the operator and other persons to an aesthetic gases and vapours, shall be in accordance with the good practice [E-4(12)]. This is inclusive of requirements for the power device, pipeline system, performance, non- interchangeability between key components and avoidance of cross connections between an aesthetic gas scavenging (AGS) disposal systems and medical gas and vacuum pipeline systems.

LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfillment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used for conformance with the requirements of the referred clauses in this NBCS.

	<i>IS No.</i>	<i>Title</i>
(1)	IS 14885 : 2022	Polyethylene pipes for the supply of gaseous fuels — Specification (<i>first revision</i>)
	IS 17544 : 2022	Multilayer (PE-AL-PE) plastic piping system for indoor gas installations with a maximum operating pressure up to and including 500 kPa (5 bar) — Specification
(2)	IS 18466 (Part 1) : 2024/ ISO 7396-1 : 2016	Medical gas pipeline systems: Part 1 Pipeline systems for compressed medical gases and vacuum (<i>first revision</i>)
	IS/ISO 7396-2 : 2007	Medical gas pipeline systems: Part 2 Anaesthetic gas scavenging disposal systems
(3)	IS 3196 (Part 1) : 2013	Welded low carbon steel cylinders exceeding 5 litres water capacity for low pressure liquefiable gases: Part 1 Cylinders for liquefied petroleum gases (LPG) — Specification (<i>sixth revision</i>)
	IS 7142 : 2025	Welded low carbon steel cylinders for low pressure liquefiable gases not exceeding 5 litre water capacity — Specification (<i>second revision</i>)
(4)	IS 8198 (Part 5) : 2022	Code of practice for steel cylinders for compressed gases: Part 5 Liquefied petroleum gas (LPG) (<i>second revision</i>)
(5)	IS 15683 : 2018	Portable fire extinguishers — Performance and construction — Specification (<i>first revision</i>)
(6)	IS 9798 : 2013	Low pressure regulators for use with liquefied petroleum gas (LPG) — Specification (<i>second revision</i>)
	IS 16280 : 2021	High pressure regulators and adaptors for use with liquefied petroleum gases (LPG) — Specification (<i>first revision</i>)
(7)	IS 6044 (Part 1) : 2018	Liquefied petroleum gas storage installations — Code of practice: Part 1 Residential, commercial and industrial cylinder installations (<i>third revision</i>)
(8)	IS 4963 : 2025	Accessibility in built environment for older adults and persons with disabilities — Requirements (<i>second revision</i>)
(9)	IS 2190 : 2024	Selection, installation and maintenance of first-aid fire extinguishers — Portable and mobile — Code of practice (<i>fifth revision</i>)
(10)	IS 6044 (Part 2) : 2021	Liquefied petroleum gas storage installations — Code of practice: Part 2 Commercial, industrial and domestic bulk storage installations (<i>second revision</i>)
(11)	IS 18466 (Part 1) : 2024/ ISO 7396-1 : 2016	Medical gas pipeline systems: Part 1 Pipeline systems for compressed medical gases and vacuum (<i>first revision</i>)
(12)	IS/ISO 7396-2 : 2007	Medical gas pipeline systems: Part 2 Anaesthetic gas scavenging disposal systems

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART F FIRE AND LIFE SAFETY

The Fire Services is a State subject and has been included as a Municipal function in the XIIth Schedule of the Constitution of India under Article 243(w). It is therefore primarily the responsibility of the State Governments to ensure safety of life and property in their area of jurisdiction. State/Union Territories (UTs) in the country are implementing fire safety measures either through the provisions as stipulated in their State Fire Services Act or through their building byelaws. The construction of any building and its fire safety thereof has to be ensured as per the norms and standards prescribed by the State Building Bye Laws/Development Control Regulations of the State/UTs.

The Department of Fire Services in the States/UTs is one of the agencies to issue NOC to the buildings from fire and life safety aspects and it is for the State/UT Governments to provide stringent norms for issuing NOC for the same.

This Part of NBCS is only for guidance and referral for the state government/local authority in respect of fire safety in buildings.

BUREAU OF INDIAN STANDARDS

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National Building Code Sectional Committee, CED 46

FOREWORD

The first version of this Part was formulated as Part 4, in the National Building Code of India 1970 (NBC 1970) and was revised in 1983. Subsequently the first revision of this Part was modified in 1997 through Amendment No. 3 to 1983 version of NBC. The modified version of this Part included a few tables for the fire resistance ratings of various building components, such as walls, columns, beams and floors. The requirements for wet riser, down-comer, automatic sprinkler installation, high velocity (10 m/s to 15 m/s) water spray or foam generating system, etc, for buildings were modified. Annex giving guidelines for selection of fire detectors had been deleted and relevant Indian Standards on fire alarm system and smoke detectors had been referred. Also, an annex for determination of fire loads and flue load density for arriving at the classification of occupancy hazard and calorific values of some common materials was included. Annex for broad classification of industrial and non-industrial occupancies into low, moderate and high hazard had also been included.

The NBC was then revised in 2005. In this revision, the significant modifications incorporated were: The text was divided into the following broad clauses: (a) *Fire prevention* – Covering aspects of fire prevention pertaining to design and construction of buildings on passive fire protection measures, also describing the various types of building materials and their fire rating; (b) *Life safety* – Covering life safety provisions in the event of fire and similar emergencies, also addressing construction and occupancy features that are necessary to minimize danger to life from fire, smoke, fumes or panic; (c) *Fire protection* – Covering the significant appurtenances and their related components and guidelines for selecting the correct type of equipment and installation meant for fire protection of the building, depending upon the classification and type of the building. Also, the classification of building based on occupancy was elaborated, wherein, starred hotels were covered as a new Subdivision A-6 under occupancy Group A Residential; heritage structures and archaeological monuments were covered under Subdivision D-3 occupancy Group D Assembly buildings; mixed assembly occupancies were covered as a new Subdivision D-6; underground elevated railways were covered as a new Subdivision D-7 under occupancy Group D Assembly buildings; and TV stations were covered under Subdivision E-5 of occupancy Group E Business buildings. Further, for the external stairs for exit requirements, the width and treads were increased to 1 250 mm and 250 mm, respectively; under the requirements for institutional buildings the clear width of all required exits which serve as egress from hospital or infirmary section was increased from 1.5 m to 2 m. Also, provision of patient-lift was included; and due cognizance of halon phase out programme was taken, while specifying provisions in this Part with respect to fire protection using fire extinguishers/systems. Based on various comments received from time-to-time, an Amendment No. 2 relating to Part 4 of the 2005 version of NBC, was issued.

In the 2016 revision of NBC, the following changes were made:

General:

- 1) The scope of application of provisions of this Part of NBC for different occupancies was clarified.
- 2) Definitions of various new terms were included and definitions of some of the existing terms were updated based on latest developments and use.

Fire Prevention:

- 3) Based on the experience in the use of the provisions of NBC in relation to various occupancies and subdivision classification under various building occupancy types, the same were reviewed and updated. Also, mixed occupancy and minor occupancy were further clarified.
- 4) Table 1 on fire resistance ratings of structural and non-structural elements was updated.
- 5) The provisions relating to fire separating walls, fire separating floors and fire partitions were detailed.
- 6) Provisions of fire safety requirements of services shafts were rationalized and updated.
- 7) A separate comprehensive clause on electrical power supply distribution for fire and life safety systems was included.
- 8) Detailed clauses on air conditioning systems towards safety and smoke control integration were provided.

- 9) Glass facade requirements were detailed towards fire protection and smoke exhaust aspects.
- 10) A separate comprehensive clause on Fire Command Centre (FCC) was introduced covering various requirements.

Life Safety:

- 11) The components of means of egress were comprehensively brought out covering specific aspects relating to exit access, exit and exit discharge. The relationship of occupant load, exit width requirements and travel distances were duly detailed enabling efficient planning for enhanced life safety provisions. The table on capacity factors was modified based on aspect of width per person approach used globally.
- 12) Requirement for displaying the occupancy load for assembly buildings and call centres, was included.
- 13) The concept of firefighting shaft for safe and efficient use by the fire fighters to access the floor on fire and also allow egress/evacuation of the occupants with simultaneous use of refuge area used as staging of the occupants, were well integrated, including in the annex for high rise buildings.
- 14) Aspects of compartmentation with fire barrier and its passive fire safety requirements were detailed for respective occupancies.
- 15) Pressurization of exits and smoke extraction requirements for respective areas including car parking were updated with an approach towards selecting alternative means therefor.
- 16) The clause on gas supply was comprehensively updated.
- 17) The clause on fire detection and alarm system was completely reviewed and updated as per the latest practices.

Fire Protection:

- 18) The table on minimum requirements for firefighting installations (erstwhile Table 23 renumbered as Table 7) was comprehensively updated.
- 19) Detailed provisions were included on fire water storage, fire pump room, sprinkler system and various alternative fire suppression systems.

Additional Occupancy-wise Requirements:

- 20) Concept of progressive evacuation in case of hospital buildings was included in detail to ensure life safety of the inmates.
- 21) Provisions relating to requirement of refuge area were updated including for D-6 occupancy and introduced for apartment buildings of height 60 m and above.
- 22) Separate provisions on atrium were included in Annex E.
- 23) Detailed separate provisions were included on commercial kitchens as an Annex.
- 24) Detailed separate provisions were included on car parking facilities in Annex G.
- 25) Separate provisions on fire and life safety requirements for metro stations and metro trainways were included in Annex J and Annex K, respectively.

In this revision, the publication is renamed as National Building Construction Standards (NBCS).

This NBCS (Part F) deals with safety from fire. It specifies the demarcation of fire zones, restrictions on construction of buildings in each fire zone, classification of buildings based on occupancy, types of building construction according to fire resistance of the structural and non-structural components and other restrictions and requirements necessary to minimize danger to life from fire, smoke, fumes, or panic before the buildings can be evacuated. The NBCS recognizes that safety of life is more than a matter of means of egress and accordingly deals with various matters which are considered essential to the safety of life. The NBCS therefore covers provisions relating to means of egress covering various components thereof namely exit access, exit, and exit discharge. It also covers provisions relating to fire protection of buildings through portable and fixed firefighting installations.

Fire protection techniques have to be based on the fire behaviour characteristics of different materials and structural elements of buildings. The activities pursued by the occupants of buildings must also be taken into consideration for assessing the extent of hazards, and method should then be devised by which the hazards could be minimized. An indefinite combination of variables is involved in the phenomenon of fire, all of which cannot be quantified. The requirements of this NBCS should, therefore, be taken as a guide and an engineering design approach should be adopted for ensuring a fire safe design for buildings. Depending upon the type and complexities in a building, qualified and trained fire protection engineers should be associated with the planning of buildings, so that adequate fire and life safety measures could be incorporated in the building design right from the beginning.

Absolute safety from fire is not attainable in practice. The objective of this Part is to specify measures that will provide that degree of safety from fire which can be reasonably achieved. The NBCS endeavours to avoid requirements that might involve unreasonable hardships or unnecessary inconvenience or interference with normal use and occupancy of buildings but insists upon compliance with minimum standards of fire safety necessary for building occupants and users. For ensuring compliance of fire protection equipment/installations to the laid down quality requirements, it is desirable to use such equipment/installation duly certified under the BIS, Certification Marks Scheme.

While providing guidelines for minimizing chances of occurrence of fire through passive fire protection measures, this Part does not intend to cover all aspects of general fire prevention including sources of ignition. Nor does it cover the prevention of accidental personal injuries from fire during the course of normal occupancy of buildings.

This Part while recognizing that panic in a building on fire may be uncontrollable, deals with the potential panic hazard through measures designed to prevent the development of panic. Experience indicates that panic seldom develops even in the presence of potential danger, so long as occupants of buildings are moving towards exits which they can see within a reasonable distance and with no obstruction or undue congestion in the path of travel. However, any uncertainty as to the location or adequacy of means of egress, the presence of smoke or fumes and the stoppage of travel towards the exit, such as may occur when one person stumbles and falls on stairs, may be conducive to panic. Danger from panic is greater when a large number of people are trapped in a confined area particularly when people are not adequately guided towards egress and safety notifications are not implemented or practiced. Consideration towards announcements and annunciations needs to be given to guide the occupants to safe egress routes and to control panic during situation of distress. These aspects are addressed in this Part.

Experience has shown that concealed spaces within a building, such as, space between ceiling and false ceiling, horizontal and vertical ducts, and shafts, etc, tend to act as flues/tunnels during a fire. Provisions therefore exist to provide fire stopping within such spaces.

Nothing in this Part of the NBCS should be construed to prohibit better types of building construction, more exits, or otherwise safer conditions than the minimum requirements specified in this Part.

Compliance with this Part should not be construed as eliminating or reducing the necessity for other provisions for safety of persons using a building or structure under normal occupancy conditions. Nor should any provision of this Part be construed as requiring or permitting any addition that maybe hazardous under normal occupancy conditions.

One of the major points brought out then in this Part (until NBC 2016) is the limitation of heights and areas of buildings based on fire safety of the occupants. Individual municipal corporations were free to alter the then Table 2 (of NBC 2016) based on local conditions, but the ratios of areas as maintained in the table for different occupancies and types of construction should be adhered to.

Advantage has been taken of the developments, particularly in fire resistance rating of materials, designating types of construction in a rational manner and relating the area limitations of different occupancies to different types of construction.

Halons (halogenated hydrocarbons) which exhibit exceptional firefighting and explosion prevention/suppression characteristics have been found to possess high ozone depleting potential. They come under Group II of Annex A of the Montreal Protocol on Substances that Deplete the Ozone Layer, the international environmental agreement for phasing out ozone depleting substances. Due to increasing evidence that the ozone layer is getting depleted at a faster rate than thought earlier, the NBCS continues to take into cognizance the need to promote use of halon alternatives as fire extinguishing media.

In this revision, the following significant changes have been made:

- a) Minor occupancies have been clarified, in [3.1.1.1](#);
- b) Mixed occupancy has been detailed in [3.1.11](#);
- c) Fire resistance rating of roofs located over 6.7 m has been updated in [Table 1](#);
- d) Occupant load factor for various occupancies has been clarified in [Table 2](#), also with respect to the net or gross floor area;
- e) Various provisions on life safety have been updated in [4](#);
- f) Travel distance values have been updated in [Table 4](#);
- g) Compartmentation related provisions have been comprehensively updated in [4.5](#);
- h) Fire detection and suppression systems based on the specific type and intensity of industrial hazards has been introduced under [5.4](#);
- j) [Table 7](#) on firefighting requirements for various occupancies has been reviewed and separately detailed (in [Table 7A](#) to [Table 7J](#)) for all the occupancies;
- k) Basis for water quantity required for fire-fighting are recommended for combined system protection (using sprinkler and hose) and similarly for hose protection system in [Table 7K](#) and [Table 7M](#), respectively;
- m) Fire protection of commercial kitchen has been elaborated through reference to the detailed Indian Standard, IS 18271 : 2023 'Fire safety in commercial kitchen — Guidelines';
- n) Water curtain requirement for basement compartmentation has been reviewed and deleted;
- p) [Annex H](#) on fire and life safety requirements for metro stations has been elaborated;
- q) [Annex J](#) on fire and life safety requirements for metro trainways has been updated;
- r) A new [Annex M](#) on performance based design has been introduced;
- s) A new [Annex N](#) on fire protection for specific industries is included as an informative guide; and
- t) Height restrictions of buildings have been eliminated.

All standards cross-referred to in the main text of this Part, are subject to revision. The parties to agreement based on this Part are encouraged to investigate the possibility of applying the most recent editions of the standards.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Part of the NBCS is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Part.

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Important Explanatory Note for Users of the NBCS

In any Part/Section of this NBCS, where reference is made to ‘**good practice**’ in relation to **design, constructional procedures or other related information**, and where reference is made to ‘**accepted standard**’ in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

- a) Good practices [F(4)] refers to the Indian Standards give at serial number (4) of the list of standards given at the end of this Part F, that is, IS 8758 : 2013 ‘Recommendations for fire precautionary measures in the construction of temporary structures and pandals (*second revision*); and
- b) Accepted standards [F(1)] refers to the Indian Standards given at serial number (1) of the list of standards given at the end of this Part F, that is, IS 3808 : 1979 ‘Method of test for non-combustibility of building materials (*first revision*)’.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART F FIRE AND LIFE SAFETY

1 SCOPE

1.1 This NBCS (Part F) covers the requirements for fire prevention, life safety in relation to fire, and fire protection of buildings. This NBCS (Part F) specifies occupancy-wise classification, constructional aspects, egress requirements and protection features that are necessary to minimize danger to life and property from fire.

1.2 The provisions of this Part are applicable to those high rise buildings having a height exceeding a certain value, or, to special buildings (having a certain area exceeded on any floor), as given below, unless otherwise mentioned specifically in this Part:

<i>Sl No.</i>	<i>Occupancy Type</i>	<i>Height beyond which Provisions of this Part are Applicable</i>	<i>Floor Area beyond which Provisions of this Part are Applicable</i>
(1)	(2)	(3)	(4)
i)	Residential	24 m	750 m ²
ii)	Educational	9 m	500 m ²
iii)	Institutional	15 m	500 m ²
iv)	Assembly	9 m	750 m ²
v)	Business	15 m	750 m ²
vi)	Mercantile	15 m	1 000 m ²
vii)	Industrial – Low hazard (G-1)	Any height and the total area of the building exceeds 2 000 m ²	
viii)	Industrial – Moderate hazard (G-2)	Any height and the total area of the building exceeds 2 000 m ²	
ix)	Industrial – High hazard (G-3)	15 m	500 m ²
x)	Storage	9 m	500 m ²
xi)	Hazardous	9 m	500 m ²
xii)	Mixed Use	15 m	1 000 m ²
NOTES 1 State/Local Authority may consider modifying the above values depending on the fire vulnerability and preparedness. 2 In the case of buildings (lesser than the height or area as above) with a single staircase, the same should be of firefighting shaft type and located only on the external periphery, with its lobby and stairwell both ventilated (having 1.6 m² minimum size opening through fixed louvres kept permanently open). A 2 h fire door for lobby and 1 h fire door for stairwell should be provided.			

1.3 Also, the provisions of this Part are applicable when buildings are altered (on an area 1 000 m² or more) or when change of occupancies are made to buildings.

1.4 The provisions of performance-based design ([Annex M](#)) are applicable only to validate the fire safety aspects for heritage buildings that are either made accessible to public or are converted to other building occupancy (and are constrained to provide exits as per the codal requirements, as that could damage the heritage nature). For the applicability of such performance-based approach to other buildings such as large assembly buildings, air traffic control towers, special museums, and existing buildings being modified for occupancy change, careful consideration of [Annex M](#) should be done by the state/local fire authorities on the basis of detailed performance-based calculations/report.

2 TERMINOLOGY

For the purpose of this Part, the following definitions will apply.

2.1 Assisted Evacuation — Strategy that exists during which a designated person or persons provide assistance, during an emergency, to another person(s) to leave a building or a specific part of the built environment and to reach a final place of safety.

2.2 Atrium — A large-volume space created by a floor opening or series of floor openings connecting two or more stories that is covered at the top of the series of openings and is used for purposes other than an enclosed stairway; lifts hoist-way; an escalator opening; or as a utility shaft used for plumbing, electrical, air conditioning, or communications facilities.

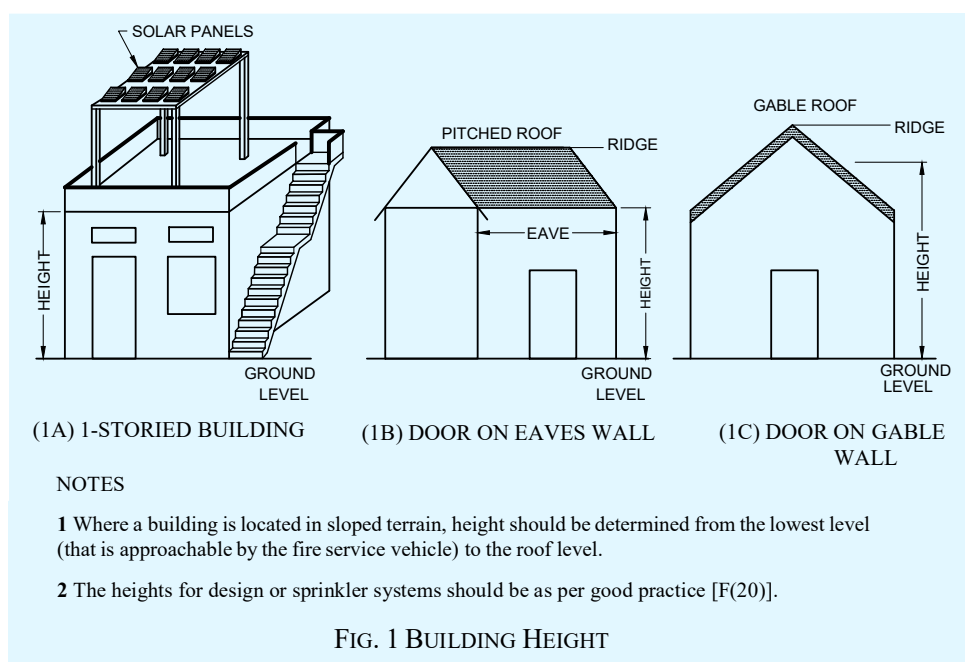
2.3 Authorities Concerned (known as Authority Having Jurisdiction) — An organization, office, or individual responsible for enforcing the requirements of a Code or standard, or for approving or accepting equipment, materials, an installation, or a procedure.

2.4 Automatic Fire Detection and Alarm System — A system comprising components and sub-systems required for automatically detecting smoke, heat or fire initiating an alarm and other actions as appropriate. This system also includes manually operated electronic fire alarm (MOEFA) system.

NOTE — MOEFA system (with or without automatic fire detection and alarm system) includes all or some of the components such as manual call stations (initiating an alarm for fire and other actions as required), talk-back system and public address system.

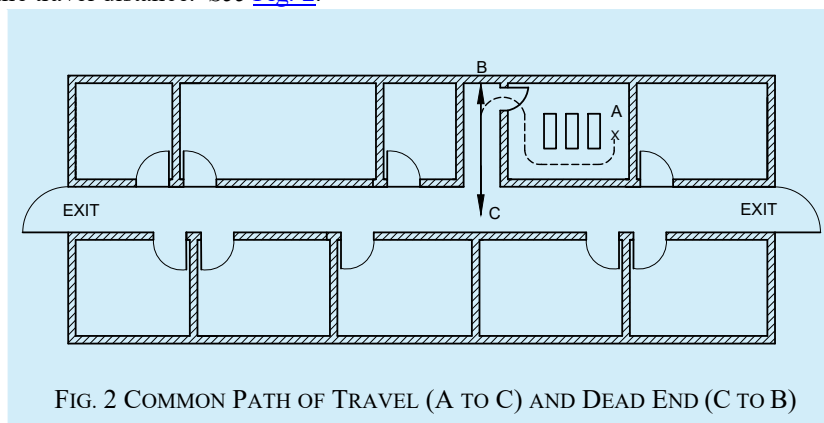
2.5 Building — Any structure for whatsoever purpose and of whatsoever materials constructed and every part thereof whether used as human habitation or not and includes foundation, plinth, walls, floors, roofs, chimneys, plumbing and building services, fixed platforms, veranda, balcony, cornice or projection, part of a building or anything affixed thereto, or any wall enclosing or intended to enclose any land or space and signs and outdoor display structures. Tents, *Shamianahs*, tarpaulin shelters, etc erected for temporary and ceremonial occasions with the permission of the Authority should not be considered as building.

2.6 Building, Height of — For the purposes of fire safety, the height of the multistoried building may be measured up to the floor level of the last occupiable floor from the grade level (where the fire tender/rescue vehicle will be located). In cases when the building height is measured up to the roof of the top-most floor, the following depictions may be considered. See [Fig. 1](#).



2.7 Combustible Material — A material which either burns itself or adds heat to a fire, when tested for non-combustibility in accordance with accepted standard [F(1)].

2.8 Common Path of Travel — That portion of the exit access which the occupants are required to traverse before two (or more) separate and distinct routes to two (or more) exits become available. Common paths of travel are part of the travel distance. See Fig. 2.



2.9 Covered Area (also Built-up Area)— Ground area covered by the building immediately above the plinth level. The area covered by the following in the open spaces is excluded from covered area:

- Garden, rockery, well and well structures, plant nursery, water pool, swimming pool (if uncovered), platform around a tree, tank, fountain, bench, *Chabutara* with open top and unenclosed on sides by walls and the like;
- Drainage culvert, conduit, catch-pit, gully pit, chamber, gutter, and the like;
- Compound wall, gate, unstoreyed porch and portico, slide, swing, uncovered staircases, ramp areas covered by *Chhajja* and the like; and
- Watchman's booth, pump house, garbage shaft, electric cabin or substations, and such other utility structures meant for the services of the building under consideration.

NOTE — For the purpose of this Part, covered area equals the plot area minus the area due for open spaces in the plot.

2.10 Down-comer — An arrangement of firefighting within the building by means of down-comer pipe connected to terrace tank through terrace pump, gate valve and non-return valve and having mains not less than 100 mm internal diameter with landing valves on each floor/landing. It is also fitted with inlet connections at ground level for charging with water by pumping from fire service appliances and air release valve at roof level to release trapped air inside.

2.11 Dry Riser — An arrangement of firefighting within the building by means of vertical rising mains not less than 100 mm internal diameter with landing valves on each floor/landing which is normally dry but is capable of being charged with water usually by pumping from fire service appliances.

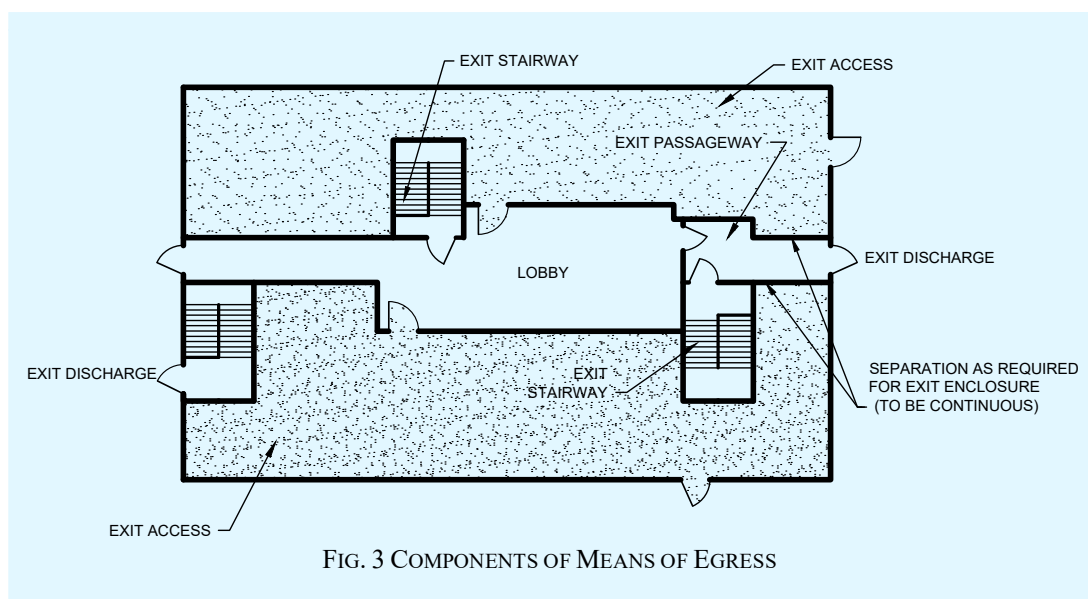
2.12 Emergency Lighting — Lighting provided for use when the supply to the normal lighting fails.

2.13 Emergency Lighting System — A complete but discrete emergency lighting installation also fed from the standby power source to the emergency lighting lamp(s), for example, self-contained emergency luminaire or a circuit from central battery (with or without monitoring system) connected through wiring to several escape lighting luminaires.

2.14 Escape Lighting — That part of the emergency lighting which is provided to ensure that the escape route is illuminated at all material times, for example, at all times when persons are on the premises, or at times the main lighting is not available, either for the whole building or for the means of egress.

2.15 Evacuation Lift — Lift that can be used, during an emergency, for self-evacuation.

2.16 Fire Exit — That unobstructed component of means of egress which is between the exit access and the exit discharge or public way. Exit components include exterior exit doors at the level of exit discharge, interior exit stairways (pressurised), exit stairway with external wall (pressurised or natural ventilated), exterior exit stairways, exit passageways, and exterior exit ramps (see Fig. 3). Fire exits should be provided with panic bar as applicable to facilitate entry of occupant into it.



2.17 Exit Access — That portion of a means of egress that leads to fire exit (for example, habitable floor area, doorways, staircase lobby, ramps, veranda, corridor, or passageway leading to an exit) (see Fig. 3).

2.18 Exit Access Corridor — A corridor in exit access which may not necessarily have the requirement of exits being met.

2.19 Exit Discharge — The component of a means of egress between the termination of an exit and a public way (see Fig. 3).

2.20 Fire Barrier (or Fire Resisting Barrier) — A fire barrier is a vertically or horizontally aligned member such as a wall or a fire curtain, or a floor. These may be with discontinuities created by openings with a specified fire resistance rating, where such members are designed and constructed with a specified fire resistance rating to limit the spread of fire that also restricts the movement of smoke.

2.21 Fire Compartment — A space within a building that is enclosed by fire barrier or fire resistant walls on all sides, including the top and bottom.

2.22 Fire Door and Fire Door Assembly — Any combination of fire door, frame, hardware and other accessories that together provide a specific fire resistant rating to the opening in terms of its stability, integrity and insulation properties, when installed in the openings in fire separation walls. Fire door {see accepted standard [F(33)]} is a component of fire door assembly.

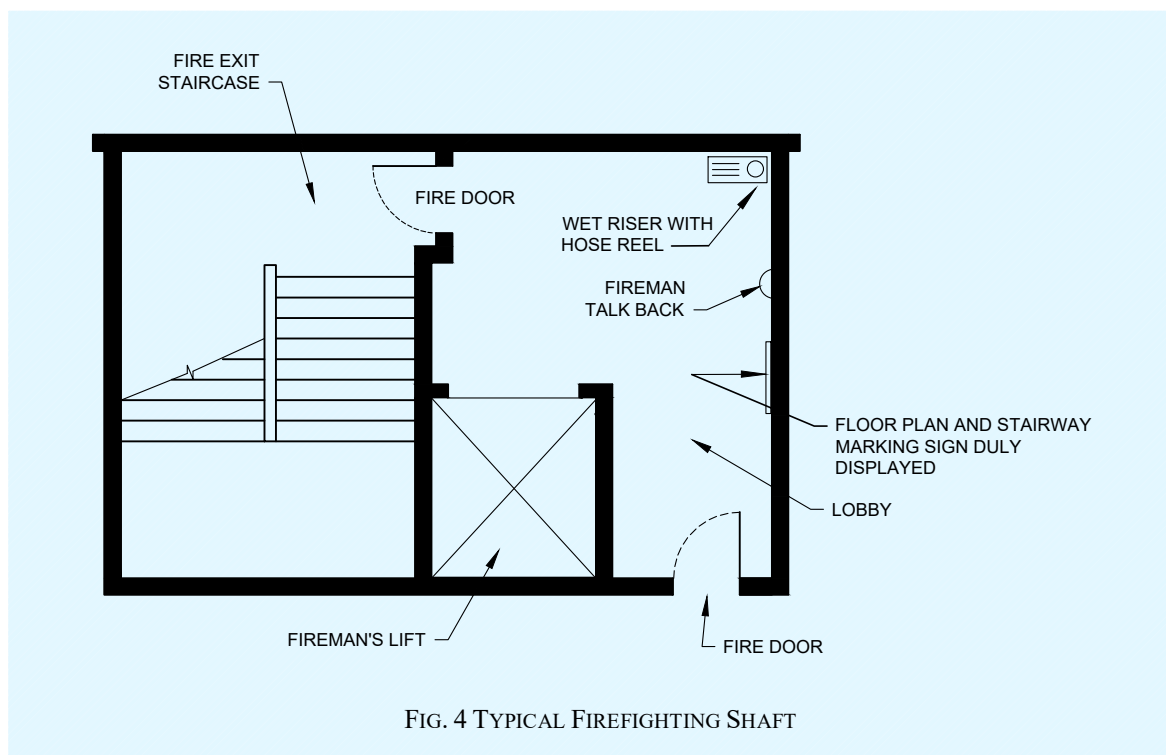
NOTES

- 1 Wherever reference has been made to fire door or fire check door in this Part, the same should be construed as fire door assembly.
- 2 Fire doors in exits should have fire rating as required in this Part to meet the requirement of integrity and stability; and the minimum insulation criteria should be 30 min.
- 3 Fire doors in exits should be provided with intumescent seal or such other globally accepted technologies/ devices (that expand upon contact with heat/hot gases, thereby forming an effective seal and aiding in integrity).
- 4 Fire doors in exits should not be allowed to be on hold open position and kept closed and to close by 'door closure - spring mechanism'.
- 5 Fire curtains should not be allowed as fire exits. If so provided for compartmentation, independent fire door should be provided meeting the requirement for fire door in exits as above (of the width as required) within the prescribed travel distance requirement.
- 6 Fire doors provided with panic bar should also be provided with the door handle on the other side on those levels to permit re-entry in designated floors.

2.23 Exit — A way out leading from exit access towards fire exit (fire exit is basically a protected staircase, see 2.16). Exits not leading to fire exits should be displayed with signage as 'NOT A FIRE EXIT'.

2.24 Firefighting Shaft (Fire Tower) — An enclosed shaft having a protected area of 120 min fire resistance rating comprising a protected lobby, staircase, and fireman's lift, connected directly to the exit discharge or

through an exit passageway with 120 min fire resistant wall at the level of exit discharge to exit discharge. These should also serve the purpose of exit requirement/strategy for the occupants. The lobby space should have arrangement to accommodate minimum two wheel chairs, and a minimum specified area for respective occupancy. The respective floors should be approachable from firefighting shaft enabling the fire fighters to access the floor and also enabling the fire fighters to assist in evacuation through fireman's lift. The firefighting shaft should be equipped with 120 min fire doors. The firefighting shaft should be equipped with firemen talk back, wet riser and landing valve in its lobby, to fight fire by fire fighters (see Fig. 4 for a typical firefighting shaft).



2.25 Fire Load — The total amount of energy that could be released during a fire in a given compartment, floor or building; usually expressed in Mega Joules (MJ) (or equivalent units).

2.26 Fire Load Density — Fire load per unit area of the compartment, floor or building concerned. It is obtained by dividing the fire load by the floor area of the concerned compartment, floor or building. It is usually expressed in megajoule (MJ) per m², or equivalent units with respect to wood, as below:

<i>Sl No.</i>	<i>Building Type</i>	<i>Fire Load Density (Expressed as Wood Equivalent kg/m²)</i>
(1)	(2)	(3)
i)	Residential (A)	25
ii)	Institutional (B) and Educational (C)	25
iii)	Assembly (D)	25 to 50
iv)	Business (E)	25 to 50
v)	Mercantile (F)	Up to 250
vi)	Industrial (G)	Up to 150
vii)	Storage (H) and Hazardous (J)	Up to 500

2.27 Fireman's Lift — A lift or a group of lifts invariably associated with all the features and requirements of a firefighting shaft. Such lift(s) are installed to enable fire services personnel to reach different floors with minimum delay, and meets the additional features as required in accordance with this Part. This lift also serves the purpose of meeting the requirement of evacuation lift for assisted evacuation.

NOTES

- 1 Lifts in firefighting shaft and fireman's lift are required to go to basements and the lobby should be pressurized.
- 2 Regular lifts serving floors above should terminate at the level of exit discharge. Lifts serving basements should be separate. In the case of lift bank, it is preferred to have all the lifts as fireman's lift instead of one or few lifts.
- 3 Lifts in the firefighting shaft should reach the terrace level, in the case of habitable spaces in terrace.

2.28 Fire Resistance — Fire resistance is a property of an element of building construction and is the measure of its ability to satisfy for a stated period, some, or all of the following criteria:

- a) *Load bearing capacity (Stability) (R)* – The ability of a load bearing element to withstand fire exposure without any loss of structural stability.
- b) *Integrity (E)* – Resistance to penetration of flame and hot gases.
- c) *Insulation (I)* – Resistance to temperature rise on the unexposed face up to a maximum of 180 °C at any single point and average temperature of 140 °C.

NOTE — Radiation (W) can be used as one of the criteria for glass used in building, which refers to the ability to resist the passage of heat from one side to another.

2.29 Fire Resistance Rating — The time that a material or construction will withstand the standard fire exposure as determined by fire test done in accordance with the standard methods of fire tests of materials/structures as per the accepted standard [F(2)].

NOTES

- 1 The requirement of rating of various building elements as given in this Part should be applicable in accordance with the provisions given in the accepted standard [F(2)].
- 2 The fire resistance rating should be specified in terms of minutes.
- 3 Fire resistance rating for non-structural material/assembly should bear a label of compliance to such rating as per the approval of competent authority based on testing and evaluation. The label should be permanently affixed to the material/assembly and may carry other relevant details such as name and type of the product, and manufacturer's details.

2.30 Fire Resistant Wall — Fire resistance rated wall, having opening(s) with specified fire resistant rating, which restricts the spread of fire from one part of a building to another part of the same building.

2.31 Fire Separation — The distance in metre, measured from the external wall of the building concerned to the external wall of any other building on the site, or from other site, or from the opposite side of street or other public space for the purpose of preventing the spread of fire.

2.32 Fire Stop — A fire resistant material, or construction, having a fire resistance rating of not less than the fire separating elements, installed in concealed spaces or between structural elements of a building to prevent the spread/propagation of fire and smoke through walls, ceilings, and the like as per the laid down criteria. See accepted standard [F(34)].

NOTES

- 1 Fire stop assembly for through penetrations is a combination of firestop compatible for use with the penetrant, penetration items such as cables, cable tray, conduits, ducts, pipes, etc, and their means of support through the wall or opening that together restores the fire resistance rating of the fire separating elements in terms of its integrity and/or insulation properties.
- 2 Fire stop assembly for joints is the one where fire stop with movement capability is used to seal the linear joints between adjacent fire separating elements, to maintain the fire resistance of the separating elements, which should be installed within its tested design limits with regard to size of the joint, type of assembly, and anticipated compression and extension of the joint.

2.33 Fire Suppression Systems

- a) *Gas Based Systems* — Systems that use gaseous agents as fire suppression media, such as, all agents alternate to halon gases, as conforming to relevant Indian Standards; other methods/types of gas based systems where their protection is equal to or better than what is suggested above for the type of application subject to the acceptance of Authorities concerned may also fall under such systems; and

- b) *Water Based Systems* — Systems that use mainly water as firefighting media such as hydrant system, sprinkler system, water spray system, foam system and water mist system.

2.34 Fire Wall or Fire Separating Wall — A fire resistance rated wall having fire protected openings, which restricts the spread of fire and extends continuously from the foundation to the roof (and through the roof at least 1 m above the roof in case of combustible roof), with sufficient structural stability under fire conditions to allow collapse of construction on one side or either side without collapse of the wall.

2.35 Floor Area (Gross) — The area of the floor within the inside perimeter of the outside walls of the floor of the building under consideration with no deductions for corridors and passage-ways, stairs, closets, thickness of interior walls, columns, lifts and building shafts or other features. Typically used for low density occupant load calculations.

2.36 Floor Area (Net) — The area of the floor within the inside perimeter of the outside walls of the floor of the building under consideration obtained after deducting the areas of corridors and passage-ways, stairs, closets (toilets and bathrooms), thickness of interior walls, columns, lifts, building shafts, services (floor mounted AHU, electrical, and the like), area of atrium, and escalator cut out. Typically used for high density occupant load calculations thus, net floor area is not the carpet area.

2.37 Floor Area Ratio (FAR) — The quotient obtained by dividing the total covered area (plinth area) on all floors by the area of the plot:

$$\text{FAR} = \frac{\text{Total covered area of all floors}}{\text{Plot area}}$$

2.38 Fire Exit Hardware — A door-latching assembly incorporating an actuating member or panic bar that releases the latch bolt upon the application of a force in the direction of egress travel, provided on exits.

2.39 High Rise Building — A building 24 m or above in height (irrespective of its occupancy).

2.40 Horizontal Exit — A defend in place or a staging arrangement, providing safety from fire and smoke originating from the area of incidence, by allowing alternative egress from a compartment to an area of refuge or another compartment at or near the same level. This also includes such egress from a compartment to an adjoining building. A horizontal exit is thus through a fire door of 120 min rating in a fire-resistant wall. Horizontal exit requires separation from the refuge area or adjoining compartment through 120 min fire barrier. The adjoining compartment of the horizontal exit should allow for unlocked and ease of egress and exits for the occupants using defend in place strategy.

2.41 Internal (Interior) Exit Stairway — A stairway that is enclosed with a fire-rated enclosure (towards building occupancy) in order to provide a protected path between the exit access and exit discharge. This enclosure must extend directly to the exterior at grade; extend through an exit passageway to grade; or comply with one of the allowances for exit discharge through a lobby, vestibule or horizontal exit.

2.42 External (Exterior) Exit Stairway — A stairway that is located on an exterior wall of the building, is open to external environment on at least one side and meets the requirements for openness as specified under [4.4.2.4.3.4](#), and is separated from the interior of the building by construction with the fire resistance rating required for enclosed stairs. The external exit stair leads directly to an open courtyard or public way at the level of exit discharge.

2.43 Lift Lobby — A space from which people directly enter a lift car(s) and into which people directly enter upon exiting a lift car(s).

2.44 Means of Egress — A continuous way of travel from any point in a building or structure to a public way, consisting of three separate and distinct parts, that is, exit access, exit, and exit discharge.

2.45 Means of Escape — A way out of a building or structure that does not conform to the strict definition of 'means of egress' but does provide an alternate way out.

2.46 Metro Station

2.46.1 Concourse — Intermediate level(s) or area(s) connecting a station platform(s) to a public way through stairs, escalators, or corridors.

2.46.2 Crush Train Load — The number of passengers inside a train when it is filled to maximum capacity permissible by rolling stock design.

2.46.3 Entraining Load — The number of passengers boarding a train at a platform.

2.46.4 Headway — The interval of time between the arrivals of consecutive trains at a platform in a station.

2.46.5 Mass Rapid Transit — Any station building or part thereof, permanent, or temporary, through which people transit for the duration of time required to enter the building and board the train to depart the station platform or to alight from the train and depart from the station building.

2.46.6 Non-transit Occupancy — Occupancy not under the control of the system operating authority.

2.46.7 Point of Safety — One of the following: (a) An enclosed exit that leads to a public way or safe location outside the station, trainway, or vehicle, (b) An at-grade point beyond the vehicle, enclosing stations, or trainway, (c) A point on open track beyond the open or enclosed station or enclosed train-way, and (d) Any other location approved by the Authorities concerned.

2.46.8 Station — A place designated for the purpose of loading and unloading passengers, including service area and ancillary spaces associated with the same structure.

2.46.8.1 Composite station — A transit station that is constructed contiguous with non-transit occupancy.

2.46.8.2 Enclosed station — A station or portion thereof that does not meet the definition of an open station.

2.46.8.3 Open station — A station that is constructed such that it is directly open to the atmosphere, and smoke and heat are allowed to disperse directly into the surrounding open atmosphere.

2.46.9 Station Platform — The area of a station immediately adjacent to a guideway, used primarily for loading and unloading passengers.

2.46.10 Link Load — The number of passengers travelling between two stations on board a train or trains during peak headway.

2.46.11 Incidental Occupancy within Station — An occupancy within a station that provides additional services which are not directly related to transportation operation or transportation services.

2.47 Mixed Occupancy — A multiple occupancy where the occupancies are intermingled.

2.48 Multiple Occupancy — A building or structure in which two or more classes of occupancy exist.

2.49 Occupancy or Use Group — The principal occupancy for which a building or a part of a building is used or intended to be used. For the purpose of classification of a building according to the occupancy, an occupancy should be deemed to include subsidiary occupancies which are contingent upon it.

2.50 Occupant Load — Maximum number of persons that might occupy a building or portion thereof at any one time.

2.51 Place of Comparative Safety — Places within a building where people can stay little longer until evacuation, for example, refuge areas, terraces, fire/smoke separated compartments, etc.

2.52 Pressurization — The establishment of a pressure difference across a barrier to protect exit, stairway, lobby, exit passageway or room of a building from smoke penetration.

2.53 Performance Based Design — An engineering approach to fire protection design based on (1) agreed upon fire safety goals and objectives, (2) deterministic and/or probabilistic analysis of fire scenarios, and (3) quantitative assessment of design alternatives against the fire safety goals and objectives using accepted engineering tools, methodologies, and performance criteria.

2.54 Pressurization Level — The pressure difference between the pressurized space and the adjoining area served by the pressurized space expressed in Pascal (Pa).

2.55 Public Way — A street, alley, or other similar parcel of land essentially open to the outside air, dedicated,

or otherwise permanently appropriated to the public for public use and having a clear width and height of not less than 3 m.

2.56 Ramp — The construction, in the form of an inclined plane that is steeper than or equal to 1 : 20 (5 percent) from the horizontal, together with any intermediate landing, that makes it possible to pass from one level to another.

2.57 Refuge Area — An area within the building for a temporary use during egress. It generally serves as a staging area which is protected from the effect of fire and smoke. The lift lobby spaces provided with adequate area (called refuge spaces) on all floors within the firefighting shaft can also serve as an alternate to refuge areas.

NOTE — Such refuge areas/spaces on every floor of firefighting shaft's lift lobby provide staging opportunity to those persons residing on any floor and are incapable of rescuing out of the building themselves.

2.58 Roof Exits — A means of escape on to the roof of a building, where the roof has access to it from the ground through alternative staircase or adjacent building.

2.59 Separated Occupancy — A multiple occupancy where the occupancies are separated by fire resistance-rated assemblies.

NOTE — Separate occupancy of a building is not incidental to the principal/main occupancy of that building.

2.60 Site (Plot) — A parcel (piece) of land enclosed by definite boundaries.

2.61 Smoke Barrier — A continuous membrane, or a membrane, where such membrane is designed and constructed to restrict the movement of smoke.

2.62 Smoke Compartment — A space within a building enclosed by smoke barriers on all sides.

2.63 Stack Pressure — Pressure difference caused by a temperature difference creating an air movement within a duct, chimney, or enclosure.

2.64 Travel Distance — The distance to be travelled from any point in a building to a protected exit or external escape route or final exit measured along the line of travel.

2.65 Ventilation — Supply of outside air into, or the removal of inside air from, an enclosed space.

2.66 Venting Fire — The process of facilitating heat and smoke to leave a building as quickly as possible by such paths so that lateral spread of fire and heat is checked, firefighting operations are facilitated, and minimum fire damage is caused.

2.67 Visual Strobes/Flashing — It is an audio-visual fire alarm for alerting persons with hearing impairment with flashing light. The strobe frequency should be from 0.5 Hz to 4.0 Hz, or as may be decided by the State Fire Authority, subject to performance required in smoke scenario.

NOTE — Care should be taken to ensure that overlapping strobes do not combine to result in a higher frequency of flashing.

2.68 Volume to Plot Area Ratio (VPR) — The ratio of volume of building measured in cubic metre to the area of the plot measured in square metre and expressed in metre.

2.69 Water Based Systems

2.69.1 Hydrant System — A distribution system having a network of piping installed underground/above-ground around and/or through inside of a building with internal and/or external hydrants fitted with landing valves at regular intervals according to the occupancy. The distribution system is connected to water supply system for firefighting.

2.69.2 Automatic Sprinkler System — A system of water pipes fitted with sprinkler heads at suitable intervals and heights and designed to actuate automatically, control and extinguish a fire by the discharge of water.

2.69.3 Automatic Water Spray Systems — A special fixed pipe system connected to a reliable source of fire protection water supply and equipped with water spray nozzles for specific water discharge and distribution over the surface or area to be protected. The piping system is connected to the water supply through an automatically actuated deluge valve which initiates flow of water. Automatic actuation is achieved by operation of automatic detecting equipment installed along with water spray nozzles. There are two types of systems namely high

velocity and medium velocity systems.

2.69.4 Water Mist Systems — A distribution system connected to a pumping and water supply system that is equipped with nozzles capable of delivering water mist to the part/entire enclosure or area, intended to control, suppress, or extinguish fire and is capable of meeting the specified performance requirements.

2.69.5 Foam Protection System — Firefighting systems where foam is made by mechanically mixing air with a solution consisting of fresh water to which a foaming agent (liquid concentrate) has been added. Firefighting foam is a stable aggregation of small bubbles of density lower than oil or water, and shows tenacious qualities for covering horizontal surfaces. There are three types of foam applications, that is, low, medium, and high expansion foams depending upon the application.

2.70 Wet Riser — An arrangement for firefighting within the building by means of vertical rising mains not less than 100 mm nominal diameter with landing valves on each floor/landing for firefighting purposes and permanently charged with water from a pressurized supply.

NOTE — For definition of other terms, reference be made to accepted standards [F(3)].

3 FIRE PREVENTION

3.1 Classification of Buildings Based on Occupancy

3.1.1 General Classification

All buildings, whether existing or hereafter erected, should be classified according to use or the character of occupancy in one of the following groups:

Group A: Residential

Group B: Educational

Group C: Institutional

Group D: Assembly

Group E: Business

Group F: Mercantile

Group G: Industrial

Group H: Hazardous

Group J: Storage

Group K: Mixed use

The details of each occupancy and example of buildings in each group are given in [3.1.2](#) to [3.1.11](#).

3.1.1.1 Minor occupancy

This is purely incidental to operations in a main occupancy, which should be considered as part of the main occupancy and should be classified under the relevant group for the main occupancy. Minor occupancy should not exceed 30 percent of the floor area. When the area exceeds 30 percent, it should be categorized as mixed occupancy ([3.1.11](#)).

3.1.1.2 Mixed occupancy

Where two or more types of occupancies intermingle in the same building, the entire building should be treated as mixed occupancy and the same should comply with [3.1.11](#).

3.1.2 Group A: Residential Buildings

These include any building in which sleeping accommodation is provided for normal residential purposes with or

without cooking or dining or both facilities, except any building classified under Group C.

Buildings and structures under Group A are subdivided as follows:

- a) *Subdivision A-I Lodging and rooming houses* — These include any building or group of buildings under the same management, in which separate sleeping accommodation on a transient or permanent basis, with or without dining facilities but without cooking facilities for individuals is provided. This includes inns, clubs, motels, and guest houses. If sleeping accommodation for more than 20 persons is provided in any one residential building, it is classified here OR as a building in Subdivision A-III as the case may be.

NOTE — A lodging or rooming house is classified as a one or two family dwelling, if no room in any of its private dwelling units is rented to more than three persons.

- b) *Subdivision A-II Dormitories* — These include any building in which group sleeping accommodation is provided, with or without dining facilities for persons who are not members of the same family, in one room or a series of closely associated rooms under joint occupancy and single management, for example, school and college dormitories, students, and other hostels and military barracks.
- c) *Subdivision A-III Apartment houses* — These include any building or structure in which living quarters are provided for three or more families, living independently of each other and with independent cooking facilities, for example, apartment houses, mansions and Chawls.
- d) *Subdivision A-IV Hotels* — These include any building or group of buildings under single management, in which sleeping accommodation is provided, with or without dining facilities for hotels classified up to Four Star Category.
- e) *Subdivision A-V Starred hotels* — These include the hotels duly approved by the concerned authorities as Five Star and above hotels.

3.1.3 Group B: Educational Buildings

These include any building used for school, college, other training institutions involving assembly for instruction, education, or recreation for not less than 20 students.

In the case of temporary buildings/structures which are utilized for educational purposes, the provisions of [3.2.3](#) apply.

If residential accommodation is provided in the schools/institutions, that portion of occupancy should be classified as a building in Subdivision A-II.

3.1.4 Group C: Institutional Buildings

These include any building which is used for purposes, such as medical or other treatment or care of persons suffering from physical or mental illness, disease, or infirmity; care of infants, convalescents, or aged persons and for penal or correctional detention in which the liberty of the inmates is restricted. Institutional buildings ordinarily provide sleeping accommodation for the occupants.

Buildings and structures under Group C are further subdivided as follows:

- a) *Subdivision C-I Hospitals and sanatoria* — This subdivision includes any building or a group of buildings under single management, which is used for housing persons suffering from physical limitations because of health or age and those incapable of self-preservation, for example, hospitals, infirmaries, sanatoria, and nursing homes;
- b) *Subdivision C-II Custodial institutions* — This subdivision includes any building or a group of buildings under single management, which is used for the custody and care of persons, such as children, convalescents and the aged who are incapable of self-preservation, for example, homes for the aged and infirm, convalescent homes and orphanages; and
- c) *Subdivision C-III Penal and mental institutions* — This subdivision includes any building or a group of buildings under single management, which is used for housing persons under restraint, or who are detained for penal or corrective purposes, in which the liberty of the inmates is restricted, for example, jails, prisons, mental hospitals, mental sanatoria and reformatories.

3.1.5 Group D: Assembly Buildings

These include any building where not less than 50 persons congregate or gather for amusement, recreation, social, religious, patriotic, civil, travel and similar purposes, for example, shopping malls, theatres/multiplexes, motion picture houses; assembly halls; auditoria; exhibition halls; museums; skating rinks; gymnasiums; restaurants/food courts; places of worship; dance halls; club rooms; passenger stations and terminals of air, surface and marine public transportation services; stadia; and underground and elevated railways.

3.1.6 Group E: Business Buildings

These include any building which is used for transaction of business for keeping of accounts and records and similar purposes, professional establishments, service facilities, etc, city halls, town halls, courthouses and libraries are classified in this group so far as the principal function of these is transaction of public business and keeping of books and records. These occupancies also include laboratories, outpatient clinics, research establishments, libraries, test houses, electronic data processing centres, computer installations, information technology parks and call centres; telephone exchanges; broadcasting stations, T.V. stations, and air traffic control towers; and datacentres.

Buildings under Group E are further subdivided as follows:

- a) Subdivision E-I Business buildings (human occupied); and
- b) Subdivision E-II Datacentres (machine environment).

3.1.7 Group F Mercantile Buildings

These include any building which is used as shops, stores, market, for display and sale of merchandise, either wholesale or retail.

Mercantile buildings are further subdivided as follows:

- a) Subdivision F-I Shops, stores, departmental stores, markets; and
- b) Subdivision F-II Underground shopping centres.

3.1.8 Group G: Industrial Buildings

These include any building or structure, in which products or materials of all kinds and properties are fabricated, assembled, manufactured, or processed, for example, assembly plants, industrial laboratories, dry cleaning plants, power plants, generating units, pumping stations, fumigation chambers, laundries, buildings or structures in gas plants, refineries, dairies and saw-mills, etc.

Buildings under Group G are further subdivided as follows:

- a) Subdivision G-I Buildings used for low hazard industries;
- b) Subdivision G-II Buildings used for moderate hazard industries; and
- c) Subdivision G-III Buildings used for high hazard industries.

The hazard of occupancy, for the purpose of this NBCS, should be the relative danger of the start and spread of fire, the danger of smoke or gases generated, the danger of explosion or other occurrences potentially endangering the lives and safety of the occupants of the buildings.

Hazard of occupancy should be determined by the Authority on the basis of the fire loads of the contents, and the processes or operations conducted in the building, provided, however, that where the combustibility of the material, the flame spread rating of the interior finish or other features of the building or structure are such as to involve a hazard greater than the occupancy hazard, the greater degree of hazard should govern the classification.

For determination of fire loads for arriving at the classification of occupancy hazard, guidance on calorific values of some common materials is given at [Annex A](#).

A broad classification of industrial occupancies based on 'fire hazard only' into low, moderate, and high hazard

classes is given at [Annex B](#), for guidance. Any occupancy not covered in [Annex B](#), should be classified in the most appropriate class depending on the degree of hazard.

NOTE — This [Annex B](#) is only based on 'fire hazard'. Other classification of industries based on chemical hazard, pollution, etc, are not addressed in this NBCS.

Where different degrees of hazard of occupancy exist in different parts of a building, the most hazardous of those should govern the classification for the purpose of this NBCS, except in cases where hazardous areas are segregated or protected as specified in this NBCS.

- a) *Subdivision G-I* — This subdivision includes any building in which the contents are of such comparative low combustibility and the industrial processes or operations conducted therein are of such a nature that there is hardly any possibility for any self-propagating fire to occur and the only consequent danger to life and property may arise from panic, fumes or smoke, or fire from some external source;
- b) *Subdivision G-II* — This subdivision includes any building in which the contents or industrial processes or operations conducted therein are liable to give rise to a fire which will burn with moderate rapidity or result in other hazardous situation and may give off a considerable volume of smoke, but from which neither toxic fumes nor explosions are to be feared in the event of fire; and
- c) *Subdivision G-III* — This subdivision includes any building in which the contents or industrial processes or operations conducted therein are liable to give rise to a fire which will burn with extreme rapidity or result in other hazardous situation or from which poisonous fumes or explosions are to be feared in the event of a fire.

3.1.9 Group H: Storage Buildings

These include any building primarily for the storage or sheltering (including servicing, processing or repairs incidental to storage) of goods, ware or merchandise (except those that involve highly combustible or explosive products or materials), vehicles or animals, for example, warehouses, cold storages, freight depots, transit sheds, storehouses, truck and marine terminals, garages, hangars, grain elevators, barns and stables. Storage properties are characterized by the presence of relatively small number of persons in proportion to the area. Any new use which increases the number of occupants to a figure comparable with other classes of occupancy should change the classification of the building to that of the new use, for example, hangars used for assembly purposes, warehouses used for office purposes, garage buildings used for manufacturing.

3.1.10 Group J: Hazardous Buildings

These include any building which is used for the storage, handling, manufacture or processing of highly combustible or explosive materials or products which are liable to burn with extreme rapidity and/or which may produce poisonous fumes or explosions for storage, handling, manufacturing or processing which involve highly corrosive, toxic or noxious alkalis, acids or other liquids or chemicals producing flame, fumes and explosive, poisonous, irritant or corrosive gases; and for the storage, handling or processing of any material producing explosive mixtures of dust which result in the division of matter into fine particles subject to spontaneous ignition. Examples of buildings in this class are those buildings which are used for,

- a) storage, under pressure of more than 0.1 N/mm² and in quantities exceeding 70 m³, of acetylene, hydrogen, illuminating and natural gases, ammonia, chlorine, phosgene, sulphur dioxide, carbon dioxide, methyl oxide and all gases subject to explosion, fume or toxic hazard, cryogenic gases, etc;
- b) storage and handling of hazardous and highly flammable liquids, liquefiable gases like LPG, rocket propellants, etc;
- c) storage and handling of hazardous and highly flammable or explosive materials (other than liquids); and
- d) manufacture of artificial flowers, synthetic leather, ammunition, explosives, and fireworks.

NOTE — A list of hazardous substances giving quantities, for which or exceeding which owners handling such substances are required to be covered under The *Public Liability Insurance Act*, 1991, has been notified under the 'Rules on Emergency Planning, Preparedness and Response for Chemical Accidents' by the Government of India, Ministry of Environment and Forests Notification No. G.S.R. 347(E) dated 01 August 1996.

3.1.11 Group K: Mixed Occupancy

3.1.11.1 These include any building having multiple occupancy where the occupancies are intermingled. The following table guides on the possible mixing of occupancies (two or more occupancies may be mixed) in a building.

Sl No.	Occupancy to be Mixed with	Minimum Required Fire Resistance Rating for the Separation between Any Two Occupancies in min										
		Residential	Educational	Institutional	Assembly	Business	Mercantile	Industrial			Storage	Hazardous
								Low (G-1)	Moderate (G-2)	High (G-3)		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
i)	Residential	–	120	120	120	120	120	120	NP	NP	NP	NP
ii)	Educational	120	–	120	120	120	120	NP	NP	NP	NP	NP
iii)	Institutional	120	120	–	120	120	120	NP	NP	NP	NP	NP
iv)	Assembly	120	120	120	–	120	120	120	120	NP	NP	NP
v)	Business	120	120	120	120	–	120	120	120	240	NP	NP
vi)	Mercantile	120	120	120	120	120	–	120	120	NP	120	NP
vii)	Industrial – Low (G-1)	120	NP	NP	120	120	120	–	120	240	120	NP
viii)	Industrial – Moderate (G-2)	NP	NP	NP	120	120	120	120	–	240	180	240
ix)	Industrial – High (G-3)	NP	NP	NP	NP	240	NP	240	240	–	240	240
x)	Storage	NP	NP	NP	NP	NP	120	120	180	240	–	240
xi)	Hazardous	NP	NP	NP	NP	NP	NP	NP	240	240	240	–
NP – Not permitted, being highly dangerous/vulnerable.												

3.1.11.2 For mixed occupancy building typology as above, if these co-exist in a building (such as large retail occupancy and business occupancy), in so far as fire protection is concerned, the entire building should be governed by the most restrictive provisions of this NBCS among those applicable for individual occupancies. However, the exits can be shared between mixed occupancies provided its aggregate provision meets the requirement of the mixed occupancy, and should be permitted if such exits are with unrestricted egress of the occupants based on the business operational hours of the occupancies sharing such exits. If such unrestricted egress of the mixed-use occupancy is not planned, each occupancy should be provided with its independent exits.

Exits in such mixed occupancy should be so arranged as to ensure that means of egress is not decreased in the direction of egress travel.

Further, in such mixed occupancies, the occupancies are also required to be separated (horizontally and/or vertically as the case may be) by fire resistance rating, see [3.1.11.1](#).

3.1.12 Where change in the occupancy of any building places it in a different group or in a different subdivision of the same group, such building should be made to comply with the requirements of this NBCS for the new group or its subdivision.

3.1.13 Where the new occupancy of a building is less hazardous, based on life and fire risk, than its existing occupancy, it should not be necessary to conform to the requirements of this NBCS for the new group or its subdivision.

3.1.14 A certificate of occupancy should be necessary, before any change is effected in the character of occupancy of any building.

3.2 Temporary Buildings or Structures

3.2.1 Temporary buildings and structures may be permitted according to the purpose for which these are to be used, by special permit from the Authority for a limited period and subject to such conditions as may be imposed in the permit.

3.2.2 Such buildings and temporary structures should be completely removed on the expiry of the period specified in the permit.

3.2.3 Adequate fire precautionary measures in the construction of temporary structures and *Pandals* should be taken in accordance with good practice [F(4)].

3.3 Existing Buildings

The existing buildings need not be required to comply with the requirements of this part unless these are altered, or in the opinion of the Local Authority, such building constitutes a hazard to the safety of the adjacent property or to the occupants of the building itself or is an unsafe building. In the event of alteration, it should be necessary to obtain permission of the Authority for such alteration consistent with fire hazard.

Alterations/modifications/renovations should be accomplished so as to ensure conformity with all the safety requirements of the new buildings. Such alterations should not in any way bring down level of fire and life safety below that which existed earlier. Any addition or alterations or construction of cubicles or partitioning (that will effect/increase the travel distance requirement), changes in the typology of the occupancy or densification of an existing occupancy for floor area exceeding 1 000 m² should be with the approval of local fire authority.

3.4 Types of Construction

3.4.1 The design of any building and the type of materials used in its construction are important factors in making the building resistant to a complete burn-out and in preventing the rapid spread of fire, smoke, or fumes, which may otherwise contribute to the loss of lives and property.

The fire resistance of a building or its structural and non-structural elements is expressed in minutes against a specified fire load which is expressed in kcal/m², and against a certain intensity of fire. The fire-resistance test for structural element should be done in accordance with accepted standards [F(2)]. For the purpose of this Part, the types of construction according to fire resistance should be classified into four categories, namely, Type 1 Construction, Type 2 Construction, Type 3 Construction and Type 4 Construction. The minimum fire resistance ratings of structural and non-structural members for various types of construction is given in [Table 1](#).

Non-combustible materials should be used for construction of buildings, and the internal walls of staircase enclosures should be of brick work or reinforced concrete or any other material of construction with minimum of 120 min rating. The walls for the chimney should be of Type 1 or Type 2 Construction depending on whether the flue gas temperature is above 200 °C or less, respectively.

3.4.2 It is required that a structural and/or non-structural element/component to have the requisite fire resistance rating as per [Table 1](#). The fire resistance rating for the structural and non-structural elements are based on guidelines as per approved and accepted standards. The fire rating should be validated and certified with a view to meeting the requirements of [Table 1](#). In the absence of any validated/certified rating, guidance may be obtained from the information available in [Annex C](#).

3.4.3 Load bearing steel beams and columns of buildings having total covered area of 500 m² and above need to be protected against failure/collapse of structure in case of fire. This could be achieved by use of appropriate

methodology using suitable fire resistance rated materials along with suppression system (see [Annex C](#), [Table 25](#), and [Table 26](#)).

3.4.4 The false ceiling, including all fixtures used for its suspension, need to be of non-combustible material. To provide adequate fire resistance to the ceiling in order to prevent spread of fire across ceiling.

3.5 General Requirements of All Individual Occupancies

3.5.1 General

All buildings should satisfy certain requirements, which contribute, individually and collectively, to the safety of life from fire, smoke, fumes, and panic arising from these or similar causes. There are, however, certain general principles and common requirements, which are applicable to all or most of the occupancies.

**Table 1 Fire Resistance Ratings of Structural and Non-Structural Elements
(in minutes)**

(Clauses [3.4.1](#), [3.4.2](#), [3.5.9.1](#), [3.5.9.2](#) and [F-2.1](#))

Sl No.	Structural/Non-Structural Element	Fire Resistance Ratings (min) for Type of Construction			
		Type 1	Type 2	Type 3	Type 4
(1)	(2)	(3)	(4)	(5)	(6)
i)	Exterior walls:				
	a) Fire separation less than 3.7 m:				
	1) Bearing	240	120	120	60
	2) Non-bearing	120	90	60	60
	b) Fire separation of 3.7 m or more but less than 9 m:				
	1) Bearing	240	120	120	60
	2) Non-bearing	90	60	60	60
	c) Fire separation of 9 m or more:				
	1) Bearing	240	120	120	60
	2) Non-bearing	60	60	60	60
ii)	Fire separation assemblies (like fire check doors)	120	120	120	120
iii)	Fire enclosures of exits	120	120	120	120
iv)	Shafts for services, lift hoistway and refuse chutes	120	120	120	120
v)	Vertical separation between adjacent tenant spaces	60	60	60	60
vi)	Dwelling unit separation:				
	a) Load bearing	120	120	60	60
	b) Non-load bearing	60	60	30	30
vii)	Interior bearing walls, bearing partitions, columns, beams, girders, trusses (other than roof trusses) and framing				
	a) Supporting more than one floor	240	120	120	120
	b) Supporting one floor only	180	90	60	60
	c) Supporting a roof only	180	90	60	60

Sl No.	Structural/Non-Structural Element	Fire Resistance Ratings (min) for Type of Construction			
		Type 1	Type 2	Type 3	Type 4
(1)	(2)	(3)	(4)	(5)	(6)
viii)	Walls supporting structural members	180	90	60	60
ix)	Floor construction	120	90	60	60
x)	Roof construction:				
	a) 5 m or less in height to lowest member	120	90	60	60
	b) More than 5 m but less than 6.7 m in height to lowest member	60	60	60	60
	c) 6.7 m or more in height to lowest member	0 60*	0 60*	0 60*	0 60*
* 60 min should be adopted when associated building services/cables are running at higher levels and supported from/near the roof, like in airports.					

3.5.2 Exceptions and Deviations

Exceptions and deviations to the general provisions of requirements of individual occupancies are given as applicable to each type of occupancy in [6.1](#) to [6.9](#). In case of practical difficulty or to avoid unnecessary hardship, without sacrificing reasonable safety, local head, fire services may consider exemptions from this Part.

3.5.3 Occupation of Buildings under Construction

3.5.3.1 A building or portion of the building may be occupied during construction, repairs, alterations, or additions only if all means of exit and fire protection measures are in place and continuously maintained for the occupied part of the building.

3.5.3.2 A high rise building during construction should be provided with the following fire protection measures, which should be maintained in good working condition at all the times:

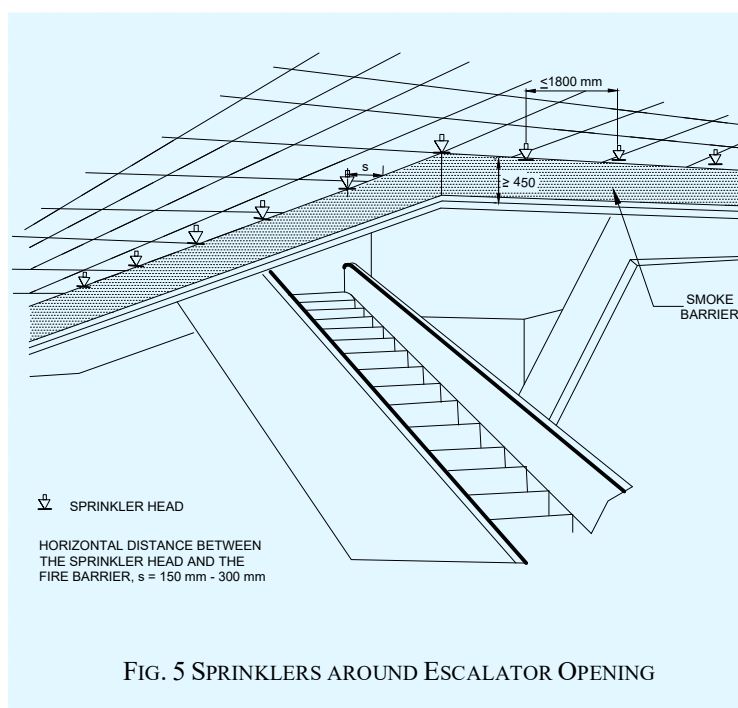
- Dry riser of minimum 100 mm diameter pipe with hydrant outlets on the floors constructed with a fire service inlet to boost the water in the dry riser and maintenance should be in accordance with good practice [F(5)].
- Drums of 2 000 litre capacity filled with water with two fire buckets on each floor;
- A water storage tank of minimum 20 000 litre capacity, which may be used for other construction purposes also.

3.5.4 Openings in Fire Resistant Walls and Floors

3.5.4.1 At the time of designing openings in fire-resistant walls and floors, particular attention should be paid to all such factors which limit fire spread through these openings and maintain fire rating of the structural members.

3.5.4.2 For Types 1 to 3 constructions, a doorway or opening in a fire resistant wall on any floor should be limited to 5.6 m² in area with a maximum height/width of 2.75 m. Every wall opening should be protected with fire-resisting doors, having the fire rating of not less than 120 min. The openings in the floors should be protected by vertical enclosures extending above and below such openings, such enclosures having a fire resistance of not less than 120 min and all openings therein being protected with a fire-resisting assembly as specified in [3.5.4.6](#).

For escalator-like openings, the smoke spill should be avoided by provision of smoke barrier (of 450 mm to 600 mm) thereby creating smoke compartment. Further, the protection should be ensured through installation of sprinklers on all sides of the cut out opening on each floor (see [Fig. 5](#) for details).



3.5.4.3 For Type 4 construction, openings in the fire separating walls or floors should be fitted with 120 min fire-resistance rated assemblies.

3.5.4.4 Service ducts and shafts

Openings in walls or floors which are necessary to be provided to allow passages of all building services like cables, electrical wirings, telephone cables, plumbing pipes, etc, should be protected by enclosure in the form of ducts/shafts having a fire resistance not less than 120 min. The inspection door for electrical shafts/ducts should be not less than 120 min, with 30 min insulation. Further, medium, and low voltage wiring running in shafts/ducts, should either be armoured type or run through metal conduits.

The space between the electrical cables/conduits and the walls/slabs should be filled in by a fire stop material {see accepted standard [F(34)]} having fire resistance rating of not less than 120 min. This should exclude requirement of fire stop sealing for low voltage services shaft.

For plumbing shafts in the core of the building, with shaft door opening inside the building, the shafts should have inspection doors having fire resistance rating not less than 30 min.

For plumbing shafts doors which open in wet areas or in naturally ventilated areas or on external wall of the building, the shafts may not require doors having any specified fire rating.

NOTE — In the case of buildings where it is necessary to lower or lift heavy machinery or goods from one floor to the other, it may be necessary to provide larger openings in the floor. Such openings should be provided with removable covers which should have the same strength and fire resistance as the floor.

3.5.4.5 Refuse chutes

Refuse chutes, if any provided in a building, should have opening at least 1 m above roof level for venting purpose and they should have an enclosure wall of non-combustible material with fire resistance of not less than 120 min. They should not be located within the staircase enclosure or service shafts, or air conditioning shafts. Refuse chutes inspection panel and doors should be tight fitting with 60 min fire resistance. Sprinkler protection system should be provided for the refuse chutes. Refuse chutes should be at least 3 m away from exits.

3.5.4.6 Vertical opening

Every vertical opening between the floors of a building should be suitably enclosed or protected, as necessary, to provide the following:

- a) Reasonable safety to the occupants while using the means of egress by preventing spread of fire, smoke,

or fumes through vertical openings from floor to floor to allow occupants to complete their use of the means of egress. Further it should be ensured to provide a clear height of 2 100 mm in the exit access.

- b) Limitation of damage to the building and its contents.

3.5.5 Electrical Installation

3.5.5.1 For requirements regarding electrical installations from the point of view of fire safety, reference may be made to good practice [F(6)] and Part D 'Building Services, Section 2 Electrical and Allied Installations' of this NBCS.

In general, it is desirable that the wiring and cabling are with flame retardant property. Medium and low voltage wiring running in shafts, and within false ceiling should run in metal conduit. Any 230 V wiring for lighting or other services, above false ceiling, should have 660 V grade insulation. See also Part D/Sec 2 for wires having options of fire resistance.

The electric distribution cables/wiring should be laid in a separate shaft. The shaft should be sealed at every floor with fire stop materials, see accepted standard [F(34)] having the same fire resistance as that of the floor. High, medium, and low voltage wiring running in shaft and in false ceiling should run in separate shaft/conduits.

Water mains, gas pipes, telephone lines, intercom lines or any other service line should not be laid in the duct for electrical cables; use of bus ducts/solid rising mains instead of cables is preferred.

All metallic items like steel structural members, etc, should be bonded properly to the earthing system.

3.5.5.2 Emergency power for fire and life safety systems

Emergency power supplying distribution system for critical requirement for functioning of fire and life safety system and equipment should be planned for efficient and reliable power and control supply to the following systems and equipment where provided:

- a) Fire pumps;
- b) Pressurization and smoke venting; including its ancillary systems such as dampers and actuators;
- c) Fireman's lifts (including all lifts);
- d) Exit signage lighting;
- e) Emergency lighting;
- f) Fire alarm system;
- g) Public address (PA) system (relating to emergency voice evacuation and annunciation);
- h) Magnetic door hold open devices; and
- j) Lighting in fire command centre and security room.

Power supply to these systems and equipment should be from normal and emergency (standby generator) power sources with changeover facility (see [Fig. 6A](#) and [Fig. 6B](#) indicating options). If power supply is from HV source and HV generation, the transformer should be planned in standby capacity to ensure continuity of power to such systems. Wherever transformers are installed at higher levels in buildings and backup DG sets are of higher voltage rating, then dual redundant cables should be taken to all transformers. The generator should be capable of taking the starting current of all the fire and life safety systems and equipment as above. Where a parallel HV/LV supply from a separate substation fed from a different grid is provided with an appropriate transformer for emergency, the provision of generator may be waived in consultation with the Authority.

The power supply to the panel/distribution board of these fire and life safety systems should be through fire proof enclosures (tested and certified) or fire survival cables or through alternate route in the adjoining fire compartment to ensure supply of power is reliable to these systems and equipment. It should be ensured that the cabling from the adjoining fire compartment is protected within the compartment of vulnerability. The location of the panel/distribution board feeding the fire and life safety system should be in fire safe zone ensuring supply of power to these systems.

Circuits of such emergency system should be protected at origin by an automatic circuit breaker with its no-volt coil removed. Master switches controlling essential service circuits should be clearly labelled.

Cables for fire alarm and PA system should be laid in metal conduits or armoured to provide physical segregation from the power cables.

3.5.5.3 Substation/Transformers

Areas in substation should not be used as storage/dump areas or for other utility purposes other than those required for the functioning of the substation.

The substation area should be adequately ventilated.

An independent, ventilated or air conditioned electrical substation and LT panel room/area should be provided on the ground level or first basement. They should be provided with access from outside (or through exit passageway accessible from outside). The electrical substation and LT panel room/area should be provided with fire resistant walls and doors of fire resistance of not less than 120 min.

If the licensees agree to provide meters on upper floors, the licensees' cables should be segregated from consumers' cables by providing a partition in the shaft. Meter rooms on upper floors should not open into staircase enclosures and should be ventilated directly to open air outside or in electrical room of 120 min fire resistant walls.

Electrical MV main distribution panel and lift panels should be provided with CO₂/inert gas flooding system for all panel compartments with a cylinder located beside the panel.

3.5.5.3.1 Oil filled substation

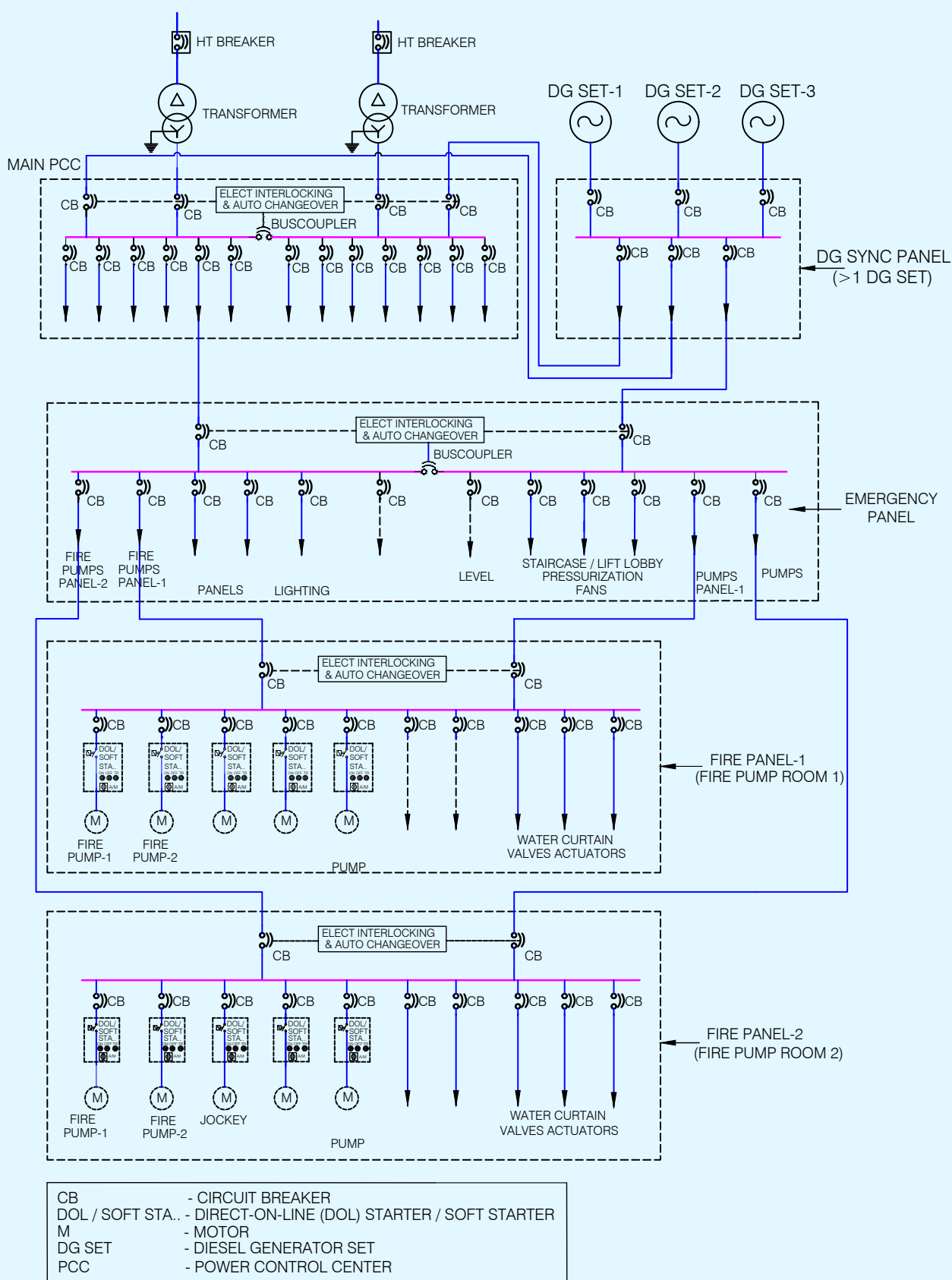
A substation or a switch-station with oil filled equipment should be limited to be installed in utility building or in outdoor location. Such substation/utility building should be at least 6 m away (or as per local requirements/actual width of fire driveway) from the adjoining building(s).

Substation equipment (exceeding aggregate oil capacity of 2 000 litre) in utility building should have fire rated baffle walls of 240 min rating constructed between such equipment, raised to at least 600 mm above the height of the equipment (including height of oil conservators/explosion vents) and exceeding 300 mm on each side of the equipment.

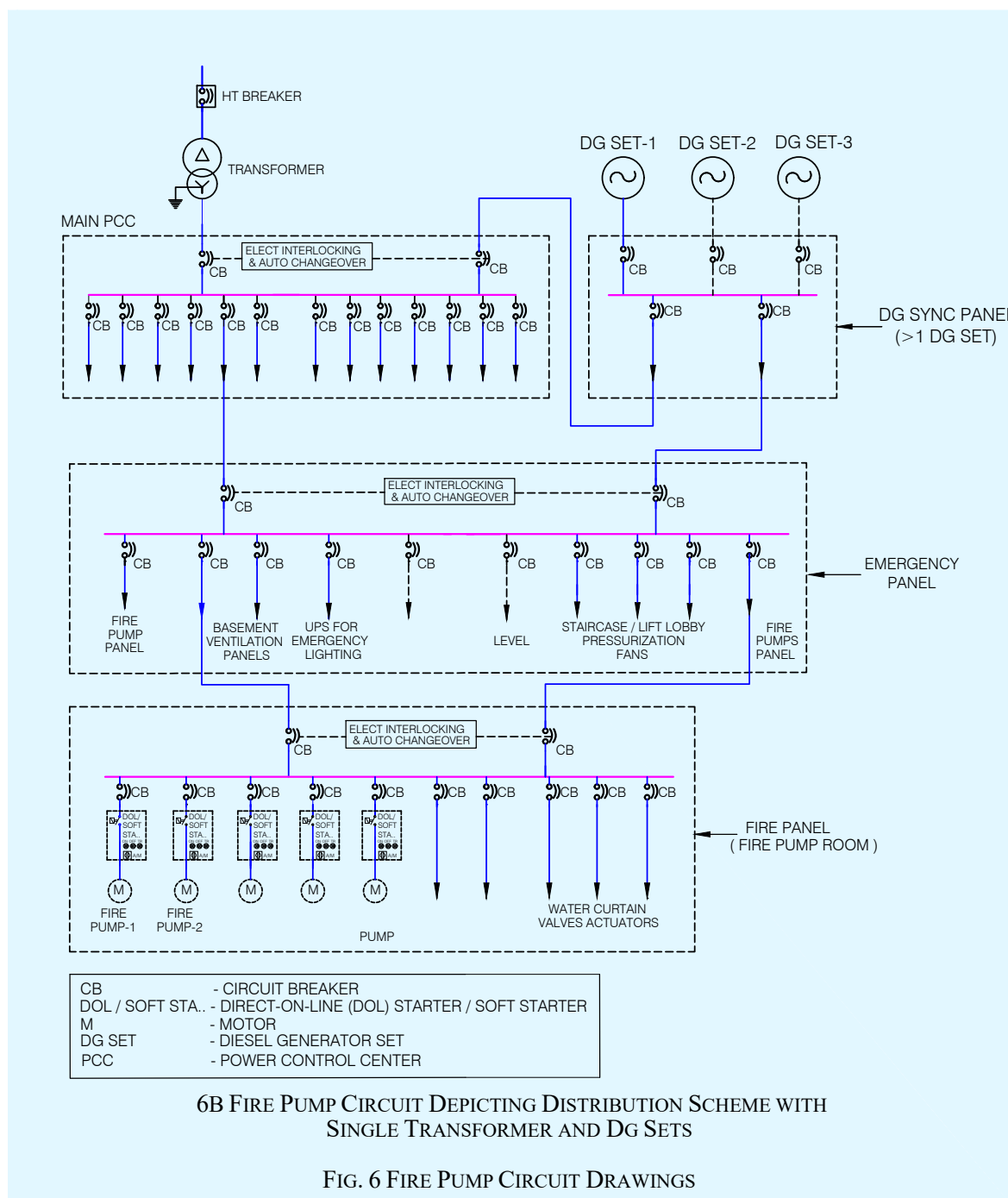
All transformers where capacity exceeds 10 MVA should be protected by high velocity water spray systems/ high pressure water mist system. In addition, nitrogen injection systems may also be provided where applicable.

3.5.5.3.2 Dry type substation

Transformers located inside a building should be of dry type and all substation/switch room walls, ceiling, floor, opening including doors should have a fire resistance rating of 120 min. Access to the substation should be provided from the nearest fire exit/exit staircase for the purpose of electrical isolation.



6A FIRE PUMP CIRCUIT DEPICTING DISTRIBUTION SCHEME WITH MULTIPLE TRANSFORMERS AND DG SETS



3.5.5.4 Standby supply

Diesel generator set(s) should not be installed at any floor other than ground/first basement. If the same are installed indoors, proper ventilation and exhaust should be planned. The DG set room should be separated by 120 min fire resistance rated walls and doors.

The oil tank for the DG sets (if not in the base of the DG) should be provided with a dyked enclosure having a volumetric capacity of at least 10 percent more than the volume of the oil tank. The enclosure should be filled with sand for a height of 300 mm.

See [Table 9](#) for specific fire protection guidelines. For detailed information regarding fire safety requirements for hazardous petroleum products, reference may be made to *The Petroleum Act, 1934* and the Rules framed thereunder.

3.5.5.5 Lightning protection of buildings

Routing of down conductors (insulated or uninsulated) of lightning protection through electrical or other service shafts are not allowed as it can create fire and explosion during lightning. For details, see Part D 'Building Services, Section 2 Electrical and Allied Installations' of this NBCS.

3.5.6 Escape Lighting and Exit Signage

Exit access, exits, and exit discharge should be properly identified, with adequate lighting maintained in the elements of the egress systems so that all occupants should be able to leave the facility safely.

3.5.6.1 Lighting

- a) The exit, exit access and exit discharge systems should be illuminated continuously. The floors of the means of egress should be illuminated at all points, including angles and intersections, in corridors and passageways, stairwells, landings of stairwells and fire exit;
- b) Emergency lighting should be powered from a source independent of that supplying the normal lighting;
- c) Escape lighting should be capable of,
 - 1) indicating clearly and unambiguously the escape routes;
 - 2) providing adequate illumination along such routes to allow safe movement of persons towards and through the fire exits; and
 - 3) ensuring that fire alarm call points and firefighting equipment provided along the escape routes can be readily located.
- d) The horizontal luminance at floor level on the centreline of an escape route should not be less than 10 lumen/m². In addition, for escape routes up to 2 m wide, 50 percent of the route width should be lit to a minimum of 5 lumen/m². In auditoriums, theatres, concert halls and such other places of assembly, the illumination of floor exit/access may be reduced during period of performances to values not less than 2 lux;
- e) Required illumination should be arranged such that the failure of any single lighting unit, such as the burning out of one luminaire, will not leave any area in darkness and does not impede the functioning of the system further;
- f) The emergency lighting should be provided to be put on within 5 s of the failure of the normal lighting supply. Also, emergency lighting should be able to maintain the required illumination level for a period of not less than 90 min in the event of failure of the normal lighting even for smaller premises;
- g) Battery pack emergency lighting, because of its limited duration and reliability, should not be allowed to be used in lieu of a diesel engine driven emergency power supply;
- h) Escape lighting luminaires should be sited to cover the following locations:
 - 1) Near each intersection of corridors,
 - 2) At exits and at each exit door,
 - 3) Near each change of direction in the escape route,
 - 4) Near each staircase so that each flight of stairs receives direct light,
 - 5) Near any other change of floor level,
 - 6) Outside each final exit and close to it,
 - 7) Near each fire alarm call point,
 - 8) Near firefighting equipment, and
 - 9) To illuminate exit and safety signs as required by the enforcing authority.

NOTE — For the purpose of this clause 'near' is normally considered to be within 2 m measured horizontally.
- j) The luminaires should be mounted as low as possible, but at least 2 m above the floor level; and
- k) Signs are required at all exits, emergency exits and escape routes, which should comply with the graphic requirements of the relevant Indian Standards.

3.5.6.2 Exit passageway and staircase lighting should also be connected to alternative supply. The alternative source of supply may be provided by battery continuously trickle charged from the electric mains. Suitable arrangements should be made by installing double throw switches to ensure that the lighting installed in the staircase and the corridor does not get connected to two sources of supply simultaneously. Double throw switch should be installed in the service room for terminating the stand-by supply.

3.5.6.3 The emergency lighting system should be well maintained by periodical inspections and tests so as to ensure their perfect serviceability at all times.

3.5.6.4 *Exit signage*

Where exit access is provided through corridors/paths, the occupants should be able to easily identify the way to exits. Exit signs should be provided such that no point in an exit access is more than 30 m from a visible exit directional sign. An exit sign indicating the direction to an exit should be provided at all changes in direction.

Exits should be clearly visible and the route to reach the exits should be clearly marked and signs posted to guide the occupants of the floor concerned. Signs should be illuminated and wired to an independent electrical circuit on an alternative source of supply. The sizes and colours of the exit signs should be in accordance with good practice [F(7)]. The colour of the exit signs should be green.

NOTE — This provision should not apply to A-II and A-IV occupancies less than 15 m in height.

The exit sign with arrow indicating the way to the escape route should be provided at a suitable height from the floor level on the wall and should be illuminated by electric light connected to corridor circuits. All exit way marking signs should be so installed that no mechanical damage should occur to them due to moving of furniture or other heavy equipment. Further, all landings of floor should have floor indicating boards prominently indicating the number of the floor.

Photo luminescent markings should be pasted at internal hydrant boxes.

3.5.7 *Air Conditioning, Ventilation and Smoke Control*

3.5.7.1 Air conditioning and mechanical ventilation requirements of different rooms or areas in any occupancy should be as given in Part D 'Building Services, Section 1 Lighting and Natural Ventilation' and 'Section 3 Air conditioning, Heating and Mechanical Ventilation' of this NBCS.

Air conditioning and ventilating systems should be so installed and maintained as to minimize the danger of spread of fire, smoke, or fumes from one floor to another or from outside to any occupied building or structure.

Many high-rise buildings integrate smoke management systems into their conventional HVAC systems. In such installation, it requires special design considerations, including safe and adequate controls, acceptable and documented testing, and regular maintenance systems.

Wherever batteries are provided, the same should be segregated by 120 min fire rated construction. Ventilation to the room should be provided as per manufacturer's instructions.

3.5.7.2 *Air handling unit*

3.5.7.2.1 From fire safety point of view, separate air handling units (AHU) for each floor should be provided so as to avoid the hazards arising from spread of fire and smoke through the air conditioning ducts. The air ducts should be separate from each AHU to its floor and in no way should interconnect with the duct of any other floor. Within a floor it would be desirable to have separate air handling unit provided for each compartment.

Air handling unit should be provided with effective means for preventing circulation of smoke through the system in the case of a fire in air filters or from other sources drawn into the system, and should have smoke sensitive devices for actuation in accordance with the accepted standard [F(8)] and control.

3.5.7.2.2 Shafts or ducts, if penetrating multiple floors, should be of masonry construction with fire damper in connecting ductwork or should have fire rated ductwork with fire dampers at floor crossing. Alternatively, the duct and equipment may be installed in room having walls, doors and fire damper in duct exiting/entering the room of 120 min fire resistance rating. Such shafts and ducts should have all passive fire control meeting 120 min fire resistance rating requirement to meet the objective of isolation of the floor from spread of fire to

upper and lower floors through shaft/duct work.

NOTE — Zoned and compartmented HVAC systems are encouraged with an approach to avoid common exhaust shafts and fresh air intake shafts which will limit the requirement of such passive measure and fire rated duct work and dampers.

3.5.7.2.3 The air filters of the air handling units should be made of non-combustible materials.

3.5.7.2.4 The air handling unit room should not be used for storage of any combustible materials.

3.5.7.3 *Duct work*

3.5.7.3.1 Air ducts serving main floor areas, corridors, etc, should not pass through the fire exits/exit passageway/ fire exit enclosure. Fire exits and lift lobbies, etc, should not be used as return air passage.

3.5.7.3.2 As far as possible, metallic ducts (of suitable rating) should be used even for the return air instead of space above the false ceiling.

3.5.7.3.3 Wherever the ducts pass through fire walls or floors, the opening around the ducts should be sealed with materials having fire resistance rating of the compartment {see accepted standard [F(34)]}. Such duct should also be provided with fire dampers at all fire walls and floors unless such ducts are required to perform for fire safety operation; and in such case fire damper may be avoided at fire wall and floor while integrity of the duct should be maintained with 120 min fire resistance rating to allow the emergency operations for fire safety requirements.

3.5.7.3.4 The ducting within compartment would require minimum fire resistance rating of 30 min. Such ducting material in substantial gauge should be in accordance with good practice [F(9)]. If such duct crosses adjacent compartment/floor and not having fire dampers in such compartment/floor, it would require fire resistance duct work rating of 120 min. The requirements of support of the duct should meet its functional time requirement as above.

3.5.7.3.5 The materials used for insulating the duct system (inside or outside) should be of non-combustible type. Any such insulating material should not be wrapped or secured by any material of combustible nature.

3.5.7.3.6 Inspection panels should be provided in the ductwork to facilitate the cleaning of accumulated dust in ducts and to obtain access for maintenance of fire dampers.

3.5.7.4 *Fire or fire/smoke dampers*

3.5.7.4.1 These dampers should be evaluated to be located in supply air ducts, fresh air and return air ducts/ passages at the following points:

- a) At the fire separation wall,
- b) Where ducts/passages enter the vertical shaft,
- c) Where the ducts pass through floors, and
- d) At the inlet of supply air duct and the return air duct of each compartment on every floor.

3.5.7.4.2 Damper should be of motorized type/fusible link. Damper should be so installed to provide complete integrity of the compartment with all passive fire protection sealing. Damper should be accessible to maintain, test and also replace, if so required. Damper should be integrated with fire alarm panel and should be sequenced to operate as per requirement and have interlocking arrangement for fire safety of the building. Manual operation facilities for damper operation should also be provided. Tested and certified dampers for dynamic condition are preferable.

3.5.8 *Heating*

3.5.8.1 Installation of chimney and heating apparatus should be in accordance with good practice [F(10)].

3.5.8.2 *Boiler rooms*

3.5.8.2.1 Provisions of boiler and boiler rooms should conform to *The Boilers Act*, 1923.

3.5.8.2.2 Further, the following additional aspects may be taken into account in the location of boiler room:

- a) The boilers should be installed in a fire resisting room of 180 min fire resistance rating;

- b) Entry to this room should be provided with a composite door of 120 min fire resistance rating;
- c) The boiler room should be provided with its dedicated natural or mechanical ventilation system. Mechanical ventilation system for the boiler room would be accepted with 120 min fire resistance rating ductwork if it has interface with other mechanical areas. Ventilation system should not be allowed to be routed through electrical room area or through exit corridor/exits; and
- d) The oil tank for the boiler should be provided with a dyked enclosure having a volumetric capacity of at least 10 percent more than the volume of the oil tank. The enclosure should be filled with sand for a height of 300 mm.

3.5.9 Glazing

3.5.9.1 The glazing should be in accordance with Part C 'Structural Design, Section 8 Glass and Glazing' of this NBCS. The entire glazing assembly should be rated to that type of construction as given in [Table 1](#). This should be applicable along with other provisions of this Part related to respective uses as specified therein. The use of glass should not be permitted for enclosures of exits and exit passageway.

3.5.9.2 Glass facade should be in accordance with the following:

- a) For fully sprinklered buildings having fire separation of 9 m or more, tempered glass in a non-combustible assembly, with ability to hold the glass in place, should be provided. It should be ensured that sprinklers are located within 600 mm of the glass facade providing full coverage to the glass. Where the fire separation is less than 9 m, additional protection such as using a minimum 900 mm spandrel panel of 1 h fire resistance rating should be provided;

NOTE — In case of all other buildings, fire resistance rating of glass facade should be in accordance with [Table 1](#).

- b) All gaps between floor-slabs and facade assembly should be sealed at all levels by approved fire resistant sealant material of equal fire rating as that of floor slab to prevent fire and smoke propagation from one floor to another. *See also* accepted standard [F(35)]; and
- c) Openable panels should be provided on each floor and should be spaced not more than 10 m apart measured along the external wall from centre-to-centre of the access openings. Such openings should be operable at a height between 1.2 m and 1.5 m from the floor, and should be in the form of openable panels (fire access panels) of size not less than 1 000 mm × 1 000 mm opening outwards. The wordings, 'FIRE OPENABLE PANEL — OPEN IN CASE OF FIRE, DO NOT OBSTRUCT' of at least 25 mm letter height should be marked on the internal side. Such panels should be suitably distributed on each floor based on occupant concentration. These should not be limited to cubicle areas and should be also located in common areas/corridors to facilitate access by the building occupants and fire personnel for smoke exhaust in times of distress.

3.5.10 Surface Interior Finishes

3.5.10.1 The use of combustible surface finishes on walls (including facade of the building) and ceilings affects the safety of the occupants of a building. Such finishes tend to spread the fire and even though the structural elements may be adequately fire resistant, serious danger to life may result. It is, therefore, essential to have adequate precautions to minimize spread of flame on wall, facade of building and ceiling surfaces. The finishing materials used for various surfaces and decor should be such that it should not generate toxic smoke/fumes.

3.5.10.2 The susceptibility of various types of wall surfaces to fire is determined in terms of the rate of spread of flame. Based on the rate of spread of flame, surfacing material should be considered as divided into four classes as follows {*see also* good practice [F(11)]}:

- a) Class 1 Surfaces of very low flame spread;
- b) Class 2 Surfaces of low flame spread;
- c) Class 3 Surfaces of medium flame spread; and
- d) Class 4 Surfaces of rapid flame spread.

3.5.10.3 The uses for which surface materials falling into various classes should be adopted in building construction are given below:

<i>Class 1</i>	<i>Class 2</i>	<i>Class 3</i>
(1)	(2)	(3)
May be used in any situation	May be used in any situation, except on walls, facade of the building, staircase, and corridors	May be used only in living rooms and bed rooms (but not in rooms on the roof) and only as a lining to solid walls and partitions; not on staircases or corridors or facade of the building.

NOTE — Panelling (lining) may be permitted in a limited area; and is not to be permitted in a vestibule.

3.5.10.4 Materials of Class 4 which include untreated wood fibreboards may be used with due fire-retardant treatment as ceiling lining, provided the ceiling is at least, 2.4 m from the top surface of the floor below, and the wall surfaces conform to requirements of the class [see Note under [3.5.10.3](#)]. Class 4 materials should not be used in kitchens, corridors, and staircases. Some materials contain bitumen and, in addition to risk from spread of fire, emit dense smoke on burning; such materials should be excluded from use under these conditions and should also not be used for construction of ceiling where the plenum is used for return air in air-conditioned buildings.

3.5.10.5 When frames, walls, partitions, or floors are lined with combustible materials, the surfaces on both sides of the materials should conform to the appropriate class, because there is considerable danger from fire starting and rapidly spreading within the concealed cavity unknown to the occupants whose escape may be hampered thereby. For detailed information on materials and details of construction with their fire resistance rating, reference may be made to good practice [F(12)].

3.5.10.6 The use of aluminium composite panels as external facade of the buildings should be avoided; if used, such panels should have a minimum of 70 percent mineral core and pass the tests as in accepted standards (see Note 2).

NOTES

1 When combustible materials are used for lining frames, walls, partitions, or floors, the surfaces on both sides of the materials must meet the required fire resistance class. This precaution helps mitigate the risk of fire spreading within concealed cavities, which may not be immediately visible to occupants and can hinder evacuation.

2 Accepted standards can be ISO 3957 or ISO 13785 (Part 1 and Part 2).

3.5.11 Fire Command Centre (FCC)

- Fire command centre should be on the entrance floor of the building having direct access. The control room should have the main fire alarm panel with communication system (suitable public address system) to aid floors and facilities for receiving the message from different floors;
- Fire command centre should be constructed with 120 min rating walls with a fire door and should be provided with emergency lighting. Interior finishes should not use any flammable materials. All controls and monitoring of fire alarm systems, pressurization systems, smoke management systems should happen from this room. Monitoring of integrated building management systems, CCTVs or any other critical parameters in building may also be from the same room.
- Details of all floor plans along with the details of firefighting equipment and installations (2 sets laminated and bound) should be maintained in fire command centre. The fire staff in charge of the fire command centre should be responsible for the maintenance of the various services and firefighting equipment and installations in coordination with security, electrical and civil staff of the building.

NOTE — Panelling (lining) should be permitted in a limited area. It should not be permitted in a vestibule.

4 LIFE SAFETY

4.1 General

Every building should be so designed, constructed, equipped, maintained, and operated as to provide adequate means of egress to avoid undue danger to the life and safety of the occupants from fire, smoke, fumes, or panic during the time period necessary for escape.

For high occupancy areas, it may be required to have annunciation, announcements, and voice guided/aided system to direct the occupants towards safe egress routes, areas of comparative safety or exits, and to avoid situation of panic during distress.

Every main occupancy may have certain occupancies which may be incidental to the main occupancy. The exit requirements pertaining to such incidental occupancies from the floor of the occupancy to the level of exit discharge should be calculated to meet the requirement of the actual occupancy of such type, to ensure adequate means of egress of the occupants.

See also good practice [F(31)] for accessibility for elderly and persons with disabilities, for various requirements for enabling a smooth and safe egress.

4.2 General Exit Requirements

4.2.1 Fire exits are unobstructed component of means of egress which is between the exit access and the exit discharge or public way. Exit components include exterior exit doors at the level of exit discharge, interior exit stairways (pressurised), exit stairway with external wall (pressurised or naturally ventilated), exterior exit stairways, exit passageways, and exterior exit ramps. Fire exits should be provided with panic bar to facilitate entry of occupant into fire exit.

An exit may also include a horizontal exit leading to an adjoining building/fire compartment having its further access to unlocked/public exit at the same level.

4.2.2 Unless otherwise specified, lifts, escalators, moving walks and revolving doors should not be considered as fire exits and should not constitute any part of the required fire exit.

4.2.3 Every fire exit, exit passageway and exit discharge should be continuously maintained free of all obstructions or impediments to full use in the case of fire or other emergency.

4.2.4 Every building having human occupancy should be provided with fire exits sufficient to permit safe egress of occupants, in case of fire or other emergency.

4.2.5 In every building or structure, fire exits should comply with the minimum requirements of this Part, except those not accessible for general public use.

4.2.6 No building should be so altered as to reduce the number, width, or protection of fire exits to less than that required.

4.2.7 For non-naturally ventilated areas (that is, pressurized areas), fire doors with 120 min fire resistance rating should be provided and particularly at the entrance to lift lobby and stair well where a 'funnel or flue effect' may be created, inducing an upward spread of fire, to prevent spread of fire and smoke.

4.2.8 Fire exits should be so arranged that they may be reached without passing through another occupied unit/passageway in others control if they pose challenge or restriction in means of egress.

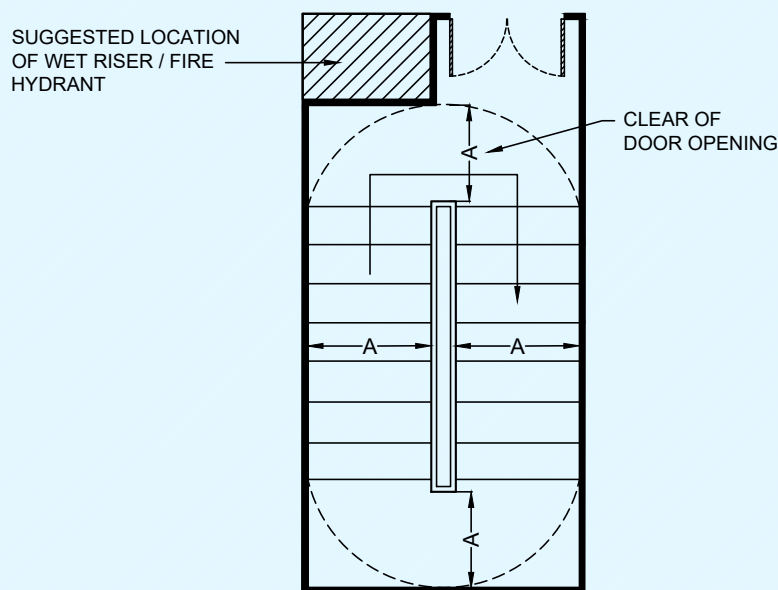
4.2.9 Doors in fire exits should open in the direction of exit. In case of assembly buildings (Group D) and institutional buildings (Group C-1), exit door should not open immediately upon a flight of stair and all such entries to the stair should be through a landing, so that such doors do not impede movement of people descending from a higher floor when fully opened (*see* [Fig. 7A](#)). While for other occupancies, such doors should not reduce the pathway in the landing by more than half the width of such staircase (*see* [Fig. 7B](#)). Over- head or sliding doors should not be installed.

4.2.10 At least half of the required exit stairs from upper floors (rounded to the next higher number) should discharge directly to the exterior or through exit passageways. For buildings less than 60 m in height, firefighting

shaft should be planned having one side on the external wall and discharging directly to the exterior. In the case of tall buildings (60 m and above), where structural design requires the location of firefighting shaft along with the structural core members, firefighting shaft(s) in the building should discharge directly to the exterior (through exit passageway).

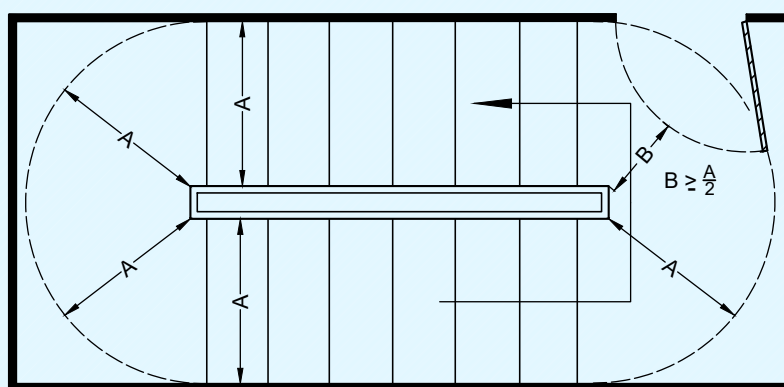
4.2.11 Unless otherwise specified, all the fire exits and exit passageways to exit discharge should have a clear ceiling height of at least 2.4 m. However, the height of fire exit door should be at least 2.0 m (see Fig. 8).

4.2.12 Where changes in elevation of more than 300 mm are encountered in the exits, ramps or sloped surfaces should be used with handrails and floor finish materials that contrast with the adjacent finish materials.



NOTE — DOOR WIDTH SHALL BE BASED ON TYPE OF OCCUPANCY

7A MINIMUM REQUIRED UNOBSTRUCTED CLEARANCE WITH DOOR LEAF ENCROACHING ON LANDING IN INSTITUTIONAL AND ASSEMBLY BUILDING



A - REQUIRED WIDTH

7B MINIMUM REQUIRED UNOBSTRUCTED CLEARANCE WITH DOOR LEAF ENCROACHING ON LANDING

FIG. 7 DOOR LOCATION AT LANDING IN FIRE EXITS

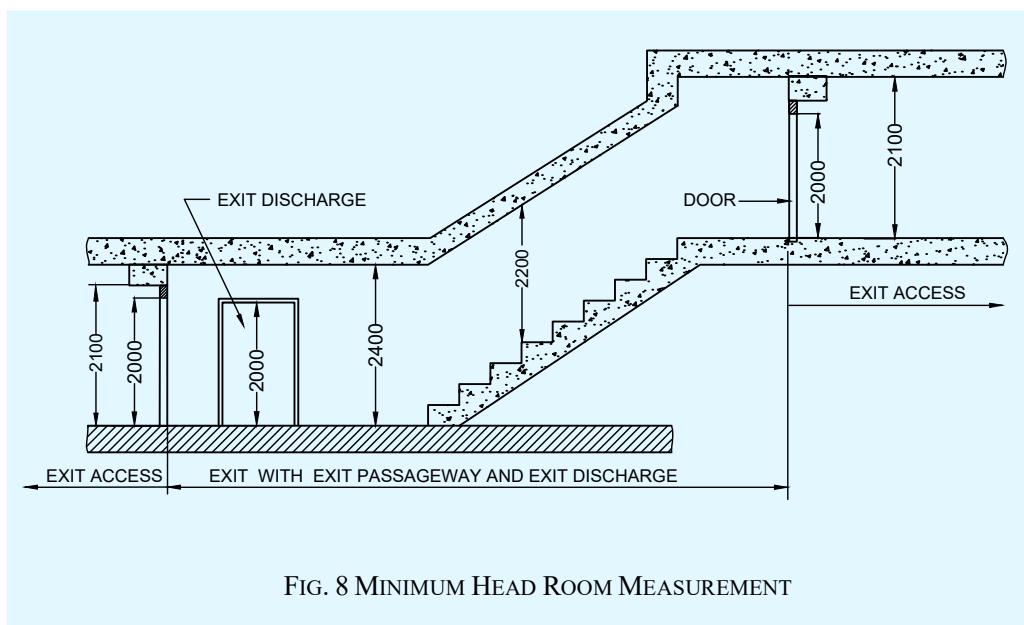


FIG. 8 MINIMUM HEAD ROOM MEASUREMENT

4.2.13 The capacity of the means of egress required from any storey of the building should not be reduced along the path of egress travel until arrival to the exit discharge.

4.2.14 The lifts, escalators, moving walks, turnstiles and revolving doors should not be considered in determining the required capacity of means of egress for the individual floor(s) or the building.

4.2.15 Turnstiles or similar devices that restrict travel to one direction or that are used to restrict unauthorized entry should not be so placed as to obstruct any required means of egress. Alternative door openings of required exit width should be available within 3 m of such devices, if installed.

4.2.16 Suitable means should be provided so that all access-controlled exit doors, turnstiles, boom barriers and other such exits should automatically operate to open mode during emergencies like fire, smoke, acts of terrorism, etc, so that people can safely and quickly egress into safe areas outside. If required, a master controlling device may be installed at a strategic location to achieve this.

4.2.17 Penetrations into and openings through fire exit are prohibited except those necessary, like for the fire protection piping, ducts for pressurization and similar life safety services. Such openings as well as vertical passage of shaft through floors should be protected by passive systems, {see accepted standard [F(34)]}.

4.2.18 Walking surfaces in exit access should comply with the following requirements for smooth exit:

- Walking surfaces should be nominally level;
- The slope of walking surface in the direction of travel should not exceed 1 in 20 unless the ramp requirements are met (see [4.4.2.4.3.5](#));
- Slope perpendicular to the direction of travel should not exceed 1 in 48; and
- Walking surfaces should be slip-resistant along the entire path of travel.

4.2.19 Basement

4.2.19.1 The basement should not be used for hazardous and vulnerable activities, like sleeping purposes, class rooms, hazardous industries, restaurant kitchen (see also [4.7.9](#)), battery rooms, UPS rooms, fuel oil storage, etc. However, the State Authorities may prescribe the list of such activities based on their own assessment. Basements should have smoke ventilation systems, high-density sprinkler systems, and redundant/enclosed exits.

4.2.19.2 The fire exits in a basement should be sufficient to provide for the capacity of the basement occupancy as determined in accordance with [4.4.2.1](#). In no case should there be fewer than two independent basement exits.

4.2.19.3 Basements with incidental occupancies to the main occupancy should be planned with exit requirements for the actual occupancy within the basement.

4.2.19.4 Where a basement is used for car parking and also there is direct approach from any occupancy above to the basement, door openings leading to the basement should be protected with fire doors with 120 min fire resistance rating, except for exit discharge doors from the basements.

4.2.19.5 Basements should have a minimum clear height of 3.4 m towards effective performance in the event of fire/smoke. Impulse or duct ventilation system needs to be designed to ensure removal of hot gases and smoke from each compartment to achieve reasonable visibility and heat reduction for firefighting operations.

4.3 Occupant Load

For determining the fire exits required, the number of persons within any floor area or the occupant load should be based on the actual number of occupants declared, but in no case less than that specified in [Table 2](#). The occupant load of a mezzanine floor discharging to a floor below should be added to that floor occupancy and the capacity of the fire exits should be designed for the total occupancy load thus established.

The occupant load of each storey considered individually should be required to be used in computing the number of means of egress at each storey, provided that the required number of means of egress is not decreased in the direction of egress travel.

Table 2 Occupant Load [Clauses 4.3 , 4.4.2.1 and H-4.1(j)]			
SI No.	Building Occupancy	Net Occupant Load Factor (m ² /person) (see Note 7)	Gross Occupant Load Factor (m ² /person) (see Note 1)
(1)	(2)	(3)	(4)
i)	Group A: Residential	12.5	-
ii)	Group B: Educational	3.0	-
iii)	Group C: Institutional (see Note 2)		
	a) Nursing homes	-	12.0
	b) Clinic (daycare)	-	12.0
	c) Inpatient area in hospital (like ward, sleeping area)	11.0	-
	d) Outpatient area in hospital	7.0	-
iv)	Group D: Assembly		
	a) Concentrated use without fixed seating	0.65	-
	b) Less concentrated use without fixed seating such as pre-function, multipurpose hall)	1.40	-
	c) Fixed seating	see Note 4	-
	d) Dining areas and restaurants with seating and table	1.80	-
	e) Gymnasium, sports room, library and alike with equipment, stacks and the like.	4.6	-
	This includes performance stages, indoor badminton court, squash court with		

Table 2 (Continued)

SI No.	Building Occupancy	Net Occupant Load Factor (m ² /person) (see Note 7)	Gross Occupant Load Factor (m ² /person) (see Note 1)
(1)	(2)	(3)	(4)
	limited number of performers and player (while audience for the same if any should be under (c) as above).		
	f) Swimming pool		
	1) Water surface	3.5	-
	2) Deck	2.0	-
	g) Mall – Basement with sales floor	2.1	-
	h) Mall – Street floor	2.1	-
	j) Mall – Upper sales floor	4.3	-
	k) Mall – Food court	-	1.8
	m) Mall – Theatre screens	-	see Note 4
	n) Storage/warehouse, receiving and the like	14.3	-
v)	Group F: Mercantile		
	a) Street floor and above	-	2.1
	b) Below street floor	-	3.0
vi)	Group E: Business	7.1	-
vii)	Group E: For Datacentres	-	20.0
	a) Admin areas	-	10.0
	b) Data hall, server room, equipment room areas	-	50.0
		see Annex G	see Annex G
viii)	Group G: Industrial	-	
	a) Low hazard	-	20.0
	b) Moderate hazard	-	15.0
	c) High hazard	-	10.0
ix)	Group H: Storage (see Note 5)	-	30.0
x)	Group J: Hazardous	-	10.0
NOTES			
1 Gross area is the floor area as defined in 2.35 .			
2 Occupant load factor in dormitory portions of homes for the aged, orphanages, insane asylums, etc, where sleeping			

Table 2 (Concluded)

SI No.	Building Occupancy	Net Occupant Load Factor	Gross Occupant Load Factor
		(m ² /person)	(m ² /person)
		(see Note 7)	(see Note 1)
(1)	(2)	(3)	(4)
	accommodation is provided, should be calculated at not less than 7.5 m² gross floor area/person. 3 Commercial kitchen in the above occupancies should be calculated at 9.3 m² net floor area/person. 4 In case of assembly occupancy having fixed seats, the occupant load should be determined by multiplying the number of seats by 1.2. 5 Car parking areas under occupancy other than storage should also be 30 m² per person. 6 Usable carpet area is the total of all corridors, passages, enclosed offices/rooms excluding rack and UPS/battery foot print 7 Net area is not the carpet area. See 2.36. * Gross area is highly recommended to be used than the net area, to avoid double counting. Net area is possible to be indicated for spaces having concentration of people is typical in certain occupancies; and other common spaces/areas in a floor plate may be used by the same set of occupants. Moreover, gross area provides a safer, more conservative occupant load for the entire floor, ensuring exits are sized for the maximum possible capacity. Further, net area is used for specific rooms (like classrooms or assembly halls) where the furniture and layout are predictable, but using it for an entire floor plate often underestimates the total population because it excludes corridors, closets, and wall thicknesses where people still circulate.		

4.4 Egress Components

Egress components to be considered are the number of exits to which access is provided, capacity of exit access, travel distance to a fire exit, the obviousness of the direction to a fire exit, and any hindrance including due to security issues involved.

4.4.1 Exit Access

- A common path of travel is desirable in exit access which leads to two independent directions to separate fire exits;
- Capacity of Exit Access* — The width of corridors, aisles or ramps required for exit access should be sufficient to ensure a smooth flow of occupants to the fire exit. Where a corridor is the only way of access to a fire exit, the corridor width should not be less than the calculated exit width;
- Objects like tables, chairs or any other temporary/permanent structures in exit access corridors should be avoided as this may result in congestion and also impeding smooth flow of personnel during emergencies;
- In order to ensure that each element of the means of egress can be effectively utilized, they should all be properly lit and marked. Lighting should be provided with emergency power back-up in case of power failures. Also, exit signs of adequate size, marking, location, and lighting should be provided so that all those unfamiliar with the location of the fire exits may safely find their way;
- Exit access to fireman's lift and refuge area/spaces on the floor should be step free and clearly signposted with the international symbol of accessibility;
- Exit access should not pass through storage rooms, closets or spaces used for similar purpose; and
- The calculation of capacity of exit access should be in accordance with [4.4.2.4](#).

4.4.2 Exits

4.4.2.1 Number of fire exits

The minimum required number of exits in a building should be determined based on occupant load (see [Table 2](#)) and width required per person (see [Table 3](#)) as appropriate to the type of exit for respective occupancies, subject to complying with maximum travel distance requirement (see [Table 4](#)).

4.4.2.2 Arrangement of fire exits

- a) Fire exits should be so located that the travel distance on the floor should not exceed the distance given in [Table 4](#).
- b) Travel distance should be measured from the most remote point within a storey or a mezzanine floor along the natural and unobstructed path of horizontal or vertical egress travel to the door of a fire exit.
- c) The dead-end corridor length in exit access should not exceed 6 m for educational, institutional and assembly occupancies. For other occupancies, the same should not exceed 15 m (see [Fig. 9](#)).
- d) Fire exits should be placed as remote from each other as possible and should be arranged to provide direct access in separate directions from any point in the area served (see [Fig. 10](#)).

4.4.2.3 Capacities of means of egress

- a) Exit capacity is the number of people that can pass through a stairway, and level components (door and corridor) and ramps. The total capacity of all the respective means of egress serving a floor should be sufficient to allow egress of the entire population of the floor.
- b) The unit of exit width, used to measure the capacity of any exit, is 500 mm. A clear width of 250 mm needs to be counted as an additional half unit. Clear widths less than 250 mm need not be counted for exit width.
- c) Width per person for stairways, and level components and ramps are to be determined using the capacity factors in accordance with [Table 3](#).

For example, if an exit doorway measures 1 000 mm in clear width its capacity factor being 6.5, it would be defined as providing exit capacity for $1\,000/6.5$ occupants, that is, 153 persons (say 150 persons) and number of such exit doorways can then be calculated depending on the occupant load.

- d) When calculating stairways, level components and ramps and other exit means, the capacity of the entire system should have to be based upon the minimum capacity available from any part of the system. The corridor, if so provided, should also to be planned with consideration of exit access adequacy for the number of occupants. Further, consider the situation of doors opening to an exit stairway. If the stairway provides an exit capacity of 150 persons, and the doors leading into the stairway provide an exit capacity of 153 persons, the overall exit system would be considered to provide the minimum exit capacity of only 150 persons afforded by the stairway. The exit planning will be limited by the most restrictive exit calculation under the means of egress.
- e) In the procedures for determining required egress capacity, the number of required means of egress is based on a floor-by-floor consideration, rather than the accumulation of the occupant loads of all the floors. However, the number of means of egress cannot decrease as an occupant proceeds along the egress path.

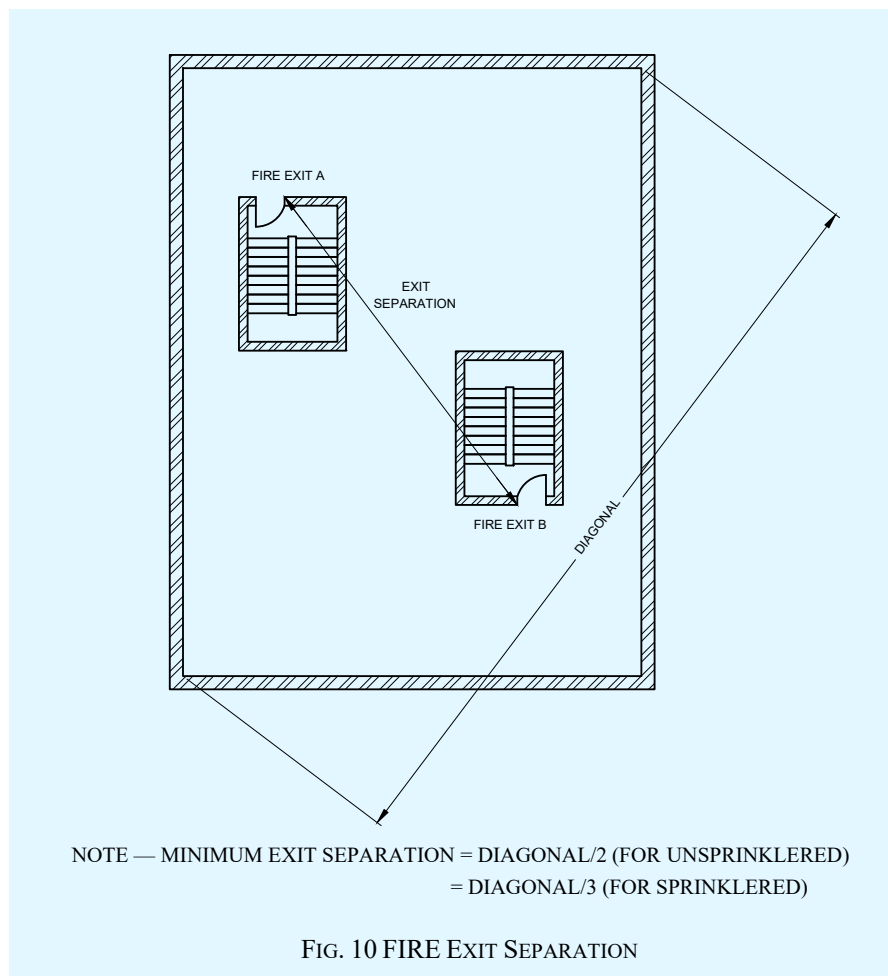
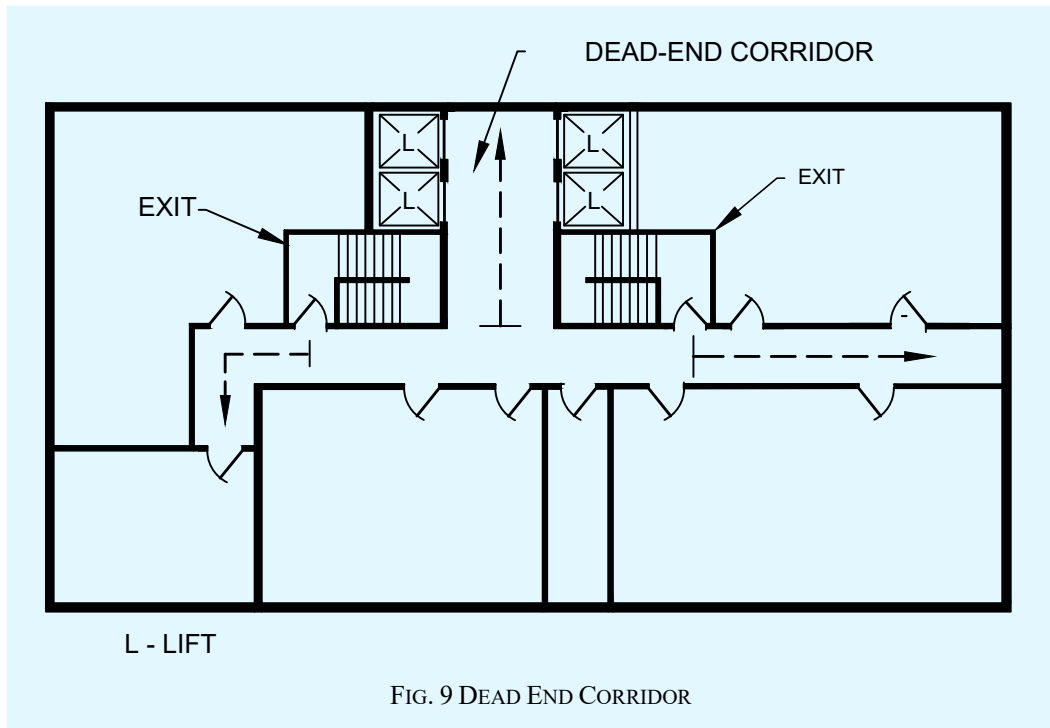


Table 3 Capacity Factors [Clauses 4.4.2.1 , 4.4.2.3(c) , 4.4.2.4.1(b) , 4.4.2.4.2(a) and 4.4.2.4.3.5(a)]			
SI No.	Occupancy Group	Width per Person	
		mm	
		Stairways	Level Components and Ramps
(1)	(2)	(3)	(4)
i)	Residential (Group A)	10	6.5
ii)	Educational (Group B)		
iii)	Institutional (Group C)	15	13
iv)	Assembly (Group D)	10	6.5
v)	Business (Group E)		
vi)	Mercantile (Group F)		
vii)	Industrial (Group G)		
viii)	Storage (Group H)		
ix)	Hazardous (Group J)	18	10

4.4.2.4 Types of exit access and exits

Various types of exit access and exits are doorways, corridors and passageways, horizontal exits, internal staircases, exit passageways, external staircases, and ramps.

4.4.2.4.1 Doorways

- Every exit doorway should open into an enclosed stairway or a horizontal exit of a corridor or passageway providing continuous and protected means of egress (see [Fig. 11](#) on unaccepted arrangement of doors in a stair).
- No exit doorway should be less than 1 000 mm in width. Door width in exits should also meet requirement of level components given in [Table 3](#) except assembly buildings, where door width should be not less than 2 000 mm (see [Fig. 12](#)). Doorways should be not less than 2 000 mm in height.
- Exit doorways should be operable from the side which they serve, without the use of a key.
- Mirrors should not be placed on exit doors and in exits, to avoid confusion regarding the direction of exit.
- Revolving doors can be accepted as a component in a means of egress where the following requirements are fully complied with:
 - Doors should be capable of collapsing to a book fold position with parallel egress paths, of width not less than 1 000 mm.
 - Doors should not be located within 3 m of the foot or top of stairs or escalators. A dispersal area should be provided between the stairs or escalators and the doors.
 - Each revolving door should be provided with a hinged door in the same wall within 3 m thereof, with same exiting capacity.
 - Each revolving door should be considered as capable of exiting only 50 persons.
- All fire rated doors and assembly should be provided with certificate and labels prominently indicating

the manufacturer's identification, door details covering door type, serial/batch number, month and year of manufacture, fire resistance rating, etc. The doors and assembly should be certified with all prescribed hardware such as hinges, locks, panic bars, door closers, and door viewers.

- g) *Access controlled doors* — Access controlled doors and electromagnetic doors should fall under this category. These should meet the following requirements:

- 1) Doors should have fire rating as per the requirements at the location of installation;
- 2) Activation of the building automatic sprinkler or fire detection system, if provided, should automatically unlock the doors in the direction of egress, and the doors should remain unlocked until the automatic sprinkler system or fire-alarm system has been manually reset;
- 3) Loss of power to the part of the access control system that locks the doors should automatically unlock the doors in the direction of egress;
- 4) A manual release device should be provided in the readily accessible vicinity of the egress door with a signage 'PUSH TO EXIT' and when the same is operated, it should result in direct interruption of power to the lock, independent of the access control system electronics.

- h) *Turnstiles* — Turnstiles or similar devices that restrict travel to one direction or are used to collect fares or admission charges should not be placed so as to obstruct any required means of egress unless door openings of required width are available within 3 m thereof.

Turnstiles or such similar devices should also be disengaged through automatic or manual intervention to allow egress in the direction of exit.

- j) Doors in folding partition should not be treated as approved means of egress.

Table 4 Travel Distance (Based on Occupancy and Construction Type)

[Clauses [4.4.2.1](#), [4.4.2.2 \(a\)](#), [4.4.2.4.2 \(a\)](#) and [6.8.1.1](#)]

Sl No.	Occupancy Group	Maximum Travel Distance m		
		Types 1 and 2 (Unsprinklered)	Types 1 and 2 (Sprinklered)	Types 3 and 4
(1)	(2)	(3)	(4)	(5)
i)	Residential (Group A)	30.00	60.00	22.5
ii)	Educational (Group B)	30.00	45.00	22.5
iii)	Institutional (Group C)	30.00	45.00	22.5
iv)	Assembly (Group D)	30.00	45.00	30.00
v)	Business (Group E)			
	Businesses (including admin area of Datacentre building)	30.00	60.00	30.00
	Datacentre building	N.A.	60.00	N.A.
vi)	Mercantile (Group F)	30.00	45.00	30.00

Table 4 (Concluded)

Sl No.	Occupancy Group	Maximum Travel Distance m		
		Types 1 and 2 (Unsprinklered)	Types 1 and 2 (Sprinklered)	Types 3 and 4
(1)	(2)	(3)	(4)	(5)
vii)	Industrial (Group G)			
	G-1, G-2	45.00	60.00	see Note 3
	G-3	22.50	45.00	
viii)	Storage (Group H)	30.00	90.00	
ix)	Hazardous (Group J)	22.50	30.00	

NOTES

- 1 The common path of travel should be not more than 50 percent of the above distances.
- 2 Ramps are not to be counted as an exit in case of basements below the first basement in car parking.
- 3 Construction of Type 3 or Type 4 is not permitted.
- 4 Utility service areas and equipment areas may be exempt from the measurement of travel distance from within the room.
- 5 For ceiling height 10 m or above at ground floor in buildings such as airports, exhibition halls, convention centres, atrium in shopping malls, entrance lobbies and the like and provided with sprinkler system, the travel distance in such areas may be increased by 100 percent of the values specified. This is not to be applied to the values given in col (4) above.
- 6 In such buildings having ceiling heights 17 m or more and would have sprinkler exemption, the travel distance of such areas may be allowed to be increased by 100 percent of the values specified. This is not to be applied to the values given in col (4) above. In such cases, to prevent heat build-up at ceiling level, suitable smoke/heat venting systems are recommended.

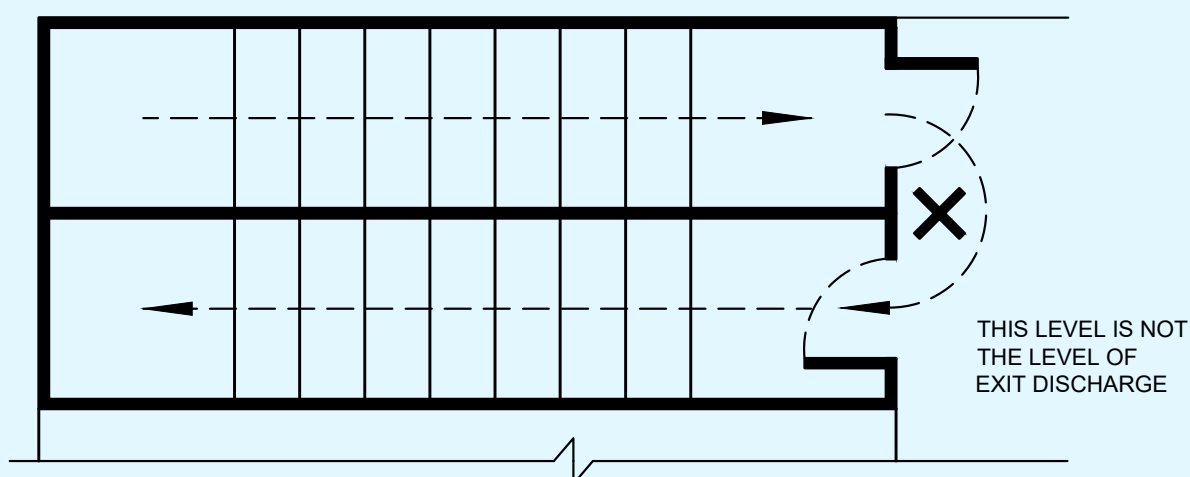
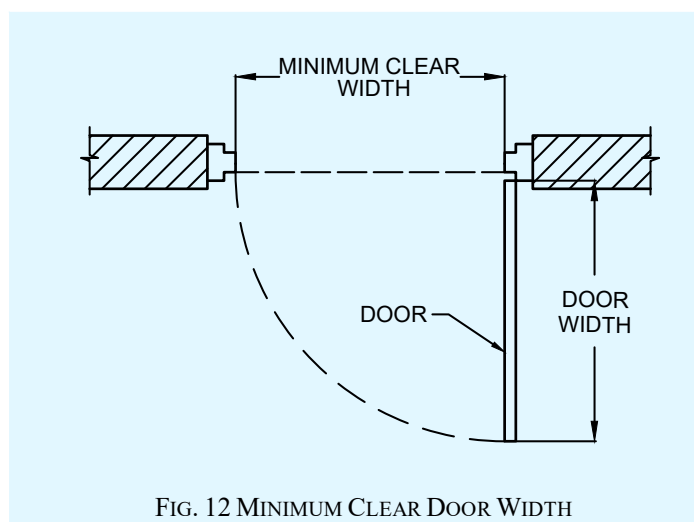


FIG. 11 UNACCEPTABLE ARRANGEMENT FOR ENCLOSING A STAIR SERVING AS A REQUIRED EXIT



4.4.2.4.2 Corridors and passageways of means of egress

- Corridors and passageways should be of width not less than the calculated aggregate width of exit doorways leading from them in the direction of travel to the exit (see [Table 3](#) and [Table 4](#)); and
- In the case of buildings where there is a central corridor, which is part of exit access, the doors of rooms (except for rooms having assembly occupancy) should open inwards to permit smooth flow of traffic in the corridor.

4.4.2.4.3 Staircases

4.4.2.4.3.1 General

The requirements of number of staircases should supplement the requirement of different occupancies in [6.1](#) to [6.9](#).

All high rises and special buildings, as mentioned in [1.2](#), should have a minimum of two staircases. The actual number of staircases should comply with the requirement of [4.4.2.1](#).

NOTE — These required fire exit staircases calculated as per the laid down provisions, are recommended to be free of FAR.

All exit staircases should discharge, at the level of exit discharge, to the exit discharge, either,

- directly, or
- through an exit passageway, or
- through a large lobby.

At least 50 percent of the staircases should discharge as per (a) and/or (b) above. For firefighting shafts, it is preferred to have the discharge through (a), or as an alternate through (b). Option (c) is not applicable for firefighting shafts. In case of residential buildings and educational buildings, firefighting shafts should discharge through (a) only.

Each compartment should be planned with its access to the firefighting shaft. In buildings/floors having three or more compartments, there should be at least two firefighting shafts to serve the compartments. In buildings/floors having three or more compartments, there should be at least two firefighting shafts to serve the compartments.

The minimum width of tread without nosing should be 250 mm for staircase of residential buildings. This should be minimum 300 mm for assembly, hotels, educational, institutional, business, and other buildings. The treads should be constructed and maintained in a manner to prevent slipping. The maximum height of riser should be 175 mm for staircase of one or two family dwelling residential buildings and 150 mm for other buildings. The number of risers in a flight is recommended as 12. However, considering the floor-to-floor height, there should not be more than 15 risers in a flight.

The staircases may be internal staircases or external staircases.

4.4.2.4.3.2 Internal staircases

The internal staircases may be constructed with an external wall, or otherwise, and should comply with the following:

- a) Internal stairs should be constructed of non-combustible materials throughout, and should have fire resistant rating of minimum 120 min.
- b) A staircase should not be arranged round a lift shaft.
- c) Fire exits should not be used as a portion of a supply, return or exhaust air system serving adjoining areas. Any opening(s) should not be permitted in walls or in doors, separating fire exits from adjoining areas.
- d) No flue chimney, electromechanical equipment, air conditioning units, gas piping or electrical panels should be allowed in the stairway.
- e) Notwithstanding the detailed provision for exits in accordance with [4.2](#) and [4.3](#), the following minimum width should be provided for staircases for respective building typology:
 - 1) Residential building (One or two family dwellings) : 1.00 m
NOTE — For row housing with 2 storeys, the minimum width should be 0.75 m.
 - 2) Residential building (A-I, A-II and A-III) : 1.25 m
 - 3) Residential hotel building (A-IV and A-V) : 1.50 m
 - 4) Assembly building : 2.00 m
NOTE — The width of stairs may be accepted to be 1.50 m in case of assembly occupancy having less than 150 persons.
 - 5) Educational building : 1.50 m
 - 6) Institutional building : 2.00 m
Maximum width of stairs in hospitals should be 2.00 m.
 - 7) All other buildings : 1.50 m
NOTE — Where staircase width is above 2.00 m, it is recommended to provide a central handrail.
- f) A handrail should be provided on one side of the staircase of width less than 1 500 mm, and on both sides of the staircase of width 1 500 mm and more. The projection of handrail(s) in the staircase width should not be more than 90 mm.
- g) Handrails may project inside the measured width by not more than 90 mm.
- h) The design of staircase should also take into account the following:
 - 1) The minimum headroom in a passage under the landing of a staircase and under the staircase should be 2.2 m.
 - 2) Access to fire sexit staircase should be through a fire door of a minimum 120 min fire resistance rating.
 - 3) No living space, store or other fire risk should open directly into staircases.
 - 4) The exit (including staircases) should be continuous from refuge floors or terrace level, as applicable, to the level of exit discharge.
 - 5) No electrical shafts/air conditioning ducts or gas pipes, etc, should pass through or open in the staircases.
 - 6) Lifts should not open in staircase (or in mid-landings).
 - 7) No combustible material should be used for decoration/wall panelling in the staircase.
 - 8) Beams/columns and other building features should not reduce the head room/width of the staircase.
 - 9) The floor indication board, indicating the location/designated number of staircase, respective floor number and direction to exit discharge should be placed inside the staircase, on the wall nearest to the fire door. It should be of size not less than 300 mm × 200 mm (see [Fig. 13](#)).
 - 10) Individual floors should be prominently indicated on the wall outside the staircase and facing it.

- 11) All staircase should terminate at the level of exit discharge. The access to the basement should be by a separate staircase.
- 12) Scissors type staircases (of type where intermingling of persons in the landing is possible or of type where both stairs use the common air space) should not be treated as part of exit.

4.4.2.4.3.3 Curved stairs

Curved stairs should not be treated as part means of egress. However, these may be used as part of exit access provided the depth of tread is not less than 280 mm at a point 350 mm from the narrower end of the tread and the smallest radius is not less than twice the stair width.

4.4.2.4.3.4 External staircases

The external staircases are the staircases provided on the external wall/facade, and should comply with the following:

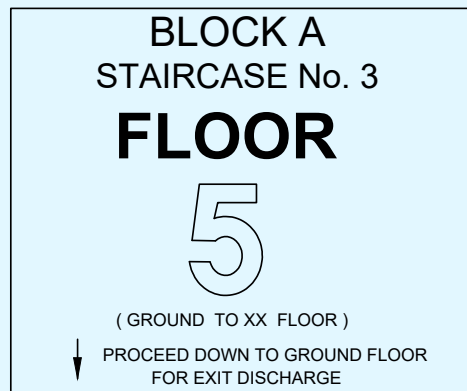
- a) External stairs should always be kept in sound and usable condition;
- b) All external stairs should be directly connected to the ground;
- c) Entrance to the external stairs should be separate and remote from the internal staircase;
- d) Where an external staircase is provided, it should be ensured that the use of it at the time of fire is not prejudiced by smoke and flame from openings (for example, windows, doors) in the external face of the building. Care should be taken to ensure that no external wall or window opening opens on to or close to an external stair. If such openings exists within 3 m from an external staircase, they should be protected with fire rated doors/window assemblies with rating of at least 60 min (*see* [Fig. 14](#));
- e) The external stairs should be constructed of non-combustible materials, and any doorway leading to it should have minimum 120 min fire resistance;
- f) No external staircase, should be inclined at an angle greater than 45° from the horizontal;
- g) External stairs should have straight flight not less than 1 500 mm wide;
- h) Handrails, to be provided on both sides, should be of a height not less than 1 000 mm and not exceeding 1 200 mm. There should be provisions of balusters with maximum gap of 150 mm; and
- j) The use of spiral staircase should be limited to low occupant load and to a building not exceeding 9 m in height. A spiral staircase should be not less than 1 500 mm in diameter and should be designed to give adequate headroom.

4.4.2.4.3.5 Ramps

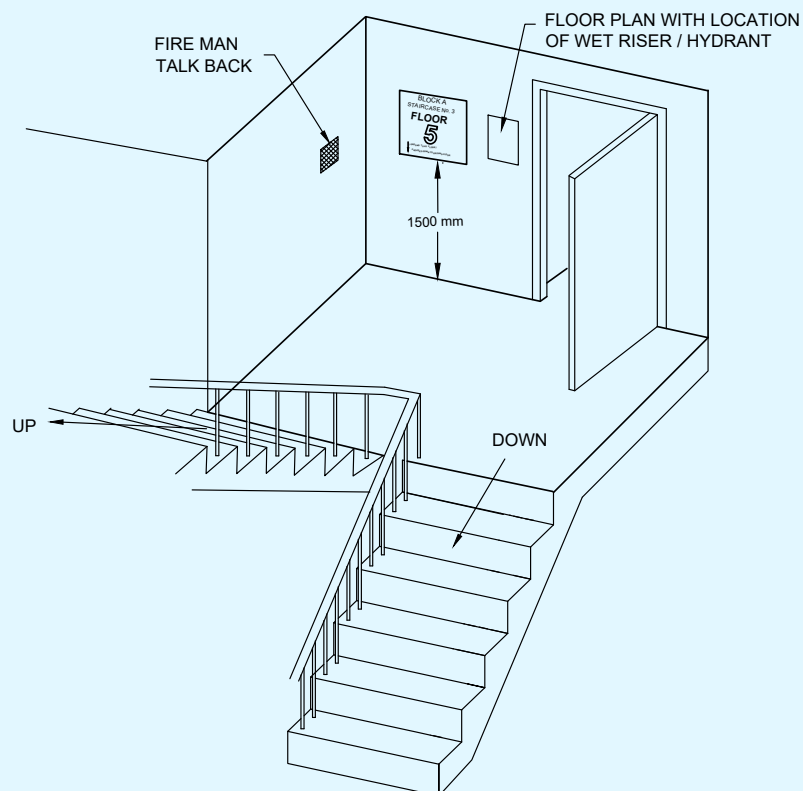
- a) Ramps should comply with all the applicable requirements for staircases regarding enclosure, capacity (*see also* [Table 3](#)) and limiting dimensions, except where specified in [6.1](#) to [6.9](#) for special uses and occupancies;
- b) The slope of a ramp should not exceed 1 in 12 (8 percent);
- c) Ramp(s) should be surfaced with approved slip resistant materials that are securely attached. No perforations are permissible on ramp floors;
- d) Any changes in travel direction in ramp should be preceded by landings of minimum 1.5 m × 1.5 m size;
- e) Ramps and intermediate landings should continue with no decrease in width along the direction of egress travel;
- f) Outside ramps and landings should be designed to minimize water accumulation on their surfaces;
- g) Ramps should have landings located at the top, at the bottom, and at doors opening onto the ramp;
- h) Every landing should be not less than 1 500 mm long in the direction of travel;
- j) Where the ramp is not part of an accessible route, the ramp landings should not be required to exceed 1 250 mm in the direction of travel, provided that the ramp has a straight run; and
- k) Handrails should be provided on all ramps on both sides (*see* [4.4.2.4.3.4](#)).

NOTE — Above requirements are not applicable to basement car parking ramps.

The ramps should, in addition, comply with the requirements given in good practice [F(31)].



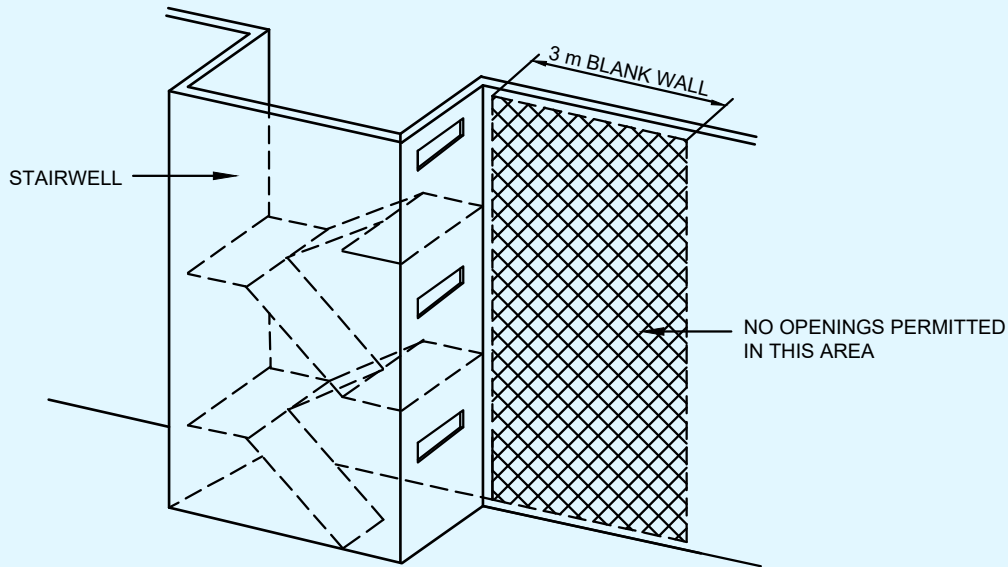
9A EXAMPLE OF A STAIRWAY MARKING SIGN (FLOOR INDICATION BOARD)



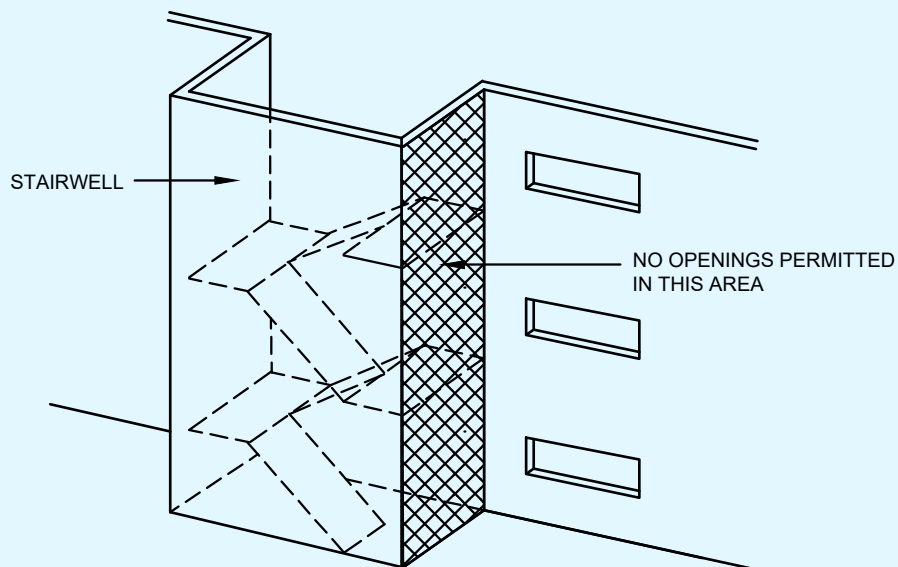
9B STAIR SIGN PLACEMENT

NOTE — BLOCK/WING/BUILDING NAME TO BE PROJECT SPECIFIC. STAIRCASE No. SHOWN AS "3" IS INTENDING TO SHOW THE NUMBER ASSIGNED TO THE STAIRCASE. ALL EXITS PREFERABLY TO BE ASSIGNED WITH NUMBER /IDENTIFICATION ENABLING OCCUPANT/FIRE MAN TO DECLARE LOCATION / POSITION.

FIG. 13 SIGN MARKING AND REQUIREMENT IN EXIT



14A LIMITATIONS ON OPENINGS NEAR STAIRWELL



14B OPENING RESTRICTIONS ON STAIRWELL WALLS

FIG. 14 OPENING RESTRICTIONS

4.4.2.5 Smoke control of exits

- a) In building design, compartmentation plays a vital part in limiting the spread of fire and smoke. The design should ensure avoidance of spread of smoke to adjacent spaces through the various leakage openings in the compartment enclosure, such as cracks, openings around pipes ducts, airflow grills and doors. In the absence of proper sealing of all these openings, smoke and toxic gases will obstruct the free movement of occupants of the building through the exits. Pressurization of staircases is of great importance for the exclusion of smoke and toxic gases from the protected exit.
- b) Pressurization is a method adopted for protecting the exits from ingress of smoke, especially in high-rise buildings. In pressurization, air is injected into the staircases, lobbies, etc, as applicable, to raise their pressure slightly above the pressure in adjacent parts of the building. As a result, ingress of smoke or toxic gases into the exits will be prevented. The pressurization of staircases and lift lobbies should be adopted as given in [Table 5](#).

The pressure difference for staircases should be 50 Pa.

Pressure differences for lobbies (or corridors) should be between 25 Pa and 30 Pa. Further, the pressure differential for enclosed staircase adjacent to such lobby (or corridors) should be 50 Pa. For enclosed staircases adjacent to non-pressurized lobby (or corridors), the pressure differential should be 50 Pa.

- c) Equipment and ductwork for staircase pressurization should be in accordance with one of the following:
 - 1) Directly connected to the stairway by ductwork enclosed in non-combustible construction.
 - 2) If ducts used to pressurize the system are passed through shafts and grilles are provided at each level, it should be ensured that hot gases and smoke from the building cannot ingress into the staircases under any circumstances.
- d) The normal air conditioning system and the pressurization system should be designed and interfaced to meet the requirements of emergency services. When the emergency pressurization is brought into action, the following changes in the normal air conditioning system should be effected:
 - 1) Any re-circulation of air should be stopped and all exhaust air vented to atmosphere.
 - 2) Any air supply to the spaces/areas other than exits should be stopped.
 - 3) The exhaust system may be continued provided,
 - i) the positions of the extraction grills permit a general air flow away from the means of egress;
 - ii) the construction of the ductwork and fans is such that, it will not be rendered inoperable by hot gases and smoke; and
 - iii) there is no danger of spread of smoke to other floors by the path of the extraction system which can be ensured by keeping the extraction fans running.
- e) For pressurized stair enclosure systems, the activation of the systems should be initiated by signalling from fire alarm panel.
- f) Pressurization system should be integrated and supervised with the automatic/manual fire alarm system for actuation.
- g) Wherever pressurized staircase is to be connected to unpressurized area, the two areas should be segregated by 120 min fire resistant wall.
- h) Fresh air intake for pressurization should be away (at least 4 m) from any of the exhaust outlets/grille.

Table 5 Pressurization of Staircases and Lift Lobbies

[Clauses [4.4.2.5 \(b\)](#) and [D-2](#)]

SI No.	Component	Height of the Building		
		Less than 15 m	15 m to 30 m	More than 30 m
(1)	(2)	(3)	(4)	(5)
i)	Internal staircase not with external wall	Pressurized except for residential buildings (apartment houses and one or two family dwellings)	Pressurized	Pressurized
ii)	Internal staircase with external wall	Pressurized except for residential buildings (Apartment houses and one or two family dwellings) or Naturally ventilated	Naturally ventilated or Pressurized	Cross-ventilated or Pressurized
iii)	Lift lobby	Not required at ground and above. However, lift lobby segregation and pressurization is required for lift commuting from ground to basement	Naturally ventilated or Pressurized ¹⁾	Cross-ventilated or Pressurized ¹⁾
NOTES 1 The natural ventilation requirement of the staircase should be achieved through opening at each landing, of an area 0.5 m² to 1.5 m² (as defined earlier) in the external wall. A cross ventilated staircase should have 2 such openings in opposite/adjacent walls or the same should be cross-ventilated through the corridor. 2 Enclosed staircase leading to more than one basement should be pressurized. ¹⁾ Lift lobby with fire doors (120 min) at all levels with pressurization of 25 Pa to 30 Pa is required. However, if lift lobby cannot be provided at any of the levels in air-conditioned buildings or in internal spaces where funnel/flue effect may be created, lift hoistway should be pressurized at 50 Pa. For building greater than 30 m, multiple point injection air inlets to maintain desired pressurization level should be provided. If the lift lobby, lift, and staircase are part of firefighting shaft, lift lobby necessarily has to be pressurized in such case, unless naturally ventilated.				

4.5 Compartmentation

4.5.1 General

It is important to limit the spread of a fire in special/high rise buildings, usually through the use fire barriers. In some instances, these barriers need to be penetrated for ductwork, plumbing and electrical systems, and in such cases, use of passive fire protection measures should be done so that the integrity of these barriers is not compromised.

4.5.2 The compartmentation can be achieved with following type of construction/type of fire barrier with protected openings:

- Type A — Masonry construction with fire rated door assembly;
- Type B — Fire rated fabric dropdown curtain assembly;
- Type C — Fire rated glazed partition and door assembly;
- Type D — Fire rated gypsum board/cementitious board assembly; and
- Type E — Masonry construction with fire rated metal sliding door.

Integrity (E) is the ability of the fire curtain assembly when exposed to fire from one side to withstand temperature up to 1 050 °C, to resist passage of flames and hot gases for a period of 120 min.

Radiation control (EW) is the ability of the fire curtain assembly when exposed to fire from one side to resist transfer of heat on the other side. When measured, it should not exceed 15 kW/m² at 1 m distance.

Insulation (EI) is the ability of the fire curtain assembly when exposed to fire from one side to limit temperature rise on the other not exceeding 140 °C above the ambient.

Smoke control (Sa) is the ability of fire curtain assembly when exposed to a fire from one side resist passage of hot/cold smoke not exceeding 3 m³/m/h at a pressure of smoke leakage rate of 25 Pa.

4.5.2.1 Type B — Fire rated fabric dropdown curtain assembly

The fire rated fabric should be tested and certified by any reputed third-party independent certification body for following parameters:

- Type 1 — Integrity (E) of 120 min + Radiation (EW) of 60 min + Smoke control (Sa);
- Type 2 — Integrity (E) of 120 min + Radiation (EW) of 120 min + Smoke control (Sa); and
- Type 3 — Integrity + Radiation (EW) of 120 min + Insulation (EI) of 60 min + Smoke control (Sa).

4.5.2.2 Type C — Fire rated glazed partition and door assembly

The fire rated glazed partition and door assembly should be tested and certified by any reputed third-party independent certification body for following parameters:

- Type 1 — Integrity (E) of 120 min + Radiation (EW) of 120 min + Insulation (EI) of 30 min + Smoke control (Sa); and
- Type 2 — Integrity (E) of 120 min + Radiation (EW) of 120 min + Insulation (EI) of 60 min + Smoke control (Sa)

4.5.3 Floor(s) should be compartmented with area as given in [Table 6](#). The maximum size of the compartment is suggested as follows. However, State/Local Authorities may prescribe their own requirements based on their risk assessment and other fire safety provisions.

Table 6 Compartmentation (Clause 4.5.3)				
Sl No.	Type of Building	Floor Area Compartmentation Area (without Sprinkler) m ²	Floor Area Compartmentation (when Sprinklered) m ²	Type of Construction/ Type of Fire Barrier with Protected Openings
(1)	(2)	(3)	(4)	(5)
i)	Enclosed basement vehicle parking in residential, educational, institutional, assembly, business, mercantile, industrial or storage occupancies	750	5 000	E - Barrier between the designated fire compart- ments/zones B - Limited to driveway not exceeding 9 m with Type-2 only.

Table 6 (Continued)

Sl No.	Type of Building	Floor Area Compartmentation on Area (without Sprinkler) m ²	Floor Area Compartmentation (when Sprinklered) m ²	Type of Construction/ Type of Fire Barrier with Protected Openings
(1)	(2)	(3)	(4)	(5)
ii)	Basements other than car parking in residential, educational, institutional, assembly, business, mercantile, industrial and storage occupancies	750	2 000	A/D - barrier between the designated fire compartments/zones
iii)	Hospitals and nursing homes with beds/sleeping patients in institutional occupancies	750	2 000	Any combination of A, B (Type-3 only), D - barrier between the designated fire compartments/zones
iv)	Buildings used for custody and care of children, convalescents, aged people who are incapable of self-preservation	750	1 500	Any combination of A, B (Type-3 only), D - barrier between the designated fire compartments/zones
v)	Mercantile buildings, multistorey mercantile buildings consisting of shops, departmental stores, markets	750	Not essential	Any combination of A, B (Type 2 only), D - barrier between the designated fire compartments/zones
vi)	Assembly buildings with/without atriums	1 500	Not essential	Any combination of A, B (Type 2 only), D - barrier between the designated fire compartments/zones
vii)	Buildings under business occupancy	750	3 000	Any combination of A, B, C (Type 1), D - barrier between the designated fire compartments/zones
viii)	Datacentre buildings	See Note 1	2 000	Any combination of A, C (Type 2) , D - barrier between the designated fire compartments/zones
ix)	Hotel building with or without dining facilities	750	3 000	Any combination of A, C (Type 1) , D - barrier between the designated fire compartments/zones
x)	Industrial buildings less than 6.7 m in height	1 500	10 000	Any combination of A, E - barrier between the designated fire compartments/zones

Table 6 (Concluded)

Sl No.	Type of Building	Floor Area Compartmentation on Area (without Sprinkler) m ²	Floor Area Compartmentation (when Sprinklered) m ²	Type of Construction/ Type of Fire Barrier with Protected Openings
(1)	(2)	(3)	(4)	(5)
xi)	Hazardous or high hazard buildings	See Note 1	500	A-barrier between the designated fire compartments/zones
xii)	Buildings with atrium with smoke management system	See Note 1	Not required	Subject to compliance of atrium as in Annex E
xiii)	Warehouses less than 5.0 m in height	See Note 1	5 000	See also good practice [F(142)]
NOTES 1 Compartmentation size is not specified, as the buildings have to be sprinklered. 2 Ceiling level partition may be considered (from ceiling level to 25 percent of floor height) to act as a smoke reservoir and venting thereof.				

4.5.3.1 For space with large heights [such as airports, exhibition halls, convention centres, atrium in shopping-malls, warehouses (of height beyond 5 m), industrial (of height beyond 6.7 m), double height entrance lobbies and the like], these may be exempted from above requirement of compartmentation.

4.5.3.2 In addition to the above floor area compartmentation, all critical utility rooms within each compartment area construction should be 2 h fire rated. Rooms such as electrical room, server room, uninterrupted power supply (UPS) room, battery room, lift machine rooms, IT networking room, main distribution frame (MDF), intermediate distribution frame (HUB/IDF rooms), main communication room (MCR room), network switch rooms, multiplexer (MUX) rooms, telecom equipment room, janitor storage room, chemical storage room, QA and QC laboratories, etc, should be enclosed from all sides and construction should be 2 h fire rated irrespective of the size/area.

4.5.3.3 Guest rooms, guest room corridors, banquets, should be 60 m fire compartmented with 30 min fire rated doors, in addition to above area of compartmentation.

4.5.3.4 In buildings with open atrium, the effect of compartmentation on the atrium should be disregarded, provided the fire safety requirements given in [Annex E](#) on atrium are complied.

4.5.3.5 In industrial and storage occupancies, all hazardous areas irrespective of area/size should be compartmented namely, vehicle charging areas, ignitable chemical storage areas, hazardous waste storage areas, flammable scrap storage areas, unless these are located 30 m away from any adjacent occupied buildings.

4.5.3.6 All live cooking areas with any type of fuel (induction, LPG, PNG, etc) in hotels, IT parks, business buildings cafeteria, hostels, educational facility, food courts, malls, any commercial kitchens should be 120 min fire rated segregated from other areas.

4.5.3.7 Pantries without live cooking, buffet setting counters (*bain-marie* serving) (with/without re-heating facility), breakout areas with microwave stations, water and dispensing areas, collaboration areas, tea coffee maker areas should be exempted from above compartmentation requirements.

4.6 Smoke Control

4.6.1 Smoke Exhaust and Pressurization of Areas Above Ground

Corridors in exit access (exit access corridor) are created for meeting the requirement of use, privacy, and layout

in various occupancies. These are most often noted in hospitality, health care occupancies and sleeping accommodations.

Exit access corridors of guest rooms and indoor patient department/areas having patients lacking self-preservation and for sleeping accommodations such as service apartments, custodial, penal, and mental institutions, etc, should be provided with 60 min fire resistant wall and 30 min self-closing fire doors along with all fire stop sealing of penetrations.

Smoke exhaust system having make-up air and exhaust air system or alternatively pressurization system with supply air system for these exit access corridors should be required.

Smoke exhaust system having make-up air and exhaust air system should also be required for theatres/auditoria.

Such smoke exhaust system should also be required for large lobbies and which have exit through staircase leading to exit discharge. This would enable eased exit of people through smoke-controlled area to exit discharge.

All exit passageway (from exit-to-exit discharge) should be pressurized or naturally ventilated. The mechanical pressurization system should be automatic in action with manual controls in addition. All such exit passageway should be maintained with integrity for safe means of egress and evacuation. Doors provided in such exit passageway should be fire rated doors of 120 min rating.

Smoke exhaust system where provided, for above areas and occupancies should have a minimum of 12 air changes per hour smoke exhaust mechanism.

Pressurization system where provided, should have a minimum pressure differential of 25-30 Pa in relationship to other areas.

The smoke exhaust fans in the mechanical ventilation system should be fire rated, that is, 250 °C for 120 min.

For naturally cross-ventilated corridors or corridors with operable windows, such smoke exhaust system or pressurization system will not be required.

4.6.2 Smoke Exhaust and Pressurization of Areas Below Ground

- a) Each basement should be separately ventilated;
- b) Vents with cross-sectional area (aggregate) not less than 2.5 percent of the floor area spread evenly round the perimeter of the basement should be provided in the form of grilles, or breakable stall board lights or pavement lights or by way of shafts.

Alternatively, a system of mechanical ventilation system may be provided. Mechanical ventilation system should be designed to permit 12 air changes per hour (ACPH) in case of fire or distress call. However, for normal operation, air changes schedule should be as given in Part D 'Building Services, Section 3 Air conditioning, Heating and Mechanical Ventilation' of this NBCS. In case of single basement, fresh-air may be considered from ramp and perimeter opening to achieve 12 ACPH ventilation requirement.

- c) In multi-level basements, independent air intake and smoke exhaust shafts (masonry or reinforced concrete) for respective basement levels and compartments therein should be planned with its make-up air and exhaust air fans located on the respective level and in the respective compartment. Alternatively, in multi-level basements, common intake masonry (or reinforced cement concrete) shaft may serve respective compartments aligned at all basement levels. Similarly, common smoke exhaust/outlet masonry (or reinforced cement concrete) shafts may also be planned to serve such compartments at all basement levels. All supply air and exhaust air fans on respective levels should be installed in fire resisting room of 120 min. Exhaust fans at the respective levels should be provided with back draft damper connection to the common smoke exhaust shaft ensuring complete isolation and compartmentation of floor isolation to eliminate spread of fire and smoke to the other compartments/floors.
- d) Due consideration should be taken for ensuring proper drainage of such shafts to avoid insanitation condition. Inlets and extracts may be terminated at ground level with stall board or pavement lights as before. Stall board and pavement lights should be in positions easily accessible to the fire brigade and clearly marked 'AIR INLET' or 'SMOKE OUTLET' with an indication of area served at or near the opening.

- e) Smoke from any fire in the basement should not obstruct any fire exit serving the ground and upper floors of the building.
- f) The smoke exhaust fans in the mechanical ventilation system should be fire rated, that is, 250 °C for 120 min.
- g) The smoke ventilation of the basement car parking areas should be through provision of supply and exhaust air ducts duly installed with its supports and connected to supply air and exhaust fans. Alternatively, a system of impulse fans (jet fans) may be used for meeting the requirement of smoke ventilation complying with the following:
 - 1) Structural aspects of beams and other down stands/services should be taken care of in the planning and provision of the jet fans.
 - 2) Fans should be fire rated, that is, 250 °C for 120 min.
 - 3) Fans should be adequately supported to enable operations for the duration as above.
 - 4) Power supply panels for the fans should be located in fire safe zone to ensure continuity of power supply.
 - 5) Power supply cabling should meet circuit integrity requirement in accordance with accepted standard [F(13)].

The smoke extraction system should operate on actuation of flow switch actuation of sprinkler system. In addition, a local and/or remote 'manual start-stop control/switch' should be provided for operations by the fire fighters.

Visual indication of the operation status of the fans should also be provided with the remote control.

No system relating to smoke ventilation should be allowed to interface or cross the transformer area, electrical switchboard, electrical rooms or exits.

Smoke exhaust system having make-up air and exhaust air system for areas other than car parking should be required for common areas and exit access corridor in basements/underground structures and should be completely separate and independent of car parking areas and other mechanical areas.

Supply air should not be less than 5 m from any exhaust discharge openings.

For specific details on smoke management using mechanical means, *see* Part D 'Building Services, Section 3 Air Conditioning, Heating and Mechanical Ventilation' of this NBCS.

4.7 Gas Supply

4.7.1 Town Gas/LPG Supply Pipes

Where gas pipes are run in buildings, the same should be run in separate shafts exclusively for this purpose and these should be on external walls, away from the staircases. Gas distribution pipes should always be below the false ceiling. The length of these pipes should be as short as possible. In the case of kitchen cooking range area, hood should have grease filters using metallic grille to trap oil vapours escaping into the fume hood.

NOTE — For detailed information on gas pipe installations, reference may be made to Part E 'Plumbing services, Section 4 Gas supply' of this NBCS.

4.7.2 Thermal Detectors

These should be installed into fume hoods of large kitchens for hotels, hospitals, and similar areas located in high rise buildings. Arrangements should be made for automatic tripping of the exhaust fan in case of fire. If gas is used, the same should be shut off. The voltage should be 24 V or 100 V d.c. operated with external rectifier. The valve should be of the hand re-set type and should be located in an area segregated from cooking ranges. Valves should be easily accessible. The hood should have manual facility for steam or suitable hood extinguishing gas released depending on duty condition.

4.7.3 Gas cylinders and manifold should need to be housed in a detached location with no other occupancy within distances prescribed in good practice [F(14)] thereof. There should be an enclosure suitably ventilated. It is desirable to provide medium velocity spray nozzles which can be operated by quick opening valve situated away

from the enclosure.

4.7.4 In the case of gas cylinders, if manifold has to be installed on podium/close to podium, the same should be away from any air intakes/smoke exhaust openings/any windows.

4.7.5 Pressure regulating stations should be designed and installed at critical locations for excess flow shut off valves. Seismic shut off valve at the main distribution point should be installed for buildings having mixed use assembly occupancy (like malls) and institutional occupancy above 15 m.

4.7.6 Gas meters should be housed in a suitably constructed metal cupboard located in a well-ventilated space, keeping in view the fact that LPG is heavier than air and natural gas is lighter than air.

4.7.7 Wherever LPG reticulation/cylinders are used in buildings above 100 m, gas leak detectors should be provided at the usage points and monitored from fire command centre. The cables used for signalling should be circuit integrity cables.

4.7.8 The gas lines should not be installed through any enclosed staircase or electrical shafts, escape routes, refuge areas/refuge floors.

4.7.9 Kitchens working on LPG fuel should not be permitted in basements. Small kitchen using other than LPG fuel, such as induction/PNG/electricity-based, may be permitted in incidental occupancies in the basement.

4.8 Hazardous Areas, Gaseous, Oil Storage Yard, etc

Rooms containing high pressure boilers, refrigerating machinery, transformers, or other service equipment subject to possible explosion should not be located directly under or adjacent to fire exits. All such rooms should be effectively cut-off from other parts of the building and should be provided with adequate vents to the outside air.

All rooms or areas of high hazard in additions to those hereinbefore mentioned, should be segregated, or should be protected with fire resistant walls having fire rating of 120 min as fire, explosion or smoke therefrom is likely to interfere with safe egress from the building. Further,

- a) each building should be provided with an approved outside gas shut-off valve conspicuously marked. The detailed requirements regarding safe use of gas should be as specified in Part E 'Plumbing services, Section 4 Gas Supply' of this NBCS; and
- b) all exterior openings in a boiler room or rooms that contain central heating equipment, if located below opening in another storey or if less than 3 m from other doors or windows of the same building should be protected by a fire assembly as in [3.5.4](#). Such assemblies should be fixed, automatic or self-closing.

4.9 Fire Detection and Alarm

- a) The requirements of fire detection and alarm systems are covered for each occupancy in [Table 7A](#) to [Table 7J](#) and under [6.1](#) to [6.9](#) for annunciation to occupants in view of the ensuing vulnerability and to warn occupants early of the existence of fire, so as to facilitate orderly and safe egress;
- b) Fire detection and alarm systems in buildings should be so planned and programmed so as to enable operations of various systems and equipment to facilitate requirements leading to life safety, compartmentation and fire protection. These systems and equipment may include electromechanical systems such as air handling units; pressurization systems; smoke management systems; creation of compartmentation through the release of fire barrier, hold-up fire doors, etc; and monitoring of fire water storage tanks and pumps, pressures in hydrant and sprinkler system, etc. These planning and requirements should be based on building occupancy and other requirements on case-to-case basis;
- c) Voice evacuation systems should employ English, Hindi and vernacular language using pre-recorded messages and integrate with fire alarm panels for alerting the zone of fire and surrounding zones/floors as required for annunciation (*see also* [Table 7A](#) to [Table 7J](#));
- d) Appropriate visual warning arrangement through visual strobes/beacons may be considered in appropriate situations particularly in public buildings, at required locations to ensure visual as well as alarm for persons with hearing impairment;
- e) For assembly buildings, institutional buildings and all buildings above 30 m in height where fire alarm system is provided in accordance to [Table 7A](#) to [Table 7J](#), detectors should also be provided inside the electrical shafts, lift machine rooms, etc, besides occupancy areas;

- f) Fire alarm panels should be connected in peer-to-peer network or with redundant cables, run in different shafts. Each panel should be able to work in standalone mode and master slave architecture may be used where required;
- g) The fire detection system should be in accordance with accepted standards [F(15)];
- h) Guidelines for selection of various types of fire detectors for different occupancies and their installation and maintenance should be in accordance with good practice [F(16)]. *See also* Handbook for 'Asset and Facility Management' for guidance;
- j) In buildings where automatic fire alarm system is provided, the following should be monitored from fire alarm panel:
 - 1) Water level in all tanks.
 - 2) Hydrant and sprinkler pressures of respective zones, as provided.
 - 3) Pump 'ON/OFF' status.
 - 4) All isolation valves, wherever provided with supervisory switch (non-padlock valves).
 - 5) Other requirements to meet electro-mechanical services interface.

Table 7A Requirements for Firefighting Installations – For Residential Buildings

[Clauses [4.9 \(a\)](#), [4.9 \(c\)](#), [4.9\(c\)](#), [5.1.1 \(a\)](#), [5.1.1\(d\)](#), [5.1.2.1](#), [5.1.3 \(a\)](#), [D-6.1](#), [G-2\(f\)](#) and [Table 31](#)]

SI No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Fire Detection and Alarm System and LPG/PNG Gas Detector	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	RESIDENTIAL BUILDINGS (A)								
a)	Lodging and Rooming Houses (A- I)	Up to 500 m² area and up to 24 m height (both conditions to be met), self-certification by a certified and State approved building professional should be treated acceptable							
1)	24 m and above in height and, having any area Or Less than 24 m in height but of area,								
i)	floor area on any single floor 501 m ² to 1 000 m ² (HL-1) See Notes 1, 2, 3 and 4	R	R	NR	NR	NR	NR	R	NR
ii)	1 001 m ² and above (HL-2) See Notes 2, 3 and 4	R	R	R	NR	NR	NR	R	R

Table 7A (Continued)

SI No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Fire Detection and Alarm System and LPG/PNG Gas Detector	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
b)	Dormitories (A-II)	Up to 500 m² area and up to 24 m height (both conditions to be met), self-certification by a certified and State approved building professional should be treated acceptable							
1)	501 m ² to 1 000 m ² (HL-1) See Notes 1, 2, 3 and 4	R	R	NR	NR	NR	NR	NR	NR
2)	1 001 m ² and above (HL-2) See Notes 2, 3 and 4	R	R	R	NR	NR	NR	R	NR
c)	Apartment Houses (A-III)	Up to 500 m² area and up to 24 m height (both conditions to be met), self-certification by a certified and State approved building professional should be treated acceptable							
1)	24 m and above but not exceeding 60 m in height; and, height less than 24 m, but of area:								
i)	501 m ² to 1 000 m ² (HL-2) See Notes 2, 3, 4, 7 and 9	R	R	R	NR	NR	NR	NR	NR

Table 7A (Continued)

SI No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Fire Detection and Alarm System and LPG/PNG Gas Detector	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ii)	1 001 m ² and above (HL-3) See Notes 2, 3, 4, 5, 7, 9 and 10	R	R	NR	R	R	NR	R	R
2)	Above 60 m height (CL-4) See Notes 3, 4, 5, 7, 8, 9, and 10	R	R	NR	R	R	R	R	R
d)	Hotels (A-IV)	Up to 500 m² area and up to 24 m height (both conditions to be met), self-certification by a certified and State approved building professional should be treated acceptable							
1)	Up to 24 m in height, but of area:								
i)	501 m ² to 1 000 m ² (HL-3) See Notes 2, 3, 4, 7 and 9	R	R	R	NR	NR	NR	R	NR
ii)	1 001 m ² and above (CL-3) See Notes 3, 4, 7 and 9	R	R	NR	R	NR	R	R	R

Table 7A (Concluded)

SI No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Fire Detection and Alarm System and LPG/PNG Gas Detector	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
2)	24 m and above but not exceeding 60 m in height OR 3/4 Star Hotels (CL-4) See Notes 3, 4, 5, 7, 8 and 9	R	R	NR	R	R	R	R	R
3)	Above 60 m – See Note 5								
e)	Starred (Franchise – 5 Star) Hotels (A-V)								
1)	Up to 60 m (CL-5) See Note 4	R	R	NR	R	R	R	R	R
2)	Above 60 m – See Note 5								

R – Required

NR – Not Required

REMARK 1: Any state or area where the municipal supply caters to municipal/public yard hydrant system with hydrant outlets provided throughout the premises with adequate water supply meeting the wet riser requirement demand flow and pressure 24×7 uninterrupted, such states/areas/ jurisdiction can replace the wet riser system and its associated pumping with water storage with a dry riser system. Dry riser system is also recommended for areas where ambient temperature below is 4°C and has facility of connecting the yard hydrant to the dry riser in any event of fire in the building. This system is to be maintained with quarterly checks to ensure the system integrity to handle the required flow and pressure. This is applicable for buildings height only up to 24 m where HL Protection is required as per above Table.

REMARK 2: Sharing of water storage tank may be permitted between the adjacent neighbouring properties including industry clusters, without any loss of hydraulic effects (which is essential to fight the fire during an emergency).

NOTES

- 1 If the building has kitchen, the protection level will be **HL-2**.
- 2 If the building has basement to be protected with sprinklers, the protection level should comply with **CL-3**.
- 3 If the building has electrical vehicles (EV) parks in podium or basements, it should be protected with sprinklers and complying to **CL-5**.
- 4 Rate of rise of heat sensor/detection system should be provided in EV (Electrical Vehicles) parks and charging areas. Smoke sensor/detectors should be provided in all basements in all areas other than internal combustion engine (ICE) vehicles parks as part of the fire alarm system. ICE vehicle parks should be provided with carbon monoxide sensors as part of fire alarm system to activate ventilation system.
- 5 If the building height exceeds 60 m the hydraulic calculations meeting recommended system architecture with multi-level pumping and storage with gravity redundant down comer may be considered. The pumps considered should not exceed 12 bar shutoff pressure.
- 6 Pump pressure and quantity should be supported by hydraulic calculations.
- 7 2,500 lpm at 10 bar common pumping and water storage of 2,55,000 litre should be provided for a cluster of maximum 5 buildings, with or without common basement. The buildings should not be closer than 9 m. If the buildings are closer than 9 m, or the basement is used for EV parking and or charging facilities, the protection level should comply with **CL-5**.
- 8 Multi-level pumping and water storage as per schematic line drawing (SLD) may be considered for cluster buildings more than 45 m in height; and water storage at every 45 m above should be provided for individual buildings.
- 9 If the cluster of buildings exceeds 5, another set of pumps and water storage of similar capacity should be provided. A maximum of two such sets is recommended, provided it satisfies the hydraulic requirements and functioning at the remotest point.
- 10 Table 7K and Table 7M suggesting the basis for water quantity required for fire-fighting, respectively for combined (sprinkler and hose) protection and hose protection be considered.

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Table 7B Requirements for Firefighting Installations – For Educational Buildings

[Clauses [4.9 \(a\)](#), [4.9 \(c\)](#), [4.9\(c\)](#), [5.1.1 \(a\)](#), [5.1.1\(d\)](#), [5.1.2.1](#), [5.1.3 \(a\)](#), [D-6.1](#), [G-2\(f\)](#) and [Table 31](#)]

SI No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	EDUCATIONAL BUILDINGS (B)	Up to 500 m² area and up to 24 m height (both conditions to be met), self-certification by a certified and State approved building professional should be treated acceptable							
1)	24 m and above but not exceeding 45 m; And, height less than 24 m, but of area:								
i)	501 m ² to 1 000 m ² (HL-3) <i>See Notes 1, 2, 3, 5, 7 and 9</i>	R	R	NR	R	R	NR	R	R
ii)	1 001 m ² and above (CL-4) <i>See Notes 3, 5, 7, 9 and 10</i>	R	R	NR	R	R	R	R	R
2)	45 m and above but not exceeding 60 m in height (CL-4) <i>(See Notes 5, 6, 7, 8, 9 and 10)</i>	R	R	NR	R	R	R	R	R
3)	Above 60 m – <i>See Note 5</i>								

R – Required

NR – Not Required

REMARK 1: Any state or area where the municipal supply caters to municipal/public yard hydrant system with hydrant outlets provided throughout the premises with adequate water supply meeting the wet riser requirement demand flow and pressure 24×7 uninterrupted, such states/areas/ jurisdiction can replace the wet riser system and its associated pumping with water storage with a dry riser system. Dry riser system is also recommended for areas where ambient temperature below is 4°C and has facility of connecting the yard hydrant to the dry riser in any event of fire in the building. This system is to be maintained with quarterly checks to ensure the system integrity to handle the required flow and pressure. This is applicable for buildings height only up to 24 m where HL Protection is required as per above Table.

REMARK 2: Sharing of water storage tank may be permitted between the adjacent neighbouring properties including industry clusters, without any loss of hydraulic effects (which is essential to fight the fire during an emergency).

NOTES

- 1 If there are laboratories, AV/computer rooms, auditoriums, they should be protected with sprinkler and the protection level should comply with **CL-3**.
- 2 If the building has basement to be protected with sprinklers, the protection level should comply with **CL-3**.
- 3 If the building has electrical vehicles (EV) parks in podium or basements, it should be protected with sprinklers and complying to **CL-5**.
- 4 Rate of rise heat sensor/detection system should be provided in Electrical Vehicles (EV) parks and charging areas. Smoke sensor/detectors should be provided in all basements in all areas other than internal combustion engine (ICE) vehicles parks as part of the fire alarm system. ICE vehicle parks should be provided with carbon monoxide sensors as part of fire alarm system to activate ventilation system.
- 5 If the building height exceeds 60 m the hydraulic calculations meeting recommended system architecture with multi-level pumping and storage with gravity redundant down comer may be considered. The pumps considered should not exceed 12 bar shutoff pressure.
- 6 Pump pressure and quantity should be supported by hydraulic calculations.
- 7 2,500 lpm at 10 bar common pumping and water storage of 2,55,000 litre should be provided for a cluster of maximum 5 buildings, with or without common basement. The buildings should not be closer than 9 m. If the buildings are closer than 9 m, or the basement is used for EV parking and or charging facilities, the protection level should comply with **CL-5**.
- 8 Multi-level Pumping and water storage as per schematic line drawing (SLD) may be considered for cluster buildings more than 45 m; and water storage at every 45 m above should be provided for individual buildings.
- 9 If the cluster of buildings exceeds 5, another set of pumps and water storage of similar capacity should be provided. A maximum of two such sets is recommended, provided it satisfies the hydraulic requirements and functioning at the remotest point.
- 10 [Table 7K](#) and [Table 7M](#) suggesting the basis for water quantity required for fire-fighting, respectively for combined (sprinkler and hose) protection and hose protection be considered.

[illegible]

Table 7C (Continued)

Sl No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
i)	501 m ² to 1 000 m ² (CL-3) See Notes 2, 4, 6, 7, 9 and 10	R	R	NR	R	R	R	R	NR
ii)	Above 1 001 m ² (CL-4) See Notes 2, 4, 6, 7, 9 and 10	R	R	NR	R	R	R	R	R
2)	Above 45 m but not exceeding 60 m in height (CL-5) See Notes 4, 6, 7, 9 and 10	R	R	NR	R	R	R	R	R
3)	Above 60 m – See Note 3								
b)	Custodial, Penal and Mental Institutions (C-II and III)	Up to 500 m ² area and up to 24 m height (both conditions to be met), self-certification by a Certified and State Approved building professional should be treated acceptable							

Table 7C (Concluded)

Sl No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
1)	For heights not exceeding 45 m in height								
i)	501 m ² to 1 000 m ² (HL-2) See Notes 1, 2, 4 and 6	R	R	R	NR	NR	NR	NR	R
ii)	1 001 m ² and above (HL-3) See Notes 1, 2, 4, 6 and 9	R	R	NR	R	R	NR	R	R
2)	Above 45 m (HL-4) See Notes 1, 2, 3, 6, 8 and 9	R	R	NR	R	R	NR	R	R

R – Required

NR – Not Required

REMARK 1: Any state or area where the municipal supply caters to municipal/public yard hydrant system with hydrant outlets provided throughout the premises with adequate water supply meeting the wet riser requirement demand flow and pressure 24×7 uninterrupted, such states/areas/jurisdiction can replace the wet riser system and its associated pumping with water storage with a dry riser system. Dry riser system is also recommended for areas where ambient temperature below is 4°C and has facility of connecting the yard hydrant to the dry riser in any event of fire in the building. This system is to be maintained with quarterly checks to ensure the system integrity to handle the required flow and pressure. This is applicable for buildings height only up to 24 m where HL Protection is required as per above Table.

REMARK 2: Sharing of water storage tank may be permitted between the adjacent neighbouring properties including industry clusters, without any loss of hydraulic effects (which is essential to fight the fire during an emergency).

NOTES

- 1 Basement area or storage of oxygen or any medical gases or any gas fuel, laundry or linen/waste, transformer/HT rooms within the building foot print or in basement should be protected with sprinkler and the protection level should comply with **CL-3**.
- 2 If the building has electrical vehicles (EV) parks in podium or basements, it should be protected with sprinklers and complying to **CL-5**.
- 3 If the building height exceeds 60 m the hydraulic calculations meeting recommended system architecture with multi-level pumping and storage with gravity redundant down comer may be considered. The pumps considered should not exceed 12 bar shutoff pressure.
- 4 Protection should comply to supplementary protection for specific hazard/critical rooms as per [Table 9/ Table 10/ Table 11](#) as applicable.
- 5 Pump pressure and quantity should be supported by hydraulic calculations.
- 6 Rate of rise heat sensor/detection system should be provided in electrical vehicles (EV) parks and charging areas. Smoke sensor/detectors should be provided in all basements in all areas other than internal combustion engine (ICE) vehicles parks as part of the fire alarm system. ICE vehicle parks should be provided with carbon monoxide sensors as part of fire alarm system to activate ventilation system. (recommendation only for parking and podium)
- 7 3785/2500 lpm at 10 bar common pumping and water storage of 3,50,000/2,55,000 litre should be provided of cluster of maximum 5 building 45 m height, with or without common basement. The buildings should not be closer than 9 m. If the buildings are closer than 9 m, or the basement is used for Electric Vehicles (EV) parking and or charging facilities, the protection level should comply with **CL-5**.
- 8 Multi-level pumping and water storage as per schematic line drawing (SLD) may be considered for cluster buildings more than 45 m; and water storage at every 45 m above should be provided for individual buildings.
- 9 If the cluster of buildings exceeds 5, another set of pumps and water storage of similar capacity should be provided. A maximum of two such sets is recommended, provided it satisfies the hydraulic requirements and functioning at the remotest point.
- 10 If the building height exceeds 30 m it should comply hydraulic calculations meeting recommended system architecture.
- 11 [Table 7K](#) and [Table 7M](#) suggesting the basis for water quantity required for fire-fighting, respectively for combined (sprinkler and hose) protection and hose protection be considered.

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Table 7D Requirements for Firefighting Installations – For Assembly Buildings

[Clauses [4.9 \(a\)](#), [4.9 \(c\)](#), [4.9\(c\)](#), [5.1.1 \(a\)](#), [5.1.1\(d\)](#), [5.1.2.1](#), [5.1.3 \(a\)](#), [D-6.1](#), [G-2\(f\)](#) and [Table 31](#)]

SI No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	ASSEMBLY BUILDINGS (D) <i>See Note 4</i>								
a)	Assembly Buildings	Up to 500 m² area and up to 9 m height (both conditions to be met), self-certification by a certified and State approved building professional should be treated acceptable							
1)	9 m and above but not exceeding 15 m in height; and, for heights less than 9 m; and area <i>See Note 6</i>								
i)	501 m ² to 1 000 m ² (HL-3) <i>See Notes 1, 2, 3, 5 and 8</i>	R	R	NR	R	NR	NR	R	NR

Table 7D (Concluded)

Sl No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ii)	1 001 m ² and above (CL-4) <i>See Notes 2, 5, 8 and 9</i>	R	R	NR	R	R	R	R	NR
2)	Above 15 m in height (CL-4) <i>See Notes 2, 5, 6, 7, 8 and 9</i>	R	R	NR	R	R	R	R	NR

R – Required

NR – Not Required

REMARK 1: Any state or area where the municipal supply caters to municipal/public yard hydrant system with hydrant outlets provided throughout the premises with adequate water supply meeting the wet riser requirement demand flow and pressure 24×7 uninterrupted, such states/areas/ jurisdiction can replace the wet riser system and its associated pumping with water storage with a dry riser system. Dry riser system is also recommended for areas where ambient temperature below is 4 °C and has facility of connecting the yard hydrant to the dry riser in any event of fire in the building. This system is to be maintained with quarterly checks to ensure the system integrity to handle the required flow and pressure. This is applicable for buildings height only up to 24 m where HL Protection is required as per above Table.

REMARK 2: Sharing of water storage tank may be permitted between the adjacent neighbouring properties including industry clusters, without any loss of hydraulic effects (which is essential to fight the fire during an emergency).

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- 1 If the building has basement to be protected with sprinklers, the protection level should comply with **CL-3**.
- 2 If the building has electrical vehicles (EV) parks in podium or basements, it should be protected with sprinklers and complying to **CL-5**.
- 3 Live cooking kitchens, boiler/furnace rooms, laundry or linen/waste rooms, transformer/HT rooms within the building foot print or in basement the area should be protected with sprinkler protection and should comply with **CL-4**.
- 4 Pump pressure and quantity should be supported by hydraulic calculations.
- 5 Rate of rise heat sensor/detection system should be provided in electrical vehicles (EV) parks and charging areas. Smoke sensor/detectors should be provided in all basements in all areas other than internal combustion engine (ICE) vehicles parks as part of the fire alarm system. ICE vehicle parks should be provided with carbon monoxide sensors as part of fire alarm system to activate ventilation system. (recommendation only for parking and podium).
- 6 2,500 lpm at 10 bar common pumping and water storage of 2,55,000 litre should be provided for a cluster of maximum 5 buildings, with or without common basement. The buildings should not be closer than 9 m. If the buildings are closer than 9 m, or the basement is used for EV parking and or charging facilities, the protection level should comply with **CL-5**.
- 7 Multi-level pumping and water storage as per schematic line drawing (SLD) may be considered for cluster buildings more than 45 m; and water storage at every 45 m above should be provided for individual buildings.
- 8 If the cluster of buildings exceeds 5, another set of pumps and water storage of similar capacity should be provided. A maximum of two such sets is recommended, provided it satisfies the hydraulic requirements and functioning at the remotest point.
- 9 [Table 7K](#) and [Table 7M](#) suggesting the basis for water quantity required for fire-fighting, respectively for combined (sprinkler and hose) protection and hose protection be considered.

Table 7E Requirements for Firefighting Installations – For Business Buildings

[Clauses [4.9 \(a\)](#), [4.9 \(c\)](#), [4.9\(c\)](#), [5.1.1 \(a\)](#), [5.1.1\(d\)](#), [5.1.2.1](#), [5.1.3 \(a\)](#), [D-6.1](#), [G-2\(f\)](#) and [Table 31](#)]

Sl No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	BUSINESS BUILDINGS (E) See Notes 5								
a)	Business Buildings (Human Occupied)	Up to 500 m² area and up to 15 m height (both conditions to be met), self-certification by a Certified and State approved building professional should be treated acceptable							

Table 7E (Continued)

Sl No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
1)	15 m and above but not exceeding 45 m in height; and, for heights less than 15 m and area								
i)	500 m ² to 1 000 m ² (CL-2) See Notes 1, 2, 3, 6, 7, 9 and 11	R	R	R	NR	NR	R	R	R
ii)	1 001 m ² to 3 000 m ² (CL-3) See Notes 2, 3, 6, 7, 9 and 11	R	R	NR	R	R	R	R	R
iii)	3 001 m ² and above (CL-4) See Notes 2, 3, 6, 7, 9 and 11	R	R	NR	R	R	R	R	R

Table 7E (Continued)

Sl No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
2)	Above 45 m but not exceeding 60 m height (CL-5) <i>See Notes 3, 4, 6, 7, 8, 9 and 11</i>	R	R	NR	R	R	R	R	R
3)	For buildings over 60 m height	<i>See Note 4</i>							
b)	Data Centres (Machine Environment) <i>See Notes 3, 12, 13 and 15</i>								
1)	Less than 15 m in height								
i)	Floor Area on any single floor up to 1 000 m ²	R	R	NR	R	R	R	R	R
ii)	1 001 m ² and above	R	R	NR	R	R	R	R	R

Table 7E (Concluded)

Sl No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
2)	Above 15 m in height								
i)	Floor area on any single floor up to 1 000 m ²	R	R	NR	R	R	R	R	R
ii)	1 001 m ² and above	R	R	NR	R	R	R	R	R

R – Required

NR – Not Required

REMARK 1: Any state or area where the municipal supply caters to municipal/public yard hydrant system with hydrant outlets provided throughout the premises with adequate water supply meeting the wet riser requirement demand flow and pressure 24×7 uninterrupted, such states/areas/jurisdiction can replace the wet riser system and its associated pumping with water storage with a dry riser system. Dry riser system is also recommended for areas where ambient temperature below is 4 °C and has facility of connecting the yard hydrant to the dry riser in any event of fire in the building. This system is to be maintained with quarterly checks to ensure the system integrity to handle the required flow and pressure. This is applicable for buildings height only up to 24 m where HL protection is required as per above table.

REMARK 2: Sharing of water storage tank may be permitted between the adjacent neighbouring properties including industry clusters, without any loss of hydraulic effects (which is essential to fight the fire during an emergency).

NOTES

1 If the building has basement to be protected with sprinklers, the protection level should comply with CL-3.

2 If the building has Electrical Vehicles (EV) parks in podium or basements, it should be protected with sprinklers and complying to CL-5.

3 Protection should comply to supplementary protection for specific hazard/critical rooms as per [Table 9/ Table 10/ Table 11](#) as applicable.

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4 If the building height exceeds 60 m the hydraulic calculations meeting recommended system architecture with multi-level pumping and storage with gravity redundant down comer may be considered. The pumps considered should not exceed 12 bar shutoff pressure.

5 Pump pressure and quantity should be supported by hydraulic calculations.

6 2500 lpm at 10 bar common pumping and water storage of 2,55,000 litre should be provided for a cluster of maximum 5 buildings, with or without common basement. The buildings should not be closer than 9 m. If the buildings are closer than 9 m, or the basement is used for EV parking and or charging facilities, the protection level should comply with **CL-5**.

7 3785 lpm at 10 bar common pumping and water storage of 3,50,000 litre should be provided of cluster of maximum 5 building, with or without common basement. The buildings should not be closer than 9 m. If the buildings are closer than 9 m, or the basement is used for Electric Vehicles (EV) parking and or charging facilities, the protection level should comply with **CL-5**.

8 Multi-level pumping and water storage as per schematic line drawing (SLD) (*see Fig. 20*) may be considered for cluster buildings more than 45 m; and water storage at every 45 m above should be provided for individual buildings.

9 If the cluster of buildings exceeds 5, another set of pumps and water storage of similar capacity should be provided. A maximum of two such sets is recommended, provided it satisfies the hydraulic requirements and functioning at the remotest point.

10 The entire building it should be protected with sprinkler and the protection level should comply with **CL-5**. IT/Data Hall should be protected with suitable clean agent gas suppression only if the racks are without BBU (not exceeding 20 kWh backup battery units).

11 All internet data centres or hyper scales or colocation data centres comply with protection level **CL6**.

12 If the IT/data hall (DC) racks are with BBUs with Lithium-ion batteries not exceeding 20 kWh/rack, wet or dry sprinkler/pre-action/water mist (3rd party tested for lithium ion BBUs) should be deployed. If the BBUs exceed 20 kWh/rack, water mist should not be installed.

13 If the cluster of data centre buildings to a maximum of 3, common pumping system and storage may be with an additional diesel engine pump, with double water storage.

14 If the any of the building houses lithium-ion energy storage systems exceeding 600 kWh/floor, the building should comply to protection level **CL-6**.

15 Lithium-ion battery energy storage systems (BESS) room should be protected with wet/single interlock pre action sprinkler only, considering entire room as “assumed area of operation”. Pump and water storage should be calculated based on maximum design and hydraulic calculations whichever is higher.

16 [Table 7K](#) and [Table 7M](#) suggesting the basis for water quantity required for fire-fighting, respectively for combined (sprinkler and hose) protection and hose protection be considered.

Table 7F Requirements for Firefighting Installations – For Mercantile Buildings

[Clauses [4.9 \(a\)](#), [4.9 \(c\)](#), [4.9\(c\)](#), [5.1.1 \(a\)](#), [5.1.1\(d\)](#), [5.1.2.1](#), [5.1.3 \(a\)](#), [D-6.1](#), [G-2\(f\)](#) and [Table 31](#)]

SI No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	MERCANTILE BUILDINGS (F) <i>See Note 4</i>								
a)	Mercantile Buildings (F)	Up to 500 m² area and up to 15 m height (both conditions to be met), self-certification by a certified and State approved building professional should be treated acceptable							
1)	Above 15 m but not exceeding 30 m in height; and, for heights less than 15 m and area								
i)	501 m ² to 1 000 m ² (HL-3) <i>See Notes 1, 2, 3, 5, 6 and 7</i>	R	R	R	NR	NR	NR	R	NR
ii)	1 001 m ² and above (CL-4) <i>See Notes 2, 3, 5, 6, 7 and 8</i>	R	R	NR	R	R	R	R	NR

Table 7F (Concluded)

SI No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
2)	Above 30 m	See Note 5							
b)	Underground Shopping Complex (CL-5) <i>See Note 3</i>	R	R	NR	R	R	R	R	R

R – Required

NR – Not Required

REMARK 1: Any state or area where the municipal supply caters to municipal/public yard hydrant system with hydrant outlets provided throughout the premises with adequate water supply meeting the wet riser requirement demand flow and pressure 24×7 uninterrupted, such states/areas/ jurisdiction can replace the wet riser system and its associated pumping with water storage with a dry riser system. Dry riser system is also recommended for areas where ambient temperature below is 4°C and has facility of connecting the yard hydrant to the dry riser in any event of fire in the building. This system is to be maintained with quarterly checks to ensure the system integrity to handle the required flow and pressure. This is applicable for buildings height only up to 24 m where HL Protection is required as per above table.

REMARK 2: Sharing of water storage tank may be permitted between the adjacent neighbouring properties including industry clusters, without any loss of hydraulic effects (which is essential to fight the fire during an emergency).

NOTES

- 1 If the building has basement to be protected with sprinklers, the protection level should comply with **CL-3**.
- 2 If the building has electrical vehicles (EV) parks in podium or basements, it should be protected with sprinklers and complying to **CL-5**.
- 3 Protection should comply to supplementary protection for specific hazard/critical rooms as per Table 9/ Table 10/ Table 11 as applicable.
- 4 Pump pressure and quantity should be supported by hydraulic calculations.
- 5 If the building height exceeds 30 m it should comply hydraulic calculations meeting recommended system architecture.
- 6 If the building height exceeds 60 m multi-level pumping and storage with gravity redundant down comer. The pumps considered should not exceed 12 bar shutoff pressure.
- 7 Rate of rise heat sensor/detection system should be provided in electrical vehicles (EV) parks and charging areas. Smoke sensor/detectors should be provided in all basements in all areas other than internal combustion engine (ICE) vehicles parks as part of the fire alarm system. ICE vehicles parks should be provided with carbon monoxide sensors as part of fire alarm system to activate ventilation system.
- 8 [Table 7K](#) and [Table 7M](#) suggesting the basis for water quantity required for fire-fighting, respectively for combined (sprinkler and hose) protection and hose protection be considered.

Table 7G Requirements for Firefighting Installations – For Industrial Buildings

(See also Annex N on Fire Protection for Specific Industries)

[Clauses [4.9 \(a\)](#), [4.9 \(c\)](#), [4.9\(e\)](#), [5.1.1 \(a\)](#), [5.1.1\(d\)](#), [5.1.2.1](#), [5.1.3 \(a\)](#), [D-6.1](#), [G-2\(f\)](#), [Annex N](#) and [Table 31](#)]

Sl No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Industrial Buildings (G) See Note 5								
a)	Low Hazard (G-1) See Notes 1 and 6								

Table 7G (Continued)

Sl No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
i)	Built up area up to 1 000 m ² (Including the MSMEs) (HL-1)	R	R	NR	NR	NR	NR	NR	NR
ii)	Built up area from 1 001 m ² to 2 000 m ² (HL-2)	R	R	R	NR	NR	NR	NR	NR
iii)	Built up area from 2 001 m ² up to 3 000 m ² (HL-3)	R	R	NR	NR	R	NR	R	NR
iv)	Above 3 001 m ² (HL-4) <i>See Note 1</i>	R	R	NR	NR	R	NR	R	NR

Table 7G (Continued)

SI No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
b)	Moderate Hazard (G-2) <i>See Notes 1, 4 and 6</i>								
i)	Built up area up to 500 m ² (HL-2)	R	R	R	NR	NR	NR	NR	NR
ii)	Built up area from 501 m ² to 2 000 m ² (HL-3)	R	R	NR	R	R	NR	R	NR
iii)	Built up area from 2 001 m ² to 3 000 m ² (HL-4)	R	R	NR	R	R	NR	R	NR
iv)	Built up area above 3 001 m ² (CL-5)	R	R	NR	R	R	R	R	NR

Table 7G (Continued)

SI No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
c)	High Hazard (G- 3)								
i)	Built up area up to 250 m ² (CL-4) <i>See Notes 1, 2, 3 and 7</i>	R	R	NR	R	R	R	NR	NR
ii)	Built up area from 251 m ² to 1 000 m ² (CL-5) <i>See Notes 1, 2, 3, 4 and 7</i>	R	R	NR	R	R	R	R	NR

Table 7G (Concluded)

Sl No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
iii)	Built up area above 1 001 m ² (CL-6) See Notes 2, 3, 4 and 7	R	R	NR	R	R	R	R	R

R – Required

NR – Not Required

REMARK 1: Any state or area where the municipal supply caters to municipal/public yard hydrant system with hydrant outlets provided throughout the premises with adequate water supply meeting the wet riser requirement demand flow and pressure 24×7 uninterrupted, such states/areas/ jurisdiction can replace the wet riser system and its associated pumping with water storage with a dry riser system. Dry riser system is also recommended for areas where ambient temperature below is 4 °C and has facility of connecting the yard hydrant to the dry riser in any event of fire in the building. This system is to be maintained with quarterly checks to ensure the system integrity to handle the required flow and pressure. This is applicable for buildings height only up to 24 m where HL Protection is required as per above Table.

REMARK 2: Sharing of water storage tank may be permitted between the adjacent neighbouring properties including industry clusters, without any loss of hydraulic effects (which is essential to fight the fire during an emergency).

86 NOTES

- 1 If the building has any type of combustible storage, combustible packing, fuel storage, boilers, furnaces such areas should be protected and should comply to protection level **CL-6**.
- 2 Alternate suppression agent should be considered as per relevant Indian Standards if water is not a suitable agent for materials protected.
- 3 Oxidizing materials, ignitable liquids, water resistive chemicals, explosives, aluminium dust, or any other explosive environment should be treated separately.
- 4 Protection should comply to supplementary protection for specific hazard/critical rooms as per [Table 9/ Table 10/ Table 11](#) as applicable.
- 5 Pump pressure and quantity should be supported by hydraulic calculations.
- 6 If the cluster of buildings exceeds 5, another set of pumps and water storage of similar capacity should be provided. A maximum of two such sets is recommended, provided it satisfies the hydraulic requirements and functioning at the remotest point.
- 7 [Table 7K](#) and [Table 7M](#) suggesting the basis for water quantity required for fire-fighting, respectively for combined (sprinkler and hose) protection and hose protection be considered.

Table 7H Requirements for Firefighting Installations – For Storage Buildings

[Clauses [4.9 \(a\)](#), [4.9 \(c\)](#), [4.9\(e\)](#), [5.1.1 \(a\)](#), [5.1.1\(d\)](#), [5.1.2.1](#), [5.1.3 \(a\)](#), [D-6.1](#), [G-2\(f\)](#) and [Table 31](#)]

SI No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Storage Buildings (H) (See Note 1, 2 and 3)								
i)	Built-up area up to 500 m ² (HL-4)	R	R	NR	R	NR	NR	R	NR
ii)	501 m ² to 1 000 m ² (HL-5)	R	R	NR	R	R	NR	R	NR

Table 7H (Concluded)

SI No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
iii)	1 001 m ² to 2 000 m ² or building ridge height up to 7.6 m (CL-6)	R	R	R	NR	R	R	R	NR
iv)	Above 2 001 m ² or Building ridge height above 7.6 m and up to 9.0 m (CL-7)	R	R	NR	R	R	R	R	R
v)	Building of any height with rack storage	R	R	NR	R	R	R	R	R

R – Required

NR – Not Required

REMARK 1: Any state or area where the municipal supply caters to municipal/public yard hydrant system with hydrant outlets provided throughout the premises with adequate water supply meeting the wet riser requirement demand flow and pressure 24×7 uninterrupted, such states/areas/ jurisdiction can replace the wet riser system and its associated pumping with water storage with a dry riser system. Dry riser system is also recommended for areas where ambient temperature below is 4 °C and has facility of connecting the yard hydrant to the dry riser in any event of fire in the building. This system is to be maintained with quarterly checks to ensure the system integrity to handle the required flow and pressure. This is applicable for buildings height only up to 24 m where HL Protection is required as per above table.

REMARK 2: Sharing of water storage tank may be permitted between the adjacent neighbouring properties including industry clusters, without any loss of hydraulic effects (which is essential to fight the fire during an emergency).

88 NOTES

1 Pump pressure mentioned in above table is considering minimum requirement, it should be supported hydraulic calculations. The above Table applies to the below Storage categories only, stored on ground, not exceeding 3.7 m, (12 feet) excluding tyre storage. In addition, they should also comply with supplementary protection for all hazard/critical rooms as per [Table 9/Table 10/ Table 11](#):

- a) Packaging materials stored on ground with or without pallets
- b) Raw materials storage, in solid pile, shelves, pallets
- c) Finished goods storage, solid pile, shelves, pallets
- d) Miscellaneous storage in and around manufacturing/assembly/packaging areas
- e) Retail storage in shelves other than catwalk storage

2 For rack storages, with plastic/combustible packaging (exposed/cartoned unexpanded or exposed/ cartoned expanded), water reactive solids/liquid storages, lithium battery storages, ignitable chemical storages, or retail cat walk, any other oxidizing agent storages, relevant Indian Standards should be followed, and pumping capacities and water storage should be justified by way of hydraulic calculations.

3 If the cluster of buildings exceeds 5, another set of pumps and water storage of similar capacity should be provided. A maximum of two such sets is recommended, provided it satisfies the hydraulic requirements and functioning at the remotest point

4 [Table 7K](#) and [Table 7M](#) suggesting the basis for water quantity required for fire-fighting, respectively for combined (sprinkler and hose) protection and hose protection be considered.

Table 7J Requirements for Firefighting Installations – For Hazardous Buildings <i>[Clauses 4.9 (a), 4.9 (c), 4.9(c), 5.1.1 (a), 5.1.1(d), 5.1.2.1, 5.1.3 (a), D-6.1, G-2(f) and Table 31]</i>									
SI No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Hazardous Buildings (J) (See Notes 1 and 2)								
1)	Up to 15 m in height								
i)	Single storey building (CL-7)	R	R	NR	R	R	R	R	NR

Table 7J (Concluded)

SI No.	Type of Building Occupancy	Type of Installation							
		Fire Extinguisher	First Aid Hose Reel	Down Comer	Wet Riser	Yard Hydrant	Automatic Wet Sprinkler System	Automatic Detection and Alarm System	Public Address and Voice Evacuation System
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ii)	More than one floor building but not exceeding 15 m (CL-8)	R	R	NR	R	R	R	R	R

R – Required

NR – Not Required

REMARK 1: Any state or area where the municipal supply caters to municipal/public yard hydrant system with hydrant outlets provided throughout the premises with adequate water supply meeting the wet riser requirement demand flow and pressure 24×7 uninterrupted, such states/areas/ jurisdiction can replace the wet riser system and its associated pumping with water storage with a dry riser system. Dry riser system is also recommended for areas where ambient temperature below is 4°C and has facility of connecting the yard hydrant to the dry riser in any event of fire in the building. This system is to be maintained with quarterly checks to ensure the system integrity to handle the required flow and pressure. This is applicable for buildings height only up to 24 m where HL Protection is required as per above Table.

REMARK 2: Sharing of water storage tank may be permitted between the adjacent neighbouring properties including industry clusters, without any loss of hydraulic effects (which is essential to fight the fire during an emergency).

NOTES

1 These buildings should deploy specific tailor-made systems and solutions in terms of water hydrant/sprinklers, dry chemical, dry powder, water mist, gas/ wet chemical suppression, foam or/and any other as per relevant Indian Standards and international standards (wherever Indian Standards are not available) in consultation with local fire services or Authority having Jurisdiction (AHJ).

2 Pump pressure and quantity should be supported by hydraulic calculations.

3 [Table 7K](#) and [Table 7M](#) suggesting the basis for water quantity required for fire-fighting, respectively for combined (sprinkler and hose) protection and hose protection be considered.

Table 7K Base Calculations for Combined Protection System (Sprinkler System and Hose Stream)

[Clause [5.1.1 \(a\)](#)]

SI No.	Combined System	Hazard Classification	Sprinkler Demand		Hose Stream Flow @ 3.5 bar	Number of Hose Streams	Pump Capacity and Quantity	Duration min
(1)	(2)	(3)	(4)		(5)	(6)	(7)	(8)
i)	Level of Protection – CL-1	Extra Low Hazard -1	2.25	84	50	2	379 lpm × 1	30 – 60
ii)	Level of Protection – CL-2	Extra Low Hazard -2	4.1	140	170	2	946 lpm × 1	30 – 60
iii)	Level of Protection – CL-3	Low Hazard	6.1	230	285	3	1 892 lpm × 1	30 – 60
iv)	Level of Protection – CL-4	Moderate Hazard -1	8.1	230	285	3	2 500 lpm × 1	60 – 120
v)	Level of Protection – CL-5	Moderate Hazard -2	12.2	230	475	3	3 785 lpm × 1	60 – 120
vi)	Level of Protection – CL-6	Extra Hazard -1	16.3	230	475	3	4 731 lpm × 1	60 – 120
vii)	Level of Protection – CL-7	Extra Hazard -2	18.3	280	670	4	7 570 lpm × 1	60 – 120
viii)	Level of Protection – CL-8	High Hazard	24.5	280	1050	4	4731 lpm x 2	60 – 120
Pump capacity = Total flow with 20 percent overflow and operating at 140 percent capacity. For example, for CL-5 , capacity will be $(12.2 \times 230) + (475 \times 3) = 4\,231$ litre; with overflow margin works to 5 077.2 litre; and pump selection will be $5\,077.2/140 \times 100 = 3\,627$ lpm (and the nearest Indian Standard pump is 3 785 lpm).								

NOTES

- 1 'Pump' implies, providing an electric pump and 'Standby pump of equivalent capacity'. For loss in pressure, jockey pump is also to be provided.
- 2 The duration of firefighting may be decided from 30 min to 120 min based on the hazard, fire preparedness, local conditions, to name a few.
- 3 The required quantity of water may be calculated as per the respective Indian Standards (where available); or based on the [Table 7K](#) and [Table 7M](#).

Table 7M Base Calculations for Hose Stream Protection System

[Clause [5.1.1 \(a\)](#) and [D-6.1](#)]

SI No.	Hose Stream Only	Hazard Classification	Hose Stream Flow	Number of Hose Streams	Pump Capacity and Quantity	Duration min
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Level of Protection – HL-1	Extra Light -1	50	4	379 lpm × 1	30 – 60
ii)	Level of Protection – HL-2	Extra Light -2	170	4	946 lpm × 1	30 – 60
iii)	Level of Protection – HL-3	Low Hazard	270	4	1 514 lpm × 1	30 – 60
iv)	Level of Protection – HL-4	Moderate -1	475	4	2 839 lpm × 1	60 – 120
v)	Level of Protection – HL-5	Moderate -2	670	5	4 731 lpm × 1	60 – 120
vi)	Level of Protection – HL-6	Extra Hazard	670	6	5 677 lpm × 1	60 – 120
Pump capacity = Total flow with 40 percent overflow and operating at 140 percent capacity. For example, for HL-3 , capacity will be $270 \times 4 = 1\,080$ litre; together with overflow margin works to 1 512 litre; and pump selection will be $1\,512/140 \times 100 = 1\,514$ lpm (nearest Indian Standard pump).						

NOTES

- 1 'Pump' implies, providing an Electric pump and 'Standby pump of equivalent capacity'. For loss in pressure, jockey pump is also to be provided.
- 2 The duration of firefighting may be decided from 30 min to 120 min based on the hazard, fire preparedness, local conditions, to name a few.
- 3 The required quantity of water may be calculated as per the respective Indian Standards (where available); or based on the [Table 7K](#) and [Table 7M](#).

4.10 Fire Officer

The essential capabilities of the Fire Officer should be to maintain fire equipment, develop and implement fire safety operational plans, train occupants on fire tools and evacuation, and coordinate with the city fire brigade to ensure constant safety compliance. It is recommended that a qualified Fire Officer should be stationed on-site for large building complexes and specific high-rise or high-risk buildings.

4.11 Fire Drills and Fire Orders

Fire notices/orders should be prepared to fulfil the requirements of firefighting and evacuation from the buildings in the event of fire and other emergency. The occupants should be made thoroughly conversant with their action in the event of emergency, by displaying fire notices at vantage points and also through regular training. Such notices should be displayed prominently in bold lettering. The State/Local Authorities may also decide on the nature and frequency of fire drills to be carried out in buildings.

5 FIRE PROTECTION

5.1 Fire Extinguishers/Fixed Firefighting Installations

5.1.1 All buildings depending upon the occupancy use and height should be protected by fire extinguishers, hose reels, wet riser, down-comer, yard hydrants, automatic sprinkler installation, deluge system, high/medium velocity water spray, foam, water mist systems, gaseous or dry powder system, manual/automatic fire alarm system, etc, in accordance with the provisions of various clauses given below, as applicable:

- a) These fire extinguishing equipment and their installation should be in accordance with accepted standards [F(17)]. The extinguishers should be mounted at a convenient height to enable its quick access and efficient use by all in the event of a fire incidence. The requirements of fire extinguishers/yard hydrant systems/wet riser/down-comer installation, etc, should be as specified in [Table 7A](#) to [Table 7J](#). See also [Table 7K](#) and [Table 7M](#) for guidance, on base calculation for capacity of water storage tanks and fire pumps. The requirements regarding size of mains/risers should be as given in [Table 8](#). The typical arrangements of down-comer and wet riser installations, as also the pumping system are shown in [Fig. 15](#), [Fig. 16](#) and [Fig. 17](#). The wet riser should be designed for zonal distribution ensuring that unduly high pressures are not developed in risers and hose-pipes.
- b) First-aid firefighting appliances should be provided and installed in accordance with good practice [F(18)]. The firefighting equipment and accessories to be installed in buildings for use in firefighting should also be in accordance with the accepted standard [F(17)] and should be maintained periodically so as to ensure their perfect serviceability at all times.
- c) Valves in fixed firefighting installations should have supervisory switch with its signalling to fire alarm panel or to have chain(s), pad lock(s), label, and tamper-proof security tag(s) with serial number to prevent tampering/unauthorized operation. These valves should be kept in their intended 'open' position.
- d) In addition to wet riser or down-comer, first-aid hose reels should be installed in buildings (where required under [Table 7A](#) to [Table 7J](#)) on all the floors, in accordance with accepted standard [F(19)]. The first-aid hose reel should be connected directly to the riser/down-comer main and diameter of the hose reel should not be less than 19 mm.
- e) Wet risers should be interconnected at terrace level to form a ring and cut-off should be provided for each connection to enable repair/ maintenance without affecting rest of the system.
- f) Pressure at the hydraulically remote hydrant and at the highest hydrant should not be less than 3.5 bar. The pressure at the hydrants should however not exceed 7.0 bar, considering the safety of operators. It may be planned to provide orifice plates for landing valves to control pressure to desired limit especially at lower levels. This could also be achieved through other suitable means of pressure reducing devices such as pressure-controlled hydrant valves.
- g) Hydrants for firefighting and hose reels should be located in the lobby in firefighting shaft. Those hydrants planned to be provided near fire exit staircase on the floor should be within 5 m from fire exit door in exit access. Such hydrant cabinet may finish with doors to meet interior finishes with requirement of glass panel to provide visibility to the installations inside and inscribed with the word: 'FIRE HOSE CABINET' of letter size 75 mm in height and 12 mm in width. Such door of the fire hose cabinet need not be fire resistant rated. The location of such cabinets should be shown on floor plan and duly displayed in the landing of the respective fire exit staircase.

5.1.2 Static Water Storage Tanks and Pump House

5.1.2.1 Static water storage tanks

A satisfactory supply of water for the purpose of firefighting should always be available in the form of underground/terrace level static storage tank with capacity specified for each building with arrangements or replenishment.

Water for the hydrant services should be stored in an easily accessible surface/underground lined reservoir or above ground tanks made of steel or concrete or masonry. The effective capacity of the reservoir above the top of the pump casing (flooded suction) for various types of occupancies should be as indicated in [Table 7A](#) to [Table 7J](#).

Water for firefighting should be stored in two or more interconnected compartments of equal size to facilitate cleaning and maintenance of the tanks without interrupting the water availability for firefighting.

To prevent stagnation of water in the static water storage tank, the suction tank of the domestic water supply should be fed only through an overflow arrangement from the fire water storage tanks to maintain the level therein at the minimum specified capacity.

Alternatively, domestic and fire water can be stored in two interconnected compartments as mentioned above. The suction inlet(s) for the domestic water pumps should be so located at an elevation that minimum water requirements for firefighting as stated in [Table 7A](#) to [Table 7J](#) will be always available for fire pumps.

The static storage water supply required for the above-mentioned purpose should entirely be accessible to the fire engines of the local fire service. Sufficient number of manholes should be provided for inspection, repairs, insertion of suction hose, etc. As an alternative to the arrangement of manholes to allow access from the top, suitable arrangement to enable efficient access to the tank by the firemen from the adjoining fire pump room having direct access from the ground level, should be made. The underground fire water storage tank(s) should not be more than 7 m in depth from the level having fire brigade draw-out connection, while the draw-out connection should not be more than 5 m away from the tank wall.

The covering slab and its supporting wall/supporting system should be able to withstand a total vehicular load of 45 tonnes (or higher as applicable) equally divided as a four-point load when the slab forms a part of fire apparatus pathway/driveway.

The static water storage tank should be provided with a fire brigade collecting head with 4 number 63 mm diameter (2 number 63 mm diameter for pump with capacity 1 400 litre/min) instantaneous male inlets arranged in a valve box at a suitable point at street level. The same should be connected to the static tank by a suitable fixed galvanized iron pipe not less than 150 mm in diameter to discharge water into the tank when required at the rate of 2 250 litre/min if tank is in the basement or not approachable for the fire engines.

Each of the static water storage tanks should also be provided with a fire brigade draw out collecting head with 63 mm diameter instantaneous male draw out arranged in a valve box at a suitable point at street level. This draw out should be connected to galvanized iron pipe of 100 mm diameter with foot valve arrangement in the tank.

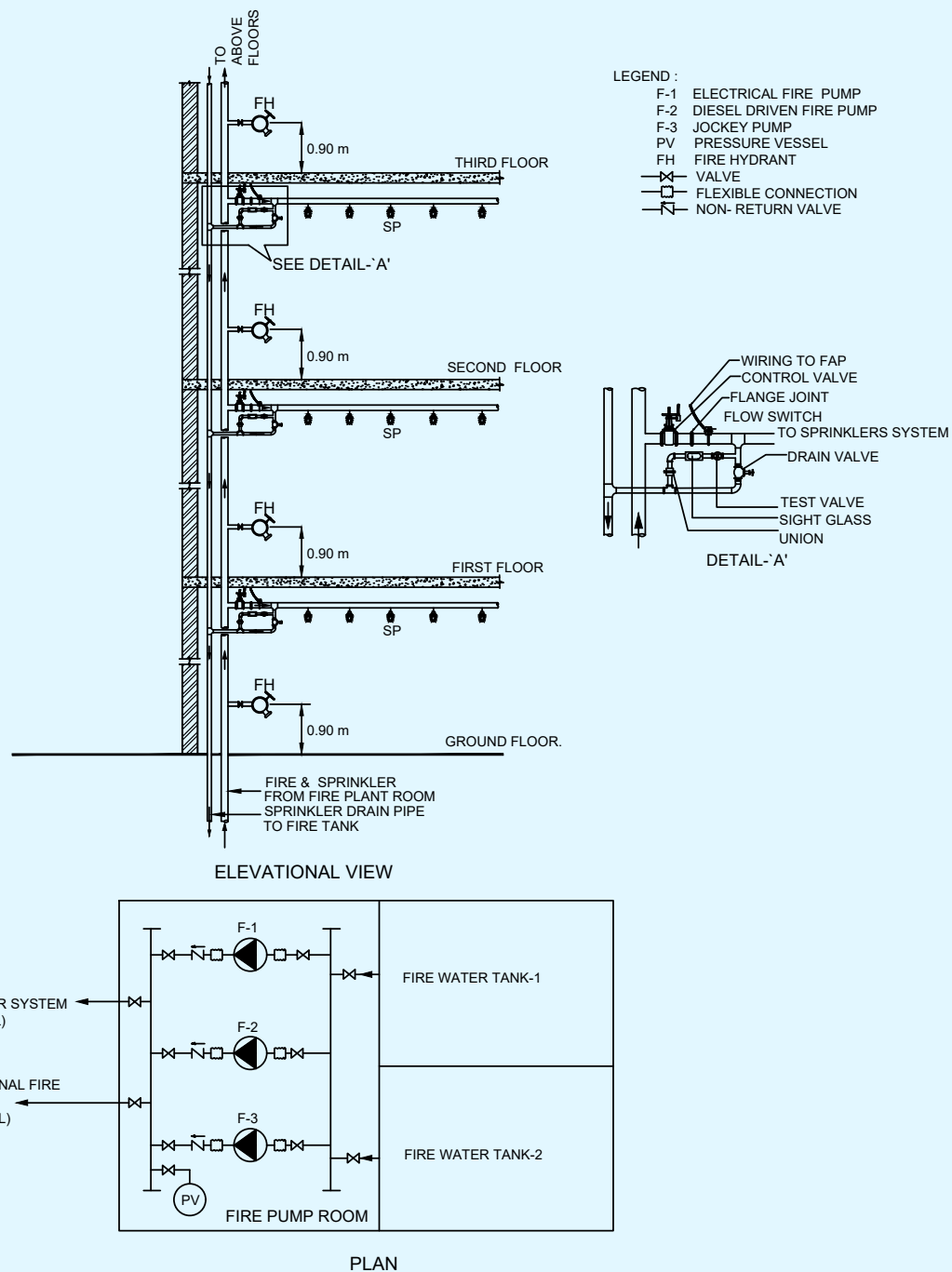


FIG. 15 TYPICAL SYSTEM OF PUMPING WITH ONE ELECTRIC, ONE DIESEL FIRE PUMP

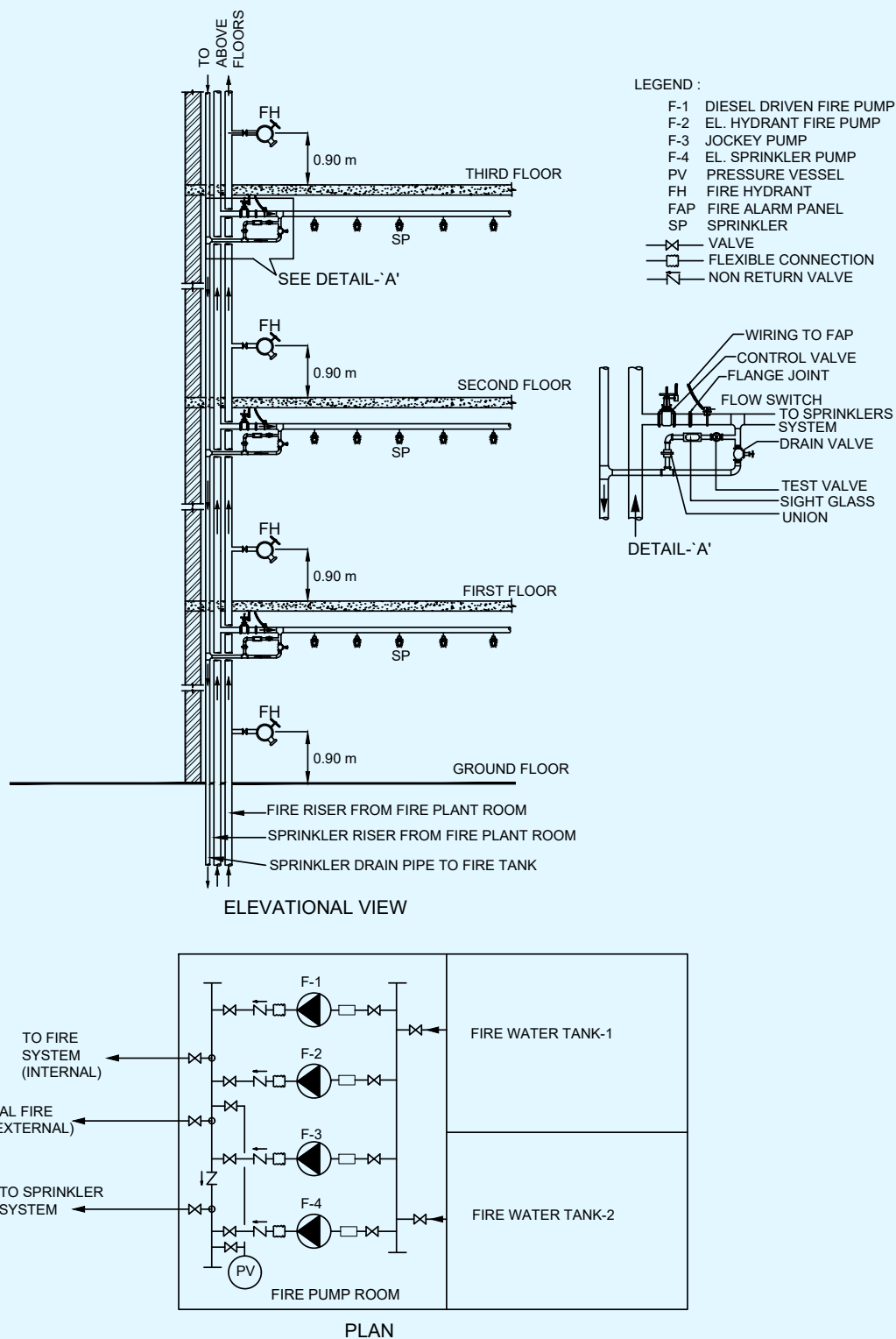


FIG. 16 TYPICAL SYSTEM OF PUMPING WITH TWO ELECTRIC, ONE DIESEL FIRE PUMP

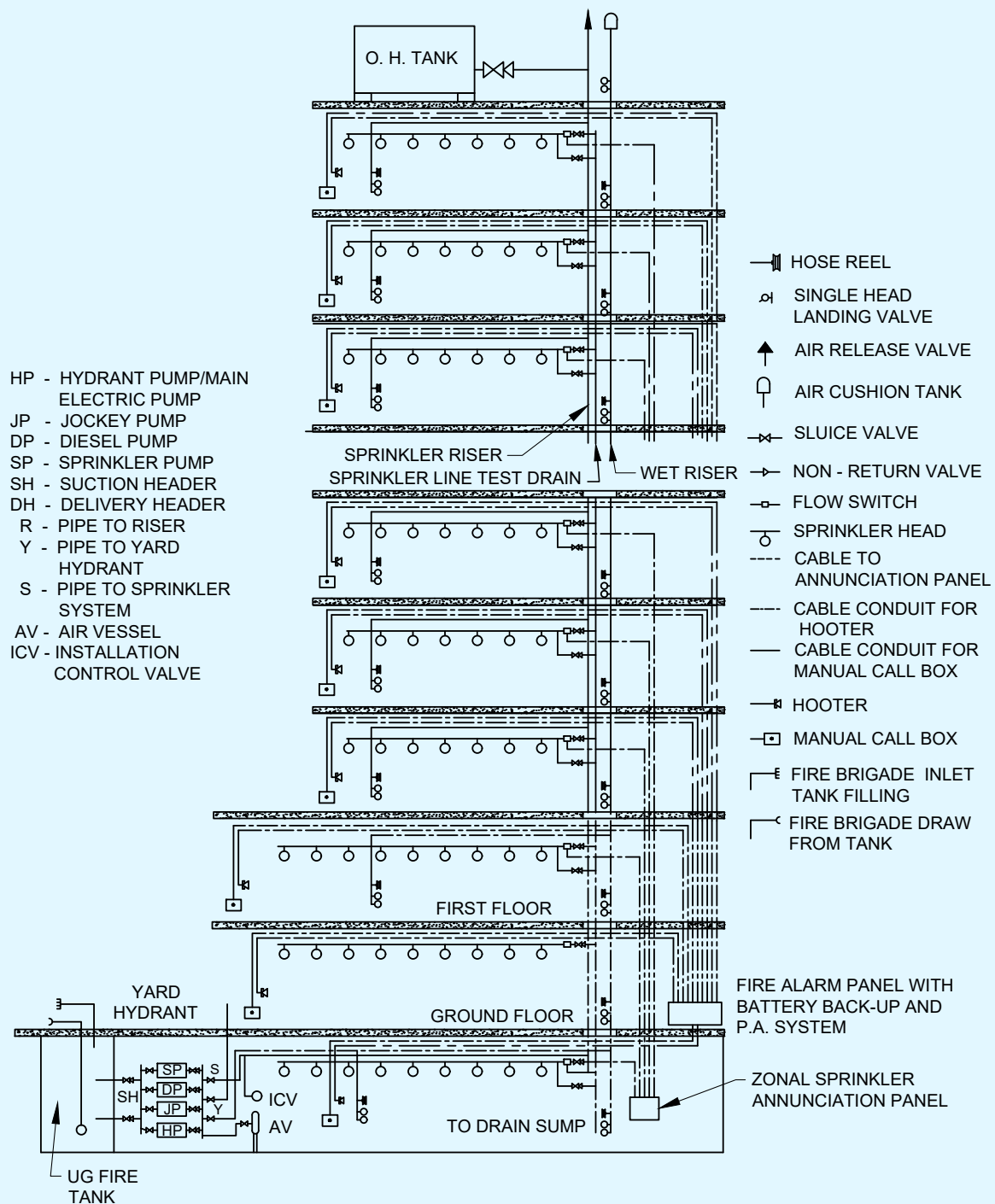


FIG. 17 TYPICAL ARRANGEMENT OF WET RISER AND TOTAL SPRINKLER SYSTEM OF BUILDING

Table 8 Size of Mains				
[Clause 5.1.1(a)]				
SI No.	Minimum Size of the Mains mm	Type of Building		Building Height m
(1)	(2)	(3)		(4)
i)	100 mm with single outlet 63 mm landing valve	a)	Residential buildings (A)	
			1) Dormitories	–
			2) Apartments	Up to 15 m
			3) Hotels	Up to 15 m
		b)	Educational buildings (B)	Up to 15 m
		c)	Institutional buildings (C)	Up to 15 m
		d)	Assembly buildings (D)	Up to 15 m
		e)	Business buildings (E)	Up to 15 m
		f)	Mercantile buildings (F)	Up to 9 m
		g)	Industrial buildings (G)	Up to 9 m
ii)	150 mm with single outlet 63 mm landing valve	a)	Hotels	Above 15 m
		b)	Starred hotels	–
		c)	Institutional buildings (C)	Above 15 m
		d)	Business buildings (E)	Above 15 m
		e)	Industrial buildings (G)	Up to 15 m
		f)	Storage buildings (H)	Up to 15 m
		g)	Hazardous buildings (J)	Up to 15 m

5.1.2.2 Firefighting pump house

The requirements should be as given below:

- Pump house should be situated so as to be directly accessible from the surrounding ground level. In the case of industrial occupancies, it is preferable to install the pump house at ground level.
- When located below ground, the pump house is much preferred in the first basement and should not be installed lower than the second basement (in which case the pumping efficiency should be ascertained). When installed in the basement, staircase with direct accessibility (or through enclosed passageway with 120 min fire rating) from the ground, should be provided. Access to the pump room should not require to negotiate through other occupancies within the basement.
- Pump house need to be separated by fire walls all around and doors to be protected by fire doors (120 min rating).

- d) Pump house should be well ventilated and due care should be taken to avoid water stagnation.
- e) No other utility equipment should be installed inside fire pump room.
- f) Insertions like flexible couplings, bellows, etc, in the suction and delivery piping should be suitably planned and installed.
- g) Installation of negative suction arrangement and submersible pumps should not be allowed.
- h) Pump house should be sufficiently large to accommodate all pumps, and their accessories like pressure reducing valves (PRVs), installation control valves (ICVs), diesel tank and electrical panel. ICV for the towers (superstructure) is allowed to be installed in the first basement, in a duly protected room of 120 min fire resistance rating.
- j) Battery of diesel engine operated fire pump should have separate charger from emergency power supply circuit.
- k) Exhaust pipe of diesel engine should be insulated as per best engineering practice and taken to a safe location at ground level, considering the back pressure.
- m) Fire pump is to be provided with soft starter or variable frequency drive starter.

5.1.3 Automatic Sprinkler Installation

The requirements should be as given below:

- a) Automatic sprinklers should be installed wherever required in terms of [Table 7A](#) to [Table 7J](#) throughout the building, for uses as specified in, and in accordance with good practice [F(20)].
- b) If selective sprinklering is adopted, there is a real danger of a fire starting in one of the unsprinklered area gathering momentum spreading to other areas and reaching the sprinklered areas as a fully developed fire. In such an event, the sprinklers can be rendered useless or ineffective.
- c) Automatic sprinklers should be installed in false ceiling voids exceeding 800 mm in height.
- d) Installation of sprinklers may be excluded in any area to be used for substation and DG set.
- e) In areas having height 17 m or above such as in atria, sprinkler installations may be rendered ineffective and hence may be avoided.
- f) Pressure in sprinkler system should not exceed 12 bar or else high pressure sprinkler to be installed for above 12 bar operations.
- g) The maximum floor area on any one floor to be protected by sprinklers supplied by any one sprinkler system riser from an installation control valve should be based on system protection area limitations considering maximum floor area on any one floor to be 4 500 m² for all occupancies except industrial and hazardous occupancies, where Authorities should be consulted for advice based on type and nature of risk.
- h) Sprinkler installation control valves, should be installed inside the fire pump room.
- j) For industrial buildings, such installation control valves may be installed outside the building and Authorities should be consulted in situations where it is not possible to locate them inside the buildings. It is advisable to provide electrically operated siren for each valve outside the buildings in addition to water gongs in such case.
- k) The sprinkler flow switches provided should be monitored by fire alarm panel.
- m) It is essential to make provisions for avoiding water from sprinkler/hydrant operation entering lifts and electrical rooms.
- n) Ramps at all levels should be protected with sprinklers.
- p) The water based sprinkler systems are not suited for specific buildings for example, areas manufacturing electronic equipment, in which case alternate type of protection should only be provided.
- q) Colour of the sprinkler piping may be matched to suit the interiors; however, in such case, the piping should be provided with red colour band of about 100 mm wide, every 3 m along the length of the pipe. In no case, the sprinkler heads should be painted.

- r) Based on interior design or interior requirements where it is necessary to provide pendent sprinkler with no false ceiling, to achieve the sprinkler response/activation, it is required to provide about 250 mm diameter circular plate around the sprinkler.

5.1.4 Automatic High Velocity and Medium Velocity Water Spray Systems

Automatic high velocity water spray or emulsifying system should be provided for protection of outdoor and/or indoor oil-cooled transformers as applicable in accordance with good practice [F(21)] where applicable (see [Annex D](#)). Also, medium velocity water spray system should be provided for tankage (where applicable), conveyors, cable galleries and other occupancies listed in good practice [F(21)].

5.1.5 Fixed Foam Installation

Fixed foam generating system should be provided for protection of oil storage area for boilers with its ancillary storage of furnace oils in basement. Fixed foam installations can be low, medium, or high expansion types, which can be provided based on the type of fire hazards identified in the facility. High expansion foams are used for cable tunnels and other confined areas. Design and installation of foam systems should be governed by good practice [F(22)].

5.1.6 Gas Based Suppression System

Gas based fire extinguishing installation should be provided in accordance with good practice on premises where water or foam cannot be used for fire extinguishing because of the special nature of the contents of the buildings/areas to be protected where either the building(s) have very limited manpower or unmanned. The protection design for fixed carbon dioxide fire extinguishing system should conform to good practice [F(23)] in all respects. For some special fire risk/essential applications, carbon dioxide may not be suitable and alternate provisions should be made as per relevant standards (see [5.1.9](#)).

NOTE — Hypoxic air technology should not be used in occupiable areas (considering the tenability aspects), and specific caution should be exercised in consultation with the Authority, regarding its applicability.

5.1.7 Firefighting equipment should be suitably located and clearly marked by luminous signs.

5.1.8 Automatic Water Mist Systems

These systems involve the use of fine water sprays for the efficient extinguishment of fires. These systems may be provided to protect areas in buildings for the uses as specified in good practice [F(24)].

5.1.9 Extinguishing Systems with Clean Agents

Alternative systems for halon gas protection systems should be provided where necessary as prescribed in this Part. These should be in accordance with the accepted standards [F(25)].

5.2 Fire Detection and Alarm System

See also [4.9](#).

5.2.1 Automatic Detection System

5.2.1.1 Pneumatic linear high-sensitivity heat detection

Pneumatic linear high-sensitivity heat detection is a gas-pressure-based linear detector. The tube material (copper/steel/Teflon) is selected based on the environmental and operational needs, copper for fast response and standard indoor use, steel for rugged corrosive areas, and Teflon for chemically aggressive or electrically sensitive spaces.

5.2.1.2 Flame/gas/spark detection

A flame detector works by sensing the electromagnetic radiation and, through algorithms and filters, discriminates it from false sources (sunlight, hot machinery). The following are the characteristic radiation emitted by the flames/gas/spark:

- a) Ultraviolet (UV) 180 nm to 260 nm, from most hydrocarbon and hydrogen flames;

- b) Infrared (IR) 2 μm to 6 μm , from hot $\text{CO}_2/\text{H}_2\text{O}$ in the flame; and
- c) Some systems also use visible light (VIS).

5.2.1.3 Video detection

Video-based fire detection (VFD) is an advanced fire detection technology that uses CCTV cameras combined with image-processing software to automatically detect visible signs of fire, such as smoke, flames, or heat distortions, and alert emergency systems for early fire response. The system is capable of simultaneous fire detection and visual verification. The system can use infrared (IR) or thermal imaging cameras for low-light or dusty conditions.

5.2.1.4 Gas detection

To detect accumulation of combustible gases or vapours (for example, methane, propane, jet fuel vapours) or toxic gases (for example, carbon monoxide, hydrogen sulphide, ammonia) before they ignite or endanger personnel. Common sensor types used are: catalytic, infrared (IR), electrochemical, or semiconductor.

5.2.1.5 Spark detection

To detect ignition sources (sparks, hot particles, embers) transported by air or material flow before they enter dust collectors, filters, or silos and ignite an explosion.

This detection uses infrared photo-sensors tuned to glowing particle radiation (approx. 0.8 μm to 1.1 μm). The same is installed in dark ducts or pneumatic conveying lines. The system should detect sparks within milliseconds (≤ 10 ms). The system should be integrated with automatic extinguishing systems (for example, water spray or deluge system) in industrial processes.

5.2.1.6 Fuel leak detection

Fuel leak detection (FLD) is a safety and monitoring system designed to identify and alert operators to the accidental release of fuel or flammable liquids from storage tanks, pipelines, or equipment. The system helps prevent fires, explosions, environmental contamination, and operational hazards by detecting leaks early.

5.2.1.7 High ceiling beam detector

The beam detector works like an electronic eye and keeps watching a clear light path; when smoke blocks the beam, it raises the alarm. A high-ceiling beam detection system is a projected or reflected IR smoke detection technology designed for large, tall spaces where spot detectors are ineffective. The system works by sensing reduction in light intensity caused by smoke particles in the beam path and provides early warning for fast-developing fires in high-volume spaces.

5.2.1.8 Linear heat sensing cable

A linear heat sensing cable (digital/analogue) is a wire-type, continuous heat detector that triggers an alarm when it gets hot at any point, making it ideal for protecting long or irregular spaces where many spot-type heat detectors would otherwise be needed.

5.2.1.9 Probe type heat detector

A probe type heat detector is essentially a temperature-sensing probe that extends into the area to detect fire or abnormal heat early, especially in confined, harsh, or high-risk industrial environments. The probe is usually metallic (steel, copper) for good thermal conduction. Its length allows it to reach inside equipment, ducts, or confined spaces where smoke detectors may not work effectively.

5.2.1.10 Aspiration high-sensitivity smoke detection

Aspiration high-sensitivity smoke detection is a pipe-based, actively sampled smoke detection system that can detect smoke at very low concentrations, giving earliest possible warning of fire, especially in high-value, high-ceiling, or challenging environments. It continuously samples air from the protected area through a network of pipes and detects smoke at very low concentrations, providing early warning of fire before visible smoke accumulates.

5.2.2 Automatic Fire Suppression Systems

5.2.2.1 Water spray system

A water spray system is a fixed fire protection system that delivers water through a network of piping and specially designed spray nozzles to cool surfaces, control, or extinguish a fire. It is similar to a sprinkler system but is designed to protect specific equipment or areas, rather than an entire room, by spraying water in a predetermined pattern and density.

5.2.2.2 Wet/Dry pre-action

A pre-action system is a special hybrid system used in areas where accidental water discharge would cause serious damage, such as data centres, archives, museums, cleanrooms. A wet sprinkler system is the simplest and most common type of sprinkler system. The pipes are always filled with water under pressure up to each sprinkler head. A dry sprinkler system is used in areas subject to freezing temperatures (for example, unheated warehouses, loading docks). The pipes are filled with pressurized air or nitrogen, and not water.

5.2.2.3 Modular panel/rack suppression system

A modular panel/rack suppression system with heat-sensitive tubing is a self-contained automatic fire protection system that uses a pressurized agent cylinder and flexible heat-sensitive tubing. The tube detects heat, ruptures at the fire spot, and releases the agent directly, providing fast, localized suppression for sensitive enclosures.

5.2.2.4 Nitrogen injection

The system is used for oil filled transformer system in which an overheating triggering the nitrogen gas displaces air/oxygen above the oil surface, preventing further combustion. The increased pressure helps to seal the transformer tank and suppress vapour formation, thereby preventing transformer fires and minimizing damage.

5.2.2.5 Water mist system

A water mist system is a specialized fire suppression system that uses very fine water droplets (mist) – typically less than 1 000 μ in diameter – to control, suppress, or extinguish fires. A water mist firefighting system suppresses fire by discharging a fine spray of water droplets that cools the fire, displaces oxygen, and reduces radiant heat, offering effective protection with minimal water use.

5.2.2.6 Foam sprinkler system

A foam sprinkler system automatically discharges a water-foam mixture through sprinkler heads to cool, smother, and suppress flammable liquid fires, providing effective fire protection for high-risk liquid storage and handling areas. It operates like a conventional sprinkler system; sprinkler heads open when heat rises above a preset temperature. It is primarily designed to suppress flammable liquid (Class B) fires, such as oil, fuel, or chemical storage, by cooling the fire and forming a foam blanket that smothers vapours.

5.2.2.7 Wet chemical kitchen suppression system

A wet chemical kitchen suppression system is a specialized fire suppression system designed to extinguish fires in commercial cooking areas, especially those involving cooking oils and grease (Class K fires). It uses a liquid wet chemical agent that reacts with hot grease to form a soapy foam layer (saponification), which cools and smothers the fire.

5.2.2.8 Dry powder flooding system

A dry powder flooding system is a fixed fire suppression system that floods a hazard area with dry chemical powder, quickly extinguishing flammable liquid, combustible metal, or electrical fires by interrupting the combustion reaction and isolating fuel from oxygen.

The dry chemical powder (for example, sodium bicarbonate, potassium bicarbonate, or special metal powders) is stored in pressurized vessels or storage tanks. Upon activation, the powder is discharged through a network of piping and nozzles into the protected space.

5.2.2.9 Clean agent total flooding

A clean agent total flooding system is a fixed fire suppression system that protects an enclosed space by discharging a gaseous extinguishing agent (clean agent) to rapidly suppress fire without leaving any residue. The term “total flooding” indicates that the agent fills the entire protected volume to extinguish the fire. The agent absorbs heat and/or interrupts the chemical reaction of the fire.

5.2.2.10 Carbondioxide flooding

A carbon dioxide flooding firefighting system is a fixed system that rapidly floods an enclosed area with carbon dioxide, extinguishing fire by removing oxygen and cooling while leaving no residue, but it requires evacuation of personnel due to asphyxiation risk. It is fast-acting and effective for flammable liquid and electrical fires.

5.2.2.11 Aerosol-based fire suppression system

An aerosol-based fire suppression system is a fixed or modular fire protection system that extinguishes fires by releasing a fine particulate aerosol a mixture of solid microscopic particles suspended in inert gas which interrupts the chemical reactions of combustion. When triggered, it rapidly vapourizes and condenses into ultra-fine particles (1 microns to 10 microns) suspended in gas. It is highly effective for enclosed spaces.

5.3 The fixed firefighting installations and systems should be maintained in accordance with good practice [F(26)]. *See also* the ‘Handbook on Asset and Facility Management’.

5.4 Fire Detection and Suppression Systems Based on the Specific Type and Intensity of Industrial Hazards

[Table 9](#) serves as a guiding tool for designers, fire safety engineers, and facility managers in selecting appropriate fire detection and suppression systems based on the specific type and intensity for G to J occupancy hazards. [Table 9](#) supports a risk-based approach, recognizing that fire protection requirements vary with operational processes, materials handled, and business continuity considerations. It moves away from a one-size-fits-all model and promotes hazard-specific fire safety planning to ensure enhanced protection, system reliability, and life safety in special industrial occupancies.

5.5 Supplementary Protection for Specific Areas/Hazards

[Table 10](#) provides a guide for the structured classification of automatic detection systems, and [Table 11](#) provides for automatic fire suppression systems, based on the specific type and intensity for G to J occupancy hazards. The tables serve as a reference for designers, fire safety engineers, and regulatory authorities to determine the appropriate supplementary protection measures based on the specific fire risks present in a building or facility.

The automatic detection systems covered in the table include high-sensitivity pneumatic linear (or tubing) heat detection, aspiration high-sensitivity smoke detection, gas/flame/visual detection, fuel leak detection, and linear heat detection (LHS). The automatic fire suppression systems include water mist, clean agent gas suppression, wet/dry pre-action sprinkler systems, water spray protection, carbon dioxide flooding, etc.

Each system is marked as either ‘S’ (Suitable) or ‘NS’ (Not Suitable) depending on its effectiveness for a given hazard scenario. The selection of a special protection system should be based on a comprehensive fire risk assessment, considering factors such as fuel type, occupancy, fire growth rate, and critical infrastructure protection. These tables provides essential guidance for ensuring compliance with fire safety regulations and enhancing fire protection strategies in buildings.

Table 9 Fire Detection and Suppression Systems
(Based on the Specific Type and Intensity of A to F Occupancies)
(Clauses [3.5.5.4](#) and [5.4](#))

Types of Special Protection System													
Sl No.	Type of Risk/ Hazard	Automatic Detection System					Automatic Fire Suppression Systems						
		Pneumatic Linear (Copper/ Steel/ Teflon Tubing) High Sensitivity Heat Detection	Aspiration High Sensitivity Smoke Detection	Gas/ Flame/ Video Detection	Fuel Leak Detection	Linear Heat Sensing Cable (LHS)	Water Mist Low/High Pressure	Clean Agent Gas Suppression	Water Pre- Action Sprinkler	Modular Panel/ Rack Suppression with Heat Sensitive Tubing	Water Spray Protection	Wet Chemical Kitchen Suppre- ssion	Aerosol Based Fire Suppre- ssion
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
i)	UPS room without batteries	NS	S	S	NS	NS	S	S	S	NS	NS	NS	NS
ii)	UPS room with batteries (except Lithium ion)	NS	S	S	NS	NS	S	S	S	NS	NS	NS	NS
iii)	Battery room with Lithium ion	S	S	S	NS	NS	NS	NS	S	NS	S	NS	NS
iv)	Server room/data halls	NS	S	NS	NS	NS	S	S	S	NS	NS	NS	NS
v)	Kitchen with any type of fuel, (live cooking)	NS	NS	S	NS	S	S	NS	NS	NS	NS	NS	NS
vi)	Network/ Electrical rooms	NS	S	NS	NS	NS	S	S	S	NS	NS	NS	NS

Table 9 (Continued)

Types of Special Protection System													
Sl No.	Type of Risk/ Hazard	Automatic Detection System					Automatic Fire Suppression Systems						
		Pneumatic Linear (Copper/ Steel/ Teflon Tubing) High Sensitivity Heat Detection	Aspiration High Sensitivity Smoke Detection	Gas/ Flame/ Video Detection	Fuel Leak Detection	Linear Heat Sensing Cable (LHS)	Water Mist Low/High Pressure	Clean Agent Gas Suppression	Water Pre- Action Sprinkler	Modular Panel/ Rack Suppression with Heat Sensitive Tubing	Water Spray Protection	Wet Chemical Kitchen Suppre- ssion	Aerosol Based Fire Suppre- ssion
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
vii)	Electrical/ MCC panels	NS	S	NS	NS	NS	NS	S	NS	S	NS	NS	S
viii)	Cable cellar and trenches and trays over 1.5 m wide	S	NS	NS	NS	S	S	NS	NS	NS	S	NS	S
ix)	Dry transformer indoor	S	NS	S	NS	NS	NS	NS	NS	NS	NS	NS	NS
x)	Oil transformer outdoor more than 2.0 MVA or 2 000 litre oil	S	NS	S	NS	NS	NS	NS	NS	NS	S	NS	NS
xi)	DG rooms	S	NS	S	S	S	S	NS	NS	NS	S	NS	S
xii)	Cold/freezer rooms	S	S	NS	NS	NS	NS	NS	S	NS	NS	NS	NS

Table 9 (Concluded)

Types of Special Protection System													
Sl No.	Type of Risk/ Hazard	Automatic Detection System					Automatic Fire Suppression Systems						
		Pneumatic Linear (Copper/ Steel/ Teflon Tubing) High Sensitivity Heat Detection	Aspiration High Sensitivity Smoke Detection	Gas/ Flame/ Video Detection	Fuel Leak Detection	Linear Heat Sensing Cable (LHS)	Water Mist Low/High Pressure	Clean Agent Gas Suppression	Water Pre-Action Sprinkler	Modular Panel/ Rack Suppression with Heat Sensitive Tubing	Water Spray Protection	Wet Chemical Kitchen Suppression	Aerosol Based Fire Suppression
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
xiii)	MRI/ scanning rooms	NS	S	NS	NS	NS	S	NS	NS	NS	NS	NS	NS
xiv)	Operation theatres	NS	S	NS	NS	NS	S	NS	S	NS	NS	NS	NS
xv)	Tunnel protection	S	NS	S	NS	S	S	NS	NS	NS	S	NS	NS
	NS = Not Suitable S = Suitable NOTES 1 Wet sprinklers are acceptable for all areas where pre-action sprinklers or closed water mist is suitable. 2 DG rooms should be protected with total flooding open nozzles where water mist is being deployed. 3 Spray system should be supplemented with suitable foam, if ignitable liquids or water miscible liquids are involved. 4 Water mist system should be selected based on 3rd party independent test carried out for the specific application, flow rate, density, etc. The products in terms of functionality and the performance should be based on actual test results and certifications, and should NOT be prescriptive. 5 Water mist solution selected, total flooding with sprinklers, total flooding with open nozzles, wet system, pre-action system should comply based on actual test results, and certifications.												

Table 10 Special Protection Systems — Automatic Detection Systems for G to J Occupancies

(Clause [5.5](#))

Sl No.	Type of Risk/Hazard	Pneumatic Linear (Copper/Steel/Teflon Tubing) High Sensitivity Heat Detection	Flame/Gas/ Spark Detection	High Ceiling Beam Detection System	Linear Heat Sensing Cable	Probe Type Heat Detector	Aspiration High Sensitivity Smoke Detection
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Generator room	S	S	NS	S	S	NS
ii)	Water reactive chemical store	S	S	NS	S	S	NS
iii)	Chemical store	S	S	NS	S	S	NS
iv)	Paint booth	S	S	NS	S	S	NS
v)	Paint/varnish storage	S	S	NS	S	S	NS
vi)	Warehouse	NS	S	S	NS	NS	S
vii)	Above ground fuel oil storage	S	S	NS	S	NS	NS
viii)	Oil transformers	S	S	NS	NS	NS	NS
ix)	Diesel/HSD tanks	S	S	NS	S	NS	NS
x)	LPG bullets/Benzene, Xylene, Toulene tanks	NS	NS	NS	NS	NS	NS
xi)	Airport hanger	S	S	S	S	S	S
xii)	Hydrogenation rooms	NS	S	NS	S	S	S
xiii)	Sulphur storage	S	S	NS	S	S	NS
xiv)	Tanker loading/unloading bay	NS	S	NS	NS	NS	NS
xv)	Kitchen	S	S	NS	S	S	NS
xvi)	Conveyors	S	S	NS	S	NS	NS

PART F FIRE AND LIFE SAFETY	Table 10 (Concluded)							
	Sl No.	Type of Risk/Hazard	Pneumatic Linear (Copper/Steel/Teflon Tubing) High Sensitivity Heat Detection	Flame/Gas/ Spark Detection	High Ceiling Beam Detection System	Linear Heat Sensing Cable	Probe Type Heat Detector	Aspiration High Sensitivity Smoke Detection
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	xvii)	QA/QC labs	S	S	NS	S	S	NS
	xviii)	Open large atriums	NS	NS	NS	NS	NS	S
	xix)	Floating roof tanks	S	S	NS	S	NS	NS
	xx)	Oil fire boiler room, oil quenching tank	S	S	NS	S	S	NS
	xxi)	Cooling tower	NS	NS	NS	S	NS	NS
	xxii)	MCC room	S	NS	NS	S	S	S
	xxiii)	Cable cellar/trenches	S	NS	NS	S	S	NS
xxiv)	Electrical panels	S	NS	NS	S	NS	NS	
NOTE								
NS = Not Suitable								
S = Suitable								

Table 11 Special Protection Systems — Automatic Fire Suppression Systems for G to J Occupancies

(Clause 5.5)

Sl No.	Type Of Risk/Hazard	Water Spray System	Wet/Dry Pre-action	Nitrogen Injection	Water Mist	Foam Sprinkler System	Dry Powder Flooding System	Clean Agent Total Flooding	Carbon-Dioxide Flooding
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
i)	Generator room	S	NS	NS	S	NS	NS	NS	NS
ii)	Water reactive chemical store	NS	NS	NS	NS	NS	S	NS	S
iii)	Chemical store	S	NS	NS	S	S	NS	NS	NS
iv)	Paint booth	S	NS	NS	S	NS	NS	NS	NS
v)	Paint/varnish storage	S	NS	NS	S	S	S	NS	NS
vi)	Warehouse	NS	NS	NS	NS	NS	NS	NS	NS
vii)	Above ground fuel oil storage	S	NS	NS	NS	S	NS	NS	NS
viii)	Oil transformers	S	NS	S	S	NS	NS	NS	NS
ix)	Diesel / HSD tanks	S	NS	NS	NS	S	NS	NS	NS
x)	LPG bullets/Benzene, Xylene, Toulene tanks	S	NS	NS	NS	NS	NS	NS	NS
xi)	Airport hanger	NS	NS	NS	NS	S	NS	NS	NS
xii)	Hydrogenation rooms	S	NS	NS	S	NS	NS	NS	NS
xiii)	Sulphur storage	NS	NS	NS	S	NS	NS	NS	NS
xiv)	Tanker loading/unloading bay	S	NS	NS	NS	NS	NS	NS	NS
xv)	Kitchen	NS	NS	NS	S	NS	NS	NS	NS
xvi)	Conveyors	S	NS	NS	S	NS	NS	NS	NS
xvii)	QA/QC labs	NS	S	NS	S	S	S	S	NS
xviii)	Open large atriums	NS	NS	NS	S	NS	NS	NS	NS
xix)	Floating roof tanks	S	NS	NS	NS	S	NS	NS	NS

Table 11 (Concluded)

Sl No.	Type Of Risk/Hazard	Water Spray System	Wet/Dry Pre-action	Nitrogen Injection	Water Mist	Foam Sprinkler System	Dry Powder Flooding System	Clean Agent Total Flooding	Carbon-Dioxide Flooding
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
xx)	Oil fire boiler room, oil quenching tank	S	NS	NS	S	S	NS	NS	NS
xxi)	Cooling tower	S	NS	NS	NS	NS	NS	NS	NS
xxii)	MCC room	NS	S	NS	S	NS	NS	S	S
xxiii)	Cable cellar/trenches	S	NS	NS	S	NS	NS	NS	S
xxiv)	Electrical panels	NS	NS	NS	NS	NS	NS	S	S

NS = Not Suitable

S = Suitable

NOTES

1 Spray system should be supplemented with suitable foam, if ignitable liquids or water miscible liquids are involved.

2 Chemical material safety data sheet (MSDS) as per manufacture should be referred to select the right suppression agent.

3 Water mist system should be selected based on third party independent test carried out for the specific application, flow rate, density, etc. The products in terms of functionality and the performance should be based on actual test results and certifications, and should not be prescriptive.

4 Water mist solution selected, total flooding with sprinklers, total flooding with open nozzles, wet system, pre-action system should comply based on actual test results, and certifications.

5 Oxygen-reduction systems, which is not listed but area proactive fire prevention solutions by maintaining the oxygen percentage below the threshold needed for ignition and sustained combustion but suitable for human occupancy. These systems should be carefully evaluated and used for normally unoccupied areas with proper monitoring of oxygen levels. Fully enclosed very sensitive and critical high valve equipment rooms, rooms in heritage buildings for storing archives or ancient manuscripts, areas where no other fire protection system are effective and feasible are a few areas where it can be implemented. The implementer should ensure all 3rd party product/system accreditations certifications for specific applications and approval from local statutory bodies and AHJs in place before use.

6 SPECIFIC OCCUPANCY WISE REQUIREMENTS

Apart from the general requirements on fire prevention given in [3](#), the life safety requirements given in [4](#), and the fire protection requirements given in [5](#), the following specific requirements should also be complied with for each type of occupancy.

For specific requirements for high rise buildings, the provisions as given in [Annex D](#) should apply.

Atriums in building occupancies should comply with the provisions as given in [Annex E](#).

For specific requirements for data centres, the provisions as given in [Annex F](#) should apply.

For fire protection requirements of commercial kitchen, cooking facilities with or without restaurants, the provisions of good practice [F(30)] should be referred.

6.1 Residential Buildings (Group A)

6.1.1 Life Safety

6.1.1.1 Subdivision A-I

- a) All locking devices, which would impede or prohibit exit, such as chain type bolts, limited opening sliding type locks and burglar locks, which are not disengaged easily by quick-releasing catches, should be prohibited.
- b) All bathroom door locks or fasteners should be designed to permit the opening of the locked or closed door from the outside in an emergency without the use of a special key.
- c) No lodging or rooming house should have its sole means of egress pass through any non-residential occupancy in the same building.
- d) Rooming and lodging houses having floor area of 500 m² on any one or more floors should have access to minimum two separate means of exits, at least one of which should be so arranged as to have direct exit discharge.

6.1.1.2 Subdivision A-II

- a) Requirements of [6.1.1.1](#) should be applicable.
- b) All sleeping rooms having occupancy of more than 10 persons should have two separate and distinct exit access in different directions.

6.1.1.3 Subdivision A-III

In case of high-rise apartments, of the minimum exits as specified in [4.4.2.4.3.1](#), the naturally ventilated/cross-ventilated exit staircases may not require the provision of fire door. However, fire door should be provided for all other staircases and pressurized staircases.

6.1.1.4 Subdivision A-IV

- a) Panic bars should be provided in the fire exits. Panic bars should be located at a height between 865 mm and 1 220 mm from the floor level.
- b) All guest rooms and suites should be protected by extended throw or quick response type sprinklers only. Also, these areas should be provided with audio-based detectors, having a sound level of at least 75 dB.
- c) Horizontal-sliding doors should not be used for door openings across corridors.

6.1.1.5 Additional Precautions

- a) Flammable liquids for household purposes should be kept in tightly stoppered or sealed containers. For the limits of quantities of flammable liquids to be allowed in various occupancies, reference may be made to appropriate regulations.
- b) No stove or combustion heater should be located directly under or immediately at the foot of stairs or otherwise so located as to block escape in case of malfunctioning of the stove or heater.

- c) All kitchen exhaust fans, where provided, should be fixed to an outside wall or to a duct of non-combustible material, which leads directly to the outside. The ducts should not pass through areas having combustible materials. However, in case of centralized ducting, the duct should be provided with adequate protection to limit the spread of fire.
- d) Stores, engineering workshops, areas of high hazard, etc used for storage of substantial amount of flammable liquids should be of 120 min fire resistance rated wall. Such areas should be provided with fire doors, to be kept closed and should be posted with a sign on each side of the door prominently in block letters stating — 'FIRE DOOR — KEEP CLOSED'.

6.2 Educational Buildings (Group B)

6.2.1 Fire Prevention

- a) Buildings intended for educational occupancy should not be used for storage of any hazardous material.
- b) Gymnasiums, indoor stadiums, and similar occupancies are permitted to have floors/running tracks of wood, cinder, synthetic or the like.

6.2.2 Life Safety

- a) Every room with a capacity of over 45 persons in area should have at least two doorways. Exit doors should be operated by panic bars except that doors leading from classrooms directly to the outside may be equipped with the same type of lock as is used on classroom doors leading to corridor, with no provision whatsoever for locking against egress from the classroom.
- b) A building, which will have only the first floor and is accessible to not more than 20 pupils at any time, may be used for school purposes with the following exceptions:
 - 1) Exterior walls or parts of walls which are less than 900 mm from adjacent property lines should have no openings therein.
 - 2) Classrooms may have only one exit not less than 900 mm wide.
- c) Rooms or areas for use by the preschool, kindergarten, Class/Grade 1 students should be located on ground floor/level of exit discharge. Rooms or areas occupied by Class/Grade II students should be located not above one floor higher than ground floor/level of exit discharge.
- d) Of the minimum exits as specified in [4.4.2.4.3.1](#), the naturally ventilated exit staircases, may not require provision of fire door. However, fire door should be provided for all other staircases and pressurized staircases.

6.2.3 Special Precautions

- a) Storage of volatile flammable liquids should be prohibited and the handling of such liquids should be restricted to science laboratories only; and
- b) All exterior openings in a boiler room or rooms containing central heating equipment, if located below an opening in another storey or if less than 3 m from other doors or windows of the same building, should be protected by a fire assembly as in [3.5.4](#). Such assemblies should be of fixed, automatic, or self-closing type.

6.3 Institutional Buildings (Group C)

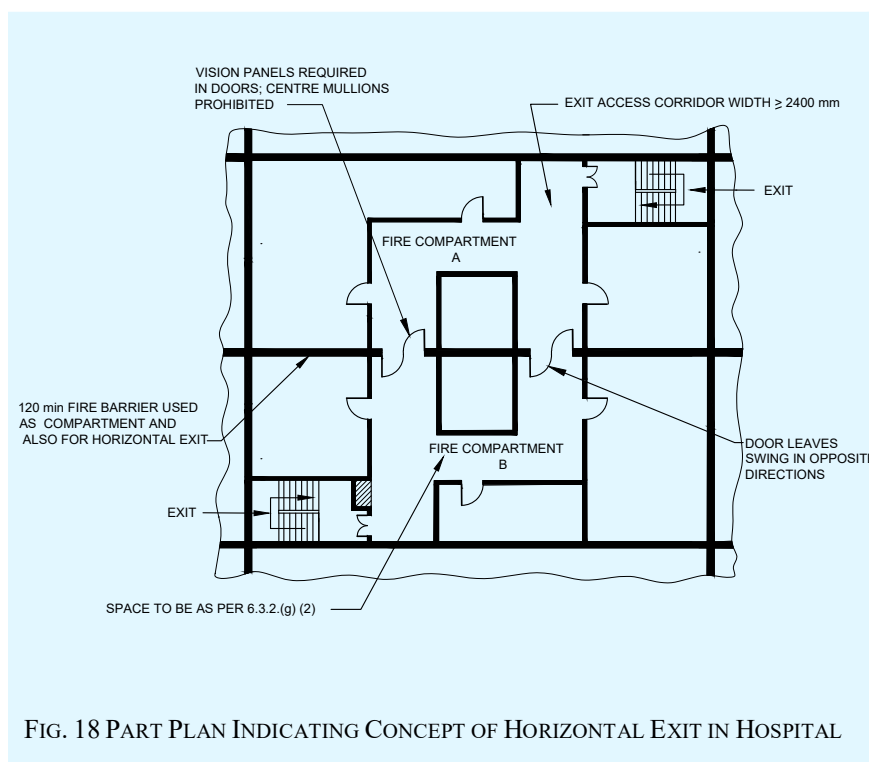
6.3.1 Fire Prevention

No combustible material of any kind should be stored or used in any building or section thereof used for institutional occupancy, except as necessary to normal occupancy and use of the building.

6.3.2 Life Safety

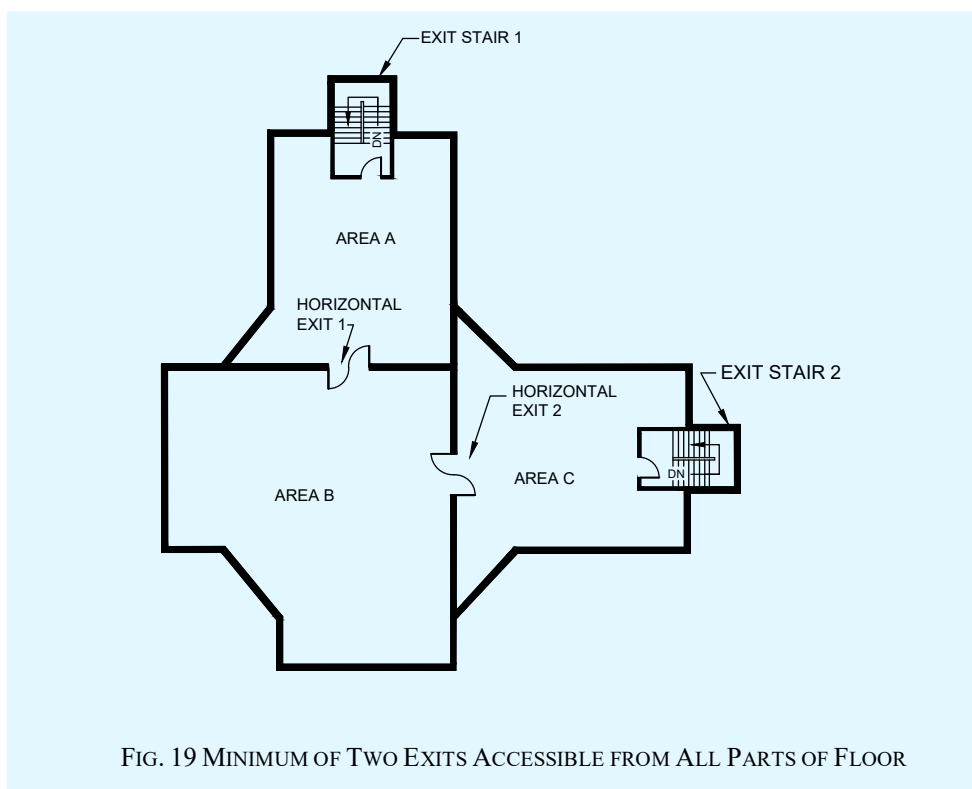
- a) The common path of travel should be up to 30 m. The maximum dead end of corridor distance should not exceed 6 m;
- b) Principle of progressive horizontal evacuation is of paramount consideration for hospital patients particularly those lacking self-preservation. This calls for moving occupants from a fire affected area to

an adjoining area at the same level through a fire resistant wall, to protect them from the immediate dangers of fire and smoke (*see* [Fig. 18](#));



- c) Progressive horizontal evacuation operates on the basis of evacuation from compartment to compartment and on use of adjacent compartments as temporary means of refuge. All compartments should be divided with self-closing (door closers) fire doors with electromagnetic hold open. A coordinator should be provided to sequence the closing of double leaf in case of emergency;
- d) Doors in fire resistant walls should be so installed that these may normally be kept in open position, but will close automatically. Corridor door openings should be not less than 2.0 m in width of double swing double leaf type door. A coordinator should be provided as above, for closing of double leaf in case of emergency;
- e) Exits and other features for penal and mental institutions, and custodial institutions should be the same as specified for hospitals [*see* [6.3.2](#) (g)], in so far as applicable. Reliable means should be provided to permit the prompt release of inmates from any locked section in case of fire or other emergency;
- f) All buildings or sections of buildings in penal and mental institution used for manufacturing, storage or office purposes should have exits in accordance with the provisions of this Part for those occupancies;
- g) For hospitals (Subdivision C-I), the following should also be complied with:
 - 1) Compartmentation should meet the requirement as per [4.5.2](#).
 - 2) Each compartment (*see* [4.5.3](#)) should be able to accommodate, in addition to its own, the patients from adjoining compartment also considering 3.5 m^2 per person. If patients are not bed-ridden, a factor of 0.6 m^2 per person is recommended. This is towards achieving horizontal evacuation.
 - 3) All critical patients incapable of self-preservation and those having physical impairment should preferably be housed within 45 m height. For hospitals exceeding 45 m in height, where such occupants are located above this level, the following non-exhaustive additional provisions should be complied with in consultation with local fire authorities:
 - i) The general life safety and protection aspects and fire protection features, as in heights up to 45 m should be provided to the entire height, including progressive horizontal evacuation.

- ii) Further, the alternate floors (particularly those housing all such patients) should have refuge terraces (basically a covered area with place of safety directly exposed to atmosphere at least one side) of sufficient area (as prescribed by the local authorities) with provisions for medical gases to support patients on ventilators temporarily during evacuation during the staging/evacuation period, to ensure uninterrupted care. The refuge terraces should have:
 - a) direct access to the fire-fighting shaft/fire tower,
 - b) access through fire rated door having a minimum 2 h fire resistant rating,
 - c) automatic sprinkler system, public announcement system, fire detection and alarm system, dedicated and separate shaft for medical gas pipelines, utility services and for electrical services,
 - d) dedicated emergency power supply back up (UPS and DG), particularly for all life-saving equipment, and
 - e) washroom facility within it or directly accessible.
- iii) A dedicated firefighting team should be available round the clock, and having relevant qualification, competence and necessary experience, as may be required by the competent authority.
- 4) Basement should not be used to store flammables or for pathological or other laboratories particularly those involving usage of inflammable chemicals. Medical health care services involving radiation facilities may be housed in the basements as per medical planning along with radiation therapy allied services like bunkers, SIM CT, brachytherapy, nuclear medicine, hot lab, patient hold and waiting area, consultant room, mould room, TPS room, subject to compliance of all requirements for fire safety.
- 5) Operation theatres, delivery rooms, intensive care units, recovery rooms, etc, that containing patients lacking self-preservation in case of emergencies should be fire/smoke separated (120 min minimum rating) from all the adjoining areas.
- 6) Aisles, corridors, ramps, etc, through which patients are moved, should have a minimum width of 2.4 m throughout. Aisles, corridors, and ramps in other areas not intended for the housing, treatment, or use of inpatients should be not less than 1.5 m in width.
- 7) All exits from hospital or infirmary sections should be not less than 2.0 m in width.
- 8) Minimum width of door of single or double occupancy patient room should be 1.25 m, while for the wards for 3 to 5 patient beds should be 1.50 m, to enable movement of patients. The minimum width of door for wards for more than 5 patient beds and for areas necessarily requiring patient evacuation on bed (such as ICU, recovery units, delivery rooms, etc), should have door width of 2.0 m. The width of 2.0 m may be reduced to minimum of 1.5 m where two such doors are provided in such areas.
- 9) Any sleeping accommodation or suite exceeding 100 m² in area should have at least two doorways leading to the exit access corridors.
- 10) Exit access corridors from a compartment to another compartment should be divided at the compartment intersection by a fire door of 120 min fire rating in the fire compartment wall.
- 11) Rooms designated for laboratory and the like should not exceed 100 m² in area, and if additional space is required, fire separation of 120 min should be provided.
- 12) A stretcher lift in a lift bank should also act as fireman's lift meeting the requirements of Part D 'Building Services, Section 5 Installation of Lifts, Escalators, Moving Walks and Power Driven Parking Systems, Subsection 5A Lifts' of this NBCS.
- h) *Progressive evacuation strategy* — See [Fig. 19](#), in which, as an example, the exit calculations at two locations should be based on requirements of total occupancy for area (A, B and C) divided by 2 considering progressive evacuation strategy.



6.3.3 Exception and Deviation

It is recognized that in institutions or part of buildings housing various types of psychiatric patients, or used as mental institutions and penal institutions, it is necessary to maintain locked doors and barred windows; and to such extent the necessary provision in other sections of this Part requiring the keeping of exits unlocked may be waived. It is also recognized that certain type of psychiatric patients are not capable of seeking safety without adequate guidance. In buildings where this situation prevails, reliable means for the rapid release of occupants should be provided, such as remote control of locks, or by keying all locks to keys commonly used by attendants.

6.4 Assembly Buildings (Group D)

6.4.1 Fire Prevention

6.4.1.1 The following should be applicable:

- a) Decorations of places of assembly should be of non-flammable materials. Fabrics and papers used for such purpose should be treated with an effective flame-retardant material. Stage settings made of combustible materials should likewise be treated with fire retardant materials of Class 1 flame spread; and
- b) Gymnasiums, indoor stadiums, and similar occupancies are permitted to have floors/running tracks of wood, cinder, synthetic or the like.

6.4.1.2 Additional fire prevention requirements for mixed use assembly buildings (including malls)

Buildings provided with multiple services/facilities like shopping, cinema theatres, multiplexes, restaurants/food court as part of mixed use assembly/mercantile occupancy should preferably limit the height of the assembly/mercantile occupancy portion of the buildings to 30 m. Above this height the buildings may be used for business or residential occupancies with 240 min separation. Independent exits should be provided for such occupancy above 30 m and should not interface with exits of assembly/mercantile occupancy.

6.4.2 Life Safety

6.4.2.1 The following should be applicable:

- a) Exit door width for assembly buildings should not be less than 2.0 m.
- b) At least four separate exits as remote from each other as practicable are required from every place of assembly in buildings primarily meant for theatrical or operatic performances and which has a stage, proscenium curtain, fixed or portable scenery or scenery loft, lights, mechanical appliances or other theatrical accessories and equipment for example, theatres, motion picture houses, auditoria, concert halls, television and radio studios admitting an audience and which are provided with fixed seats for over 1 000 persons.
- c) At least two separate exits as remote from each other as practicable are required in buildings primarily meant for or theatrical or operatic performances and which has a stage, proscenium curtain, fixed or portable scenery or scenery loft, lights, mechanical appliances or other theatrical accessories and equipment for example, theatres, motion picture houses, auditoria, concert halls, television and radio studios admitting an audience and which are provided with fixed seats. Those building having a capacity from 600 and up to 1 000 persons, at least three exits should be provided with each exit not less than of 2.0 m width.
- d) Clear aisles not less than 1.2 m in width should be formed at right angles to the line of seating in such number and manner that no seat should be more than seven seats away from an aisle. Rows of seats opening on to an aisle at one end only should have not more than seven seats. Under the conditions, where all these aisles do not directly meet the exit doors, cross-aisles should be provided parallel to the line of seating so as to provide direct access to the exit, provided that not less than one cross aisle for every 10 rows should be required. The width of cross-aisles should be minimum of 1 m. Steps should not be placed in aisles to overcome differences in levels unless the gradient exceeds 1 in 10.
- e) The fascia of boxes, balconies and galleries should have substantial railings not less than 1 000 mm high above the floor. The railings at the end of aisles extending to the fascia should be not less than 1 100 mm high for the width of the aisle or 1.2 m high at the foot of steps.
- f) Cross-aisles, except where the backs of seats on the front of the aisle project 600 mm or more above the floor of the aisle, should be provided with railings not less than 900 mm high.
- g) Turnstiles or other devices to restrict the movement of persons should not be installed in any place of assembly in such a manner as to interfere in any way with the required exit facilities.
- h) In theatres and similar places of public assembly where persons are admitted to the building at a time when seats are not available for them and are allowed to wait in a lobby or similar space until seats are available, such use of lobby or similar space should not encroach upon the required clear width of exits. Such waiting should be restricted to areas separated from the required exit ways by substantial permanent partitions or fixed rigid railing not less than 1 050 mm high. Exit should be provided for such waiting spaces on the basis of maximum capacity of two largest screen of multiplex (considering the staggered movie timings). Such exits should be in addition to the exits specified for the main auditorium area and should conform in construction and arrangement to the general rules of exits given above.
- j) Display or exhibit should not be so installed or operated as to interfere in any way with access to any required exit, or with any required exit sign.
- k) All displays or exhibits of combustible material or construction and all booths and temporary construction in connection therewith should be so limited in combustibility or protected as to avoid any undue hazard of fire which might endanger occupants before they have opportunity to use the available exits, as determined by the authority.
- m) Places of assembly in buildings of other occupancy may use exits common to the place of assembly and the other occupancy, provided the assembly area and the other occupancy are considered separately, and each has exits sufficient to meet the requirements of this Part.
- n) Exits should be sufficient for simultaneous occupancy of both the places of assembly and other parts of the building.
- p) For detailed information regarding cinema buildings, reference may be made to good practice [F(27)].
- q) Seats in places of public assembly, accommodating more than 300 persons, should be securely fastened

to the floor, except as permitted in (r) below. All seats in balconies and galleries should be securely fastened to the floor, except that in nailed-in enclosures, boxes with level floors and having not more than 14 seats, the seats need not be fastened.

- r) Chairs not secured to the floor may be permitted in restaurants, night clubs and other occupancies where the fastening of seats to the floor may not be practicable, provided that in the area used for seating, excluding dance floor, stage, etc, there should be not more than one seat for each 1.4 m² of floor area and adequate aisles to reach exits should be maintained at all times. The arrangements should be as follows in general:
 - 1) Rows of seats between aisles should have not more than 14 seats.
 - 2) Rows of seats opening on to an aisle at one end only should have not more than 7 seats.
 - 3) Seats without dividing arms should have their capacity determined by allowing 450 mm per person.
- s) The spacing of rows of seats from back to back should be neither less than 850 mm nor less than 700 mm plus the sum of the thickness of the back and inclination of the back. There should be a space of not less than 350 mm between the back of one seat and the front of the seat immediately behind it as measured between plumb lines.
- t) *Lighting* — No open flame lighting devices should be used in any place of assembly, except in the following cases:
 - 1) Where necessary for ceremonial purposes, the enforcing Authority may consider permitting open flame lighting under such restrictions as are necessary to avoid danger of ignition of combustible materials or injury to occupants.
 - 2) Candles may be used on restaurant tables if securely supported on non-combustible bases and so located as to avoid danger of ignition of combustible materials.
 - 3) Open flame devices may be used on stages where they are a necessary part of theatrical performance, provided adequate precautions, satisfactory to the Authority are taken to prevent ignition of combustible materials.

6.4.2.2 Life safety provisions for mall-like (Assembly, Mercantile, Cinema hall, Restaurant) occupancy

The following additional requirements should be applicable:

- a) Assembly occupancies of theatres, cinema halls and multiplexes should be so located in the mall building that their exits will be separate and lead the occupant directly to exit discharge.
- b) The common path of travel should be 30 m. The maximum dead end of corridor distance should not exceed 6 m.
- c) The minimum width of an exit passageway should be 2.0 m.
- d) Where wheeled carts or buggies are used by customers, adequate provision should be made for the transit and parking of such carts to minimize the possibility that they might obstruct means of egress. Any other storage or hindrances causing obstruction in exits should be avoided.
- e) Car parking facilities should comply with [Annex G](#). Car parking areas at upper levels adjacent to shops, food courts or multiplex should be separated by 120 min fire rated construction and building elements;
- f) 50 percent lifts in common areas in the mall-like occupancies should be with features and requirement of fireman's lift.
- g) The manual call points should be break glass and not pull stations.
- h) Photoluminescent markings should be done along the width and length of treads in all enclosed exits staircases. Exit directional arrow on the wall (in the direction of egress) should be 175 mm × 50 mm in size.
- j) Refuge area to be provided on the floor at or immediately above 18 m should be not less than 10 percent of gross area of floor. Next refuge area to be at/on the floor immediately above 24 m. The refuge area should be 10 percent of the respective floor, which may be divided into two or more separate refuge areas at each of the respective floors, with each being not less than 100 m². Refuge area should also meet all the requirements of life safety as per [4](#).

- k) When refuge spaces are resorted to, every floor should be provided with space for 20 percent of the respective occupants of that floor.

6.4.3 Fire Protection

- a) Every stage equipped with fly galleries, grid irons and rigging for movable theatre type scenery, should have a system of automatic sprinklers over and under such stage areas or spaces and auxiliary spaces, such as dressing rooms, store rooms and workshops, and the proscenium opening should be provided with a fire resisting curtain, capable of withstanding a lateral pressure of 4 kN/m² over the entire area. The curtain should have an emergency closing device capable of causing the curtain to close without the use of power and when so closed, it should be reasonably tight against the passage of smoke.
- b) The stage roof of every theatre using movable scenery or having a motion picture screen of highly combustible construction should have a ventilator or ventilators in or above it, openable from the stage floor by hand and also opening by fusible links or some other approved automatic heat/smoke actuated device, to give a free opening equal to at least one-eighth the area of the floor of the stage.
- c) The proscenium wall of every theatre using movable scenery of decorations should have, exclusive of the proscenium opening, not more than two openings entering the stage, each not to exceed 2 m² and fitted with self-closing fire resistant doors.
- d) Every place of assembly in which projection of motion pictures by light is made should have the projection apparatus enclosed in a fire resisting fixed booth in accordance with good practice [F(27)], except that such booth should not be required where no nitrocellulose motion picture film is used.

6.4.4 Fire Protection and Firefighting System for Metro Stations and Metro Trainways Occupancy

Fire and life safety requirements for metro stations and metro trainways should be in accordance with [Annex H](#) and [Annex J](#), respectively. In case of any ambiguity with respect to provisions in Part F or any other Part of this NBCS, the respective provisions given in [Annex H](#) and [Annex J](#) should be applicable.

6.5 Business Buildings (Group E)

6.5.1 Life Safety

Not less than two exits should be provided for every floor, including basements occupied for office purposes or uses incidental thereto.

6.5.2 Fire Protection

- a) For business buildings (namely laboratories, outpatient clinics, research establishments, libraries, and test houses) the requirement should be provision of automatic fire detection alarm system, while for fire protection, CO₂ and/or foam-based installation to be planned based on the requirements; and
- b) For business buildings (namely electronic data processing centres, computer installations, information technology parks and call centres, telephone exchanges, broadcasting stations, T.V. stations, and air traffic control towers) the requirement should be provision of automatic fire detection alarm system, while for fire protection, any or combination of clean agents, mist technologies, etc, may be planned for appropriate/special situations/locations. Electrical panels may be provided with CO₂/inert gas flooding system based on the requirement and reliability of power for the functional requirement and performance.

6.6 Mercantile Buildings (Group F)

6.6.1 Life Safety

6.6.1.1 Open air mercantile operations, such as open air markets, petrol filling stations, roadside stands for the sale of a farm produce and other outdoor mercantile operations should be so arranged and conducted as to maintain free and unobstructed ways of travel at all times to permit prompt escape from any point of danger in case of fire or other emergency, but having no dead-ends in which persons might be trapped due to display stands, adjoining buildings, fences, vehicles or other obstructions.

6.6.1.2 If such mercantile operations are conducted in roofed-over areas, these should be treated as mercantile buildings, provided canopies over individual small stands to protect merchandise from the weather should not be

construed to constitute buildings for the purpose of this Part.

6.6.2 Exception and Deviation

Any mercantile occupancy, where goods of a highly hazardous nature are predominant, should be considered under Group J occupancy for the purpose of this Part.

6.7 Industrial Buildings (Group G)

6.7.1 Fire Prevention

6.7.1.1 Fire separating walls, fire separating floors and fire partitions

Fire separating walls should be provided between two buildings or between two blocks inside a building, having different fire hazards in accordance with the provisions of this Part. The areas having storage, manufacturing, hazardous activities such as paint store, oil storage, spray booths, etc, should be separated from non-hazardous areas like administrative office, staff canteen, etc by fire rated walls/doors of 120 min fire resistance rating. The fire resistance rating of high hazardous areas like petrochemical, explosives should be 240 min.

The fire separating wall where provided should comply with the following requirements:

- a) The separating wall should be carried through the roof. This portion of the wall extending above the roof, known as 'screen wall' should be of such a height (in no case less than 600 mm) that the horizontal distance at the level of the top of the screen wall between the roofs of the buildings/compartments being segregated is at least 6 m.
- b) The screen wall should be of not less than 230 mm in thickness if it is an extension of a masonry wall, and not less than 150 mm in thickness if it is an extension of a reinforced concrete wall.
- c) The separating wall need not be extended as a screen wall if the roof of one or both of the buildings/blocks being segregated is of reinforced concrete construction (RCC).
- d) In the case of buildings of unequal height, windows, or other openings in the wall of the higher building overlooking the roof of the lower building and within 6 m, thereof, should be protected by fire resistant glass assembly or by approved type fire doors unless the roof of the lower building is of reinforced concrete.
- e) In case the eaves of the higher building fall within 6 m of the roof of the lower building such eaves should be cut-off and the screen wall raised as a parapet, 600 mm high over the roof of the higher building, unless the roof of the lower building is of RCC construction.
- f) Fire separating walls should also be extended outwards on both sides by at least 450 mm.
- g) Doors and window openings in external walls within 3 m of the fire separating walls should be protected by fire doors having a rating of at least 60 min and window openings may be protected by fire resistant glass assembly having same fire rating.
- h) In the case of buildings/compartments having north-light roofs when a separating wall runs parallel to the axis of the north-light opening, the screen wall should be carried through and 600 mm above the ridge of the north light. If, however, the separating wall is at right angles to the axis of the north-light opening, the saw tooth gaps should be bricked up and the screen wall extended 600 mm above the ridge of the north light as well as beyond the extreme north-light opening.
- j) Similarly, the thickness of the floor slabs in case of buildings having upper levels should be designed to provide fire rating as mentioned above.
- k) Storage areas should be separated from the remainder of the building/block by fire walls. Moderate and high hazard areas in industries should have two fire doors each having 180 min fire resistance rating.

6.7.2 Life Safety

6.7.2.1 In buildings used for aircraft assembly or other occupancy requiring undivided floor areas so large that the distances from points within the area to the nearest outside walls where exit doors could be provided are in excess of 45 m, requirements for distance to exits may be satisfied by providing stairs leading to exit tunnels or to overhead passageways. In cases where such arrangements are not practicable, the Authority may, by special

ruling, permit other exit arrangements for single storeyed buildings with distances in excess of the maximum distances specified in 4, if completely automatic sprinkler protection is provided and if the heights of ceiling curtain boards and roof ventilation are such as to minimize the possibility that employees will be overtaken by the spread of fire or smoke within 1.8 m of the floor level before they have time to reach exits, provided, however, that in no case may the distance of travel to reach the nearest exit exceed 65 m where smoke venting is required as a condition for permitting distances of travel to exits in excess of the maximum otherwise allowed.

6.7.2.2 The following applies to special purpose industrial occupancies:

- a) Exits need be provided only for the persons actually employed; spaces not subject to human occupancy because of the presence of machinery or equipment may be excluded from consideration.
- b) Where unprotected vertical openings are necessary to manufacturing operations, these may be permitted beyond the limits specified for industrial occupancy, provided every floor level has direct access to one or more enclosed stairways or other exits protected against obstruction by any fire in the open areas connected by the unprotected vertical openings or smoke therefrom.

6.7.2.3 The following applies to high hazard industrial occupancies:

- a) From every point in every floor area, there should be at least two exits accessible in different directions; where floor areas are divided into rooms, there should be at least two ways of escape from every room, however small rooms, except toilet rooms, so located that the points of access thereto are out of or suitably shielded from areas of high hazard.
- b) In addition to types of exits for upper floors specified for Group G occupancies, slide escapes may be used as required exits for both new and existing buildings.

NOTE — All high hazard industrial occupancies should have automatic sprinkler protection or such other protection as may be appropriate to the particular hazard, including explosion venting for any area subject to explosion hazard, designed to minimize danger to occupants in case of fire or other emergency before they have time to utilize exits to escape.

6.7.3 Additional Precautions

- a) In any room in which volatile flammable substances are used or stored, no device generating a glow or flame capable of igniting flammable vapour should be installed or used, such a room should be provided with a suitably designed exhaust ventilation system (*see Annex K*).
- b) For detailed information on fire safety of certain individual (specific) industrial occupancies, reference may be made to good practice [F(28)].
- c) Fire protection considerations for venting industrial occupancies should be as in *Annex K*.

6.8 Storage Buildings (Group H)

6.8.1 Life Safety

6.8.1.1 Every area used for the storage of hazardous commodities should have an exit within 30.0 m (*see Table 4*) of any point in the area where persons may be present or 35 m where automatic sprinkler protection is provided.

6.8.1.2 Every storage area or space exceeding 1 400 m² gross area, or where more than 10 persons may be normally present should have at least two exit access doors leading to the corridors in exit access, which can be readily opened. This should not be subject to locking so long as any persons are inside and should not depend on power operation. Exits in such cases should be as remote from each other as practicable.

For warehouses, natural draft smoke venting should utilize roof vents or vents in walls at or near the ceiling level; such vents should be normally open, or, if closed, should be designed for automatic opening in case of fire, by release of smoke sensitive devices.

6.8.1.3 The following special provisions should apply to aircraft hangers:

- a) Exits from aircraft hangers (storage or servicing areas) should be provided at intervals of not more than 45 m on all exterior walls of aircraft hangers. There should be a minimum of two exits serving each aircraft storage or servicing areas. Horizontal exits through interior fire walls should be provided at intervals of not more than 30 m. 'Dwarf' or 'Smash' doors accommodating aircraft may be used to comply with these requirements. All doors designated as exits should be kept unlocked in the direction

of exit travel while the area is occupied; and

- b) Exits from mezzanine floors in aircraft storage or servicing areas should be so arranged that the maximum travel to reach the nearest exits from any point on the mezzanine should not exceed 22.5 m. Such exits should lead directly to a properly enclosed stairwell discharging directly to the exterior or to a suitably cut-off area or to outside fire escape stairs.

6.8.1.4 The following special provisions should apply to grain elevators:

- a) There should be at least one stair tower from basement to first floor and from the first floor to the top floor of workhouse which is enclosed in a dust tight non-combustible shaft.
- b) Non-combustible doors of self-closing type should be provided at each floor landing.
- c) An exterior fire escape of the stair or basket ladder type should be provided from the roof of the workshop to ground level or the roof of an adjoining annexe with access from all floors above the first.
- d) An exterior fire escape of either the stair or basket ladder type should be provided from the roof of each storage annexe to ground level.

6.8.1.5 For provisions relating to car parking facilities, see [Annex G](#).

6.8.2 Additional Precautions

Requirements specified in [6.7.3 \(a\)](#) should apply to Group H occupancies also.

6.9 Hazardous Uses (Group J)

6.9.1 Life Safety

Requirements specified in [6.7.2.3](#) should apply to Group J occupancies also.

6.9.2 Additional Precautions

The following requirements should apply to all Group J occupancies, as applicable:

- a) Hazardous buildings should have vapour/flame/ember/spark detectors and explosion suppression systems depending on the type of fire hazard involved;
- b) Each building where gas is employed for any purpose should be provided with an approved outside gas shut-off valve conspicuously marked. The detailed requirements regarding safe use of gas should be as specified in Part E 'Plumbing Services, Section 4 Gas Supply' of this NBCS;
- c) Each boiler room or room containing a heating plant should be separated from the rest of the building by a separating wall;
- d) In any room in which volatile flammable substances are used or stored, no device generating a spark, or glow flame capable of igniting flammable vapour should be installed or permitted unless it is enclosed in a flameproof enclosure;
- e) The use, handling, storage and sale of gasoline, fuel oil and other flammable liquids should not be permitted in Group J occupancies unless such use, handling, storage, and sale is in accordance with appropriate legislation in force;
- f) All openings in exterior walls except wall vents should be protected by a fire stop assembly as in [4](#) and they should be fixed, automatic or self-closing. Wall vents having an area of not less than 100 cm² each should be placed in the exterior walls near the floor line, not more than 1 800 mm apart horizontally. Each building should be provided with a power driven fan exhaust system of ventilation which should be arranged and operated so as to produce a complete change of air in each room every 3 min;
- g) Each machine in dry-cleaning establishments which uses flammable liquid should have an adequate steam line or any other suitable extinguishing agent directly connected to it, so arranged as to have the agent automatically released to the inside of each machine should an explosion occur in the machine; and
- h) Equipment or machinery which generates or emits combustible or explosive dust or fibres should be provided with an adequate dust collecting and exhaust system.

ANNEX A

(Clause 3.1.8)

CALORIFIC VALUES OF COMMON MATERIALS

A-1 The calorific values of some common materials are given in [Table 12](#) for guidance.

Table 12 Calorific Values of Common Materials (Clause A-1)			
Sl No.	Material	Calorific Value (10 ³ kJ/kg) ¹⁾	Wood Equivalent (kg/kg)
(1)	(2)	(3)	(4)
i)	Solid Fuels		
	a) Anthracite	28.6	1.66
	b) Bituminous coal	30.8	1.75
	c) Charcoal	28.4	1.61
	d) Coke (average)	27.5	1.56
	e) Peat	20.9	1.19
	f) Sub-bituminous coal	22.0	1.25
	g) Woods (hard or softwood)	17.6	1.00
ii)	Hydrocarbons		
	a) Benzene	39.6	2.25
	b) Butane	47.1	2.68
	c) Ethane	49.1	2.79
	d) Ethylene	47.7	2.71
	e) Fuel oil	41.6	2.36
	f) Gas oil	42.9	2.44
	g) Hexane	44.9	2.55
	h) Methane (natural gas)	52.8	3.00
	j) Octane	45.3	2.58
	k) Paraffin	39.6 to 44.0	2.3 to 2.5
	m) Pentane	46.0	2.61
	n) Propane	47.3	2.69
	p) Propylene	46.2	2.63
iii)	Alcohols		
	a) Ethyl alcohol	28.4	1.61
	b) Methyl alcohol	21.1	1.20

Table 12 (Continued)

SI No.	Material	Calorific Value (10 ³ kJ/kg) ¹⁾	Wood Equivalent (kg/kg)
	c) Propyl alcohol	31.9	1.81
iv)	Polymers		
	a) Casein	23.1	1.31
	b) Cellulose	16.5	0.94
	c) Cellulose acetate	17.8	1.01
	d) Polyethylene	48.4	2.75
	e) Polypropylene	48.4	2.75
	f) Polystyrene	41.8	2.38
	g) Polyvinylchloride	20.9	1.19
	h) Polymethyl methacrylate	24.6	1.40
	j) Polyurethane	35.2	2.00
	k) Polyamide (nylon)	22.0	1.25
	m) Polyester	22.0	1.25
v)	Common Solids		
	a) Asphalt	38.3	2.13
	b) Bitumen	33.4	1.90
	c) Carbon	32.1	1.83
	d) Cotton (dry)	15.8	0.90
	e) Flax	14.3	0.81
	f) Furs and skins	18.7	1.06
	g) Hair (animal)	20.9	1.19
	h) Leather	17.6	1.00
	j) Ozokerite (wax)	43.5	2.46
	k) Paper (average)	15.4	0.88
	m) Paraffin wax	40.9	2.33
	n) Pitch	33.0	1.88
	p) Rubber	37.4	2.13
	q) Straw	13.2	0.75
	r) Tallows	37.6	2.14
	s) Tan bark	20.9	1.19

Table 12 (Concluded)

SI No.	Material	Calorific Value (10 ³ kJ/kg) ¹⁾	Wood Equivalent (kg/kg)
	t) Tar (bituminous)	35.2	2.00
	u) Wool (raw)	21.6	1.23
	w) Wool (scoured)	19.6	1.11
vi)	Foodstuffs		
	a) Barley	14.1	0.80
	b) Bran	11.0	0.63
	c) Bread	9.9	0.56
	d) Butter	29.5	1.68
	e) Cheese (Cheddar)	18.1	1.03
	f) Corn meal	14.1	0.80
	g) Flour	14.1	0.80
	h) Margarine	29.5	1.68
	j) Oatmeal	15.8	0.90
	k) Rice	13.9	0.79
	m) Soya bean flour	16.1	0.91
	n) Sugar	15.4	0.88
	p) Whole wheat	14.3	0.81
vii)	Miscellaneous		
	a) Acetone	29.7	1.69
	b) Acetaldehyde	25.1	1.43
	c) Formaldehyde	17.6	1.00
	d) Hydrogen	134.2	7.63
	e) Magnesium	24.0	1.36
¹⁾ 1 kJ is approximately equal to 1 Btu, so the figures in the tables are also equivalent to Btu/kg.			

ANNEX B

(Clause [3.1.8](#))

BROAD CLASSIFICATION OF INDUSTRIAL OCCUPANCIES INTO DIFFERENT DEGREE OF FIRE HAZARD

G1 – Low Hazard	G2 – Moderate Hazard	G3 -High Hazard
Abrasive Manufacturing Premises	Aluminium Factories	SUB-CATEGORY (A)
Aerated Water Factories	Atta and Cereal Grinding units	Aircraft Hangers units
Agarbatti Manufacturing	Auto Components (excluding plastics and foam components) manufacturing	Aluminium/Magnesium Powder Plants
Areca nut slicing and/or Betel nut Factories	Bakeries and Biscuit Factories	Auto Components (including Plastics and Foam Components)
Analytical and/or Quality Control Laboratories	Beedi Factories	Auto Mobiles and other Motor Vehicles powered by Lithium or Lithium-Ion Batteries manufacturing
Athletic Equipment manufacturing	Bobbin Factories	Battery Charging/Battery Service Stations
Asbestos Steam Packing and Lagging Manufacturing	Bookbinders, Envelopes and Paper bag Manufacturing	Battery Manufacturing
Auto mobiles and other motor vehicles	Cable manufacturing	Bituminized Paper and/or Hessian Cloth/Tar Felt Manufacturing
Breweries	Camphor Boiling facilities	Chemical Store (water reactive)
Brick Works	Candle Works	Cotton Waste Factories
Canning Factories	Carbon Paper/Typewriter Ribbon Manufacturing	Celluloid Goods Manufacturing
Cardamom Factories	Cardboard Box Manufacturing	Chemical Manufacturing using raw materials having flash points below 23 °C
Cement Factories and/or Asbestos or Concrete Products Manufacturing	Carpenters, Wood wool and Furniture Manufacturing	Cigarette Filter Manufacturing
Ceramic Factories and Crockery and Stoneware Pipe Manufacturing	Carpet and Durries Factories	Diesel Generator Room
Clay Works	Cashewnut Factories	Fire-works Manufacturing
Clock and Watch Manufacturing	Chemical Manufacturing using raw materials having flash points above 23 °C	Foam Plastics Manufacturing and/or Converting Plants
Coffee Curing Roasting and grinding Premises	Chemical Store (non-water reactive)	Godowns and Warehouses (storing combustible/flammable goods)
Condensed Milk Factories, Milk Pasteurizing Plant and Dairies	Cinema Films and T.V. Production Studios	Grass, Hay, Fodder and Bhoosa (chaff) Pressing Factories
Confectionery Manufacturing	Cigar and Cigarette Factories	Hydrogenation rooms (involving Flammable Hydrogen)
Cooling Towers	Coal and/or Coke and/or Charcoal Ball and Briquettes Manufacturing	Industrial Gas Manufacturing (other than Inert/Halogenated Hydrocarbon Gases)
Electric Generating Houses (Hydroelectric)	Coir Factories	

G1 – Low Hazard	G2 – Moderate Hazard	G3 -High Hazard
Incandescent and Fluorescent and TV Picture Tube Manufacturing	Coir Carpets, Rugs, Tobacco, Hides and Skin Presses	Jute mills and jute presses
Electro-plating Works	Cold storage premises	Linoleum Factories
Engineering Workshops	Cold Freezer Room	Lithium Ion Battery Manufacturing Plants
Fruits and Vegetables Dehydrating and Drying Factories	Collieries	LPG Bottling Plants (Mini)
Fruit Products and Condiment Factories	Conveyors	Man-made Fibres (Acrylic fibres/yarn manufacturing)
Garages	Cork products manufacturing	Match Factories
Glass and Glass Fibre Manufacturing	Cotton Seed Cleaning or De-linting Factories	Metal or Tin Printers (where more than 50 percent of floor area is occupied as Engineering Workshop; this may be taken as Ordinary Hazard Risk)
Godowns and Warehouses storing non-combustible goods only	Distilleries	Oil Mills
Green houses	Dry Cleaning, Dyeing and Laundries	Oil Extraction Plants
Gold Thread/Gilding Factories	Duplicating/Stencil Paper Manufacturing	Oil Fire Boiler Room/ Oil Quenching Tank
Gum and/or Glue and Gelatine Manufacturing	Electric substations/Distribution stations	Oil Terminals/Depots handling flammable liquids having flash point of 23 °C and below
Ice, Ice Candy, and Ice-cream Manufacturing	Electric Generating stations (other than Underground power houses)	Oil Transformer Room
Ink (excluding Printing Ink) Factories	Electric Lamps Manufacturing	Paints and Varnish Factories
Mica Products Manufacturing	Electrical Appliances Manufacturing	Paint Booths
Multiple block apartment buildings	Electrical/Motor Control Centre (MCC) Panel Rooms	Paint/Varnish Storage
Open Large Atriums	Electronic Products Manufacturing	Paper and Cardboard Mills having raw material yards
Pottery Works	Enamelware Factories	Piers, Wharfs and Jetties – Handling Extra Hazardous Materials
Poultry Farms	Filter and Wax paper Manufacturing	Printing Ink Manufacturing.
QA/QC Laboratories	Flour Mills	Rosin Lamp black and Turpentine Factories
Salt Crushing Factories and Refineries	Food Processing establishments	Saw Mills
Stables	Garment Makers	Spray painting facilities
Sugar Candy Manufacturing	Ghee Factories (other than vegetable)	Server room
Tanneries/Leather Goods Manufacturers	Godowns and Warehouses (other than those under light and high hazard A categories)	Surgical Cotton Manufacturing
Umbrella Assembling Factories		Tanker Loading/Unloading Bay
Vermicelli Factories		Tarpaulin and Canvas Proofing Factories

G1 – Low Hazard	G2 – Moderate Hazard	G3 -High Hazard
Water Treatment/Filtration Plants and Water Pump Houses	Grain and/or Seeds Disintegrating and/or Crushing Factories	Turpentine and Rosin Distilleries
Zinc/Copper Factories	Grease Manufacturing	Tyre Retreading and Resoling Factories
	Hosiery, Lace, Embroidery and Thread Factories	Tunnel Protection
	Incandescent Gas Mantle Manufacturing	UPS Room with batteries
	Indoor Dry Transformer Room	Battery Room with Lithium-ion batteries
	Industrial Gas Manufacturing (Inert/halogenated hydrocarbon gases)	SUB-CATEGORY (B)
	Kitchen with any type of fuel (Live cooking)	Ammonia and Urea Synthesis Plants
	MRI/scanning room	Benzene/ Xylene/ Toluene Tanks
	Man-made Yarn/Fibre Manufacturing (other than acrylic fibres/yarn Manufacturing)	CNG Compressing and Bottling Plants
	Manure and Fertilizer Works (Blending, Mixing, and granulating)	Coal based methane plants
	Mattress and Pillow Making	LPG bullets
	Mineral Oil Blending and Processing	Explosive Factories
	Multiple block business buildings	Oil Storage (above ground)
	Multiple block star hotel buildings	Outdoor Oil Transformer Room
	Musical Instruments Manufacturing	
	Network/electrical room	
	Oil and Leather Cloth Factories	
	Oil Terminals/Depots other than those categorized under High hazard A	
	Open storage of flammable liquids in drums, cans, etc	
	Operation Theatre Rooms	
	Optical Goods facilities	
	Oxygen Plants	
	Paper and Cardboard Mills without Raw Material Yards	
		NOTE — In case of complexes having separate plants having varying degrees of hazard, authority having jurisdiction should be consulted to decide on level of protection to be provided.

G1 – Low Hazard	G2 – Moderate Hazard	G3 -High Hazard
	Pharmaceuticals Piers, Wharves, Jetties and Dockyards other than those categorized under High hazard A Plastic Goods Manufacturing Plywood/Wood Veneering Factories Printing Press Premises Pulverising and Crushing Mills Rice Mills Rope Works Rubber Goods Manufacturing Rubber Tyres and Tubes Manufacturing Semiconductors Manufacturing Sulphur Storage Sugar Factories and Refineries Shellac Factories Silk Filatures Soaps and Glycerine Factories Solar Cells and Modules manufacturing Sponge Iron Steel Plants (Gas Based) Starch Factories Tea Factories Textile Mills Tobacco (Chewing) and Pan-Masala Making facilities Tobacco Grinding and Crushing Tobacco Redrying Factories UPS Room without batteries Woollen Mills	

NOTES

1 Any occupancy that is not covered in this Annex should be classified in the most appropriate class which resembles the proposed occupancy.

2 For classification of industries based on 'chemical hazard', pollution, etc, the respective state/central guidelines/documents/rules may be referred.

ANNEX C

(Clauses 3.4.2, 3.4.3 and F-2.1)

AVAILABLE DATA REGARDING FIRE RESISTANCE RATING OF VARIOUS BUILDING COMPONENTS

C-1 The following tables (see Table 13 to Table 29) may be used, in the absence of any validated/certified rating regarding the materials used in the structural and/or non-structural element/component.

Table 13 Masonry Walls: Solid (Required to Resist Fire from One Side at a Time)											
(Clause C-1)											
Sl No.	Nature of Construction and Materials	Minimum Thickness (mm), Excluding any Finish, for a Fire Resistance (<i>min</i>) of									
		Load Bearing					Non-Load Bearing				
		60	90	120	180	240	60	90	120	180	240
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
i)	Reinforced cement concrete ¹⁾	120 (25) ²⁾	140 (25) ²⁾	160 (25) ²⁾	200 (25) ²⁾	240 (25) ²⁾	—	—	—	—	—
ii)	Unreinforced cement concrete	150	175	—	—	—	—	—	—	—	—
iii)	No-fines concrete with:										
	a) 13 mm cement/sand or gypsum/sand	—	—	—	—	—	150	150	150	150	150
	b) 13 mm light weight aggregate gypsum plaster	—	—	—	—	—	150	150	150	150	150
iv)	Bricks of clay:										
	a) Without finish	90	100	100	170	170	75	90	100	170	170
	b) With 13 mm light weight aggregate gypsum plaster	90	90	90	100	100	75	90	90	90	100
v)	Bricks of sand lime:										
	a) Without finish	90	100	100	190	190	75	90	100	170	170
	b) With 13 mm light weight aggregate gypsum plaster	90	90	90	100	100	75	90	90	90	100
vi)	Blocks of concrete:										
	a) Without finish	90	100	100	—	—	75	90	100	140	150

Table 13 (Concluded)

Sl No.	Nature of Construction and Materials	Minimum Thickness (mm), Excluding any Finish, for a Fire Resistance (<i>min</i>) of									
		Load Bearing					Non-Load Bearing				
		60	90	120	180	240	60	90	120	180	240
	b) With 13 mm light weight aggregate gypsum plaster	90	90	90	100	100	75	75	75	90	100
	c) With 13 mm cement/sand or gypsum/sand	—	—	—	—	—	75	90	90	100	140
vii)	Blocks of light weight concrete:										
	a) Without finish	90	100	100	140	150	75	75	75	125	140
	b) With 13 mm light weight aggregate gypsum plaster	90	90	90	100	100	50	63	75	75	75
	c) With 13 mm cement/sand or gypsum/sand	—	—	—	—	—	75	75	75	90	100
viii)	Blocks of aerated concrete:										
	a) Without finish	90	100	100	140	180	50	63	63	75	100
	b) With 13 mm light weight aggregate gypsum plaster	90	90	100	100	150	—	—	—	—	—
1) Walls containing at least 1 percent of vertical reinforcement.											
2) Minimum thickness of actual cover to reinforcement.											

Table 14 Masonry Walls: Hollow (Required to Resist Fire from One Side at a Time)

(Clause [C-1](#))

Sl No.	Nature of Construction and Materials	Minimum Thickness (mm), Excluding any Finish, for a Fire Resistance (<i>min</i>) of										
		Load Bearing						Non-load Bearing				
		60	90	120	180	240	30	60	90	120	180	240
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
1)	Bricks of clay:											
	a) Without finish	170	170	170	200	200	75	75	90	100	170	170

Table 14 (Concluded)

SI No.	Nature of Construction and Materials	Minimum Thickness (mm), Excluding any Finish, for a Fire Resistance (<i>min</i>) of										
		Load Bearing						Non-load Bearing				
		60	90	120	180	240	30	60	90	120	180	240
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
	b) With 13 mm lightweight aggregate gypsum plaster	170	170	170	200	200	75	75	90	100	170	170
ii)	Blocks of concrete:											
	a) Without finish	—	—	—	—	—	90	125	125	140	140	150
	b) With 13 mm cement/sand or gypsum/ sand	—	—	—	—	—	90	125	125	140	140	140
	c) With 13 mm lightweight aggregate gypsum plaster	190	200	200	—	—	75	90	90	100	125	125
iii)	Blocks of lightweight concrete:											
	a) Without finish	100	100	100	—	—	75	90	90	100	140	150
	b) With 13 mm cement/sand or gypsum/sand	—	—	—	—	—	75	75	75	100	140	140
	c) With 13 mm lightweight aggregate gypsum plaster	—	—	—	—	—	63	63	63	75	90	100

Table 15 Framed Construction, Load Bearing (Required to Resist Fire from One Side at a Time)
(Clause [C-1](#))

SI No.	Nature of Construction and Materials/Timber Studs at Centres not Exceeding 600 mm, Faced on Each Side with	Minimum Thickness (mm) of Protection for a Fire Resistance of 60 min
(1)	(2)	(3)
i)	Plasterboard layers with joints staggered, joints in outer layer taped and filled — Total thickness for each face	25
ii)	One layer of 12.7 mm plasterboard with a finish of lightweight aggregate gypsum plaster	13
iii)	Metal lath and plaster, thickness of plaster:	
	a) Sanded gypsum plaster (metal lathing grade)	22
	b) Lightweight aggregate gypsum plaster	13

Table 16 Framed Construction, Non-Load Bearing (Required to Resist Fire from One Side at a Time)
(Clause [C-1](#))

SI No.	Nature of Construction and Materials/Steel or Timber Frame at Centres not Exceeding 600 mm, Facings on Both Sides of	Stud Construction	Minimum Thickness (mm) of Protection for a Fire Resistance (min)			
			30	60	90	120
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Dry lining with materials fixed direct to studs, without plaster finish					
	a) One layer of plasterboard with taped and filled joints	Timber or steel	12.7	–	–	–
	b) Two layers of plasterboard with joints staggered, joints in outer layer taped and filled – Total thickness for each face	Timber or steel	19	25	–	–
	c) One layer of asbestos insulating board with transverse joints backed by fillers of asbestos insulating board not less than 9 mm thick, or by timber	Timber or steel	9	12	–	–
	d) One layer of wood wool slabs	Timber	25	–	–	–
	e) One layer of chipboard or of plywood	Timber or steel	18	–	–	–
ii)	Lining with materials fixed direct to studs, with plaster finish					
	a) Plasterboard of thickness:					
	1) With not less than 5 mm gypsum plaster finish	Timber or steel	9.5	–	–	–
	2) With not less than 13 mm gypsum plaster finish	Timber or steel	–	12.7	–	–
iii)	Wet finish					
	a) Metal lath and plaster, thickness of plaster:					
	1) Sanded gypsum plaster	Timber or steel	13	–	–	–
	2) Lightweight aggregate gypsum plaster	Timber	–	13	19	25
		Steel	–	13	–	–

Table 17 Framed External Walls Load Bearing (Required to Resist Fire from One Side at a Time)

(Clause [C-1](#))

SI No.	Nature of Construction and Materials	Minimum Thickness (mm) of Protection for a Fire Resistance of 60 min
(1)	(2)	(3)
i)	Timber studs at centres not exceeding 600 mm with internal linings of: Plasterboard layers with joints in outer layer taped and filled, total thickness of plasterboard	31

Table 18 Framed External Walls Non-Load Bearing [Required to Resist Fire from Inside the Building (A)]

(Clause [C-1](#))

SI No.	Nature of Construction and Materials	Minimum Thickness (mm) of Protection for a Fire Resistance (min) of					
		30	60	90	120	180	240
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Steel frame with an external cladding of non-combustible sheets (excluding sheet steel), with a steel supporting framework and internal lining of:						
	a) Metal lath and plaster, thickness of plaster:						
	1) Sanded gypsum plaster (metal lathing grade)	13	13	—	—	—	—
	2) Lightweight aggregate gypsum plaster	10	13	15	15	15	19
	b) Two layers of plasterboard with joints staggered joints in outer layer taped and filled — Total thickness	21	32	—	—	—	—
	c) Plasterboard of thickness:						
	1) With not less than 5 mm gypsum plaster finish	12.7	—	—	—	—	—
	2) With not less than 13 mm gypsum plaster finish	9.5	—	—	—	—	—
	3) With not less than 10 mm lightweight aggregate gypsum plaster	9.5	—	—	—	—	—
	d) One layer of asbestos insulating board with transverse joints backed by fillers of asbestos insulating	9	9	12	12	12	12

Table 18 (Concluded)

SI No.	Nature of Construction and Materials	Minimum Thickness (mm) of Protection for a Fire Resistance (min) of					
		30	60	90	120	180	240
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	board not less than 9 mm thick, or by timber						
	e) One layer of wood/wool slabs without finish	—	50	—	—	—	—
	f) One layer of compressed straw building slabs :						
	1) Without finish	50	—	—	—	—	—
	2) With not less than 5 mm gypsum plaster finish	—	50	—	—	—	—
	g) Aerated concrete blocks	50	50	63	63	75	100
	h) Bricks of clay:						
	1) Without finish	75	75	90	90	100	100
	2) With not less than 13 mm lightweight aggregate gypsum plaster	—	—	75	75	90	90

Table 19 Framed External Walls Non-Load Bearing [(Required to Resist Fire from Inside the Building (B))]

(Clause [C-1](#))

SI No.	Nature of Construction and Materials	Minimum Thickness (mm) of Protection to Provide Sufficient Insulation to Achieve a Modified Fire Resistance of Up to 240 min
(1)	(2)	(3)
i)	Steel frame with an external cladding of sheet steel fully lapped, steel bolted and fixed to steel sheeting rails, with timber or steel supporting framework and internal lining of:	
	a) Metal lath and plaster, thickness of plaster:	
	1) Sanded gypsum plaster (metal lathing grade)	13
	2) Lightweight aggregate gypsum plaster	10

Table 19 (Concluded)

SI No.	Nature of Construction and Materials	Minimum Thickness (mm) of Protection to Provide Sufficient Insulation to Achieve a Modified Fire Resistance of Up to 240 min
(1)	(2)	(3)
	b) One layer of plasterboard with joints taped and filled	12.7
	c) Plasterboard of thickness with not less than 5 mm gypsum plaster finish	9.5
	d) One layer of asbestos insulating board with transverse joints backed by fillers of asbestos insulating board not less than 9 mm thick, or by timber	9
	e) One layer of wood/wool slabs	25
	f) One layer of compressed straw building slabs	50
	g) One layer of chipboard or of plywood	18
	h) Aerated concrete blocks	50
	j) Bricks of clay	75
	k) Any internal decorative lining with a cavity fill independently supported and retained in position of mineral fibre insulating material (excluding glass) at a density of 48 kg/m ³	50

Table 20 Framed Walls Non-Load Bearing [Required to Resist Fire from Inside the Building (C)]
(Clause [C-1](#))

SI No.	Nature of Construction and Materials	Minimum Thickness (mm) of Protection for a Fire Resistance of 90 min
(1)	(2)	(3)
i)	Timber frame with external cladding of weather boarding or external plywood, 9.5 mm with an internal lining of :	
	a) Plasterboard not less than 9.5 mm thick, finished with:	
	1) Gypsum plaster	13
	2) Lightweight aggregate gypsum plaster	10
	b) Plasterboard not less than 12.7 mm thick, finished with:	
	1) Gypsum plaster	10

Table 20 (Concluded)

Sl No.	Nature of Construction and Materials	Minimum Thickness (mm) of Protection for a Fire Resistance of 90 min
(1)	(2)	(3)
	2) Lightweight aggregate gypsum plaster	10
	c) One layer of asbestos insulating board with transverse joints backed by fillers of asbestos insulating board not less than 9 mm thick, or by timber	9 12

Table 21 Reinforced Concrete Columns

(Clause [C-1](#))

Sl No.	Nature of Construction and Materials		Minimum Dimensions (mm) Excluding any Finish, for a Fire Resistance (min) of					
			30	60	90	120	180	240
(1)	(2)		(3)	(4)	(5)	(6)	(7)	(8)
i)	Fully exposed	a) Width	150	200	250	300	400	450
		b) Cover	40	40	40	40	40	40
ii)	50 percent exposed	a) Width	125	160	200	200	300	350
		b) Cover	40	40	40	40	40	40
iii)	One face exposed	a) Thickness	100	120	140	160	200	240
		b) Cover	40	40	40	40	40	40

Table 22 Concrete Beams

(Clause [C-1](#))

Sl No.	Nature of Construction and Materials		Minimum Dimensions (mm) Excluding any Finish, for a Fire Resistance (min) of					
			30	60	90	120	180	240
(1)	(2)		(3)	(4)	(5)	(6)	(7)	(8)
i)	Reinforced concrete (simply supported)	a) Width	200	200	200	200	240	280
		b) Cover	20	20	20	40	60 ¹⁾	70 ¹⁾
ii)	Reinforced concrete (continuous)	a) Width	200	200	200	200	240	280
		b) Cover	20	20	20	30	40	50 ¹⁾
iii)	Prestressed concrete (simply supported)	a) Width	100	120	150	200	240	280
		b) Cover	25	40	55 ¹⁾	70 ¹⁾	80 ¹⁾	90 ¹⁾

Table 22 (Concluded)

SI No.	Nature of Construction and Materials		Minimum Dimensions (mm) Excluding any Finish, for a Fire Resistance (min) of					
			30	60	90	120	180	240
(1)	(2)		(3)	(4)	(5)	(6)	(7)	(8)
iv)	Prestressed concrete (continuous)	a) Width	80	100	120	150	200	240
		b) Cover	20	30	40	55 ¹⁾	70 ¹⁾	80 ¹⁾
¹⁾ Require attention to the additional measures necessary to reduce the risk of spalling.								

Table 23 Concrete Floors

(Clause [C-1](#))

SI No.	Nature of Construction and Materials		Minimum Dimensions (mm) Excluding any Finish, for a Fire Resistance (min) of					
			30	60	90	120	180	240
(1)	(2)		(3)	(4)	(5)	(6)	(7)	(8)
i)	Reinforced concrete (simply supported)	a) Thickness	75	95	110	125	150	170
		b) Cover	20	20	25	35	45 ¹⁾	55 ¹⁾
ii)	Reinforced concrete (continuous)	a) Thickness	75	95	110	125	150	170
		b) Cover	20	20	20	25	35	45 ¹⁾
1) Require attention to the additional measures necessary to reduce the risk of spalling.								

Table 24 Concrete Floors: Ribbed Open Soffit

(Clause [C-1](#))

SI No.	Nature of Construction and Materials		Minimum Dimensions (mm) Excluding any Finish, for a Fire Resistance (min) of					
			30	60	90	120	180	240
(1)	(2)		(3)	(4)	(5)	(6)	(7)	(8)
i)	Reinforced concrete (simply supported)	a) Thickness of floor	75	95	110	125	150	170
		b) Rib width	125	125	125	125	150	175
		c) Cover	20	20	35	45 ¹⁾	55 ¹⁾	65 ¹⁾
ii)	Reinforced concrete (continuous)	a) Thickness	75	95	110	125	150	170
		b) Width	125	125	125	125	150	175
		c) Cover	20	20	20	35	45 ¹⁾	55 ¹⁾
1) Require attention to the additional measures necessary to reduce the risk of spalling.								

Table 25 Encased Steel Columns, 203 mm × 203 mm

(Clause 3.4.3 and C-1)

SI No.	Nature of Construction and Materials	Minimum Dimensions (mm) Excluding any Finish, for a Fire Resistance (min) of				
		60	90	120	180	240
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Hollow protection (without an air cavity over the flanges):					
	a) Metal lathing with trowelled lightweight aggregate gypsum plaster ¹⁾	13	15	20	32	—
	b) Plasterboard with 1.6 mm wire binding at 100 mm pitch, finished with lightweight aggregate gypsum plaster not less than the thickness specified:					
	1) 9.5 mm plaster board	10	15	—	—	—
	2) 19 mm plaster board	10	13	20	—	—
	c) Asbestos insulating boards, thickness of board:					
	1) Single thickness of board, with 6 mm cover fillets at transverse joints	—	19	25	—	—
	2) Two layers of total thickness	—	—	—	38	50
	d) Solid bricks of clay, composition, or sand lime, reinforced in every horizontal joint, ulastered	50	50	50	75	100
	e) Aerated concrete blocks	60	60	60	—	—
ii)	f) Solid blocks of lightweight concrete Hollow protection (with an air cavity over the flanges)	50	50	50	60	75
	Asbestos insulating board screwed to 25 mm asbestos battens	12	19	—	—	—
iii)	Solid protections					
	a) Concrete, not leaner than 1 : 2 : 4 mix (unplastered):					
	1) Concrete not assumed to be load bearing, reinforced ²⁾	25	25	25	50	75
	2) Concrete assumed to be load bearing	50	50	50	75	75
	b) Lightweight concrete, not leaner than 1 : 2 : 4 mix (unplastered): concrete not assumed to be load bearing, reinforced ²⁾	25	25	25	40	60
¹⁾ So fixed or designed, as to allow full penetration for mechanical bond. ²⁾ Reinforcement should consist of steel binding wire not less than 2.3 mm in thickness, or a steel mesh weighing not less than 0.5 kg/m ² . In concrete protection, the spacing of that reinforcement should not exceed 200 mm in any direction.						

Table 26 Encased Steel Beams, 406 mm × 176 mm

(Protection Applied on Three Sides)

(Clause 3.4.3 and C-1)

Sl No.	Nature of Construction and Materials	Minimum Thickness (mm) of Protection for a Fire Resistance (min) of					
		30	60	90	120	180	240
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Hollow protection (without an air cavity beneath the lower flange):						
	a) Metal lathing with trowelled lightweight aggregate gypsum plaster ¹⁾	13	13	15	20	25	—
	b) Plasterboard with 1.6 mm wire binding ²⁾ at 100 mm pitch, finished with lightweight aggregate gypsum plaster not less than the thickness specified:						
	1) 9.5 mm plaster board	10	10	15	—	—	—
	2) 19 mm plaster board	10	10	13	20	—	—
	c) Asbestos insulating boards, thickness of board :						
	1) Single thickness of board, with 6 mm cover	—	—	19	25	—	—
	2) Two layers of total thickness	—	—	—	—	38	50
ii)	Hollow protection (with an air cavity below the lower flange):						
	a) Asbestos insulating board screwed to 25 mm asbestos battens	9	12	—	—	—	—
iii)	Solid protection :						
	a) Concrete, not leaner than 1 : 2 : 4 mix (unplastered):	25	25	25	25	50	75
	1) Concrete not assumed to be load bearing, reinforced ³⁾	50	50	50	50	75	75
	2) Concrete assumed to be load bearing						
	b) Lightweight concrete ⁴⁾ , not leaner than 1 : 2 : 4 (mix) unplastered	25	25	25	25	40	60
¹⁾ So fixed or designed, as to allow full penetration for mechanical bond. ²⁾ Where wire binding cannot be used, expert advice should be sought regarding alternative methods of support to enable the lower edges of the plasterboard to be fixed together and to the lower flange, and for the top edge of the plasterboard to be held in position. ³⁾ Reinforcement should consist of steel binding wire not less than 2.3 mm in thickness or a steel mesh weighing not less than 0.5 kg/m ² . In concrete protection, the spacing of that reinforcement should not exceed 200 mm in any direction. ⁴⁾ Concrete not assumed to be load bearing, reinforced.							

Table 27 Timber Floors — Tongued and Grooved Boarding, or Sheets of Tongued and Grooved Plywood or Wood Chipboard, of not Less than 21 mm Finished Thickness <i>(Clause C-1)</i>				
SI No.	Nature of Construction and Materials	Minimum Thickness (mm) of Protection for a Fire Resistance (min) of		
		30	60	120
(1)	(2)	(3)	(4)	(5)
i)	37 mm (minimum) timber joists with a ceiling of:			
	a) Timber lathing and plaster, plaster of thickness	15	—	—
	b) Metal lathing and plaster, thickness of plaster:			
	1) Sanded gypsum plaster (metal lathing grade)	15	—	—
	2) Lightweight aggregate gypsum plaster	13	13	25
	c) One layer of plasterboard with taped and filled joints	12.7	—	—
	d) Two layers of plasterboard with joints staggered, joints in outer layer taped and filled total thickness	19	31	—
	e) One layer of plasterboard not less than 9.5 mm thick, finished with:			
	1) Gypsum plaster	5	—	—
	2) Sanded gypsum plaster	13	—	—
	3) Lightweight aggregate gypsum plaster	13	—	—
	f) One layer of plasterboard not less than 12.7 mm thick, finished with:			
	1) Gypsum plaster	5	—	—
	2) Lightweight aggregate gypsum plaster	10	—	—
	g) One layer of asbestos insulating board with any transverse joints backed by fillets of asbestos insulating board not less than 9 mm thick, or by timber	9	12	—

Table 28 Timber Floors — Tongued and Grooved Boarding, or Sheets of Tongued and Grooved Plywood or Wood Chipboard, of not Less than 15 mm Finished Thickness <i>(Clause C-1)</i>				
SI No.	Nature of Construction and Materials	Minimum Thickness (mm) of Protection for a Fire Resistance (min) of		
		30	60	120
(1)	(2)	(3)	(4)	(5)
i)	37 mm (minimum) timber joists with a ceiling of:			
	a) Timber lathing and plaster, plaster of thickness	15	—	—

Table 28 (Concluded)

Sl No.	Nature of Construction and Materials	Minimum Thickness (mm) of Protection for a Fire Resistance (min) of		
		30	60	120
(1)	(2)	(3)	(4)	(5)
	b) Metal lathing and plaster, thickness of plaster for:			
	1) Sanded gypsum plaster (metal lathing grade)	15	—	—
	2) Lightweight aggregate gypsum plaster	13	13	25
	c) One layer of plasterboard with taped and filled joints	12.7	—	—
	d) Two layers of plasterboard with joints staggered, joints in outer layer taped and filled total thickness	22	31	—
	e) One layer of plasterboard not less than 9.5 mm thick, finish with:			
	1) Gypsum plaster	5	—	—
	2) Sanded gypsum plaster	15	—	—
	3) Lightweight aggregate gypsum plaster	13	—	—
	f) One layer of plasterboard not less than 12.7 mm thick, finished with:			
	1) Gypsum plaster	5	—	—
	2) Lightweight aggregate gypsum plaster	10	—	—
	g) One layer of asbestos insulating board, with any transverse joints backed by fillets of asbestos insulating board not less than 9 mm thick, or by timber	9	12 ¹⁾	—
¹⁾ Finished on top with 25 mm minimum thick glass fibre or mineral wool laid between joints.				

Table 29 Timber Floors — Any Structurally Suitable Flooring of Timber or Lignocelluloses Boards

(Clause [C-1](#))

Sl No.	Nature of Construction and Materials	Minimum Thickness (mm) of Protection for a Fire Resistance (min) of	
		30	60
(1)	(2)	(3)	(4)
i)	37 mm (minimum) timber joists with a ceiling of:		
	a) Timber lathing and plaster, plaster of thickness	15	—

Table 29 (Concluded)

SI No.	Nature of Construction and Materials	Minimum Thickness (mm) of Protection for a Fire Resistance (min) of	
		30	60
(1)	(2)	(3)	(4)
	b) Metal lathing and plaster, thickness of plaster for :		
	1) Sanded gypsum plaster (metal lathing grade)	15	—
	2) Lightweight aggregate gypsum plaster	13	19
	c) One layer of plasterboard with joints taped and filled and backed by timber	12.7	—
	d) Two layers of plasterboard with joints staggered, joints in outer layer taped and filled total thickness	25	—
	e) Two layers of plasterboard, each not less than 9.5 mm thick, joints between boards staggered and outer layer finished with gypsum plaster	5	—
	f) One layer of plasterboard not less than 9.5 mm thick, finish with :		
	1) Sanded gypsum plaster	13	—
	2) Lightweight aggregate gypsum plaster	15	—
	g) One layer of plasterboard not less than 12.7 mm thick, finished with :		
	1) Sanded gypsum plaster	15	—
	2) Lightweight aggregate gypsum plaster	13	—
	h) One layer of asbestos insulating board with any transverse joints backed by fillets of asbestos insulating board not less than 9 mm thick, or by timber	12	—

ANNEX D

(Clauses [5.1.4](#) and [6](#))

ADDITIONAL REQUIREMENTS FOR HIGH RISE BUILDINGS

D-1 GENERAL

High rise buildings should receive special attention with respect to fire and life safety particularly with regard to planning, design, execution, maintenance and training so that the intended provisions of this NBCS are well implemented. These get further accentuated as the buildings go taller; some of the key aspects are as follows:

- a) Staging and evacuation requirements of occupants;
- b) Stack effect posing challenges towards pressurization and smoke exhaust;
- c) Zoning of firefighting system to meet functional requirements of hydraulic pressure and flow; and
- d) Challenges experienced by fire personnel in reaching the place of fire and towards evacuation.

Aspects to mitigate these challenges require innovative approach, interaction with local fire authorities and meaningful strategic planning towards maintenance and fire drills.

D-2 EGRESS AND EVACUATION STRATEGY

One firefighting shaft should be planned for each residential building/tower, in an educational building/block, and for each compartment of institutional, assembly, business and mercantile occupancy types. In buildings/floors having three or more compartments, there should be at least two firefighting shafts to serve the compartments.

For other occupancy types, requirement of firefighting shaft should be ascertained in consultation with the local fire authority. The firefighting shaft should necessarily have connectivity directly to exit discharge or through exit passageway (having 120 min fire resistance walls) to exit discharge.

Staircase and fire lift lobby of a firefighting shaft should be smoke controlled as per [4.4.2.5](#) and [Table 5](#).

It is recommended that the pressurization requirement for staircase in firefighting shaft and for other fire exit staircases in buildings greater than 60 m in height be evaluated to limit the force required to operate the door assembly (in the direction of door opening) to not more than 133 N to set the door leaf in motion. The aspect of pressurization, door area/width and door closure should be planned in consideration to the above.

D-3 FIRE SAFETY REQUIREMENTS FOR LIFTS

The provisions as given in [7.1](#) to [7.2.4](#) under fire safety requirements of lifts in high rise buildings in Part D 'Building Services, Section 5 Installation of Lifts, Escalators, Moving Walks and Power-Driven Parking Systems, Subsection 5A Lifts' of this NBCS should be applicable.

D-4 HORIZONTAL EXITS AND REFUGE AREA

D-4.1 Horizontal Exit

A horizontal exit should be through a fire door of 120 min rating in a fire resistant wall. Horizontal exit require separation with the adjoining compartment through 120 min fire barrier. The adjoining compartment of the horizontal exit should allow unlocked and ease of egress and exits for the occupants using defend in place strategy.

Requirements of horizontal exits are as under:

- a) Width of horizontal exit doorway should be suitable to meet the occupant load factor for egress;
- b) Doors in horizontal exits should be openable at all times from both sides; and
- c) All doors should swing in the direction of exit travel. For horizontal exits, if a double leaf door is used, the right hand door leaf should swing in the direction of exit travel.

D-4.2 Refuge Space/Area

- a) Refuge spaces and refuge areas are provided for occupants staging requirements and also enabling assisted evacuation. The aspect of staging and refuge requirements should be planned in the building for people with disabilities and ailments, including specific assistance required for certain section of occupants based on their age and other challenges for self-evacuation.
- b) Refuge spaces should be achieved by adequate planning and consideration of space provided in the lift lobby of firefighting shaft, enabling the staging for such occupants and further assisted evacuation through fireman lift in the firefighting shaft. This should enable these occupants to be assisted evacuated by the lift and further to be brought to exit discharge through the well-planned evacuation strategy through exit passageway on the ground or level of exit discharge. Provision for such staging of the occupants should be minimum 12 m² or 5 percent of calculated occupants of the floor (served by that firefighting shaft) with a refuge area space of 0.45 m² per person in the firefighting shaft lift lobby, whichever is higher.
- c) Alternate to refuge spaces, refuge areas should be provided in buildings of height more than 24 m. Refuge area provided should be planned to accommodate the occupants of two consecutive floors (this should consider occupants of the floor where refuge is provided and occupants of floor above) by considering area of 0.3 m² per person for the calculated number of occupants and should include additionally to accommodate one wheelchair space of an area of 0.9 m² for every 200 occupants and portion thereof, based on the occupant load served by the area of refuge or a minimum of 15 m², whichever is higher, should be provided as under:

- 1) The refuge area should be provided on the periphery of the floor and open to air at least on one side protected with suitable railings.
- 2) Refuge area(s) should be provided at/or immediately above 24 m and thereafter at every 15 m or so.

The above refuge area requirement for mall-like (mix of assembly, mercantile, cinema screen and restaurant) occupancy requirement should however be in accordance with [6.4.2.2](#).

- 3) A prominent sign bearing the words 'REFUGE AREA' should be installed at the entry of the refuge area, having height of letters of minimum 75 mm, and also containing information about the location of refuge areas on the floors above and below this floor. The same signage should also be conspicuously located within the refuge area.
- 4) Each refuge area should be ventilated and provided with first aid box, fire extinguishers, public address speaker, fire man talk back, and adequate emergency lighting as well as drinking water facility.
- 5) Refuge areas should be approachable from the space they serve by an accessible means of egress.
- 6) Refuge areas should connect to firefighting shaft (comprising fireman's lift, lobby and staircase) without having the occupants requiring to return to the building spaces through which travel to the area of refuge occurred.
- 7) The refuge area should always be kept clear. No storage of combustible products and materials, electrical and mechanical equipment, etc should be allowed in such areas.
- 8) Refuge area should be provided with adequate drainage facility to maintain efficient storm water disposal.
- 9) Entire refuge area should be provided with sprinklers.
- 10) Where there is a difference in level between connected areas for horizontal exits, ramps of slope not steeper than 1 in 12 should be provided (and steps should be avoided).

NOTE — Refuge area provided in excess of the requirements should be counted towards FAR.

High rise apartment buildings with apartments having balcony, need not be provided with refuge area; however, apartment buildings without balcony should provide refuge spaces or refuge areas as given above. When refuge areas are preferred over refuge spaces in apartment buildings of height above 60 m (and having balconies), the same should be provided around 60 m level and thereafter at levels around every 30 m. The refuge area should be an area equivalent to 0.3 m² per person for accommodating occupants of two consecutive floors, where occupant load should be derived on basis of 12.5 m² of gross floor area and additionally 0.9 m² for accommodating wheelchair requirement or should be 15 m², whichever is higher.

D-5 ELECTRICAL SERVICES

The specific requirements for electrical installations in multi-storeyed buildings given in Part D 'Building Services, Section 2 Electrical and Allied Installations' of this NBCS and Section 7 of National Electrical Code of India 2023 should be followed.

Wherever transformers are planned at higher floors, the HT cables should be routed through a separate shaft having its own fire resistance rating of 120 min. Wherever HT generators are planned centrally at ground or first basement level, redundant transformers and HT cables should be planned for buildings above 60 m in height.

D-6 FIRE PROTECTION

D-6.1 For residential occupancies above 120 m in height and other occupancies above 60 m in height, the sprinklers should be fed from the main and an alternate/standby riser with suitable isolation valves. The entire sprinkler system should be designed in accordance with good practice [F(20)].

D-6.2 Where the height of the building exceeds 150 m to 175 m, fire water static storage and pumps should be required to be provided at 160 m to 180 m and thereafter at intermediate floors at higher levels enabling efficient and functional firefighting installations. The static fire water storage tanks located at such levels should have capacity at minimum half of the storage of underground static water storage tank as in. Such tanks should be supplemented with water supplies through one working and one standby pump of capacity 2 850 litre/min with two risers at alternate locations feeding to such fire water static storage tanks. The fire pump's requirement and capacity should also be derived for occupancy type as per [Table 7A](#) and [Table 7J](#) substituting the diesel pump with electrical pump. The fire pump room at such level should have dedicated connectivity through passageway (with 120 min integrity) from the firefighting shaft. Such fire pump room should have 120 min fire resisting wall and provided with adequate ventilation with talk-back connectivity to the main fire pump room and Fire Command Centre.

D-6.3 As an alternative to bottom-fed and standpipe and sprinkler that rely on pumps, in high rise buildings, water can be fed by gravity based tanks located at such levels along the height of the building designed to cater to the stories located 55 m to 60 m below. See [Fig. 20](#) and [Fig. 21](#) for guidance/details regarding sizing of pipes and achieving the desired pressure and capacity/volume {as suggested in [F(29)]} to cater respectively to the hydrants and the sprinklers. It is recommended to have the top-most tank having dual risers each connecting to different tank. A cross connection with isolation valve at the top and bottom of the dual risers is recommended.

Repeater tanks provided at higher stories can also be designed to fill the tank below, should the need arise. The top-most tank should have additional capacity (50 percent) to cater to the tank below. Overflow pipes can feed the tank below and designed accordingly. This system is particularly preferred in the case of single tower high rise building. See [Fig. 20](#) and [Fig. 21](#) for details.

D-6.4 For high rise buildings, seismic bracings should be considered for firefighting installations depending on seismic vulnerability of the region and the type of occupancy.



EQUIPMENT DATA:

PUMP	P-1	P-2	P-3
DESIGNATION	COMMON STANDBY PUMB	COMMON MAIN PUMP	COMMON JOCEKY PUMP
CAPACITY	2 850 lpm	2 850 lpm	180 lpm
HEAD	10 bar	10 bar	10 bar
MODE OF OPERATION	AUTO START and MANUAL START/STOP	AUTO START and MANUAL START/STOP	AUTO START/STOP
DS	DIESEL STAND BY		
SE	STAND BY ELECTRICAL PUMP		
ME	MAIN ELECTRICAL PUMP		
JE	JOCKEY ELECTRICAL		
B	BOOSTER PUMP		

PUMP	P-4	P-5	P-6
DESIGNATION	COMMON STANDBY PUMB	COMMON MAIN PUMP	COMMON JOCEKY PUMP
CAPACITY	2 850 lpm	2 850 lpm	180 lpm
HEAD	15 bar	15 bar	15 bar
MODE OF OPERATION	AUTO START and MANUAL START/STOP	AUTO START and MANUAL START/STOP	AUTO START/STOP
DS	DIESEL STAND BY		
SE	STAND BY ELECTRICAL PUMP		
ME	MAIN ELECTRICAL PUMP		
JE	JOCKEY ELECTRICAL		
B	BOOSTER PUMP		

PUMP	B
DESIGNATION	BOOSTER PUMP
CAPACITY	900 LPM
HEAD	4.5 BAR

FIG. 20 DETAILS OF EQUIPMENT FOR FIGURE 55 M TO 90 M ABOVE

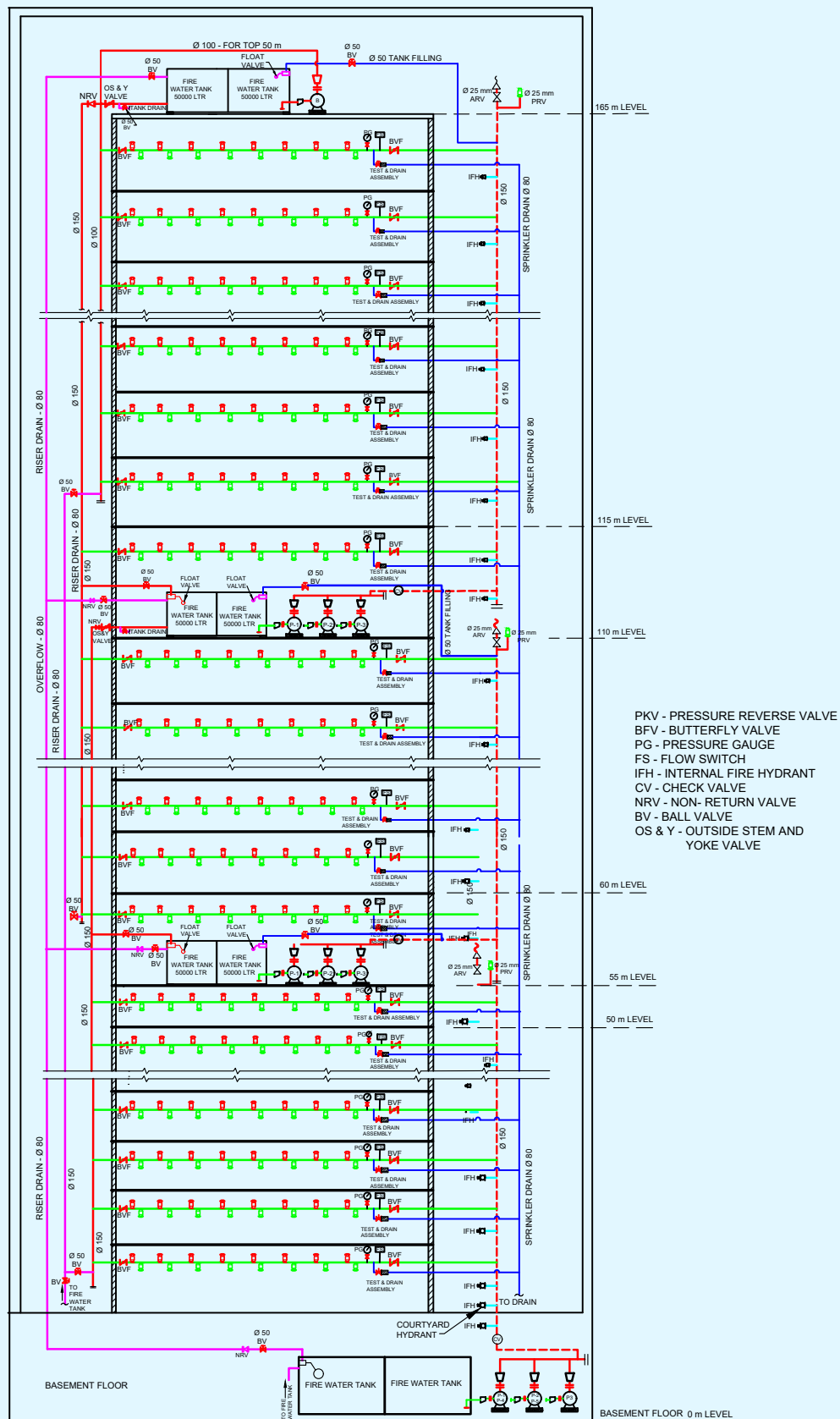


FIG. 21 SCHEMATIC OF GRAVITY BASED FEED FOR 90 M AND ABOVE HIGH RISES (CONTINUED)

EQUIPMENT DATA:

PUMP	P-1	P-2	P-3
DESIGNATION	COMMON STANDBY PUMB	COMMON MAIN PUMP	COMMON JOCEKY PUMP
CAPACITY	2 850 lpm	2 850 lpm	180 lpm
DUTY HEAD	10 bar	10 bar	10 bar
SHUT OF HEAD	12 bar	12 bar	12 bar
MODE OF OPERATION	AUTO START and MANUAL START/STOP	AUTO START and MANUAL START/STOP	AUTO START/STOP

PUMP	B
DESIGNATION	BOOSTER PUMP
CAPACITY	900 LPM
HEAD	4.5 BAR

FIG. 21 SCHEMATIC OF GRAVITY BASED FEED FOR 90 M AND ABOVE HIGH RISES

D-7 FIRE AND LIFE SAFETY AUDIT

D-7.1 The following should be applicable:

- Fire and life safety audit should be carried out for all high rise buildings;
- Such audits should preferably be conducted by a third party auditor having requisite experience in fire and life safety inspections;
- Frequency of such audits should be once in two years.

DI-7.2 Good practices towards fire and life safety audit should be as given below:

- Objective is to ensure that a building's fire prevention, fire protection and life safety measures are evaluated against established Indian Standards;
- Methodology includes: Audit planning, on-site audit execution, documentation and reporting;
- Audit objectives to cover: compliance verification, risk identification, performance evaluation, corrective recommendations, documentation and process review;
- Fundamental principles should be: objectivity, consistency, risk-based approach, comprehensive coverage, continuous improvement, transparency and documentation, and stakeholder involvement;
- In the audit methodology and process the following should be adhered: planning and preparation (Pre-audit data collection, Risk assessment and prioritization techniques, Audit team composition and qualifications), Audit execution (On-site inspection procedure, documentation review and interview, checklist and performance indicators), and Audit reporting (structure and content of the report, documentation of non-compliances and recommendations, methods for risk communication);
- Audit criteria for fire safety should be well established; covering respectively for fire prevention namely:
 - Building design and construction features
 - Structural fire resistance
 - Fire compartmentation and separation

- iii) Material selection and non-combustibility
- iv) Means of egress and access routes
- v) Integration of fire safety systems
- vi) Compliance with regulatory requirements
- vii) Documentation and verification
- 2) Maintenance of fire prevention systems and housekeeping practices
 - i) Regular maintenance and servicing
 - ii) Housekeeping practices
 - iii) Documentation and verification
 - iv) Training and accountability;
- g) Audit criteria for fire safety should be well established; covering respectively for fire protection systems namely:
 - 1) Assessment of fire detection and alarm systems
 - i) Design and installation compliance
 - ii) Function performance and testing
 - iii) Maintenance and calibration
 - iv) Integration with emergency systems
 - v) Documentation and compliance verification
 - 2) Inspection of fire suppression systems
 - i) Design and installation compliance
 - ii) Operational testing and performance
 - iii) Maintenance and inspection records
 - iv) Integration with fire safety systems
 - v) Physical condition and accessibility
 - vi) Compliance with regulatory and manufacturer specifications
 - 3) Evaluation of emergency lighting signage
 - i) Design and installation compliance
 - ii) Visibility and illumination levels
 - iii) Operational performance
 - iv) Maintenance and inspection
 - v) Integration with overall safety systems
 - vi) Documentation and compliance verification
 - 4) Evaluation of life safety components
 - i) Assessment of exit routes, corridors and stairways
 - Design and layout
 - Unobstructed passage
 - Dimensions and capacity
 - Signage and emergency lighting
 - Physical condition and maintenance

- Special provisions
- Documentation and verification
- ii) Verification of occupant load, exit capacities and travel distances
 - Occupant load verification
 - Assessment of exit capacities
 - Measurement of travel distances
 - Integration and documentation
 - Assessment of portable and wheeled fire extinguisher;
- h) Audit criteria for life safety should be well established to cover:
 - 1) Structural and environment safety
 - i) Evaluation of structural fire resistance and compartmentation
 - ii) Assessment of emergency communication systems and evacuation aids
 - 2) Accessibility and special needs
 - i) Consideration for persons with disabilities
 - ii) Audit of assisted evacuation procedures;
- j) Performance indicators and compliance assessment – To address:
 - 1) Scoring and rating methodologies for audit findings
 - 2) Risk evaluation and prioritization of deficiencies
 - 3) Criteria for Compliance/Partial Compliance/Non-Compliance;
- k) Corrective actions and follow-up are essential so as to provide opportunity to the building owner/resident to address the shortfall so as to make the premises safe; and
- m) Reporting and record keeping – To have standard formats/templates to record observations, measurements including photographs stored in compatible formats, all with a view to enable suitable reviewing for consistency.

D-8 HELIPAD

For high rise buildings above 200 m in height, provision for helipad is recommended for specific requirements like landing of fire equipment, and support facilities or other emergencies.

Suitable fire protection is required for the helipad areas to enable their safe usage at all times. The firefighting facilities for rooftop helipad includes fire detection, fixed or portable firefighting foam system (oscillating foam monitor, deck integrated foam nozzles or portable foam units), fire hydrants and trained response team.

Low expansion aqueous film forming foam (AFFF) can be applied to the helideck from either fixed nozzles or an oscillating nozzle, particularly on the landing pad, at the rate of 10 lpm per square foot for a duration of 10 min.

Without the requirement for any additional water, buildings having helipad should use the water available for wet riser, sprinkler system, etc for foam based fire protection. This can be achieved by a suitable tee connection in the standpipe by attaching a 100 mm or 150 mm deluge valve (to control the flow of water) and extending a 100 mm or 150 mm water line from that valve, to the foam producing equipment located near (outside) the helipad area.

ANNEX E

(Clauses [4.5.3.4](#), [6](#) and [Table 6](#))

ATRIUM

E-1 ATRIUM REQUIREMENTS

- a) In order for an atrium to be permitted in buildings, the following should be complied with:
 - 1) Atrium should be permitted in buildings of Type 1 and Type 2 construction only.
 - 2) The use of combustibles furnishings and decorations on the floor of the atrium should be limited and sparsely distributed.
- b) Smoke detectors (preferably beam type or aspirating type) should be provided on the underside of each floor protruding into the atrium, at the atrium roof and adjacent to each return air intake from the atrium. Within atrium space, beam type or aspirating type smoke detectors should be used to ensure detection of smoke, considering factors such as stratification of smoke.
- c) Where the ceiling of the atrium is more than 17 m above the floor, water based protection (automatic sprinklers) at the ceiling of atrium is not required. In such case, travel distance allowance should be as applicable for sprinkler spaces.
- d) Hydrants should be available at the floor of the atrium and also at the adjoining upper spaces/floors of the atrium. Sprinklers are required to be installed for coverage of glass areas of retail, tenant and other areas adjoining the exit access corridor and atrium. Sprinklers should be at a distance of 450 mm to 600 mm enabling cooling of such glass and limiting the extent of fire and smoke to the atrium (*see Fig. 22*). This provision does not allow similar sprinkler installation arrangement to offset fire compartmentation requirements, in which case fire barrier is required as per relevant provisions of this Part.
- e) Atrium in business occupancy should be planned with 6 air changes per hour (ACPH) while atrium in hotels and assembly occupancy should be planned with 8 ACPH smoke extraction system.
- f) Such air changes should be planned in atrium for a height of 15 m from the top.
- g) Smoke exhaust fans should be capable of operating effectively at 250 °C for 120 min.
- h) Makeup air supply points should be located beneath the smoke layer and on the lower levels connected by the atrium.
- j) Makeup air should be provided by fans, openings to outside to allow infiltration, or the combination thereof.
- k) It is recommended that makeup air be designed at 85 percent to 95 percent of the exhaust flow rate, not including the leakage through these small paths.
- m) The makeup air should not cause door-opening force to exceed allowable limits.
- n) The makeup air velocity should not exceed 1.02 m/s is where the makeup air could come into contact with the plume unless a higher makeup air velocity is supported by engineering analysis;
- p) Atrium smoke management system fans should be provided with emergency power.
- q) If so, required by the Authority, an engineering analysis should be performed which demonstrates that the smoke system for the atrium is designed to keep the smoke layer interface 1 800 mm above the highest occupied floor level of exit access, open to the atrium, for a period equal to 1.5 times the calculated egress time or 20 min, whichever is greater.

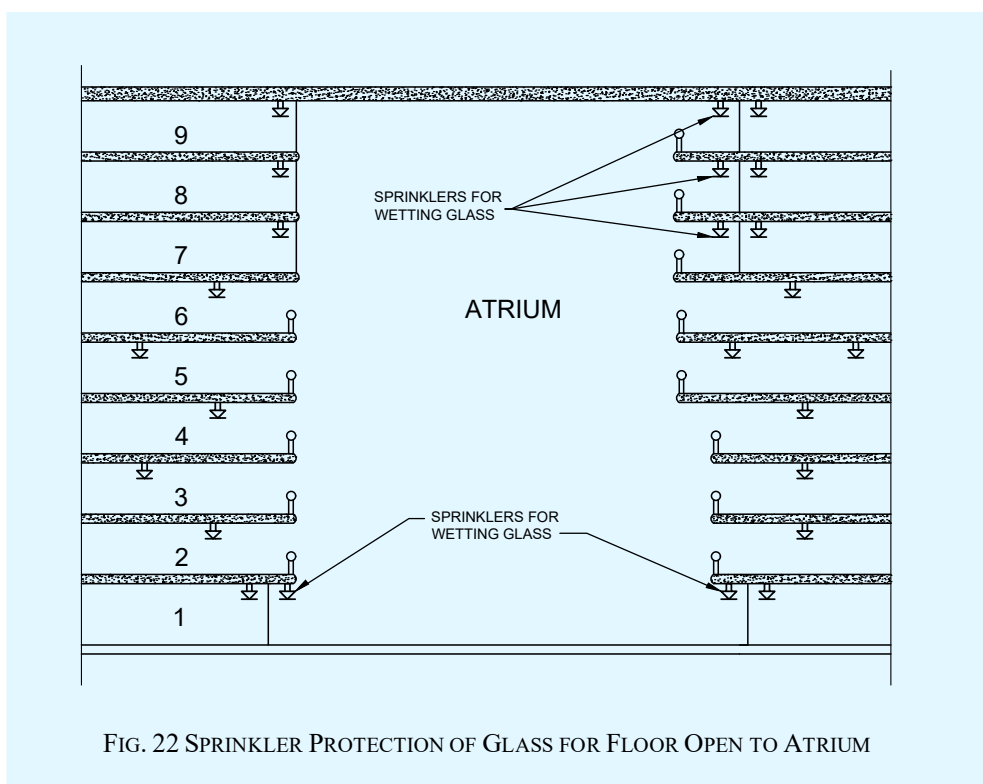


FIG. 22 SPRINKLER PROTECTION OF GLASS FOR FLOOR OPEN TO ATRIUM

ANNEX F

(Clause 6)

REQUIREMENTS FOR DATA CENTRES

F-1 GENERAL

Data centres are classified under Group E Business occupancy and in particular as E-II sub-occupancy considering its primary use, scale, and presence of process equipment. A comprehensive fire risk assessment should be conducted at the planning stage and periodically during operation. Assessments must include ignition sources, fire load, electrical risks, HVAC systems, etc. For guidance on the facilities and infrastructure of datacentres, refer to the following good practices.

IS/ISO/IEC 22237-1 : 2021	Information technology – Data centre facilities and infrastructures: Part 1 General concepts
IS/ISO/IEC 22237-3 : 2021	Information technology – Data centre facilities and infrastructures: Part 3 Power distribution
IS/ISO/IEC 22237-4 : 2021	Information technology – Data centre facilities and infrastructures: Part 4 Environmental control

F-2 BUILDING DESIGN AND CONSTRUCTION

F-2.1 The fire resistance rating for the structural and non-structural elements should be based on guidelines as per approved and accepted standards. The fire rating should be validated and certified with a view to meeting the requirements of [Table 1](#). In the absence of any validated/certified rating, guidance may be obtained from the information available in [Annex C](#).

F-2.2 An anti-static floor system is highly recommended to prevent electrostatic discharge and support cable management. The data centre must be completely isolated from dust, water, vibration, and humidity. All external openings should be sealed, and no water sources or pipelines should be routed through or above the data centre,

UPS room, or communications room to eliminate risks of leakage or moisture intrusion. However, any location of chillers at higher heights be opted considering the operational efficiency and the same should be adequately considered in the structural design; Authorities may consider exempting the increased building height in such a case. Considering the sensitivity and fire hazards, such floors should have no other occupancy or else this should be considered in the calculation of floor area ratio and height.

F-2.3 Server rooms, battery banks, fuel storage, and other high-risk areas should be physically separated using fire-rated assemblies. Cable trays and ducts openings for cable trays and ducts should be sealed with materials having fire resistance rating of the compartment.

F-2.4 The number of basements should be kept as minimum and it should not be more than 2. The basement(s) should be within the building line and their total area should not exceed the area of ground floor.

F-2.5 The parking requirement should be arrived at considering the requirement for administrative building (at 1 ECS per 100 m²). The parking requirement for the data centre/utility building should be based on the actual required number and area (of trucks) for which additional considerations of loading and unloading spaces for services, including rack placement/replacement as advised by the OEMs, DG set, utilities, etc should be made. In addition, consideration for the fire truck and its movement should be ensured in consultation with the local Fire Authority.

F-2.6 Lifts, when planned, consider respectively for administrative building as passenger lift as per Part D/ Section 5A of this NBCS should, and for the datacentre building as special goods lifts and special service lifts, for which OEMs should be consulted considering the size, speed and weight of payload; and any required special requirement.

F-2.7 Heights of stories should be arrived at based on the size of racks opted and the associated services (like cooling/ducting) and the same is effectively addressed in the structural design. The total height of the datacentre building should be arrived based on the FAR and only Type 1 construction. Ground-coverage should ideally not exceed 60 percent considering the peculiar requirements and the essential nature of these buildings which are required in the society in the current times, and the associated moment of heavy vehicles to install/maintain/repair the racks/occupant.

NOTE — Authorities to consider also the possible change of occupancy from data centre to other permitted business as well; and the ground-coverage.

F-3 FIRE DETECTION SYSTEMS

F-3.1 Use of aspirating smoke detection system in server areas is recommended for very early detection.

F-3.2 The system should incorporate heat sensitivity and high-sensitivity smoke detectors (HSSD) to enable early fire detection and prompt activation of suppression mechanisms.

F-3.3 For other requirements on fire detection and alarm system [4.9](#) be referred.

F-4 FIRE SUPPRESSION SYSTEMS

F-4.1 The fire suppression system in data centres should be designed to ensure both operational continuity and protection of sensitive equipment. The preferred suppression technologies and clean agent extinguishers (such as heptafluoro propane, fluorinated ketone based system or equivalent) should be available throughout data hall areas. These systems should be effective in suppressing fires without leaving any residue, making them safe for electronic and electrical installations.

F-4.2 All critical areas within the data centre, including server rooms, network rooms, UPS and battery areas, and under-floor and overhead spaces, must be covered comprehensively by the fire suppression system.

F-4.3 The suppression agents should be ozone-friendly and compliant with international environmental norms. The recharge process must be easy, with minimal downtime, and the equipment should be easy to test periodically.

F-4.4 An early warning fire detection and suppression system should be integrated with the fire alarm and building management system (BMS) for real-time monitoring and response. The system should be engineered to offer reliable and quick suppression to minimize damage and downtime.

F-5 ELECTRICAL AND HVAC FIRE SAFETY (*See also* Part D/Sec 2 and Part D/Sec 3 of this NBCS)

F-5.1 UPS, transformer, and generator rooms should be separated with fire-rated partitions/doors and have independent suppression systems. Should the need arise, the generators can be placed one over the other/stacked (suggested upto 5 numbers) parallel to the data centre building, but adequate fire safety should be ensured for which OEM's advice and specialist literature should be referred.

F-5.2 Wherever batteries are provided, the same should be segregated by 120 min fire rated construction. Ventilation to the room should be provided as per manufacturer's instructions.

F-5.3 Air conditioning and ventilating systems should be so installed and maintained as to minimize the danger of spread of fire, smoke or fumes from one floor to other or from outside to any occupied building or structure.

F-5.4 Supply air ducts, fresh air and return air ducts/passages must include fire dampers and smoke detectors (*see* [3.5.7.4](#)).

F-5.5 Insufficient power capacity design for a data centre can result in various issues such as system downtime, data loss, increased cost, reduced performance, regulatory non-compliance, etc. Ensuring a robust power capacity design is crucial for the reliable and efficient operation of data centres.

F-6 MEANS OF EGRESS

F-6.1 Exit Access

Exits should be provided on each floor or compartment in accordance with [4.4.2](#). All exits should be unobstructed, illuminated, and clearly marked.

F-6.2 Staircases and Corridors

Staircases must be enclosed, fire-rated, and positively pressurized to prevent smoke ingress. Staircases should be of minimum 1.5 m in width, and 2 numbers of it will be required in the datacentre building. Staircase for administrative building should be arrived at using [4](#); in any case, minimum 2 staircases are required.

F-6.3 Emergency Lighting and Signage

Emergency lighting consistent with emergency system for the building should be provided. Addressable emergency exit lighting system is recommended, adaptive type exit signages can also be used.

F-7 MEP, FIREFIGHTING AND SEISMIC SUPPORT REQUIREMENT

F-7.1 Earthquakes can cause downtime, data loss, service disruptions, fire, etc. Seismic bracing is essential to secure MEP equipment and systems, safeguarding them against the threats posed by earthquakes.

F-7.2 Non-structural measures focus on protecting operational components. These include seismic-rated mounting systems for equipment racks, secure cable pathways, MEP and firefighting services to avoid disconnections, and reinforced architectural features like ceilings and raised floors. These elements must meet seismic standards to ensure systems stay operational during and after an earthquake.

F-7.3 Seismic analysis/calculations should be carried out based on Part C 'Structural Design, Section 1 Loads and Load Effects' of this NBCS and good practice [F(32)]. Each non-structural component's seismic interactions with all other connected components and with the supporting structure should be accounted for in the design. The component should accommodate drifts, deflections, and relative displacements determined in accordance with the applicable seismic requirements of standards.

F-7.4 Each straight pipe/duct/cable containment run with two or more supports requires a minimum of two transverse braces (perpendicular to the run) and one longitudinal brace (parallel to the run).

F-7.5 Wire rope angular bracing should be seismic certified/tested by third party accredited lab as per method of testing for rating seismic and wind restraints. Rigid angular bracing and modular support system should be

analyzed for both tensile and compressive loads for strength and serviceability for the maximum length of element being used in worst case condition as per load combination as mentioned in standards.

F-7.6 Calculate the static and dynamic loading due to wind forces that is required to select/design vibration isolators, bases and seismic and wind restraints for outdoor and roof top equipment's/services. The calculation of wind load should be as per Part C 'Structural Design, Section 1 Loads and Load Effects' of thi NBCS. Worst case between seismic loads and wind loads should be considered for supporting and vibration isolation.

F-8 SECURITY

F-8.1 It is recommended that all doors should be access controlled or should be as per best suitable secure design requirement.

F-8.2 Doors should send an alarm if left open.

F-8.3 In line with safety requirements, access should be open for the fire affected area.

F-9 OPTIONAL PERFORMANCE INDICATORS

For guidance on the performance indicators of datacentres, the following good practices may be referred.

IS/ISO/IEC 30134-1 : 2016	Information technology — Data centres — Key performance indicators: Part 1 Overview and general requirements
IS/ISO/IEC 30134-2 : 2016	Information technology — Data centres — Key performance indicators: Part 2 Power usage effectiveness (PUE)
IS/ISO/IEC 30134-3 : 2016	Information technology — Data centres — Key performance indicators: Part 3 Renewable energy factor (REF)
IS/ISO/IEC 30134-4 : 2017	Information technology — Data centres — Key performance indicators: Part 4 IT equipment energy efficiency for servers (ITEEsv)
IS/ISO/IEC 30134-5 : 2017	Information technology — Data centres — Key performance indicators: Part 5 IT equipment utilization for servers (ITEUsv)
IS/ISO/IEC 30134-6 : 2021	Information technology — Data centres — Key performance indicators: Part 6 Energy reuse factor (ERF)
IS/ISO/IEC 30134-8 : 2022	Information technology — Data centres — Key performance indicators: Part 8 Carbon usage effectiveness (CUE)
IS/ISO/IEC 30134-9 : 2022	Information technology — Data centres — Key performance indicators: Part 9 Water usage effectiveness (WUE)

ANNEX G

[Clauses [6.4.2.2\(e\)](#) and [6.8.1.5](#)]

CAR PARKING FACILITIES

G-1 The provisions given in [G-2](#) to [G-5](#) should apply to parking structures of the closed or open type, within buildings above or below grade.

G-2 GENERAL

- Where both parking and repair operations are conducted in the same building, the entire building should comply with the requirements for Group G occupancies, unless the parking and repair sections are effectively separated by separation walls of 120 min fire rating.
- Floor surfaces should be non-combustible, sloping towards drains to remove accumulation of water.
- Those parts of parking structures located within, immediately above or below, attached to, or less than

3 m away from a building used for any other purpose should be separated by fire resistant walls and floors having fire resistance rating of not less than 120 min. This should exclude those incidental spaces which are occupied by cashier, attendant booth or those spaces used for toilets, with a total area not exceeding 200 m².

- d) Vehicle ramps should not be considered as exits unless pedestrian facilities are provided.
- e) Other occupancies like fuel dispensing, should not be allowed in the building. Car repair facilities, if provided, should be separated by 120 min fire resistant construction.
- f) In addition to fire protection requirements as per [Table 7A](#) to [Table 7J](#), appropriate fire detection and suppressions systems should be provided for the protection of hydraulic oil tank and pumps located below ground level for operation of car lifts.
- g) Means of egress should meet the requirements specified in [4](#).

G-3 OPEN PARKING STRUCTURES (INCLUDING MULTI-LEVEL PARKING AND STILT PARKING)

- a) The term open parking structure specifies the degree to which the structure's exterior walls must have openings. Parking structures that meet the definition of the term open parking structure provide sufficient area in exterior walls to vent the products of combustion to a greater degree than an enclosed parking structure.
- b) A parking structure having each parking level wall openings open to the atmosphere, for an area of not less than 0.4 m² for each linear metre of its exterior perimeter should be construed as open parking structure. Such openings should be distributed over 40 percent of the building perimeter or uniformly over two opposing sides. Interior wall lines should be at least 20 percent open, with openings distributed to provide ventilation, else, the structure should be deemed as enclosed parking structures.

NOTE — A car park located at the stilt level of a building (not open to sky) can be considered an open or an unenclosed car park if any part of the car park is within 30 m of a permanent natural ventilation opening and any one of the following is complied with towards the permanent natural ventilation requirement:

- 1) 50 percent of the car park perimeter should be open to permanent natural ventilation.
- 2) At least 75 percent of the car park perimeter is having the 50 percent natural ventilation opening.
- c) All stilt parking is required to be provided with sprinkler system where such buildings are required to be sprinklered.
- d) Open parking structures are not required to be provided with compartmentation.
- e) Open car parking (open to sky) within building complex having fire hydrant system should also need to be protected with yard hydrant installation system in accordance with good practice [F(29)].

G-4 ENCLOSED PARKING STRUCTURES

- a) Those car parking structures which are enclosed on all sides and on top, not falling within the definition of open car parking [*see* G-3(b)] and also those situated in the basements should be known as enclosed car parking structures.
- b) All sprinklers in car parking should be standard response type with minimum K-factor of 80, area coverage of 9 m² and designed as per good practice [F(20)].
- c) For smoke ventilation requirement of car parking, *see* [4.6](#).
- d) All fire exit doors from the car parking to exits should be painted green and should display exit signage.

G-5 AUTOMATED CAR PARKING UTILIZING MECHANICAL OR COMPUTERIZED/ ROBOTIC MEANS

- a) Automated car parking structure can be of open parking type or enclosed types.
- b) Automated car parking facilities pose more hazard compared to manual parking due to following reasons:
 - 1) High density of cars due to close stacking one over another.
 - 2) Lack of provision on fire separation/compartmentation horizontal or vertical leading to rapid fire spread.

- 3) Non-availability of any person to notice/control the fire in initial stages.
 - 4) Limited access to firefighting personnel.
 - 5) Extensive height and depth involved with highly combustible load.
- c) All basements should deploy sprinkler system with not less than K 115 quick response sprinklers, with a minimum of 2 sprinkler for each vehicle with not more than 7.2 m² coverage per sprinkler. The area of operation for basement sprinkler should not be less than 230 m². This should be applicable to both internal combustion engines (ICE) and electric battery operated or hybrid vehicles. All the sprinklers installed should be protected with a sprinkler guard.
 - d) All basements that deploy stack parking with one car above the other, where 2 sprinklers cannot be installed for the car below, a quick response extended coverage horizontal side wall of K 115 should be deployed. The pressure rating for this sprinkler should be based on original equipment manufacturers (OEM) recommendation based on the length and width covered. This sprinkler coverage should be within the OEM listing and approvals. Electric or hybrid car should not be parked on stack parking at any level. All the sprinklers installed should be protected with a sprinkler guard.
 - e) All buildings with automated puzzle parking without human interface should deploy sprinklers as applicable for normal parking. However, the riser should be limited to 24 m height, and additional risers with isolation valves should be added for each riser of 15 m thereon. This is to facilitate maintenance and isolation in case of false activation of sprinklers. All the sprinklers installed should be protected with a sprinkler guard.
 - f) Fire escape staircases, at least 1 250 mm wide should be provided at appropriate locations so that no place is more than 45 m from the nearest staircase. Horizontal walkways, at least 1 000 mm wide for access to all the areas should be provided at every parking level.
 - g) Travel distance and means of egress should be governed by the respective sections of this NBCS.
 - h) The hazardous areas like DG sets, transformers, HT/LT panels for the parking lot should be suitably segregated from other areas as per requirements given in this NBCS and all such areas should be protected by suitable automatic fire suppressions systems.

G-6 REQUIREMENTS FOR ELECTRIC VEHICLES (EV) CAR PARKING

See also 10.7 of Part D 'Building Services, Section 2 Electrical and Allied Installation' of this NBCS, including regarding electrical safety and location.

EV car parking are best planned in outdoors in well ventilated condition or in separate parking building at/above ground level having naturally cross-ventilated sides. Wherever EVs are designated in the regular car parking they should be segregated from other vehicles using both passive means and active measures of fire protection. The preferred locations are in the ground level and in no case below the first basement. Charging points for EVs may not be permitted below the ground level; and when permitted the same should be restricted to first basement. To enable immediate attention, the EV charging points for the EVs should be available at the entrance/first opportunity of the basement. Isolation cut-off switches (and fireman's emergency switch) should preferably be of the automated type or when manually operated should be located at the ground level also.

Active protection measure includes one or more of: water mist sprinkler system, and smart detection/monitoring system that enables early detection of thermal runaway issues that can flare up the fire should it happen. Fire suppressing materials using vermiculite/clay/mica based encapsulating agents that provide satisfactory performance may also be used.

The passive protection measures include fire rated walls (preferred over fire curtains/smoke curtains), fire doors, etc and additional protection measures to protect the structural elements from possible rising heat due to fire/hot gases. In the case of basements, maximum sizes of such a compartment for the EV parking (and charging) stations should be 1 000 m².

NOTE — The above provisions may be modified based on the availability of more suitable/better extinguishing media that proves effective against such fires.

ANNEX H

(Clause [6.4.4](#))

FIRE AND LIFE SAFETY REQUIREMENTS FOR METRO STATIONS

H-1 APPLICATION AND SCOPE

The provisions of this Annex relate to the buildings constructed as part of the metro stations/metro rail systems.

H-2 TERMINOLOGY — See [2.46](#).

H-3 GENERAL

H-3.1 Classification

Metro stations can be classified under ‘Assembly occupancies’. These include any station building or part thereof, permanent or temporary, through which people transit for the duration of time required to enter the building and board the train to depart the station platform or to alight from the train and depart from the station building.

H-3.2 Sub-classification

Metro rail system buildings can be further classified under the following headings:

- a) Elevated and enclosed stations,
- b) Elevated and open stations,
- c) Underground (enclosed) stations, and
- d) Train depots.

NOTE – The above includes all open and enclosed metro stations along with their associated ancillary structures and train depots. Elevated should mean to include ‘at-grade’ stations as well.

H-3.3 Fire Zones

Metro stations, that is, mass rapid transit, may be constructed in any Fire Zone No. 1, 2 or 3, as per the transportation requirement/policy.

H-3.4 Type of Construction

Metro stations should conform to Type 1 or Type 2 or combinations of Types 1 and 2 non-combustible constructions, as defined in [3.4](#).

H-4 LIFE SAFETY REQUIREMENTS

H-4.1 Occupant Load

- a) Main occupant load in a transit station is the platform occupant load, on which basis the life safety provisions of transit stations are designed. Occupant loads in transit stations are mainly a function of the train carrying capacities rather than the areas of a station.
- b) For calculating platform occupant loads for a multiline, multilevel or multiplatform station, maximum occupant load for each platform should be considered separately for determining the egress capacity from that platform.
- c) At levels where egress routes from separate platforms converge, occupant loads of all platforms should be considered to calculate egress capacity from that level. Simultaneous loads should be considered for all egress routes passing through each level of that station.

- d) Platform occupant load should be based on the greater of the AM or PM ‘peak hour loads’ generated by the system and train loads.
- e) Peak hour load should be converted to peak minute loads by dividing by 60 and multiplying further by a system surge factor varying from 1.3 to 1.5, as defined by the transit system authority. Surge factor may require to be further enhanced where increased footfalls are anticipated, like stations catering to sports complex, transit oriented development, and stations serving two or more lines.

NOTE — Surge factors should be based on actual calculations by transit system authority and in no case it should be less than 1.3.

- f) Platform occupant load should be determined by factoring in the peak hour entraining loads at platform edges and train loads, in emergency evacuation scenario, as defined here:
 - 1) *Train on Fire* – It is assumed that a train loaded to crush capacity/link load as may be decided by the transit system authority, travelling in peak direction at peak hour has caught fire, and as per the operating procedure it is brought to the next station, whereupon it should require to be evacuated.
 - 2) The train in this case being called the ‘incident train’ and the platform on which it is arriving being called ‘incident platform’ of the ‘incident station’.
 - 3) Passengers waiting on all the platforms of the ‘incident station’ should require to be evacuated as well.
 - 4) Number of passengers should be determined as defined in (g) below.
 - 5) Only one source of fire, that is, train fire should be assumed at a time.
 - 6) It is assumed that the train on non-incident platform should not stop at the incident station and should stop only at previous/next station.
- g) Platform occupant load for emergency evacuation scenario, should be the addition of two headway entraining loads for peak direction platform, one headway entraining load for off-peak direction platforms and one crush train load/link load.

NOTE — See also [H-4.1\(b\)](#) and [H-4.1\(c\)](#).

- h) *Occupancy in Non-public Areas:*
 - 1) Use of designated non-public areas by station staff whose work assignments require their presence in the station structures should be permitted.
 - 2) Unmanned plant rooms and any other areas where usual occupancy is less than 10 persons at any given time should be treated as unoccupied areas.
- j) *Segregation with Non-transit Occupancy:*

Non-transit (and incidental) occupancies, of area up to 300 m² (total area of all floors), should be permitted to be accommodated within the station building. Occupant load of areas under non-transit occupancies should be in accordance with [Table 2](#).

H-5 EGRESS PROVISIONS

H-5.1 Evacuation Time

- a) *Means of Egress–public Areas* — Egress from metro station should be designed to facilitate the evacuation of a pre-defined platform occupant load to a designated point of safety in pre-defined emergency scenario(s), as defined by the transit system authority.
- b) *Evacuation Time from Platform* — Enough egress capacity should be provided to evacuate the platform occupant load within 4 min from enclosed stations and within 5.5 min from open stations.

- c) *Evacuation Time to Point of Safety* — Further, station design should permit evacuation of the remotest person on platform to a point of safety within 6 min in case of enclosed stations and within 8 min in case of open stations.

NOTE – Evacuation time specified in (b) and (c) may be modified based on actual engineering analysis by evaluating material heat release rates, station geometry and emergency ventilation systems.

- d) For open stations where the concourse is below or protected from the platform by distance or materials as determined by an appropriate engineering analysis, that concourse should be permitted to be defined as a point of safety.
- e) For enclosed stations equipped with an emergency ventilation system and where the emergency ventilation system provides protection for the concourse from exposure to the effects of a train fire at the platform as confirmed by engineering analysis, that concourse is permitted to be defined as a point of safety.

H-5.2 General Arrangement

- a) Means of egress from each station platform should be provided so that they are equally distributed as far as possible, in capacity and placement, throughout the length of platform.
- b) There should be at least two means of egress remote from each other.
- c) Means of egress may however, be allowed to converge at concourse or subsequent levels with sufficient capacity to achieve the required evacuation time.
- d) As far as possible, all the egress capacity required for emergency evacuation should be provided with the unenclosed stairs and escalators which are used for normal circulation.

H-5.3 Travel Distance

The maximum travel distance on the platform to a point at which a means of egress route leaves the platform should not be more than 100 m. This should, however be subject to fulfilling the egress requirements specified in [H-5.2](#).

H-5.4 Means of Egress

H-5.4.1 Non-public Areas

Means of egress from unoccupied non-public areas should be permitted to be merged into public means of egress.

H-5.4.2 Non-transit Occupancies

Separate means of egress should be provided for such non-transit occupancies which are segregated from the station's public area to ensure independent evacuation from either occupancy. Such means of egress may be allowed to converge beyond station concourse area with the approval of the competent authority. Travel distance and other egress provisions of such non-transit occupancies should be in accordance with [4.4](#).

H-5.4.3 Platform, Corridors, Ramps

Platforms, corridors and ramps serving as means of egress should be designed as per the following criteria:

- a) A minimum clear width of 1 200 mm should be provided along all platforms, corridors, and ramps serving as means of egress
- b) In computing the means of egress capacity available on platforms, corridors, and ramps, 300 mm should be deducted at each side wall and 450 mm should be deducted at platform edges that are open to the trainway.
- c) The maximum means of egress capacity of platforms, corridors, and ramps should be computed at 0.0820 people/mm-min.

- d) The maximum means of egress travel speed along platforms, corridors, and ramps should be computed at 38.0 m/min.
- e) The means of egress travel speed for concourses and other areas where a lesser pedestrian density is anticipated should be computed at 60.0 m/min.

H-5.4.4 Stairways

- a) Unenclosed stairs and escalators should be permitted to be counted as contributing to the means of egress capacity in stations.
- b) Minimum widths of such unenclosed staircases should be 1 800 mm for unidirectional stairs, and 2 400 mm for bidirectional stairs.
- c) Wall or floor-mounted railings on both sides should be permitted on stairs without affecting the widths stated in (b) above by more than 300 mm.
- d) Enclosed stairs in the means of egress should be minimum 1 200 mm wide.
- e) Capacity and travel speed for stairs should be computed at 0.0555 people/mm-min and 14.6 m/min, respectively.

H-5.4.5 Escalators

- a) Escalators should not account for more than one-half of the means of egress capacity at any one level.
- b) In calculating the egress capacity of escalators,
 - 1) one escalator at each level should be considered as being out of service, and
 - 2) the escalator chosen should be the one having the most adverse effect upon egress capacity.
- c) Where escalators are permitted as a means of egress in stations, the following criteria should be complied with:
 - 1) The escalators should be constructed of non-combustible materials (exceptions like rollers and handrails should be permitted with intimation to the Competent Authority).
 - 2) Escalators running in the direction of egress should be permitted to remain operating.
 - 3) Escalators running in reverse to the direction of egress should be capable of being stopped either locally by a manual stopping device at the escalator or remotely by a manual stopping device at a remote location or remotely as part of a pre-planned evacuation response. A stopped escalator should be counted as equivalent to 1 m wide staircase.

- d) Where provision is made for remote stopping of escalators counted as means of egress, one of the following should apply:

The stop should be delayed until it is preceded by a minimum 15 s audible signal or warning message sounded at the escalator; wherein (1) the signal or message should have a sound intensity that is at least 15 dBA above the average ambient sound level for the entire length of the escalator, (2) the signal should be distinct from the fire alarm signal, and (3) the warning message should meet audibility and intelligibility requirements.

OR

Where escalators are equipped with the necessary controls to decelerate in a controlled manner under the full rated load, the stop should be delayed for at least 5 s before beginning deceleration, and the deceleration rate should be no greater than 0.052 m/s².

- e) Escalators with or without intermediate landings should be acceptable as a means of egress, regardless of vertical rise.
- f) Escalators exposed to the outdoor environment should be provided with slip-resistant landing and floor plates.
- g) Stopped escalators should be permitted to be started in the direction of egress in accordance with the requirements for stopping of escalators described in [H-5.4.5\(c\)\(3\)](#) and [H-5.4.5\(d\)](#).

- h) Capacity and travel speed for escalators should be computed at 120 people/min and 18.5 m/min (vertical component of travel speed), respectively.

H-5.4.6 Lifts

Lifts meeting the following requirements should be counted as one of the means of egress in stations.

H-5.4.6.1 Capacity of lifts

Where lifts are counted as contributing to the means of egress capacity;

- a) lifts should have a minimum of 60 min fire resistance rating;
- b) they should account for no more than 50 percent of the required egress capacity;
- c) at least one lift should be considered out of service, and one lift should be reserved for fire service; and
- d) the capacity of each lift should be the carrying capacity of the lifts within 30 min.

H-5.4.6.2 Holding area for lifts

Lifts counted as one of the means of egress from any level of a station should be accessed through holding areas or lobbies at that level, which should be designed as follows:

- a) The holding areas or lobbies should be separated from the platform by a smoke-tight fire separation having a fire resistance rating of at least 60 min but not less than the time required to evacuate the holding area occupant load.
- b) At least one stair should be accessible from the holding area.
- c) The holding area should be sized to accommodate one person per 0.2 m².
- d) If the holding area includes portions of the platform, the area within 600 mm of the trainway should not be considered in the calculation;
- e) Upon activation of smoke control in the platform or adjacent trainway areas, the holding area should be pressurized to a minimum of 25 Pa.
- f) The holding area should be provided with emergency voice alarm devices with two-way communication to the system operations control centre.

H-5.4.6.3 Design features of lifts

Lifts counted as one of the means of egress should be designed as follows:

- a) Shaft enclosures should be constructed as fire separations having a 120 min fire resistance rating.
- b) The design should limit water flow into the shaft.
- c) No more than two lifts used for means of egress or fire department access should share the same machine room or lift pit.
- d) Machine rooms should be separated from each other by fire separation having a minimum fire resistance rating of 120 min.
- e) The lifts should be connected to back-up power supply.
- f) During emergency evacuation, the lifts should travel only between the incident level and a point of safety.

H-5.4.7 Doors and Gates

The egress capacity for doors and gates in a means of egress serving public areas should be computed as,

- a) 60 people per minute (p/min) for single leaf doors and gates;

- b) 0.082 people/mm-min for bi-parting multi-leaf doors and gates measured for the clear width dimension; and
- c) gates in a means of egress should be designed in accordance with the requirements for doors serving as a means of egress.

H-5.4.8 Fare Barriers

- a) Fare barriers complying with the following provisions should be permitted in the means of egress serving the stations. Such barriers should be designed to release, permitting unimpeded travel in the direction of egress under all the following conditions:
 - 1) Power failure or ground fault condition;
 - 2) Activation of the station fire alarm signal; and
 - 3) Manual activation from a switch in a constantly attended location in the station or operations control centre.
- b) Fare barriers that do not comply with the requirements of [H-5.4.8\(a\)](#) should be permitted in the means of egress where barriers in the equipment are designed to provide egress when a horizontal force not exceeding 66 N is applied in the egress direction.
- c) Gate-type fare barriers in the means of egress should meet the following criteria:
 - 1) Each unit should provide a minimum of 450 mm clear width at and below a height of 1 000 mm and 530 mm clear width above that height.
 - 2) Each unit should be credited with a capacity of 50 p/min for egress calculations.
 - 3) Fare barriers should be designed so that their failure to operate properly will not prohibit the movement of passengers in the direction of emergency egress.

H-5.4.9 Horizontal Exits

Horizontal exits should be permitted for up to 100 percent of the number of horizontal exits and required egress capacity, provided that not more than 50 percent of the number and required capacity is into a single building and provided they comply with the following provisions:

- a) Width of the horizontal exit should at least be the same as that of the exit doorways.
- b) A horizontal exit should be equipped with at least one fire/smoke door of a minimum 60 min fire resistance, of self-closing type. Further, it should have direct connectivity to the egress staircase for evacuation.
- c) Where there is a difference in level between connected areas for horizontal exits or ramps, slope not more than 1 in 12 should be provided.
- d) Doors in horizontal exits should be able to open at all times from both sides.

H-5.4.10 Platform Screen and Edge Door

Horizontal sliding platform screen or platform edge doors should be permitted to separate the platform from the trainway in stations, provided that the doors permit emergency egress from the train to the platform regardless of the stopping position of the train; and the doors provide egress when a force not exceeding 220 N is applied from the train side of the doors. The doors should be designed to withstand positive and negative pressures caused by passing trains.

H-5.4.11 Access for Fire Brigade Personnel

A dedicated access staircase for firemen should be provided in each underground station. This access should be arranged so as to provide uninterrupted access from ground level to station concourse and platform levels. If continuous access is not feasible, smoke sealed lobbies should be provided at concourse or other intermediate

levels and such lobbies should be protected with fire doors of adequate rating. Firemen staircases should be maintained under positive pressure at all times.

H-5.4.12 Emergency and escape lighting and illumination of means of all exits should comply with the various provisions under [3.5.6](#) and [H-9](#).

H-6 FIRE SEPARATION AND COMPARTMENTATION

H-6.1 Fire compartments should be provided in transit stations in accordance with the provisions of this section. Fire ratings of various occupancies within open stations and enclosed stations should be as indicated in [Table 30](#).

H-6.2 Further transit station should be divided into fire compartments by means of compartment walls and compartment floors by a fire separation of at least 120 min between the following occupancies:

- a) Public areas and non-public areas,
- b) Transit and non-transit areas, and
- c) Ancillary areas located beneath and within 3 m of the trainway in open stations.

H-6.3 No fire separation should be required for occupancies like ticketing offices, toilets, other offices and the like.

H-6.4 Incidental kiosks inside stations for other purposes like commercial use, etc should be provided with detection and a suitable automatic suppression system, if areas occupied by such occupancies exceed 10 m².

H-6.5 Fire Doors

Fire doors should comply with the following requirements:

- a) Fire doors should be constructed of non-combustible material having appropriate fire resistance, and two fire doors may be fitted in an opening if each door by itself is capable of closing the opening and the two doors together achieve the required level of fire resistance.
- b) All fire doors should be fitted with an automatic self-closing device, of same fire rating as the door, which is capable of closing the door from any angle and against any latch fitted to the door.
- c) Any fire door fitted within an opening which is provided as a means of escape should be capable of being opened manually or by an electromagnetic or electro-mechanical device which can be activated by the presence of smoke and/or the fire alarm system, including the case of fire doors opening into pressurised exit staircases.

H-6.6 Smoke Compartmentation

- a) Smoke compartments should be created to ensure that in case of a fire scenario, the escape routes are not full of smoke and a safe passage is provided to firefighters to reach the fire location.
- b) Smoke barriers should be provided within the ceiling at platform and concourse levels at locations and spacing as determined by the engineering analysis and should be designed to withstand temperatures up to 250 °C for 60 min.
- c) Smoke barriers should also be provided around all openings containing staircases, escalators and lifts in public areas connecting platform to levels above or below them.
- d) Multiple smoke compartments catered by a combined ductwork should be provided with fire dampers for a fire barrier.

Table 30 Fire Ratings for Mass Rapid Transit Station Occupancies

(Clause [H-6.1](#))

Sl No.	Occupancies	Fire Rating of Enclosures h		Fire Rating of Doors h	
		Open Station	Enclosed Station	Open Station	Enclosed Station
(1)	(2)	(3)	(4)	(5)	(6)
i)	Auxiliary substation, electrical UPS/battery room	2	2	2	2
ii)	Signalling equipment room, telecom equipment room, S&T (signalling and train control) UPS/battery room, electrical cable shafts, S&T cable shafts	2	2	1.5	1.5
iii)	Environmental control system (ECS) plant room, tunnel ventilation room, firemen's staircase, emergency equipment store, CDMA room, GSM room, sewage ejector room, sump pump room, chiller plant room, pump room, DG panel room	2	2	1.5	1.5
iv)	Station manager room, security room, stores, cleaners room, refuse storage, plumbing shafts, staff mess rooms, distribution board room and the like.	1	2	0.5	1.5

H-7 OTHER SPECIFICS REGARDING SAFETY REQUIREMENTS

- Materials used as interior wall and ceiling and other decorative features in enclosed stations should be non-combustible.
- For protection against intrusion of flammable and combustible liquids and flooding of underground transit systems, any structure including emergency egress or access stairs or vent or fan shaft structure utilized for ventilation of underground system should be permitted to be terminated at grade, provided that the level at which it is terminated is at least 450 mm above the surrounding ground level/footpath level.
- Emergency ventilation fans, their motors, and all related components exposed to the exhaust airflow should be rated for 250 °C for a minimum of 120 min. Emergency ventilation fans should be connected to a secondary power supply.
- Operation of the emergency ventilation system components should be initiated from the central supervising station.
- All types of kiosks like ticketing booth, information counter, commercial, if any, etc, should be constructed with non-combustible materials.
- Stations should be provided with power supply as per [H-8](#).
- Emergency lighting should be provided throughout the station as per [3.5.6.1](#) and [H-9](#).

- h) Voice evacuation/public address system should be provided throughout the station, and ancillary buildings as per requirements.
- j) Equipment rooms should be protected by fire suppression systems as per relevant standards.
- k) Emergency command centres and emergency response procedures should be in place.
- m) Evacuation should take place under the guidance and control of authorized, trained system employees or other authorized personnel as warranted under an emergency situation.
- n) Adequate warning signs, directional signs, exit signs should be provided throughout the facility.

H-8 POWER SUPPLY SYSTEM

H-8.1 General

The auxiliary substation at metro stations can be provided at any location, for example, ground level, concourse level, platform level, etc. The access for dismantling and replacement in future and ease of operation and maintenance of the equipment are to be ensured.

H-8.2 Uninterrupted Power Supply (UPS) System

The UPS should be an online parallel redundant system with battery backup for emergency power supply. The batteries should provide backup for a minimum 90 min. If, DG set power backup to UPS is provided, the battery backup time should be suitably selected.

H-8.3 Back-up or Emergency Power Supply

- a) The back-up/emergency power supply should have a capacity to cater all critical loads such as emergency lighting, protective signalling system, communication system, station control room/fire command centre, lifts providing required egress capacity, etc.
- b) Power supply to these systems and equipment should be from normal and emergency (standby generator power sources with changeover facility).
- c) If the power supply is taken from HT source and HT generation, the transformer should be selected with standby capacity to ensure continuity of power supply to such systems. Where parallel HV/LV supply from a separate independent source is provided with appropriate transformer for emergency, the provision of diesel generator for emergency supply may be waived off at the discretion of metro Authority. In such a scenario, in case of failure of mains power, the transfer to emergency power supply should be automatic and within 10 s for ensuring uninterrupted supply to emergency systems.

H-8.4 Cables

- a) Fire resistant/survival, low smoke zero halogen, copper conductor, armoured, cross linked polyethylene (XLPE) insulated cables should be used at enclosed stations for fire and life safety equipment like fire alarm system (FAS), tunnel ventilation fans, smoke extraction fans, staircase pressurization fans, fire pumps, emergency lighting, etc.
- b) All cables and wiring used for fire and life safety equipment should have a 3 h fire survival capability at 950 °C.
- c) For all other essential services at enclosed stations like chillers, AHU, normal lighting, chilled water pumps etc. flame retardant, low smoke zero-halogen (LSZH), armoured, XLPE insulated aluminium/copper conductor cables should be used.
- d) At open stations, flame retardant, low smoke halogen free (FRLSH), armoured, XLPE insulated aluminium/copper conductor cables should be used for power supply.

H-9 EMERGENCY LIGHTING SYSTEM

- a) LED lighting fixtures should be used for lighting at metro stations.
- b) Emergency illumination should include illumination of means of egress, illuminated exit signs, and all other luminaires specified as necessary to provide required illumination. Emergency lighting systems should be designed and installed so that the failure of any illumination source cannot leave in total

darkness any space that requires emergency illumination. Emergency lighting should not provide less than average of 10 lux. Minimum illumination of 10 lux should be maintained on egress route.

- c) Fire resistant, low smoke and zero halogen (FS) copper conductor wires should be used for wiring of emergency lighting system at underground stations. Emergency lighting system circuit should have 3 h fire survival capability at 950 °C.
- d) Out of total light fixtures installed at any station, at least 30 percent of the total lights, all uniformly distributed should be connected to emergency power source, that is, UPS.
- e) Emergency lighting should be powered from a source independent of that supplying the normal lighting.

H-10 FIRE PROTECTION FACILITIES

The following requirement should apply to all open (elevated and at-grade), and enclosed metro stations and depots.

H-10.1 The details of the protection and facilities required for open (elevated and at-grade) stations and depots are given hereunder.

H-10.1.1 Pumping Arrangements

Each station should be provided with two pumps having a minimum discharge capacity of 1 620 litre/min with different prime movers. It is possible to provide two electrically driven pumps, if assured back-up power is available for driving the pumps. Jockey pumps having a capacity of 180 litre/min should also be provided to take care of minor system leakages, (*Also refer to [H-10.1.9](#)*).

H-10.1.2 Fire Extinguishers

- a) Water CO₂ type, CO₂/ABC type fire extinguisher(s) should be provided in each platform and distributed in such a way that an the extinguisher is available within a travel distance of 30 m from any point;
- b) Water CO₂ type, CO₂/ABC type fire extinguisher(s) should be provided in each concourse for every 300 m² floor area and distributed in such a way that an extinguishers are available within a travel distance of 30 m from any point; and
- c) At least one extinguisher should be provided for each equipment room and other enclosure for every 300 m² floor area.

H-10.1.3 Small Bore Hose Reels

- a) One hose reel should be provided for every 1 000 m² floor area of station platform or part thereof, subject to a travel distance of 30 m to the nearest hose reel.
- b) One hose reel should be provided for every 1 000 m² floor area, concourse or part thereof subject to a travel distance of 30 m to the nearest hose reel.
- c) Where equipment rooms not covered by hose reel's reach from FHCs located at concourse and platform locations, additional hose reel points should be provided to ensure its coverage.

H-10.1.4 Fire Alarm System

- a) Fire detection system should be of the addressable type. Detectors are required to be provided only in areas where there is false ceiling(s)/ false floor(s) and areas for equipment rooms. Wherever there are false ceilings/false floors with perforation less than 70 percent, the detectors should be provided both above and below false ceiling/false floor, giving due consideration to the depth of false ceiling/flooring. In open stations, in areas having high heights and cross-ventilation, detectors will not be effective and hence it can be dispensed. Detector(s) should be provided in lift shaft(s) and escalator pit(s) also.
- b) Spacing of detectors and devices should be as per [4(26)].
- c) Manual call stations(s) should be provided at central location(s) on each platform (near emergency stop plunger) and at least two on the concourse, on each sidewall. When the concourse is in two halves, at least one manual call station should be provided on each side.

H-10.1.5 Inert Gas/CO₂/ based fire Suppression System

AIS HT, Main LT and panel supplying power to UPS, DG change-over, fire pump panel and transformers at the stations should be protected with the inert gas/ CO₂ based fire suppression system in accordance with IS 15506/ IS 15525/IS 15501/IS 15528. A suitable detection mechanism should be provided for triggering the suppression system automatically.

H-10.1.6 Provision of Hydrants and Wet Risers

- a) Minimum two hydrants should be provided at each platform with a hose box containing two lengths of hose pipes (15 m each with a nozzle) subject to a travel distance of 30 m to reach the nearest hydrant. Location of hydrants should be staggered for better coverage. The hydrants should cover the entire length of the platform when two hose pipes are coupled to the landing valves and used. If not fully covered, additional hydrant(s) should be provided.
- b) Minimum two hydrants should be provided for each concourse, including commercial area, if any, with a hose box containing two lengths of hose pipes (15 m each with a nozzle), subject to a travel distance of 30 m to reach the nearest hydrant.
- c) Hydrants should be provided at each entry to the station at ground and upper levels near the staircase. At least two hydrants for parking area should be provided subject to a travel distance of 30 m to the nearest hydrant.
- d) In depots, fire piping should be provided along the depot boundary as ring main. The ring main should be provided with a yard hydrant every 60 m. The building should be provided with a yard hydrant, preferably at its entry, through a branch from the ring main. The inside of the building should be provided with a number of FHCs required for coverage and maintaining a maximum travel distance of 30 m to the nearest FHC. Yard hydrant in the depot should be provided as per [Table 31](#) through branches from ring main. Other provision(s) of fire detection and suppression in the depot should be as per [Table 31](#).

H-10.1.7 Automatic Sprinkler System

Sprinklers are required to be provided only in the commercial areas and any combustible storage area within the station (also refer [H-10.1.9](#)). Alternatively, automatic fire suppression system instead of water-based sprinkler system, as per application, is also acceptable.

H-10.1.8 Water Requirement

- a) For stand-alone open metro stations, exclusive water requirement for the fire protection system for the installed pumping capacity, as stated in [H-10.1.1](#) should be at least 50 000 litre.
- b) However, when commercial areas are present within open metro stations, firefighting facilities should need augmentation as in [H-10.1.9](#).

H-10.1.9 In case of other types of occupancy like commercial areas inside metro stations and combustible storage areas, if any, within open stations, the pumping/water requirement should be as per [Table 32](#).

The volume of water stated in Table 32 should be stored in more than one interconnected compartments/tanks. Wherever existing structures are upgraded to include a commercial area and additional water requirement as per [Table 32](#) if any, may be met by additional tank or overhead tanks as permissible by structure. Size of the main header should not be less than 150 mm diameter in all cases. Entire fire protection system should be designed and installed as per the relevant Indian Standards.

Table 31 Requirement of Firefighting Installations in Depot Building

(Clause [H-10.1.6](#))

SI No.	Name/Part of the Building	Occupancy	Fire Extinguisher	First Aid Hose Reel	Wet Riser	Down Comer	Yard Hydrant	Automatic Sprinkler System Installation	Manually Operated Electronic Fire Alarm System	Automatic Detection and Alarm System	Underground Static Water Storage Tank Combined Capacity for Wet Riser, Yard Hydrant and Sprinklers per Set of Pumps	Terrace tank over Respective Tower Terrace	Pump near Underground Static Water Storage Tank (Fire Pump) with minimum pressure of 3.5 Kg/cm ² at Remotest Location	Pump at the Terrace Tank level with minimum pressure of 3.5 Kg/cm ²	Remarks
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
i)	Live OHE areas/third rail areas (like inspection bay, workshop bay, stabling lines, auto coach washing plant, covered electrical substation, etc)	Assembly	R	R	R	NR	R	NR	R	NR	2 00 000 litre	NR	2 850 lpm main and standby electric pumps, and 180 lpm jockey pump	NR	
ii)	Areas having engineering workshops and offices [Inspection offices, workshop offices, (ETU) workshop, P-way offices, depot control centre, PPIO, other administrative offices, etc]		R	R	R	NR	R	NR	R	R		NR		NR	
iii)	Storage areas (DCOS, Store rooms, scrap yards within buildings, lubricant and any other combustible storage areas)		R	R	R	R	R	R	R	R		R (10 000 litre) For main store building		NR	

Table 31 (Concluded)

SI No.	Name/Part of the Building	Occupancy	Fire Extinguisher	First Aid Hose Reel	Wet Riser	Down Comer	Yard Hydrant	Automatic Sprinkler System Installation	Manually Operated Electronic Fire Alarm System	Automatic Detection and Alarm System	Underground Static Water Storage Tank Combined Capacity for Wet Riser, Yard Hydrant and Sprinklers per Set of Pumps	Terrace tank over Respective Tower Terrace	Pump near Underground Static Water Storage Tank (Fire Pump) with minimum pressure of 3.5 Kg/cm ² at Remotest Location	Pump at the Terrace Tank level with minimum pressure of 3.5 Kg/cm ²	Remarks
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
iv)	Miscellaneous areas (canteen, time office, security rooms, open scrap yards, sewage treatment plants, contractor rooms, training establishments, commercial areas, etc)		(See Note 2)				R	(See Note 2)							
R – Required NR – Not Required															
NOTES															
1 Automatic sprinkler system is required in every storage areas and scrap yard inside the buildings. The same wet riser pumps can be used for feeding sprinkler system, if any.															
2 Miscellaneous areas or any other building structure within the depot premise (separated from other depot establishments) can be considered an individual occupancy and minimum requirements for firefighting provisions for these occupancies should be adopted as given in Table 7A to Table 7J .															

Table 32 Water Capacity

(Clause [H-10.1.9](#))

Sl No.	Area Occupied by Commercial Occupancies	Hydrant Pump Capacity (Minimum Pressure of 3.5 kg/cm ² at the Remotest Location)	Sprinkler Pump Capacity	Jockey Pump	Common Standby Pumps	Water Capacity
	m ²	litre/min	litre/min	litre/min	litre/min	litre
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Up to and including 300	2 280	Common pump	180	2 280	75 000
ii)	More than 300 and up to 1 000	2 850	Common pump	180	2 850	1 00 000
iii)	More than 1 000	2 850	2 850	180 each	2 850	2 00 000

H-10.2 Details of Protection and Facilities Required for Enclosed Metro Stations

H-10.2.1 Hydrant System

- Pumping Arrangements* — Each station should be provided with two pumps having a minimum discharge flow rate of 2 850 litre/min with different prime movers. It is possible to provide two electrically driven pumps, if assured back-up power is available for driving the pumps. Jockey pump having a capacity of 180 litre/min should also be provided to take care of minor system leakages.
- Flow of Hydrant Pump* — For hydrant system, having three or more hydrants on any floor, the minimum flow rate for the most demanding horizontal hydrant should be 2 850 litre/min.
- Requirement of Residual Pressure at Remotest Hydrant* — Hydraulically designed hydrant system should be designed to provide the water flow rate required at a minimum residual pressure of 7 bar at the remotest located hydrant.
- Flow of each Hydrant* — A hydraulically designed hydrant system supplying three or more hydrants on any floor, hydraulic calculations and pipe sizes should be calculated by providing 946 litre/min at the three most remote hydrants and at the minimum residual pressure of 7 bar.
- Water Requirement for Hydrant System* — The minimum water supply should be capable of providing the system demand for at least 60 min.
- Orifice plate* — Required to be installed when the static pressure at the hydrant exceeds 12.1 kg/cm².
- Distance of Fire Hydrant (FHC)* — FHC should be provided with hose box containing two lengths of hose pipes (15 m each with a nozzle). The travel distance to any hydrant should not exceed 30 m.

H-10.2.2 Manual Medium Velocity Spray Nozzle Systems

- Design Density* — A general range of water spray application rates that should apply to most ordinary combustible solids or liquids should be from 6.1 lpm/m² to 20.4 lpm/m² of protected surface.
- The horizontal distance between spray nozzles/open head sprinklers should not exceed 2.5 m.
- Water Requirement for Spray System* — The water supply duration for an exposure protection sprinkler system should be for a minimum of 60 min.

H-10.2.3 Water Requirement

The enclosed station should be provided with water storage of 2 00 000 litre capacity. The underground fire water storage tanks should not be more than 7 m in depth from the level having fire brigade draw-out connection, while the draw-out connection should not be more than 5 m away from tank wall. In case any of the above criteria could not be met, one additional booster pump of suitable rating should be provided to support fire brigade vehicle for proper suction of water from the tank.

H-10.2.4 Fire Extinguishers — See [H-10.1.2](#)

H-10.2.5 Inert Gas/CO₂ based Fire Suppression System — See [H-10.1.5](#)

H-10.2.6 Automatic Sprinkler System — See [H-10.1.7](#)

H-10.2.7 Station Control Room/Fire Command Centre

- a) The station control room should have the main fire alarm panel with communication system (suitable public address system) to aid floors and facilities for receiving the message from different floors.
- b) Fire command centre (FCC)/station control room (SCR) should be provided with emergency lighting. All controls and monitoring of fire alarm systems, pressurization systems, smoke management systems should be done from FCC/SCR. Monitoring of integrated building management systems, CCTVs or any other critical parameters in building may also be from the FCC/SCR.
- c) Details of all floor plans along with the details of firefighting equipment and installations (2 sets laminated and bound) should be maintained in fire command centre.

H-10.2.8 Fire Alarm System

An addressable fire detection and alarm system should be provided. Major components of fire alarm system should be as per IS 2189 and as specified herein :

a) FACP Battery Backup

FACP should have provision of battery for power backup in case of power failure.

The secondary power supply should have sufficient capacity to operate the system under quiescent load (system operating in a non-alarm condition) for a minimum of 24 h and, at the end of that period, should be capable of operating all alarm notification appliances used for evacuation or to direct aid to the location of an emergency for 5 min.

Battery calculations should include a minimum 20 percent safety margin above the calculated required amp-hour capacity.

In case of availability of emergency generator or two reliable sources of power supply from two separate transformers, 4 h of power backup may be taken in place of 24 h.

b) Response Indicator

Generally, response indicators are provided to identify the location of detectors which are installed at unapproachable/concealed locations. Response indicator should not be provided with addressable type fire alarm system.

H-11 STATION SMOKE MANAGEMENT SYSTEM

H-11.1 The subsequent clauses give the requirement of smoke extraction and smoke purging system required in enclosed metro stations. Smoke management system is not required for open (elevated and at-grade) stations.

H-11.2 Station Public Area Smoke Extraction Systems

Station public area should be provided with active smoke extraction system to maintain tenable environment for the time required for safe evacuation of passenger(s).

The smoke extraction system should be an engineered smoke control system and its design should be based on the following:

a) *Fire size*

The capacity of the engineered smoke extraction system should be calculated based on the incidence of a likely minimum baggage fire size of 1 MW.

b) *Smoke clear height*

The design smoke clear height should be above the heads of people escaping beneath it. The minimum height should be 2.0 m.

c) *Perforated ceiling*

For cases where the smoke reservoir is above the false ceiling, the ceiling should be of perforated type with at least 70 percent opening.

d) *Makeup air*

Makeup air should be drawn directly from the external space/air shaft, through inlet air ventilators or door-ways.

e) *Mode of activation*

1) *Automatic activation*

The engineered smoke extraction system should have provision of automatic activation by smoke detectors located in the fire zone. Use of smoke detectors for activation should be carefully designed, so that accidental or premature activation of smoke detectors in a non-fire zone (due to smoke spills or spread from other areas) are avoided.

2) *Manual activation*

A remote manual activation and control switches, as well as visual indication of the operation status of the smoke ventilation system, should also be provided at the station control room (SCR).

f) *Interlocking with other systems*

1) Except for smoke extraction system, all other air conditioning and mechanical ventilation systems within the areas served should be shut down automatically upon activation of the smoke extraction system.

g) *Standby fans/multiple fans*

Either a standby fan or multiple fans with excess capacity should be provided for each mechanical smoke extraction system, such that in the event the duty fan or the largest capacity fan fails, the required smoke extraction rate will still be met. The standby fan should be automatically activated in the event the duty fan fails.

h) Emergency ventilation fans, their motors, and all related components exposed to the exhaust airflow should be rated for 250 °C for a minimum of 120 min. Emergency ventilation fans should be connected to a secondary power supply.

H-11.3 Smoke Purging System

a) For extraction of cold smoke after the extinguishing of fire, smoke purging system should be provided.

b) Smoke purging system should be provided in the following areas of underground station:

- 1) Station control room;
- 2) S and T rooms;
- 3) Fire pump rooms;
- 4) UPS rooms; and
- 5) Auxiliary sub-station.

For other back of house rooms, normal ventilation system may be utilized to remove the cold smoke.

c) *Design of smoke purging system*

Smoke purging system, where provided in station and ancillary building, should comply with all the following requirements:

- 1) The smoke purging system need not be a dedicated system.
- 2) The purging system should have purge rate of not less than 9 air changes per hour (ACPH) or room ventilation ACPH, whichever is higher.
- 3) Makeup air should be drawn from adjacent area or corridor free from smoke or can be supplied by a separate mechanical system.
- 4) Smoke purging fans, their motors, and all related components exposed to the exhaust airflow should be fire rated and designed to operate in smoke of 250°C for a minimum of 120 min. Smoke purging fans should be connected to a secondary power supply.
- 5) In case of smoke/fire in entries subway, it should be rendered out of use and evacuation should be dealt procedurally.

H-11.4 Tenability Criteria

A tenable environment should be maintained in the path of egress. The tenability criteria considerations are given hereunder:

- a) Smoke obstruction:
 - 1) Internally illuminated signage (80 lux) discernible at 30 m.
 - 2) Walls and doors or externally illuminated signage discernible at 10 m.
- b) Heat effects:
 - 1) Exposure temperature (occupied zone) $\leq 60^{\circ}\text{C}$ for 10 min.
 - 2) Radiant heat exposure $\leq 2.5 \text{ kW/m}^2$
- c) Carbon monoxide – The smoke extraction system to be designed to maintain fractional effective dose (FED) of less than 0.3. The permissible exposure to carbon monoxide should be as per following table:

<i>Sl No.</i>	<i>Time min</i>	<i>Maximum Carbon Monoxide Exposure ppm</i>
(1)	(2)	(3)
i)	4	1 706
ii)	6	1 138
iii)	10	683
iv)	15	455
v)	30	228
vi)	60	114
vii)	240	28

- d) Velocity – Air velocity should be greater than or equal to 0.75 m/s and less than or equal to 11 m/s along the path of egress.
- e) Noise – Noise level along the path of egress should not exceed 85 dBA at any point 1.5 m above walking surface.

ANNEX J

(Clause [6.4.4](#))

FIRE AND LIFE SAFETY REQUIREMENTS FOR METRO TRAINWAYS

J-1 APPLICATION

Provisions of this Annex should apply to all portions of underground, elevated and at-grade metro trainway including tail buffer tracks and sidings not intended to be occupied by the passengers. These provisions may be amended, if required, as per the approval of the Competent Authority.

J-2 USE AND OCCUPANCY

J-2.1 Passengers should be allowed to enter the trainway only in the case it becomes necessary to evacuate a train.

J-2.2 Passenger evacuation from train onto the trainway should take place only under the guidance and control of authorized, trained system employee or other authorized personnel as warranted under an emergency situation.

J-2.3 Warning signs in accordance with [J-4.4.1](#) should be posted at locations where unauthorized personnel might trespass.

J-3 CONSTRUCTION

J-3.1 Construction Type

J-3.1.1 *Cut and Cover*

Where trainways are to be constructed by cut and cover method, perimeter walls, base and roof slabs and other related constructions should be of not less than Type 1 or Type 2 or combination of Types 1 and 2 non-combustible constructions as defined in this Part. The type of construction should be further determined by an engineering analysis of potential fire exposure hazards to the structure.

J-3.1.2 *Bored Tunnels*

Where trainway sections are to be constructed by a tunnelling method through earth, unprotected steel liners, reinforced concrete, shotcrete, or equivalent should be permitted to be used.

J-3.1.3 *Rock Tunnels*

Where trainway sections are to be constructed by a tunneling method through rock, steel bents with concrete liner, if lining is required, should be permitted to be used.

J-3.1.4 *Underwater Tunnels*

Underwater tunnels should be not less than Type 2 construction.

J-3.1.5 *Mid Tunnel Ventilation Shaft and Trainway Exit structures*

Mid tunnel ventilation shafts and trainway emergency egress or access stairs should not be less than Type 1 construction.

J-3.1.6 *At-Grade Construction*

Materials used for at-grade construction should not be less than Type 2 materials. The type of construction should be further determined by an engineering analysis of potential fire exposure hazards to the structure.

J-3.1.7 *Elevated Viaduct or Portals*

All elevated structures necessary for trainway support and all structures and enclosures on or under the trainway should be of not less than Type 1 or Type 2 or combination of Type 1 and Type 2 construction. The type of construction should be further determined by an engineering analysis of potential fire exposure hazards to the structure.

J-3.2 Protection against Intrusion of Flammable and Combustible Liquids and Flooding of Enclosed Trainway

Any structure including emergency egress or access stairs or vent or fan shaft structure utilized for ventilation of underground system should be permitted to be terminated at grade, provided that the level at which it is terminated is at least 450 mm above the surrounding ground level/footpath level.

J-3.3 Compartmentation

J-3.3.1 Ancillary areas should be separated from trainway areas within underwater trainway sections by construction having a minimum 180 min fire resistance rating.

J-3.3.2 Ancillary areas should be separated from trainway areas within enclosed trainway sections by construction having a minimum 120 min fire resistance rating.

J-3.4 Combustible Components

Where combustible components not specifically mentioned here are installed in a trainway, a fire hazard analysis should be conducted to determine that the level of occupant fire safety is not adversely affected by the contents.

J-3.5 Walking Surfaces

Walking surfaces designated for evacuation of passengers should be constructed of non-combustible materials.

J-4 EMERGENCY EGRESS

J-4.1 Location of Egress Routes

J-4.1.1 The system should incorporate a walking surface or other approved means for passengers to evacuate a train at any point along the trainway so that they can proceed to the nearest station or other point of safety.

J-4.1.2 Walkway continuity should be maintained at special track sections like cross over or pocket tracks. Walkway continuity should be provided by cross walks at track level.

J-4.1.3 Within enclosed trainways, the maximum distance between emergency exits/exit stairways should not exceed 750 m.

J-4.1.4 For trainways in twin bore tunnels or trainways in a tunnel divided by a minimum 120 min fire rated wall separation, cross-passageways should be permitted to be used in-lieu of emergency exit stairways.

J-4.1.5 Where cross passageways are utilized in-lieu of emergency exit stairways, the following requirements should apply:

- a) First cross-passageway should be provided not farther than 500 m from the end of platform of an enclosed station or from the tunnel portal or from a mid-tunnel exit way.
- b) Distance between any two cross-passageways should not be more than 250 m.
- c) The crossover in an enclosed trainway should be considered as cross-passageway. The distance between crossover and subsequent cross-passageway should not be more than 250 m.
- d) Cross-passageways should be separated from each trainway with separate self-closing fire door assemblies having a fire protection rating of minimum 90 min.
- e) A tenable environment should be maintained in the portion of the trainway that is not involved in an emergency and that is being used for evacuation.
- f) A ventilation system for the incident trainway should be designed to control smoke in the vicinity of the passengers.
- g) Provisions should be made for evacuating passengers via the non-incident trainway to a nearby station or other emergency exit.
- h) The provisions should include measures to protect passengers evacuating from non-incident tunnel from oncoming traffic and from other hazards.

- j) Where cross-passageways are used in-lieu of emergency exit stairways, the interior of the cross-passage should not be used for any purpose other than as an area of refuge or for access/egress to the opposite tunnel, except for installation of non-combustible equipment inside the cross passageways, provided that such equipment does not infringe the required clear space of the cross-passage.

J-4.1.6 Level of Cross Passage

The level of cross-passage inside the tunnel should be designed in accordance to the following requirements:

- a) In trainways where passengers have to walk on the track bed during evacuation, the floor of the cross-passages in such trainways should be kept close to the level of the track bed. The maximum permitted level difference between the level of track bed and the floor of the cross-passage should be 250 mm. Further, a smooth surface for transition of level difference should be provided at such cross-passages.
- b) In trainways where passengers have to walk on a walkway mounted at the level of the train floor during evacuation, the floor of the cross-passages in such trainways should be kept in level with the walkway. Further, such walkways should always be provided on the cross-passageway side of the trainway for unobstructed access to the cross-passageway.

J-4.1.7 For open-cut trainways, an engineering analysis should be conducted to evaluate the impact of the trainway configuration on safe egress from a train fire to a point of safety.

J-4.1.8 Where the engineering analysis indicates that the configuration will impact tenability beyond the immediate vicinity of the fire, egress routes should be provided such that the maximum distance from any point within the open-cut section to a point of egress from the trainway should not be more than 380 m.

J-4.1.9 All the distances mentioned in [J-4.1.3](#), [J-4.1.5](#) and [J-4.1.8](#) should be permitted with a maximum variation of ± 10 percent.

J-4.2 Size of Egress Routes

J-4.2.1 Cross-passageways should be a minimum of 1 200 mm in clear width and 2 100 mm in height.

J-4.2.2 The width of exit stairs should not be required to exceed 1 200 mm for enclosed trainways.

J-4.2.3 If double leaf doors wider than 1 200 mm are provided in egress routes serving trainways, then size of active leaf should not be less than 810 mm.

J-4.3 Egress Components

J-4.3.1 Walking surfaces serving as egress routes within guide ways should have a uniform, slip-resistant design except as permitted herein:

- a) Where the trainway track bed serves as the emergency egress pathway, it should be nominally level and free of obstructions.
- b) Walkways that are more than 750 mm above the floor or grade below should be provided with a continuous handrail along the side opposite the trainway.
- c) Raised walkways that are greater than 1 200 mm wide and located between two trainways should not be required to have a handrail.
- d) Doors in the means of egress, except cross-passageway doors, should open in the direction of exit travel.
- e) Doors in the means of egress should,
- 1) open fully when a force not exceeding 220 N is applied to the latch side of the door, and
 - 2) be adequate to withstand positive and negative pressures caused by passing trains and the emergency ventilation system.

J-4.3.2 Horizontal sliding doors should be permitted in cross-passageways.

J-4.4 Signage, Illumination and Emergency Lighting

J-4.4.1 Warning signs posted on entrances to the trainway and on fences or barriers adjacent to the trainway

should clearly state the hazard (for example, DANGER: HIGH VOLTAGE, 25 000 V) with letter sizes and colours in conformance with requirements of *Metro Railway Rules*.

J-4.4.2 System egress points should be illuminated.

J-4.4.3 Points of exit from elevated and enclosed trainways should be marked with internally or externally illuminated signs.

J-4.4.4 Tunnel lighting should be 100 percent on emergency power source, that is, UPS. Minimum illumination of 10 lux should be maintained on tunnel egress routes. Emergency lighting should be powered from a source independent of that supplying the normal lighting.

J-4.4.4 Identification

Emergency exit facilities should be identified and maintained to allow for their intended use.

J-4.4.5 Enclosed trainways greater in length than the minimum length of one train should be provided with directional signs as appropriate for the emergency procedures developed for the metro system.

J-4.4.6 Directional signs indicating station or portal directions should be installed at maximum 25 m intervals on either side of the enclosed trainways.

J-4.4.7 Directional signs should be readily visible by passengers for emergency evacuation.

J-5 FIRE PROTECTION AND LIFE SAFETY SYSTEMS

J-5.1 Emergency Access

J-5.1 Except as described herein, points of egress and exits from the guide way should serve as emergency access routes.

J-5.2 If security fences are used along the trainway, access gates should be provided in security fences.

J-5.3 Access gates should be a minimum 1 200 mm wide and should be of the hinged or sliding type.

J-5.4 Access gates should be placed as close as practicable to the portals to permit easy access to tunnels.

J-5.5 Information that clearly identifies the route and location of each gate should be provided on the gates or adjacent thereto.

J-5.6 Access to the elevated trainway should be from stations or by mobile ladder equipment from roadways adjacent to the track way.

J-5.7 If no adjacent or crossing roadways exist for the elevated trainway, access roads at a maximum of 750 m intervals should be required.

J-5.8 Where the configuration of an open-cut trainway prevents or impedes access for firefighting, provisions should be made to permit fire fighter access to that section of trainway at intervals not exceeding 750 m.

J-5.9 All the distances mentioned in [J-5.7](#) and [J-5.8](#) above should be permitted with a maximum variation of $\begin{smallmatrix} +10 \\ 0 \end{smallmatrix}$ percent.

J-5.10 Distance between Tunnel Hydrants

Double headed type hydrants should be provided inside tunnels.

- a) The travel distance to tunnel hydrant should not be more than 30 m.
- b) Tunnel hydrant should be installed on a pipe having diameter of not less than 100 mm.

J-6 EMERGENCY VENTILATION SYSTEM FOR TRAINWAY

J-6.1 General

J-6.1.1 The following defines the requirements for the mechanical ventilation systems used in passenger trainway area.

J-6.1.2 Mechanical emergency ventilation should be provided in underground or enclosed trainways exceeding 300 m.

J-6.1.3 Exemption

- a) A mechanical emergency ventilation system is not required for underground or enclosed trainways at most 60 m in length.
- b) An engineering analysis is required for underground or enclosed trainways more than 60 m in length but less than 300 m to determine if a mechanical emergency ventilation is required; and
- c) Emergency ventilation meeting the tenability criteria for occupied areas should not be required in storage track area like stabling line, depot line, Y-siding, etc where the storage track has no openings along its length to passenger trainway areas and where a fire hazard analysis indicates that a fire on a train in the storage track area will not impact passengers or passenger areas.

J-6.1.4 Engineering Analysis

The analysis should take into consideration all the factors that affect fire safety. A report of the analysis should be submitted. If the engineering analysis is not carried out, then a mechanical ventilation system should be provided.

J-6.2 The engineering analysis of the emergency ventilation system should include the following:

- a) A validated trainway analytical simulation program that provides a quantitative analysis of airflow dynamics produced in the fire scenario, a validated computational fluid dynamic (CFD) technique, or both, as approved; and
- b) The no-fire (or cold) air velocities that can be measured during commissioning to confirm that a mechanical ventilation system as built meets the requirements determined by the analysis.

J-6.2.1 Where required by [J-6.1.2](#), the mechanical ventilation system should make provision for the protection of passengers, employees, and emergency personnel from fire and smoke during a fire emergency.

J-6.3 Design

J-6.3.1 The emergency ventilation system should be designed to cater the following:

- a) Provide a tenable environment along the path of egress from a fire incident in enclosed station and enclosed trainways;
- b) Produce sufficient airflow rates within enclosed trainways to meet critical velocity;
- c) Be capable of reaching full operational mode within 180 s;
- d) Address the maximum number of trains that could be between ventilation shafts during an emergency; and
- e) Maintain the required airflow rates for a minimum of 2 h.

J-6.3.1.1 Where the airflow rates required to accomplish [J-6.3.1 \(a\)](#), [J-6.3.1 \(b\)](#), or approved alternative performance criteria are dependent upon the unimpaired function of the air distribution system, that system should be designed to continue operation when exposed to the conditions generated during the design incident for the duration of 2 h.

J-6.3.1.2 Where the airflow rates required to accomplish [J-6.3.1\(a\)](#), [J-6.3.1\(b\)](#), or approved alternative performance criteria are dependent upon the continued integrity of structural and architectural features, those features should be designed to remain intact when exposed to the conditions generated during the design incident for the duration determined as per [J-6.3.1\(e\)](#).

J-6.3.2 Additional Design Requirements

The design should encompass the following:

- a) The heat release rate from a vehicle and any other combustible materials (if permitted) in the trainways that could contribute to the fire load at the incident site, subject to the approval of the Metro Rail Administration;
- b) The rate of fire growth
- c) Fire scenarios and fire profiles.
- d) Station and trainway geometries.
- e) The effects of elevation, elevation differences, ambient temperature differences and ambient wind.
- f) A system of fans, shaft and devices for directing airflow in stations and trainways.
- g) Predetermined procedures for initiating quick response from the OCC/SCR during fire.
- h) A ventilation system reliability engineering analysis that, as a minimum considers the following subsystems:
 - 1) Electrical;
 - 2) Mechanical; and
 - 3) Supervisory control.
- j) Criteria for the system reliability analysis in [6.3.2 \(h\)](#) should be established and approved.
- k) The analysis should consider as a minimum of the following events:
 - 1) Fire in the trainway or station;
 - 2) Local incident within the electrical utility that interrupts power to the emergency ventilation system;
 - 3) Derailment; and
 - 4) The loss of a fan that results in the most adverse effect on the ventilation system performance.

J-6.3.3 The design and operation of the signalling system, and ventilation system should be coordinated to match the total number of trains that could be between ventilation shafts during an emergency.

J-6.3.4 Emergency ventilation air distribution system should be permitted to serve more than one trainway.

J-6.4 Emergency Ventilation Fans

- a) The emergency ventilation system fans that are designated for use in fire and similar emergency should be capable of satisfying the emergency ventilation requirements to move train-way air in directions to provide the needed ventilation response as per engineering analysis.
- b) Individual emergency ventilation fan motors should be designed to achieve their full operating speed in no more than 30 s from a stopped position when started across the line and in no more than 60 s for variable speed motors.
- c) Emergency ventilation fans, their motors, and all related components exposed to the exhaust airflow should be designed to operate at 250 °C for a minimum of 2 h.
- d) Local fan motor starter and related operating control device should be located away from the direct airstream of the fans to the greatest extent practical.
- e) Thermal overload protective devices in fan motors, damper motors, or on motor controls used for emergency ventilation should not be permitted.
- f) Non-emergency ventilation airflows that do not impact the emergency ventilation airflows should be permitted to be left operational where identified in the engineering analysis.
- g) The ventilation system designated for use in emergencies should be capable of operating at full capacity in either the supply mode or exhaust mode to provide to provide the needed ventilation response where dilution of noxious products is to be maximized.

- h) The ventilation system designated for use in emergencies should be capable of being turned off and dampers closed to provide the needed ventilation response where dispersion of noxious products is to be minimized.
- j) Fan should be rated in accordance with the ANSI/AMCA 300, AMCA 250/ISO 13350, ASHRAE Handbook-Fundamentals, ISO 5801, and ASHRAE 149.
- k) Fans that are associated only with the passengers or employee comfort and that are not designed to function as a part of emergency ventilation system should shut down automatically on identification and initiation of a fire emergency ventilation program so as not to jeopardize or conflict with emergency airflows.

J-6.5 Airflow Control Devices

- a) Devices that are interrelated with the emergency ventilation system and that are required to meet the emergency ventilation system airflow should be structurally capable of withstanding both maximum repetitive and additive piston pressure of moving trains and emergency airflow velocities.
- b) Devices in the emergency ventilation system that are exposed to the exhaust airflow and are critical to the system's effective functioning in the event of an emergency should be constructed of materials suitable for operation in an ambient atmosphere at the design condition of 250 °C for 2 h.
- c) Finishes applied to non-combustible devices should not be required to meet the provisions of [J-6.5\(b\)](#).
- d) Other devices should be designed to operate throughout the anticipated temperature range. Over current elements in devices or on device controls required to support the emergency ventilation should not be permitted where such over current elements are subject to false operation due to exposure to elevated temperatures during a fire emergency.

K-6.6 Testing

- a) Equipment used for emergency ventilation (including fans, dampers and airflow control devices and associated electrical equipment) should be approved in accordance with the requirements of a recognized standard for the type of equipment to be installed and should be factory acceptance tested.
- b) The non-fire (or cold) airflows provided by the installed mechanical ventilation system should be measured during commissioning to confirm that the airflow meets the requirement determined by the engineering analysis.

L-6.7 Emergency Ventilation Openings

- a) Emergency ventilation openings should be positioned or designed to minimize recirculation of smoke into the station or enclosed trainway through any openings, such that a tenable environment is maintained along the path of egress for the time of tenability and as required for designated points of safety.
- b) Adjacent structures and property uses should also be considered.

M-6.8 Emergency Ventilation System

- a) Operation of the emergency ventilation system should be initiated from the operations control centre.
- b) The operations control centre should receive verification of proper response by emergency ventilation fan(s) and an interrelated device(s).
- c) Local controls should be permitted to override the operations control centre becomes inoperative or where the operation of the emergency ventilation system is specifically redirected to another site.

N-6.9 Power Supply for Emergency Ventilation System

- a) The design of the power for the emergency ventilation system should comply with the requirements of [H-8](#) of this Part.
- b) Overcurrent elements that are designed to protect conductors serving motors for both emergency fans and related emergency devices should not be permitted where such overcurrent elements are subject to false operation due to exposure to elevated temperatures during a fire emergency. All other motor and

fan protection device should be bypassed during a fire emergency, except for motor overcurrent and excessive vibration.

- c) The emergency ventilation circuits routed through the station public areas and trainway should be protected from physical damage by fixed guideway transit or passenger rail vehicles or other normal operations and from fire.

J-6.10 Critical Velocity

- a) The minimum steady-state velocity of the ventilation airflow moving toward the fire within an enclosed trainway or enclosed passageway that is required to control back layering at the fire site, such that a tenable environment is maintained along the path of egress upstream of the fire, and as required for designated points of safety.
- b) The equation for critical velocity is as follows:

$$V_c = K_1 K_g \left(\frac{gHQ}{\rho C_p A T_i} \right)^{\frac{1}{3}}$$

$$T_f = \left(\frac{Q}{\rho C_p A V_c} \right) + T$$

$$K_g = \{1 + 0.0374 (G)^{0.8}, 1, \frac{G < 0}{G \geq 0}$$

where

- V_c = critical velocity (m/s or fpm)
- A = the annulus area (not including the area of blockages, such as vehicles) perpendicular to the flow (m² or ft²)
- C_p = the specific heat of the air (kJ/kg/K or Btu/lb/R)
- g = acceleration due to gravity (m/s² or ft/s²)
- G = absolute value of tunnel grade, as a percent
- K_g = grade factor
- H = height from the base of the fire to the tunnel ceiling at the fire site (not tunnel height) (m or ft)
- K_1 = Critical Froude number factor with a constant value of 0.606 (critical Froude number of the negative one-third power)
- ρ = average density of the approach (upstream) air (kg/m³ or lb/ft³)
- Q = convective heat the fire adds directly to the air at the fire site (kW or Btu/s)
- T = temperature of the approach air (K or R)
- T_f = Average temperature of the fire site gases (K or R)

ANNEX K

(Clause [6.7.3](#))

FIRE PROTECTION CONSIDERATIONS FOR VENTING IN INDUSTRIAL BUILDINGS

K-1 APPLICATION

K-1.1 The provisions given below are applicable only to single storey industrial buildings (factories and storage buildings) covering large floor areas without sub-dividing/separating walls which are usually designed to meet modern production methods.

K-1.2 The requirements of fire and explosion venting of industrial buildings, as dealt with in this section, fall under two categories:

- a) Smoke and fire venting; and
- b) Explosion relief vents.

K-2 SMOKE AND FIRE VENTING

K-2.1 The basic considerations for formulating the design and other requirements for smoke and fire vents are as given in [K-2.1.1](#) to [K-2.1.20](#).

K-2.1.1 The smoke and hot combustion products from a fire, being lighter than the surrounding air, tend to rise, and on reaching the roof or ceiling spread out (mushroom) on all sides and form a layer which floats on top of the cold air beneath. In the absence of vents, this layer becomes progressively deeper until the whole building is filled with hot smoky gases. The time consumed for this to happen may be only a few minutes, depending on variables like, type of materials on fire, process/storage conditions involved, etc.

K-2.1.2 The hot gases at the roof level moved by convection currents contribute to rapid lateral spread of fire.

K-2.1.3 The provision of properly designed and suitably located vents in adequate number helps the speedy removal of smoke and hot gases, thereby preventing spread of fire, besides reducing risks of explosion of unburnt gases, and reducing damage to the contents and structure of the building by heat and smoke. In addition, they facilitate firefighting operations, and minimize personal hazards to the firemen.

K-2.1.4 The time taken for accumulation of smoke and hot gases within a building on fire being very short, the venting devices installed should be designed to operate in the early stage of the fire and must be automatic so as to ensure speed and efficiency in their operation.

K-2.1.5 The smoke and fire venting system should be designed in such a manner as to keep the temperature of the combustion products from the fire as low as possible, preferably below approximately 150 °C.

K-2.1.6 Automatic venting systems are complementary to the fire extinguishing systems, and automatic sprinklers, where provided, should operate before the operation of the vents; otherwise, venting may delay sprinkler operation.

K-2.1.7 It is easier to vent a building of smoke than clear it of smoke once it has been filled.

K-2.1.8 Venting is particularly desirable in large area industrial buildings or warehouses, windowless buildings, underground structures or in areas housing hazardous operations. Automatic fire vents should be provided for all industrial occupancies (including storage buildings) classified as medium hazard or above having floor areas exceeding 750 m², irrespective of whether they are compartmentalized or not.

K-2.1.9 These provisions do not cover other aspects, of ventilation (or lighting) designed for regulation of temperature within a building for personal comfort or meeting process needs.

K-2.1.10 Similarly, fire and smoke venting requirements as given here under are also not applicable to multi-storeyed buildings, as their requirements are different and more complex.

K-2.1.11 It is difficult to determine precise venting requirements on account of the many variables involved. For instance, the rate of combustion varies appreciably according to the nature, shape, size, and packaging of the combustible materials as well as the size, height, and disposition of the stacks of materials.

K-2.1.12 In industrial buildings of floor area less than 750 m² and used as low fire hazard occupancies, conventional ventilators fitted high up near the eaves of the external walls may serve as vents for smoke and hot gases, provided care is taken to ensure that they are kept open at all times or are designed to open automatically in case of fire.

K-2.1.13 Extinction of fires by closing the doors and windows is not likely in the case of industrial buildings because of their large size, where sufficient air to sustain the fire at least in the initial stages can be expected to be present.

K-2.1.14 Of the two types of building ventilation, namely, vertical, and horizontal, vertical ventilation is the one commonly adopted in the case of single storey industrial buildings.

K-2.1.15 Since 70 percent to 80 percent of heat produced in a fire is convective heat, the ventilation system has to be suitably designed to ensure early outflow of the heat and thereby minimize fire spread.

K-2.1.16 Combustible roof linings should be avoided, as they themselves will contribute to the spread of fire, thereby multiplying the venting problems.

K-2.1.17 A wind blowing across a flat roof or a roof with a pitch under 40° produces a negative pressure, that is, it tends to draw gases out of the building and so aids venting of hot gases. Wind blowing across a roof of pitch greater than 40° will draw gases out on the leeward side, but oppose outward flow on the windward side of the roof.

K-2.1.18 For vents to work at full efficiency, the area of the inlets for cold air entering the compartment must equal at least the total area of the vents. Ideally, the inlets should be as close to the ground as possible.

K-2.1.19 Where roof vents are installed in a single-storey building any neighbouring buildings, particularly those of more than one storey, will be subject to some degree of exposure hazard either from flying birds or radiation, or both, as a result.

K-2.1.20 If vents are to be installed, the size, design, number and disposition of the vents and the associated roof screens/curtain boards have to be assessed after careful analysis of the various factors stated under [K-2.1.11](#), as well as other related factors like type of building construction, nature and height of roof, process hazards, exposure hazard, etc.

K-2.2 Venting Area

K-2.2.1 The estimated requirements for ventilation are largely based on the assumed build-up of the fire from the time of initial outbreak to the time of effective firefighting action by fire brigade.

K-2.2.2 The vent area required to be provided should be approximately proportional to the perimeter of the fire area, because the entrained air forms the bulk of the vented gases.

K-2.2.3 The effective area should be the minimum cross-sectional area through which the hot gases must flow out to the atmosphere.

K-2.2.4 No consideration should be given to the increased air movement obtained by power operated fans, since it must be assumed that in the event of fire, power will be interrupted, or fans damaged by heat.

K-2.2.5 The total vent areas to be provided should be as per the following ratios of effective area of vent openings to floor area for various occupancy classifications indicated:

- a) Low heat release content (Sub-division G-1) — 1 : 150
- b) Moderate heat release content (Sub-division G-2) — 1 : 100
- c) High heat release content (Sub-division G-3) — 1 : 30 to 1 : 50

K-2.3 Types of Vents

K-2.3.1 Venting should be accomplished by any of the types such as monitors, continuous gravity vents, until type vents or sawtooth roof skylights.

K-2.3.2 Where monitor type vents are installed, wire glass or metal panels should be used only if the sash is arranged to open automatically.

K-2.3.3 The use of plain thin glass for venting should be avoided on account of its unpredictable behaviour during fire. However, if glass or other suitable plastic sheet materials with early disintegration characteristics are used, they should be designed for automatic operation.

K-2.3.4 Where monitors or unit type vents are used, the panels should be hinged at the bottom and designed to open automatically. Both sides of the vents should be designed to vent simultaneously to ensure that their effectiveness at the time of fire is not in any way impeded by wind direction.

K-2.3.5 Where movable shutters are provided for continuous gravity vents, these should open automatically in the event of fire.

K-2.3.6 Unit type vents should be of relatively small area, ranging between 1 m² and 9 m², having light weight metal frames and housing with hinged dampers which should be designed for both manual and automatic operation.

K-2.3.7 Sawtooth roof skylight should be considered as satisfactory for venting purposes only when designed for automatic operation.

K-2.3.8 Likewise, exterior wall windows should not be reckoned as satisfactory means for venting of fire gases and smoke in industrial buildings. However, they may be reckoned as additional means of venting when, they are located close to the eaves and are provided with ordinary glass or movable sash arranged for both manual and automatic operation.

K-2.3.9 Baffles should not be installed inside vents, as they greatly reduce the effective area for venting.

K-2.4 Vent Operation

K-2.4.1 The vents should be automatic in operation, unless where specified in these provisions that they should be designed for both manual and automatic operation.

K-2.4.2 The release mechanism should be simple for operation and independent of electrical power since electrical services may be interrupted by fire.

K-2.4.3 The automatic operation of vents should be achieved by actuation of fusible links or other types of heat and smoke detectors, or by interlocking with operation of sprinkler system or any other automatic fire extinguishing system covering the area. Following their release, the vents should be designed to open by a system of counterweights and associated equipment utilizing the force of gravity or spring loaded levers.

K-2.4.4 Automatic fire alarm system, where installed, should be coupled to the automatic vents to ensure simultaneous operation.

K-2.4.5 Automatic sprinklers, where installed, should operate before the vents open in order to avoid any likely delay in sprinkler operation. However, heat actuated devices used for vent release should be suitably shielded from sprinkler discharge so that water does not delay their action. Further, provision of operating the vents manually may also be provided.

K-2.4.6 Premises where height of roof apex is 10 m or more or where the materials handled or stored have high smoke producing characteristics, in addition to fusible links, the vent release mechanism should be interlinked to smoke actuated automatic fire detectors to ensure early operation of vents.

K-2.4.7 Non-corrosive materials should be used for hinges, hatches and other related parts to ensure long fail-safe operation of the vents.

K-2.4.8 In case of any doubts regarding the types of vents required to be installed for any particular occupancy, authorities having jurisdiction should be consulted.

K-2.5 Size, Spacing and Disposition of Vents

K-2.5.1 Vents should be correctly sited to ensure their functional efficiency. Ideally, they should be sited at the highest point in each area to be covered.

K-2.5.2 They should, as far as possible, be located immediately above the risk to be protected so as to allow free and speedy removal of smoke and other combustion products in the event of fire.

K-2.5.3 The minimum dimension for an effective vent opening should be not less than 1.25 m in any direction.

K-2.5.4 The spacing of the individual vent should be based on the principle that more number of well distributed smaller vents are more effective than fewer number of badly located larger vents.

K-2.5.5 The maximum spacing between vents for the three occupancy classifications should be as follows:

- a) Low heat release content : 45 m between content centres
- b) Moderate heat release content : 36 m to 37 m between release content centres
- c) High heat release content : 22.5 m to 30 m between content centres, depending on the severity of fire potential.

K-2.5.6 Vents should be placed in a sheltered situation where advantage can be taken of the prevailing wind. The design of the vent should be such as to produce a suction effect. A wind blowing across a flat roof or one with a pitch of 40° produces a negative pressure, that is, it tends to draw gases out of the building and so aids venting of hot gases. Wind blowing across a roof of pitch greater than 40° will draw gases out of the leeward side, but oppose outward flow on the windward side of the roof.

K-2.5.7 Low level inlets, with total area not less than the total area of vents, should be provided to permit outside air to be drawn in to aid automatic venting. These inlets, which may be in the form of doors, windows, or such other openings, should be designed for manual operation when desired.

K-2.6 Roof Screens or Curtain Boards

K-2.6.1 Industrial buildings with large areas and having no subdivision/separating walls limiting the area of individual compartments to 750 m² or less, should be provided with roof screens or curtain boards.

These screens which extend from the roof downwards at specific intervals not only prevent lateral spread of heat and smoke in the event of fire below, but substantially assist in early operation of automatic sprinklers and vents.

K-2.6.2 They should be of sheet metal or any other substantially non-combustible material strong enough to withstand damage by heat or impact.

K-2.6.3 They should be reasonably gas-tight, although small openings for passage of pipes, conduits, etc, should be permitted.

K-2.6.4 They should extend down from the roof/ceiling for a minimum depth of 2.2 m. Around specific hazards, the depth should be 4 m. Where roof/ceiling height exceeds 15 m they should extend down to within 3 m of the floor. For pitched sawtoothed roofs, they should extend down to truss level dividing the roof into compartments.

K-2.6.5 In moderate hazard occupancies, the distance between the screens/curtain boards should not exceed 75 m and the curtained areas should be limited to a maximum of 4 500 m².

K-2.6.6 In high hazard occupancies, the distance between screens should not exceed 30 m and the curtained area should be limited to 750 m².

K-2.6.7 The curtained roof area should be so arranged that they effectively aid in the venting of smoke and hot gases through the automatic vents provided in each area.

K-2.6.8 In sprinklered buildings, the screens should preferably be so located as to coincide with the individual sprinkler system areas.

K-3 EXPLOSION RELIEF VENTS

K-3.1 Industrial premises where combustible dusts can accumulate or where flammable gases, vapours or mists in explosive concentrations may be present are constantly exposed to explosion hazards. Pressures developed by such explosions may be of the order of 7×10^5 Pa and ordinary buildings will not be able to withstand the shock of such pressures. Hence, such buildings require explosion relief vents for preventing structural damage.

K-3.2 Basic Principle/Considerations

K-3.2.1 Most ordinary building walls will not withstand a sustained internal pressure as great as 6.9×10^3 Pa. Hence, explosion relief vents for buildings must be designed to operate at pressures well below those at which the building walls will fail.

K-3.2.2 There is a rise in pressure during an explosion within an enclosure even with open, unobstructed vents, and any delay in opening the venting devices increases that pressure.

K-3.2.3 Structural damage can be minimized by locating hazardous operations or equipment outside buildings and cut-off from other operations by a pressure resisting wall. Such isolated processes or equipment should be housed in single-storey buildings properly vented and a device provided at the inlet of the collector which will prevent an explosion from blowing back through the duct work and into the building.

K-3.2.4 Where highly hazardous operations cannot be located outside of main buildings, they should be segregated by pressure resisting walls and each such unit should be ventilated outdoors. External walls may be of heavy construction, if equipped with suitable vents or high weight panels which blow out easily.

K-3.2.5 Operations or equipment involving explosion hazards should not be permitted in basements or areas partially below grade.

K-3.2.6 Fire can be expected to follow an explosion in most occupancies, so that any fixed fire extinguishing equipment, like sprinklers, if installed, should be such that only the minimum damage is caused to it.

K-3.2.7 For a given material, the finer the particle size of the dust, the more violent is the explosion. Some materials, such as aluminium powder, hydrogen, and acetylene, are difficult to vent effectively due to the rapid rate of pressure rise. Some slow burning materials, such as coal dust in a confined space, may do much damage because of the longer duration of their presence. Some dusts, such as magnesium, titanium and zirconium and several metal hydrides may react with and ignite in some common inert gases, such as nitrogen and carbon dioxide.

K-3.2.8 The maximum explosion pressure in a vented structure decreases as the size of the vent increases but is independent of the rupturing pressure of a diaphragm.

K-3.2.9 The most effective vent for the release of explosion pressures is an unobstructed vent opening.

K-3.2.10 Pressure required to rupture diaphragms of the same area and material directly varies with the thickness of the material.

K-3.2.11 The slower the rate of pressure rise, the more easily can the explosion be vented.

K-3.2.12 The degree of venting required is directly proportional to the degree of explosion hazard.

K-3.2.13 Experience has shown that most explosions of dusts, vapours and gases do not involve a large part of the total volume of the enclosure, and frequently occur near the upper or lower limits of the explosive range. Consequently, such explosions are relatively weak compared with the optimum.

K-3.2.14 Rectangular unrestricted vents are as effective as square vents of equal area.

K-3.3 Types of Explosion Relief Vents

K-3.3.1 The explosion relief vents should be any one or more of the following types, depending on individual requirements as assessed by the Authority. Open or unobstructed vents, louvres, open roof vents, hanger type doors, building doors, windows, roof or wall panels or movable fixed sash.

K-3.3.2 The effect of external wind pressure or suction on these devices should be taken into consideration while designing and selecting the type of vents, since wind pressures may reach over 2×10^5 Pa in severe windstorms.

K-3.3.3 The type of vent for explosion relief for any occupancy should be selected with life safety as the primary aim followed by minimum damage to property.

K-3.3.4 Where large hanger type doors of metal curtain doors inside walls are used as vents, care should be taken to ensure that they are kept wide open during operations.

K-3.3.5 Where weather hoods are used to cover roof vents, they should be as light as possible and lightly attached so as to enable them to be blown off quickly when an explosion occurs.

K-3.3.6 Doors and windows when used as explosion vents should be installed to swing outwards. Doors should have friction, spring or magnetic latches that will function automatically to permit the door to open under slight internal pressure.

K-3.3.7 Movable sash should be of the top or bottom hinged or protected type. These should be equipped with a latch or friction device to prevent accidental opening due to wind action or intrusion. Such latches or locks should be well maintained.

K-3.3.8 Fixed sash should be set in place with very light wall anchorages, or, if tight, should be securely fitted and glazed with plastic panes in plastic putty.

K-3.3.9 Where the process is such that the whole of a building or a room may be desirable to arrange for a lightly constructed wall or roof to collapse and thus avert the worst effects of an explosion.

K-3.4 Design, Size and Disposition of Vents

K-3.4.1 The required area of explosion vents should ordinarily depend on the expected maximum intensity of an explosion in the occupancy, the strength of the structure, the type of vent closure and other factors.

K-3.4.2 Venting should be planned in such a manner as to prevent injury to personnel and damage to exposure. In congested locations, substantial ducts or diverters should be provided to direct the blast.

K-3.4.3 When ductwork is used, the ducts should be of sufficient strength to withstand the maximum expected explosion pressure.

K-3.4.4 Where explosions are likely within duct and piping systems, they should be vented by the use of suitable diaphragms designed to blow out at a predetermined pressure. There should be no physical connection between ductwork system for more than one collector.

K-3.4.5 In large structures, the position of vents should be relative to the point of origin of explosion, when it can be determined.

K-3.4.6 Where relatively slow explosions involving coal dust, chlorinated solvents, etc are involved, light, hinged swinging panels may be preferred to diaphragm type of vents.

K-3.4.7 Obstructions of any kind blocking the vents from the risk covered should be avoided, particularly where risks of rapid violent explosions are present.

K-3.4.8 Counterweights add to the inertia of the vents and so should be avoided.

K-3.4.9 Various relieving devices, including devices actuated by detonators, should start to open at as low a pressure as possible. They should be of light construction, so that full opening can be quickly attained.

K-3.4.10 Vents should be of such size and design as to prevent rupture of the protected device or apparatus.

K-3.4.11 Skylights or monitors with movable sash that will open outwards, or fixed sash containing panes of glass or plastic that will blow out readily under pressure from within, can be used to supplement wall vents or windows, provided resistance to their displacement or opening is kept as low as consistent with the requirements for structural strength.

K-3.4.12 Flexible plastic sheets when used for vent closures should be installed in slotted frames in such a way that pressure from within bulges the sheets and releases them from the holding frame.

K-3.4.13 Fragile sheets made of plastic, when used for vent closures, should be thin sheets that will crack or rupture under less pressure, than single strength glass. For this reason, use of transparent or translucent plastic sheets is more advantageous than use of glass in window sash.

K-3.4.14 If closed vents are used, they should be larger in area than unenclosed vents to provide equivalent explosion pressure relief.

K-3.4.15 Small enclosures, such as machines, should be vented more generously than buildings, because if an explosion occurs in a machine, its entire volume may be involved.

K-3.4.16 Vents for the protection of buildings and equipment should be installed on the following basis:

- a) Small enclosures of less than 30 m³, machines and ovens of light construction: 1 000 cm² for each 0.3 m³ to 0.9 m³.
- b) For small enclosures of more substantial construction having reasonably high bursting strength: 1 000 cm² for each 0.9 m³.
- c) Fairly large enclosures of 30 m³ to 700 m³, such as bins, silos, rooms, storage tanks, etc: 1 000 cm² for each 0.9 m³ to 1.5 m³. In these cases, attempt should be made to the extent possible to predict the likely point of origin of the explosion in relation to the vent.
- d) Large rooms and buildings over 700 m³ containing hazardous equipment comprising a small fraction of the entire volume:
 - 1) For heavy reinforced concrete, walls: 100 cm² for each 2.25 m³.
 - 2) For light reinforced concrete, brick, or wood construction: 1 000 cm² for each 1.65 m³ to 2.25 m³.
 - 3) For lightweight construction such as prefabricated panels: 1 000 cm² for each 1.5 m³ to 1.65 m³.
- e) Large rooms or building over 700 m³ containing hazardous equipment comprising a large part of the entire volume of a room or building should be vented as generously as possible: 1 000 cm² for each 0.3 m³ to 1.05 m³.
- f) In order to obtain these ratios, the size of the building or room must be limited. For some hazardous materials, such as hydrogen, acetylene, carbon disulphide, etc, these limits are extremely low.
- g) Emphasis should always be placed on segregating hazardous areas by means of firewalls or separating walls to prevent spread of fire.
- h) Interior walls of light construction, such as tile, should be avoided in hazardous locations since they can cause injuries to personnel in the event of an explosion.

ANNEX M

(Clause [1.4](#))

PERFORMANCE BASED DESIGN

M-1 GENERAL

M-1.1 Performance based design may be applicable under following conditions to any occupancy, subject to:

- a) Approval from state/local authority, where they have resource (local/shared/outsourced) to evaluate a performance-based design for any building;
- b) The performance-based design should meet the goals and objectives of this Annex and should be carried out by a registered professional;
- c) The local authority having jurisdiction should make the final determination whether the performance objectives have been met; and
- d) All necessary documentation as per this annex pre- and post-construction should be part of the submission and should be retained during the entire life of the building.

M-1.2 The provisions of this Annex provides performance based design (PBD) which is applicable only to validate the fire safety aspects for heritage buildings (constructed well before the first version of NBC) that are either made available to public or are converted to other building occupancy (and constrained to add additional staircase/refuge that damages its heritage nature).

The Authority may exempt compliance (of such heritage buildings as above constructed before the publication of the first NBC) with the relevant requirements in this NBCS, if the alternate design(s) proposed based on performance-based solutions duly validated by engineering calculations/computational fluid dynamics (CFD)

modelling, achieve the objectives.

PBD is an engineering approach to fire protection design based on (1) agreed upon fire safety goals and objectives, (2) deterministic and/or probabilistic analysis of fire scenarios, and (3) quantitative assessment of design alternatives against the fire safety goals and objectives using accepted engineering tools, methodologies, and performance criteria.

M-1.3 The application of performance-based fire engineering is highly individualized, excluding its universal adoption across different structures. Due to its reliance on custom-built environmental and occupancy variables, any implementation should undergo a unique validation process by the relevant jurisdiction.

Implementing a performance-based approach necessitates a rigorous assessment of site-specific data to ensure all fire dynamics calculations reflect the unique hazards of the structure. The proponent is responsible for establishing safety parity with prescriptive standards through quantitative evidence. Due to the sophisticated nature of computational fluid dynamics (CFD) modelling, an independent technical audit is recommended to validate the integrity of the design's assumptions and outcomes.

The implementation of PBD requires a multilateral agreement on risk thresholds involving all key stakeholders, including the authority having jurisdiction (AHJ). Due to the sophisticated nature of these designs, specialized fire science expertise is mandatory to ensure that the transition from a theoretical model to a specific environment is technically sound.

NOTE — Computational models, particularly CFD simulations, should be recalculated to reflect the unique spatial geometry and airflow dynamics of the site. Also, the analysis must incorporate local emergency response variables, ensuring the design aligns with the actual equipment and arrival times of the area's fire services.

M-2 MINIMUM CRITERIA

M-2.1 The minimum criteria to accept the PBD solutions should be the compliance to the following:

- a) Safe egress ($ASET > RSET$) through validated evacuation modelling.
- b) Controlled fire and smoke spread, maintaining tenable conditions; and
- c) Reliable fire suppression and detection exceeding the minimum requirements of this Part.

NOTE — $ASET$ = Available Safe Egress Time; and $RSET$ = Required Safe Egress Time

M-2.2 Tenability refers to the conditions within a space that allow occupants to survive and potentially evacuate safely during a fire. Various factors contribute to tenability, and recommendations suggested is limiting the exposure to the most life-threatening hazards. Thus, to ensure occupant safety in fire scenarios, maintaining tenability criteria at a height of 1.8 m from the floor level is a crucial consideration.

Where possible, the toxicity effects (life threatening hazard) is arrived by obtaining the amount of fractional effective dose (FED) of asphyxiant gases (like, carbon monoxide, hydrogen cyanide); and fractional effective concentration (FEC) of the irritant gases (like, HCl, NO, SO₂). The requirements are:

- a) Maintaining $FED < 0.3$ (recommended to prevent incapacitation of sensitive individuals); and
- b) Maintaining $FEC < 0.3$ (recommended).

FEC is used to assess the combined effect of multiple irritant gases (present in the fire effluent) expressed as:

$$FEC = \sum(C_i/F_i)$$

where

C_i = the concentration of each individual irritant gas (i).

F_i = the concentration of that specific irritant gas (i) that is expected to cause a significant reduction in the possibility of escape and survivability. These F_i values are typically based on experimental data and toxicological studies.

NOTE — Specialist literature be referred to obtain the values of F_i .

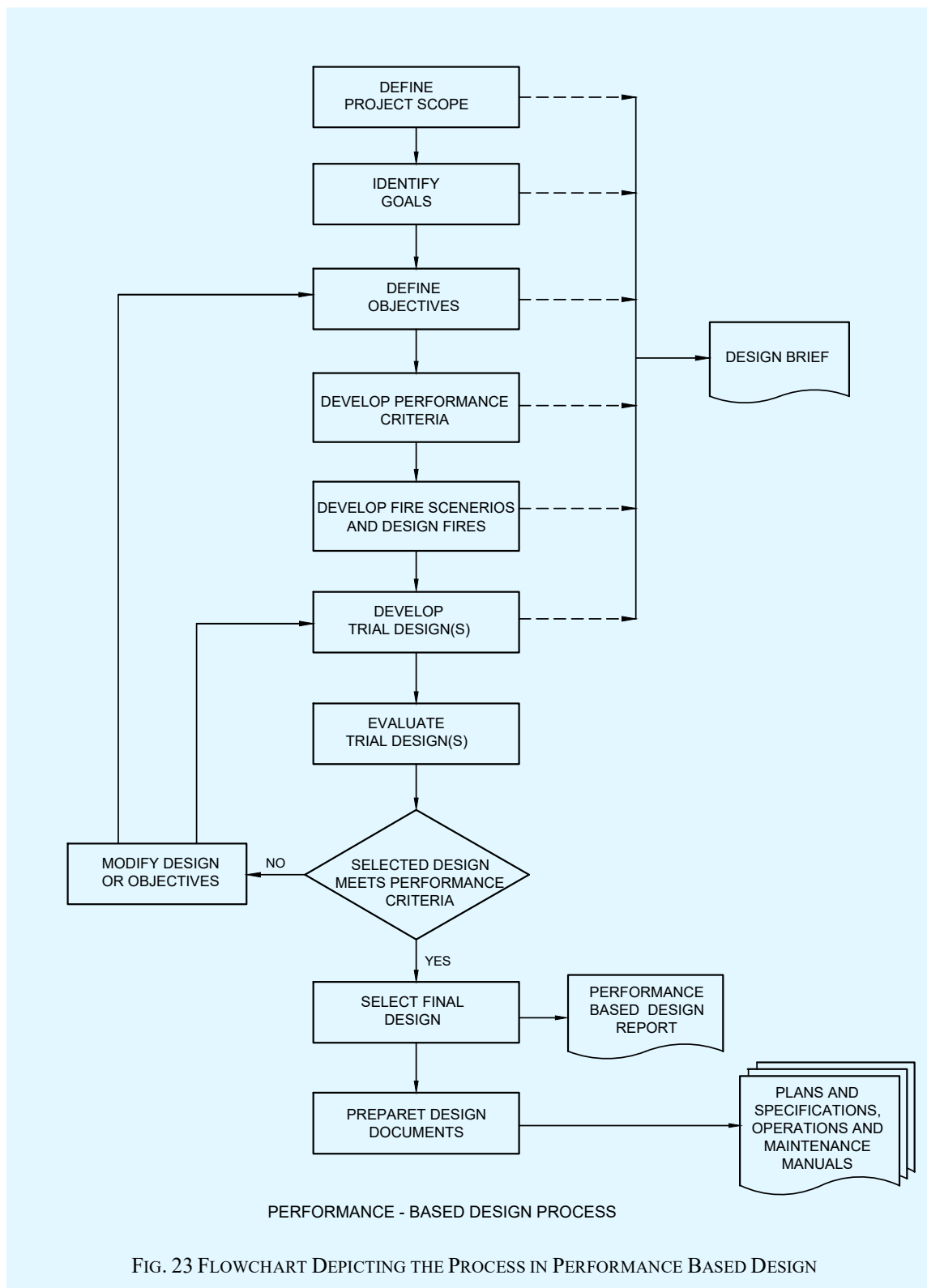
M-2.3 In addition, the following thermal exposure limits are recommended:

- a) *Convective Heat Exposure* — Air temperature should not exceed 60 °C to prevent heat stress; and

b) *Radiant Heat Exposure* — Maintain below 2.5 kW/m² to avoid skin burns and incapacitation.

M-2.4 Also, the visibility through smoke criteria is essential and the same is suggested as minimum visibility of 5 m to 10 m should be maintained for safe evacuation and wayfinding.

M-2.5 The PBD involves iterative process in arriving at the desired outcome. See [Fig. 23](#) for the typically accepted flowchart.



ANNEX N

([Table 7G](#))

(Informative)

SUGGESTED REQUIREMENTS OF FIRE PROTECTION FOR SPECIFIC INDUSTRIES
(Supplementary to [Table 7G](#))

Sl No.	Type of Factory	Type of Installation								Recommended Water Supply		Recommended Pump Capacity
		Extinguisher	Small Bore Hose Reels	Wet Riser	Down Comer	Yard Hydrant	Automatic Sprinkler System	Manual Fire Alarm System	Automatic Fire Detection & Alarm System	Static Tank at Ground Level	Overhead / Terrace Tank	litre per minute
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Minimum Firefighting Requirements for Factories/Industries – LOW HAZARD (Notes 1, 2a, 3, 4, 5 and 13)												
i)	Artisans workshops, village and cottage industries, tiny sector industries (duly licensed by the Govt. of India)	R	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
ii)	Aggregate built-up area less than 500 m ²	R	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
iii)	Aggregate built-up area from 500 m ² to less than 1 500 m ²	R	R	NR	NR	NR	NR	NR	NR	15 000	5 000 (Note 12)	R (Note 7a)
iv)	Aggregate built-up area from 1 500 m ² to less than 2 500 m ²	R	R	NR	NR	NR	NR	NR	NR	25 000	5 000 (Note 12)	R (Note 7a)

Sl No.	Type of Factory	Type of Installation								Recommended Water Supply		Recommended Pump Capacity
		Extinguisher	Small Bore Hose Reels	Wet Riser	Down Comer	Yard Hydrant	Automatic Sprinkler System	Manual Fire Alarm System	Automatic Fire Detection & Alarm System	Static Tank at Ground Level	Overhead / Terrace Tank	Fire Pump Ground Level
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
v)	Aggregate built-up area more than 2 500 m ²	Pl. see Appendix "1" and also Notes 8, 9, 10 and 11										
Minimum Firefighting requirements for Factories/Industries – ORDINARY HAZARD (Notes 1, 2b, 3, 5 and 13)												
vi)	Aggregate Built-up area less than 500 m ²	R	R	NR	NR	NR	NR	NR	NR	20 000	5 000 (Note 12)	R (Note 7a)
vii)	Aggregate Built-up area from 500 m ² to less than 1 500 m ²	R	R	NR	NR	R	NR	NR	NR	40000	10 000 (Note 12)	R (Note 7b)
viii)	Aggregate Built-up area from 1 500 m ² to less than 2 500 m ²	R	R	R	R	R	NR	R	R	60 000	15000 (Note 12)	R (Note 7c)
ix)	Aggregate Built-up area more than 2 500 m ²	Pl. see Appendix "2" and also Notes 8, 9, 10 and 11										
Minimum Firefighting requirements for Factories/Industries – High Hazard ‘A’ (Notes 2c, 5 and 13)												
x)	Aggregate built-up area less than 1 000 m ²	R	R	R	R	R	R	R	R	75 000	10 000 (Note 12)	R (Note 7d)

Sl No.	Type of Factory	Type of Installation								Recommended Water Supply		Recommended Pump Capacity
		Extinguisher	Small Bore Hose Reels	Wet Riser	Down Comer	Yard Hydrant	Automatic Sprinkler System	Manual Fire Alarm System	Automatic Fire Detection & Alarm System	Static Tank at Ground Level	Overhead / Terrace Tank	Fire Pump Ground Level
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
xi)	Aggregate built-up area from 1 000 m ² to less than 2 500 m ²	R	R	R	R	R	R	R	R	1 00 000	15 000 (Note 12)	R (Note 7d)
xii)	Aggregate built-up area more than 2 500 m ²	Pl. <i>see</i> Appendix '3' and also Notes 8, 9, 10 and 11										
Minimum Firefighting requirements for Factories/Industries – High Hazard "B" (Notes 2c and 6)												
Please <i>see</i> Appendix '4' and also Notes 8, 9, 10 and 11												
Firefighting provisions should comply with various provisions contained in Note 8. In addition, all factories/industries should be protected as per the applicable standards like oil industry safety directorate (OISD) or others as specified in Note 6.												
R - Required. NR - Not Required.												

NOTES

1 The aggregate built up area of all floors of the facility could be a process building or a storage, a utility building or open storage and any other facilities as the case may be.

2a Buildings above 9 m are not permitted for factories/industries mentioned in SI No. 1, 2 and 3 of low hazard of Annex 'A' Table.

2b Buildings/Plants above 18 m in height are not permitted for factories/industries mentioned in SI No. 1 to 4 of ordinary hazard of Annex 'A' Table. However, where nature of process within the Plants demand heights beyond 18 m (for example, where there are tall machinery, vessels, columns, reactors, etc are involved) and occupant load is less (for example, power plants, boiler installations, etc), increase in heights can be allowed. Non-industrial occupancies within the complex but beyond operational area such as offices, residential and the like should comply with respective provisions in this NBCS according to the occupancy.

2c Buildings/plants above 15 m in height are not permitted for high hazard factories/industries. However, where there are vessels, columns, reactors, etc, are involved and specifically related to the process requirements, the heights can be more than 15 m. However, where nature of process within the building/plants demand heights beyond 15 m (for example, where there are tall machinery, vessels, columns, reactors, etc are involved) and occupant load is less (for example, power plants, boiler installations, etc), increase in heights can be allowed. Non-industrial occupancies within the complex but beyond operational area such as offices, residential and the like should comply with respective provisions in this NBCS according to the occupancy.

3 In case, the factories/industries complex consists of building(s) with occupancies having mixed hazards, fire protection systems should be governed by the highest hazard.

4 Those factories/industries which were defined or licensed as "Artisans workshops, Village and Cottage Industries, Tiny sector industries" by the Government of India should need to provide only extinguishers and buckets, their quantity and distribution should comply with IS 2190.

5 Multiple occupancy industrial estates (all in one building) should be protected according to the requirements for 'Ordinary' hazard industries. No high hazard occupancies should be allowed in such buildings.

6 In case of high hazard industries-A (where applicable) and high hazard industries-B like Petrochemical industries, Refineries and the like, compliance with the requirements of other agencies like Oil Industry Safety Directorate (OISD), Petroleum and Explosives Safety Organisation (PESO), Petroleum and Natural Gas Regulatory Board (PNGRB), Oil Mines Regulations (OMR), Central Electricity Authority (CEA) should be necessary in addition to those from Annex 'A' Table.

7a Entire occupancy should be protected by one electric driven main pump of at least 450 lpm discharge capacity at 50 m head, one standby pump of similar capacity (with reliable back-up power supply like DG Set).

7b Entire occupancy should be protected by one electric driven main pump of at least 900 lpm discharge capacity at 50 m head, one standby pump of similar capacity (with reliable back-up power supply like DG Set).

7c Entire occupancy should be protected by one electric driven main pump of at least 1 620 lpm discharge capacity at 70 m head, one standby diesel engine driven pump of similar capacity and one jockey pump of capacity of at least 180 lpm.

7d Entire occupancy should be protected by one electric driven main pump of at least 2 280 lpm discharge capacity at 70 m head, one standby diesel engine driven pump of similar capacity and one jockey pump of capacity of at least 180 lpm.

8 Entire occupancy should be protected by hydrant, sprinkler, water spray, water mist, clean agent, foam, fire alarm systems, etc as applicable in accordance with the current versions of IS 13039, IS 15105, IS 15325, IS 15519, IS 12385, IS 15693, IS 2189, IS 15908, etc in all respects that include design and installation of pumps, pipelines, etc.

9 Certain occupancies may also be protected by automatic water mist systems with appropriate (pressure) applications. Installation and design of such systems should comply to IS 15519.

10 For all occupancies, pumping capacities and water requirements should need to be provided wherever indicated in respective columns in Annex 'N' Table. However, where pumping capacities and water requirements are not indicated in the respective columns, reference should be made to the concerned Indian Standard codes of practices for hydrant/sprinkler/water spray, etc, for details. In either case, design and installation of systems like hydrant, sprinkler, spray systems, etc, should strictly be carried out as per provisions in respective Indian Standards stated in Note 8 above.

11 Where application of water as extinguishing medium is not appropriate due to the presence of water-reactive materials or other valid acceptable reasons, a suitable alternative extinguishing system and method should be provided in accordance with relevant Indian Standards.

12 Overhead fire water tank should be connected to hose reels and also sprinkler system (where applicable).

13 Classifications of various occupancies shown in Annex 'B' is only indicative for ready reference and understanding. While designing hydrant and sprinkler systems, classifications for the purpose of design and installation should strictly adhere to those available in IS 13039 and IS 15105 respectively.

Appendix '1' to Annex N								
Low Hazard Factories/Industries with Aggregate Built-up area more than 2 500 m ²								
Sl No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	Abrasive manufacturing premises	R	R	R	NR	Required for packing material/other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
ii)	Aerated water factories, mineral water factories	R	R	R	NR	Required for packing material/ bottles and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
iii)	Agarbatti manufacturing	R	R	R	NR	Required for packing material/Finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
iv)	Analytical and/or quality control laboratories (without pilot plant)	R	R	R	Required if the combined floor area of the building is more than 5 000 m ²	NR	Suitable detection system backed up by manual call stations should be provided for the entire factory where combined area of the process/Storage blocks	

Appendix '1' to Annex N								
Low Hazard Factories/Industries with Aggregate Built-up area more than 2 500 m ²								
Sl No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
							exceeds 5 000 m ²	
v)	Areca nut slicing and/or betel nut factories	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
vi)	Asbestos steam packing and lagging manufacturers	R	R	R	NR	NR	Suitable detection system backed up by manual call stations should be provided for the entire factory where combined area of the process/storage blocks exceeds 5 000 m ²	
vii)	Battery charging/battery service station	R	R	R	R	R	Suitable detection system backed up by manual call stations should be provided for the entire factory where combined area of the process/storage blocks exceeds 5 000 m ²	Water mist systems as per IS 15519

Appendix '1' to Annex N								
Low Hazard Factories/Industries with Aggregate Built-up area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
viii)	Battery manufacturing	R	R	R	R	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Suitable detection system backed up by manual call stations should be provided for the entire factory where combined area of the process/storage blocks exceeds 5 000 m ²	Water mist systems in battery charging stations as per IS 15519
ix)	Breweries	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
x)	Brick works	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xi)	Canning factories	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	

Appendix '1' to Annex N								
Low Hazard Factories/Industries with Aggregate Built-up area more than 2 500 m ²								
Sl No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xii)	Cardamom factories	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xiii)	Cement factories and/or asbestos or concrete products manufacturing	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	Clean agent-inert gas systems (CO ₂ , Nitrogen) for bag filters, mill house, coal bins
xiv)	Ceramic factories and crockery and stoneware pipe manufacturing	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xv)	Clay works	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xvi)	Clock and watch manufacturing	R	R	R	NR	Raw Material godowns should be sprinklered if the value of plastic/rubber and other similar commodities in the storage area exceeds 20 percent of the total	Manual fire alarm system should be provided throughout, operations like spray painting, wood working, solvent cleaning etc, should need to be spot-protected if combined area	

Appendix '1' to Annex N								
Low Hazard Factories/Industries with Aggregate Built-up area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
						value of the storage building	occupied by such operations exceed 10 percent of the total manufacturing area	
xvii)	Coffee curing, roasting and grinding premises	R	R	R	Required if the area occupied by process building exceeds 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	If combined area of the entire factory is more than 2 000 m ² , manual fire alarm system should be required, Suitable detection system backed up by Manual call stations should be provided for the entire factory where combined area of the process/storage blocks exceeds 5 000 m ²	If combined area of process and storage blocks in the same building exceeds 2 000 m ² , sprinkler installation will be required in the entire area
xviii)	Condensed milk factories, milk pasteurising plant and dairies	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xix)	Confectionery manufacturing	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	

Appendix '1' to Annex N

Low Hazard Factories/Industries with Aggregate Built-up area more than 2 500 m²

SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xx)	Crockery and stoneware pipe manufacturing	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxi)	Electric generating houses (Hydro electric)	R	R	R	NR	Sprinkler protection for storage areas not forming part of main occupancy if area exceeds 2 000 m ²	Auto fire alarm system for all areas except detached storage buildings and utility blocks where manual alarm systems should be provided	Water spray system for transformers, FDS/NIFPS system, cable galleries, conveyors including transfer towers, etc as per IS 3034 and IS 15325
xxii)	Electric Lamps (Incandescent and Fluorescent) and TV Picture Tube Manufacturing	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxiii)	Electric Sub-Station/ Distribution Station	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	Water spray system for transformers if capacity thereof is more than 10 MVA or oil capacity exceeds 2 000 litre.

Appendix '1' to Annex N								
Low Hazard Factories/Industries with Aggregate Built-up area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
								(As per CEA Regulations and IS 3034, IS 15325)
xxiv)	Electro Plating Works	R	R	R	NR	NR	NR	
xxv)	Electronic and/or Computer equipments Assemble and Manufactures	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxvi)	Engineering workshops heavy (Structural Steel Fabricators, Sheet Metal Fabricators (Tin Printing excluded), Hot or Cold Rolling, Pipe Extruding, Stamping, Pressing, Drawing (other than wire Drawing) or Forging Mills, Ore Crushing Mills, Ferrous and non Ferrous Metal Smelting or Extraction Works and Foundries (allowing storage of Wooden	R	R	R	NR	Raw material godowns should be sprinklered if the value of plastic/rubber and other similar commodities in the storage area exceeds 20 percent of the total value of the storage building.	Manual fire alarm system should be provided throughout, operations like manufacture of plastic/rubber components, powder coating/spray painting, drying of painted goods, use or storage of flammable liquids having flash point of 23 °C and below, wood working, upholstering, engine testing and flammable gas plant should need to be spot-protected by appropriate means if combined area	

Appendix '1' to Annex N

Low Hazard Factories/Industries with Aggregate Built-up area more than 2 500 m²

SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Patterns only) and galvanizing works and the like)						occupied by such operations exceed 10 percent of the total manufacturing area. Else such areas should be fire separated from the main operations	
xxvii)	Engineering workshops Light (Blacksmiths, Electroplating, Anodizing or Lead, Zinc or other Metal Coating Works, Machine Shops, Heat Treatment shops, Electro Magnets manufacturing, Electrical and Mechanical Machinery Assemblers/Manufacturers, General Engineering Goods and Merchandise Manufacturers, Maintenance and Repair Workshops, Mfrs. of Welding Electrodes and Wire drawing, Automobile factories and the like)							

Appendix '1' to Annex N								
Low Hazard Factories/Industries with Aggregate Built-up area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xxviii)	Fruits and vegetables dehydrating/drying factories	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxix)	Fruit products and condiment factories	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxx)	Glass and glass Fibre manufacturing	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxxi)	Gold thread factories/gilding factories	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxxii)	Granite manufacturing	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	If inflammable liquids having a flash point of less than 65° are used, manual foam spray

Appendix '1' to Annex N

Low Hazard Factories/Industries with Aggregate Built-up area more than 2 500 m²

SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
								system should be used in the areas concerned
xxxiii)	Gum and/or Glue and Gelatine Manufacturing	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxxiv)	Ice Candy and Ice-cream Manufacturing	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxxv)	Ice Factories	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxxvi)	Ink (excluding Printing Ink) Factories	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	

Appendix '1' to Annex N								
Low Hazard Factories/Industries with Aggregate Built-up area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xxxvii)	Laundries	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxxviii)	Mica products manufacturing	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxxix)	Pottery works	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xl)	Poultry farms	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xli)	Salt Crushing Factories and Refineries	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	

Appendix '1' to Annex N								
Low Hazard Factories/Industries with Aggregate Built-up area more than 2 500 m ²								
Sl No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xlii)	Sugar Candy Manufacturing	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xliii)	Sugar Factories and Refineries	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xliv)	Tanneries/Leather Goods Manufacturers	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xlvi)	Tea Blending and Tea Packing Factories	R	R	R	Required if the area occupied by process building exceeds 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	If combined area of the entire factory is more than 2 000 m ² , manual fire alarm system should be required, suitable detection system backed up by manual call stations should be provided for the entire factory where combined area of the process/storage blocks exceeds 5 000 m ²	If combined area of process and storage blocks in the same building exceeds 2 000 m ² , sprinkler installation will be required in the entire area

Appendix '1' to Annex N								
Low Hazard Factories/Industries with Aggregate Built-up area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xlvi)	Umbrella Assembling Factories	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xlvii)	Vermicelli Factories	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xlvihi)	Water Treatment/Water Filtration Plants and Water Pump House	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xlxi)	Zinc/Copper/Aluminium Factories	R	R	R	NR	NR	Suitable detection system backed up by manual call stations should be provided for the entire factory where combined area of the process/storage blocks exceeds 5 000 m ²	

R – Required. NR – Not Required.

Appendix '2' to Annex N								
Ordinary Hazard Factories/Industries with Aggregate Built-up Area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	Airport and other transportation terminal building	R	R	R	To be fully sprinklered	To be fully sprinklered (cargo)	To be fully protected by automatic intelligent fire alarm system	Kitchen hoods and ducts should be protected with wet chemical systems as per this NBCS. Also, systems as per ICAO standards.
ii)	Aluminium factories	R	R	R	NR	NR	NR	
iii)	Bakeries	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
iv)	Beedi factories	R	R	R	NR	NR	NR	
v)	Biscuit factories	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
vi)	Bobbin factories	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	

Appendix '2' to Annex N								
Ordinary Hazard Factories/Industries with Aggregate Built-up Area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
vii)	Bookbinders, envelopes and paper bag manufacturers	R	R	R	Required when a single un-compartmented area is more than 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
viii)	Cable manufacturing	R	R	R	Required when a single un-compartmented area is more than 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
ix)	Carbon paper/typewriter ribbon manufacturers	R	R	R	Required when a single un-compartmented area is more than 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
x)	Cardboard box manufacturing	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xi)	Carpenters, wood wool and furniture manufacturers. Basket weavers, cane furniture manufacturing	R	R	R	Required when a single un-compartmented area is more than 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	

Appendix '2' to Annex N								
Ordinary Hazard Factories/Industries with Aggregate Built-up Area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xii)	Carpet and drugget factories	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xiii)	Chemical manufacturing involving chemicals, dyes, intermediates, pigments, acids, medicinal/ pharmaceuticals formulation and toilet preparation including soap manufacturing, chemical mixing and blending factories using no chemicals/ solvents having flash point below 23 °C	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	If the quantity of chemicals/ solvents having flash point below 23 °C is less than 200 litre in process and there is no storage of such solvents/chemicals more than 200 litre then such industries can be considered under this table. Else, <i>see</i> High hazard
xiv)	Cigar and cigarette factories	R	R	R	Required in full	Required in full	Automatic fire alarm system should be provided for godown buildings, manual system for process blocks	

Appendix '2' to Annex N								
Ordinary Hazard Factories/Industries with Aggregate Built-up Area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xv)	Coffee grinding premises	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xvi)	Coir, carpets, rugs and tobacco, hides and skin presses	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xvii)	Cork products manufacturing	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xviii)	Dry cleaning, dyeing, laundries	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xix)	Electric generating stations (other than hydel)	R	R	R	NR	Sprinkler protection for storage areas not forming part of main occupancy if area exceeds 2 000 m ²	Automatic fire alarm system for all areas except detached storage buildings and utility blocks where manual alarm systems should be provided	Water spray system for transformers, FDS/NIFPS system, cable galleries, conveyors including transfer towers, etc as per IS 3034 and IS 15325

Appendix '2' to Annex N
Ordinary Hazard Factories/Industries with Aggregate Built-up Area more than 2 500 m²

SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xx)	Enamelware factories	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxi)	Filter and wax paper manufacturing	R	R	R	Required when a single un-compartmented area is more than 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxii)	Flour mills	R	R	R	Required if combined floor plate area exceeds 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxiii)	Garment makers	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxiv)	Ghee factories (other than vegetable)	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	

Appendix '2' to Annex N								
Ordinary Hazard Factories/Industries with Aggregate Built-up Area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xxv)	Grain and/or seeds disintegrating and/or crushing factories	R	R	R	Required if combined floor plate area exceeds 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxvi)	Grease manufacturing	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxvii)	Hat and topee factories	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxviii)	Hosiery, lace, embroidery and thread factories	R	R	R	Required if combined floor plate area exceeds 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxix)	Incandescent gas mantle manufacturers	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	

Appendix '2' to Annex N								
Ordinary Hazard Factories/Industries with Aggregate Built-up Area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xxx)	Industrial gas manufacturing Including halogenated hydrocarbon gases	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2000 m ²	
xxxi)	Linoleum factories	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxxii)	Man-made yarn/fibre manufacturing (except acrylic)	R	R	R	Required if combined floor plate area exceeds 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Automatic fire alarm system should be provided when combined floor plate exceeds 5 000 m ² , else manual systems should be provided	
xxxiii)	Manure and fertiliser works. (blending, mixing and granulating only)	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxxiv)	Mineral oil blending and processing	R	R	R	Required when a single un-compartmented area is more than 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding	Manual fire alarm system for entire unit if area of single largest building	

Appendix '2' to Annex N								
Ordinary Hazard Factories/Industries with Aggregate Built-up Area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
						2 000 m ² in area	exceeds 2 000 m ²	
xxxv)	Oil and leather cloth factories	R	R	R	Required when a single un-compartmented area is more than 2 000 m ²	Required for Packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxxvi)	Oxygen plants	R	R	R	Required for manufacturing blocks	Required for storage blocks	Automatic fire alarm system should be provided throughout	
xxxvii)	Plywood manufacturing/wood veneering factories	R	R	R	Required when a single un-compartmented area is more than 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xxxviii)	Paper and cardboard mills	R	R	R	NR	Raw material and Finished goods storage should be provided with sprinklers	Automatic fire alarm system should be provided throughout	
xxxix)	Plastic goods manufacturing polytetra fluoroethylene, flourinated ethylene-	R	R	R	Required if combined floor plate area exceeds 2 000 m ²	Raw material and finished goods storage should be provided with sprinklers if area exceeds	Manual fire alarm system for entire unit if area of single largest building	

Appendix '2' to Annex N								
Ordinary Hazard Factories/Industries with Aggregate Built-up Area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	propylene, poly-chlorotrifluoroethylene, polyvinyl chloride, polyvinyl acetate, polyvinylidene chloride, vinyl chloride-vinylidene, chloride copolymer, polyvinylidene fluoride, polycarbonate, phenylene oxide polymer, polysulfone, chlorinated polyether, phenolics, melamine formaldehyde, urea formaldehyde, silicones, phenol furfural, phenol formaldehyde, tetrafluoroethylene, polyvinyl fluoride, cresol formaldehyde, polyamide (nylon). rope works, assembling of plastic goods such as toys and the like					2 000 m ²	exceeds 2 000 m ²	
xl)	Plastic goods manufacturing polyethylene,	R	R	R	Required if combined floor plate area exceeds	Raw material and finished goods storage should be provided with	Manual fire alarm system for entire unit if area of building(s) exceeds	Many chemicals are water reactive with water and alternative systems

Appendix '2' to Annex N								
Ordinary Hazard Factories/Industries with Aggregate Built-up Area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	polypropylene, polybutylene, ethylenemethacrylic acid ionomer, polyvinyl alcohol, polyvinyl butyrate, polyvinyl formaldehyde, polystyrene, styrene-acrylonitrile copolymer, acrylonitrile- butadiene-styrene, polymethyl methacrylate, cellulose acetate, cellulose acetate butyrate, cellulose propionate, Cellulose triacetate, ethyl cellulose, acetal, epoxy, polyesters, polyurethanes, acid monomer, acrylate, allyl, methyl methacrylate, polythene				2 000 m ²	sprinklers	2 000 m ²	(not involving water) should be used
xli)	Plastic goods manufacturing foamed plastics	R	R	R	R	Raw material and finished goods storage should be provided with sprinklers	Manual fire alarm system for entire unit	

Appendix '2' to Annex N								
Ordinary Hazard Factories/Industries with Aggregate Built-up Area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xlii)	Printing press premises	R	R	R	Required if area of building(s) exceeds 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system required if area of building(s) exceeds 2 000 m ²	
xliii)	Pulverising and crushing mills of metal and metallic ores (excluding aluminium and magnesium metals)	R	R	R	NR	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system is required if the floor area exceeds 2 000 m ²	
xliv)	Pulverising and crushing mills of metallic ores of aluminum and magnesium and coal	R	R	R	Required if area of building(s) exceeds 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Automatic fire alarm systems if the floor area exceeds 2 000 m ²	Depending on the circumstances like water reacting chemicals, open-sided structure, etc, sprinklers may be replaced with water mist systems
xlvi)	Rice mills	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
xlvii)	Rope works (other than plastic)	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	

Appendix '2' to Annex N								
Ordinary Hazard Factories/Industries with Aggregate Built-up Area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xlvi)	Rubber tyres and tubes manufacturing	R	R	R	Required if area of building(s) exceeds 2 000 m ²	Required for packing material/finished goods godowns	Automatic fire alarm systems if the floor area exceeds 2 000 m ²	
xlvi)	Shellac factories	R	R	R	NR	NR	Manual fire alarm system required	Only local protection, that is, where chemicals used, have a flash point below 23 °C, with medium velocity water spray for manufacturing blocks, similar protection for the area where materials are stored (raw material as well as finished goods)
xlii)	Silk filatures and cocoon stores	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
i)	Spray painting	R	R	R	R	NR	NR	If area of the building exceeds 500 m ² , protection systems like sprrinklers, inert gas agents should be installed

Appendix '2' to Annex N								
Ordinary Hazard Factories/Industries with Aggregate Built-up Area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
li)	Soaps and glycerine factories	R	R	R	NR	Raw material and finished goods godowns should be sprinklered if the storage area is more than 2 000 m ²	Automatic fire alarm systems if the floor area exceeds 2 000 m ²	
lii)	Starch factories	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
liii)	Steel plants	R	R	R	NR	NR	Automatic fire alarm systems if the floor area exceeds 2 000 m ²	
liv)	Tanneries/leather goods manufac- turers	R	R	R	Required if area of building(s) exceeds 2 000 m ²	Required for packing material/finished goods and other raw materials godowns if exceeding 2 000 m ² in area	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
lv)	Tank farms other than those categorised under high hazard 'A'	R	R	R	Water spray/foam system as applicable should be provided for all tanks			

Appendix '2' to Annex N								
Ordinary Hazard Factories/Industries with Aggregate Built-up Area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
lvi)	Textile mills	R	R	R	Required for all process blocks from mixing operations till weaving shed	Required for raw material and finished goods godowns	Automatic fire alarm systems should be provided for raw materials and finished goods godowns	Hazardous areas like singeing, raising, polymerizing, etc should be fire separated (2 h) from adjoining blocks. Concealed spaces/return air ducts/plenum in all process blocks upto weaving should be provided with quick response sprinklers. Diversion systems should be provided where the chute feed machines are installed between blow and carding sections
lvii)	Tea factories	R	R	R	Required if area of building(s) exceeds 2 000 m ²	Required if area exceeds 2 000 m ²	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
lviii)	Tobacco (chewing) and pan-masala making	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	

Appendix '2' to Annex N								
Ordinary Hazard Factories/Industries with Aggregate Built-up Area more than 2 500 m ²								
SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
lix)	Tobacco grinding and crushing	R	R	R	NR	Required if the area of building exceeds 2 000 m ²	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
lx)	Tobacco redrying factories	R	R	R	Required if the area of building exceeds 2 000 m ²	Required if the area of building exceeds 2 000 m ²	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
lxi)	Woollen mills	R	R	R	Required for all processing areas	Required for raw material/finished goods godowns	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	
R – Required. NR – Not Required.								

Appendix '3' to Annex N

High Hazard "A" Factories/Industries with Aggregate Built-up Area more than 2 500 m²

SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	Aircraft hangers	R	R	R	Required in full	Required in full	Automatic intelligent fire alarm system with flame detectors	High expansion foam systems should need to be installed. Also, systems as per ICAO standards
ii)	Aluminium/magnesium powder plants	R	R	R	NR	NR	Manual fire alarm system for entire unit if area of single largest building exceeds 2 000 m ²	Special dry powder systems
iii)	Bituminized paper and/or hessian cloth manufacturing including tar felt manufacturing	R	R	R	Required if area of the building exceeds 2 000 m ²	Raw material and finished goods godowns should be sprinklered	Manual fire alarm system for entire unit	
iv)	Chemical manufacturing involving chemicals, dyes, intermediates, pigments, acids, medicinal/ pharmaceuticals formulation and toilet preparation including soap manufacturing, chemical mixing and blending factories using chemicals/ solvents having flash point below 23 °C and metallic	R	R	R	MV spray systems and/or foam system should be provided locally as applicable for vessels, pumps, compressors, reactors, columns, etc where chemicals having flash point below 23 °C are in process. (Suitable care should taken to check whether chemicals are water reacting)	MV spray systems and gaseous protection systems as applicable for godowns storing materials with flash points below 23 °C. (Suitable care should taken to check whether chemicals are water reacting)	Manual fire alarm systems should be provided for entire unit	All tanks having a diameter of 9 m or more, storing Class A or B chemicals should be provided with medium velocity spray systems and foam systems as applicable backed up with suitable detection systems

Appendix '3' to Annex N

High Hazard "A" Factories/Industries with Aggregate Built-up Area more than 2 500 m²

SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	nitrates, coal tar and wood distillation plant							
v)	Cotton waste factories	R	R	R	Entire area should be sprinklered if the combined area of all buildings exceed 2 000 m ²		Manual fire alarm systems should be provided for entire unit	
vi)	Coal and/or coke and/or charcoal ball briquettes manufacturing	R	R	R	NR	Raw materials and finished goods godown should be sprinklered when area is more than 2 000 m ²	Manual fire alarm systems should be provided for entire unit	
vii)	Celluloid goods manufacturing	R	R	R	NR	NR	Manual fire alarm systems should be provided for entire unit	Alcohol resistant foam, carbon dioxide (CO ₂), dry chemicals where applicable for all process areas and godowns
viii)	Cigarette filter manufacturing	R	R	R	Required for all process areas	Raw materials and Finished goods godown should be sprinklered when area is more than 2 000 m ²	Manual fire alarm systems should be provided for entire unit	
ix)	Cinema films and t.v. production studios	R	R	R	Required for all process areas	Godowns should be sprinklered if area is more than 2 000 m ²	Automatic fire alarm system should be provided	
x)	Cotton seed cleaning or de-linting factories, cotton ginning and pressing units	R	R	R	Required for all process areas	Godowns should be sprinklered if area is more than 2000 m ²	Manual fire alarm systems should be provided for entire unit	

Appendix '3' to Annex N

High Hazard "A" Factories/Industries with Aggregate Built-up Area more than 2 500 m²

SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xi)	Distilleries	R	R	R	Local protection with medium velocity spray/foam systems as applicable should be provided for process areas	NR	Automatic fire alarm system should be provided	
xii)	Duplicating and stencil paper manufacturing	R	R	R	Required for all process areas	Godowns should be sprinklered if area is more than 2 000 m ²	Manual fire alarm systems should be provided for entire unit	
xiii)	Fire-works manufacturing	R	R	R	Required for all process areas	Required for all raw material and finished goods godowns	Manual fire alarm systems should be provided for entire unit	
xiv)	Foamed plastics manufacturing and/or converting plants	R	R	R	Required for all process areas	Required for all raw material and finished goods godowns	Manual fire alarm systems should be provided for entire unit	
xv)	Jute mills and jute presses	R	R	R	Required for all process areas	Required for all raw material and finished goods godowns	Manual fire alarm systems should be provided for entire unit	
xvi)	LPG bottling plants	R	R	R	Required for all process areas	Required for all storage and decantation, truck loading areas	Manual fire alarm systems should be provided for entire unit	LPG storage tankage like bullets and spheres should be protected with medium velocity water spray systems backed up with suitable detection systems as per IS 15325.

Appendix '3' to Annex N

High Hazard "A" Factories/Industries with Aggregate Built-up Area more than 2 500 m²

SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
								Also, systems as per OISD- STD-144 and other applicable OISD standards
xvii)	Match factories	R	R	R	Required for all process areas	Required for all raw material and finished goods godowns	Manual fire alarm systems should be provided for entire unit	
xviii)	Man made fibres (acrylic fibres/yarn making)	R	R	R	Required for all process areas	Required for all raw material and finished goods godowns	Automatic fire alarm systems should be provided for entire unit	
xix)	Mattress and pillow making	R	R	R	Required for all process areas if area of the building is more than 2 000 m ²	Required for all raw material and finished goods godowns having area more than 2 000 m ²	Manual fire alarm systems should be provided for entire unit	
xx)	Metal or tin printers	R	R	R	Required for all process areas if area of the building is more than 2 000 m ²	Required for all raw material and finished goods godowns having area more than 2 000 m ²	Manual fire alarm systems should be provided for entire unit	
xxi)	Oil mills	R	R	R	Required for all process areas if area of the building is more than 2 000 m ²	Required for all raw material and finished goods godowns having area more than 2 000 m ²	Automatic fire alarm systems should be provided for entire unit	Local protections such as medium velocity spray or foam systems as applicable should be provided in process and storage areas

Appendix '3' to Annex N

High Hazard "A" Factories/Industries with Aggregate Built-up Area more than 2 500 m²

SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xxii)	Oil extraction plants (other than those forming part of ghee factories and oil refining factories)	R	R	R	Required for all process areas if area of the building is more than 2 000 m ²	Required for all raw material and finished goods godowns having area more than 2 000 m ²	Automatic fire alarm systems should be provided for entire unit	Local protections such as medium velocity spray or foam systems as applicable should be provided in process and storage areas
xxiii)	Oil terminals/depots handling flammable liquids having flash point of 23 °C and below	R	R	R	Automatic medium velocity water spray and /or foam systems should be provided as applicable for all tanks backed up with suitable automatic detection systems. Also, systems as per OISD-STD-117 and other applicable OISD standards			
xxiv)	Paints and varnish factories	R	R	R	Required for all process areas if area of the building is more than 2 000 m ²	Required for all raw material and finished goods godowns having area more than 2 000 m ²	Automatic fire alarm systems should be provided for entire unit	Local protections such as medium velocity spray or foam systems as applicable should be provided in process and storage areas
xxv)	Printing ink manufacturing	R	R	R	Required for all process areas if area of the building is more than 2 000 m ²	Required for all raw material and finished goods godowns having area more than 2 000 m ²	Automatic fire alarm systems should be provided for entire unit	Local protections such as medium velocity spray or foam systems as applicable should be provided in process and storage areas
xxvi)	Saw mills	R	R	R	NR	NR	Manual fire alarm system should be provided for entire unit	

Appendix '3' to Annex N

High Hazard "A" Factories/Industries with Aggregate Built-up Area more than 2 500 m²

SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xxvii)	Solvent extraction plants	R	R	R	NR	NR	Automatic fire alarm systems should be provided for entire unit	Entire factory and storage tanks to be protected by medium velocity water spray system and/or foam system
xxviii)	Surgical cotton manufacturers	R	R	R	Required for all process areas	Required for all raw material and finished goods godowns	Automatic fire alarm systems should be provided for entire unit	
xxix)	Tank farms storing flammable liquids having flash point of 23 °C and below	R	R	R	Automatic medium velocity water spray and /or Foam systems should be provided as applicable for all tanks backed up with suitable automatic detection systems. Also, systems as per OISD-STD-116 and other applicable OISD standards			
xxx)	Tarpaulin and canvas proofing factories	R	R	R	Required for all process areas where building area is more than 2 000 m ²	Required for all raw material and finished goods godowns where the building area is more than 2 000 m ²	Automatic fire alarm systems should be provided for entire unit	
xxxi)	Turpentine and rosin distilleries	R	R	R	NR	NR	Automatic fire alarm systems should be provided for entire unit	Local protections such as medium velocity water spray or foam systems as applicable should be provided in process and storage areas

Appendix '3' to Annex N

High Hazard "A" Factories/Industries with Aggregate Built-up Area more than 2 500 m²

SI No.	Description	Extinguisher	Hose Reel	Hydrant System	Sprinkler System		Fire Alarm System	Special Protection
					Process Blocks	Storage Blocks		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
xxxii)	Tyre retreading and resoling factories	S	R	R	Required for all process areas where building area is more than 2 000 m ²	Required for all Raw material and finished goods godowns where the building area is more than 2 000 m ²	Automatic fire alarm systems should be provided for entire unit	
R – Required. NR – Not Required. S – Suitable								

Appendix '4' to Annex N

High Hazard 'B' Factories/Industries (All Chemicals, Petroleum, Gas Processing, Petrochemicals, Fertilizer Plants, Thermal Power Plants, etc)

SI No.	Description	Fire Extinguishers, Firefighting Equipment	Hydrant System, Fire Monitors, Hose Reels	Fire Alarm System	Sprinkler System	High Volume Long Range (HVLR) Monitors	Gas/HC/Fire/Heat Detection System	Automatic Heat/Fire Detection and Suppression System	Remarks
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
i)	Process units (columns, boilers, furnaces, reactors, vessels, drums, etc)	R	R (Note 1)	R	R (Note 2)	Refer Note 3	R (Note 4)	NR	
ii)	Tank farm areas (storage of liquid/gas flammable)	R	R	R	R	R	R	R	High volume long range monitors of capacity 500 to

Appendix '4' to Annex N

High Hazard 'B' Factories/Industries (All Chemicals, Petroleum, Gas Processing, Petrochemicals, Fertilizer Plants, Thermal Power Plants, etc)

Sl No.	Description	Fire Extinguishers, Firefighting Equipment	Hydrant System, Fire Monitors, Hose Reels	Fire Alarm System	Sprinkler System	High Volume Long Range (HVLR) Monitors	Gas/HC/Fire/Heat Detection System	Automatic Heat/Fire Detection and Suppression System	Remarks
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	products Class A and cluster of Class A and B products)				(Note 5, 6)	(Note 7, 8)		(Note 9)	4 000 gpm (variable flow) as required as per Standards; HVLR monitors of 10 500 gpm should be considered to fight fire in large dia tanks. Fire water monitors around the tank farm area as per IS/OISD standards
iii)	Pump house area (Pumps and Compressors - rotary equipment)	R	R	R	R	NR	R (Note 4)	NR	
iv)	Loading/unloading gantry (tank wagon and tank truck loading/unloading)	R	R	R	R	R	R (Note 4)	NR	Water sprinkler throughout the length of Gantry. Hydrocarbon (HC) detectors at drains
v)	LPG filling stations	R	R	R	R	NR	R	NR	LPG storage – Mounded storage, Horton spheres, Bullets should have water sprinklers with medium velocity spray

Appendix '4' to Annex N

High Hazard 'B' Factories/Industries (All Chemicals, Petroleum, Gas Processing, Petrochemicals, Fertilizer Plants, Thermal Power Plants, etc)

Sl No.	Description	Fire Extinguishers, Firefighting Equipment	Hydrant System, Fire Monitors, Hose Reels	Fire Alarm System	Sprinkler System	High Volume Long Range (HVLR) Monitors	Gas/HC/Fire/Heat Detection System	Automatic Heat/Fire Detection and Suppression System	Remarks
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
vi)	Conveyor belt areas	R	R	R	R	NR	NR	R (Note 10)	Handling flammable dusts like coal / lignite dust, etc
vii)	Sub-station and electrical panels	R	NR	R	NR	NR	NR	R (Note 11, 12)	Substation to be air pressurized
viii)	Chlorination plant	R	NR	NR	R	NR	R (Note 13)	NR	
ix)	Transformers	R	R	R	R	NR	NR	R (Note 14)	
x)	Cable trenches, racks and cable galleries	R	NR	R	NR	NR	R	R (Note 11 and 12)	Adequate ventilation and sealing in cable gallery housing
xi)	Quality control lab	R	R	R (Note 15)	R	NR	R	NR	

Appendix '4' to Annex N

High Hazard 'B' Factories/Industries (All Chemicals, Petroleum, Gas Processing, Petrochemicals, Fertilizer Plants, Thermal Power Plants, etc)

Sl No.	Description	Fire Extinguishers, Firefighting Equipment	Hydrant System, Fire Monitors, Hose Reels	Fire Alarm System	Sprinkler System	High Volume Long Range (HVL R) Monitors	Gas/HC/Fire/Heat Detection System	Automatic Heat/Fire Detection and Suppression System	Remarks
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
xii)	Admin. building	R	R	R (Note 15)	R	NR	NR	NR	
xiii)	Control room	R	NR	R	R	NR	NR	R (Note 16)	
xiv)	Firefighting pumps and fire water system	Electric and diesel/HSD driven pumps of similar capacity and same head. Jockey pump with 3 percent to 5 percent capacity of main pumps. Overhead fire water tanks and reservoir with adequate capacity to fight two simultaneous fires.							
xv)	Fire tenders	Independent fire station equipped with water, foam, dry chemical power tenders and rescue vehicles - response time on fire call must be within 05 min.							
R – Required. NR – Not Required.									

NOTES

- 1 Fire water network with fire hydrants, monitors all around the process unit and tank farm areas should be provided in accordance with relevant Indian Standards, OISD. Fire hydrants should be at different platforms/levels in TPH/boilers/furnaces/columns/reactors.
- 2 Water sprinklers in technological platforms for all static equipment normally reflux drums, air-fin fan coolers, etc should be provided in accordance with relevant Indian Standards, OISD.
- 3 Elevated fire water monitors (variable flow) to fight fires at tall columns / boiler house / reactors / technological platforms, etc should be provided in accordance with relevant Indian Standards, OISD.
- 4 Hydrocarbon detectors, gas detectors, heat sensors at strategic / critical locations normally drain points of vessels, columns, pump drain, tank drain points, etc should be provided in accordance with relevant Indian Standards, OISD.
- 5 Fixed foam pourer system for Class-A products storage tanks should be provided in accordance with relevant Indian Standards, OISD.
- 6 Water Sprinklers ring at top and middle of tanks for water cooling in all storage tanks should be provided in accordance with relevant Indian Standards, OISD.
- 7 HVLR - Remote/manual operated variable flow monitors having discharge of 500 gpm to 4 000 gpm capacity should be provided as per requirement. HVLR Monitor of 10 500 gpm should be considered for large dia tanks (for > 60 M and above). Fire robot may be considered for fighting multiple tank/area fires as per the relevant Indian Standards.
- 8 Medium expansion foam generators (MEFG) may be provided to fight fire due to accidental spillage of flammable materials in tank dykes/spill fires.
- 9 Automatic site configurable linear heat detection and foam suppression system should be provided for EFR tanks (rim seal areas) Class A products.
- 10 Linear heat detection system, site configurable type with water mist/spray system should be provided in all conveyor belts for early detection of rise in temperature.
- 11 Linear heat detection system, site configurable type should be provided in all cable cellars, trenches, for early detection.
- 12 All electrical panels should be provided with linear heat detection and inert gas suppression system (environment friendly halon free).
- 13 Chlorine leakage kit, chlorine protective suit, sensors, neutraliser, etc should be provided.
- 14 Water mist /deluge system and nitrogen injection fire prevention system (NIFPS) with site configurable linear heat sensing detector should be provided for oil filled transformers.
- 15 Wireless heat and smoke detection with alarm and notification to concerned should be provided.
- 16 Clean agent inert gas fire suppression system (environment friendly halon free) should be provided in all control rooms.

LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfilment of the requirements of this Part. The latest version of a standard should be adopted at the time of enforcement of this Part. The standards listed may be used by the Authority for conformance with the requirements of the referred clauses in this Part.

In the following list, the number appearing in the first column within parentheses indicates the number of the reference in this Part.

<i>IS No.</i>	<i>Title</i>
(1) IS 3808 : 1979	Method of test for non-combustibility of building materials (<i>first revision</i>)
(2) IS/ISO 834-1 : 1999	Fire resistance tests — Elements of building construction: Part 1 General requirements
IS/ISO 834-4 : 2000	Fire Resistance tests — Elements of building construction: Part 4 Specific requirements for load bearing vertical separating elements
IS/ISO 834-5 : 2000	Fire resistance tests — Elements of building construction: Part 5 Specific requirements for load bearing horizontal separating elements
IS/ISO 834-7 : 2000	Fire-resistance tests — Elements of building construction: Part 7 Specific requirements for columns
IS/ISO 834-8 : 2003	Fire-Resistance tests — Elements of building construction: Part 8 Specific requirements for non-load bearing vertical separating elements
IS/ISO 834-9 : 2003	Fire-resistance tests — Elements of building construction: Part 9 Specific requirements for non-load bearing ceiling elements
(3) IS 8757 : 2021	Glossary of terms associated with fire safety (<i>second revision</i>)
IS 7673 : 2004	Glossary of terms for fire fighting equipment (<i>first revision</i>)
(4) IS 8758 : 2013	Recommendations for fire precautionary measures in the construction of temporary structures and pandals (<i>second revision</i>)
(5) IS 9668 : 1990	Code of practice for provision and maintenance of water supplies and fire fighting
IS 3844 : 1989	Code of practice for installation and maintenance of internal fire hydrants and hose reels on premises (<i>first revision</i>)
(6) IS 1646 : 2015	Fire safety of buildings (General): Electrical installations — Code of practice(<i>third revision</i>)
(7) IS 9457 : 2025	Safety colours, safety signs and accident prevention tags — Code of practice (<i>second revision</i>)
IS 12349 : 1988	Fire protection — Safety sign
IS 12407 : 1988	Graphic symbols for fire protection plan
(8) IS/ISO 7240-7 : 2023	Fire detection and alarm systems Part 7: Point-type smoke detectors using scattered light, transmitted light, or Ionization (<i>first revision</i>)

<i>IS No.</i>	<i>Title</i>
IS/ISO 7240-12 : 2022	Fire detection and alarm systems: Part 12 Line type smoke detectors using a transmitted optical beam (<i>first revision</i>)
(9) IS 655 : 2006	Specification for air ducts (<i>third revision</i>)
(10) IS 1649 : 1962	Code of practice for design and construction of flues and chimneys for domestic heating appliances (<i>first revision</i>)
IS 1642 : 2013	Code of practice for safety of buildings (general): Details of construction (<i>second revision</i>)
(11) IS 12777 : 1989	Fire safety — Flame spread of products — Method for classification
(12) IS 1642 : 2013	Code of practice for safety of buildings (general): Details of construction (<i>second revision</i>)
(13) IS 16246 : 2015	Elastomer insulated cables with limited circuit integrity when affected by fire — Specification
(14) IS 6044 (Part 1) : 2018	Liquefied petroleum gas storage installations — Code of practice: Part 1 Residential commercial and industrial cylinder installations (<i>third revision</i>)
IS 6044 (Part 2) : 2021	Code of practice for liquefied petroleum gas storage installations: Part 2 Commercial, industrial and domestic bulk storage installations (<i>second revision</i>)
(15) IS 2175 : 1988	Specification for heat sensitive fire detectors for use in automatic fire alarm system (<i>second revision</i>)
IS/ISO 7240-5 : 2018	Fire detection and alarm systems: Part 5 Point-type heat detectors (<i>second revision</i>)
IS/ISO 7240-7 : 2023	Fire detection and alarm systems: Part 7 Point-type smoke detectors using scattered light, transmitted light, or Ionization (<i>first revision</i>)
IS/ISO 7240-11 : 2011	Fire detection and alarm systems: Part 11 Manual call points
IS/ISO 7240-15 : 2014	Fire detection and alarm systems: Part 15 Point-type fire detectors using scattered light, transmitted light, or ionization sensor in combination with a heat sensor (<i>first revision</i>)
IS/ISO 7240-12 : 2022	Fire detection and alarm systems: Part 12 Line type smoke detectors using a transmitted optical beam (<i>first revision</i>)
(16) IS 2189 : 2026	Selection, installation and maintenance of automatic fire detection and alarm system — Code of practice (<i>third revision</i>)
(17) IS 636 : 2018	Non-percolating flexible fire fighting delivery hose (<i>fourth revision</i>)
IS 884 : 2025	First-aid hose-reel for fire fighting — Specification (<i>second revision</i>)

<i>IS No.</i>	<i>Title</i>
IS 901 : 1988	Specification for couplings, double male, and double female instantaneous pattern for fire fighting (<i>third revision</i>)
IS 902 : 1992	Specification for suction hose couplings for fire fighting purposes (<i>third revision</i>)
IS 903 : 2025	Fire hose delivery couplings, branch pipe, nozzles and nozzle spanner — Specification (<i>fifth revision</i>)
IS 904 : 1983	Specification for two-way and three-way suction collecting heads for fire fighting purposes (<i>second revision</i>)
IS 905 : 1980	Specification for delivery breechings, dividing and collecting, instantaneous pattern for fire fighting purposes (<i>second revision</i>)
IS 906 : 2024	Specification for revolving branch pipe for fire fighting (<i>fourth revision</i>)
IS 907 : 1984	Specification for suction strainers, cylindrical type for fire fighting purpose (<i>second revision</i>)
IS 908 : 1975	Specification for fire hydrant, stand post type (<i>second revision</i>)
IS 909 : 1992	Specification for underground fire hydrant: Sluice valve type (<i>third revision</i>)
IS 910 : 1980	Specification for combined key for hydrant, hydrant cover and lower valve (<i>second revision</i>)
IS 926 : 1985	Specification for fireman's axe (<i>second revision</i>)
IS 927 : 1981	Specification for fire hooks (<i>second revision</i>)
IS 928 : 1984	Specification for fire bells (<i>second revision</i>)
IS 937 : 1981	Specification for washers for water fittings for fire fighting purposes (<i>second revision</i>)
IS 939 : 1977	Specification for snatch block for use with fibre rope for fire brigade use (<i>first revision</i>)
IS 941 : 1985	Specification for blowers and exhauster for fire fighting (<i>second revision</i>)
IS 942 : 1982	Functional requirements for 275-l/min portable pump set for fire fighting (<i>second revision</i>)
IS 943 : 1979	Functional requirement for 680-l/min trailer pump for fire brigade use (<i>second revision</i>)
IS 944 : 1979	Functional requirement for 1800-l/min trailer pump for fire brigade use (<i>second revision</i>)
IS 947 : 1985	Functional requirement for towing tender for trailer fire pump for fire brigade use (<i>first revision</i>)

<i>IS No.</i>	<i>Title</i>
IS 948 : 1983	Functional requirement for water tender, Type A, for fire brigade use (<i>second revision</i>)
IS 949 : 2012	Functional requirement for emergency (rescue) tender for fire brigade use (<i>second revision</i>)
IS 950 : 2012	Functional requirements for water tender, Type B for fire brigade use (<i>third revision</i>)
IS 952 : 1986	Specification for fog nozzle for fire brigade use (<i>second revision</i>)
IS 955 : 1980	Functional requirements for dry power tender for fire-brigade use (150 kg capacity) (<i>first revision</i>)
IS 957 : 1967	Specification for control van for fire brigade
IS 1941 (Part 1) : 1976	Functional requirements for electric motor sirens: Part 1 AC, 3-phase 50 Hz, 415 Volts type (<i>second revision</i>)
IS 2097 : 2024	Specification for foam making branch pipe (<i>third revision</i>)
IS 2175 : 1988	Specification for heat sensitive fire detectors for use in automatic detectors for use in automatic fire alarm system (<i>second revision</i>)
IS 2546 : 1974	Specification for galvanized mild steel fire bucket (<i>first revision</i>)
IS 2696 : 1974	Functional requirements for 1125 l/min light fire engine (<i>first revision</i>)
IS 2745 : 1983	Specification for non-metal helmet for firemen and civil defence personnel (<i>second revision</i>)
IS 2871 : 2012	Specification for branch pipe, universal for fire fighting purposes (<i>second revision</i>)
IS 2930 : 1980	Functional requirements for hose laying tender for fire brigade use (<i>first revision</i>)
IS 3582 : 1984	Specification for basket strainers for fire fighting purposes (cylindrical type) (<i>first revision</i>)
IS 4308 : 2019	Specification for dry powder for fire fighting (<i>third revision</i>)
IS 4571 : 1977	Specification for aluminium extension ladders for fire brigade use (<i>first revision</i>)
IS 4643 : 1984	Specification for suction wrenches for fire brigade use (<i>first revision</i>)
IS 4927 : 1992	Specification for unlined flax canvas hose for fire fighting (<i>first revision</i>)
IS 4928 : 1986	Specification for delivery valve for centrifugal pump outlets (<i>first revision</i>)
IS 4947 : 2006	Specification for gas cartridges for use in fire extinguishers (<i>third revision</i>)

<i>IS No.</i>	<i>Title</i>
IS 4989 : 2018	Foam concentrate for producing mechanical foam for fire fighting — Specification (<i>fourth revision</i>)
IS 4989 (Part 4) : 2003	Multipurpose aqueous film forming foam liquid concentrate for extinguishing hydrocarbon and solvent fires
IS 5131 : 2002	Specification for dividing breeching with control, for fire brigade use (<i>second revision</i>)
IS 5290 : 2025	Specification for landing valve (<i>fourth revision</i>)
IS 5486 : 1985	Specification for quick release knife (<i>first revision</i>)
IS 5505 : 1985	Specification for multi-edged rescue axe (non-wedging) (<i>first revision</i>)
IS 5612	Specification for hose-clamps and hose-bandages for fire brigade use
(Part 1) : 1977	Hose clamps (<i>first revision</i>)
(Part 2) : 1977	Hose bandages (<i>first revision</i>)
IS 5714 : 1981	Specification for hydrant, stand-pipe for fire fighting (<i>first revision</i>)
IS 6026 : 1985	Specification for hand operated sirens (<i>first revision</i>)
IS 6067 : 1983	Functional requirements for water tender, Type 'X' for fire brigade use (<i>first revision</i>)
IS/ISO 7240-5 : 2018	Fire detection and alarm systems: Part 5 Point-type heat detectors (<i>second revision</i>)
IS/ISO 7240-7 : 2023	Fire detection and alarm systems: Part 7 Point-type smoke detectors using scattered light, transmitted light, or ionization (<i>first revision</i>)
IS/ISO 7240-11 : 2011	Fire detection and alarm systems Part 11 Manual call points
IS 8090 : 1976	Specification for couplings, branch pipe, nozzle, used in hose reel tubing for fire fighting (<i>first revision</i>)
IS 8096 : 2005	Rubber or plastics coated fabrics for water resistant clothing — Specification (<i>second revision</i>)
IS 8423 : 1994	Specification for controlled percolating hose for fire fighting (<i>first revision</i>)
IS 8442 : 2024	Specification for stand post type water monitor for fire fighting (<i>second revision</i>)
IS 9972 : 2023	Specification for automatic sprinkler heads (<i>second revision</i>)
IS 10460 : 1983	Functional requirements for small foam tender for fire brigade use

<i>IS No.</i>	<i>Title</i>
IS 10993 : 1984	Functional requirements for 2 000 kg dry powder tender for fire brigade use
IS 11101 : 1984	Specification for extended branch pipe for fire brigade use
IS 11108 : 1984	Specification for portable fire extinguisher halon —1211 type
IS 11833 : 1986	Specification for dry powder fire extinguisher for metal fires
IS 12717 : 1989	Functional requirements of fire fighting equipment — High capacity portable pumpset (1 100-1 600 L/min)
IS 12796 : 1989	Specification for fire rake
IS 13039 : 2014	Code of practice for provision and maintenance of external hydrant system (<i>first revision</i>)
IS 14933 : 2001	Specification for high pressure fire fighting hose
IS 14951 : 2001	Specification for fire extinguisher, 135 l capacity mechanical foam type
IS 15051 : 2001	Specification for high pressure fire hose delivery couplings
IS 15105 : 2021	Design and installation of fixed automatic sprinkler fire extinguishing system (<i>first revision</i>)
IS 15220 : 2002/ ISO 7201-1	Specification for halon 1211 and halon 1301 — Fire extinguishing media for fire protection
IS 15683 : 2018	Portable fire extinguishers — Performance and construction — Specification (<i>first revision</i>)
IS 15493 : 2021	Gaseous fire extinguishing systems — General requirements (<i>first revision</i>)
IS 15505 : 2004/ ISO 14520-6	Gaseous fire extinguishing systems: HCFC Blend A extinguishing systems
IS 15506 : 2004/ ISO 14520-14	Gaseous fire extinguishing systems — IG 55 extinguishing systems
IS 15525 : 2004/ ISO 14520-13	Gaseous fire extinguishing systems — IG 100 extinguishing systems
IS 15501 : 2004/ ISO 14520-15	Gaseous fire extinguishing systems — IG 541 extinguishing systems
IS 15497 : 2004/ ISO 14520-12	Gaseous fire extinguishing systems — IG 01 extinguishing systems
IS 15519 : 2020	Code of practice for water mist fire protection systems — System design, installation, and commissioning (<i>first revision</i>)
IS 15528 : 2004	Gaseous fire extinguishing systems — Carbon dioxide, total flooding and

<i>IS No.</i>	<i>Title</i>
	local application including in cabinet subfloors systems
IS 15517 : 2004/ ISO 14520-9	Gaseous fire extinguishing systems — HFC 227ea (Hepta Fluoro Propane) extinguishing system
IS 16018 : 2012	Wheeled fire extinguisher — Performance and construction — Specification
(18) IS 2190 : 2024	Code of practice for selection, installation, and maintenance of portable first-aid fire (<i>fifth revision</i>)
(19) IS 884 : 2025	Specification for first aid hose reel for fire fighting (<i>second revision</i>)
(20) IS 15105 : 2021	Design and installation of fixed automatic sprinkler fire extinguishing system — Code of practice (<i>first revision</i>)
(21) IS 15325 : 2020	Design and installation of fixed automatic high and medium velocity water spray system — Code of practice (<i>first revision</i>)
(22) IS 12835 (Part 1) : 2021	Design and installation of fixed foam fire extinguishing system — Code of practice Part 1 Low expansion foam (<i>first revision</i>)
(23) IS 15517 : 2004	Gaseous fire extinguishing systems — Carbon dioxide, total flooding and local application including in cabinet subfloors systems
(24) IS 15519 : 2020	Water mist fire protection systems — System design, installation, and commissioning — Code of Practice (<i>first revision</i>)
(25) IS 15493 : 2021	Gaseous fire extinguishing systems — General requirements
IS 15505 : 2004	Gaseous fire extinguishing systems — HCFC blend A
IS 15506 : 2004	Gaseous fire extinguishing systems — IG 55 extinguishing systems
IS 15525 : 2004	Gaseous fire extinguishing systems — IG 100 extinguishing systems
IS 15501 : 2004	Gaseous fire extinguishing systems — IG 541 extinguishing systems
IS 15497 : 2004	Gaseous fire extinguishing systems — IG 01 extinguishing systems
IS 15517 : 2004	Gaseous fire extinguishing systems — Carbon dioxide, total flooding and local application including in cabinet subfloors systems
IS 9668 : 1990	Code of practice for provision and maintenance of water supplies and fire fighting
IS 3844 : 1989	Code of practice for installation and maintenance of internal fire hydrants and hose reels on premises (<i>first revision</i>)
(26) IS 2189 : 2026	Selection, installation and maintenance of automatic fire detection and alarm system — Code of practice (<i>second revision</i>)

<i>IS No.</i>	<i>Title</i>
IS 2190 : 2024	Selection, installation, and maintenance of portable first-aid fire extinguishers — Code of practice (<i>fifth revision</i>)
IS 13039 : 2014	External hydrant systems — Provision and maintenance — Code of practice (<i>first revision</i>)
(27) IS 4878 : 1986	Byelaws for construction of cinema buildings (<i>first revision</i>)
(28) IS 2726 : 1988	Code of practice for fire safety of industrial buildings: Cotton ginning and pressing (including cotton seed delinting) factories (<i>first revision</i>)
IS 3034 : 1993	Code of practice for fire safety of industrial buildings: Electrical generating and distributing stations (<i>second revision</i>)
IS 3058 : 1990	Code of practice for fire safety of industrial buildings: Viscose rayon yarn and/or staple fibre plants (<i>first revision</i>)
IS 3079 : 1990	Code of practice for fire safety of industrial buildings: Cotton textile mills (<i>first revision</i>)
IS 3594 : 2024	Code of practice for fire safety of industrial buildings: General storage and warehousing including cold storage (<i>second revision</i>)
IS 3595 : 2002	Code of practice for fire safety of industrial buildings: Coal pulverizers and associated equipment (<i>second revision</i>)
IS 3836 : 2000	Code of practice for fire safety of industrial buildings: Jute mills (<i>second revision</i>)
IS 4209 : 2013	Code of safety in chemical laboratories (<i>second revision</i>)
IS 4226 : 1988	Code of practice for fire safety of industrial buildings: Aluminium/ Magnesium powder factories (<i>first revision</i>)
IS 4886 : 1991	Code of practice for fire safety of industrial buildings: Tea factories (<i>first revision</i>)
IS 6329 : 2000	Code of practice for fire safety of industrial buildings: Saw mills and wood works (<i>first revision</i>)
IS 9109 : 2000	Code of practice for fire safety of industrial buildings: Paint and varnish factories
IS 11457 (Part 1) : 1985	Code of practice for fire safety of chemical industries: Part 1 Rubber and plastic
(29) IS 13039 : 2014	External hydrant systems — Provision and maintenance — Code of practice (<i>first revision</i>)
(30) IS 18271 : 2023	Fire safety in commercial kitchen — Guidelines
(31) IS 4963 : 2025	Accessibility in built environment for older adults and persons with disabilities — Requirements (<i>second revision</i>)

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(32) IS 16700 : 2023	Criteria for structural safety of tall concrete buildings (<i>first revision</i>)
(33) IS 3614 : 2021	Fire doors and doorsets — Specification (<i>first revision</i>)
(34) IS 12458 : 2019	Fire resistance of through penetration firestops — Method of test (<i>first revision</i>)
(35) IS 18190 : 2023	Fire resistance of perimeter fire barrier joint system — Method of tests

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- Access Point* — Part D/Section 6
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